Observation and Recovery

Geometry, Mixtures, and Spectral Inversion

A Unified Edition of the Ten Subsequent Papers

Following the 223-page treatise

KEIO

16 October 2026

This volume collects the ten subsequent research manuscripts in one continuous document. The earlier 223-page integrated treatise is not reproduced. The original arguments, hypotheses, proofs, research ledgers, and limitations are retained, with unified navigation and references. This is an editorial consolidation of AI-assisted research manuscripts, not an independent certification of their mathematical claims.

Abstract

This volume assembles ten successive investigations of observation and recovery in arithmetic, geometry, probability, and statistics. Its central objects are adaptive continuous measurements, repeated-observation mixture experiments, spectral and zero-set inverse maps, and factorization boundaries for probability distributions. The sequence develops recovery widths and restricted information classes; population estimation under replication, sampling budgets, and contamination; Gaussian and stable mixture identification; prime and exponential-sum zero phenomena; continuous spectral inversion; and Bessel–gamma positivity and extraction problems. Complete source arguments are retained in chronological parts, so that assumptions, intermediate advances, counterexamples, and remaining obstructions can be read together. A common table of contents, globally distinct theorem and equation numbers, cross-links, and a consolidated bibliography make the sequence accessible as one document.

Scope of this edition. The source interval begins with *Exact Recovery, Optimal Replication, and Bessel Classification* and ends with *Observation Budgets and Hidden Parameters*. It contains all ten later manuscripts available in this research sequence as of 16 October 2026. The 223-page *Observation and Recovery in Arithmetic and Probability: Prime Flows, Rare Events, and Infinite Mixtures* remains an external predecessor. References to that work are retained, but none of its chapters is imported.

How to read the chronology. The parts preserve the source order and the original mathematical text. Each begins with its source abstract and keeps its stated hypotheses and claim accounting. Statements about what is new, previously known, or still open are dated statements within that part; later parts may extend the range, weaken an assumption, or settle a question left there. Counts of targets are local research ledgers, not additive counts of independent discoveries or historical conjectures. Repeated statements are retained when they support a later proof, comparison, or extension.

Numbering and references. Number $i.j.k$ denotes item k in section j of Part i. Machine-linked references have been renumbered throughout. Unprefixed section or theorem numbers inside bibliographic annotations refer to the original cited publication or manuscript. Companion-paper bibliography entries link to the corresponding part when the source is included here. The source abstracts are historical abstracts, not a fresh claim of priority by this editorial edition. Symbols are defined within their respective parts; a symbol reused in another model has its local meaning.

Proof and verification status. This compilation preserves the status of the source manuscripts. It adds no new proof of a named conjecture and does not turn a conditional result into an unconditional one. Its production checks concern source coverage, reference integrity, compilation, and presentation. Mathematical assessment should use the full hypotheses and proofs, rather than the source titles or target counts alone.

A specific overlap in the budget results. The general regular-component budget theorem in Part I, Theorem 1.7.1, already contains the $B^{-1/3}$ order for Bernoulli components. Part X, Theorem 10.3.1, revisits this rate with a direct Bernoulli proof and an explicit finite linear-program implementation. These are not two independent minimax-rate discoveries. The source text is retained so the two arguments can be compared.

Progression and model distinctions. The restricted likelihood ranges in Parts III–IV precede the complete curve stated in Part VI. The prime-zeta endpoint discussion in Part I precedes the counting results in Parts VI–VII. Part VII treats the constant-one coefficient conjecture separately from the still-distinct optimal-exponent question. Part V already gives logarithmic constructions for convex polynomial and analytic measurements; Part X adds fixed two-sided Hessian bounds in the smooth class. Its remaining analytic questions should not be read as withdrawing Part V's upper bound. The positive finite-query noise gap of Part IX is quantified in Part X; matching decay rates remain a separate question.

The zero-set counterexamples with additive frequency relations coexist with the finite, internally rationally independent inverse theorem of Part X. Similarly, separate terminal laws under a two-sided scale-moment condition, separate terminal laws without moments, and joint replicates are different experiments. Their identification thresholds should not be combined without preserving those assumptions. Five or six sufficient replicates are not claimed to be minimal for every fully mixed model.

Meaning of a single-file source. The accompanying TEX file contains the complete typesetting source and requires no external chapter files, images, or bibliography database. Mathematical dependencies on the excluded 223-page predecessor are still dependencies; compilability does not assert that every foundational result has been reproved here.

The ten constituent papers

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I	Exact Recovery, Optimal Replication, and Bessel Classification	14 Oct. 2026	1
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III	Binomial Mixture Likelihood and Exact Observation Thresholds	14 Oct. 2026	91
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V	Adaptive Analytic Information and Sharp Recovery Boundaries	14 Oct. 2026	170
VI	Boundary Laws for Observation and Recovery	15 Oct. 2026	219
VII	Exact Recovery Laws and a Coefficient Conjecture	16 Oct. 2026	263
VIII	Geometry and Rigidity of Inverse Observation	16 Oct. 2026	309
IX	Sharp Boundaries for Adaptive Recovery and Mixture Identification	16 Oct. 2026	345
X	Observation Budgets and Hidden Parameters	16 Oct. 2026	383

Suggested thematic routes. For measurement complexity, follow the information sections in Parts IV–X. For likelihood, corruption, and budget allocation, follow the statistical sections in Parts I–VII and IX–X. For joint and separate mixture observations, follow the Gaussian and stable sections throughout the volume. For zero dynamics and inverse spectra, follow the arithmetic and spectral sections; for distribution factorization, follow the Bessel, Thorin, and gamma sections. The detailed contents below provides the precise section locations.

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Part I

Exact Recovery, Optimal Replication, and Bessel Classification

Extensions of Observation and Recovery, with an Audit of Fifteen Published Questions

Source manuscript: 14 October 2026.

Source abstract. We develop several extensions of an integrated study of arithmetic and probabilistic observation maps. For repeated observations with an arbitrary mixing distribution on an interval, regular component estimation and a local quadratic relative-entropy bound imply the sharp total-budget Wasserstein rate $B^{-1/3}$, attained by a fixed unequal allocation and unimprovable by adaptive policies. In a factorial Legendre experiment this strictly improves every equal-group allocation. For each fixed stable shape, we obtain necessary and sufficient conditions on the observed partial-sum indices; the converse constructs one signed mixing density annihilating all selected observations simultaneously. We classify exact real Mellin zeros and integer moment-sign patterns, identify the precise quasianalytic boundary of sparse-moment recovery, and use quadrature zeros to separate individual identification from uniform statistical learning. For the published McKay classification problem, complex Bessel zeros exclude hyperbolic complete monotonicity for four families, while the second family admits an exact self-decomposability threshold and is never a generalized gamma convolution. Further results concern correlated blind calibration and uniformly translated prime-flow zero counts. An explicit ledger treats fifteen questions from published papers or public preprints. It separates inherited proofs, classical consequences, partial answers, and unresolved targets. In particular, no Riemann-zeta conjecture is claimed solved, and no numerical count of new independent conjecture resolutions is asserted.

1.1 Scope, principal results, and the problem ledger

This paper is a sequel to *Observation and Recovery in Arithmetic and Probability: Prime Flows, Rare Events, and Infinite Mixtures* [1], a 223-page integrated working manuscript. The question motivating this sequel is concrete: which externally stated problems can its methods actually settle, and which internal sufficient conditions can be sharpened to exact boundaries? We give complete proofs of the principal extensions rather than infer additional conclusions from the length or ambition of the earlier manuscript.

Here "new extension" means new relative to that manuscript. It does not establish historical priority. The literature audit uses specific primary-source formulations and the follow-ups located as of 14 October 2026; it is not an exhaustive certification that no other resolution exists. The mathematical proofs and the priority assessment are separate questions. The manuscript has received internal independent proof checks, but not external peer review or formal verification.

1.1.1 The most substantial extensions

1. **A complete allocation rate.** Theorem 1.7.1 proves $\mathfrak{R}_B^{\mathrm{ad}} \asymp \mathfrak{R}_B^{\mathrm{fix}} \asymp B^{-1/3}$ under explicit component-estimation and local-KL assumptions. The factorial example gives a strict improvement over the lower bound $(B\log(eB))^{-1/4}$ for every equal-group design (Corollary 1.7.3). The multilevel idea itself has established predecessors.

2. **Exact sparse stable sampling.** For fixed stable index $\alpha \neq 1$, arbitrary mixing laws are recovered from the separate partial-sum laws at indices $\mathcal{N}$ exactly when $\sum_{n\in\mathcal{N}} n^{-|1-\alpha|} =$

∞. At index one with nonzero skewness, the exact condition is $\sum_{n\in\mathcal{N}}(1+(\log n)^2)^{-1}=\infty$. Theorem 1.5.1 includes genuine probability-law counterexamples whenever the appropriate sum converges.

3. **A concrete published classification.** Corollary 1.2.3 answers the HCM portion of the McKay question negatively throughout all four stated families. For $\psi_{\mu,a,b}$, Theorem 1.2.4 proves

$$\text{SD} \quad\Longleftrightarrow\quad (2\mu+3)\sqrt{b^2-a^2}\geq b, \qquad \text{GGC never holds when } a>0.$$

This is a complete answer to those subquestions, not to every clause of the source's multipart problem.

4. **Exact recovery boundaries beyond a single example.** Sections 1.8, 1.9, and 1.6 give exact moment-sign conditions, a Denjoy–Carleman criterion, and density constraints that restore identification. The last result also proves that individual identification alone does not imply uniform learnability.

1.1.2 Fifteen externally stated targets

The entries below are *targets*, not fifteen claims of new solutions. Three entries are distinct numbered prime-zeta conjectures; the multipart McKay question is kept as one entry. The stable, regularity, calibration, and allocation extensions are not counted as extra published conjectures. The two inherited resolutions are proved again in Section 1.4.

No.	Published target	Outcome and exact boundary
1	Lin: arbitrary normal directing laws [7, §2.6]	Positive answer without variance moments. Inherited proof rechecked; Proposition 1.4.1. No new priority claim.
2	Le et al.: converse to the differential barrier [8, §6]	Negative for the two stated finite operator classes. Inherited entire kernel counterexample rechecked; Proposition 1.4.2.
3	Le et al.: constraints restoring estimation [8, §6]	Partial answer for finite polynomial defects: prescribed quadrature zeros give identification; a common smoothness bound gives uniform learning. The general research program remains open.
4	Bürgin–Epperlein–Wirth, Conjecture 17 [17]	Correct, but already a consequence of Luxemburg–Korevaar (1971). Alternative proof and stronger formulations; not a new historical closure.
5	Belovas et al., prime-zeta Conjecture 1 [21]	Sharp right boundary already follows from Sepulcre–Vidal (2022). Not a new closure.
6	Belovas et al., prime-zeta Conjecture 2 [21]	Uniform normalized ordinates follow from classical almost-periodic zero density in the specified strip. A uniform twist/translate strengthening is proved here.
7	Belovas et al., prime-zeta Conjecture 3 [21]	Positive linear counting follows in $1<\gamma<S$. Classical consequence; the endpoint $\gamma=1$ is not settled.
8	Ford–Zaharescu / FSZ: secondary interval term [25, 26]	Unresolved here. An $o(N(T))$ error does not give the required $o(T)$ error for sharp interval tests.
9	Çiçek–Gonek: average-gap normalization [27]	Unresolved here. The Fourier test requires a growing frequency $x=T^k$, outside the fixed-frequency estimates.
10	Odlyzko / Arias de Reyna: Gram-phase distribution [28]	Unresolved here. Nonlinear phase and growing local frequency leave an unproved exponential-sum estimate.

No.	Published target	Outcome and exact boundary
11	Baricz et al., §4.2: noncentral chi-square HCM [3]	HCM classification already answered by Salazar (2026). An elementary GGC obstruction is also proved here, without a priority claim.
12	Baricz et al., §4.3: McKay classification [3]	HCM subquestion closed negatively for all four families; exact SD/GGC classification of ψ. Other SD/GGC cases and the extra Theorem 12 question remain unresolved.
13	Baricz et al., §4.4: SD of three further laws [3]	Unresolved here. One case reduces to the explicit Bessel heat-trace inequality (1.2.13).
14	Lau–Scarlett: interaction in one-bit mean estimation [15]	Unresolved here. Repeated observations of one latent individual are unavailable in that observation model.
15	Vinayak et al.: larger-range binomial MLE bound [13, Remark 3.2]	Unresolved here. The unequal-allocation theorem is not an equal-group MLE theorem; verified 2025 progress is cited explicitly.

There are six targets unresolved here (8–10 and 13–15), two partly answered targets (3 and 12), two inherited proof resolutions (1 and 2), and five targets treated as classical consequences or previously answered questions (4–7 and 11). This bookkeeping is intentionally conservative: the exact new McKay subcase results and internal sharp classifications have mathematical value without being relabeled as fifteen independent open-problem resolutions.

1.1.3 Notation and observation models

The Wasserstein distance W_1 is taken with the usual Euclidean cost; on a real interval it equals the L^1 distance between distribution functions. Total variation uses the probability convention $d_{\mathrm{TV}}(P,Q) = \sup_A |P(A) - Q(A)|$, while the variation norm of a signed measure is its total mass after absolute variation. Constants in $\lesssim$ and $\asymp$ may depend on a fixed model or interval, but not on the sample size or budget. “Separate partial-sum laws” do not mean jointly observed sums. “Two replicates” means their joint distribution under a shared latent parameter. The distinction is essential.

1.2 Bessel laws: published classification questions and exact answers

The three problems in Sections 4.2–4.4 of Baricz, Prabhu K, Singh, and Vijesh V [3] provide external, explicitly stated targets. Their published theorem numbers are 1–4, 12, 14, 18, and 22; the corresponding arXiv numbers are 2.1–2.4, 3.12, 3.14, 3.18, and 3.22. We use the published numbering. The first problem concerns hyperbolically completely monotone densities for noncentral chi-square laws, with generalized gamma convolutions discussed immediately before the question. The second is a multipart classification of McKay laws. The third asks for self-decomposability of three further Bessel laws. They are three source problems, not a separate conjecture for every parameter or every distribution appearing in them.

The first problem has a recent HCM solution in Salazar [5]; we make no priority claim for that answer. Below we prove the negative HCM answer for all four McKay families, and an exact self-decomposability and GGC classification for the second family. These conclusions answer explicit parts of the second source problem. They do not settle its remaining SD/GGC cases or its additional question about the density in Theorem 12. We also give the elementary GGC obstruction for noncentral chi-square laws and a precise reduction of one case of the third problem. The latter remains unresolved here. The source and follow-up checks were made on

14 October 2026; the classification proofs are independent of any claim of historical priority.

1.2.1 A zero-free obstruction for HCM functions

A positive function f is *hyperbolically completely monotone* (HCM) when $f(uv)f(u/v)$ is completely monotone as a function of $v+v^{-1}$ for every $u>0$. A *generalized gamma convolution* (GGC) is a law whose Laplace exponent has a Thorin representation

$$h(s)=ds+\int_{(0,\infty)}\log(1+s/t)\,U(dt),\qquad d\ge 0, \tag{1.2.1}$$

with a nonnegative measure U and the integrability needed for a finite exponent. In particular,

$$h'(s)=d+\int_{(0,\infty)}\frac{U(dt)}{s+t} \tag{1.2.2}$$

is a Stieltjes function and has nonpositive imaginary part in the upper half-plane. The classical implication from an HCM density to a GGC law is stronger than infinite divisibility and will not be reversed.

Lemma 1.2.1 (Zero-free continuation of HCM functions)**.** *A strictly positive HCM function has a holomorphic, nowhere vanishing extension to $D=\mathbb{C}\setminus(-\infty,0]$. Consequently, if a positive real function already has a holomorphic continuation to D with a zero in D, it is not HCM.*

Proof. The classical representation recalled in [4, eq. (7)] is

$$f(x)=Cx^{\beta-1}h_1(x)h_2(1/x),\qquad -(\log h_i)' \text{ is Stieltjes},$$

where $C>0$ and $\beta\in\mathbb{R}$ in the nonzero case. Each Stieltjes function S_i extends holomorphically to D: the usual integral representation converges uniformly on its compact subsets. Since D is simply connected, it has an analytic primitive. Thus

$$\widetilde{h}_i(z)=h_i(1)\exp\left\{-\int_1^z S_i(w)\,dw\right\}$$

is holomorphic and nowhere zero on D, and agrees with h_i on the positive axis. Both $z\mapsto 1/z$ and the principal power $z^{\beta-1}$ are holomorphic on D, and the latter never vanishes. Their product with these two functions gives the claimed extension. Uniqueness of holomorphic continuation proves the last assertion. □

For $\mu>-1$ define the normalized entire Bessel factor

$$F_{\mu,a}(z)=\sum_{k=0}^{\infty}\frac{(a^2z^2/4)^k}{k!(\mu+1)_k}=\Gamma(\mu+1)(2/(az))^{\mu}I_{\mu}(az). \tag{1.2.3}$$

The series is the definition at the origin and removes any ambiguity from fractional powers. If $j_{\mu,n}$ is a positive zero of J_μ, then $F_{\mu,a}(ij_{\mu,n}/a)=0$. Existence of these positive zeros and the Bessel connection formulas are classical [6, §§10.21,10.27,10.42].

Theorem 1.2.2 (Finite products of modified Bessel factors)**.** *Let $r\ge 1$, $a_j>0$, $\mu_j>-1$, $m_j\in\mathbb{N}$, and $\beta\in\mathbb{R}$. Suppose*

$$\beta+\sum_{j=1}^{r}m_j\mu_j>-1,\qquad b>\sum_{j=1}^{r}m_ja_j.$$

The normalized probability density proportional to

$$x^{\beta}e^{-bx}\prod_{j=1}^{r}I_{\mu_j}(a_jx)^{m_j},\qquad x>0, \tag{1.2.4}$$

is not HCM.

Proof. The local Bessel series and its exponential asymptotic at infinity show integrability under the displayed assumptions. Every factor is positive for $x>0$, so the normalizing constant is positive and finite. On D the density extends as

$$C'z^{\beta+\sum_j m_j\mu_j}e^{-bz}\prod_{j=1}^{r}F_{\mu_j,a_j}(z)^{m_j},$$

using the principal power. All factors are holomorphic there. At $z=ij_{\mu_1,1}/a_1$ the first entire factor vanishes; every other factor is finite, and the power and exponential are nonzero. Hence the extension has a zero in D. Lemma 1.2.1 applies. □

Corollary 1.2.3 (The HCM part of the McKay question)**.** *Each of the following densities, with its positive normalizing constant suppressed, fails to be HCM throughout the indicated range:*

$$\begin{array}{lllll}
\varphi_{\mu,a,b}(x) & \propto x^{\mu}e^{-bx}I_{\mu}(ax), & \mu>-\tfrac12, & b>a>0,\\
\psi_{\mu,a,b}(x) & \propto x^{\mu+1}e^{-bx}I_{\mu}(ax), & \mu>-1, & b>a>0,\\
\varphi_{\mu,\nu,a,b}(x) & \propto x^{\nu-1}e^{-bx}I_{\mu}(ax), & \mu>-1, & \mu+\nu>0, \quad b>a>0,\\
\xi_{\mu,a,b}(x) & \propto x^{2\mu}e^{-bx}I_{\mu}(ax)^2, & \mu>-\tfrac14, & b>2a>0.
\end{array}$$

In particular, this answers the HCM question for all parameters covered by Theorems 1–4 of [3]. The last conclusion holds beyond the infinite-divisibility range proved in that source.

Proof. Apply Theorem 1.2.2 with (β,m) equal to $(\mu,1)$, $(\mu+1,1)$, $(\nu-1,1)$, and $(2\mu,2)$, respectively. □

The obstruction concerns the density, not its Laplace transform. The first McKay law is nevertheless a GGC, as already proved in [3, Theorem 1]. Thus its non-HCM classification is consistent with its positive GGC classification. Salazar's recent result treats a gamma density multiplied by ${}_0F_1$ with argument linear in x; the factors here have argument proportional to x^2. The above proof does not use a purported identification of these two families.

1.2.2 An exact SD/GGC classification for the second McKay family

Theorem 1.2.4 (Sharp classification of $\psi_{\mu,a,b}$)**.** *Let $\mu>-1$ and $b>a>0$, and let X have density*

$$\psi_{\mu,a,b}(x)=\frac{\sqrt{\pi}(b^2-a^2)^{\mu+3/2}}{2b(2a)^{\mu}\Gamma(\mu+3/2)}x^{\mu+1}e^{-bx}I_{\mu}(ax),\qquad x>0. \tag{1.2.5}$$

This law is infinitely divisible, is never a GGC, and is self-decomposable if and only if

$$(2\mu+3)\sqrt{b^2-a^2}\geq b. \tag{1.2.6}$$

Its density is never HCM. The infinite-divisibility assertion is already present in [3, Theorem 2]; the point here is the exact additional classification.

Proof. Put $\alpha = \mu + 3/2 > 1/2$. Termwise integration of the nonnegative series for I_μ and the gamma duplication formula give, for $c > a$,

$$\int_0^\infty x^{\mu+1} e^{-cx} I_\mu(ax)\,dx = \frac{2(2a)^\mu \Gamma(\alpha)}{\sqrt{\pi}} c(c^2 - a^2)^{-\alpha}.$$

Consequently the Laplace transform and exponent satisfy

$$L(s) = (1 + s/b)\left(\frac{b^2 - a^2}{(s+b)^2 - a^2}\right)^\alpha, \tag{1.2.7}$$

$$h'(s) = \frac{\alpha}{s+b-a} + \frac{\alpha}{s+b+a} - \frac{1}{s+b} = \int_0^\infty e^{-st} k(t)\,dt, \tag{1.2.8}$$

$$k(t) = e^{-bt}(2\alpha\cosh(at) - 1). \tag{1.2.9}$$

Here $k > 0$ and $k(t)/t$ is a Lévy density. Integration in s gives $h(s) = \int_0^\infty (1 - e^{-st}) k(t)\,dt/t$, proving infinite divisibility directly.

If the law were GGC, its logarithmic derivative would have the Stieltjes property (1.2.2). At $z = -b + i\eta$,

$$\operatorname{Im} h'(z) = \frac{1}{\eta} - \frac{2\alpha\eta}{a^2 + \eta^2} > 0 \quad \text{when} \quad 0 < \eta^2 < \frac{a^2}{2\alpha - 1}.$$

Analytic continuation of (1.2.8) is unique in the upper half-plane, so this contradicts the Stieltjes property.

For this smooth, exponentially decreasing k, self-decomposability is equivalent to k being nonincreasing. We recall the argument to make the necessity as well as sufficiency explicit. If k is nonincreasing and $0 < c < 1$, then

$$h(s) - h(cs) = \int_0^\infty (1 - e^{-st}) \frac{k(t) - k(t/c)}{t}\,dt$$

is a Bernstein function vanishing at zero. Hence $L(s)/L(cs)$ is a Laplace transform, which is exactly self-decomposability. Conversely, if the law is self-decomposable, each $R_c(s) = L(s)/L(cs)$ is a Laplace transform. Thus $(1 - R_c)/(1 - c)$ is Bernstein, and its limit as $c \uparrow 1$ is $sh'(s)$. Closure of Bernstein functions under finite pointwise limits shows that $sh'(s)$ is Bernstein. Integration by parts gives

$$sh'(s) = \int_0^\infty (1 - e^{-st})(-k'(t))\,dt.$$

The right side is a finite signed Lévy representation, and its limit at infinity is $k(0)$. The positive Bernstein representation therefore has zero drift and a finite measure. Uniqueness of Laplace transforms of finite measures forces that measure to be $-k'(t)\,dt$, so $k' \le 0$.

Finally,

$$k'(t) = e^{-bt}\{b - 2\alpha[b\cosh(at) - a\sinh(at)]\}.$$

The minimum of the bracketed hyperbolic expression on $t \ge 0$ is $\sqrt{b^2 - a^2}$, attained at $t = a^{-1}\operatorname{arctanh}(a/b)$. Therefore $k' \le 0$ is equivalent to $2\alpha\sqrt{b^2 - a^2} \ge b$, which is (1.2.6). The non-HCM assertion follows from Corollary 1.2.3. □

The exclusion of GGC holds even in the nonempty SD region. For example, $\mu = 0$, $b = 2$, and $a = 1$ satisfy (1.2.6). This explicitly prevents the invalid inference that failure of a Stieltjes test proves failure of self-decomposability. In the limiting case $a \downarrow 0$, the law converges to a gamma law of shape $2\mu + 2$ and rate b; the strict assumption $a > 0$ in the negative GGC and HCM assertions is essential.

1.2.3 Noncentral chi-square laws and the unresolved third problem

Proposition 1.2.5 (The noncentral chi-square GGC obstruction)**.** *A noncentral chi-square law with degrees of freedom $\kappa > 0$ and noncentrality $\lambda \geq 0$ is GGC if and only if $\lambda = 0$. Consequently its density is HCM if and only if $\lambda = 0$. The latter classification is already given by [5]; the following is a short independent GGC argument, without a priority claim.*

Proof. Its Laplace transform is $L(s) = (1+2s)^{-\kappa/2} \exp\{-\lambda s/(1+2s)\}$, so

$$h'(s) = \frac{\kappa}{1+2s} + \frac{\lambda}{(1+2s)^2}.$$

At $z = -1/2 - \varepsilon + i\varepsilon$ one has

$$\operatorname{Im} h'(z) = -\frac{\kappa}{4\varepsilon} + \frac{\lambda}{8\varepsilon^2} > 0 \quad \text{if } 0 < \varepsilon < \frac{\lambda}{2\kappa}.$$

For $\lambda > 0$ this violates the Stieltjes condition necessary for GGC. An HCM density would imply GGC and is therefore impossible. For $\lambda = 0$ the law is gamma, whose density is HCM and whose Laplace exponent has a one-atom Thorin representation. □

The third problem in [3, §4.4] concerns self-decomposability of the laws in its Theorems 14, 18, and 22. One can make a concrete reduction for Theorem 18. With $\mu, \nu > 1/2$ and $a, b > 0$, its Laplace transform is

$$L(s) = e^{-(a+b)\sqrt{s}}\, {}_0F_1(;\mu+1;a^2 s/4)\, {}_0F_1(;\nu+1;b^2 s/4). \tag{1.2.10}$$

The Bessel product formula gives

$$h'(s) = \frac{a+b}{2\sqrt{s}} - \sum_{n\geq 1} \frac{1}{s + j_{\mu,n}^2/a^2} - \sum_{n\geq 1} \frac{1}{s + j_{\nu,n}^2/b^2}. \tag{1.2.11}$$

Hence the Lévy spectral function is

$$k(t) = \frac{a+b}{2\sqrt{\pi t}} - \sum_{n\geq 1} e^{-j_{\mu,n}^2 t/a^2} - \sum_{n\geq 1} e^{-j_{\nu,n}^2 t/b^2}. \tag{1.2.12}$$

The series and all their derivatives converge locally uniformly for $t > 0$, by $j_{\eta,n} \sim \pi n$. Their Laplace transforms may be integrated termwise for $s > 0$, since $\sum_n (s + j_{\eta,n}^2/c^2)^{-1} < \infty$. Thus (1.2.12) is the inverse Laplace transform of (1.2.11). The already proved infinite divisibility in the source ensures that it is nonnegative, by uniqueness of the Bernstein measure. The standard Lévy criterion then says that self-decomposability is equivalent to

$$\sum_{n\geq 1} \frac{j_{\mu,n}^2}{a^2} e^{-j_{\mu,n}^2 t/a^2} + \sum_{n\geq 1} \frac{j_{\nu,n}^2}{b^2} e^{-j_{\nu,n}^2 t/b^2} \leq \frac{a+b}{4\sqrt{\pi}\, t^{3/2}} \qquad (t > 0). \tag{1.2.13}$$

This identifies the missing heat-trace inequality exactly. Neither the negative residues in (1.2.11) nor positivity of k proves it. No proof of (1.2.13), and no classification for all three families in the third source problem, is asserted here.

1.3 Blind calibration with arbitrary local correlations

The scalar calibration argument in the integrated rare-event theory uses signs of mixed second derivatives that can fail for correlated local vectors. There is another exact route: use first

derivatives at several points instead of insisting on a second-order calculation at one point. This is an extension of that manuscript, not a claim to resolve an externally documented longstanding conjecture.

Let $Y = (Y_1, \dots, Y_m)$ be a bounded nonnegative random vector such that

$$p_0 = \mathbb{P}(Y = 0) > 0, \qquad \mathbb{P}(Y_j > 0) > 0 \quad (1 \le j \le m). \tag{1.3.1}$$

Its coordinates may have arbitrary dependence. For $p_j > 1$ put

$$h(v) = -\log \mathbb{E}e^{-v\cdot Y}, \qquad H(s) = \int_0^\infty h(s_1 x^{-p_1}, \dots, s_m x^{-p_m})\, dx. \tag{1.3.2}$$

For an original counting exponent $\rho > 0$ and channel powers F_j, the substitution $x = y^\rho$ gives precisely $p_j = F_j/\rho$. Thus these ratios, rather than an unnormalized ρ, are the identifiable parameters in the next theorem.

Lemma 1.3.1 (Analyticity and visible axes)**.** *Under (1.3.1), H is finite and continuous on $[0,\infty)^m$, real analytic on $(0,\infty)^m$, and satisfies*

$$H(c^{p_1}s_1, \dots, c^{p_m}s_m) = cH(s), \qquad c > 0, \tag{1.3.3}$$

$$H(te_j) = C_j t^{1/p_j}, \qquad t \ge 0, \quad 0 < C_j < \infty. \tag{1.3.4}$$

Proof. The estimates $0 \le h(v) \le -\log p_0$ and $h(v) \le \sum_j v_j \mathbb{E}Y_j$ control $x \le 1$ and $x \ge 1$, respectively. They give convergence and, uniformly for bounded s, an integrable majorant; continuity follows by dominated convergence. The substitution $x = cy$ proves (1.3.3), and the substitution $x = t^{1/p_j} y$ on the jth axis proves (1.3.4). Nontriviality of Y_j gives $C_j > 0$.

For analyticity fix a positive s^0, put $S = \sum_j s_j^0 x^{-p_j} Y_j$, and allow complex perturbations satisfying $|s_j - s_j^0| \le \delta s_j^0$. If $\Delta = \sum_j (s_j - s_j^0) x^{-p_j} Y_j$, then $|\Delta| \le \delta S$. Writing $L_0 = \mathbb{E}e^{-S} \ge p_0$, we have

$$|\mathbb{E}e^{-(S+\Delta)} - L_0| \le \delta \mathbb{E}[Se^{-(1-\delta)S}] \le \frac{\delta}{e(1-\delta)}.$$

Choose δ so that the last bound is below $p_0/2$. A single local analytic logarithm is then defined by expansion about L_0, uniformly in x. For $x \le 1$ its absolute value is bounded by a constant depending on p_0 and δ. For large x, boundedness of Y gives the sharper bound $O(\sum_j x^{-p_j})$, uniformly in this complex polydisc. Holomorphic parameter integration proves analyticity of H. □

Theorem 1.3.2 (Finite-point blind calibration without covariance signs)**.** *For every nonempty open set $U \subset (0,\infty)^m$, the functions*

$$\phi_j(s) = s_j \partial_j H(s), \qquad 1 \le j \le m,$$

are linearly independent on U. Consequently, for an open dense set of m-tuples $(s^{(1)}, \dots, s^{(m)}) \in U^m$, whose complement has Lebesgue measure zero, the matrix and vector

$$A_{\ell j} = s_j^{(\ell)} \partial_j H(s^{(\ell)}), \qquad b_\ell = H(s^{(\ell)}) \tag{1.3.5}$$

satisfy

$$\det A \ne 0, \qquad (p_1, \dots, p_m)^T = A^{-1} b. \tag{1.3.6}$$

In particular, the unknown power ratios are determined by the exact first derivatives and values of H at m generic points of any open observation patch.

Proof. Differentiating (1.3.3) at $c = 1$ gives

$$\sum_j p_j \phi_j(s) = H(s). \tag{1.3.7}$$

Suppose $\sum_j v_j \phi_j = 0$ on U for constants v_j. Analyticity extends this identity to the connected positive orthant. For every $s > 0$ the derivative of $t \mapsto H(e^{tv_1} s_1, \dots, e^{tv_m} s_m)$ is zero for all $t \in \mathbb{R}$. Hence this function is constant. Passing to the jth axis using continuity and (1.3.4) yields

$$C_j (e^{tv_j} s_j)^{1/p_j} = C_j s_j^{1/p_j} \qquad (t \in \mathbb{R}).$$

Since $C_j > 0$, one has $v_j = 0$ for every j.

The vectors $(\phi_1(s), \dots, \phi_m(s))$, $s \in U$, therefore span $\mathbb{R}^m$. One may choose m of them as a basis, proving that the determinant in (1.3.5) is not identically zero as an analytic function on $((0,\infty)^m)^m$. Its zero set has empty interior and Lebesgue measure zero, which proves the generic assertion on U^m. Applying (1.3.7) at these points gives $Ap = b$ and the formula follows. □

No pairwise distinction among the exponents is needed in this theorem. Its visibility condition is essential: if $Y_j = 0$ almost surely, the response is independent of the jth channel and cannot determine its power. The result provides exact identifiability, not a lower bound for $|\det A|$ and not stability under measurement noise.

Corollary 1.3.3 (Calibration from a smooth active rate patch)**.** *Suppose $-\nabla^2 H$ is positive definite on an open positive patch, and let $e = \nabla H(s)$ there. For the lower-tail dual $I(e) = \sup_{u \ge 0}\{H(u) - u \cdot e\}$, an open active patch determines all p_j. Generically m first jets of I suffice: use*

$$A_{\ell j} = -e_j^{(\ell)} \partial_j I(e^{(\ell)}), \qquad b_\ell = I(e^{(\ell)}) - e^{(\ell)} \cdot \nabla I(e^{(\ell)}). \tag{1.3.8}$$

A sufficient condition for the required strict Hessian inequality on the whole positive orthant is that the powers p_j be pairwise distinct, in addition to (1.3.1).

Proof. The inverse function theorem and strict concavity give $s = -\nabla I(e)$, $\nabla H(s) = e$, and $H(s) = I(e) - e \cdot \nabla I(e)$. Substitution in (1.3.5) proves the assertion.

For the sufficient condition, differentiation under the integral gives

$$-v^T \nabla^2 H(s) v = \int_0^\infty \mathrm{Var}_{s,x}\left(\sum_j v_j x^{-p_j} Y_j \right) dx,$$

where the tilt has density proportional to $\exp(-\sum_j s_j x^{-p_j} Y_j)$. The derivatives are integrable by the analytic bounds in Lemma 1.3.1. If the integral is zero, the variance vanishes for almost every x. The tilted law is equivalent to the original law and retains a positive atom at $Y = 0$, so the almost sure constant is zero. Fubini's theorem then shows, for almost every Y, that $\sum_j v_j Y_j x^{-p_j} = 0$ for almost every $x > 0$. This continuous function vanishes everywhere, and linear independence of distinct powers gives $v_j Y_j = 0$ almost surely for every j. The coordinate visibility assumptions imply $v = 0$. □

Corollary 1.3.4 (Some finite jet at one point determines the powers)**.** *At every fixed $s^0 > 0$ there is a finite integer $r = r(H, s^0)$ such that the derivatives of H at s^0 through order r determine all power ratios. No uniform bound on r is asserted.*

Proof. If the vectors $(\partial^\alpha \phi_1(s^0), \dots, \partial^\alpha \phi_m(s^0))$ did not span $\mathbb{R}^m$, a nonzero vector would annihilate all of them. The Taylor series and analytic continuation would give a nontrivial constant linear relation among the ϕ_j, contrary to Theorem 1.3.2. Select finitely many multi-indices giving a basis and differentiate (1.3.7) at those multi-indices. This gives a nonsingular linear system for p, involving only finitely many derivatives of H. □

Connection with exact recovery of the local law. In the repeated-coordinate design of the integrated manuscript, use $Y = (X_1, \dots, X_d, X_1)$ and assign the last coordinate a power different from the first. For a law with a positive atom at zero and remaining support in a compact box $[a, b]^d$, $a > 0$, the preceding calibration identifies all the power ratios. The known-power repeated-coordinate Mellin inversion can then be applied to the exact response. Thus arbitrary local correlations are no obstruction to blind calibration when an open response patch is available. This combination still requires a function-valued observation to recover the entire microscopic law; the finite calibration data alone do not identify that law. The unavoidable common rescaling of all original powers and the counting exponent is also unchanged.

1.4 Exchangeable mixtures and differential obstructions: a source audit

This branch concerns three externally posed targets. Two are precise yes-or-no questions; the third is a broad question about restrictions that restore statistical estimation. They must not be counted as four questions by splitting ordinary and shifted differential operators.

1. **Normal directing distributions.** Jeff Lin, *A Negative Answer to a Problem of Aldous on Determination of Exchangeable Sequences*, arXiv:1507.00678v1, Section 2.6, printed page 17 (PDF page index 16)[7], explicitly leaves open whether the class of all normal distributions is good. Proposition 2.6.4 gives a positive answer when the random variance has an exponential moment. The exact source inspected is `https://arxiv.org/pdf/1507.00678v1`. The journal article is *Electronic Journal of Probability* 21 (2016), DOI `10.1214/16-EJP4403`; its publisher PDF could not be inspected in this audit. The complete no-moment solution is already in the integrated draft, and is rederived below. It is not a new result of the present attack.

2. **The converse to the differential barrier.** Le, Bariletto, Rinaldo, and Ho, *Partial Differential Equation Barriers to Identifiability in Infinite Mixture Models*, author manuscript dated 8 August 2026, remark following Proposition 6 and the first question of Section 6, printed page 13 (PDF page index 12)[8], ask whether absence of the stated differential structure implies identifiability. The inspected source is `https://nhatptnk8912.github.io/Identifiability_Infinite_Mixture_final.pdf`; the current arXiv record is `https://arxiv.org/abs/2608.08597`. The integrated draft already contains a complete negative answer for both finite operator classes in Definitions 1 and 4. Its proof is rechecked below. Arbitrary nonlocal operators are outside the claim.

3. **Restrictions restoring consistent estimation.** The second question in Section 6 of the same source asks which model or truth constraints rule out invisible perturbations and permit consistent estimation. This is a research program, not one precise universal conjecture. The quadrature-node restriction proved below gives a new special-case answer for kernels with a finite polynomial defect. It does not close the full question for all kernel families.

Targeted web and SciSpace searches located the two source papers and nearby finite-mixture literature but no subsequent resolution of these exact formulations. This bounded search does not establish priority or the present status of every version. Lin's general Aldous question, its unrestricted continuous-time analogue, and the unrestricted random moment question already receive negative answers in his paper; they are not counted as new open targets here.

1.4.1 Independent rederivation of the two precise draft closures

Proposition 1.4.1 (The normal case without exponential moments)**.** *Let (A, V) have any Borel probability law on $\mathbb{R} \times [0, \infty)$ and, conditionally on it, let $X_j = A + \sqrt{V} Z_j$ with independent standard normal Z_j. If $\mathcal{N} \subset \mathbb{N}$ satisfies $\sum_{n \in \mathcal{N}} 1/n = \infty$, the separate laws of $S_n = \sum_{j=1}^{n} X_j$, $n \in \mathcal{N}$, determine the joint law of (A, V).*

Proof. For the signed difference ρ of two candidate laws and fixed $t \neq 0$, define

$$H_t(z) = \int e^{ita - zv} \, d\rho(a, v), \qquad \operatorname{Re} z > 0.$$

It is bounded by $\|\rho\|_{\mathrm{TV}}$ and holomorphic: on $\operatorname{Re} z \geq \epsilon > 0$, every derivative of the integrand is bounded by a constant times $v^k e^{-\epsilon v}$. Equality of the laws of S_n/n gives $H_t(t^2/(2n)) = 0$. A nonzero bounded holomorphic function in the right half-plane has zeros satisfying $\sum_j \operatorname{Re} z_j/(1 + |z_j|^2) < \infty$. Indeed, the map $w = (z-1)/(z+1)$ and Jensen's formula give the disk condition $\sum_j (1 - |w_j|) < \infty$, and $1 - |w|^2 = 4 \operatorname{Re} z/|z+1|^2$ makes the two conditions equivalent. Here the sum diverges, so H_t is zero. Continuity in t includes $t = 0$. Consequently the finite measure $e^{-v}\rho(da, dv)$ has Fourier transform $H_t(1 - iu) = 0$ for every t, u. Fourier uniqueness gives this measure, and hence ρ, equal to zero. No extension through the boundary point $z = 0$ and no variance moment were used. □

Proposition 1.4.2 (A counterexample to the finite differential converse)**.** *For $x, \theta \in [0, 1]$, let $L_n(t) = P_n(2t - 1)$ be shifted Legendre polynomials and set*

$$k(x \mid \theta) = 1 + \sum_{n=2}^{\infty} \frac{L_n(x) L_n(\theta)}{n!}.$$

This is a normalized, strictly positive, jointly entire kernel. Its signed mixing nullspace is exactly $\{cL_1(\theta)d\theta : c \in \mathbb{R}\}$, yet no nonzero finite point-derivative functional annihilates all its parameter sections. In particular it has neither a finite-order parameter differential annihilator nor a finite-shift, finite-order differential annihilator as defined in the cited source.

Proof. On the unit square $|L_n| \leq 1$, so $k \geq 1 - (e - 2) = 3 - e > 0$. Legendre orthogonality gives normalization. On a complex compact set, $|L_n|$ grows at most exponentially in n, so the factorial series and every fixed derivative converge locally uniformly. For a finite signed measure ν, coefficient extraction gives its mass and every Legendre moment except the first. Thus $A\nu = 0$ implies

$$\int p \, d\nu = 3 \left(\int L_1 \, d\nu \right) \int_0^1 p L_1 \quad \text{for every polynomial } p.$$

Polynomial density identifies the asserted nullspace. Conversely orthogonality proves that every displayed measure is invisible. The distinct positive mixing densities $1 \pm cL_1$, $0 < c < 1$, therefore produce the same uniform observation density.

Suppose $T(p) = \sum_{i=1}^{J} \sum_{j=0}^{r_i} b_{ij} p^{(j)}(\theta_i)$, with distinct nodes, annihilates $k(x \mid \cdot)$ for almost every x. Coefficient extraction, including parameter differentiation, gives $T(1) = 0$ and $T(L_n) = 0$ for $n \geq 2$. Hence $T(p) = 3T(L_1) \int p L_1$ on polynomials. With $Q(t) = \prod_i (t - \theta_i)^{r_i + 1}$,

$$0 = T(Q^2 L_1) = 3T(L_1) \int_0^1 Q^2 L_1^2.$$

The integral is strictly positive, so $T(L_1) = 0$. Hermite interpolation then forces every $b_{ij} = 0$. Complex coefficients are handled by real and imaginary parts. At each fixed base parameter, a finite differential or finite shifted differential relation is such a functional, after combining coincident nodes. Its coefficients must therefore vanish. The nullspace above remains an explicit nonlocal integral obstruction; it is not being denied. □

No fatal mathematical gap was found in these two arguments. Their historical-priority status remains separate from their validity.

1.5 A new sharpening: exact sparse sampling for every fixed stable shape

The integrated draft proves the sufficient conditions in the following theorem and proves the converse only for the Gaussian case. The necessity for every other fixed stable shape is the new extension here. It addresses a natural sharpening of the normal-case target, not an additional independently posed published conjecture.

For $0 < \alpha \le 2$, $-1 \le \beta \le 1$, put

$$\psi_{\alpha,\beta}(t) = \begin{cases} |t|^{\alpha}\{1 - i\beta \tan(\pi\alpha/2) \operatorname{sgn} t\}, & \alpha \neq 1, \\ |t|\{1 + i(2\beta/\pi) \operatorname{sgn} t \log |t|\}, & \alpha = 1, \end{cases} \qquad \psi_{\alpha,\beta}(0) = 0.$$

Write $K_{a,b}$ for the stable law with characteristic function $\exp\{ita - b\psi_{\alpha,\beta}(t)\}$, where $(a, b) \in \mathbb{R} \times [0, \infty)$. Conditional on one draw (A, B) from the mixing law, draw the entire iid sequence from $K_{A,B}$. The index and skewness are fixed throughout a model.

Theorem 1.5.1 (Necessary and sufficient sparse stable sampling)**.** *Let $\mathcal{N} \subset \mathbb{N}$. If $\alpha \neq 1$, the separate partial-sum laws $(\mathcal{L}(S_n))_{n\in\mathcal{N}}$ identify every Borel mixing probability on $\mathbb{R} \times [0, \infty)$ if and only if*

$$\sum_{n\in\mathcal{N}} n^{-|1-\alpha|} = \infty. \tag{1.5.1}$$

If $\alpha = 1$ and $\beta \neq 0$, the corresponding necessary and sufficient condition is

$$\sum_{n\in\mathcal{N}} \frac{1}{1 + (\log n)^2} = \infty. \tag{1.5.2}$$

For $(\alpha, \beta) = (1, 0)$, no choice of $\mathcal{N}$ gives injectivity, even when all positive integers are observed.

Whenever the appropriate sum converges in either of the first two cases, nonidentifiability has witnesses that are absolutely continuous, have all mixed polynomial moments finite and equal, and have the same B marginal. For $\alpha \neq 1$ they also have the same A marginal. Any prescribed lower bound $B \ge b_0 > 0$ can be imposed. No moment assumption is required in the positive directions.

1.5.1 A regularized Blaschke-product construction

Lemma 1.5.2 (Half-line Schwartz realizations of the required zeros)**.** *Each of the following zero sets admits a nonzero real Schwartz function f on $\mathbb{R}$, supported in $[0, \infty)$, whose Laplace transform F vanishes on that set and satisfies $F(0) = 0$:*

1. *$\{c\lambda_n, \bar{c}\lambda_n : n \in \mathcal{N}\}$, where $\operatorname{Re} c = 1$, and either $0 < \lambda_n \le 1$, $\sum_n \lambda_n < \infty$, or $\lambda_n \ge 1$, $\sum_n \lambda_n^{-1} < \infty$;*
2. *$\{1 + iy_n, 1 - iy_n : n \in \mathcal{N}\}$, where $\sum_n (1 + y_n^2)^{-1} < \infty$.*

Finite or empty sets are allowed.

Proof. For a zero ζ in the right half-plane let $b_\zeta(z) = (z - \zeta)/(z + \overline{\zeta})$. Its modulus is at most one there and exactly one on the imaginary axis. Pair conjugate zeros and form

$$P(z) = \prod_n b_{\zeta_n}(z) b_{\overline{\zeta}_n}(z) = \prod_n \frac{z^2 - 2(\operatorname{Re}\zeta_n)z + |\zeta_n|^2}{z^2 + 2(\operatorname{Re}\zeta_n)z + |\zeta_n|^2}.$$

For the three cases of the statement, the difference of each factor from one on a fixed compact subset of $\operatorname{Re} z > 0$ is, respectively, $O(\lambda_n)$, $O(\lambda_n^{-1})$, or $O((1+y_n^2)^{-1})$. Thus the product converges locally uniformly, is not identically zero, has the required zeros, obeys $|P| \le 1$, and has the conjugation symmetry $P(\overline{z}) = \overline{P(z)}$.

We justify the boundary regularity needed for a joint signed-density construction. Put $Q = P'/P$ away from the zeros. For one factor,

$$\frac{b_\zeta'(z)}{b_\zeta(z)} = \frac{1}{z-\zeta} - \frac{1}{z+\overline{\zeta}} = \frac{2\operatorname{Re}\zeta}{(z-\zeta)(z+\overline{\zeta})}.$$

If $\zeta = (1+i\gamma)\lambda$, then

$$|i\xi - \zeta| = |i\xi + \overline{\zeta}| = \{\lambda^2 + (\xi - \gamma\lambda)^2\}^{1/2} \ge C_\gamma |\xi|, \qquad |i\xi - \zeta| \ge \lambda.$$

Differentiating the rational expression therefore gives, for every $r \ge 0$, an upper bound $C_r \lambda |\xi|^{-r-2}$ in the small λ case, and $C_r \lambda^{-r-1}$ in the large λ case. Summation is valid locally uniformly on the imaginary axis away from zero in the first case, and on the whole axis in the second. It follows that every fixed boundary derivative of Q has at most polynomial growth at zero and infinity.

For $\zeta = 1 + iy$, the same derivative is at most $C_r/(1+(\xi-y)^2)$, because both denominator moduli are at least one. The elementary inequality

$$\frac{1}{1+(\xi-y)^2} \le \frac{2(1+\xi^2)}{1+y^2}$$

therefore bounds each derivative of the sum by $C_r(1+\xi^2)\sum_n (1+y_n^2)^{-1}$, including the conjugate terms. The recursion $P^{(j+1)}/P = (P^{(j)}/P)' + Q(P^{(j)}/P)$ shows that boundary derivatives of P have at most polynomial growth in the same sense; we use $|P(i\xi)| = 1$ wherever the boundary product is defined.

With principal square roots set

$$F(z) = P(z)\exp\{-\sqrt{z} - 1/\sqrt{z}\}, \qquad \operatorname{Re} z > 0.$$

The boundary function $H(\xi) = F(i\xi)$, extended by $H(0) = 0$, is Schwartz and flat at zero, since

$$\operatorname{Re}\sqrt{i\xi} = \sqrt{|\xi|/2}, \qquad \operatorname{Re}(i\xi)^{-1/2} = 1/\sqrt{2|\xi|}$$

dominate all the preceding polynomial derivative bounds. Define

$$f(b) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{i\xi b} H(\xi)\, d\xi.$$

Fourier inversion makes f real, nonzero and Schwartz. For $b < 0$, shift the vertical contour to $R + i\mathbb{R}$. The horizontal sides vanish because, for $x \ge 0$, $\operatorname{Re}\sqrt{x+i\xi} \ge \sqrt{|\xi|/2}$. Starting at $\epsilon + i\mathbb{R}$ and then taking $\epsilon \downarrow 0$ justifies the boundary contour. Hence

$$|f(b)| \le \frac{e^{bR}}{2\pi}\int_{\mathbb{R}} e^{-\sqrt{|\xi|/2}}\, d\xi.$$

Letting $R \to \infty$ proves $f(b) = 0$ for $b < 0$. Fubini and Cauchy's formula on right half-disks give

$$\int_0^\infty e^{-sb} f(b)\, db = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{H(\xi)}{s - i\xi}\, d\xi = F(s), \qquad s > 0.$$

The large semicircle vanishes by the square-root exponential bound; the origin can be removed by a small semicircle, since F tends to zero there. Finally $F(s) \to 0$ as $s \downarrow 0$, so dominated convergence gives $\int f = 0$. The formula also holds for complex s with positive real part by holomorphy. □

1.5.2 Proof of the exact stable criteria

Proof of Theorem 1.5.1. For the difference ρ of two mixing laws, retain $H_t(z) = \int e^{ita-zb}\,d\rho(a,b)$. The law of S_n/n supplies the value at $z_n(t) = n\psi_{\alpha,\beta}(t/n)$. If $\alpha \neq 1$, write $c_t = 1 - i\beta\tan(\pi\alpha/2)\operatorname{sgn} t$. Then

$$z_n(t) = |t|^\alpha c_t n^{1-\alpha}, \qquad \frac{\operatorname{Re} z_n(t)}{1+|z_n(t)|^2} \asymp_t n^{-|1-\alpha|}.$$

These points are distinct and lie in the open right half-plane. The half-plane zero condition in Proposition 1.4.1, followed by Fourier–Laplace uniqueness, proves sufficiency of (1.5.1). For $\alpha = 1$, put $\kappa = 2\beta/\pi$. When $\kappa \neq 0$,

$$z_n(t) = |t| + i\kappa t\log|t| - i\kappa t\log n, \qquad \frac{\operatorname{Re} z_n(t)}{1+|z_n(t)|^2} \asymp_t \frac{1}{1+(\log n)^2}.$$

The same proof gives sufficiency of (1.5.2).

For necessity when $\alpha \neq 1$, set $\lambda_n = n^{1-\alpha}$ and $c = 1 - i\beta\tan(\pi\alpha/2)$. Convergence of the appropriate sum is exactly one of the first two hypotheses of Lemma 1.5.2. Obtain its real function f and Laplace transform F with zeros $c\lambda_n$ and $\bar{c}\lambda_n$. Choose a nonzero real even $h \in C_c^\infty(\{t : 1 < |t| < 2\})$ and put

$$G(t,b) = h(t)|t|^\alpha f(b|t|^\alpha), \qquad g(a,b) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-ita}G(t,b)\,dt.$$

All derivatives of G are rapidly decreasing in b, uniformly on its compact t-support. Fourier integration by parts makes $g \in \mathcal{S}(\mathbb{R}^2)$. It is real, nonzero, supported in $b \geq 0$, and has total integral zero because G vanishes near $t = 0$. Its Fourier–Laplace transform is

$$\int e^{ita-sb}g(a,b)\,da\,db = h(t)F(s/|t|^\alpha), \qquad t \neq 0, \quad \operatorname{Re} s > 0. \tag{1.5.3}$$

Substituting $s = z_n(t)$ makes this zero for every selected n.

For $\alpha = 1$, $\kappa \neq 0$, use part (2) of the lemma with $y_n = \kappa\log n$, and define instead

$$G(t,b) = h(t)|t|f(b|t|)\exp\{i\kappa tb\log|t|\}, \qquad g(a,b) = \frac{1}{2\pi}\int e^{-ita}G(t,b)\,dt.$$

Here the displayed formula is used for $t \neq 0$, and G is set equal to zero near $t = 0$, as permitted by the support of h. The modulation introduces only polynomial factors in b upon differentiation, so the same Schwartz and support conclusions hold. Moreover $G(-t,b) = \overline{G(t,b)}$, which makes g real. Now

$$\int e^{ita-sb}g(a,b)\,da\,db = h(t)F\left(s/|t| - i\kappa\operatorname{sgn}(t)\log|t|\right), \quad t \neq 0, \quad \operatorname{Re} s > 0. \tag{1.5.4}$$

At $s = z_n(t)$ the argument is $1 - i\kappa\operatorname{sgn}(t)\log n$, one of the prescribed zeros. This frequency-dependent modulation is needed: simply reusing the homogeneous scaling would not cancel the logarithmic location term.

In both constructions write $g = g_+ - g_-$ and $m = \int g_+ = \int g_- > 0$. The normalized Jordan parts $m^{-1}g_+\,da\,db$ and $m^{-1}g_-\,da\,db$ are distinct absolutely continuous mixing probabilities with identical selected partial-sum laws. Schwartz decay gives every polynomial moment. For each fixed b,

$$\int a^j g(a,b)\,da = i^{-j}\partial_t^j G(0,b) = 0 \qquad (j \geq 0).$$

Thus the B marginals and all mixed polynomial moments agree. In the homogeneous case, $\int G(t,b)db = h(t)\int f = 0$, so the A marginals also agree by Fourier uniqueness. That extra

marginal claim is not asserted for the modulated index-one construction. Replacing B by $B+b_0$ multiplies the signed transform by $e^{-b_0 s}$ and preserves its zeros, the equalities of moments, and the equal coordinate marginals that have been established.

For the remaining symmetric Cauchy case choose $b, c > 0$. The mixing law of $(A, B) = (cZ, b)$, with Z standard symmetric Cauchy, and the point mass at $(0, b+c)$ give the same law of S_n: both are centered Cauchy with scale $n(b+c)$. The laws of the exchangeable sequences differ, because the within-sequence difference of the first two coordinates has scales $2b$ and $2(b+c)$, respectively. This completes the proof. □

Remark 1.5.3 (Full sampling and examples). For $\alpha \neq 1$, observing every positive integer satisfies the criterion because $0 < |1-\alpha| \leq 1$. For $\alpha = 1$, $\beta \neq 0$, the logarithmic criterion also diverges on all positive integers. At index $\alpha = 3/2$, observations at $n = k^2$ suffice, because $\sum_k (k^2)^{-1/2}$ diverges, whereas observations at $n = k^3$ do not. For a skew index-one family, a sampling sequence with $\log n_k \asymp k^a$ identifies all mixing laws exactly when $a \leq 1/2$. These are consequences of the new necessity proof, not additional externally posed problems.

1.5.3 A separate observation: two joint replicates always suffice

Proposition 1.5.4 (Exact replication threshold for fixed stable shapes)**.** *For every fixed α, β above, two conditionally independent observations sharing (A, B) identify its arbitrary Borel mixing law. One observation does not. Thus the exact shared-latent replication threshold is two, including the symmetric Cauchy family.*

Proof. Equality of two-observation laws gives

$$H_u(\psi(t) + \psi(u-t)) = 0 \qquad (t, u \in \mathbb{R})$$

for the signed mixing transform H_u. Fix $u \neq 0$ and let $t > \max\{u, 0\} + 1$. The real part of the argument is $t^\alpha + (t-u)^\alpha$, a strictly increasing positive function of t. Thus the zeros include distinct points accumulating at an interior point of the right half-plane. The identity theorem gives $H_u = 0$. Continuity includes $u = 0$, and Fourier–Laplace uniqueness finishes the argument. For one observation, fix $b, c > 0$, take A with law $K_{0,c}$ and $B = b$, or instead take $(A, B) = (0, b+c)$. The convolution-semigroup identity makes the observed law $K_{0,b+c}$ in both cases, although the mixing laws are distinct. This proves both assertions. □

This proposition is an elementary transform consequence and its historical novelty has not been established. It also explains why the Cauchy counterexample for separate partial-sum marginals does not contradict identification from a joint pair of observations.

1.6 A new constraint mechanism: prescribed density zeros repair finite defects

This section gives a concrete special-case answer to the second question in Section 6 of Le et al. The restriction is on the unknown mixing density, not an extra observable moment. Its statistical effect depends on whether a common smoothness bound is imposed.

Let $S \subset \mathbb{N}$ be finite and nonempty, put $M = \max S$, and let

$$k_S(x \mid \theta) = 1 + \sum_{\substack{n \geq 1 \\ n \notin S}} 2^{-n^2-2} L_n(x) L_n(\theta), \qquad x, \theta \in [0, 1].$$

This is a positive jointly entire kernel. Write A for its one-observation mixing operator and $\mathcal{H}_S = \operatorname{span}\{L_s : s \in S\}$. Coefficient extraction and polynomial density, exactly as in Proposition 1.4.2, give

$$A\mu = A\nu \quad \Longleftrightarrow \quad \nu - \mu = h(\theta)d\theta \quad \text{for some } h \in \mathcal{H}_S. \tag{1.6.1}$$

Every $h \in \mathcal{H}_S$ has degree at most M and integral zero.

Put $J = \lceil (M+1)/2 \rceil$ and let $z_1, \dots, z_J$ be the roots of L_J in $(0,1)$. These nodes admit positive quadrature weights w_i such that

$$\int_0^1 p(t)dt = \sum_{i=1}^{J} w_i p(z_i) \qquad (\deg p \le 2J-1), \qquad w_i > 0. \tag{1.6.2}$$

For completeness, orthogonality proves that L_J has J distinct interior roots: multiplying by the polynomial formed from its sign-changing roots would otherwise contradict orthogonality. For any p of degree at most $2J-1$, division by L_J gives $p = qL_J + r$ with $\deg q, \deg r < J$. Integrate and interpolate r at the nodes to obtain the formula with $w_i = \int \ell_i$, where ℓ_i is the ith Lagrange polynomial. Applying the formula to ℓ_i^2 gives $w_i = \int \ell_i^2 > 0$.

Theorem 1.6.1 (Quadrature zeros identify a density among all competitors)**.** *If $g \in C^1([0,1])$ is a nonnegative probability density satisfying $g(z_i) = 0$ for $1 \le i \le J$, then $\mu = g(t)dt$ is identifiable among all Borel mixing probabilities from its one-observation law under k_S. In particular the polynomial density*

$$g_*(t) = (2J+1)L_J(t)^2$$

is an individually identifiable absolutely continuous mixing law inside this globally nonidentifiable model.

Proof. If $A\nu = A\mu$, (1.6.1) says $\nu = (g+h)dt$ for $h \in \mathcal{H}_S$. Since $g+h$ is continuous, nonnegativity almost everywhere implies nonnegativity everywhere. Thus $h(z_i) \ge 0$. But (1.6.2) gives

$$0 = \int_0^1 h = \sum_{i=1}^{J} w_i h(z_i),$$

so all these values are zero. Both g and $g+h$ have differentiable local minima at each interior node, hence $g'(z_i) = 0$ and $h'(z_i) = 0$. The polynomial h has at least $2J$ zeros counted with multiplicity and degree at most $M < 2J$, so $h = 0$. The example is normalized by Legendre orthogonality. □

Theorem 1.6.2 (The same density zeros do not suffice for uniform learning)**.** *Let $\mathcal{D}$ be the class of entire nonnegative probability densities on $[0,1]$ vanishing at all z_i. Every member is identified among all Borel competitors by one observation, but there is a constant $c_S > 0$ such that*

$$\inf_{\widehat{\mu}} \sup_{g \in \mathcal{D}} \mathbb{E}_{(A(gdt))^{\otimes N}} W_1(\widehat{\mu}, gdt) \ge c_S \qquad (N \ge 1).$$

In contrast, for every fixed $L < \infty$ for which the following class is nonempty, uniform W_1 consistency holds on

$$\mathcal{D}_L = \left\{ g \in C^2([0,1]) : g \ge 0,\ \int_0^1 g = 1,\ g(z_i) = 0\ (1 \le i \le J),\ \|g\|_{C^2} \le L \right\}.$$

The last assertion requires no analyticity of g.

Proof. For $a \in [0,1] \setminus \{z_i\}$, normalize

$$g_{a,t}(x) = C_{a,t} L_J(x)^2 e^{-(x-a)^2/t}, \qquad t > 0.$$

It is an entire probability density with the prescribed zeros and converges weakly to δ_a as $t \downarrow 0$. The same holds at endpoints, using one-sided concentration. Nodes themselves are approximated by points outside the finite node set. Finite convex combinations stay in $\mathcal{D}$. Since finite atomic probabilities are weakly dense in all probabilities on the interval, the weak closure of $\{g\,dt : g \in \mathcal{D}\}$ is all of $\mathcal{P}([0,1])$.

Choose $s \in S$ and the distinct mixing laws $\mu_\pm = (1 \pm \frac{1}{2} L_s)dt$, which have equal observed laws. Approximate both weakly by members $\mu_{\pm,j}$ of the class. The kernel is continuous on a compact rectangle, so weak convergence of mixing laws gives uniform convergence of their observed densities and hence total-variation convergence. For each fixed N,

$$\mathrm{TV}\left((A\mu_{+,j})^{\otimes N}, (A\mu_{-,j})^{\otimes N}\right) \longrightarrow 0.$$

The two-point bound follows by integrating the triangle inequality for W_1 against the overlap of the two data laws. Letting $j \to \infty$ yields the constant $c_S = \frac{1}{2} W_1(\mu_+, \mu_-) > 0$. For $S = \{1\}$ it is $1/12$.

For $\mathcal{D}_L$, the uniform second-derivative bound makes its density functions precompact in C^1 by Arzelà–Ascoli. Every weak limit consequently has a C^1 nonnegative density with the same prescribed zeros and integral one. The previous theorem therefore makes A injective on the weak closure K of this class. That closure is compact; continuity of A makes its inverse on $A(K)$ uniformly continuous for weak metrics. To see uniform estimation directly, take finite $1/N$-nets of K in W_1, estimate the observed law by its empirical measure, and choose a net element minimizing bounded-Lipschitz distance to that empirical law. The bounded-Lipschitz unit ball on $[0,1]$ has finite uniform nets, so the expected empirical distance tends to zero uniformly over all observation laws. Uniform continuity of A and of its inverse, together with the bounded diameter of the parameter interval, gives the required uniform expected W_1 convergence. Fixed tie-breaking makes the finite-net estimator measurable. □

The quadrature-zero argument and its consistency distinction are new extensions of the integrated draft. They do not solve the broad constraint question for arbitrary standard families, and no historical novelty claim is made. A fixed smoothness bound alone is insufficient in a nonidentifiable family; the prescribed zeros remove the finite polynomial ambiguity, while the common bound prevents that ambiguity from reappearing in the closure.

1.7 Optimal replication budgets through multilevel distribution recovery

An equal number of repeated observations for every individual need not be an efficient use of a total observation budget. We prove a sharp separation for the factorial Legendre experiment: an unequal, predetermined allocation attains a Wasserstein risk of order $B^{-1/3}$, and no adaptive allocation can improve this order. The previously established equal-group lower bound is of order $(B \log(eB))^{-1/4}$. Thus the change of allocation improves a power of the budget, not merely a logarithm.

The upper bound uses the classical multilevel Monte Carlo telescoping principle. Distribution-function versions of that principle were developed by Giles, Nagapetyan and Ritter [9]; nested conditional expectations are treated in [10]. The proof below uses integrated distribution-function error, needs no density or smoothness of the mixing law, and is paired with an adaptive statistical lower bound. Lee and Valiant [11] already use unequal coin-sampling depths and

telescoping corrections in a population problem; their target is the fraction of coins above a known gap, rather than the complete mixing law in W_1. See also Lee's thesis [12], Section 4.3, pp. 65–67. The generic multilevel technique and the idea of combining unequal replication depths are not claimed as new.

1.7.1 The experiment and a general allocation theorem

Let $(P_\theta)_{\theta\in[0,1]}$ be a known family of probability laws. Starting a new group draws a fresh latent parameter independently from an unknown $G \in \mathcal{P}([0,1])$. Every observation within that group has law P_θ, independently conditional on its latent parameter. A policy may start new groups, revisit previous groups, randomize, and adapt every choice to the entire observed history. Each observation costs one unit. The total number of observations is at most the deterministic integer budget B. The latent parameters themselves are never observed. Write $\mathfrak{R}_B^{\mathrm{ad}}$ for the minimax expected W_1 risk over all such policies and estimators, and $\mathfrak{R}_B^{\mathrm{fix}}$ when all group sizes are chosen in advance, but need not be equal.

Theorem 1.7.1 (Sharp adaptive budget under regular component estimation)**.** *Suppose there are an integer $b_0 \geq 1$ and estimators T_r valued in $[0,1]$, based on $b_0 r$ observations from a single group, such that*

$$\sup_{\theta\in[0,1]} \mathbb{E}_\theta|T_r - \theta| \leq Ar^{-1/2}, \qquad r \geq 1. \tag{1.7.1}$$

The estimators can be randomized, with their randomization independent of the unknown parameter. Then

$$\mathfrak{R}_B^{\mathrm{ad}} \leq \mathfrak{R}_B^{\mathrm{fix}} \leq CB^{-1/3}, \tag{1.7.2}$$

where C depends only on A, b_0.

Suppose additionally that on some nondegenerate closed interval $J \subset (0,1)$ there is a finite D with

$$\mathrm{KL}(P_\theta \| P_{\theta'}) \leq D|\theta - \theta'|^2 \qquad (\theta, \theta' \in J). \tag{1.7.3}$$

Then constants $c, C > 0$ satisfy

$$cB^{-1/3} \leq \mathfrak{R}_B^{\mathrm{ad}} \leq \mathfrak{R}_B^{\mathrm{fix}} \leq CB^{-1/3} \tag{1.7.4}$$

for all sufficiently large B. Thus adaptive group selection cannot improve the minimax order attained by a fixed unequal allocation.

Proof of the upper bound. Put $r_\ell = 2^\ell$. For a fresh group at level $\ell \geq 1$, compute T_{r_ℓ} from its $b_0 r_\ell$ observations and compute $T_{r_{\ell-1}}$ from the first $b_0 r_{\ell-1}$ observations. Independent auxiliary randomization may be used for the two statistics if needed. They share the same latent parameter, and both have the required marginal laws. Consequently

$$\mathbb{E}_G|T_{r_\ell} - T_{r_{\ell-1}}| \leq A(1+\sqrt{2})r_\ell^{-1/2}. \tag{1.7.5}$$

At level zero use T_1. Use N_ℓ independent groups at each level, and use disjoint groups across levels. Define the random signed distribution function

$$\begin{aligned} H_L(t) = \frac{1}{N_0}\sum_{i=1}^{N_0} \mathbf{1}_{\{T_{1,i}\leq t\}} \\ + \sum_{\ell=1}^{L} \frac{1}{N_\ell}\sum_{i=1}^{N_\ell} \left(\mathbf{1}_{\{T_{r_\ell,i}\leq t\}} - \mathbf{1}_{\{T_{r_{\ell-1},i}\leq t\}}\right). \end{aligned} \tag{1.7.6}$$

Within a parenthesis the two estimates come from the same group. Let $F_{G,L}$ be the distribution function of T_{r_L} when its latent parameter has law G. Telescoping gives $\mathbb{E}H_L = F_{G,L}$. No independence between the two estimates in a parenthesis is needed.

For any $u, v \in [0,1]$,

$$\int_0^1 |\mathbf{1}_{\{u\le t\}} - \mathbf{1}_{\{v\le t\}}|^2 \, dt = |u - v|.$$

The independence of the groups and levels, followed by (1.7.5), therefore yields

$$\int_0^1 \mathrm{Var}(H_L(t))\, dt \le C_A \sum_{\ell=0}^{L} \frac{r_\ell^{-1/2}}{N_\ell}, \qquad C_A = \max\{1, A(1+\sqrt{2})\}.$$

Cauchy–Schwarz first in probability and then in t gives

$$\mathbb{E}\|H_L - F_{G,L}\|_{L^1[0,1]} \le \left(C_A \sum_{\ell=0}^{L} \frac{r_\ell^{-1/2}}{N_\ell} \right)^{1/2}. \tag{1.7.7}$$

Coupling T_{r_L} with its own latent parameter and using the one-dimensional distribution-function formula for W_1 also gives

$$\|F_{G,L} - F_G\|_1 \le A r_L^{-1/2}. \tag{1.7.8}$$

Here is a deterministic allocation with explicit integer rounding. Put $b = \lfloor B/b_0 \rfloor$ and, for $b \ge 64$, choose $L \ge 0$ so that

$$R := 2^L \le b^{2/3}/16 < 2^{L+1}, \qquad S_L = \sum_{\ell=0}^{L} r_\ell^{1/4}, \qquad N_\ell = \left\lceil \frac{b}{2S_L} r_\ell^{-3/4} \right\rceil.$$

The allocation costs at most B, since

$$\sum_{\ell=0}^{L} r_\ell N_\ell \le b/2 + \sum_{\ell=0}^{L} r_\ell < b/2 + 2R \le b.$$

Furthermore,

$$\sum_{\ell=0}^{L} \frac{r_\ell^{-1/2}}{N_\ell} \le \frac{2S_L^2}{b}, \qquad S_L \le \frac{R^{1/4}}{1 - 2^{-1/4}}, \qquad b^{2/3}/32 < R \le b^{2/3}/16.$$

Combining these bounds with (1.7.7) and (1.7.8) shows

$$\sup_G \mathbb{E}\|H_L - F_G\|_1 \le C b^{-1/3}. \tag{1.7.9}$$

Finally convert H_L, which need not be monotone, to a probability law. Enumerate a countable W_1-dense family (Q_j) of rational atomic probabilities on $[0,1]$. Select the first index satisfying

$$\|F_{Q_j} - H_L\|_1 \le \inf_k \|F_{Q_k} - H_L\|_1 + B^{-1},$$

and output $\widehat{G} = Q_j$. The objective functions and their countable infimum are measurable, so this defines a measurable estimator. Density of the family implies that the infimum is at most $\|F_G - H_L\|_1$. Hence

$$W_1(\widehat{G}, G) \le 2\|H_L - F_G\|_1 + B^{-1}.$$

Equation (1.7.9) proves the claim. For the finitely many smaller budgets, increase the constant and use any fixed law. □

Proof of the adaptive lower bound. Write $J = [a, a+s]$ with $s > 0$. For an integer $K \geq 1$, set

$$a_j = a + \frac{s(j-1/2)}{K}, \qquad h = \frac{s}{4K}, \qquad G_v = \frac{1}{K}\sum_{j=1}^{K} \delta_{a_j + h v_j}, \qquad v \in \{-1,1\}^K.$$

The intervals containing different atoms are disjoint and ordered. Matching sorted atoms therefore gives the exact formula

$$W_1(G_v, G_w) = \frac{2h}{K}\, d_H(v,w). \tag{1.7.10}$$

Consider a more informative oracle: it reveals the locations a_j and the common weights $1/K$, and at each step allows the statistician to choose any index j and receive an independent observation with law $P_{a_j + h v_j}$. The statistician still has at most B observations. Every original grouped policy can be simulated in this oracle experiment: when the policy starts a group, draw an independent uniform index in $\{1, \dots, K\}$, and query that same index whenever the group is revisited. The simulator may conceal the index from the original policy. Thus a lower bound for this oracle experiment is also a lower bound for the original one.

For any adaptive randomized oracle policy, let $\mathbb{P}_v$ denote its full transcript law, including its random seed, and let N_j be the number of queries to index j. If $v^{(j)}$ flips only the jth sign, the chain rule for relative entropy and (1.7.3) imply

$$\mathrm{KL}(\mathbb{P}_v \| \mathbb{P}_{v^{(j)}}) \leq 4Dh^2 \mathbb{E}_v N_j. \tag{1.7.11}$$

Indeed the conditional decision rule is identical in the two models at every fixed past transcript; only observations at index j contribute. Stopping before B causes no difficulty, since subsequent dummy observations can be identical under all models. Since $\sum_j N_j \leq B$, averaging (1.7.11) over v, j gives

$$\frac{1}{K 2^K} \sum_v \sum_j \mathrm{KL}(\mathbb{P}_v \| \mathbb{P}_{v^{(j)}}) \leq \frac{4Dh^2 B}{K}.$$

Pinsker's inequality and Jensen's inequality consequently imply

$$\frac{1}{K 2^K} \sum_v \sum_j d_{\mathrm{TV}}(\mathbb{P}_v, \mathbb{P}_{v^{(j)}}) \leq \left(\frac{2Dh^2 B}{K}\right)^{1/2}. \tag{1.7.12}$$

For any sign estimator $\widehat{v}$, the binary testing inequality for each pair $v, v^{(j)}$, averaged over all such pairs, gives

$$\frac{1}{2^K}\sum_v \mathbb{E}_v d_H(\widehat{v}, v) \geq \frac{K}{2}\left(1 - \frac{1}{K 2^K}\sum_v \sum_j d_{\mathrm{TV}}(\mathbb{P}_v, \mathbb{P}_{v^{(j)}})\right).$$

Choose $K = \max\{1, \lceil (2Ds^2 B)^{1/3} \rceil\}$. The right side of (1.7.12) is at most $1/4$, so the average Hamming risk is at least $3K/8$.

Given any probability-valued estimator $\widehat{G}$, choose a nearest $G_{\widehat{v}}$ among this finite family, using a fixed tie rule. The triangle inequality and nearest-point property imply

$$W_1(\widehat{G}, G_v) \geq \frac{1}{2} W_1(G_{\widehat{v}}, G_v) = \frac{h}{K} d_H(\widehat{v}, v).$$

Averaging shows that its worst-case risk is at least $3h/8$, which is bounded below by $cB^{-1/3}$ for all sufficiently large B. The argument includes randomized estimators and every adaptive policy. □

1.7.2 Application to the factorial Legendre kernel

Let L_j denote the shifted Legendre polynomials and let

$$k_*(x \mid \theta) = 1 + \sum_{j\ge 2} \frac{L_j(x)L_j(\theta)}{j!}, \qquad x, \theta \in [0,1].$$

This is the strictly positive entire kernel already considered in the replication analysis. Its one-observation mixture map has a nontrivial null direction, whereas pairs identify arbitrary mixing laws.

Lemma 1.7.2 (Equal-group obstruction for the factorial kernel)**.** *Let $R_{N,q}$ be the minimax expected W_1 risk from N independent groups of q observations each in the factorial experiment. Uniformly over integers $N, q \ge 1$,*

$$R_{N,q} \ge c \max\left\{ N^{-1/2}, \min\left\{ \frac{1}{\log(eN)}, \frac{1}{\sqrt{q\log(eN)}} \right\} \right\}.$$

Consequently $\inf_{Nq\le B} R_{N,q} \ge c\{B\log(eB)\}^{-1/4}$ *for sufficiently large B.*

Proof. An oracle revealing the N latent parameters is more informative. The laws $G_\pm = (1/2 \pm a)\delta_1 + (1/2 \mp a)\delta_0$ with $a = 1/(16\sqrt{N})$ have transport distance $2a$ and N-sample relative entropy bounded by an absolute constant below $1/2$. Pinsker's inequality and the two-point testing bound give $R_{N,q} \ge cN^{-1/2}$.

For the other term put $\theta_0 = 1/2$, $\mathcal{H} = L^2(P_{\theta_0}; \mathbb{C})$, and $f_z(x) = k_*(x \mid \theta_0 + z)/k_*(x \mid \theta_0)$. The factorial series converges locally uniformly for complex parameters, and the denominator is at least $3 - e$. Thus for constants $R \in (0, 1/4]$ and $C_0 \ge 1$, f_z is $\mathcal{H}$-holomorphic and

$$\|f_z - 1\|_{\mathcal{H}} \le \sqrt{C_0}|z|, \qquad \int f_z \, dP_{\theta_0} = 1 \quad (|z| \le R).$$

The last identity cancels the cross term in the squared Hilbert norm. Writing $F_z = f_z^{\otimes q} = \sum_{j\ge 0} T_j z^j$ in $L^2(P_{\theta_0}^{\otimes q}; \mathbb{C})$ gives, for $0 < h \le R/2$,

$$\|F_z\| \le e^{C_0 q|z|^2/2}, \qquad \|T_j\| \le e^{2C_0 q h^2}(2h)^{-j},$$

by Cauchy's coefficient estimate on $|z| = 2h$.

For an integer $m \ge 1$, set $n = m + 1$ and $\theta_j = \theta_0 - h + 2hj/n$. Define

$$G = 2^{1-n} \sum_{j \ meven} \binom{n}{j} \delta_{\theta_j}, \qquad H = 2^{1-n} \sum_{j \ modd} \binom{n}{j} \delta_{\theta_j}, \qquad 0 \le j \le n.$$

Finite differences show that their moments agree through degree m. On the jth grid interval the difference of their cumulative distribution functions is $2^{1-n}(-1)^j \binom{n-1}{j}$. Its absolute values sum to one, so $W_1(G, H) = 2h/(m+1)$. Writing $Q_q(G) = \int P_\theta^{\otimes q} \, dG(\theta)$ and using $|\int (\theta - \theta_0)^j \, d(G - H)| \le 2h^j$ yields

$$\left\| \frac{dQ_q(G) - dQ_q(H)}{dP_{\theta_0}^{\otimes q}} \right\|_2 \le 2e^{2C_0 q h^2} \sum_{j>m} 2^{-j} = 2e^{2C_0 q h^2} 2^{-m}.$$

Hence the one-group total variation distance is at most $e^{2C_0 q h^2} 2^{-m}$. Take

$$m = \left\lceil \frac{4}{3\log 2} \log(4N) \right\rceil, \qquad h = \min\left\{ R/2, \sqrt{\frac{m \log 2}{8C_0 q}} \right\}.$$

The one-group distance is at most $1/(4N)$, so the N-group distance is at most $1/4$. The two-point bound implies

$$R_{N,q} \geq c\frac{h}{m+1} \geq c' \min\{\log(eN)^{-1}, (q\log(eN))^{-1/2}\}.$$

Finally set $L = \log(eN)$. If $q \geq L$, the maximum of the two lower bounds is at least $(NqL)^{-1/4} \geq \{B\log(eB)\}^{-1/4}$, up to a constant. If $q < L$, the bound $c/L \geq c/\log(eB)$ is larger than the asserted bound for all sufficiently large B. □

Corollary 1.7.3 (Unequal replication is strictly better than equal groups)**.** *For the factorial Legendre experiment,*

$$\mathfrak{R}_B^{\mathrm{ad}} \asymp \mathfrak{R}_B^{\mathrm{fix}} \asymp B^{-1/3}.$$

If $\mathfrak{R}_B^{\mathrm{eq}}$ denotes the corresponding infimum restricted to equal groups, then

$$\mathfrak{R}_B^{\mathrm{eq}} \geq c(B\log(eB))^{-1/4}, \qquad \frac{\mathfrak{R}_B^{\mathrm{fix}}}{\mathfrak{R}_B^{\mathrm{eq}}} \leq CB^{-1/12}(\log(eB))^{1/4} \longrightarrow 0.$$

Thus no equal-group allocation is order optimal under the unrestricted total-budget criterion. Adaptive selection is unnecessary for the optimal order, but unequal group sizes are essential for attaining it.

Proof. Put $A_j(x) = j!(2j+1)L_j(x)$. The bounded pair statistic

$$Y(x,y) = \frac{1}{2} + \frac{35}{18}\left(A_2(x)A_3(y) - \frac{10}{21}A_5(x) - \frac{4}{15}A_3(x)\right)$$

satisfies $\mathbb{E}_\theta Y(X_1, X_2) = \theta$ and $|Y| < 2100$. For completeness, $\mathbb{E}_\theta A_j(X) = L_j(\theta)$ and

$$L_2L_3 = \frac{10}{21}L_5 + \frac{4}{15}L_3 + \frac{9}{35}L_1, \qquad L_1(\theta) = 2\theta - 1,$$

prove unbiasedness. The bounds $|L_j| \leq 1$ give $|A_2A_3| \leq 420$, $|A_5| \leq 1320$, and $|A_3| \leq 42$, which prove the stated numerical bound. Average Y over r disjoint pairs and clip to $[0,1]$. The resulting T_r satisfies (1.7.1) with $b_0 = 2$ and $A = 2100$.

The kernel is bounded below by $c_0 = 3 - e > 0$. Uniform convergence of its derivative on the compact square gives $M = \sup_{x,\theta} |\partial_\theta k_*(x \mid \theta)| < \infty$. Therefore

$$\mathrm{KL}(P_\theta \| P_{\theta'}) \leq \chi^2(P_\theta \| P_{\theta'}) \leq \frac{M^2}{c_0}|\theta - \theta'|^2.$$

Theorem 1.7.1 applies. Taking the ratio of its upper bound with Lemma 1.7.2 proves the strict separation. □

Remark 1.7.4 (What this closes)*.* The complete total-budget minimax order is now determined for arbitrary unequal and adaptive allocations in this kernel model. This closes the allocation problem left open by the equal-group analysis. It does not determine the sharp growing-replication curve $R_{N,q}$ for each fixed pair (N,q), and it does not close the logarithmic gap within the smaller equal-group class itself. The result is a proved resolution of that internally posed extension, not a claim to have resolved a named published conjecture or established historical priority.

1.7.3 An exact Bernoulli reduction and the remaining equal-group gap

Proposition 1.7.5 (Bounded unbiased statistics transfer population bounds)**.** *Suppose a statistic Y based on b_0 observations has $|Y| \le M$ and $\mathbb{E}_\theta Y = \theta$ for $\theta \in [0,1]$. With an independent uniform U, define*

$$Z = \mathbf{1}_{\{U \le (Y+M)/(2M)\}}.$$

Then conditionally on θ, Z is Bernoulli with parameter $p(\theta) = (\theta + M)/(2M)$. Disjoint blocks produce conditionally independent Bernoulli variables. Consequently every distribution-learning upper bound for N groups of t Bernoulli observations transfers to N groups of $b_0 t$ observations from the original experiment, with the W_1 risk multiplied by at most $2M$.

Proof. Conditional expectation gives the Bernoulli parameter, and independent blocks and auxiliary uniforms give conditional independence. If a Bernoulli-population estimator produces a law $\widehat{H}$ on $[0,1]$, push it forward by the clipped inverse map $u \mapsto \min\{1, \max\{0, 2Mu - M\}\}$. This map is $2M$-Lipschitz and is the true inverse on $p([0,1])$, so contraction of transport under Lipschitz maps gives the stated risk bound. Randomization causes no statistical difficulty; it is part of the estimator, not additional data. □

Vinayak, Kong, Valiant and Kakade [13] prove a sharp binomial-mixture rate $(t \log N)^{-1/2}$ in the range $t \in [\Omega(\log N), O(N^{2/9-\varepsilon})]$; see their Section 1.1 and Theorem 3.2. Their Remark 3.2 explicitly conjectures extending the MLE bound to $t = O(N^{2/3-\varepsilon})$. That theorem cannot be extrapolated to $t \asymp N/\log N$, which is the range relevant to the optimal equal-group total budget. The exact reduction above is valid, but an appropriate all-regime upper bound or a direct proof would still be needed to close that logarithmic gap. Musco, Musco, Rosenblatt and Singh [14], Theorem 20, improve the proved MLE range to $t = O(N^{1/4-\varepsilon})$. Their discussion of conjectured improvements (published Appendix C.2; Section 6.2 of the cited arXiv version) still requires sharper coefficient and likelihood bounds. Neither proved range reaches the budget endpoint. No closure of that gap is claimed here.

1.7.4 Published questions checked against the new result

Lau and Scarlett's 2026 COLT open problem [15] asks whether arbitrary nonadaptive one-bit quantizers can attain the optimal adaptive mean-estimation rate over nonparametric finite-moment classes, or whether one adaptive transition is necessary. This is an explicit published open question; the present replication theorem does not resolve it. The observation is different: their quantizer receives one fresh unbounded sample, while the present experiment can repeatedly observe the same latent individual. The bounded Bernoulli reduction also does not remove that distinction or supply a bounded unbiased statistic for their unrestricted class.

The second explicit published question is Vinayak et al.'s Remark 3.2 conjecture just described. It concerns the particular MLE under equal group sizes, and neither our reduction nor our different allocation proves it. Their Remark 3.3 and supplementary Section A.6 also outline local moment matching for a broader range, but the supplement calls this a high-level construction and does not state a proved all-regime bound sufficient for the present equal-budget endpoint. The 2025 paper just cited provides a verified later improvement and still discusses the stronger assertions as conjectural. No subsequent resolution was verified in the targeted search. In particular, the coefficient assertion $|C(t,m,j)| \le \exp(m^2/t)$, where $C(t,m,j)$ are the degree-t Bernstein coefficients of the shifted degree-m Chebyshev polynomial, is a precise related approximation question; we have not proved it and do not count it as an additional resolution. Finally, Lee and Kennedy's October 2026 preprint [16] already treats heterogeneous trial profiles for pointwise estimation of smooth binomial mixing densities. Its loss and smoothness class

differ from the arbitrary-law W_1 budget problem proved here; trial heterogeneity by itself cannot be described as an untouched research direction.

1.8 Exact Mellin zeros, moment comparisons, and the tail-order conjecture

Problem and priority ledger. The explicitly published unresolved assertion addressed in this section is Conjecture 17 of Bürgin, Epperlein, and Wirth [17]: on a prescribed positive compact interval, reciprocal-summable integer exponents should admit a nonzero continuous signed density annihilating those moments. The conjecture is located on printed page 15 of the cited arXiv version. Its assertion is correct, but it already follows by a logarithmic change of variables from Luxemburg–Korevaar's 1971 construction [18, Theorem 5.2], restated with its product formula in Wellershoff [19, Theorem 21, printed p. 18]. Accordingly, this is an explicit deduction and an alternative proof, not a claim of a historically new resolution. The exact real zero and sign classifications below are further deductions from that classical construction. Their precise formulations were not located in the targeted search, which does not establish priority. They are not counted as three additional published open problems.

For a finite signed measure ν supported on a compact subinterval of $(0,\infty)$, write

$$\mathcal{M}_\nu(z) = \int x^z \, d\nu(x), \qquad z \in \mathbb{C},$$

where $x^z = e^{z\log x}$. This is an entire function. For an integrable function h, write $\mathcal{M}_h = \mathcal{M}_{h(x)dx}$. A function is in $G^\sigma(\mathbb{R})$ if it is smooth and satisfies $\|h^{(k)}\|_\infty \le CB^k(k!)^\sigma$ for some $C, B > 0$ and every integer $k \ge 0$.

1.8.1 An alternative proof of the published conjecture

Proposition 1.8.1 (Localized smoothing of real Müntz annihilators)**.** *Let $0 < a < b$, let I be a nonempty open interval with $\overline{I} \subset (a,b)$, and let $0 = \lambda_0 < \lambda_1 < \lambda_2 < \cdots$ tend to infinity. For every $\sigma > 1$ the following are equivalent:*

1. *$\sum_{j\ge1} \lambda_j^{-1} < \infty$;*
2. *there exists a nonzero real $h \in G^\sigma(\mathbb{R})$ with compact support in I and $\mathcal{M}_h(\lambda_j) = 0$ for every j.*

The same assertion holds for a finite sequence, with the sum interpreted as finite. The annihilator space in the second assertion is infinite dimensional.

Proof. Choose a nondegenerate interval $[c,d] \subset I$. The Müntz theorem on positive intervals [20, Theorem 3.2] gives a nonzero real signed measure ν supported on $[c,d]$ that annihilates all the specified powers when the reciprocal sum converges. Choose a nonnegative, nonzero $\varphi \in G^\sigma(\mathbb{R})$ supported in a sufficiently small neighborhood of 1, and set

$$h(x) = \int t^{-1}\varphi(x/t)\, d\nu(t). \tag{1.8.1}$$

The neighborhood can be chosen so that $\operatorname{supp} h \Subset I$. Moreover

$$\|h^{(k)}\|_\infty \le \|\nu\|_{\mathrm{TV}} c^{-k-1} \|\varphi^{(k)}\|_\infty,$$

which proves the Gevrey bound. Fubini's theorem gives the entire identity

$$\mathcal{M}_h(z) = \mathcal{M}_\nu(z)\mathcal{M}_\varphi(z). \tag{1.8.2}$$

For every real s, the second factor is strictly positive. A nonzero compactly supported signed measure has a nonzero integer moment by polynomial density, so $h \neq 0$. For necessity, the divergent Müntz condition makes the specified powers dense in $C[a,b]$. An annihilator must therefore vanish. Disjoint subintervals of I give infinitely many linearly independent annihilators.

For completeness, the required Gevrey bump can be obtained by multiplying two translates of $e^{-t^{-1/(\sigma-1)}}\mathbf{1}_{t>0}$. Cauchy's estimate on disks of radius proportional to $t > 0$, followed by maximizing $t^{-k}e^{-ct^{-1/(\sigma-1)}}$, gives the stated Gevrey order; extension by zero is smooth. □

Taking integer $\lambda_j = n_j$ proves the precise assertion of the published conjecture, with arbitrary interior localization and every Gevrey order above one. There is no obstruction arising from total mass: adding the exponent zero costs only a finite extra constraint.

1.8.2 The complete positive-real zero set and its multiplicities

We use the following classical external input. In the notation adapted to the present proof, Luxemburg–Korevaar's theorem states that if $d_j \neq 0$ and $\sum_j |d_j|^{-1} < \infty$, then for any $u < v$ and any integer $m \geq 0$ there is a nonzero $g \in C_c^\infty([u,v])$ with

$$\widehat{g}(z) = C_* z^m e^{-i\tau z} \prod_j (1 - z^2/d_j^2) \prod_k \cos(\epsilon_k z), \qquad \widehat{g}(z) = \int g(t) e^{-itz}\, dt, \tag{1.8.3}$$

where $C_* \neq 0$, $\epsilon_k > 0$, $\sum_k \epsilon_k < \infty$, and τ is real. Repetitions of d_j encode multiplicity. The theorem and factorization used here were checked in the cited 2021 author-hosted restatement; the original 1971 publisher PDF was not retrievable in this verification pass. The existence theorem itself is an imported classical result, not proved anew here.

Theorem 1.8.2 (Exact real Mellin zeros with multiplicity)**.** *Let $\Lambda \subset (0,\infty)$ have no finite accumulation point, let $m_\lambda \in \mathbb{N}$ for $\lambda \in \Lambda$, and fix an integer $m_0 \geq 1$. The following conditions are equivalent:*

1. $\displaystyle\sum_{\lambda\in\Lambda} m_\lambda/\lambda < \infty$;
2. *for every $0 < a < b$, every nonempty open I with $\overline{I} \subset (a,b)$, and every $\sigma > 1$, there is a nonzero real $h \in G^\sigma(\mathbb{R})$ compactly supported in I such that $\mathcal{M}_h$ has a zero of order m_0 at zero, and its positive-real zeros are exactly Λ, with multiplicities m_λ.*

The construction may additionally arrange that its negative-real zeros are exactly $-\Lambda$, with the same multiplicities.

Proof. For necessity, if $\operatorname{supp} h \subset [a,b]$, then

$$F(z) = b^{-z}\mathcal{M}_h(z) = \int_a^b (x/b)^z h(x)\, dx$$

is bounded and holomorphic on $\operatorname{Re} z > 0$ and is not identically zero. The disk map $w = (z-1)/(z+1)$ and the Blaschke zero condition imply

$$\sum_{\lambda\in\Lambda} m_\lambda \frac{\lambda}{1+\lambda^2} < \infty.$$

Since the zero set has no finite accumulation and is bounded away from zero, this is equivalent to the required reciprocal sum. One can obtain the disk condition directly from Jensen's formula after removing a zero at the center: boundedness bounds the sums $\sum_{|w_j|<r} \log(r/|w_j|)$ uniformly as $r \uparrow 1$, and hence $\sum_j (1 - |w_j|) < \infty$.

For sufficiency, first suppose Λ is infinite. Choose $[c,d] \Subset I$, put $u = \log c$, $v = \log d$, and apply (1.8.3) to the sequence consisting of $i\lambda$ repeated m_λ times, and to $m = m_0$. Multiplying g by a nonzero constant permits the normalization $C_* = i^{-m_0}$. All other factors have the conjugation symmetry $\widehat{g}(-\overline{z}) = \overline{\widehat{g}(z)}$, so uniqueness of the Fourier transform shows that g is real. Set $h_0(x) = x^{-1} g(\log x)$ on $(0,\infty)$ and zero elsewhere. It is real and smooth, with compact support in $[c,d]$, and

$$\mathcal{M}_{h_0}(s) = s^{m_0} e^{\tau s} \prod_{\lambda \in \Lambda} (1 - s^2/\lambda^2)^{m_\lambda} \prod_k \cosh(\epsilon_k s), \qquad s \in \mathbb{R}. \tag{1.8.4}$$

The last product is finite and strictly positive for every real s: its logarithm converges because $\sum_k \epsilon_k^2 < \infty$. The middle product converges locally uniformly, and away from its specified zeros it is nonzero. Indeed, all sufficiently late factors are near one and $\sum_\lambda m_\lambda/\lambda^2 < \infty$. Thus the real zero set and all its multiplicities are exactly those asserted.

If Λ is finite, start instead with a nonzero nonnegative smooth bump g_0 supported in (u,v). The constant-coefficient differential operator corresponding to the polynomial $s^{m_0} \prod_{\lambda\in\Lambda}(1 - s^2/\lambda^2)^{m_\lambda}$ produces g whose bilateral Laplace transform is that polynomial times the strictly positive real transform of g_0. Specifically, replace s by $-d/dt$ in this polynomial. This also treats $\Lambda = \varnothing$.

Finally apply (1.8.1) to the signed measure $h_0(x)\,dx$, choosing the bump support near 1 sufficiently small. The result belongs to G^σ, is still supported in I, and its Mellin transform is multiplied by an entire function strictly positive on the real axis. Hence no real zero is created or removed, and the multiplicities are unchanged. This proves the assertion. □

Corollary 1.8.3 (Exact real moment coincidences with shape constraints)**.** *Under the summability condition in Theorem 1.8.2, there are distinct probability densities $g_+, g_- \in G^\sigma(\mathbb{R})$ with support $[a,b]$, positive and strictly log-concave on (a,b), such that their real-order moments agree at a positive exponent s if and only if $s \in \Lambda$. The multiplicity of that agreement as a zero of the moment-difference function is m_s. Their difference is supported in any prescribed interior interval.*

Proof. Take the mass-zero h of Theorem 1.8.2. With $p = (\sigma - 1)^{-1}$, normalize the probability density

$$g_0(x) = C \exp(-(x-a)^{-p} - (b-x)^{-p}) \mathbf{1}_{a<x<b}.$$

It lies in G^σ and has strictly negative logarithmic second derivative on (a,b). On the compact support of h, both positivity and the inequality $g_0 g_0'' - (g_0')^2 < 0$ have a strict margin. Consequently $g_\pm = g_0 \pm \varepsilon h$ preserve them for sufficiently small $\varepsilon > 0$. Outside that compact set the densities equal g_0. Their masses are one, and their moment difference is $2\varepsilon \mathcal{M}_h$. □

1.8.3 A complete classification of integer moment sign patterns

This records more than equality or inequality in the eventual tail order. It specifies every comparison of the two integer moment sequences.

Theorem 1.8.4 (Exact signs of moment differences)**.** *Let $(\eta_n)_{n\ge1}$ be a sequence in $\{-1,0,1\}$ that is not identically zero. Let $Z = \{n \ge 1 : \eta_n = 0\}$, and enumerate its complement as $n_1 < n_2 < \cdots$ when the complement is infinite. The following are equivalent:*

1. *There are two distinct probability measures supported in a compact subinterval of $(0,\infty)$ such that*

$$\operatorname{sgn}\left(\int x^n \, d\mu_+(x) - \int x^n \, d\mu_-(x) \right) = \eta_n \quad (n \ge 1).$$

2. *The set Z satisfies $\sum_{n\in Z} n^{-1} < \infty$ (so its complement is infinite), and*

$$\sum_{j:\,\eta_{n_j}\neq\eta_{n_{j+1}}} \frac{1}{n_{j+1}} < \infty. \tag{1.8.5}$$

3. *For every $0 < a < b$ and $\sigma > 1$, such a pair can be chosen with strictly log-concave G^σ probability densities supported on $[a, b]$, positive on (a, b), whose difference is supported in any prescribed nonempty interior interval.*

Proof. Let $F(z) = \int x^z \, d(\mu_+ - \mu_-)(x)$. It is nonzero, real on real arguments, and its positive zeros obey the Blaschke summability condition proved above. In particular $\sum_{n\in Z} 1/n < \infty$. If consecutive nonzero samples have opposite signs, the intermediate value theorem gives a zero $r_j \in (n_j, n_{j+1})$. These intervals are disjoint, and $1/n_{j+1} \le 1/r_j$. Summing gives (1.8.5). This proves necessity, including the fact that a finite nonzero complement cannot occur.

Conversely, put a root of multiplicity two at every integer in Z. For each sign transition put one simple root at $r_j = n_{j+1} - 1/2$. This point lies strictly between n_j and n_{j+1}, is not an integer, and satisfies $1/r_j \le 2/n_{j+1}$. The combined root multiset therefore satisfies Theorem 1.8.2. Use that theorem with a simple root at zero. Its transform has exactly those positive roots. Even roots at Z do not change the sign; the selected simple roots change it exactly once between consecutive nonzero samples when required. Multiplying h by -1 if necessary matches the first prescribed nonzero sign. Thus every sign is correct. The perturbation in Corollary 1.8.3 produces the required probability densities. The last assertion plainly implies the first. □

Example 1.8.5 (Sparse and dense switching behave differently)**.** If no zero moments are prescribed, signs can switch precisely at indices $2, 4, 8, 16, \ldots$, or at $2^2, 3^2, 4^2, \ldots$, because the reciprocal sums converge. They cannot switch at every integer, or at all prime indices, because the corresponding sums diverge. Thus smooth strict log-concavity permits infinitely many alternating moment blocks, but it does not permit arbitrary switching frequency. This is an exact restriction, not merely an upper bound on the number of changes. It refines the smooth alternating examples in the published tail-order literature, rather than treating those already established examples as unresolved problems.

1.8.4 Which ultrafilters preserve all moment information?

For $S \subset \mathbb{N}_0$ set $w(S) = \sum_{n\in S\setminus\{0\}} 1/n$, and let

$$\mathcal{F}_{\mathrm{M}} = \{A \subset \mathbb{N}_0 : w(\mathbb{N}_0 \setminus A) < \infty\}.$$

For a free ultrafilter $\mathcal{U}$, two real sequences are identified if their equality set belongs to $\mathcal{U}$. Let $\Phi_{\mathcal{U}}$ map a measure to the equivalence class of its integer moment sequence. Bürgin–Epperlein–Wirth's Theorem 6 gives the sufficiency of $\mathcal{F}_{\mathrm{M}} \subset \mathcal{U}$. The next result records the converse, including a strong regularity restriction on the test class.

Theorem 1.8.6 (Exact ultrafilter criterion)**.** *Fix $0 < a < b$ and $\sigma > 1$, and let $\mathcal{L}_{a,b,\sigma}$ be the probability measures with $G^\sigma(\mathbb{R})$ densities supported on $[a, b]$, positive and strictly log-concave on (a, b). For a free ultrafilter $\mathcal{U}$, the following are equivalent:*

1. *$\mathcal{F}_{\mathrm{M}} \subset \mathcal{U}$;*
2. *$\Phi_{\mathcal{U}}$ is injective on all finite positive measures with compact support in $(0, \infty)$;*
3. *$\Phi_{\mathcal{U}}$ is injective on $\mathcal{L}_{a,b,\sigma}$.*

Proof. Assume the first condition. If two measures have equality set E in $\mathcal{U}$ and $w(E) < \infty$, then $\mathbb{N}_0 \setminus E$ belongs to $\mathcal{F}_{\mathrm{M}} \subset \mathcal{U}$, a contradiction. Thus $w(E) = \infty$, and Müntz uniqueness gives equality of the measures. The second condition plainly implies the third.

If the first condition fails, choose $A \in \mathcal{F}_{\mathrm{M}}$ with $A \notin \mathcal{U}$. Then $S = \mathbb{N}_0 \setminus A$ belongs to $\mathcal{U}$ and satisfies $w(S) < \infty$. Since the ultrafilter is free, S is infinite. Corollary 1.8.3, with $\Lambda = S \setminus \{0\}$, gives distinct measures in $\mathcal{L}_{a,b,\sigma}$ whose moments agree at least on S. Their equality set belongs to $\mathcal{U}$, so the third condition fails. This proves the equivalence. □

What is and is not closed. The mathematical assertion called Conjecture 17 in the 2022 paper is proved, with a clear classical antecedent dating to 1971. Exact real Mellin-zero sets, multiplicities, and the full sign-pattern criterion supply useful stronger formulations. The ultrafilter theorem completes a published one-way criterion, but the cited source did not pose that converse as a separate open conjecture. No claim is made here that three or four distinct published open problems have been resolved.

1.9 A complete regularity criterion for sparse-moment identification

The Gevrey boundary in the integrated manuscript is one slice of a more general dichotomy. The ingredients below are classical Denjoy–Carleman and Müntz theorems; the conclusion is their application to the same inverse observation problem. It is an extension of an internal research question, not the resolution of a separately identified famous conjecture.

Let $M = (M_k)_{k \ge 0}$ be a positive log-convex sequence with $M_0 = 1$, $M_k \ge k!$, and $M_k^{1/k} \to \infty$. Write $\mathcal{C}_M[a, b]$ for the smooth real functions on the closed interval such that

$$\|f^{(k)}\|_\infty \le A B^k M_k \qquad (k \ge 0)$$

for some $A, B > 0$. We use the classical Denjoy–Carleman criterion: this class has uniqueness from the full jet at one point precisely when

$$\sum_{k \ge 1} \frac{M_{k-1}}{M_k} = \infty. \tag{1.9.1}$$

When this sum converges, nonzero nonnegative compactly supported $\mathcal{C}_M$ functions exist in every open interval. These standard facts are stated, with proofs and the relevant endpoint interpretation, in [2, Theorem 3.6 and Corollary 3.12].

Theorem 1.9.1 (Quasianalyticity is the exact sparse-moment boundary)**.** *Fix $0 < a < b$ and an unbounded set $T \subset \mathbb{N}_0$ containing 0 with $\sum_{n \in T \setminus \{0\}} 1/n < \infty$. The following conditions are equivalent:*

1. *The sum in (1.9.1) diverges.*
2. *The T-moments identify probability densities within $\mathcal{C}_M[a, b]$.*

If the sum in (1.9.1) converges, the signed kernel of the T-moment map contains infinitely many linearly independent $\mathcal{C}_M$ functions supported in any prescribed nonempty interior interval. In particular nonidentification persists among strictly positive, strictly log-concave densities on $[a, b]$.

Proof. Suppose first that (1.9.1) holds. If f, g have equal T-moments, put $h = f - g$. If some derivative at b is nonzero, let m be its least order. Taylor's formula and $t = s(1 - x/b)$ give

$$\int_a^b x^s h(x)\, dx \sim (-1)^m h^{(m)}(b) \frac{b^{s+m+1}}{s^{m+1}} \qquad (s \to \infty). \tag{1.9.2}$$

Indeed $|h(x)| \le C(b-x)^m$ throughout $[a,b]$. After multiplication by $s^{m+1}b^{-s-m-1}$, the transformed integrand is dominated by $Ct^m e^{-t}$ and converges to $(-1)^m h^{(m)}(b)t^m e^{-t}/m!$. Dominated convergence proves the displayed asymptotic. It precludes moment zeros along an unbounded set. Hence every derivative of h at b is zero. Extend h by zero just to the right of b; it remains smooth and obeys the same derivative bounds. Quasianalytic uniqueness on this larger interval gives $h = 0$.

Conversely suppose the sum in (1.9.1) converges. Choose a closed interval $[c,d]$ strictly inside the requested support interval I. Müntz non-density on $[c,d]$, Hahn–Banach, and Riesz representation give a nonzero finite real signed measure ν annihilating x^n for $n \in T$. Choose a nonnegative nonzero $\mathcal{C}_M$ bump φ supported sufficiently near 1 that $[c,d]\operatorname{supp}\varphi \subset I$. Set

$$h(x) = \int t^{-1}\varphi(x/t)\,d\nu(t).$$

Its derivatives satisfy

$$\|h^{(k)}\|_\infty \le \|\nu\|_{\mathrm{var}} c^{-k-1}\|\varphi^{(k)}\|_\infty \le A'(B/c)^k M_k.$$

The moment identity

$$\int x^s h(x)\,dx = \left(\int t^s\,d\nu(t)\right)\left(\int u^s\varphi(u)\,du\right)$$

annihilates T and also proves $h \neq 0$: a nonzero compactly supported measure has a nonzero integer moment, and the second factor is positive. Using disjoint subintervals of I gives linear independence.

For the density conclusion take the normalized analytic baseline $g_0(x) = C\exp(-x^2)$ on $[a,b]$. It belongs to $\mathcal{C}_M$ because $M_k \ge k!$. For small $\varepsilon > 0$, $g_\pm = g_0 \pm \varepsilon h$ are positive and integrate to one. On the compact support of h, $g_0 g_0'' - (g_0')^2 = -2g_0^2$ is bounded strictly below zero, so a sufficiently small C^2 perturbation preserves strict log-concavity; outside the support there is no change. These densities are smooth up to the endpoints; no globally smooth extension by zero is claimed. □

If the reciprocal sum for T diverges, classical Müntz completeness identifies all finite measures, regardless of regularity. Thus there are two independent ways to obtain uniqueness: completeness of the observations, or quasianalyticity of the unknown density.

Corollary 1.9.2 (Fixed derivative bounds and uniform learning)**.** *Let $S \subset \mathbb{N}_0$ be unbounded and contain 0. For the interval kernel*

$$k_S(x \mid \theta) = 1 + \sum_{s \in S\setminus\{0\}} 2^{-s^2-s-2}L_s(x)\theta^s, \quad x \in [0,1], \quad \theta \in [1,2],$$

where L_s is the shifted Legendre polynomial normalized by $L_s(1) = 1$, consider the probability densities satisfying fixed bounds $\|g^{(k)}\|_\infty \le AB^k M_k$ for every k. If (1.9.1) holds, there are estimators from independent one-view observations whose worst-case expected W_1 error tends to zero. If (1.9.1) fails and $\sum_{s\in S\setminus\{0\}} 1/s < \infty$, there exist fixed finite A, B for which the worst-case risk is bounded away from zero for every sample size. No rate for the first assertion is claimed.

Proof. Legendre orthogonality gives $\mathbb{E}_\theta L_s(X) = 2^{-s^2-s-2}\theta^s/(2s+1)$ for $s \in S$; absolute uniform convergence of the series gives the converse. Thus equality of one-view laws is exactly equality of the S-moments. If the reciprocal sum over $S \setminus \{0\}$ diverges, classical Müntz completeness gives injectivity. Otherwise Theorem 1.9.1 gives injectivity in the quasianalytic case.

The set K of densities with the fixed bounds is compact in $C[1,2]$: successive Arzelà–Ascoli subsequences give convergence of every derivative, and the derivative bounds and probability

constraints are closed under these limits. The observation map $A : K \to \mathcal{P}([0,1])$ is continuous. Choose a countable dense family (v_j) in the unit ball of $C[0,1]$, and metrize weak convergence by

$$d(P,Q) = \sum_{j\ge 1} 2^{-j} |Pv_j - Qv_j|.$$

For the empirical law P_N, boundedness gives $\sup_P \mathbb{E} d(P_N, P) \le N^{-1/2}$. A measurable approximate minimum-distance estimator over a countable dense subset of K satisfies $d(A\widehat{g}_N, Ag) \le 2d(P_N, Ag) + N^{-1}$. Compactness and injectivity make A^{-1} uniformly continuous on $A(K)$. Its inverse modulus, Markov's inequality, and boundedness of W_1 on $[1,2]$ yield uniform expected consistency.

In the nonquasianalytic case, the two distinct densities supplied by Theorem 1.9.1 satisfy common finite derivative constants A, B and induce identical observation laws. The triangle inequality forces minimax risk at least half their positive W_1 distance for all N. □

For example, $M_k = (k!)^\sigma$ gives the familiar threshold $\sigma = 1$. The criterion also handles scales arbitrarily close to analyticity: if $M_k = k! \prod_{j=1}^{k} \log(e+j)^\beta$, then $M_{k-1}/M_k = 1/[k \log(e+k)^\beta]$. Identification within the corresponding class therefore holds for $\beta \le 1$, whereas localized ambiguities exist for $\beta > 1$ whenever the observed moments are Müntz-sparse. Here $\beta \ge 0$ ensures the stated log-convex and factorial lower-bound hypotheses.

1.10 Prime-zeta questions: a prior-art audit and a uniform counting theorem

Belovas, Čepaitytė, and Sabaliauskas [21, Section 3] formulate three conjectures in the absolutely convergent half-plane. The first asks whether their right boundary $S = 1.7795446535\ldots$ is sharp. The second asks whether the ordinates, normalized by the observation height, become uniform when zeros are restricted to $\Re s > \gamma$, with $1 < \gamma < S$. The third asks for two-sided linear bounds for their number. We treat these as three separate targets, not as three newly solved open problems: the first is already implied by Sepulcre–Vidal [22, Theorem 4.3], and the classical Jessen–Tornehave theory [23, Sections 25–26 and Theorems 5–7] already supplies the density mechanism behind the other two. The following direct proof records the additional uniform quantifiers and removes an ambiguity about counting distinct or multiple zeros.

For positive coefficients a_p satisfying $0 < a_- \le a_p \le a_+ < \infty$, put

$$F_\omega(s) = \sum_p a_p \omega_p p^{-s}, \qquad \Omega = \prod_p \{z \in \mathbb{C} : |z| = 1\}.$$

Let $S > 1$ be the unique solution of

$$a_2 2^{-S} = \sum_{p\ge 3} a_p p^{-S}.$$

For $1 < \gamma < S$, denote by $N_\omega(\gamma; A, T)$ the number of *distinct* zeros in $\Re s > \gamma$, $A < \Im s \le A + T$, and by $N_\omega^{\mathrm{mult}}(\gamma; A, T)$ the count with multiplicity.

Theorem 1.10.1 (Uniform linear density and negligible multiple zeros)**.** *For every fixed $1 < \gamma < S$ there is $c_\gamma > 0$, depending on the coefficients but not on their phases, such that*

$$\sup_{\omega\in\Omega,\, A\in\mathbb{R}} \left| \frac{N_\omega(\gamma; A, T)}{T} - c_\gamma \right| \longrightarrow 0 \qquad (T \longrightarrow \infty). \tag{1.10.1}$$

The same limit holds for N_ω^{mult}. In particular the multiplicity excess is $o(T)$ uniformly in the twist and initial height. For each fixed $x \in (0,1)$,

$$\frac{N_\omega(\gamma; A, xT)}{N_\omega(\gamma; A, T)} \longrightarrow x \tag{1.10.2}$$

uniformly in ω, A.

Proof. We give the compact-hull argument, including the exceptional-set check.

Uniform local zero bounds. The family F_ω is compact for locally uniform convergence on $\Re s > 1$: coordinatewise convergence of phases and the summable majorant give convergence on each compact set. No hull member is identically zero, since $2^s F_\omega(s) \to a_2\omega_2 \neq 0$ along the positive real axis. Consequently every fixed compact rectangle in this half-plane has a uniform bound on the number of zeros, counted with multiplicity. Indeed, otherwise a convergent sequence of hull members would have unboundedly many zeros in that rectangle; choose finitely many disks covering it whose boundaries avoid the zeros of the nonzero limit, and apply Rouché on those boundaries. This gives a contradiction. The same bound holds after any vertical translation, which merely changes the phases. The triangle inequality shows that every hull member is zero-free in $\Re s > S$.

Uniform bounded densities. Fix a compact real interval $I \subset (1,\infty)$ and give Ω product Haar probability. The random variable $F_\omega(\sigma)$ has a planar density bounded uniformly for $\sigma \in I$. To see this, retain any five primes. The characteristic function of their sum is $\prod_{j=1}^{5} J_0(a_{p_j} p_j^{-\sigma}|u|)$, where $J_0(v) = (2\pi)^{-1}\int_0^{2\pi} e^{iv\cos\theta}\,d\theta$. The elementary estimate $|J_0(v)| \le C(1+|v|)^{-1/2}$ makes this product integrable on $\mathbb{R}^2$, uniformly in $\sigma \in I$. For completeness, the estimate follows by splitting the integral into intervals of length $|v|^{-1/2}$ around the stationary points of $\cos\theta$ and integrating by parts on their complement. Fourier inversion gives a bounded density; convolution with the independent remaining sum does not increase its supremum.

The pair $(F_\omega(\sigma), F_\omega'(\sigma))$ likewise has a bounded density on $\mathbb{R}^4$, uniformly on I. Use ten fixed primes and put $\lambda_j = \log p_j$. Its characteristic function, with dual variables $u, v \in \mathbb{C}$, is bounded by

$$\prod_{j=1}^{10} \left| J_0\Big(a_{p_j} p_j^{-\sigma}|u - \lambda_j v|\Big)\right|.$$

There is $b > 0$ such that at least nine of the ten quantities $|u - \lambda_j v|$ are at least $b(|u|^2+|v|^2)^{1/2}$. Otherwise two of them would both be smaller than this bound; invertibility of the two distinct linear forms, and a sufficiently small common b, would contradict the norm of (u,v). Thus the product is $O((1+\|(u,v)\|)^{-9/2})$, which is integrable in dimension four. Fourier inversion and convolution again give the claimed bounded density. Haar invariance makes both density bounds uniform also in the ordinate.

Boundary and multiple zeros are Haar-null events. All derivatives of the hull are uniformly bounded on a fixed compact rectangle and a neighborhood of it. On any fixed line segment, if a zero exists, a mesh of spacing h contains a point where $|F_\omega| \le Ch$. The planar density bound and the union bound give probability $O(h^{-1}h^2) = O(h)$; letting $h \downarrow 0$ proves that the probability of a zero on that segment is zero.

Similarly, if a multiple zero exists in a compact rectangle, a square mesh of size h contains a point where simultaneously $|F_\omega| \le Ch^2$ and $|F_\omega'| \le Ch$. The four-dimensional density bound gives probability $O(h^6)$ at one mesh point. There are $O(h^{-2})$ points, so the probability is $O(h^4)$ and tends to zero. We used the second-derivative bound and $F' = 0$ at the multiple zero in obtaining the stronger h^2 bound for F. Thus almost every hull member has only simple zeros in every fixed compact rectangle.

Counting as an integrable observable. Choose $R > S$ and let $n(\omega)$ count distinct zeros in $\gamma < \Re s < R$, $0 < \Im s \le 1$; define n_{mult} analogously. Both are uniformly bounded by the local zero bound. The possible discontinuities of n are contained in the phases having a boundary zero or a multiple zero in the closed rectangle; this is a closed Haar-null subset by the preceding argument. Rouché shows local constancy off it. The discontinuities of n_{mult} only require the boundary exceptional set. Consequently both observables admit continuous upper and lower approximations whose Haar integrals differ arbitrarily little. This last assertion follows by taking a small open neighborhood of their closed discontinuity set, covering its complement by finitely many neighborhoods of constancy, and using a partition of unity.

The flow $\Phi_t\omega = (\omega_p p^{-it})_p$ has uniform Haar averages for every continuous function on Ω. For finite characters this follows directly from $\sum k_p \log p \neq 0$ for every nonzero finite integer vector, by unique factorization. Finite character polynomials are uniformly dense in $C(\Omega)$, so the conclusion extends to continuous functions. Applying the continuous upper and lower approximations now gives, uniformly in the initial phase,

$$\frac{1}{T}\int_0^T n(\Phi_t\omega)\,dt \longrightarrow \int n\,d\mu, \qquad \frac{1}{T}\int_0^T n_{\mathrm{mult}}(\Phi_t\omega)\,dt \longrightarrow \int n_{\mathrm{mult}}\,d\mu.$$

The two integrals are equal because Haar-almost every hull member has only simple zeros in the rectangle. Each zero with ordinate between 1 and T contributes exactly one to the first sliding-window integral. Only the two end windows contribute a discrepancy, bounded uniformly by the local zero bound. The same assertion holds with multiplicity. Thus both counting functions divided by T converge uniformly to the common value $c_\gamma = \int n\,d\mu$.

Positivity. The frozen phases $\omega_2 = 1$, $\omega_p = -1$ for $p \ge 3$ give a simple zero at S, since the normalized germ has derivative $\sum_{p\ge3}(a_p/a_2)(2/p)^S \log(p/2) > 0$. A vertical phase translation places this zero at ordinate $1/2$. It is inside the rectangle because $\gamma < S < R$. Rouché then gives a nonempty open neighborhood of phases for which the rectangle contains a zero. Every nonempty open subset of the product torus has positive Haar measure, so $c_\gamma > 0$. Initial height is absorbed into the phase; this proves all uniformities in (1.10.1). Applying the same limit to T and xT proves (1.10.2). □

Corollary 1.10.2 (Status of the three published prime-zeta targets)**.** *For the ordinary prime zeta function, the assertions of [21, Conjectures 1–3] hold, with Conjectures 2 and 3 restricted to the nonempty strips $1 < \gamma < S$. The source first specifies $0 < \gamma < S$ and then imposes $\gamma > 1$, so this is precisely its stated admissible range. For $\gamma \ge S$ the ordinary prime zeta function has no zeros in $\Re s > \gamma$ and a positive linear lower bound would be false. Conjecture 1 follows already from [22, Theorem 4.3]; Conjectures 2 and 3 follow from Theorem 1.10.1. Their density mechanism is classical, so these implications are literature reconciliation rather than a claimed new major theorem.*

Proof. For the first assertion, the frozen simple zero and Kronecker approximation produce zeros of the ordinary prime zeta function arbitrarily near S, while the triangle inequality excludes the half-plane to its right. At $\Re s = S$, equality would force $-e^{-it\log(p/2)} = 1$ for every odd prime. The coordinates $p = 3, 5$ already contradict unique factorization, so the boundary itself is not attained. The first conjecture follows. The other two assertions are respectively (1.10.2) and $N(\gamma; 0, T) = c_\gamma T + o(T)$ with $c_\gamma > 0$. □

The unresolved endpoint is not absorbed by this argument. The additional statement $M_1(T) = O(T)$ in the discussion of the third conjecture is not proved here. The constants above depend on a compact rectangle strictly inside $\Re s > 1$; taking $\gamma \downarrow 1$ after $T \to \infty$ does not provide a bound at $\gamma = 1$. In particular, a bound for limiting densities alone would not

justify interchanging these limits. This is an explicit failed endpoint attempt, with the missing uniformity identified.

1.11 An internal extension: logarithmic weights at the kernel-budget boundary

This section addresses an internal quantitative question, not a separately verified published conjecture. The power-weight calculation in the integrated manuscript has a transition at exponent 3. Logarithmic weights reveal a second transition at logarithmic exponent -2, including an iterated logarithm exactly at that boundary.

For positive summable weights, define

$$\Psi_w(B) = \inf_{\substack{D_p\in\mathbb{Z}_{\ge 0},\ D \text{ finite} \\ \sum_p D_p \log p \le B}} \sum_p w_p \left(1 - \cos\frac{\pi}{D_p+2}\right).$$

The rectangular positive-kernel extremality theorem in the integrated manuscript identifies this with the least possible mean energy among nonnegative normalized kernels with that degree budget.

Theorem 1.11.1 (Complete logarithmic transition at exponent three)**.** *If $w_p \asymp p^{-3}(\log p)^{\kappa}$, then, as $B \to \infty$,*

$$\Psi_w(B) \asymp B^{-2} \begin{cases} (\log B)^{\kappa+2}, & \kappa > -2, \\ (\log\log B)^3, & \kappa = -2, \\ 1, & \kappa < -2. \end{cases} \tag{1.11.1}$$

For $\kappa < -2$, writing $L = \sum_p w_p^{1/3}(\log p)^{2/3} < \infty$, the stronger limit holds:

$$\lim_{B\to\infty} B^2 \Psi_w(B) = \frac{\pi^2}{2} L^3.$$

Proof. Put $c(d) = 1 - \cos(\pi/(d+2))$ and $L(x) = \sum_{p\le x} w_p^{1/3}(\log p)^{2/3}$. The elementary bounds $2/(d+2)^2 \le c(d) \le \pi^2/(2(d+2)^2)$ and Hölder give

$$\sum_p w_p c(D_p) \ge \frac{2L(x)^3}{(B + 2\sum_{p\le x}\log p)^2}. \tag{1.11.2}$$

Indeed, write each summand of $L(x)$ as $(w_p/(D_p+2)^2)^{1/3}((D_p+2)\log p)^{2/3}$ before applying Hölder. Prime partial summation yields

$$L(x) \asymp \begin{cases} (\log x)^{(\kappa+2)/3}, & \kappa > -2, \\ \log\log x, & \kappa = -2, \end{cases}$$

and $L(\infty) < \infty$ exactly when $\kappa < -2$. Taking $x = B$ in (1.11.2) proves the first two lower estimates; a single prime proves the remaining B^{-2} lower estimate.

Suppose first $\kappa > -2$, and set

$$A = \frac{B}{C(\log B)^{(\kappa+2)/3}}, \qquad D_p = \left\lfloor Ap^{-1}(\log p)^{(\kappa-1)/3} \right\rfloor.$$

The active range is $p \le x$ up to a fixed finite initial adjustment, where $x(\log x)^{-(\kappa-1)/3} = A$ and $x \asymp B/\log B$. The budget satisfies

$$\sum_p D_p \log p \le A \sum_{p \le x} \frac{(\log p)^{(\kappa+2)/3}}{p} \ll A(\log x)^{(\kappa+2)/3} \le B$$

for sufficiently large fixed C. As $D_p + 1$ is at least the quantity rounded down, the active energy is bounded by

$$CA^{-2} \sum_{p \le x} \frac{(\log p)^{(\kappa+2)/3}}{p} \ll B^{-2}(\log B)^{\kappa+2}.$$

The inactive tail is $O(x^{-2}(\log x)^{\kappa-1}) = O(B^{-2}(\log B)^{\kappa+1})$, which is smaller. The finitely many initially adjusted coordinates obey the same bound after increasing the constant.

For $\kappa = -2$, take

$$A = \frac{B}{C \log\log B}, \qquad D_p = \left\lfloor \frac{A}{p \log p} \right\rfloor.$$

Now $x \log x = A$ and $x \asymp B/(\log B \log\log B)$. The budget is $O(A \log\log x) \le B$, while the active energy is $O(A^{-2} \log\log x) = O(B^{-2}(\log\log B)^3)$. The tail is $O(x^{-2}(\log x)^{-3}) = O(B^{-2}(\log\log B)^2/\log B)$. This proves the matching upper estimate at the boundary.

Finally suppose $L < \infty$ and put $t_p = w_p^{1/3}(\log p)^{-1/3}$, $D_p = \lfloor Bt_p/L \rfloor$. These degrees have finite support and budget at most B. For each fixed prime,

$$B^2 w_p c(D_p) \longrightarrow \frac{\pi^2}{2} L^2 w_p^{1/3} (\log p)^{2/3}.$$

The same expression without $\pi^2/2$ and with an absolute constant is a summable majorant, since $D_p + 1 \ge Bt_p/L$. Dominated convergence proves the upper limiting constant. For the lower limit, fix finitely many primes. Along any sequence with bounded scaled energy, their degrees tend to infinity, so $c(D_p) = (\pi^2/2 + o(1))/(D_p + 2)^2$. The Hölder argument then gives the lower constant $(\pi^2/2)L(x)^3$ on that fixed set. Let the set increase to all primes. This proves the asserted exact limit. □

Corollary 1.11.2 (Critical-weight recurrence bounds)**.** *For exact prime frequencies, or relative prime frequencies with a fixed prime omitted, let $\mathcal{R}_w(\varepsilon)$ be the uniform energy covering time and $\ell = \log(1/\varepsilon)$. Under the preceding weight assumption,*

$$\log \mathcal{R}_w(\varepsilon) \ll \varepsilon^{-1/2} \begin{cases} \ell^{(\kappa+2)/2}, & \kappa > -2, \\ (\log \ell)^{3/2}, & \kappa = -2, \\ 1, & \kappa < -2. \end{cases}$$

Together with the small-ball lower bound $\log \mathcal{R}_w(\varepsilon) \gg \varepsilon^{-1/2} \ell^{(\kappa-3)/2}$, this yields

$$\lim_{\varepsilon \downarrow 0} \frac{\log\log \mathcal{R}_w(\varepsilon)}{\log(1/\varepsilon)} = \frac{1}{2} \qquad \textit{for every real } \kappa.$$

Proof. Use product extremizers at a budget for which $\Psi_w(B) \le \varepsilon/4$. Their inactive tail has mass at most $\varepsilon/4$, their Fourier norm has logarithm $O(B)$, and the linear certificate $(\varepsilon/2 - \mathcal{E}_Q)K$ has arithmetic height $O(B)$. The elementary rational-separation argument in the integrated manuscript therefore gives a covering time $\exp(CB)$. Invert the three orders in (1.11.1). The lower bound is the logarithmic-weight small-ball avoidance estimate. All displayed logarithmic factors are subpowers, proving the final limit. □

This sharp optimization is a statement about the specified positive Fourier certificate. It does not close the logarithmic discrepancy between the actual arithmetic covering time and the entropy lower bound. No additional conjecture is declared solved by that optimization.

1.12 Zeta-phase targets: a critical audit and three unresolved transfers

This section attacks three published distribution problems using the prime-phase framework. None of the three is claimed to be solved. The useful output is an independent audit of the elementary and arithmetic mechanisms, together with an exact obstruction to deducing equidistribution from phase domination alone. The global constant

$$c = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.672500703679\ldots$$

is the previously published Montgomery–Taylor value in the unconditional theorem of Alpöge–Furman, not a new global bound. Their Theorem A and Sections 2–5 are the relevant primary comparison [24]. The checks below do not independently validate a full formalization or all estimates in the earlier research drafts.

1.12.1 The published targets and their exact scales

Target Z1: interval statistics at the secondary scale. Ford–Zaharescu, Conjecture A, and Ford–Soundararajan–Zaharescu, Conjecture 1, ask for

$$\#\{0 < \gamma \le T : \{\alpha\gamma\} \in I\} = |I|N(T) + T\int_I g_\alpha(u)\,du + o(T), \tag{1.12.1}$$

uniformly in circular intervals I, for fixed $\alpha > 0$. The zeros are counted with multiplicity. For example, at $\alpha = \log p/(2\pi)$,

$$g_\alpha(u) = -\frac{\log p}{\pi} \sum_{k\ge 1} p^{-k/2} \cos(2\pi k u).$$

The primary locators are [25, Conjecture A, pp. 5–6] and [26, Conjecture 1, p. 2; Theorems 1–3]. The latter also distinguishes the sharp discrepancy conjecture from ordinary uniform distribution. Our shrinking-box assertions concern relative errors against $|I|N(T)$; even for a fixed interval, $o(N(T))$ is insufficient for the $o(T)$ remainder in (1.12.1).

Target Z2: distribution at the average-gap normalization. Çiçek–Gonek explicitly state, immediately before Conjecture 1, that uniform distribution of $\gamma \log T/(2\pi)$ for $0 < \gamma \le T$ is not known. Their Theorem 1.3 proves it under RH and Gonek's exponential-sum conjecture

$$\sum_{0<\gamma\le T} x^{i\gamma} \ll_\varepsilon Tx^{-1/2+\varepsilon} + T^{1/2}x^\varepsilon \qquad (x, T \ge 2). \tag{1.12.2}$$

See [27, Section 1, Conjecture 1 and Theorem 1.3]. The kth Weyl test of the normalized sequence is $\sum_{0<\gamma\le T}(T^k)^{i\gamma}$, for fixed integer $k \ne 0$. Thus a fixed-prime phase theorem is not the required assertion. Neither (1.12.2) nor its normalized-zero consequence is proved by the arguments audited here.

Target Z3: distribution modulo Gram points. Arias de Reyna records Odlyzko's conjecture that

$$x_n = \frac{\vartheta(\gamma_n)}{\pi}$$

is uniformly distributed modulo one; here ϑ is the Riemann–Siegel theta function. See [28, Section 1 and the questions at the end of Section 1.1]. This nonlinear normalization is distinct

from that in Target Z2. Its local frequency is $2k\vartheta'(T) = k\log(T/(2\pi)) + O(T^{-2})$. The paper proves a restricted range of complex-zero exponential tests in Theorem 3.1, which must not be confused with all integer Weyl tests for the real ordinates. No completion of the latter tests follows here.

The literature search performed for this audit found no primary source resolving Z1–Z3. This is a search outcome, not a proof of historical priority or an exhaustive current-status certificate.

1.12.2 What survives the audit of the localization mechanism

The following ingredients can be checked without presuming that a new zeta-distribution theorem has already been established.

Lemma 1.12.1 (Rank and inertia)**.** *Suppose $G = P + Q$ is Hermitian, $P \succeq 0$,* $\operatorname{rank} P \le s$, $\operatorname{tr} P \le s$, *$n_+(Q) \le b$, and $s + 2b \le M$. Then*

$$s \ge 4\operatorname{tr} G - \operatorname{tr} G^2 - 2M.$$

Proof. Let $\lambda_1 \ge \cdots \ge \lambda_r > 0$ be the positive eigenvalues of G. Since $G \preceq P + Q_+$, min–max gives $\lambda_{b+j}(G) \le \lambda_j(P)$. Thus the positive tail after the first b entries has at most s entries and sum at most s. For $q(x) = 4x - x^2$, use $q(x) \le 0$ on $x \le 0$, $q(x) \le 4$ everywhere, and $q(x) \le 2x + 1$ everywhere. Consequently $\operatorname{tr} q(G) \le 4b + 3s \le 2M + s$. □

Lemma 1.12.2 (Nontrivial rational resonances are sparse)**.** *Let S be a finite set of primes and let $q \ne 1$ be a positive rational supported on S. If $n = p^j$, $m = r^h$, with $j, h \ge 1$, and $qn^\epsilon m^\eta = 1$, where $\epsilon, \eta \in \{-1, 1\}$, then both p and r belong to S. Consequently*

$$\sum_{qn^\epsilon m^\eta = 1} \frac{\Lambda(n)\Lambda(m)}{\sqrt{nm}} \le \left(\sum_{p \in S} \frac{\log p}{\sqrt{p} - 1} \right)^2.$$

Proof. If $p \notin S$, its valuation must be cancelled by r, so $r = p$. The remaining exponent must vanish, whence $q = 1$, a contradiction. The same argument applies to r. Dropping the equation then factorizes the positive sum; for each prime, $\sum_{j \ge 1} (\log p) p^{-j/2} = (\log p)/(\sqrt{p} - 1)$. □

For a real even f supported strictly inside $(-1/2, 1/2)$, put $\phi(u) = f(u/L)$ and $\alpha_k = T + 2\pi k/L$. Direct Fourier inversion and Poisson summation give, for fixed λ and sufficiently large L,

$$\sum_{k \in \mathbb{Z}} e^{i\lambda\alpha_k} \widehat{\phi}(t - \alpha_k) \widehat{\phi}(t' - \alpha_k) = L e^{i\lambda(t+t')/2} \int \phi(u - \lambda/2)\phi(u + \lambda/2) e^{-i(t-t')u}\, du. \tag{1.12.3}$$

There are no aliases because the sum of the two frequency supports, translated by λ, contains no nonzero multiple of L. At $\lambda = 0, t = t'$, the normalized full-grid squared norm is one. This confirms the sign, sampling normalization, and location of the rational twist used in the weighted compression.

Writing $q = b/c$ in lowest terms, the difference $bn - cm$ is a nonzero integer in a nonresonant opposite-sign term. With $b, c \le H$ and $n, m \le X$, the relevant frequencies have separation $\gg (HX)^{-1}$. In an equal-sign term, closeness of qnm to one instead implies $bnm \asymp c \le H$, which gives an adequate lower bound as well. Thus the stated rational separation has no missing prime-pair hypothesis. This verification applies to rational twists; it does not extend automatically to an arbitrary real frequency.

The exact zero-side decomposition also passes the elementary check: an off-line reflected pair contributes

$$uu^{\mathsf{T}} + \bar{u}\bar{u}^{\mathsf{T}} = 2vv^{\mathsf{T}} - 2ww^{\mathsf{T}}, \qquad u = v + iw,$$

and has at most one positive direction. A small operator norm does not imply a small positive index. The localization argument must therefore retain the deep zeros and charge them by their count, while subtracting the shallow exterior contribution exactly. Selberg's density bound permits a depth $d_0 = B \log\log T / \log T$ at a cost $O(N(T, 2T)(\log T)^{-B/4})$; this estimate is stated in [25, Lemma 2]. For a fixed phase buffer, repeated integration by parts gives arbitrary inverse powers of the physical separation times $\log T$, enough for the corresponding trace and Hilbert–Schmidt tail errors.

Proposition 1.12.3 (The exact remaining implication)**.** *Let B be a fixed Haar-continuity phase region and let K be a smooth inner cutoff with $0 \le K \le 1$. Suppose the exact retained Weil matrix H_K satisfies*

$$\operatorname{tr} H_K = (\mu(K) + o(1))N, \qquad \operatorname{tr} H_K^2 \le (R\mu(K) + o(1))N,$$

and its simple-zero count s admits the rank–inertia decomposition of Lemma 1.12.1 with budget $M = (\mu(B) + o(1))N$. Then

$$\liminf \frac{s}{N} \ge (4 - R)\mu(K) - 2\mu(B).$$

Taking inner cutoffs and phase buffers to their limits gives $\liminf s/N \ge (2 - R)\mu(B)$.

Proof. Substitute the stated traces and budget in Lemma 1.12.1. All cutoffs and buffers are fixed before the height limit. Inner approximation then gives the last assertion. □

No contradiction or sign error was found in these central joints. That is a narrower conclusion than a new independent certification of the complete uniform arithmetic proposition and complex Gaussian box sandwich. In particular, subsequent closures should state explicitly whether they need the fixed-region theorem, the stronger shrinking-region theorem, or only the unconditional linear algebra. The results in the next subsection require none of the zeta estimates.

1.12.3 A sharp Fourier obstruction to an equidistribution upgrade

Theorem 1.12.4 (Sharp character bound under measure domination)**.** *Let μ be Haar probability measure on a finite or countable product of circles, and let χ be a nontrivial continuous character. Suppose*

$$c\mu \le \nu \le \mu, \qquad 0 < c < 1, \qquad m = \nu(\Omega), \qquad t = \frac{m - c}{1 - c} \in [0, 1].$$

Then

$$\left| \int \chi \, d\nu \right| \le \frac{1 - c}{\pi} \sin(\pi t). \tag{1.12.4}$$

For every prescribed $m \in [c, 1]$ equality is attainable. In particular, domination by fixed positive constants does not imply vanishing Fourier coefficients or equidistribution after normalization.

Proof. Write $d\nu = (c + (1 - c)g)d\mu$, where $0 \le g \le 1$ and $\int g \, d\mu = t$. Haar invariance gives $\int \chi \, d\mu = 0$. The pushforward of μ by χ is uniform on the circle: a nontrivial character of a product of circles is onto. Rotate by a unit complex number so that the integral to be bounded is real and nonnegative. Conditional expectation on the character coordinate reduces the problem to maximizing

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} g_0(\theta) \cos\theta \, d\theta, \quad 0 \le g_0 \le 1, \quad \frac{1}{2\pi} \int g_0 = t.$$

Let $A = [-\pi t, \pi t]$ and $a = \cos(\pi t)$. Pointwise, $(g_0 - 1_A)(\cos\theta - a) \le 0$. The integral of $g_0 - 1_A$ vanishes, so the displayed objective is at most $\sin(\pi t)/\pi$. Equality holds for $g_0 = 1_A$, including the endpoint cases by continuity. Pulling this density back through χ proves attainability. □

For the normalized probability ν/m, the sharp upper bound is

$$\frac{(1-c)\sin(\pi t)}{\pi(c+(1-c)t)}.$$

At the Montgomery–Taylor value of c, its maximum over t is approximately 0.1256444247, strictly positive. This number is an information bound for the abstract dominated class; it is not a claim that zeta phases attain it. The argument is an elementary rearrangement principle, and no historical novelty is asserted for that principle.

Corollary 1.12.5 (Lower domination leaves room for fixed angular bias)**.** *Even if every weak limit of a family of simple-zero phase submeasures satisfies $c\mu \le \nu \le \mu$, this property alone cannot prove uniform distribution of the full selected family.*

Proof. Take $0 < t < 1$ and the attaining density from Theorem 1.12.4. Its nonzero character coefficient remains nonzero after normalization. Hence the desired conclusion is false for the class specified solely by the measure inequalities. This counterexample is an obstruction to inference, not a counterexample involving actual zeta zeros. □

1.12.4 Why the three attacks stop at different boundaries

For Z1, the boundary smoothing and the depth scale are incompatible with simply upgrading the current proof to its secondary term. A cutoff boundary layer of width η has a Haar error of order η. Its Haar-predicted count error is of order $N\eta$; making this $o(T)$ in the sandwich requires $\eta = o(1/\log T)$. The existing complex-box argument removes deep zeros at depth $d_0 \asymp \log\log T/\log T$ and uses an analytic smoothing width s with $d_0/s = o(1)$ and s smaller than the boundary buffer. These conditions force $\eta \gg d_0$, the opposite range. This is an obstruction to this particular smoothing proof, not a lower bound on the true discrepancy or a disproof of Z1.

For Z2, the rational-character proposition requires $\log H = o(\log T)$ if its bandwidth is to remain asymptotically $\log T$. The Weyl test $x = T^k$ lies outside that range. Even when T^k happens to be rational, its rational height is of order T^k, not $T^{o(1)}$. Treating the associated phase as another fixed prime coordinate would therefore be an invalid substitution.

For Z3, linearizing the theta phase over a short height interval does not remove the same obstruction: its slope still has order $\log T$. The derivative varies slowly, but slow variation is not small frequency. A short-interval argument would also need a new uniform zero-count and twisted-moment input at that frequency.

Finally, the abstract fourth-moment certificate remains valid without giving a zeta closure. The polynomial identities

$$1 + \frac{4x}{\theta} - q_\theta(x) = \frac{(x^2 - 2x + \theta)^2}{\theta^2}, \qquad \frac{8}{\theta} - q_\theta(x) = \frac{(x-2)^2(x^2+2\theta)}{\theta^2},$$

with

$$q_\theta(x) = \frac{8\theta x - (4+2\theta)x^2 + 4x^3 - x^4}{\theta^2},$$

yield $s \ge (L_+)^2/D$ in the rank–inertia model, where $L = 4\,\mathrm{tr}\,G - \mathrm{tr}\,G^2 - 2N$ and $D = \mathrm{tr}(G(2I-G))^2$. The input moment vector $(1, 4/3, 2, 13/4)$ would give $16/21$, but the identities do not establish that vector for a zeta compression. None of Z1–Z3 is closed by postulating it.

Outcome of this branch. There are three documented published targets and zero claimed closures. The rigorously established addition is the sharp domination-to-Fourier bound, which identifies exactly what one cannot infer from the phase-measure comparison. The fixed-rational resonance argument and elementary localization reduction survive the present audit; their strongest analytic extensions still need a complete independent verification before being used to advertise a solution to an external open problem.

Part II

Observation Geometry and Exact Recovery

Replication Rank, Thorin Measures, and Zero Patterns

Source manuscript: 14 October 2026.

Source abstract. We extend two earlier manuscripts through several exact recovery and classification problems. For arbitrary Gaussian mixing laws with a linear covariance constraint, the sharp replication threshold is the smallest rank at which a projected Gram map has interior, plus one for an unknown mean. For covariance graphs this is the generic completion rank plus one. A multilevel Fourier construction gives the sharp adaptive total-budget Wasserstein rate $B^{-1/(d+2)}$ for regular d-dimensional components, including dimension two without a logarithm; support-dependent uniform components give $B^{-1/(d+1)}$ for $d \geq 2$. Symmetry and forced atoms of Thorin measures yield complete GGC classifications for two generalized McKay families. A recently published Bessel-zero order theorem implies self-decomposability in all three families of a stated external problem. Additional results include exact early-time heat sampling criteria, Cantor families of shape-constrained phase ambiguities, an exact random recurrence entropy constant, and inverse recovery of all prime phases from the right-edge critical-point divisor of an Euler product. For finite exponential sums we prove equidistribution of zero ordinates modulo every length outside a countable resonance set, answering a published existence question. We explicitly separate these conclusions from remaining conjectures and from established inputs. Twenty targets are assessed: fifteen have complete answers in their specified scope, one has a special-case answer, and four remain open here. These counts refer to research targets, not to twenty independent historically open conjectures.

2.1 The research question and the scope of the answers

The starting point is the 223-page integrated manuscript *Observation and Recovery in Arithmetic and Probability* [1], followed by the 42-page manuscript *Exact Recovery, Optimal Replication, and Bessel Classification* [29]. The present paper is a separate, self-contained sequel. Its purpose is to take the tools that survived those drafts, formulate further questions with explicit quantifiers, and determine which implications can actually be proved.

The common object is an observation map from a hidden parameter or law to a transform, a zero pattern, or a finite collection of experiments. Three mechanisms recur. Algebraic equations satisfied by attainable transform arguments produce compactly supported invisible perturbations. Analytic uniqueness turns a sufficiently rich observation set into identification. Positive representing measures turn classification into support and mass constraints. These mechanisms then transfer to graph completion, statistical allocation, heat observations, and zero dynamics. No claim that all these applications are instances of one universal theorem is needed.

2.1.1 What changes from the two predecessors

The previous scalar replication result does not decide arbitrary random covariance models. Theorems 2.2.3 and 2.3.1 replace a scalar count by the interior rank of a projected Gram map. This yields an exact combinatorial interpretation for graph-constrained covariances, and counterexamples that survive smoothness, strong log-concavity, and finitely many moment constraints. Theorem 2.2.4 further distinguishes a fixed experiment from changing terminal lengths.

The previous budget exponent $1/3$ becomes $1/(d+2)$ for general regular d-dimensional components, with matching lower bounds against every adaptive policy. The proof requires a transport-controlling Fourier norm in place of the one-dimensional distribution function. The different exponent $1/(d+1)$ for uniform endpoints exposes the role of component information, rather than dimension alone.

For the two remaining McKay GGC classifications, we give both positive Thorin measures and necessary mass inequalities. This supplies the missing directions of the classification, not just further sufficient examples. For the separate SD question we use an external order theorem posted on 7 October 2026 [39]; that recent theorem is an input, and its authors receive explicit credit.

The recurrence theorem identifies a leading constant for rapidly decreasing weights, a stronger conclusion in a different spectral class from the earlier order bounds. The zero-divisor theorem is an inverse statement: even discarding the values of the Euler logarithmic derivative retains every prime phase if its near-edge zeros are kept. Finally, the modular-ordinate theorem answers an external existence question for finite exponential sums and supplies equidistribution for generic lengths.

2.1.2 Twenty targets, with no conjecture-count inflation

In the following register, **E** means a question or a clause located in an external primary source; **N** means a question formulated within this project. A complete entry means that the precise statement in its row is proved below, including the negative direction when claimed. It does not certify historical novelty. In particular E1–E3 are clauses of one multipart source problem, and E4 contains three distributional cases of another source problem.

ID	Precisely scoped target	Result and proof location
E1	GGC classification of the third McKay density over its full density domain.	Complete: $\nu \le \mu + 1$ is necessary and sufficient. Theorem 2.5.4.
E2	GGC classification of the squared-Bessel fourth McKay density.	Complete: $-1/4 < \mu \le 1/2$ is exactly the GGC range. Theorem 2.5.5.
E3	HCM status of the additional density in source Theorem 12.	Complete: no $\mu > 0$ is HCM. Proposition 2.5.6.
E4	SD of the three families asked about in source §4.4.	Complete on all stated source ranges, using the credited Bessel-zero order theorem. Theorem 2.6.4.
E5	Existence of lengths giving dense zero ordinates modulo the length for rationally independent exponential sums.	Complete, with a stronger result: equidistribution for every nonresonant length. Theorem 2.10.7.
N6	Fixed joint replication needed for arbitrary Gaussian mean/covariance mixing.	Complete: $d+1$ observations, or d for a known mean. Theorems 2.2.3 and 2.2.6.
N7	Exact replication count for a linear covariance space or covariance graph.	Complete: $r(V)+1$, respectively $\operatorname{gcr}(G)+1$, with unknown mean. Theorem 2.3.1 and Corollary 2.3.3.
N8	Minimal vector-output count and terminal-length criterion with bounded covariance.	Complete: d outputs; an infinite set of terminal lengths is necessary and sufficient. Theorems 2.2.4 and 2.2.5.
N9	Optimal total observation budget for regular multivariate mixing laws.	Complete under stated component assumptions: $B^{-1/(d+2)}$ for every d. Theorems 2.4.1–2.4.2.
N10	Optimal grouped budget for unknown uniform-rectangle endpoint populations.	Complete for $d \ge 2$: $B^{-1/(d+1)}$. Corollary 2.4.8.
N11	Whether strong log-concavity, Gevrey regularity, and finitely many moments eliminate continuous Fourier-intensity ambiguity.	Complete negative answer: a compact Cantor family remains, even after translation and reflection are removed. Theorem 2.7.1.

ID	Precisely scoped target	Result and proof location
N12	Exact identification from early-time heat observations with derivative multiplicities.	Complete: $\sum_j r_j t_j = \infty$; localized shape-constrained counterexamples otherwise. Theorem 2.7.2.
N13	Exact leading logarithmic covering constant for rapidly decreasing weights and independent random frequencies.	Complete under the stated width condition: covering entropy divided by Haar small-ball entropy tends to one. Theorem 2.8.5.
N14	Whether one exact radial law determines a summable weighted product of circles.	Complete: every positive weight and its multiplicity are recoverable. Theorem 2.9.1.
N15	Whether the near-edge Euler critical-point divisor determines the entire prime twist.	Complete: coordinatewise inverse recovery and topological conjugacy. Theorem 2.10.3.
E16	Limiting zero density for arbitrary atomic stationary Gaussian analytic spectra.	Special case: an explicit conditional density for three frequencies whose two nonzero values have irrational ratio. The general program remains open here. Theorem 2.10.8.
N17	Matching entropy-order covering bounds for the exact prime flow with power exponent $\sigma > 3$.	Open here. One anchor prime's exponential degree cost can be removed, but the exponent gap remains. Proposition 2.8.8.
N18	SD classification on the extended density domains of the third and fourth McKay laws.	Open outside the source's affirmative parameter ranges. GGC classification does not decide every non-GGC SD case.
E19	Whether bounded maximum likelihood threshold forces bounded generic completion rank.	Open here. The replication interpretation does not supply the missing graph rank bound; see [34, p. 4].
E20	The original equal-group binomial population-MLE range question.	Open here. Optimal unequal allocation does not settle the prescribed equal-group estimator; see [13, Remark 3.2].

The fifteen completed rows consist of five external-question clauses across three source groups and ten newly formulated targets. They are not fifteen independent previously recognized conjectures. Among the three external source groups, the SD question in [3, §4.4] and the modular-density existence question in [48, Section 1] receive affirmative answers throughout their stated scope. The broader McKay program retains the SD extension in N18. E16 is only a special-case result, and rows 17–20 are not counted as completed.

The source audit was targeted, with a cutoff of 14 October 2026. Some arguments may already be implicit in older general results. In particular, Wishart deconvolution, hypergeometric ratio measures, multilevel Monte Carlo, almost-periodic spectra, and convolution phase ambiguities are established theory. Their use here is distinguished from the additional statements and proofs. A claim of an Annals-level breakthrough, historical priority, or the resolution of a famous longstanding conjecture would require evidence beyond this manuscript.

2.1.3 Reading map and notation

Sections 2.2 and 2.3 develop exact observability and its graph interpretation. Section 2.4 studies finite-budget statistical recovery. Sections 2.5 and 2.6 concern positive distributional representations. Section 2.7 treats exact ambiguity and heat observations. Sections 2.8 and 2.9 connect recurrence and radial geometry. Section 2.10 studies zero patterns and the external exponential-sum question.

All observation-identification assertions concern exact probability laws, not one finite sample. Statistical assertions explicitly use the total number B of observations and expected Wasserstein loss. Zero divisors always retain multiplicity. Constants may depend on fixed model parameters but not on the asymptotic budget or tolerance. Detailed assumptions appear in each theorem.

2.2 Multivariate Gaussian environments: replication, rank, and changing time scales

The fixed univariate stable-shape results do not describe an arbitrary random covariance matrix. In dimension d, the relevant observation geometry is the rank of a matrix of Fourier frequencies. This section derives three classifications: the exact fixed-group threshold $d+1$ for unknown mean and covariance; the threshold d when changing terminal-time designs are available; and the threshold d for centered covariance mixtures, equivalently integer-degree Wishart mixtures. These are internally posed extensions. No source inspected in the priority audit explicitly posed these exact classifications as open conjectures, and they are not counted as additional published open problems. The scalar normal question is not counted again.

Let $\mathbb{S}_d$ be the real symmetric matrices, with positive semidefinite and positive definite cones $\mathbb{S}_d^+$ and $\mathbb{S}_d^{++}$. Draw (A,Σ) from an arbitrary Borel probability μ on $\mathbb{R}^d \times \mathbb{S}_d^+$ and, conditional on this draw, let X_j be iid $N_d(A,\Sigma)$. Degenerate covariance matrices are permitted in the positive results. Write $S_n = \sum_{j=1}^n X_j$. Equality of observation laws always means equality of their complete distributions, rather than finitely many moments or one finite dataset.

2.2.1 The Fourier–Laplace transform and its attainable arguments

For a finite signed mixing measure ρ, define

$$H_u(Z) = \int \exp\{iu^T a - \operatorname{tr}(Z\Sigma)\}\, d\rho(a,\Sigma), \qquad u \in \mathbb{R}^d, \quad \operatorname{Re} Z \in \mathbb{S}_d^{++}. \tag{2.2.1}$$

For each fixed u, this is bounded and holomorphic on the indicated tube domain. Indeed, on a compact subset of the domain, $\operatorname{tr}((\operatorname{Re} Z)\Sigma) \geq c \operatorname{tr}\Sigma$; every derivative is bounded by a polynomial in $\operatorname{tr}\Sigma$ times $e^{-c\operatorname{tr}\Sigma}$. No moment of A or Σ is required.

We use the following elementary uniqueness fact. If $H_u(T) = 0$ for every real $T \in \mathbb{S}_d^{++}$ and every u, then $\rho = 0$. Holomorphic uniqueness first gives $H_u(I - iV) = 0$ for all $V \in \mathbb{S}_d$. These are all Fourier coefficients of the finite signed measure $e^{-\operatorname{tr}\Sigma}\rho$ on the Euclidean space $\mathbb{R}^d \times \mathbb{S}_d$. Fourier uniqueness and the strict positivity of the tilt prove the assertion. Holomorphic uniqueness on a real open set can be justified by iterating the one-variable identity theorem in a small coordinate box and then continuing on the connected tube domain.

Lemma 2.2.1 (Weighted Gram realization)**.** *Let $w_1,\dots,w_q > 0$ and $\tau = \sum_j w_j$. The possible pairs*

$$u = \sum_{j=1}^q w_j t_j, \qquad T = \frac{1}{2}\sum_{j=1}^q w_j t_j t_j^T, \qquad t_j \in \mathbb{R}^d,$$

are exactly those for which

$$C = T - \frac{uu^T}{2\tau} \succeq 0, \qquad \operatorname{rank} C \leq q-1. \tag{2.2.2}$$

Proof. Put $v = (\sqrt{w_1},\dots,\sqrt{w_q})^T$ and let the jth column of Y be $\sqrt{w_j}t_j$. Then $Yv = u$. The orthogonal decomposition

$$Y = \frac{uv^T}{\tau} + Z, \qquad Zv = 0, \qquad YY^T = \frac{uu^T}{\tau} + ZZ^T$$

proves necessity. Conversely, factor $2C = LL^T$ with L having $r = \operatorname{rank} C$ columns. Choose a $q \times r$ matrix V with orthonormal columns in $v^\perp$ and set $Z = LV^T$. The resulting Y gives the required frequencies. The case $r = 0$ uses the zero matrix Z. □

For a fixed group $(X_1, \dots, X_q)$ the joint characteristic function is $H_u(T)$ with $w_j = 1$. For independent Gaussian blocks of lengths w_j the same formula holds with the displayed weights: conditionally a block has mean $w_j A$ and covariance $w_j \Sigma$.

2.2.2 Determinantal differential operators give local counterexamples

Coordinates on $\mathbb{S}_d$ are σ_{ij}, $i \le j$, with Lebesgue measure in these independent coordinates. Define the symmetric matrix of commuting differential operators

$$(\nabla_\Sigma)_{ii} = \partial_{\sigma_{ii}}, \qquad (\nabla_\Sigma)_{ij} = \tfrac{1}{2}\partial_{\sigma_{ij}} \quad (i < j), \qquad (\nabla_\Sigma)_{ji} = (\nabla_\Sigma)_{ij}.$$

Integration by parts against the transform in (2.2.1) replaces ∇_Σ by T, and replaces $\nabla_a \nabla_a^T$ by $-uu^T$. Consequently

$$\mathcal{D}_\tau = \det(2\tau \nabla_\Sigma + \nabla_a \nabla_a^T), \qquad \mathcal{E} = \det(\nabla_\Sigma) \tag{2.2.3}$$

have transform multipliers $\det(2\tau T - uu^T)$ and $\det T$, respectively. Every determinant here is a polynomial in commuting constant coefficient operators; it is not a determinant of noncommuting operators. In particular, these are nonzero operators. For example, at $u = 0$ the first multiplier is $(2\tau)^d \det T$.

Lemma 2.2.2 (Local smooth perturbations with protected marginals)**.** *Let $\mathcal{P}$ be any nonzero real constant coefficient differential operator on $\mathbb{R}^d \times \mathbb{S}_d$. For every nonempty open set $U \subset \mathbb{R}^d \times \mathbb{S}_d^{++}$ and every integer $K \ge 0$, there are distinct compactly supported C^∞ probability densities p_+, p_- inside U whose difference is*

$$p_+ - p_- = 2\epsilon\, \partial_{a_1}^{K+1} \partial_{\sigma_{11}}^{K+1} \mathcal{P} f$$

for a real $f \in C_c^\infty(U)$ and $\epsilon > 0$. They have the same full A marginal, the same full Σ marginal, and the same joint polynomial moments of total degree at most K. They are strictly positive on a neighborhood of the support of their nonzero difference.

Proof. Choose a nonzero real bump f in a sufficiently small ball with closure in U and denote the displayed differentiated function by g. A nonzero constant coefficient operator cannot kill a nonzero compactly supported smooth function: its Fourier transform would give $P(i\xi)\widehat{f}(\xi) = 0$, and the complement of the zero set of the nonzero polynomial is dense. Continuity would imply $\widehat{f} = 0$. Hence $g \neq 0$. Choose a smooth nonnegative probability density p supported in a slightly larger ball in U, with p bounded below by a positive constant on $\operatorname{supp} g$. For sufficiently small $\epsilon > 0$, put $p_\pm = p \pm \epsilon g$. Integration of a compactly supported derivative shows $\int g = 0$, so these are probabilities. Integrating first in a_1 or in σ_{11} shows that both coordinate marginals of g vanish. Integration by parts $K + 1$ times in a_1 kills every polynomial of total degree at most K. The same argument shows that ϵ can be made arbitrarily small while the two densities remain distinct. □

Thus the negative results below hold in every small interior parameter neighborhood, with covariance uniformly bounded and bounded away from singularity. They do not arise from heavy tails, zero covariance, or a failure to know the separate parameter marginals. Compact support precludes matching all polynomial moments for distinct measures; only an arbitrarily prescribed finite degree is asserted.

2.2.3 Exact fixed-group replication threshold

Theorem 2.2.3 (Unknown mean and full covariance require exactly $d + 1$ views)**.** *The map*

$$\mu \longmapsto \int N_d(a, \Sigma)^{\otimes q}\, d\mu(a, \Sigma)$$

is injective on all Borel probabilities on $\mathbb{R}^d \times \mathbb{S}_d^+$ if and only if $q \geq d+1$. For every $q \leq d$, every neighborhood U and every finite moment degree K as in Lemma 2.2.2, the nonidentifiability witnesses can have all the properties in that lemma.

Proof. For $q \geq d+1$, Lemma 2.2.1 gives, for every fixed u, all T in the real open set $T - uu^T/(2q) \succ 0$. Equality of observed laws therefore makes H_u vanish on this open set. Holomorphic uniqueness gives its vanishing on the whole tube domain, and Fourier–Laplace uniqueness gives equality of the mixing laws.

For $q \leq d$, every attainable pair satisfies $\det(2qT - uu^T) = 0$, because the centered Gram matrix has rank at most $q - 1 < d$. Apply Lemma 2.2.2 with $\mathcal{P} = \mathcal{D}_q$. The transform of the difference of its two densities vanishes at every attainable pair. Thus all joint characteristic functions of the q observations agree, although the mixing densities are distinct. □

This gives a finite certificate for equality of entire Gaussian exchangeable laws: equality of their $(d+1)$-coordinate marginals forces equality of their directing parameter laws and hence of all their finite-dimensional distributions. No smaller fixed marginal has that property, even under the local smooth restrictions above. The number $d+1$ counts observations sharing one latent parameter; it does not count the number of independently sampled groups used in statistical estimation.

2.2.4 Changing the terminal length lowers the dimension threshold

For $n \geq d$ define the d-vector observation group

$$\mathcal{O}_n = (X_1, \ldots, X_{d-1}, S_n). \tag{2.2.4}$$

When $d = 1$, this means only S_n. The separate laws of these groups for different n are given; no joint coupling between experiments with different terminal lengths is assumed observed.

Theorem 2.2.4 (A rank-completing sequence of partial-sum experiments)**.** *Let $\mathcal{N} \subset \{d, d+1, \ldots\}$.*

1. *If $\sum_{n\in\mathcal{N}} n^{-1} = \infty$, the laws $\{\mathcal{L}(\mathcal{O}_n) : n \in \mathcal{N}\}$ identify every Borel mixing law on $\mathbb{R}^d \times \mathbb{S}_d^+$, without moments.*
2. *On the class whose covariance support is bounded, the same design is injective if and only if $\mathcal{N}$ is infinite. No tail restriction on the mean is needed for this assertion.*
3. *For every finite $\mathcal{N}$, there are local smooth nonidentifiability witnesses with identical parameter marginals and moments through any fixed finite degree, as in Lemma 2.2.2.*

Proof. An invertible linear transformation turns $\mathcal{O}_n$ into the independent conditional blocks

$$X_1, \ldots, X_{d-1}, \quad S_n - S_{d-1},$$

with positive lengths $1, \ldots, 1, n-d+1$ and total length n. Fix $u \neq 0$ and a positive semidefinite matrix C of rank $d-1$ such that u is not orthogonal to $\ker C$. Lemma 2.2.1 shows that equality of the observed laws gives

$$H_u\left(C + \frac{uu^T}{2n}\right) = 0 \qquad (n \in \mathcal{N}).$$

The function

$$F(z) = H_u(C + zuu^T), \qquad \operatorname{Re} z > 0,$$

is bounded and holomorphic, because $C + (\operatorname{Re} z)uu^T$ is positive definite. A nonzero bounded holomorphic function in a half-plane has zeros satisfying the Blaschke sum $\sum \operatorname{Re} z_j/(1+|z_j|^2) < \infty$. Applying this condition to $z_n = 1/(2n)$ proves $F = 0$ under the condition in (1).

Every $T \in \mathbb{S}_d^{++}$ is reached in this way. Indeed, put

$$\lambda = (u^T T^{-1} u)^{-1}, \qquad C = T - \lambda uu^T.$$

Conjugation by $T^{-1/2}$ shows that $C \succeq 0$ and has rank $d-1$. Its kernel is spanned by $T^{-1}u$, whose inner product with u is positive. Since $T = C + \lambda uu^T$, we get $H_u(T) = 0$. Continuity in u includes $u = 0$, and transform uniqueness proves (1).

If both candidate covariance supports are bounded, then F extends to an entire function of z: the quantity $u^T \Sigma u$ is uniformly bounded, while $e^{iu^T a}$ has modulus one. An infinite $\mathcal{N}$ gives zeros accumulating at zero, so the ordinary identity theorem suffices. The same argument proves the positive half of (2).

For a finite nonempty $\mathcal{N}$, use in Lemma 2.2.2 the nonzero operator

$$\mathcal{P} = \prod_{n \in \mathcal{N}} \mathcal{D}_n.$$

At the Fourier–Laplace arguments of experiment n, its factor $\det(2nT - uu^T)$ vanishes by the weighted Gram relation. The same single pair of compactly supported densities is therefore invisible to every experiment in the finite list. For the empty list take $\mathcal{P}$ to be the identity. This proves (3) and the negative half of (2). □

The harmonic condition in (1) is only asserted to be sufficient for unrestricted covariance laws. The exact sparse condition in that unrestricted multivariate setting is not settled here. The "infinite if and only if" statement is restricted to bounded covariance support.

Theorem 2.2.5 (Fewer than d vector outputs cannot recover an arbitrary covariance)**.** *Suppose $d \geq 2$ and $q < d$. Allow every experiment consisting of q vector outputs*

$$Y_\ell = \sum_{j=1}^{m} w_{\ell j} X_j, \qquad 1 \leq \ell \leq q,$$

with arbitrary finite m and arbitrary known real scalar weights $w_{\ell j}$. Even the collection of all these experiment laws does not identify the Gaussian mixing law. A single pair of local smooth mixing densities, with identical separate parameter marginals and any prescribed finite degree of equal joint moments, is invisible to every one of these experiments simultaneously. In particular, all separate partial-sum laws fail to identify the mixing law in every dimension $d \geq 2$.

Proof. For output frequencies $t_1, \ldots, t_q$, put $v_j = \sum_\ell w_{\ell j} t_\ell$. The transform is evaluated at

$$u = \sum_j v_j, \qquad T = \tfrac{1}{2} \sum_j v_j v_j^T.$$

Every v_j lies in the span of $t_1, \ldots, t_q$, so $\operatorname{rank} T \leq q < d$ and $\det T = 0$. Apply the local perturbation lemma with $\mathcal{P} = \mathcal{E}$. Its transform vanishes at every such argument, independently of the number or values of the weights. For separate partial sums, take $q = 1$ and all weights equal to one. □

The weights in this theorem are scalar and each output is a d-dimensional vector. Matrix-valued weights can alter the rank geometry and are not covered. Combining the last two theorems gives the sharp output count d for this class of changing-length experiments, compared with $d+1$ for one fixed iid group.

2.2.5 Centered mixtures and singular Wishart observations

Theorem 2.2.6 (Exact centered covariance and integer Wishart thresholds)**.** *If the mean is known and fixed, q shared-latent Gaussian replicates identify an arbitrary covariance mixing law on $\mathbb{S}_d^+$ if and only if $q \geq d$. For covariance laws supported in $\mathbb{S}_d^{++}$, the same threshold holds if only*

$$W_q = \sum_{j=1}^{q} (X_j - A)(X_j - A)^T$$

is observed, that is, for mixtures of Wishart laws with known integer degrees of freedom q. When $q < d$, compactly supported smooth covariance mixing densities already give counterexamples.

Proof. Subtract the known mean. For the full group the characteristic transform of a signed covariance mixing measure is $L(T) = \int e^{-\operatorname{tr}(T\Sigma)} d\rho(\Sigma)$ with $T = \frac{1}{2}\sum_j t_j t_j^T$. If $q \geq d$, all positive definite T are attainable, and Laplace uniqueness proves identification. If $q < d$, take a nonzero real bump f compactly supported in any small positive definite neighborhood. The nonzero function $g = \det(\nabla_\Sigma) f$ has total integral zero and transform $\det(T) L_f(T)$, which vanishes on every rank-q probe. Adding and subtracting a sufficiently small multiple of g to a smooth probability bump positive on its support produces the claimed covariance densities. This already makes their W_q laws equal.

It remains to prove sufficiency from W_q for $q \geq d$ and positive definite covariance. Its conditional density, with respect to Lebesgue measure on the positive definite cone, is

$$c_{d,q} |W|^{(q-d-1)/2} |\Sigma|^{-q/2} e^{-\operatorname{tr}(\Sigma^{-1}W)/2}.$$

Equality of two mixture laws therefore gives a zero difference of these densities almost everywhere. The integral after dividing by the positive factor in W is continuous on the positive definite cone: on a compact set with $W \succeq cI$, the quantity $|\Sigma|^{-q/2} e^{-c\operatorname{tr}(\Sigma^{-1})/2}$ is uniformly bounded. Thus the equality holds everywhere.

Push the signed covariance difference forward under $B = \Sigma^{-1}$, and fix $W_0 \succ 0$. The signed measure

$$d\eta(B) = |B|^{q/2} e^{-\operatorname{tr}(BW_0)/2} \, d\rho_*(B)$$

is finite, by the same eigenvalue bound. Its Laplace transform at $V \succ 0$ is the vanished mixture difference at $W_0 + 2V$. Laplace uniqueness gives $\eta = 0$, and strict positivity of the weight implies $\rho_* = 0$ and $\rho = 0$. □

The sufficiency side belongs naturally to classical Wishart deconvolution, rather than constituting a newly solved conjecture. Haff, Kim, Koo, and Richards develop nonparametric estimation of Wishart mixing densities on the positive definite cone using Helgason–Fourier analysis[30]. The rank argument above explicitly exhibits the obstruction in the singular integer-degree range.

2.2.6 A transfer to random anisotropic diffusions

Corollary 2.2.7 (Changing observation times versus one fixed time grid)**.** *Let $B_t = tA + \Sigma^{1/2} W_t$, with standard d-dimensional Brownian motion independent of an arbitrary random pair (A, Σ). The collection of joint laws*

$$\{\mathcal{L}(B_h, B_{2h}, \ldots, B_{dh}) : h > 0\}$$

identifies the environment law of (A, Σ). No collection of observations at fewer than d time points per experiment, even allowing all possible positive times, can have this property. On one fixed time grid, exactly $d + 1$ time points are necessary and sufficient.

Proof. Successive increments on a time grid are independent conditional on (A, Σ), with lengths equal to the time increments. For a fixed total length τ and q increments, the weighted Gram lemma applies verbatim. Thus one fixed grid has an open set of transform arguments exactly when $q \geq d + 1$; the proof of Theorem 2.2.3 applies, using $\mathcal{D}_\tau$ for the negative direction.

For the varying equal-step grid with $q = d$, fix $u \neq 0$ and $T \succ 0$. Set

$$\tau = \tfrac{1}{2} u^T T^{-1} u, \qquad h = \tau/d.$$

Then $T - uu^T/(2\tau)$ is positive semidefinite of rank $d - 1$, so $H_u(T)$ is directly observed at this time grid. Continuity includes $u = 0$ and transform uniqueness gives the positive claim. For $q < d$, every grid produces a covariance frequency matrix of rank at most q, so the single covariance-determinant perturbation used in Theorem 2.2.5 is invisible on all the grids simultaneously. □

This separates recovery of a random matrix environment from recovery of its one-time predictive distributions. In particular, for $d \geq 2$ all one-time laws of such random-drift diffusions can coincide while their environment laws and full process laws differ.

2.2.7 Why finite Gaussian mixtures behave differently

Proposition 2.2.8 (The classical finite-mixture contrast)**.** *If both candidate mixing laws have finite support on distinct Gaussian parameters, one observation already identifies them. This is identification within the finite-support class, not among all Borel competitors.*

Proof. For the finitely many distinct parameter pairs choose a vector v for which the projected pairs $(v^T a_j, v^T \Sigma_j v)$ are distinct. Such a v exists: for each distinct pair of parameters, at least one of the linear mean difference or quadratic covariance difference is a nonzero polynomial, and finitely many of their common zero sets cannot cover $\mathbb{R}^d$.

It remains to prove linear independence of the finitely many functions $e^{im_j t - v_j t^2/2}$ with distinct (m_j, v_j) and $v_j \geq 0$. In a zero linear combination, let v_* be the smallest variance and multiply by $e^{v_* t^2/2}$. The terms of larger variance tend to zero as $t \to \infty$. The remaining trigonometric polynomial $P(t) = \sum_{v_j = v_*} c_j e^{im_j t}$ therefore tends to zero. Its mean square on $[0, R]$ tends, as $R \to \infty$, to $\sum_{v_j = v_*} |c_j|^2$, because the means in this group are distinct. Since P tends to zero and is bounded, this mean square tends to zero. All coefficients in the smallest-variance group vanish. Iterating proves the claim. □

The proposition is classical; it is associated with the multivariate Gaussian case of Yakowitz and Spragins[32] and is rederived only to prevent a finite-versus-arbitrary-mixture confusion. A single Gaussian can already have a nontrivial continuous location-mixture representation if arbitrary competitors are allowed: mix $N_d(a, \Sigma/2)$ over $a \sim N_d(0, \Sigma/2)$ to obtain $N_d(0, \Sigma)$.

2.2.8 Source status and limits of the priority check

Vandermeulen and Scott's grouped-sample paper[31], Theorem 4 on printed page 5 of the inspected 16-page preprint, establishes a sharp universal $2m - 1$ group-size bound for mixtures with m arbitrary component measures. That finite-component statement neither asserts nor contradicts the present thresholds for arbitrary mixing laws restricted to a Gaussian parameter family. Haff et al.[30], Sections 2 and 3 (reprint pages 2–6), already analyze invertible Wishart convolution on the positive definite cone. These are related published results, not open problems claimed solved here.

Searches also located Barndorff-Nielsen's classical paper on mixtures of exponential families[33], but its full text could not be retrieved in this audit. We therefore do not claim that the fixed-group threshold is historically new or absent as a corollary of that literature. Searches of the exact rank and varying-terminal-length formulations did not locate an exact prior statement. This bounded negative search is not a proof of priority. The statements and counterexamples above are proved here independently; historical novelty remains a separate question.

2.3 Covariance replication thresholds are generic completion ranks

The finite differential obstructions in the integrated manuscript lead to a different question: can the exact number of observations needed for an arbitrary Gaussian mixing law be read from an algebraic invariant of its covariance constraints? The answer is affirmative. For a graph constraint, the invariant is the classical generic completion rank, not the maximum likelihood threshold. This is a transfer between two observation problems; it does not solve the open combinatorial problem of computing these ranks for every graph.

Let V be a linear subspace of the real symmetric $d \times d$ matrices containing a positive definite matrix. Equip it with the Frobenius inner product and let π_V be orthogonal projection. Put $C = V \cap \mathbb{S}^d_+$ and $C^\circ = V \cap \mathbb{S}^d_{++}$. For $r \geq 0$ define the polynomial Gram map

$$\Gamma_r : \mathbb{R}^{d\times r} \longrightarrow V, \qquad \Gamma_r(Y) = \pi_V(YY^T), \qquad \Gamma_0 = 0,$$

and let $r(V)$ be the least r for which its image has nonempty interior in V. Equivalently, its Jacobian is surjective at some point. In particular $1 \leq r(V) \leq d$.

Conditionally on (A, Σ), the observations are iid $N_d(A, \Sigma)$, and the mixing law is an arbitrary Borel probability on $\mathbb{R}^d \times C^\circ$. A q-view law is the joint law of q such observations sharing (A, Σ).

Theorem 2.3.1 (Exact threshold for a linear covariance model)**.** *The q-view law identifies every such mixing probability if and only if*

$$q \geq r(V) + 1.$$

If the mean is fixed and known, the exact threshold is instead $q \geq r(V)$. Below either threshold, distinct compactly supported smooth mixing densities with identical observations exist in every open ball whose closure lies inside the parameter domain. They can be chosen strongly log-concave on their common convex support and can agree on all polynomial moments up to any prescribed finite degree. The positive assertions require no moment or support assumptions.

Lemma 2.3.2 (The Gram image and polynomial annihilators)**.** *If $r < r(V)$, a nonzero real homogeneous polynomial P of positive degree on V vanishes on $\Gamma_r(\mathbb{R}^{d\times r})$. If $r \geq r(V)$, that image contains a nonempty open subset of the interior of the dual cone $C^* = \{Z \in V : \langle Z, \Sigma\rangle \geq 0$ for all $\Sigma \in C\}$.*

Proof. The image is semialgebraic by the real projection theorem. A polynomial map whose image has empty interior has image of dimension less than $\dim V$; its Zariski closure is then proper, so a nonzero polynomial vanishes on it. Equivalently one can use the characteristic-zero Jacobian criterion for algebraic dependence of the coordinates of Γ_r. Since the image is a cone, every homogeneous part of an annihilating polynomial also annihilates it. Its constant part is zero, giving the first assertion. These elementary real-algebraic facts are used only for polynomial maps, where dimension and Jacobian rank agree.

Every Gram image lies in C^*, because $\langle \pi_V(YY^T), \Sigma\rangle = \operatorname{tr}(Y^T\Sigma Y) \ge 0$. An open subset of this image is therefore in $\operatorname{int} C^*$. Surjectivity at one point gives such an open subset by the submersion theorem. The reverse implication follows either from Sard's theorem or the polynomial dimension statement. Finally, Γ_d has surjective differential at every invertible Y, proving $r(V) \le d$. □

Proof of the identification direction. Choose orthonormal coordinates on V. For the signed difference ρ of two candidate mixing laws and fixed $u \in \mathbb{R}^d$, put

$$H_u(Z) = \int e^{iu\cdot a - \langle Z,\Sigma\rangle}\, d\rho(a,\Sigma), \qquad \operatorname{Re} Z \in \operatorname{int} C^*.$$

This is bounded and holomorphic on that tube. Indeed, on a compact subset of its real base there is $c > 0$ with $\langle \operatorname{Re} Z, \Sigma\rangle \ge c\|\Sigma\|$ for $\Sigma \in C$. All covariance derivatives are dominated by polynomials times $e^{-c\|\Sigma\|}$. The mean factor has modulus one.

The joint characteristic function at $t_1, \dots, t_q$ gives $H_u(Z)$ with

$$u = \sum_i t_i, \qquad Z = \frac{1}{2}\pi_V\left(\sum_i t_i t_i^T\right) = \frac{1}{2q}\pi_V(uu^T) + \frac{1}{2}\Gamma_{q-1}(Y).$$

Here Y ranges over all $d \times (q-1)$ real matrices: write the centered columns $t_i - u/q$ in an orthonormal basis of the sum-zero subspace of $\mathbb{R}^q$. For each fixed u, if $q - 1 \ge r(V)$, these probes contain a real open subset of $\operatorname{int} C^*$. Translation by $\pi_V(uu^T)/(2q) \in C^*$ preserves that interior. Equality of the observations makes H_u zero on that open subset. The identity theorem, first in a complex polydisc and then on the connected tube, makes H_u identically zero.

Fix $Z_0 \in \operatorname{int} C^*$. The Fourier transform of the finite signed measure $e^{-\langle Z_0,\Sigma\rangle}\rho$ is $H_u(Z_0 - iW)$, zero for all $u \in \mathbb{R}^d$ and $W \in V$. Fourier uniqueness proves $\rho = 0$. With known mean, no sum constraint is present and the covariance probes are $\frac{1}{2}\Gamma_q(Y)$; the same argument gives the threshold $r(V)$. □

Proof of the sharp counterexamples. Suppose $q - 1 < r(V)$ and take P from Lemma 2.3.2. The real polynomial

$$Q(u, Z) = P\left(2Z - \frac{1}{q}\pi_V(uu^T)\right)$$

is nonzero, vanishes on all q-view probes, is even in u, and satisfies $Q(0,0) = 0$. In orthonormal covariance coordinates θ, define the constant-coefficient operator

$$D = Q(i\partial_a, \partial_\theta).$$

All its coefficients are real: every monomial has even total degree in u. It is nonzero. For a nonzero real bump ϕ supported in a smaller ball inside the requested parameter ball, set $h = D\phi$. Integration by parts gives

$$\int e^{iu\cdot a - \langle Z,\Sigma\rangle} h(a,\Sigma)\, da\, d\Sigma = Q(u, Z) \int e^{iu\cdot a - \langle Z,\Sigma\rangle} \phi(a,\Sigma)\, da\, d\Sigma.$$

Thus h has mass zero and annihilates the entire observation map. Moreover $h \ne 0$. A nonzero constant-coefficient operator cannot annihilate a nonzero compactly supported smooth function: on taking the ordinary Fourier transform, its nonzero polynomial symbol is nonzero on a real open set, forcing an entire Fourier transform to vanish there and hence everywhere if the result were zero.

To make all moments through degree K vanish, replace D by $D\partial_{a_1}^{2M}$ with $2M > K$; its symbol remains nonzero and its observation zeros persist. Integration by parts annihilates each polynomial of degree at most K.

Choose a slightly larger ball of radius R still inside the requested parameter ball, with Euclidean coordinates x about its center, and use the normalized baseline

$$g_0(x) = c \exp\left(-\frac{1}{1 - \|x\|^2/R^2}\right) \mathbf{1}_{\{\|x\|<R\}}.$$

It is globally smooth, positive in the ball, and its log-Hessian is bounded above by $-2R^{-2}I$. For sufficiently small $\varepsilon > 0$, $g_0 \pm \varepsilon h$ are distinct normalized positive densities with the same support and observations. On the compact support of h, smallness in C^2 preserves a strictly negative log-Hessian; outside it the baseline is unchanged. The moment constraints persist.

For known mean use $P(2\partial_\theta)$ and, for moment matching, an additional even high derivative in one covariance coordinate. This proves the stated negative conclusions in both models. □

2.3.1 An exact graph-theoretic interpretation

For a finite simple graph G on d vertices, let

$$V_G = \{\Sigma \in \mathbb{S}^d : \Sigma_{ij} = 0 \text{ when } i \neq j \text{ and } ij \notin E(G)\}.$$

This is a covariance-zero model; it is not the precision-zero graphical model used to define the usual maximum likelihood threshold. The same graph nevertheless indexes the same matrix-entry projection. Generic completion rank is classically defined as the least rank at which this projection of the symmetric determinantal variety is dominant; see [34, Definition 1.1].

Corollary 2.3.3 (Replication equals generic completion rank, plus one)**.** *For arbitrary Gaussian mixtures with covariance in $V_G \cap \mathbb{S}^d_{++}$, the exact joint replication thresholds are*

$$q_{\text{unknown mean}}(G) = \operatorname{gcr}(G) + 1, \qquad q_{\text{known mean}}(G) = \operatorname{gcr}(G).$$

In particular, with unknown mean the thresholds are 2 for an edgeless graph, 3 for a forest containing an edge, 4 for a cycle, and $\omega(G) + 1$ for a chordal graph. For the complete graph they are $d + 1$ and d, respectively.

Proof. Over the complex numbers every symmetric matrix of rank at most r can be written YY^T. Dominance of the projection is therefore equivalent to a nonzero maximal Jacobian minor of Γ_r. Its coefficients are real, so a nonzero minor is also nonzero at some real Y. Thus the classical generic completion rank equals $r(V_G)$. Apply Theorem 2.3.1.

Here are elementary checks of the displayed graph classes. A clique of size k forces $r(V_G) \geq k$, because its specified principal block must have rank at most r and cannot contain an open set if $r < k$. If G has an elimination ordering with at most k later neighbors per vertex, take $r = k+1$ and choose row vectors y_i such that every set of at most r of them is linearly independent. A linear dependence among the differentials of the specified Gram entries is a stress with vertex equations

$$2a_i y_i + \sum_{j:ij\in E(G)} b_{ij} y_j = 0.$$

At the first vertex in the ordering, independence of its vector and its remaining neighbors forces every incident coefficient to zero. Remove it and continue. All stress coefficients vanish, so the Jacobian is surjective. Consequently $r(V_G) \leq \operatorname{degeneracy}(G) + 1$. This proves the forest claim and, by a perfect elimination ordering, the chordal claim. An edgeless graph has independent positive diagonal entries already at rank one. For a cycle, degeneracy gives $r \leq 3$; at rank two the symmetric rank-two variety on its $d \geq 3$ vertices has dimension $2d - 1$, whereas d diagonals and d edges give $2d$ specified entries. Hence $r \geq 3$. The complete graph follows from the clique bound. □

The graph invariant is classical; the asserted transfer concerns the identification of an entire probability law on mean and covariance space. Generic matrix completion, finite Gaussian-mixture identifiability, and arbitrary mixing-law identification have different quantifiers. In particular the counterexamples above are smooth continuous mixtures, not finite collections of Gaussian components.

2.3.2 A second transfer: likelihood existence does not measure recovery

Blekherman–Sinn prove that $\mathrm{mlt}(K_{5,5}) = 4$ whereas $\mathrm{gcr}(K_{5,5}) = 5$ and, more generally, $\mathrm{gcr}(K_{m,m}) = m$ for $m \geq 3$ while $\mathrm{mlt}(K_{m,m})$ has order $\sqrt{m}$ [34, Corollary 2.11 and Theorems 2.1, 2.7]. The preceding corollary therefore gives an exact threshold of six joint replicates in the unknown-mean $K_{5,5}$ covariance mixture, and a threshold $m+1$ for $K_{m,m}$. The gap from the likelihood-existence threshold is unbounded. This is a transfer of their established graph computations, not a new computation of those graph invariants.

The conjectures in [35, Section 2] concerning equality of MLT and GCR for sparse random graphs remain combinatorial questions. Theorem 2.3.1 gives a new observation-theoretic interpretation of GCR but supplies no proof that it equals MLT on those graphs. Similarly, the general question in [34, p. 4] asking whether bounded MLT forces bounded GCR is not settled here. These are explicit external targets checked against the transfer; the missing step is a bound on the Gram-map rank, not an additional transform uniqueness argument.

2.3.3 Quantitative consequence of a fixed replication obstruction

Corollary 2.3.4 (A sample-size-independent lower bound below the threshold)**.** *Fix any of the subthreshold models in Theorem 2.3.1, and any finite number of prescribed moment constraints. There is a compact class of smooth, strongly log-concave mixing densities inside the parameter domain and a constant $c > 0$ such that the minimax expected W_1 error is at least c, regardless of the number of independent groups, provided every group has at most q observations. The conclusion also allows randomized adaptive choice of group sizes up to this same cap.*

Proof. Take the two densities supplied by the theorem. Their q-view laws agree, hence so do all their lower-view marginals. Each adaptive group experiment can be simulated from an independent full q-tuple for each fresh individual by revealing coordinates only when requested. The transcript laws therefore agree for the two models. The triangle inequality gives a worst-case risk at least half the positive W_1 distance between their mixing laws. The two-point class itself is compact. □

This lower bound concerns a fixed cap on within-individual replication. It is compatible with the polynomial total-budget rates obtained later by allowing the replication depths to grow.

2.4 Dimension and component regularity in optimal replication budgets

The scalar allocation theorem extends to an arbitrary latent dimension, but its distribution-function proof does not. A smoothed Fourier construction provides the missing replacement. It gives the exact total-budget rate $B^{-1/(d+2)}$ for regular d-dimensional components, including $d = 2$ without a logarithmic loss. A directed relative-entropy argument also covers support-dependent components: mixtures of d-dimensional uniform rectangles have rate $B^{-1/(d+1)}$ for

$d \geq 2$. Both statements allow every adaptive randomized replication policy in their lower bounds and use predetermined unequal group sizes in their upper bounds.

Multilevel telescoping, negative Sobolev bounds for transport, and population learning are established methods; see [9, 10, 36, 11]. The results below are complete deductions for the stated grouped experiments. Historical priority for this general combination is not asserted. The loss is Wasserstein distance between arbitrary mixing laws, including singular laws. No density, separation, or finite-support assumption on the mixing law is used.

2.4.1 A general rate theorem

Let $\Theta = [0,1]^d$ with its Euclidean metric, and let $(P_\theta)_{\theta\in\Theta}$ be a known measurable family of probability laws on a standard Borel observation space. A new individual draws a latent θ independently from an unknown $G \in \mathcal{P}(\Theta)$. Repeated observations of that individual are independent with law P_θ, conditional on its latent parameter. A policy can start individuals, revisit them, and randomize using the entire observed history. At most B observations are allowed, with B deterministic. Write $\mathcal{R}_B^{\mathrm{ad}}$ for the minimax expected W_1 risk over all policies and probability-valued estimators, and $\mathcal{R}_B^{\mathrm{fix}}$ for the restriction to predetermined, possibly unequal group sizes.

We impose the following component-estimation assumption: for an integer $b_0 \geq 1$ and constants $A, \alpha > 0$, there are measurable estimators $T_r \in \Theta$, using $b_0 r$ observations from one individual, such that

$$\sup_{\theta\in\Theta} \mathbb{E}_\theta \|T_r - \theta\|^2 \leq A^2 r^{-2\alpha}, \qquad r \in \mathbb{N}. \tag{2.4.1}$$

Independent auxiliary randomization is allowed.

Theorem 2.4.1 (Dimension-dependent multilevel upper bound)**.** *Under* (2.4.1)*, for all sufficiently large B,*

$$\mathcal{R}_B^{\mathrm{ad}} \leq \mathcal{R}_B^{\mathrm{fix}} \leq C \begin{cases} B^{-\alpha/(1+d\alpha)}, & d \geq 2, \\ B^{-\alpha/(1+\alpha)}, & d = 1,\ 0 < \alpha < 1, \\ B^{-1/2}\log(eB), & d = 1,\ \alpha = 1, \\ B^{-1/2}, & d = 1,\ \alpha > 1. \end{cases} \tag{2.4.2}$$

The constants depend only on d, α, A, b_0. In particular, root-mean-square component estimation, $\alpha = 1/2$, gives $CB^{-1/(d+2)}$ in every dimension.

For the lower bound, a one-sided condition suffices. Suppose some cube $J = a + [0,s]^d \Subset \Theta$ and constants $D, \kappa > 0$ satisfy

$$\mathrm{KL}(P_\theta \| P_{\theta+te_1}) \leq Dt^\kappa \quad \text{whenever } t > 0 \text{ and } \theta, \theta + te_1 \in J. \tag{2.4.3}$$

The reverse relative entropy can be infinite.

Theorem 2.4.2 (Adaptive lower bound from directed local information)**.** *Under* (2.4.3)*,*

$$\mathcal{R}_B^{\mathrm{ad}} \geq c \max\{B^{-1/2}, B^{-1/(d+\kappa)}\}. \tag{2.4.4}$$

Consequently, if (2.4.1) *holds with $\alpha = 1/\kappa$, the following rates are exact for both adaptive and fixed unequal allocations:*

$$\mathcal{R}_B^{\mathrm{ad}} \asymp \mathcal{R}_B^{\mathrm{fix}} \asymp \begin{cases} B^{-1/(d+\kappa)}, & d + \kappa > 2, \\ B^{-1/2}, & d + \kappa < 2. \end{cases} \tag{2.4.5}$$

At $d + \kappa = 2$, necessarily $d = 1, \kappa = 1$, the proved bounds are $cB^{-1/2}$ and $CB^{-1/2}\log(eB)$. No sharp claim is made at that critical point.

The reciprocal relation $\alpha = 1/\kappa$ in the last assertion is an assumption tying component estimation to local information. Without it, the separately stated upper and lower bounds still hold, but need not match.

2.4.2 A Fourier transport bound and the critical dimension

Embed $[0,1]^d$ in the flat torus $\mathbb{T}_d = \mathbb{R}^d/(4\mathbb{Z})^d$. The torus distance agrees with Euclidean distance between any two points of this cube. For a mass-zero finite signed measure ρ on the torus, define

$$\|\rho\|_{\mathrm{KR}} = \sup_{\mathrm{Lip}(f)\le 1} \left| \int f \, d\rho \right|.$$

Constants in f do not affect this expression. For probability measures it is the torus W_1 distance. In particular it is the original Euclidean W_1 when both laws are supported on Θ. For $k \in \mathbb{Z}^d$, put

$$\widehat{\rho}(k) = \int e^{-\pi i k\cdot x/2} \, d\rho(x), \qquad \|\rho\|_{-1}^2 = \sum_{k\neq 0} \frac{|\widehat{\rho}(k)|^2}{\|k\|^2}.$$

Lemma 2.4.3 (Smoothed point masses in a transport-controlling Hilbert norm)**.** *There is $C_d < \infty$ such that every smooth mass-zero signed density satisfies*

$$\|\rho\|_{\mathrm{KR}} \le C_d \|\rho\|_{-1}. \tag{2.4.6}$$

Let K_h be the law on $\mathbb{T}_d$ of hZ modulo $4\mathbb{Z}^d$, where $Z \sim N(0, I_d)$ and $0 < h \le 1$. Then

$$W_1(K_h * \mu, \mu) \le \sqrt{d}\, h \quad \text{for every probability law } \mu.$$

For any coupled X, Y with $\mathbb{E}\|X - Y\|^2 \le a^2$, where the norm uses representatives in Θ,

$$\mathbb{E}\|K_h * (\delta_X - \delta_Y)\|_{-1}^2 \le C_d \sum_{k\neq 0} e^{-ch^2\|k\|^2} \|k\|^{-2} \min\{1, a^2\|k\|^2\}. \tag{2.4.7}$$

Proof. For a smooth 1-Lipschitz periodic function f, Parseval's identity and its gradient bound imply $\sum_{k\neq 0} \|k\|^2 |\widehat{f}(k)|^2 \le C_d$, with Fourier coefficients of functions normalized by torus volume. Fourier duality and Cauchy–Schwarz prove (2.4.6). A general Lipschitz function is uniformly approximated by its convolution with smooth probability kernels, without increasing its Lipschitz constant, so the same conclusion holds for the supremum defining $\|\cdot\|_{\mathrm{KR}}$.

Couple $X \sim \mu$ to $X + hZ$ modulo the torus. Their torus distance is at most $h\|Z\|$, which proves the smoothing bound. The Fourier coefficient of K_h is $\exp(-\pi^2 h^2 \|k\|^2/8)$. Also

$$|e^{-\pi i k\cdot X/2} - e^{-\pi i k\cdot Y/2}|^2 \le \min\{4, \tfrac{\pi^2}{4}\|k\|^2 \|X - Y\|^2\}.$$

Taking expectations and using Tonelli's theorem proves (2.4.7). □

Lemma 2.4.4 (The allocation sum)**.** *Set $r_\ell = 2^\ell$, $R = 2^L$ and $h = R^{-\alpha}$. Consider a same-individual coupling of T_{r_ℓ} and $T_{r_{\ell-1}}$, using a prefix for the coarser statistic. The smoothed signed differences have second moments in $\|\cdot\|_{-1}$ bounded by $C v_\ell$, where one may choose*

$$v_\ell = \begin{cases} r_\ell^{-\alpha}, & d = 1, \\ 1 + \log(R/r_\ell), & d = 2, \\ R^{\alpha(d-2)}, & d \ge 3. \end{cases} \tag{2.4.8}$$

At level zero the centered random measure $K_h * (\delta_{T_1} - \mathcal{L}(T_1))$ obeys the same bound. Writing $S_L = \sum_{\ell=0}^{L} \sqrt{r_\ell v_\ell}$ gives

$$S_L \le C \begin{cases} R^{(1-\alpha)/2}, & d = 1,\ \alpha < 1, \\ 1 + \log R, & d = 1,\ \alpha = 1, \\ 1, & d = 1,\ \alpha > 1, \\ R^{[1+\alpha(d-2)]/2}, & d \ge 2. \end{cases} \tag{2.4.9}$$

All bounds are uniform in G.

Proof. The coupling shares its latent parameter, and (2.4.1) gives

$$\mathbb{E}_G \|T_{r_\ell} - T_{r_{\ell-1}}\|^2 \le 2A^2(1 + 2^{2\alpha}) r_\ell^{-2\alpha}.$$

Independence between these two statistics is neither required nor used. The number of lattice points with $n \le \|k\| < n + 1$ is at most $C_d n^{d-1}$ for $n \ge 1$. Lemma 2.4.3 reduces the second-moment estimate, after changing constants, to

$$\sum_{n \ge 1} n^{d-3} e^{-ch^2 n^2} \min\{1, n^2 r_\ell^{-2\alpha}\}. \tag{2.4.10}$$

In dimension one, split at $n = r_\ell^\alpha$: the initial sum is at most $C r_\ell^{-\alpha}$ and the tail of $\sum n^{-2}$ has the same bound. In dimension two, split at r_ℓ^α and $h^{-1} = R^\alpha$. The first and last pieces are bounded, and the middle piece is at most $C \log(R^\alpha / r_\ell^\alpha)$. For $d \ge 3$ drop the minimum and bound the remaining Gaussian sum by $Ch^{2-d} = CR^{\alpha(d-2)}$. For level zero, use $\mathbb{E}|\widehat{\delta_{T_1}}(k) - \mathbb{E}\widehat{\delta_{T_1}}(k)|^2 \le 1$ and the same sums; in dimension one their bound is constant.

The one-dimensional assertions about S_L are geometric sums. For $d = 2$, write $j = L - \ell$ to obtain

$$S_L = \sqrt{R} \sum_{j=0}^{L} 2^{-j/2} \sqrt{1 + j \log 2} \le C\sqrt{R}.$$

For $d \ge 3$, factor out $R^{\alpha(d-2)/2}$ and sum $\sum_{\ell=0}^{L} 2^{\ell/2} \le C\sqrt{R}$. This proves (2.4.9). In particular the logarithm in the two-dimensional variance does not survive the allocation sum. □

2.4.3 Construction, integer costs, and probability-valued output

Proof of Theorem 2.4.1. Use N_ℓ fresh independent individuals at each level, disjoint across levels. Each level-ℓ individual supplies $b_0 r_\ell$ observations. For $\ell \ge 1$, the two statistics are the fine statistic and its coarse prefix version. Define the smooth signed measure of total mass one

$$\begin{aligned} H_L = {} & \frac{1}{N_0} \sum_{i=1}^{N_0} K_h * \delta_{T_{1,i}} \\ & + \sum_{\ell=1}^{L} \frac{1}{N_\ell} \sum_{i=1}^{N_\ell} K_h * (\delta_{T_{r_\ell,i}} - \delta_{T_{r_{\ell-1},i}}). \end{aligned} \tag{2.4.11}$$

Let G_R be the law of T_R when its latent parameter has law G. Correct marginal laws of every prefix give $\mathbb{E}H_L = K_h * G_R$. For a mean-zero random vector in a Hilbert space, second moments of independent sums add. Lemma 2.4.4 therefore gives

$$\mathbb{E}\|H_L - \mathbb{E}H_L\|_{-1}^2 \le C \sum_{\ell=0}^{L} \frac{v_\ell}{N_\ell}.$$

This identity can alternatively be checked for every finite Fourier truncation and passed to the limit by monotone convergence, avoiding any integrability ambiguity. Coupling T_R with its latent parameter and applying Lemma 2.4.3 gives

$$\mathbb{E}\|H_L - G\|_{\mathrm{KR}} \le C\left(\sum_{\ell=0}^{L} \frac{v_\ell}{N_\ell}\right)^{1/2} + AR^{-\alpha} + \sqrt{d}\,h. \tag{2.4.12}$$

Put $b = \lfloor B/b_0 \rfloor$. For $R = 2^L \le b/4$, set

$$N_\ell = \left\lceil \frac{b}{2S_L}\sqrt{\frac{v_\ell}{r_\ell}} \right\rceil. \tag{2.4.13}$$

The deterministic number of blocks is bounded by

$$\sum_{\ell=0}^{L} r_\ell N_\ell \le b/2 + \sum_{\ell=0}^{L} r_\ell < b/2 + 2R \le b,$$

and $\sum_\ell v_\ell/N_\ell \le 2S_L^2/b$. Thus (2.4.12) is at most $C(S_L/\sqrt{b} + R^{-\alpha})$. Choose R to be the largest dyadic integer at most the corresponding quantity below:

$$\frac{1}{16}\begin{cases} b^{1/(1+d\alpha)}, & d \ge 2, \\ b^{1/(1+\alpha)}, & d = 1,\ \alpha < 1, \\ b^{1/2}, & d = 1,\ \alpha = 1, \\ b^{1/(2\alpha)}, & d = 1,\ \alpha > 1. \end{cases}$$

For all sufficiently large b these quantities exceed one and satisfy $R \le b/4$. Equation (2.4.9) proves the rates in (2.4.2) for the signed output.

To obtain a probability-valued estimator, enumerate a countable W_1-dense collection (Q_j) of rational atomic probabilities on Θ. Choose the first j satisfying

$$\|Q_j - H_L\|_{\mathrm{KR}} \le \inf_k \|Q_k - H_L\|_{\mathrm{KR}} + B^{-1}.$$

This is measurable. Indeed, after fixing $f(0) = 0$, the Lipschitz unit ball on the compact torus has a countable dense subset in the uniform norm; its supremum represents each displayed norm. The objective and its countable infimum are therefore measurable. Density implies the infimum is no larger than $\|G - H_L\|_{\mathrm{KR}}$. For $\widehat{G} = Q_j$, the triangle inequality gives

$$W_1(\widehat{G}, G) = \|\widehat{G} - G\|_{\mathrm{KR}} \le 2\|H_L - G\|_{\mathrm{KR}} + B^{-1}.$$

This proves the theorem. No assertion of computational optimality is needed for this statistical minimax construction. □

2.4.4 An adaptive lower bound with only directed finite entropy

Proof of Theorem 2.4.2. Partition $J = a + [0, s]^d$ into $K = m^d$ congruent cubes and write $a_1, \dots, a_K$ for their centers. Set $h = s/(8m)$ and

$$G_v = K^{-1}\sum_{j=1}^{K} \delta_{a_j + h v_j e_1}, \qquad v \in \{-1, 1\}^K.$$

All these points are in J. Distances between points from different cells are at least $6h$. Within each cell the two possible points are $2h$ apart. Consequently

$$W_1(G_v, G_w) = \frac{2h}{K} d_H(v, w). \tag{2.4.14}$$

The upper bound matches atoms within cells. For the lower bound assign the values $h, -h$ to the two points in every differing cell, with h assigned to the G_v point; assign zero in unchanged cells. These assignments are 1-Lipschitz on the finite set, because cross-cell distances are at least $6h$. Their McShane extension is a valid Kantorovich test function and proves equality.

Give the statistician a stronger oracle: the cells, centers, and equal weights are known, and a query to any chosen cell j returns a fresh observation from $P_{a_j+hv_je_1}$. Signs remain unknown. An ordinary grouped policy can be simulated exactly: every time it starts an individual, independently draw a uniform cell index; query that cell on every revisit, and conceal the index from the original policy. Thus any lower bound for this oracle is a lower bound for the original experiment.

Fix an adaptive randomized oracle policy, and include its parameter-free random seed in the transcript. Write $\mathbb{P}_v$ for its transcript law and N_j for its number of queries to cell j. Only use oriented edges with $v_j = -1$ and let $v^{(j)}$ flip that sign to $+1$. The chain rule for relative entropy and (2.4.3) give

$$\mathrm{KL}(\mathbb{P}_v \| \mathbb{P}_{v^{(j)}}) \le D(2h)^\kappa \mathbb{E}_v N_j.$$

At every fixed past transcript the policy's conditional choice is the same in both models; only queries to j contribute entropy. An early stop can be padded with common dummy observations. Since $\sum_j N_j \le B$,

$$\frac{1}{K2^{K-1}} \sum_{j=1}^{K} \sum_{v: v_j=-1} \mathrm{KL}(\mathbb{P}_v \| \mathbb{P}_{v^{(j)}}) \le \frac{2D(2h)^\kappa B}{K}.$$

Pinsker and Jensen therefore bound the average total variation over these edges by

$$\left(\frac{D(2h)^\kappa B}{K}\right)^{1/2} = \left(\frac{D(s/4)^\kappa B}{m^{d+\kappa}}\right)^{1/2}.$$

Choose $m = \max\{1, \lceil (16D(s/4)^\kappa B)^{1/(d+\kappa)} \rceil\}$. The average is at most $1/4$. Binary testing on each edge then gives, for every sign estimator,

$$2^{-K} \sum_v \mathbb{E}_v d_H(\widehat{v}, v) \ge 3K/8.$$

For an arbitrary probability-valued estimate $\widehat{G}$, choose a nearest law $G_{\widehat{v}}$ in the finite family. The triangle inequality and (2.4.14) imply

$$W_1(\widehat{G}, G_v) \ge \tfrac{1}{2} W_1(G_{\widehat{v}}, G_v) = \frac{h}{K} d_H(\widehat{v}, v).$$

Its maximum risk is at least $3h/8 \ge cB^{-1/(d+\kappa)}$. The argument uses finite entropy only in the displayed orientation.

For the $B^{-1/2}$ term, an oracle that reveals B independent latent parameters can simulate any allowed grouped experiment: use the next latent draw whenever a previously unobserved individual is first queried, and generate its conditional observations as required. At most B individuals can be queried. For distinct fixed $u, w \in \Theta$, consider $G_\pm = (1/2 \pm \varepsilon)\delta_u + (1/2 \mp \varepsilon)\delta_w$ with $\varepsilon = 1/(16\sqrt{B})$. Their B-sample relative entropy is bounded by an absolute constant below $1/2$, and their W_1 separation is $2\varepsilon\|u - w\|$. The two-point testing inequality proves the remaining term. Combining with Theorem 2.4.1 yields the stated exact rates and critical bounds. □

2.4.5 Regular observation channels: three concrete applications

Corollary 2.4.5 (Gaussian random effects and joint Bernoulli traits)**.** *For each of the following experiments, with arbitrary latent mixing law on* $[0,1]^d$*,*

$$\mathcal{R}_B^{\mathrm{ad}} \asymp \mathcal{R}_B^{\mathrm{fix}} \asymp B^{-1/(d+2)} :$$

1. $P_\theta = N_d(\theta, \Sigma)$ *with a known positive-definite covariance matrix* Σ*;*
2. $P_\theta = \bigotimes_{j=1}^{d} \mathrm{Bernoulli}(\theta_j)$.

An observation is one d-vector. If individual coordinate measurements are charged separately, the fixed factor d changes constants, not the exponent. No independence of the latent coordinates under G is assumed.

Proof. For the Gaussian experiment, project the sample mean onto $[0,1]^d$. Euclidean projection cannot increase its distance to θ, and its mean squared error is at most $\mathrm{tr}(\Sigma)/r$. The Gaussian relative entropy is $\frac{1}{2}(\theta-\theta')^T\Sigma^{-1}(\theta-\theta')$. Thus the two main theorems apply with $\alpha = 1/2, \kappa = 2$.

For the Bernoulli experiment the vector of coordinate sample means has mean squared error at most $d/(4r)$. On $J = [1/4, 3/4]^d$, relative entropy is bounded above by chi-squared divergence coordinate by coordinate, giving

$$\mathrm{KL}(P_\theta \| P_{\theta'}) \le \sum_{j=1}^{d} \frac{(\theta_j - \theta'_j)^2}{\theta'_j(1-\theta'_j)} \le \frac{16}{3}\|\theta - \theta'\|^2.$$

The same theorems apply. □

Corollary 2.4.6 (The tensor factorial kernel)**.** *Let*

$$k_*(x \mid t) = 1 + \sum_{j \ge 2} \frac{L_j(x)L_j(t)}{j!}, \qquad P_\theta(dx) = \prod_{a=1}^{d} k_*(x_a \mid \theta_a)\, dx,$$

where $L_j(x) = P_j(2x-1)$ *are shifted Legendre polynomials. Then arbitrary joint mixing laws on* $[0,1]^d$ *have exact adaptive and fixed unequal budget risk* $B^{-1/(d+2)}$*.*

Proof. For $A_j(x) = j!(2j+1)L_j(x)$, define the scalar pair statistic

$$Y(x,y) = \frac{1}{2} + \frac{35}{18}\left(A_2(x)A_3(y) - \frac{10}{21}A_5(x) - \frac{4}{15}A_3(x)\right).$$

Legendre orthogonality gives $\mathbb{E}_t A_j(X) = L_j(t)$, and

$$L_2L_3 = \frac{10}{21}L_5 + \frac{4}{15}L_3 + \frac{9}{35}L_1$$

implies $\mathbb{E}_t Y = t$. Since $|L_j| \le 1$ on $[0,1]$, $|Y| < 2100$. Apply this statistic coordinatewise to r disjoint pairs and project its average onto the cube. Its mean squared error is at most $d\,2100^2/r$, so (2.4.1) holds with $b_0 = 2$.

The scalar density is at least $3 - e > 0$, and its parameter derivative is bounded on the compact square by uniform convergence of the factorial series. If this derivative bound is M, then

$$\mathrm{KL}(P_t^* \| P_{t'}^*) \le \chi^2(P_t^* \| P_{t'}^*) \le M^2(3-e)^{-1}|t-t'|^2.$$

Relative entropy adds over coordinates. Hence $\kappa = 2$, and the claimed rate follows. □

2.4.6 Unknown Gaussian covariance: statistical dimension versus observability

Let V be a p-dimensional linear subspace of the symmetric $m \times m$ matrices, with fixed basis $S_1, \dots, S_p$. Consider an injective affine parametrization of a full-dimensional compact box of mean and covariance parameters,

$$a = a(u), \qquad \Sigma = \Sigma_0 + \sum_{j=1}^{p} v_j S_j, \qquad (u, v) \in [0,1]^{m+p},$$

where the mean map is nonsingular and every covariance satisfies $\lambda_0 I \preceq \Sigma \preceq \lambda_1 I$ for fixed $0 < \lambda_0 \le \lambda_1 < \infty$. Rescaling box coordinates is absorbed into the basis and mean map.

Corollary 2.4.7 (Budget law for structured Gaussian mixing)**.** *For $P_{u,v} = N_m(a(u), \Sigma(v))$ and arbitrary mixing laws on the parameter box, put $k = m + p$. Then*

$$\mathcal{R}_B^{\mathrm{ad}} \asymp \mathcal{R}_B^{\mathrm{fix}} \asymp B^{-1/(k+2)}. \tag{2.4.15}$$

The loss may use either Euclidean box coordinates or the parameter metric $[\|a - a'\|^2 + \|\Sigma - \Sigma'\|_F^2]^{1/2}$, since these fixed metrics are equivalent.

For a covariance graph H with m vertices and free diagonal and edge entries, $p = m + |E(H)|$ and the exponent is $1/(2m + |E(H)| + 2)$. This depends on the number of covariance parameters, whereas the exact finite-replication observability threshold depends on the rank geometry of the projected Gram map in Theorem 2.3.1. Here the graph constrains covariance entries, not precision-matrix entries. The increasing replication depths used in (2.4.15) also explain why this result is consistent with the risk floor under a fixed cap below the observability threshold.

Proof. Use $n = 2r$ observations from one individual. The sample mean $\overline{X}$ and unbiased sample covariance $\widehat{\Sigma}$ satisfy

$$\mathbb{E}\|\overline{X} - a\|^2 = \frac{\operatorname{tr}\Sigma}{n}, \qquad \mathbb{E}\|\widehat{\Sigma} - \Sigma\|_F^2 = \frac{(\operatorname{tr}\Sigma)^2 + \operatorname{tr}(\Sigma^2)}{n-1}.$$

For the second identity, an orthogonal change of the sample indices expresses $(n-1)\widehat{\Sigma}$ as the sum of $n-1$ independent matrices ZZ^T with $Z \sim N_m(0, \Sigma)$. The Gaussian fourth-moment identity gives $\operatorname{Var}(Z_i Z_j) = \Sigma_{ii}\Sigma_{jj} + \Sigma_{ij}^2$; summing over i, j proves the formula. Thus both errors are uniformly $O(r^{-1})$.

Project $\widehat{\Sigma} - \Sigma_0$ orthogonally onto V, recover its basis coefficients, apply the inverse affine map to the sample mean, and project the resulting parameter vector onto $[0,1]^k$. All intervening linear maps have fixed finite norms, so this gives (2.4.1) with $d = k, \alpha = 1/2, b_0 = 2$.

The Gaussian entropy formula is

$$\mathrm{KL}(N(a,\Sigma)\|N(a',\Sigma')) = \frac{1}{2}\left\{\operatorname{tr}(\Sigma'^{-1}\Sigma) - m - \log\det(\Sigma'^{-1}\Sigma) + (a-a')^T\Sigma'^{-1}(a-a')\right\}.$$

The eigenvalues of $\Sigma'^{-1/2}\Sigma\Sigma'^{-1/2}$ lie in a fixed compact subinterval of $(0,\infty)$. On that interval, $x - 1 - \log x \le C(x-1)^2$. The entropy is therefore at most $C(\|a-a'\|^2 + \|\Sigma - \Sigma'\|_F^2)$, hence at most a constant times squared parameter distance. Apply the two main theorems with $\kappa = 2$. The graph dimension count is immediate. □

2.4.7 A nonregular application: populations of unknown endpoints

Corollary 2.4.8 (Exact budget for uniform-rectangle mixtures)**.** *For an endpoint vector $\theta \in [1,2]^d$, let the component law be*

$$P_\theta = \bigotimes_{j=1}^{d} \mathrm{Uniform}[0, \theta_j].$$

For every $d \ge 2$ and arbitrary $G \in \mathcal{P}([1,2]^d)$,

$$\mathcal{R}_B^{\mathrm{ad}} \asymp \mathcal{R}_B^{\mathrm{fix}} \asymp B^{-1/(d+1)}. \tag{2.4.16}$$

In dimension one the present argument gives only $cB^{-1/2} \le \mathcal{R}_B^{\mathrm{ad}} \le \mathcal{R}_B^{\mathrm{fix}} \le CB^{-1/2}\log(eB)$.

Proof. Let $M_{r,j}$ be the maximum of the r observations of coordinate j. Its normalized distribution satisfies $\mathbb{P}_\theta(M_{r,j}/\theta_j \le t) = t^r$ for $0 \le t \le 1$. Direct integration gives

$$\mathbb{E}_\theta\left(\frac{r+1}{r}M_{r,j}\right) = \theta_j, \qquad \operatorname{Var}_\theta\left(\frac{r+1}{r}M_{r,j}\right) = \frac{\theta_j^2}{r(r+2)}.$$

Clip these coordinate estimates to $[1,2]$ and subtract one to work on the unit cube. Their mean squared error is at most $4d/r^2$. Thus $\alpha = 1$, with $b_0 = 1$.

If only the first endpoint increases by $t > 0$, then

$$\mathrm{KL}(P_\theta \| P_{\theta + te_1}) = \log\left(1 + \frac{t}{\theta_1}\right) \le t.$$

This is a directed bound with $\kappa = 1$. In the opposite direction the relative entropy is infinite, because the larger rectangle has mass outside the smaller rectangle. The orientation in Theorem 2.4.2 is therefore necessary for this application. Since $d + 1 > 2$ precisely when $d \ge 2$, the theorem proves the exact rate in those dimensions and the stated bounds when $d = 1$. □

The mixture density for a single observation is

$$p_G(x) = \int \prod_{j=1}^{d} \frac{\mathbf{1}_{\{0 \le x_j \le \theta_j\}}}{\theta_j}\, dG(\theta),$$

the classical scale-mixture-of-uniforms model. Pavlides and Wellner [37, Section 5, p. 88] posed density-MLE rate conjectures for that model. Kim, Kur and Guntuboyina [38, Corollary 4.4] later proved the proposed cube-root Hellinger rate up to logarithms under an additional lower-bound condition. Our experiment permits repeated measurements of the same endpoint vector, and the target is the latent G in W_1, rather than p_G in Hellinger distance. Corollary 2.4.8 therefore solves a specified replication extension, not those original MLE conjectures.

2.4.8 What the dimensional result determines

For regular components the allocation mechanism has no logarithmic exception at $d = 2$: its level variances grow logarithmically, but their cost-weighted square roots sum geometrically. This is a statement about the grouped experiment, and does not settle critical-dimensional questions for direct empirical measures or an equal-group restriction. For Gaussian covariance mixing, the latent dimension $k = m + \dim V$ controls the statistical budget, while algebraic rank controls exact observability from a fixed number of replicates. For uniform endpoints, the change from a quadratic to a linear local information exponent improves the budget exponent from $1/(d+2)$ to $1/(d+1)$.

The one-dimensional endpoint logarithm is left open. Likewise, the equal-group binomial MLE range conjecture of Vinayak et al. [13, Remark 3.2], and its later refinements in [14, Theorem 20 and Section 6.2 of the author version], are not resolved by switching to a different allocation class. The completed results here are the multidimensional arbitrary-allocation minimax theorems and their fully specified applications.

2.5 Thorin symmetry and the remaining McKay classifications

Two externally posed problems guide this section: the multipart McKay classification in [3, §4.3], and the self-decomposability problem in [3, §4.4]. The published theorem numbers 3, 4, 12, 14, 18, and 22 correspond to arXiv numbers 2.3, 2.4, 3.12, 3.14, 3.18, and 3.22. The previous

manuscript treated the HCM obstruction for the four McKay densities and classified the second family. Those results are not counted as new here. We now classify the GGC property of the third and fourth families over their entire admissible parameter ranges, answer the additional HCM question for Theorem 12, and prove SD for all three families singled out in the third source problem.

The McKay argument uses the classical positive hypergeometric ratio measure, followed by a new application of its inverse first moment. The necessity argument extracts the symmetry, total mass, and forced atoms of a possible Thorin measure. The SD argument transfers the order monotonicity theorem of Baricz and Prabhu K [39] to a heat-trace bound. That theorem was posted on 7 October 2026, and is explicitly credited; its proof and discussion contain no SD classification of the three families below. These are two source problems with several cases, not a separate conjecture for every case.

2.5.1 A positive Thorin measure from a hypergeometric ratio

Write $F(A,B;C;z) = {}_2F_1(A,B;C;z)$. We use the following classical representation, stated explicitly as equation (5-1) in the published version of [3] and credited there to Küstner. For $0 < A, B \le C$ there is a probability measure q on $[0,1]$ with

$$\frac{F(A+1,B+1;C+1;z)}{F(A,B;C;z)} = \int_{[0,1]} \frac{q(dt)}{1-tz}, \qquad z \in \mathbb{C} \setminus [1,\infty). \tag{2.5.1}$$

This representation is a classical input, not a new assertion of the present work.

Lemma 2.5.1 (The inverse first moment of the ratio measure)**.** *Under these hypotheses, with* $m = \min(A,B)$*,*

$$q(\{0\}) = 0, \qquad \int_{(0,1]} \frac{q(dt)}{t} = \frac{C}{\max(A,B)}. \tag{2.5.2}$$

Proof. By symmetry suppose $A \le B$. If $A = C$, then $A = B = C$ and the identity $F(A,A;A;z) = (1-z)^{-A}$ proves the claim directly. Otherwise $A < C$, and Euler's integral gives

$$F(A,B;C;-R) = D \int_0^1 t^{A-1}(1-t)^{C-A-1}(1+Rt)^{-B}\,dt, \qquad D = \frac{\Gamma(C)}{\Gamma(A)\Gamma(C-A)}.$$

For $B > A$, substitution $u = Rt$ shows

$$F(A,B;C;-R) \sim DR^{-A}\mathrm{B}(A,B-A).$$

For completeness, dominated convergence applies on $u \le R/2$; the contribution from $u > R/2$, after multiplication by R^A, is $O(R^{A-B})$, by integrability of $(1-t)^{C-A-1}$ at one. Applying the same argument to the differentiated integral gives

$$\frac{RF'(A,B;C;-R)}{F(A,B;C;-R)} \longrightarrow A. \tag{2.5.3}$$

Indeed, the ratio of its two limiting integrals is $B\mathrm{B}(A+1,B-A)/\mathrm{B}(A,B-A) = A$. When $B = A$, the denominator after the substitution is asymptotic to $\log R$. In the numerator the extra factor is $Au/(1+u)$; replacing it by A changes the integral by a bounded amount, since the integrand then gains a factor $(1+u)^{-1}$. The contribution near $u = R$ is bounded in the same way as above. Thus (2.5.3) remains valid.

Using $F' = (AB/C)F(A+1,B+1;C+1;\cdot)$ in (2.5.1), monotone convergence now gives

$$\int \frac{q(dt)}{t} = \lim_{R\to\infty} \int \frac{R}{1+Rt}\,q(dt) = \frac{C}{AB}A = \frac{C}{B}.$$

Its finiteness also rules out an atom at zero. □

Theorem 2.5.2 (A symmetric hypergeometric Thorin measure)**.** *Let $b > d > 0$, $0 < A, B \le C$, and $M \ge 2m$, where $m = \min(A, B)$. Normalize*

$$L(s) = \left(\frac{b}{s+b}\right)^{M} \frac{F(A, B; C; d^2/(s+b)^2)}{F(A, B; C; d^2/b^2)}, \qquad s \ge 0. \tag{2.5.4}$$

Then L is the Laplace transform of a GGC. Its Thorin measure is

$$U = (M - 2m)\delta_b + (T_-)_{\#}w + (T_+)_{\#}w, \qquad w(dt) = \frac{AB}{C}\frac{q(dt)}{t}, \quad T_\pm(t) = b \pm d\sqrt{t}. \tag{2.5.5}$$

It has total mass M and is supported in $[b - d, b + d]$. Here $(T)_{\#}w$ denotes the pushforward measure.

Proof. Put $u = s + b$ and $h = -\log L$. Differentiation and (2.5.1) give

$$h'(s) = \frac{M}{u} + \frac{2ABd^2}{Cu}\int \frac{q(dt)}{u^2 - d^2 t}.$$

For $t > 0$ the elementary identity

$$\frac{2d^2}{u(u^2 - d^2 t)} = \frac{1}{t}\left\{\frac{1}{u - d\sqrt{t}} + \frac{1}{u + d\sqrt{t}} - \frac{2}{u}\right\}$$

and Lemma 2.5.1 yield

$$h'(s) = \frac{M - 2m}{s + b} + \int \left\{\frac{1}{s + b - d\sqrt{t}} + \frac{1}{s + b + d\sqrt{t}}\right\} w(dt).$$

All measures and coefficients are nonnegative, $w((0, 1]) = m$, and all rates are bounded below by $b - d > 0$. Integrating from zero to s therefore gives the Thorin representation $h(s) = \int \log(1 + s/r)\, U(dr)$ with (2.5.5). This is the defining GGC representation and proves all assertions. □

2.5.2 A necessary symmetry and atom budget

Lemma 2.5.3 (Symmetry forced by an odd logarithmic derivative)**.** *Let a probability density f have exponential decay rate $r_0 = b - d > 0$, in the sense that its Laplace integral converges throughout $\operatorname{Re} s > -r_0$ and is analytic there. Suppose its logarithmic derivative $g(s) = -L'(s)/L(s)$ has, for sufficiently large $|s + b|$, a convergent Laurent expansion*

$$g(s) = \frac{M}{s + b} + \sum_{n \ge 1} \frac{c_n}{(s + b)^{2n+1}}, \qquad M > 0, \tag{2.5.6}$$

with real coefficients. If the law is GGC, its Thorin measure U has total mass M, is symmetric about b, and is supported in $[b - d, b + d]$. If, in addition,

$$f(x) \sim C x^{q-1} e^{-(b-d)x}, \qquad C > 0, \quad q > 0, \tag{2.5.7}$$

then

$$U(\{b - d\}) = U(\{b + d\}) = q. \tag{2.5.8}$$

Finally, whenever $\lim_{\eta \downarrow 0} i\eta g(-b + i\eta)$ exists, that limit is $U(\{b\})$.

Proof. For a GGC one has $g(s) = a + \int (s+r)^{-1}\, U(dr)$ with $a \geq 0$. The expansion at positive infinity gives $a = 0$, and monotone convergence in $s/(s+r)$ gives $U((0,\infty)) = M$. Since L is positive on the real interval $s > -r_0$, g is locally analytic and real across that interval. Stieltjes inversion implies $U((0, r_0)) = 0$.

The Laurent series is a single-valued analytic continuation to the exterior of a sufficiently large circle centered at $-b$, and is real on its real part. In the upper half-plane it agrees with the Stieltjes representation by the identity theorem. Stieltjes inversion on the sufficiently negative real axis therefore implies that U has bounded support. Consequently all centered moments exist, and expansion of $\int (s + b + (r-b))^{-1}\, U(dr)$ shows from (2.5.6) that every odd centered moment vanishes. The reflection of U about b has the same moments as U; uniqueness of moments for compactly supported measures makes the two measures equal. Reflection of the lower support bound $r \geq b - d$ gives the upper bound $r \leq b + d$.

Assumption (2.5.7), applied also to $xf(x)$, gives $L(-r_0+\varepsilon) \sim C\Gamma(q)\varepsilon^{-q}$ and $-L'(-r_0+\varepsilon) \sim C\Gamma(q+1)\varepsilon^{-q-1}$. These elementary Abelian limits follow by scaling $y = \varepsilon x$ and bounding the tail above and below by $(1 \pm \delta)Cx^{q-1}e^{-r_0 x}$. Thus $\varepsilon g(-r_0 + \varepsilon) \to q$. In the Stieltjes representation the same limit, by dominated convergence and the lower support bound, is $U(\{r_0\})$. Symmetry gives the other endpoint atom. For the center, $|i\eta/(r - b + i\eta)| \leq 1$, and dominated convergence directly gives $i\eta g(-b + i\eta) \to U(\{b\})$. □

Theorem 2.5.4 (Complete GGC classification of the generalized McKay law)**.** *Let $\mu > -1$, $\mu + \nu > 0$, and $b > a > 0$. The law with density*

$$f_{\mu,\nu,a,b}(x) = C_{\mu,\nu,a,b}\, x^{\nu-1} e^{-bx} I_\mu(ax), \qquad x > 0, \tag{2.5.9}$$

where $C_{\mu,\nu,a,b} > 0$ normalizes the integral, is GGC if and only if

$$\nu \leq \mu + 1. \tag{2.5.10}$$

It is therefore self-decomposable throughout this range, which is exactly the range covered by [3, Theorem 3]. Its Thorin measure there has total mass $\mu + \nu$, is symmetric about b, has no central atom, and has endpoint atoms of mass $(\nu - 1/2)_+$ each at $b \pm a$.

Proof. The density is integrable by its Bessel asymptotics at zero and infinity. Termwise integration of its nonnegative Bessel series gives (2.5.4) with

$$M = \mu + \nu, \quad A = M/2, \quad B = (M+1)/2, \quad C = \mu + 1, \quad d = a.$$

The condition $\nu \leq \mu + 1$ is precisely $B \leq C$; then $0 < A < B \leq C$ and $M = 2A$. Theorem 2.5.2 proves sufficiency and the stated total mass and absence of a central atom.

For necessity the hypergeometric function is analytic and nonzero near zero, regardless of (2.5.10). Its logarithmic derivative consequently has an odd Laurent expansion (2.5.6) with leading coefficient M. At infinity the density satisfies

$$f_{\mu,\nu,a,b}(x) \sim C' x^{\nu - 3/2} e^{-(b-a)x}.$$

If $\nu > 1/2$, Lemma 2.5.3 forces two atoms of mass $\nu - 1/2$. Their total cannot exceed M, so $2\nu - 1 \leq \mu + \nu$, or $\nu \leq \mu + 1$. If $\nu \leq 1/2$, the admissibility condition $\mu + \nu > 0$ implies $\mu + 1 > 1 - \nu \geq \nu$, so the same necessary inequality is automatic. This proves the equivalence.

The endpoint mass is $\nu - 1/2$ when positive, as just proved. For $\nu < 1/2$ the transform stays finite at its exponential boundary, so $\varepsilon g(-b + a + \varepsilon) \to 0$; for $\nu = 1/2$ its growth is logarithmic and the same limit is zero. This proves the remaining endpoint claims. Every GGC is SD, either by the classical inclusion or by its decreasing Lévy spectral function $k(t) = \int e^{-rt}\, U(dr)$. □

Theorem 2.5.5 (Complete GGC classification of the squared-Bessel McKay law)**.** *Let $\mu > -1/4$ and $b > 2a > 0$. The density*

$$f_{\mu,a,b}(x) = C_{\mu,a,b} x^{2\mu} e^{-bx} I_\mu(ax)^2 \tag{2.5.11}$$

defines a GGC if and only if

$$-\tfrac{1}{4} < \mu \le \tfrac{1}{2}. \tag{2.5.12}$$

Hence every law covered by [3, Theorem 4] is SD. Its Thorin measure in this range has total mass $4\mu + 1$, central atom $2\max(\mu, 0)$ at b, and endpoint atoms $2\max(\mu, 0)$ each at $b \pm 2a$.

Proof. The source's Laplace calculation, or multiplication and integration of the Bessel series, gives (2.5.4) with

$$M = 4\mu + 1, \quad A = \mu + \tfrac{1}{2}, \quad B = 2\mu + \tfrac{1}{2}, \quad C = \mu + 1, \quad d = 2a.$$

For $-1/4 < \mu \le 1/2$ one has $0 < A, B \le C$. If $\mu \le 0$, $M = 2B = 2\min(A, B)$; if $\mu \ge 0$, $M - 2\min(A, B) = 2\mu$. Theorem 2.5.2 therefore proves sufficiency and the central mass formula.

Suppose now $\mu > 1/2$ and the law were GGC. Its odd Laurent expansion has total Thorin mass $M = 4\mu + 1$. Its tail is $C' x^{2\mu-1} e^{-(b-2a)x}$, so Lemma 2.5.3 forces endpoint atoms of mass 2μ each. There is also a forced central atom. Indeed, put $u = i\eta$ and $z = d^2/u^2 = -d^2/\eta^2$. The logarithmic derivative satisfies

$$ug(u - b) = M + 2z \frac{F'(A, B; C; z)}{F(A, B; C; z)}.$$

Here $0 < A < C$ and $B > A$, so the Euler-integral proof of (2.5.3), which does not require $B \le C$, gives $zF'/F \to -A$. The displayed expression tends to $M - 2A = 2\mu$. The analytic continuation along the imaginary u-axis is legitimate: its hypergeometric argument is negative, and Euler's integral is strictly positive there; continuation from large $|u|$ agrees with the Stieltjes transform by uniqueness. The last clause of Lemma 2.5.3 identifies the limit with $U(\{b\})$.

The two endpoint atoms and this central atom are at three distinct points and have total mass 6μ. Positivity of U would imply $6\mu \le 4\mu + 1$, contrary to $\mu > 1/2$. This proves necessity. The endpoint masses in the positive range follow from the tail in the same way as in Theorem 2.5.4; they vanish at and below $\mu = 0$. The SD conclusion again follows from the GGC representation. □

At $\mu = 1/2$ the fourth family has the especially transparent Thorin measure $\delta_{b-2a} + \delta_b + \delta_{b+2a}$: it is a sum of three independent exponential variables with those rates. The necessity proof identifies why extending the GGC range beyond $1/2$ is impossible: the forced masses at the center and both endpoints would exceed the entire available Thorin mass. A failure of GGC outside the stated ranges is not a proof of failure of SD; the latter remains a separate question outside the parameters in the source theorems.

2.5.3 The additional HCM density in the source problem

Proposition 2.5.6 (Theorem 12 has no HCM parameter)**.** *For every $\mu > 0$, the density in [3, Theorem 12], namely*

$$\varsigma_\mu(x) = \frac{\mu}{x} e^{-1/(2x)} I_\mu(1/(2x)), \qquad x > 0, \tag{2.5.13}$$

is not HCM.

Proof. The classical HCM representation $f(x) = Cx^{\beta-1}h_1(x)h_2(1/x)$ with $-(\log h_i)'$ Stieltjes implies that a positive HCM function has a zero-free holomorphic continuation to $D = \mathbb{C}\backslash(-\infty, 0]$: integrate the two Stieltjes functions on this simply connected domain and exponentiate. See [4, eq. (7)]. On the other hand, setting $\mathcal{F}_\mu(z) = {}_0F_1(;\mu+1;z^2/4)$, the density in (2.5.13) extends to

$$C'x^{-\mu-1}e^{-1/(2x)}\mathcal{F}_\mu(1/(2x))$$

on D, with the principal power. The Bessel zeros give

$$\mathcal{F}_\mu(\pm ij_{\mu,n}) = 0, \qquad x = \pm i/(2j_{\mu,n}) \in D.$$

The continuation vanishes at these points. Its other factors are holomorphic and nonzero there, contradicting the HCM continuation property. □

2.6 A heat-trace transfer closes the three Bessel SD cases

Let $j_{\nu,n}$ be the positive zeros of J_ν and put $Q_\nu(t) = \sum_{n\ge1} j_{\nu,n}^2 e^{-tj_{\nu,n}^2}$. The following recent theorem is an external input, not a new solution of the Ismail–Miller conjecture in this manuscript:

$$Q_\nu(t) \le Q_\mu(t) \quad \text{for } t > 0, \quad \nu \ge \mu \ge 0. \tag{2.6.1}$$

It is Theorem 1.1 of Baricz and Prabhu K [39]. The primary proof derives a positive integral representation for an order derivative of $I_{\nu+1}/I_\nu$, identifies a completely monotone function by Bessel products, and uses a probability-tail representation to obtain (2.6.1). We apply it to a different question, the SD problem recorded in [3, §4.4].

Lemma 2.6.1 (A sharp continuum bound for the sine heat trace)**.** *For every $t > 0$,*

$$Q_{1/2}(t) = \pi^2 \sum_{n\ge1} n^2 e^{-\pi^2 n^2 t} < \frac{1}{4\sqrt{\pi}\, t^{3/2}}. \tag{2.6.2}$$

The ratio of the two sides tends to one as $t \downarrow 0$.

Proof. The Jacobi transformation, obtained by Poisson summation for a Gaussian, reads

$$\sum_{n\ge1} e^{-\pi^2 n^2 t} = \frac{1}{2\sqrt{\pi t}} - \frac{1}{2} + \frac{1}{\sqrt{\pi t}} \sum_{n\ge1} e^{-n^2/t}.$$

Termwise differentiation, justified locally uniformly for $t > 0$, gives

$$Q_{1/2}(t) = \frac{1}{4\sqrt{\pi}\, t^{3/2}} - \frac{1}{2\sqrt{\pi}\, t^{5/2}} \sum_{n\ge1} (2n^2 - t) e^{-n^2/t}.$$

For $0 < t \le 1$ every term in the last sum is positive. This proves the strict inequality there, as well as the asymptotic at zero. For $t \ge 1$, the ratio of successive summands of $\sum n^2 e^{-\pi^2 n^2 t}$ is at most $4e^{-3\pi^2 t}$, and therefore

$$4\sqrt{\pi}\, t^{3/2} Q_{1/2}(t) \le \frac{4\pi^{5/2} t^{3/2} e^{-\pi^2 t}}{1 - 4e^{-3\pi^2 t}} < 1.$$

For an explicit elementary bound, $t^{3/2}e^{-\pi^2 t}$ decreases on $[1, \infty)$, the denominator is greater than $1/2$, and the last expression is less than $8\pi^{5/2}e^{-\pi^2} < 256e^{-9} < 1$. □

Theorem 2.6.2 (A self-decomposable normalized I-Bessel factor)**.** *For $\nu \geq 1/2$ and $a > 0$, the function*

$$P_{\nu,a}(s) = e^{-a\sqrt{s}}{}_0F_1(;\nu+1;a^2s/4), \qquad s \geq 0, \tag{2.6.3}$$

is the Laplace transform of a self-decomposable probability law. Its Lévy spectral function is

$$k_{\nu,a}(t) = \frac{a}{2\sqrt{\pi t}} - \sum_{n\geq 1} e^{-j_{\nu,n}^2 t/a^2}, \tag{2.6.4}$$

and $k'_{\nu,a}(t) < 0$ for every $t > 0$.

Proof. The Bessel product formula gives, for $h = -\log P_{\nu,a}$,

$$h'(s) = \frac{a}{2\sqrt{s}} - \sum_{n\geq 1} \frac{1}{s + j_{\nu,n}^2/a^2} = \int_0^\infty e^{-st} k_{\nu,a}(t)\, dt.$$

The separate Laplace transforms are finite for $s > 0$, so termwise integration is justified. Differentiating (2.6.4) and using (2.6.1) and Lemma 2.6.1 gives

$$k'_{\nu,a}(t) = -\frac{a}{4\sqrt{\pi}\, t^{3/2}} + \frac{1}{a^2} Q_\nu(t/a^2) < 0.$$

The spectral function tends to zero as $t \to \infty$, so its strict decrease implies $k_{\nu,a}(t) > 0$. Moreover $k_{\nu,a}(t) \leq a/(2\sqrt{\pi t})$. Thus $k(t)/t$ integrates $\min(1,t)$, and integrating the displayed formula for h' yields

$$h(s) = \int_0^\infty (1 - e^{-st}) \frac{k_{\nu,a}(t)}{t}\, dt,$$

since $P_{\nu,a}(0) = 1$. This proves infinite divisibility without assuming it in advance. For $0 < c < 1$, monotonicity of k gives

$$h(s) - h(cs) = \int_0^\infty (1 - e^{-st}) \frac{k(t) - k(t/c)}{t}\, dt,$$

again a Bernstein exponent. Hence $P_{\nu,a}(s)/P_{\nu,a}(cs)$ is a Laplace transform, which proves self-decomposability. □

We next record the two K-Bessel factors used to transfer this result.

Lemma 2.6.3 (Two classical GGC K-factors)**.** *The function*

$$G_{\nu,b}(s) = \frac{2^{1-\nu} b^\nu}{\Gamma(\nu)} s^{\nu/2} K_\nu(b\sqrt{s}) \tag{2.6.5}$$

is GGC for $\nu > 0$, $b > 0$. The reciprocal factor

$$R_{\nu,b}(s) = 2^{\nu-1}\Gamma(\nu) b^{-\nu} s^{-\nu/2} \frac{e^{-b\sqrt{s}}}{K_\nu(b\sqrt{s})} \tag{2.6.6}$$

is GGC for $\nu \geq 1/2$, $b > 0$, with the convention $R_{1/2,b} = 1$.

Proof. The first is the transform of an inverse gamma density, proportional to $x^{-\nu-1}e^{-b^2/(4x)}$. The product of its values at uv and u/v is a positive constant times $\exp\{-(b^2/(4u))(v + v^{-1})\}$, so that density is HCM and hence GGC.

For the second, write $D_\nu(t) = J_\nu(t)^2 + Y_\nu(t)^2$. The classical Stieltjes representation of $K_{\nu-1}/K_\nu$, also used in the proof of Theorem 22 of [3], yields

$$-\frac{d}{ds} \log R_{\nu,b}(s) = \frac{1}{\pi} \int_0^\infty \frac{1 - 2/(\pi t D_\nu(t))}{s + t^2/b^2}\, dt. \tag{2.6.7}$$

Here the first contribution is $b/(2\sqrt{s})$ and the subtracted one is $bK_{\nu-1}(b\sqrt{s})/(2\sqrt{s}K_\nu(b\sqrt{s}))$; the recurrence for K'_ν cancels the apparent $\nu/(2s)$ term. Nicholson's integral [6, eq. 10.9.30]

$$D_\nu(t) = \frac{8}{\pi^2}\int_0^\infty K_0(2t\sinh u)\cosh(2\nu u)\,du$$

shows that $D_\nu(t) \geq D_{1/2}(t) = 2/(\pi t)$ for $\nu \geq 1/2$. Thus the numerator in (2.6.7) is nonnegative. The standard small- and large-argument Bessel asymptotics ensure convergence; alternatively it follows directly from the convergent Stieltjes representation before subtraction. At $s = 0$ the normalization in (2.6.6) has limit one. Integrating the positive Stieltjes derivative from zero proves the GGC representation. For $\nu = 1/2$ the numerator vanishes identically. □

Theorem 2.6.4 (The three SD cases in the third source problem)**.** *All three laws asked about in [3, §4.4] are self-decomposable throughout the parameter ranges of the source theorems. More precisely:*

1. *The law in Theorem 14 is SD for $a, b, \nu > 0$ and $\mu > 1/2$.*
2. *The law in Theorem 18 is SD for $a, b > 0$ and $\mu, \nu > 1/2$.*
3. *The law in Theorem 22 is SD for $a, b > 0$ and $\mu, \nu > 1/2$.*

The same assertions extend to the non-strict endpoints of any of these order restrictions where the displayed factors are defined.

Proof. With the normalizations (2.6.3)–(2.6.6), the three Laplace transforms in the source factor exactly as

$$L_{14}(s) = P_{\mu,a}(s)G_{\nu,b}(s), \qquad L_{18}(s) = P_{\mu,a}(s)P_{\nu,b}(s), \qquad L_{22}(s) = P_{\mu,a}(s)R_{\nu,b}(s). \tag{2.6.8}$$

For example, the first product is

$$\frac{2^{\mu-\nu+1}\Gamma(\mu+1)b^\nu}{a^\mu\Gamma(\nu)}e^{-a\sqrt{s}}s^{(\nu-\mu)/2}I_\mu(a\sqrt{s})K_\nu(b\sqrt{s}),$$

which is precisely the source's normalized transform. The other two products follow by the same cancellation of powers and constants. Theorem 2.6.2 treats every P factor, and Lemma 2.6.3 treats every G and R factor. A GGC is SD, and a product of SD Laplace transforms is SD: for each $c \in (0, 1)$ its ratio at s and cs is a product of Laplace transforms. This proves all three statements, including their defined endpoint extensions. □

What is closed and what is not. For the parameter sets actually stated in Theorems 1–4 and 12 of the source, the preceding manuscript and this section together give the full HCM/GGC/SD answers requested in its second problem. In addition, Theorems 2.5.4 and 2.5.5 classify GGC on the larger full domains where those densities exist. SD outside the source's ranges for the third and fourth McKay families has not been classified here. Theorem 2.6.4 answers the third source problem on all its stated ranges, using the explicitly credited recent order theorem. The classical ratio representation, HCM zero-free principle, and order theorem themselves are not claimed as new discoveries. Historical priority for the consequences proved here still requires specialist literature review; their full proofs and precise scope do not depend on a priority claim.

2.7 Moment cancellation transferred to diffraction and heat observation

This section treats two different observation maps. Its first result constructs a compact Cantor family of strongly log-concave objects with the same complete diffraction data and any prescribed finite number of common moments. Its second result gives an exact criterion for observing a one-sided heat source through time jets at a single sensor. The underlying convolution ambiguity, marginal log-concavity, and entire-function construction are classical; the simultaneous constraints and the precise transfer statements are proved below. Historical priority for these formulations is not asserted.

2.7.1 A Cantor family invisible to complete Fourier intensity

For a probability density f on $\mathbb{R}$, put $\widehat{f}(\xi) = \int e^{-i\xi x} f(x)\,dx$. We call f κ-strongly log-concave if $e^{\kappa x^2/2} f(x)$ is log-concave, with zero values allowed outside its interval of support. For smooth positive densities this means $(\log f)'' \le -\kappa$. Write G^σ for the global Gevrey class defined by $\|f^{(r)}\|_\infty \le AB^r (r!)^\sigma$ for some $A, B > 0$.

Theorem 2.7.1 (A compact ambiguity family under simultaneous shape priors)**.** *Fix $L > 0$, $\kappa > 0$, $\sigma > 1$, and an integer $K \ge 1$. There is a family*

$$\{f_\omega : \omega \in \{-1,1\}^{\mathbb{N}},\ \omega_1 = 1\} \subset G^\sigma(\mathbb{R})$$

of probability densities with all of the following properties:

1. *Every density has support exactly $[-L, L]$, is positive on its interior, has mean zero, and is κ-strongly log-concave. All members obey common Gevrey constants A, B.*
2. *The Fourier magnitude $|\widehat{f_\omega}(\xi)|$ is independent of ω for every real ξ. The moments $\int x^r f_\omega(x)\,dx$, $0 \le r \le K$, are also independent of ω.*
3. *Distinct members are not translates or reflected translates of one another. The displayed parametrization is a homeomorphism from the Cantor space onto its image in $C^\infty(\mathbb{R})$.*

In particular, the ambiguity is uncountable even after the usual translation, reflection, and constant-phase symmetries are removed.

Proof. Choose $a_j > 0$ with $\sum_j a_j = L$, and $c_j > 0$ with $\sum_j c_j^{-1} \le \kappa^{-1}$; for example, $a_j = L2^{-j}$ and $c_j = \kappa 2^j$. Choose increasing odd integers $m_j > K$. We first construct a density q_j on $[-a_j, a_j]$ which is c_j-strongly log-concave and whose first nonzero odd cumulant has order m_j.

Put $p = (\sigma - 1)^{-1}$ and normalize the even density

$$b_j(x) = C_j \exp\Big(-c_j x^2 - (a_j - x)^{-p} - (a_j + x)^{-p}\Big)\,\mathbf{1}_{|x|<a_j}.$$

It belongs to G^σ, is positive on its interior, and satisfies $(\log b_j)'' \le -2c_j$. Let ψ_j be an even nonnegative nonzero Gevrey bump supported in $(-a_j/2, a_j/2)$, and set $h_j = \psi_j^{(m_j)}$. Integration by parts gives

$$\int x^r h_j(x)\,dx = 0 \quad (0 \le r < m_j), \qquad \int x^{m_j} h_j(x)\,dx = (-1)^{m_j} m_j! \int \psi_j \ne 0. \tag{2.7.1}$$

The function h_j is odd. For a sufficiently small nonzero ε_j, the density $q_j = b_j + \varepsilon_j h_j$ remains positive and c_j-strongly log-concave: on the compact perturbation support this follows by continuity in C^2, and elsewhere nothing changes. It has mass one and mean zero. Its odd

moments below m_j vanish, whereas its m_jth moment does not. The moment–cumulant identities therefore show that its odd cumulants below m_j vanish and that its m_jth cumulant is nonzero. Write this last cumulant as $\gamma_j \neq 0$.

Let X_j be independent with density q_j. For each sign sequence ω the series $S_\omega = \sum_j \omega_j X_j$ converges absolutely since $|X_j| \le a_j$. Denote its density by f_ω. This density exists because it is the convolution of $q_1(\omega_1 \cdot)$ with the law of the remaining sum. Hence

$$\|f_\omega^{(r)}\|_\infty \le \|q_1^{(r)}\|_\infty,$$

which gives common global Gevrey constants. The support of the remaining sum is its full interval $[-\sum_{j\ge 2} a_j, \sum_{j\ge 2} a_j]$: finite sums have interval support, and the omitted tails are uniformly bounded by $\sum_{j>n} a_j \to 0$. Convolving with q_1 therefore gives support $[-L, L]$ and strict positivity inside it.

We recall the precise curvature rule for convolution. If q, r are c, d-strongly log-concave, write $q(x) = e^{-cx^2/2}Q(x)$ and $r(y) = e^{-dy^2/2}R(y)$ with Q, R log-concave. Completing the square gives

$$e^{\frac{cd}{c+d}z^2/2}(q * r)(z) = \int e^{-\frac{c+d}{2}(x-\frac{d}{c+d}z)^2} Q(x)R(z-x)\,dx.$$

The integrand is jointly log-concave in (z, x). The marginal theorem of Prékopa [40, Theorem 6] shows that $q * r$ is $(c^{-1} + d^{-1})^{-1}$-strongly log-concave. Consequently every finite partial convolution here is κ-strongly log-concave. Those convolutions converge uniformly to f_ω by uniform continuity of q_1 and the deterministic tail bound. Passing the log-concavity inequality to the limit proves the same curvature bound for f_ω.

Reflection conjugates each factor's Fourier transform on the real axis, so

$$|\widehat{f}_\omega(\xi)| = \lim_{n\to\infty} \prod_{j=1}^{n} |\widehat{q}_j(\xi)|$$

is independent of ω. Equation (2.7.1) says that $q_j(x)$ and $q_j(-x)$ have the same moments through K. Expanding moments of each finite sum and then passing to the bounded limit gives the common moments asserted in the theorem.

For distinct sign sequences let j be their first different index. Cumulants add for independent bounded random variables. This also holds for the infinite sum: bounded partial sums converge uniformly, and their moment-generating functions and logarithms converge holomorphically on a common neighborhood of zero. At the odd order m_j, all later factors have zero cumulant, while all earlier factors agree between the two sequences. The difference between the two cumulants is therefore $2\omega_j\gamma_j \neq 0$. This proves injectivity. All means are zero, so a translating parameter would have to vanish. Reflection changes every sign; injectivity on the full sign space shows that no two sequences with first sign $+1$ represent reflections of one another. Positive mass one excludes nontrivial constant phases.

Finally, if two sequences with fixed first sign agree through index n, couple all X_j identically. Their remaining sums differ by at most $2\sum_{j>n} a_j$. For every $r \ge 0$ the mean-value theorem applied to $q_1^{(r)}$ yields

$$\|f_\omega^{(r)} - f_\eta^{(r)}\|_\infty \le 2\|q_1^{(r+1)}\|_\infty \sum_{j>n} a_j.$$

The parametrization is continuous into C^∞; a continuous injection from a compact space into a Hausdorff space is a homeomorphism onto its image. □

Published target and exact scope. The 2025 *Phasebook* survey poses questions about inferring object complexity and certifying reconstructions from diffraction data [41, Questions 3.1

and Section 3.4]. Theorem 2.7.1 supplies a rigorous obstruction in the one-dimensional continuous setting even after substantial smoothness, curvature, support, and moment information is supplied. It does not resolve the survey's broad algorithmic or higher-dimensional questions. Nonnegativity alone was already shown insufficient in the discrete setting by Beinert [42]; reflection of convolution factors is also classical. The theorem's proved content is the simultaneous Cantor-family construction under the stated constraints, whose historical priority has not been established.

The obstruction applies to uncertainty quantification as well. If a data distribution depends on f only through its Fourier intensity, choose two members a positive distance δ apart in L^1 (or in W_1). Under their common data law every estimator has worst-case expected error at least $\delta/2$. Every confidence set covering each object with probability at least $1-\alpha$ has diameter at least δ with probability at least $1-2\alpha$. These statements follow respectively from the triangle inequality and the union bound; they remain true with arbitrarily many or noiseless intensity data.

2.7.2 An exact criterion for early-time heat jets at one sensor

Let $0 < a < b$ be known. For a finite signed measure μ supported on $[a,b]$, its heat evolution observed at the origin is

$$U_\mu(t) = \frac{1}{\sqrt{4\pi t}} \int_a^b e^{-x^2/(4t)}\, d\mu(x), \qquad t > 0.$$

The one-sided support assumption removes the elementary ambiguity between sources at x and $-x$. Let $t_j \downarrow 0$ be distinct and let $r_j \geq 1$ be integers. The observations are the time jets $U_\mu^{(\ell)}(t_j)$ for $0 \leq \ell < r_j$.

Theorem 2.7.2 (Weighted sampling criterion and exact hidden coincidences)**.** *The following statements are equivalent:*

1. $\sum_j r_j t_j = \infty$.
2. *The displayed observations identify every finite signed measure supported on* $[a,b]$.
3. *They identify all compactly supported, smooth, strongly log-concave probability densities with support* $[a,b]$.

If the sum is finite, then for every $\sigma > 1$, every $\kappa > 0$, and every nonempty open interval $I \Subset (a,b)$, there are distinct κ-strongly log-concave probability densities $f_+, f_- \in G^\sigma(\mathbb{R})$, positive on (a,b), whose difference is supported in I, such that

$$U_{f_+}(t) = U_{f_-}(t) \quad \Longleftrightarrow \quad t \in \{t_j : j \geq 1\}, \qquad t > 0.$$

At t_j this difference has a zero of exactly order r_j.

Proof. Write $s = (4t)^{-1}$ and $H_\mu(s) = \int e^{-sx^2}\, d\mu(x)$. Multiplication by the nonzero factor $(4\pi t)^{-1/2}$ and the analytic change of variable $t \mapsto (4t)^{-1}$ preserve zero multiplicities. A difference of two observation sequences vanishes precisely when H_μ has zeros of orders at least r_j at $s_j = (4t_j)^{-1}$.

The function $e^{a^2 z} H_\mu(z)$ is bounded and holomorphic on $\operatorname{Re} z > 0$. If it is nonzero, its zeros obey the half-plane Blaschke condition

$$\sum_j r_j \frac{s_j}{1+s_j^2} < \infty.$$

Since $s_j \to \infty$, this is equivalent to $\sum_j r_j t_j < \infty$. Divergence therefore forces $H_\mu \equiv 0$. Differentiating at zero gives all polynomial moments of the pushforward under $x \mapsto x^2$; polynomial density and injectivity of this map on $[a,b]$ give $\mu = 0$.

For the converse, we use the exact-real-zero consequence of the Luxemburg–Korevaar construction, as verified in Wellershoff's explicit restatement [19, Theorem 21]. For clarity the precise consequence needed here is the following: if $s_j > 0$ tend to infinity and $\sum_j r_j/s_j < \infty$, every positive interval J contains the support of a nonzero real $q \in G^\sigma$ such that

$$\int q(y)\,dy = 0, \qquad Z_{(0,\infty)}\left(s \mapsto \int y^s q(y)\,dy\right) = \{s_j\},$$

with the prescribed multiplicities r_j. To see the exactness, the classical product construction, on a logarithmic interval, has bilateral Laplace transform

$$se^{\tau s}\prod_j (1 - s^2/s_j^2)^{r_j}\prod_k \cosh(\epsilon_k s), \qquad \epsilon_k > 0, \quad \sum_k \epsilon_k < \infty.$$

The last product is finite and strictly positive for real s; the first product has exactly the claimed real zeros. The logarithmic pushforward gives a smooth q. Multiplicative convolution with a nonnegative Gevrey bump supported sufficiently near 1 makes it Gevrey while preserving its localization and every real zero, since that convolution multiplies its Mellin transform by a strictly positive real factor. This is the classical imported existence input, not a newly proved entire-function theorem.

Choose $J \Subset \{e^{-x^2} : x \in I\}$ and take such a q. Set

$$h(x) = 2xe^{-x^2} q(e^{-x^2}) \quad (x > 0), \qquad h(x) = 0 \quad (x \le 0).$$

It is real, compactly supported in I, and Gevrey of order σ; Gevrey classes are preserved by multiplication and analytic changes of variable on compact sets, as follows directly from the Faà di Bruno formula. Substitution gives

$$\int_a^b e^{-sx^2} h(x)\,dx = \int y^s q(y)\,dy. \tag{2.7.2}$$

In particular $\int h = 0$, and the heat transform has exactly the desired positive-time zeros and multiplicities.

Finally normalize

$$f_0(x) = C\exp\Big(-\kappa x^2 - (x-a)^{-p} - (b-x)^{-p}\Big)\,\mathbf{1}_{a<x<b}, \qquad p = (\sigma-1)^{-1}.$$

Its logarithmic second derivative is at most -2κ. For sufficiently small $\varepsilon > 0$, the densities $f_\pm = f_0 \pm \varepsilon h$ are positive on (a,b), κ-strongly log-concave, and of mass one, by the same compact C^2 perturbation argument used above. Equation (2.7.2) proves every required coincidence and excludes all others. □

For values alone, $t_j = j^{-\alpha}$ identifies all one-sided sources exactly when $\alpha \le 1$. Geometric sampling $t_j = 2^{-j}$ does not identify even the strongly log-concave Gevrey class. If time jets are available, $r_j = \lceil 2^j/j \rceil$ restores identification at these same geometric times. These are consequences of one weighted criterion, not separate problem resolutions.

Proposition 2.7.3 (Quasianalytic source priors override sparse times)**.** *Let $M_0 = 1$ and M be positive and log-convex, with $M_k \ge k!$, and suppose $\sum_{k\ge1} M_{k-1}/M_k = \infty$. Among densities smooth on $[a,b]$ satisfying $\|f^{(k)}\|_\infty \le AB^k M_k$ for some finite A, B, the values $U_f(t_j)$ identify f for every sequence $t_j \downarrow 0$, independently of $\sum_j t_j$. No smooth extension by zero at the endpoints is assumed.*

Proof. Let h be the difference of two such densities. If its first nonzero derivative at a has order m, elementary endpoint Laplace asymptotics give

$$\int_a^b e^{-sx^2} h(x)\,dx \sim \frac{h^{(m)}(a) e^{-sa^2}}{(2a)^{m+1} s^{m+1}} \qquad (s \to \infty).$$

Indeed, after $u = s(x-a)$ the normalized integrand is bounded by $Cu^m e^{-2au}$ and converges to $h^{(m)}(a) u^m e^{-2au}/m!$. This rules out zeros along an unbounded sequence. Thus all derivatives of h at a vanish. Its zero extension just to the left of a obeys the same derivative bounds. Denjoy–Carleman uniqueness on the enlarged interval [2, Theorem 3.6 and Corollary 3.12] forces $h = 0$. □

Published target and exact scope. Discrete recovery of heat sources and its connection with exponential systems are established subjects; Choulli's paper gives the relevant primary literature and analyzes point measurements through Dirichlet series [43, Sections 1.1–1.3]. Its bounded-domain stability results are not claimed to be replaced or improved here. Our target is the explicitly formulated whole-line, one-sided-source, early-time jet problem, with an exact sampling criterion and localized shape-constrained counterexamples. No exact published open question with this complete formulation was found in the targeted search. The result is therefore reported as a proved transfer theorem, not as a newly closed named open problem. Neither theorem in this section adds another resolution of the old moment-tail conjecture.

2.8 From arithmetic certificates to exact recurrence entropy

This section separates three questions suggested by the 223-page integrated manuscript and its 42-page sequel. First, can the cost of a fixed prime coordinate be removed from the exponential arithmetic estimate? Second, can a random-frequency theorem identify the leading constant of the covering time, rather than only its order? Third, how much of a weighted phase space can be recovered from its one-dimensional radial distribution? We give precise answers below. None is presented as the solution of a separately verified longstanding conjecture. In particular, the exact-prime exponent gap for power weights with exponent greater than three remains open.

The finite-dimensional connection between ergodization and small divisors is classical; see Bounemoura [44]. Small deviations for rapidly decreasing weighted sums are also an established subject; for geometric weights the constant used below is already a special case of Aurzada's Theorem 4 [45], and broader fast-weight results appear in Rozovsky [46]. Our small-ball calculation is included for self-containment. The additional conclusion is an almost-sure equality, to leading constant, between that entropy and the logarithm of a uniform covering time. The proof uses a sum of inverse small divisors, rather than the product of a worst small divisor and a Fourier norm.

Let $\Omega = \mathbb{T}^{\mathbb{N}}$, where angles have period 2π, and let μ be product Haar probability. For positive weights with $W = \sum_j w_j < \infty$, put

$$E_w(\theta) = \sum_j w_j(1 - \cos\theta_j), \qquad \Phi_t\theta = (\theta_j - t\lambda_j)_j.$$

The associated product metric is $d_w(\theta,\varphi)^2 = 2E_w(\theta - \varphi)$. Write $R_{w,\lambda}(\varepsilon)$ for the infimum of the times T such that every starting point enters $\{E_w < \varepsilon\}$ during $[0,T]$. Translation invariance makes the same bound uniform in the target and in the initial time of an observation interval.

2.8.1 A summed-divisor certificate on compact abelian groups

Lemma 2.8.1 (Averaging the actual inverse divisors)**.** *Let G be a compact abelian group with Haar probability, let $t \mapsto a_t$ be a continuous one-parameter subgroup, and write $\chi(a_t) = e^{-it\omega_\chi}$ for its characters. Let F be a real trigonometric polynomial with positive mean $m = \widehat{F}(1)$, and suppose $\omega_\chi \neq 0$ at every nontrivial character in its Fourier support. Set*

$$S(F) = \sum_{\chi \neq 1} \frac{|\widehat{F}(\chi)|}{|\omega_\chi|}.$$

Every orbit segment of length $T > 2S(F)/m$ meets $\{F > 0\}$.

Proof. Termwise integration of the finite Fourier expansion gives, for every $x \in G$ and every initial time A,

$$\left| \frac{1}{T} \int_A^{A+T} F(xa_t)\, dt - m \right| \leq \frac{2}{T} \sum_{\chi \neq 1} \frac{|\widehat{F}(\chi)|}{|\omega_\chi|}.$$

The average is positive under the stated inequality, so at least one point of the segment has $F > 0$. □

The lemma applies to nonproduct Fourier supports without alteration. It is an elementary finite Fourier identity, not a new general ergodization theorem. The following probabilistic use of the identity is what improves the leading constant.

Lemma 2.8.2 (Fractional moments avoid the worst-divisor loss)**.** *Let $\lambda_1, \dots, \lambda_n$ be independent, with densities bounded by $L \geq 1$. Let $\mathcal{K} \subset \mathbb{Z}^n \setminus \{0\}$ be a finite deterministic set of cardinality at most N, and let $|c_k| \leq B$. For $0 < q < 1$,*

$$\mathbb{E}\left(\sum_{k \in \mathcal{K}} \frac{|c_k|}{|k \cdot \lambda|} \right)^q \leq C_q B^q N L^q. \tag{2.8.1}$$

The same conclusion holds for $k \cdot \lambda - \lambda_0 \sum_j k_j$, if λ_0 is any additional independent random variable.

Proof. For each nonzero k, choose j with $k_j \neq 0$ and condition on every coordinate except λ_j. The conditional density of the linear form is at most $L/|k_j| \leq L$. Consequently

$$\mathbb{P}(|k \cdot \lambda| < u) \leq \min(1, 2Lu).$$

Integration of this bound, splitting at $u = (2L)^{-1}$, gives $\mathbb{E}|k \cdot \lambda|^{-q} \leq C_q L^q$. For nonnegative numbers, $(\sum a_k)^q \leq \sum a_k^q$; applying this inequality proves (2.8.1). In the relative model condition also on λ_0. No independence between distinct linear forms is asserted or needed. □

2.8.2 Rapidly decreasing spectra: small balls with the exact constant

Fix $a, b > 0$ and assume

$$\log w_j = -aj^b + o(j^b) \qquad (j \to \infty). \tag{2.8.2}$$

The weights need not be monotone. Set

$$\ell = \log(1/\varepsilon), \qquad H(\varepsilon) = \frac{b}{2(b+1)} a^{-1/b} \ell^{1+1/b}. \tag{2.8.3}$$

Lemma 2.8.3 (Elementary spectral sums)**.** *Under* (2.8.2)*,*

$$\log \sum_{j>n} w_j = -an^b + o(n^b), \tag{2.8.4}$$

$$\frac{1}{2}\sum_{j\ge 1}(L + \log w_j)_+ \sim \frac{b}{2(b+1)} a^{-1/b} L^{1+1/b} \qquad (L \to \infty). \tag{2.8.5}$$

If $n(\varepsilon)$ is the least integer with $\sum_{j>n} w_j \le \varepsilon/8$, then $n(\varepsilon) \sim (\ell/a)^{1/b}$.

Proof. For any $0 < \delta < a$, all sufficiently late weights lie between $e^{-(a+\delta)j^b}$ and $e^{-(a-\delta)j^b}$. The lower bound for the tail follows from its first term. Integral comparison bounds the upper tail by a polynomial factor times $e^{-(a-\delta)n^b}$, with a harmless adjustment of that polynomial when $b > 1$. Divide logarithms by n^b and let $\delta \downarrow 0$.

For the second assertion, the same comparison reduces the calculation to $w_j = e^{-cj^b}$. Summing through $n = (L/c)^{1/b} + O(1)$ and using $\sum_{j\le n} j^b = n^{b+1}/(b+1) + o(n^{b+1})$ gives

$$\frac{1}{2}\sum_{j\le n}(L - cj^b) = \frac{b}{2(b+1)} c^{-1/b} L^{1+1/b}(1 + o(1)).$$

Finitely many initial weights contribute only $O(L)$, which is smaller than the displayed scale. Letting $\delta \downarrow 0$ proves the formula for the original weights. The last assertion follows from (2.8.4), including the minimality of $n(\varepsilon)$. □

Proposition 2.8.4 (The Haar small-ball constant)**.** *For weights satisfying* (2.8.2)*,*

$$-\log \mu\{E_w \le \varepsilon\} \sim H(\varepsilon). \tag{2.8.6}$$

Proof. If Θ is uniform on $[-\pi, \pi]$, then $\mathbb{P}(1 - \cos\Theta \le u) \asymp \sqrt{u}$ for $0 < u \le 1$. Also

$$\mathbb{E}e^{-t(1-\cos\Theta)} = e^{-t} I_0(t), \qquad -\log(e^{-t} I_0(t)) \ge \tfrac{1}{2}\log t - C \quad (t \ge 1).$$

The last bound follows directly by comparing $1 - \cos\theta$ with θ^2 on $[-\pi, \pi]$ and integrating a Gaussian. Exponential Markov with $t = 1/\varepsilon$ therefore gives

$$-\log \mu\{E_w \le \varepsilon\} \ge \frac{1}{2}\sum_j (\ell + \log w_j)_+ - C\#\{j : w_j \ge \varepsilon\} - 1.$$

The count is $O(\ell^{1/b}) = o(H)$, so Lemma 2.8.3 proves the lower asymptotic bound.

For the other direction take $n = n(\varepsilon)$. The inactive tail contributes at most $2\sum_{j>n} w_j \le \varepsilon/4$, deterministically. For each $j \le n$ require

$$|\Theta_j| \le \min\{1, \sqrt{\varepsilon/(nw_j)}\}.$$

The active energy is at most $\varepsilon/2$. By independence, the negative logarithm of the probability of this event is at most

$$Cn + \frac{1}{2}\sum_{j\le n}(\ell + \log n + \log w_j)_+.$$

Extending the sum to all j and applying (2.8.5), with $\log n = o(\ell)$, gives $(1 + o(1))H(\varepsilon)$. This proves (2.8.6). □

2.8.3 An exact leading constant for random-frequency covering

Let $\lambda_j = c_j + \xi_j$, where the ξ_j are independent uniforms on $[-r_j, r_j]$, with $0 < r_j \le 1$. Assume that the centers have at most polynomial growth and that

$$L_n := \max\{1, (2r_j)^{-1} : 1 \le j \le n\}, \qquad \log L_n = o(n^{b+1}). \tag{2.8.7}$$

The conclusion also holds for the relative frequencies $\lambda_j - \lambda_0$, where λ_0 is an independent bounded random variable. All assertions are for a fixed deterministic weight sequence satisfying (2.8.2).

Theorem 2.8.5 (Exact recurrence entropy)**.** *With probability one,*

$$\log R_{w,\lambda}(\varepsilon) \sim -\log \mu\{E_w \le \varepsilon\} \sim \frac{b}{2(b+1)} a^{-1/b} (\log(1/\varepsilon))^{1+1/b}. \tag{2.8.8}$$

The covering estimate is uniform in starting points, targets, and initial times. The same asymptotic holds for the supremum of the lengths of intervals avoiding $\{E_w < \varepsilon\}$ along any one orbit.

Proof. We first construct a certificate with the correct number of modes. For an integer $D \ge 0$ put

$$u_k = \sqrt{\frac{2}{D+2}} \sin \frac{(k+1)\pi}{D+2} \quad (0 \le k \le D), \qquad K_D(\theta) = \left| \sum_{k=0}^{D} u_k e^{ik\theta} \right|^2 .$$

The elementary sine eigenvector identity gives

$$\int K_D = 1, \quad K_D \ge 0, \quad \int (1 - \cos\theta) K_D = 1 - \cos \frac{\pi}{D+2} \le \frac{\pi^2}{2(D+2)^2}. \tag{2.8.9}$$

Indeed $\sum u_k^2 = 1$ and $\sum_{k=0}^{D-1} u_k u_{k+1} = \cos(\pi/(D+2))$; expanding the square proves the identities. Its Fourier support is $[-D, D]$, each coefficient has modulus at most one, and $\|K_D\|_A \le (\sum |u_k|)^2 \le D + 1$.

Take $n = n(\varepsilon)$ and, for $1 \le j \le n$, set

$$D_j = \left\lceil \sqrt{64\pi^2 n w_j / \varepsilon} \right\rceil, \quad K = \prod_{j \le n} K_{D_j}, \quad F = (\varepsilon/2 - E_{w,\le n}) K.$$

Then $\widehat{F}(0) \ge \varepsilon/2 - \varepsilon/128 > \varepsilon/3$, and $F > 0$ implies $E_w < \varepsilon$. The Fourier support is contained in $|k_j| \le D_j + 1$, so its cardinality N_ε satisfies

$$\begin{aligned} \log N_\varepsilon &\le \sum_{j \le n} \log(2D_j + 3) \\ &\le Cn + \frac{1}{2} \sum_{j \le n} (\ell + \log n + \log(64\pi^2) + \log w_j)_+ \le (1 + o(1)) H(\varepsilon). \end{aligned} \tag{2.8.10}$$

Moreover $|\widehat{F}(k)| \le \varepsilon/2 + 2W$, by convolution with the Fourier coefficients of the energy and $|\widehat{K}| \le 1$. In particular the individual coefficient bound stays bounded as $\varepsilon \downarrow 0$.

Use the sequence $\varepsilon_m = e^{-m}$. Since $n(\varepsilon_m) \sim (m/a)^{1/b}$, assumption (2.8.7) yields $\log L_{n(\varepsilon_m)} = o(H(\varepsilon_m))$. For any fixed $\eta > 0$, choose $q < 1$ so close to one that $q(1+\eta) > 1$. Lemma 2.8.2 and Markov's inequality show that

$$\mathbb{P}\{S(F) > e^{(1+\eta)H(\varepsilon_m)}\} \le \exp\{[1 - q(1+\eta) + o(1)] H(\varepsilon_m)\}.$$

These probabilities are summable because $H(\varepsilon_m) \asymp m^{1+1/b}$. Borel–Cantelli and Lemma 2.8.1 give almost surely

$$\log R_{w,\lambda}(\varepsilon_m) \le (1+\eta)H(\varepsilon_m) + m + O(1).$$

Here $m = o(H(\varepsilon_m))$. Intersecting over positive rational η gives the upper constant one. Monotonicity of the covering time and $H(e^{-(m+1)})/H(e^{-m}) \to 1$ extend it to every small ε. The relative-frequency version follows from the last assertion of Lemma 2.8.2.

For the lower bound, polynomial growth of the frequencies and (2.8.2) ensure $V = \sum_j w_j|\lambda_j| < \infty$. Hence $E_w(\Phi_t\theta)$ is uniformly V-Lipschitz in t. If every starting point has energy below ε at some time in $[0,T]$, a time mesh of spacing ε/V covers Ω by at most $2+TV/\varepsilon$ translates of $\{E_w \le 2\varepsilon\}$. Haar measure implies

$$1 \le (2+TV/\varepsilon)\mu\{E_w \le 2\varepsilon\}.$$

By Proposition 2.8.4, taking logarithms gives $\log T \ge (1-o(1))H(\varepsilon)$, since $\log(1/\varepsilon) = o(H)$ and $H(2\varepsilon) \sim H(\varepsilon)$. This proves (2.8.8).

Almost surely all finite integer relations among the chosen frequencies are nonzero, by conditioning on one coordinate and then taking a countable union. Every orbit is therefore dense by the finite-dimensional Kronecker theorem. To obtain long avoidance intervals on every orbit, apply the preceding mesh argument at tolerance 2ε and $T = \exp((1-\eta)H(\varepsilon))$. For all small ε its union has measure less than one, so there is a segment whose energy stays above 2ε. Compactness in time and continuity give an open neighborhood of starting points whose segments stay above ε. Every dense orbit meets that neighborhood. The uniform upper covering bound supplies the opposite bound, and then $\eta \downarrow 0$ proves the final assertion. □

Example 2.8.6 (Arbitrarily small perturbations of prime logarithms)**.** Enumerate the primes as p_j, put $c_j = \log p_j$, and let $r_j = \rho p_j^{-A}$ with any fixed $A > 0$ and $0 < \rho \le 1$. The width hypothesis holds because $\log L_n = O(\log n)$. Thus (2.8.8) holds for weights satisfying (2.8.2), for both full and relative prime frequencies. For $w_j = e^{-aj}$ the leading constant is $1/(4a)$; for $w_j = e^{-aj^2}$ it is $1/(3\sqrt{a})$. This does not imply the same statement at the unperturbed vector: a full-measure set need not contain that prescribed point.

Proposition 2.8.7 (Perturbation widths can change the leading constant)**.** *Let $w_j = e^{-aj^b}$ and suppose $|\lambda_j| \le e^{-cj^{b+1}}$ for all j, with $a, b, c > 0$. Then*

$$\liminf_{\varepsilon\downarrow 0} \frac{\log R_{w,\lambda}(\varepsilon)}{(\log(1/\varepsilon))^{1+1/b}} \ge c\, a^{-(b+1)/b}.$$

Consequently, if $c > ab/[2(b+1)]$, the ratio-one conclusion in (2.8.8) fails for every such frequency vector. This includes independent uniform perturbations of the zero vector with these radii, which are nonresonant with probability one.

Proof. Choose the largest integer j for which $w_j \ge 8\varepsilon$. Then $j \sim (\ell/a)^{1/b}$. Start its phase at π. During $0 \le t \le \pi/(2|\lambda_j|)$ that coordinate stays in the opposite closed semicircle, so its energy is at least w_j and the target cannot be reached. If $\lambda_j = 0$ this gives an infinite obstruction. Otherwise

$$\log R_{w,\lambda}(\varepsilon) \ge cj^{b+1} + \log(\pi/2).$$

Divide by $\ell^{1+1/b}$ and compare the result with (2.8.3). Nonresonance in the uniform model again follows by conditioning on one coordinate for each of the countably many nonzero integer characters. □

Thus control of concentration of the frequency law is substantive. The proposition demonstrates a sharp relevant growth scale for that control; it does not determine the full transition at the boundary $\log L_n \asymp n^{b+1}$, and it concerns zero-centered rather than prime-centered perturbations.

Precise improvement over the two previous manuscripts. The integrated manuscript proved matching upper and lower orders for randomly perturbed prime flows with polynomial-logarithmic weights, and explicitly did not claim a limiting leading constant. The sequel optimized a deterministic rectangular budget at its logarithmic transition. Theorem 2.8.5 treats a different, rapidly decreasing spectral class and proves a ratio-one limit for the actual uniform covering time. The fractional-moment argument is essential to this constant: bounding all divisors by their worst value and then multiplying by the Fourier norm can pay the mode entropy twice. We do not claim a ratio-one limit for polynomial weights, where the $O(n)$ terms in the mode count are of leading size.

2.8.4 One exact anchor and the remaining arithmetic obstruction

The original actual-support obstruction used a pure high-frequency prime mode to force a large arithmetic height. Such a pure mode has a large, rather than small, temporal frequency. The following elementary refinement makes this distinction quantitative.

Proposition 2.8.8 (An anchor prime has no exponential degree cost)**.** *Fix a prime q. For a finite prime-indexed integer vector k put $v(k) = \sum_p k_p \log p$. If $\sum_{p\neq q} |k_p| \log p \le B$, then every $k \neq 0$ satisfies*

$$|v(k)| \ge e^{-1-2B}, \tag{2.8.11}$$

independently of the size of k_q. Consequently a real finite polynomial F with positive mean m, whose Fourier modes all obey that tail-height bound, gives a uniform prime-flow hitting time

$$T > 2e^{1+2B}\|F\|_A/m \tag{2.8.12}$$

for $\{F > 0\}$.

Proof. If $|v(k)| > 1$ there is nothing to prove. Otherwise $|k_q| \log q \le 1 + B$, and therefore $\sum_p |k_p| \log p \le 1+2B$. Write $v(k) = \log(A/C)$ with coprime positive integers A, C. Unique prime factorization gives $A \neq C$, and $\max(A, C) \le e^{1+2B}$. Since $|\log(A/C)| \ge |A - C|/\max(A, C)$, the separation follows. Apply Lemma 2.8.1 and $S(F) \le e^{1+2B}\|F\|_A$ for the second assertion. □

For a product of the kernels in (2.8.9), the norm cost of the anchor is at most $D_q + 1$. Thus resolving that coordinate to energy $O(\varepsilon)$ costs only $O(\log(1/\varepsilon))$ in the logarithm of (2.8.12), even though its degree is of order $\varepsilon^{-1/2}$. This is a strict removal of one artificial height cost; it is not a change in the known $\varepsilon^{-1/2}$ power for a full power-weight spectrum.

Proposition 2.8.9 (Quantitative return-section reduction)**.** *Index one distinguished coordinate by 0, suppose $\lambda_0 > 0$, and assume $V_2^2 = \sum_{j\ge1} w_j\lambda_j^2 < \infty$. Let $P = 2\pi/\lambda_0$, and let $Q(\varepsilon)$ be the least integer N such that the discrete tail translation with angles $(2\pi\lambda_j/\lambda_0)_{j\ge1}$ hits its tail energy ball from every starting point at some iterate $0 \le n \le N$. Then*

$$R_{w,\lambda}(\varepsilon) \le P\,[Q(\varepsilon) + 1], \tag{2.8.13}$$

$$Q(C\varepsilon) \le \lceil R_{w,\lambda}(\varepsilon)/P\rceil + 1, \qquad C = \left(1 + \frac{\pi V_2}{2\lambda_0\sqrt{w_0}}\right)^2. \tag{2.8.14}$$

The inequalities are interpreted by approximation if an infimum is not attained, and are trivial if the relevant time is infinite.

Proof. From any starting point the distinguished coordinate first reaches zero at a time $t_0 \in [0, P)$. Its subsequent exact returns are $t_0 + nP$, and the tail at those times follows the stated discrete translation. This proves (2.8.13).

For the other direction start the distinguished coordinate at zero, with an arbitrary tail point, and take a continuous-time hit at t. Choose the nearest nonnegative integer n to t/P. The elementary circle estimate $\theta^2 \le (\pi^2/2)(1-\cos\theta)$ gives

$$|t - nP| \le \frac{\pi}{\sqrt{2}\lambda_0}\sqrt{\varepsilon/w_0}.$$

In the tail metric $d^2 = 2E$, translation over a time u moves a point by at most $|u|V_2$. The triangle inequality therefore bounds the energy at nP by

$$\frac{1}{2}\left(\sqrt{2\varepsilon} + \frac{\pi V_2}{\sqrt{2}\lambda_0}\sqrt{\varepsilon/w_0}\right)^2 = C\varepsilon.$$

The index is no larger than the stated integer bound. Taking the supremum over tail starting points proves the result. □

Why this does not close the exact-prime gap. Two free anchor degrees cannot both be removed from (2.8.11). Indeed $\log 2/\log 3$ is irrational by unique factorization, and the pigeonhole proof of Dirichlet approximation produces nonzero integers r, s with $|r\log 2 - s\log 3| \to 0$. If both primes were omitted from the height, the proposed bound would be a positive constant for these forms, which is false. The return section likewise replaces one continuous coordinate by a simultaneous discrete approximation problem; it does not make the remaining coordinates independent. Removing finitely many primes leaves the power-weight tail in the same asymptotic class. Therefore neither proposition establishes

$$\log R_{w,\log p}(\varepsilon) \asymp \varepsilon^{-1/(\sigma-1)}(\log(1/\varepsilon))^{-\sigma/(\sigma-1)} \quad \text{for } w_p \asymp p^{-\sigma},\ \sigma > 3.$$

That matching entropy estimate remains an arithmetic question.

2.9 Recovering the weighted phase geometry from one radial law

The leading small-ball entropy in the preceding theorem records two tail parameters. The full radial distribution contains strictly more information: it determines every weight, including finite multiplicities. This is an inverse statement about the product geometry, not about its temporal frequencies.

Theorem 2.9.1 (Exact radial rigidity of weighted circle products)**.** *Let (w_j) and (v_j) be finite or countably infinite positive summable sequences. If*

$$\sum_j w_j(1-\cos\Theta_j) \quad \text{and} \quad \sum_j v_j(1-\cos\Theta_j)$$

have the same distribution under independent uniform angles, then their multisets of weights agree, including multiplicities. Thus their weighted product-circle metric spaces are isometric by a coordinate permutation. In fact, equality of all even cumulants already forces equality of the weight multisets.

Proof. Put $X_w = \sum_j w_j \cos\Theta_j$. This is bounded, and its moment generating function is

$$M_w(t) = \prod_j I_0(tw_j).$$

The classical Bessel product [6, eq. 10.21.15] gives

$$I_0(z) = \prod_{r\ge 1}(1 + z^2/j_{0,r}^2), \qquad \log I_0(z) = \sum_{m\ge 1} \frac{(-1)^{m+1}}{m} \left(\sum_{r\ge 1} j_{0,r}^{-2m}\right) z^{2m}$$

near zero. All coefficients in the latter series are nonzero. Local uniform convergence, justified by $\sum_j w_j^2 < \infty$, therefore gives

$$\kappa_{2m}(X_w) = (2m)! \frac{(-1)^{m+1}}{m} \left(\sum_{r\ge 1} j_{0,r}^{-2m}\right) \sum_j w_j^{2m}. \tag{2.9.1}$$

To see that the interchange is legitimate, take $|t| < j_{0,1}/(2\sup_j w_j)$; the logarithmic products converge absolutely there and are dominated by a constant times $\sum_j w_j^2$. Centering and changing sign do not change even cumulants of order at least two. Hence the radial law, or its even cumulants alone, determines every number $P_m = \sum_j w_j^{2m}$, $m \ge 1$.

Now define the finite positive measure

$$\nu_w = \sum_j w_j^2 \, \delta_{w_j^2}.$$

Its moments of orders $0, 1, \ldots$ are $P_1, P_2, \ldots$. The measures associated with two candidate sequences are supported in one common compact interval. Polynomial approximation of continuous functions shows that equality of these moments implies equality of the measures. At a positive atom x, its multiplicity as a squared weight is exactly $\nu_w\{x\}/x$. Zero has no atom because the weights are positive. This recovers the full multiset. A coordinate permutation preserving weights preserves the metric, which proves the geometric assertion. □

Corollary 2.9.2 (A constructive descending recovery rule)**.** *From the power traces in* (2.9.1)*, the largest weight u and its multiplicity r are*

$$u = \lim_{m\to\infty} P_m^{1/(2m)}, \qquad r = \lim_{m\to\infty} P_m/u^{2m}.$$

Subtracting ru^{2m} from each P_m and repeating recovers every distinct positive weight and its multiplicity.

Proof. Summability implies that the largest positive weight is attained and has finite multiplicity. The inequalities $u^{2m} \le P_m \le u^{2m-2}P_1$ prove the first limit. For the second, dominated convergence applies to $\sum_j (w_j/u)^{2m}$, dominated for $m \ge 1$ by the summable series $\sum_j (w_j/u)^2$. After deleting the largest weights the same argument applies. There are only finitely many weights above any positive threshold, so descending recovery reaches every weight. □

Proposition 2.9.3 (Leading entropy is not a complete invariant)**.** *Changing finitely many weights in a sequence satisfying* (2.8.2) *does not change the leading small-ball constant* (2.8.6)*. Such a change can nevertheless change the radial law and the metric space.*

Proof. The changed sequence still satisfies (2.8.2) with the same a, b, so Proposition 2.8.4 applies with the same constant. Increase, for example, one largest weight strictly. The multiset changes, and so does P_1 and the variance in (2.9.1). Thus the radial law changes. The diameter squared of the weighted product metric is $4\sum_j w_j$, which also increases, so the two metric spaces are not isometric. □

This inverse theorem extends the earlier manuscript's use of small-ball entropy as a coarse scale. It needs the full law, or infinitely many exact cumulants. It provides no finite-data stability estimate and does not identify the frequencies: Haar radial laws are unchanged when the temporal frequency vector changes.

2.10 What a zero pattern remembers: phases, spectra, and modular ordinates

The objects in this section are zero divisors in a region of absolute convergence, with vertical translation as the dynamics. They are not the nontrivial zeros of the Riemann zeta function. In particular, the Euler product below is zero-free; its critical points are the zeros under study. Three questions organize the argument.

1. *An inverse question formulated here.* Does the critical-point pattern near the right edge retain the infinitely many prime phases, or does passing from a function to its zeros destroy some of this information? Theorem 2.10.3 proves complete recovery and identifies the entire dynamical system, not just a factor of it.

2. *A published modular-density question.* Podesta [48, Section 1, paragraphs following the average vertical zero density] asks whether, for an exponential sum with rationally independent positive exponents, there exist lengths L for which the imaginary parts of its zeros are dense modulo L. The paper states that a solution will be presented in a future publication. We prove the stronger equidistribution statement for every L outside an explicit countable resonance set. This answers the existence question as printed; it does not classify the resonant lengths.

3. *A special case of an external research program.* Feldheim's research statement [52, p. 1, paragraph beginning with the unresolved atomic case] asks for an understanding of the random limiting zero density when a stationary Gaussian analytic function has atoms in its spectral measure. Her paper [50, Theorem 1 and Remark 2.3] proves randomness of that limit and describes the two-atom example. We compute the conditional density explicitly for three frequencies with irrational ratio, and express the random limit and its quadratic variance coefficient through three independent Rayleigh variables. This is a special-case computation, not a solution for arbitrary atomic spectra.

The general relation between almost periodicity and pure point spectra is classical; see [49]. The results below are proved directly to isolate the additional inverse statement and the precise scope of the applications. No historical priority claim is made for the elementary torus arguments or the three-term intensity formula. A targeted search through 14 October 2026 located the cited modular question but no subsequent primary-source resolution of that exact formulation; this is a bounded search, not a priority certificate.

2.10.1 Compact analytic families and their zero divisors

Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ and $\Omega = \prod_p \mathbb{T}$, with product topology and Haar probability m. Put

$$R_t\omega = (\omega_p p^{-it})_p, \qquad Q_\omega(s) = \sum_p \frac{(\log p)\omega_p p^{-s}}{1-\omega_p p^{-s}} = -\frac{Z'_\omega(s)}{Z_\omega(s)}, \qquad \Re s > 1.$$

The product $Z_\omega = \prod_p (1-\omega_p p^{-s})^{-1}$ never vanishes there. All sums and their derivatives converge locally uniformly, uniformly in ω. Thus $\omega \mapsto Q_\omega$ is continuous for locally uniform convergence, and $Q_{R_t\omega}(s) = Q_\omega(s+it)$. No Q_ω is identically zero: as $\sigma \to +\infty$, its first Dirichlet term is $(\log 2)\omega_2 2^{-\sigma}$ and the remaining terms are uniformly smaller in relative size.

For a vertical open strip D compactly contained horizontally in $\{\Re s > 1\}$, denote the zero divisor by

$$\mathcal{Z}_D(\omega) = \sum_{\rho \in D:\ Q_\omega(\rho)=0} \mathrm{mult}_{Q_\omega}(\rho)\, \delta_\rho.$$

We use the vague topology of locally finite measures on D, and let T_t translate a divisor down by it. A point carries its real coordinate as a mark. Forgetting that mark is a separate operation.

Lemma 2.10.1 (Continuity and uniform local zero bounds)**.** *The map $\mathcal{Z}_D$ is continuous and satisfies $\mathcal{Z}_D(R_t\omega) = T_t\mathcal{Z}_D(\omega)$. For every compact real interval $I \subset (1,\infty)$ there is $C_I < \infty$ such that, uniformly in ω and $u \in \mathbb{R}$, the number of zeros in $I + i[u, u+1]$, counted with multiplicity, is at most C_I. The same statements hold for any compact continuous family of nonzero holomorphic functions with the stated translation covariance.*

Proof. Suppose $f_j \to f \not\equiv 0$ locally uniformly. Around the finitely many zeros relevant to a compactly supported continuous test function, choose disjoint small disks with zero-free boundaries. Rouche's theorem preserves the number of zeros, with multiplicity, in each disk. On the remaining compact set the limit has a positive lower modulus, so no additional zeros occur. Uniform continuity of the test function then gives convergence of its integrals against the divisors. This argument does not require simple zeros. It proves continuity of $\mathcal{Z}_D$; covariance follows from the analytic identity.

For a fixed compact rectangle, continuity and compactness give a uniform bound on zero counts: use a nonnegative compactly supported test function equal to one on the rectangle. A slightly larger open strip may be used when the rectangle touches the boundary of the original D. Translation covariance gives the same bound at every height. □

2.10.2 The edge is a complete phase code

Let $E > 1$ be the unique solution

$$\frac{\log 2}{2^E + 1} = \sum_{p \geq 3} \frac{\log p}{p^E - 1}, \qquad z_2^* = -1, \quad z_p^* = 1 \ (p \geq 3), \qquad w_p = (\log p)p^{-E}. \tag{2.10.1}$$

For the untwisted product $E = 2.813014\ldots$ is the classical right edge for zeros of ζ'; its value and qualitative phase freezing appear in [47, Theorems 5.1 and 6.1]. The following short argument also establishes the uniform input needed for the inverse problem.

Lemma 2.10.2 (Uniform edge recovery estimate)**.** *There exist $\epsilon_0, C > 0$ such that every zero $\rho = \sigma + i\gamma$ of Q_ω with $E - \epsilon_0 < \sigma \leq E$ satisfies*

$$\sum_p w_p \left|\omega_p - z_p^* p^{i\gamma}\right|^2 \leq C(E - \sigma). \tag{2.10.2}$$

There are no zeros with $\sigma > E$. For every ω, there exist zeros with real parts tending to E and ordinates tending to $+\infty$.

Proof. For real $\sigma > 1$, write $r_p = p^{-\sigma}$ and

$$c_p = \frac{(\log p)r_p^2}{1 - r_p^2}, \quad R_p = \frac{(\log p)r_p}{1 - r_p^2}, \quad \xi_p = \frac{z_p - r_p}{1 - r_p z_p}.$$

For $|z_p| = 1$ we have $|\xi_p| = 1$ and $(\log p)r_p z_p/(1 - r_p z_p) = c_p + R_p\xi_p$. Set $C_0 = \sum_p c_p$, $b_0 = R_2$, $R_0 = \sum_{p \geq 3} R_p$, and

$$\Delta(\sigma) = C_0 + R_0 - b_0 = \sum_{p \geq 3} \frac{\log p}{p^\sigma - 1} - \frac{\log 2}{2^\sigma + 1}.$$

The ratio of the first sum to the final positive term is strictly decreasing from infinity to zero: the logarithmic derivative of its pth summand is $(\log 2)2^\sigma/(2^\sigma + 1) - (\log p)p^\sigma/(p^\sigma - 1) < 0$.

This proves uniqueness of E and negativity of Δ to its right. At a zero use $z_p = \omega_p p^{-i\gamma}$. The vector equation

$$C_0 + b_0\xi_2 = -\sum_{p\ge 3} R_p\xi_p$$

rules out $\Delta < 0$. Close to E it implies

$$C_0 b_0|\xi_2+1|^2 \le \Delta(\sigma)(R_0+b_0-C_0), \qquad \sum_{p\ge 3} R_p|\xi_p-1|^2 = 2\Delta(\sigma) + b_0|\xi_2+1|^2.$$

Here C_0, b_0, R_0 and $b_0 - C_0$ stay positive and bounded away from zero. Since $\Delta(\sigma) = O(E-\sigma)$, and the circle maps satisfy

$$|\xi - 1| = \frac{(1+r)|z-1|}{|1-rz|}, \qquad |\xi+1| = \frac{(1-r)|z+1|}{|1-rz|},$$

uniform comparison and $R_p \ge w_p$ give (2.10.2).

For existence, the frozen function is $Q_{z^*}(s) = \Delta(s)$. Writing $h = (\log 2)/(2^E+1)$ and $a_p = (\log p)/(p^E-1)$ gives

$$\Delta'(E) = h\frac{(\log 2)2^E}{2^E+1} - \sum_{p\ge 3} a_p \frac{(\log p)p^E}{p^E-1} < h(\log 2 - \log 3) < 0.$$

Thus E is a simple frozen zero. Integer independence of the prime logarithms makes every forward R_t orbit dense in Ω. Choose $t_j \to \infty$ with $R_{t_j}\omega \to z^*$. Local uniform convergence and Rouche's theorem produce zeros $\rho_j = E + o(1) + it_j$ of Q_ω. Their real parts cannot exceed E. At equality the displayed defects force $R_{\Im\rho_j}\omega = z^*$; there is at most one such ordinate, since two would give a nonzero common period for all prime logarithms. Hence the sequence can be chosen with $\Re\rho_j < E$. □

Theorem 2.10.3 (Inverse phase recovery and exact zero-hull conjugacy)**.** *Fix $0 < \epsilon < E - 1$ and $\delta > 0$, and put $D = (E-\epsilon, E+\delta) + i\mathbb{R}$. The map $\omega \mapsto \mathcal{Z}_D(\omega)$ is a homeomorphism of Ω onto its image. It conjugates R_t with vertical translation of the zero pattern. Its image is the translation hull of the divisor of the zeros of ζ' in D.*

More explicitly, any sequence of zeros $\rho_j = \sigma_j + i\gamma_j$ with $\sigma_j \to E$ recovers the phase by

$$\omega_p = z_p^* \lim_{j\to\infty} p^{i\gamma_j} \qquad (p \text{ prime}). \tag{2.10.3}$$

A single such zero recovers it with the weighted error (2.10.2). If two twists have sequences of zeros with real parts tending to E and differences of ordinates tending to zero, then the twists are identical.

Proof. Since each $w_p > 0$, (2.10.2) implies (2.10.3), independently of the chosen sequence. If two divisors in D coincide, apply this formula to a sequence given by Lemma 2.10.2; their phases coincide coordinatewise. The divisor map is therefore continuous and injective. Its domain is compact and the vague measure space is Hausdorff, so it is a homeomorphism onto its image. Covariance proves conjugacy. The orbit of $\omega = (1)_p$ is dense, so its divisor has precisely this image as its translation hull. Finally $p^{i(\gamma_j - \gamma_j')} \to 1$ for each fixed prime, which proves the last assertion by the same recovery formula. □

Remark 2.10.4 (Prime sums and the unavoidable scalar ambiguity). For $F_\omega(s) = \sum_p a_p\omega_p p^{-s}$ with positive bounded amplitudes, assume that there exists $S > 1$ satisfying the edge equation $a_2 2^{-S} = \sum_{p\ge 3} a_p p^{-S}$. The same argument recovers the relative phases $\eta_p = -\omega_p/\omega_2$, $p \ge 3$:

$$\eta_p = \lim_j (p/2)^{i\gamma_j}.$$

Indeed, normalization gives $1-\sum_{p\ge 3} b_p\eta_p(p/2)^{-(s-S)}$ with $\sum b_p=1$. At a zero $\sigma+i\gamma$ near S, taking real parts and using a first-order expansion in $S-\sigma$ bounds $\sum b_p|\eta_p(p/2)^{-i\gamma}-1|^2$ by $C(S-\sigma)$. The frozen zero is simple, and relative prime logarithms are integer independent, so the existence and compactness proof applies verbatim. A common scalar multiplying all ω_p leaves the divisor unchanged and cannot be recovered. The resulting hull is the relative phase torus, with frequencies generated by $\log(p/2)$, not the full Euler phase torus.

2.10.3 Exact spectra, correlations, and what hyperuniformity still needs

Write $k=(k_p)_p\in\mathbb{Z}^{(\mathcal{P})}$ for a finitely supported integer vector, and put

$$\chi_k(\omega)=\prod_p \omega_p^{k_p}, \qquad \lambda_k=\sum_p k_p\log p.$$

The characters form an orthonormal basis of $L^2(\Omega,m)$ and $\chi_k(R_t\omega)=e^{-i\lambda_k t}\chi_k(\omega)$. Distinct k give distinct λ_k, by unique factorization.

Theorem 2.10.5 (The spectrum of the marked edge-zero process)**.** *The Haar random marked divisor of Theorem 2.10.3 is stationary and ergodic, with simple pure point dynamical spectrum*

$$\{\log q : q\in\mathbb{Q}_{>0}\}.$$

It is not mixing. For any continuous observable A of that divisor, write

$$a_k=\int A(\mathcal{Z}_D(\omega))\overline{\chi_k(\omega)}\,dm.$$

Its centered covariance is exactly

$$C_A(t)=\sum_{k\neq 0}|a_k|^2e^{-i\lambda_k t}. \tag{2.10.4}$$

This series converges absolutely and uniformly. The same correlations are deterministic vertical averages for every initial twist:

$$C_A(t)=\lim_{T\to\infty}\frac{1}{T}\int_0^T (A(\mathcal{Z}_D(R_{u+t}\omega))-a_0)\,\overline{(A(\mathcal{Z}_D(R_u\omega))-a_0)}\,du.$$

In particular, randomizing a long vertical shift of the actual ζ' edge divisor converges to this Haar law.

Proof. The inverse theorem transfers the character basis to the divisor space, giving exactly the stated spectrum. No nonconstant character has zero frequency, proving ergodicity. A character has correlation $e^{-i\lambda_k t}$, which does not tend to zero, so mixing fails. Parseval gives (2.10.4) and $\sum_{k\neq 0}|a_k|^2<\infty$, proving uniform absolute convergence.

For a continuous function on Ω, finite character sums approximate it uniformly by Stone–Weierstrass. The average of every nontrivial character along R_u tends to zero uniformly in its starting point. Thus all continuous observables have uniform Haar time averages. Apply this first to the product in the displayed correlation, and then to an arbitrary continuous test on the divisor hull. □

Proposition 2.10.6 (Bragg weights and an exact variance test)**.** *Fix a compact real interval $I\subset(1,\infty)$ and project the zeros in $I+i\mathbb{R}$ to their ordinates, retaining multiplicity. Let this*

random measure be ν_ω, and write $N_T = \nu_\omega([0,T])$, $\rho = \mathbb{E}N_1$. There are constants $c_k \in \mathbb{C}$, with $c_0 = \rho$, such that

$$\operatorname{Var}\left(\int \psi \, d\nu_\omega\right) = \sum_{k\neq 0} |c_k|^2 |\widehat{\psi}(\lambda_k)|^2, \qquad \widehat{\psi}(\lambda) = \int e^{-i\lambda t}\psi(t)\, dt, \tag{2.10.5}$$

for compactly supported bounded measurable ψ. Consequently

$$\operatorname{Var} N_T = \sum_{k\neq 0} |c_k|^2 \frac{4\sin^2(\lambda_k T/2)}{\lambda_k^2}. \tag{2.10.6}$$

Define $B(r) = \sum_{0<|\lambda_k|\le r} |c_k|^2$. Then

$$\operatorname{Var} N_T = o(T) \quad \Longleftrightarrow \quad B(r) = o(r) \quad (r \downarrow 0), \tag{2.10.7}$$

$$\sup_{T>0} \operatorname{Var} N_T < \infty \quad \Longleftrightarrow \quad \sum_{k\neq 0} \frac{|c_k|^2}{\lambda_k^2} < \infty. \tag{2.10.8}$$

The same statements hold for a bounded nonnegative real-coordinate weight of compact support in place of 1_I.

Proof. Uniform local zero bounds imply that ν_ω is uniformly translation bounded. Define the complex locally finite measures

$$\nu_k(B) = \int_\Omega \nu_\omega(B)\overline{\chi_k(\omega)}\, dm(\omega).$$

Covariance and Haar invariance give $\nu_k(B+u) = e^{-i\lambda_k u}\nu_k(B)$. Multiplication by $e^{i\lambda_k t}$ makes this a translation-invariant measure, so $d\nu_k(t) = c_k e^{-i\lambda_k t} dt$. Thus the kth Fourier coefficient of $\int \psi\, d\nu_\omega$ is $c_k\widehat{\psi}(\lambda_k)$. Parseval proves (2.10.5), including indicator functions, and then (2.10.6).

The measure $\sum_{k\neq 0}|c_k|^2\delta_{\lambda_k}$ is finite near zero, because the Fourier transform of a sufficiently narrow positive test function is bounded away from zero there. Also

$$\sum_{|\lambda_k|\ge 1} \frac{|c_k|^2}{\lambda_k^2} < \infty.$$

Indeed, integrate (2.10.6) over $0 < T < 1$; its left side is uniformly bounded, whereas the integral of $4\sin^2(\lambda T/2)$ is bounded below by a positive constant for $|\lambda| \ge 1$.

For $|\lambda|T \le 1$ the sine factor bounds the variance below by $cT^2B(1/T)$. This proves necessity in (2.10.7). Conversely suppose $B(r) = o(r)$. The part $|\lambda| \le 1/T$, divided by T, is at most $TB(1/T)$. For $1/T < |\lambda| < \delta$, use dyadic shells and $B(r) \le \eta r$ for $r \le 2\delta$ to bound the same quotient by $C\eta\sum_{j\ge 0} 2^{-j}$. The part $|\lambda| \ge \delta$ tends to zero by the established inverse-square summability away from zero. Let $\eta \downarrow 0$.

Inverse-square summability directly implies bounded variance. Conversely, if the variance is bounded by M, average over $0 < T < R$. For every nonzero fixed λ the average of $4\sin^2(\lambda T/2)$ tends to 2. Fatou's lemma gives $2\sum_{k\neq 0}|c_k|^2/\lambda_k^2 \le M$. □

The measure in (2.10.5), including the atom $\rho^2\delta_0$, is the pure point diffraction measure in the Fourier convention used there. Individual c_k may vanish. Complete dynamical spectrum does not assert a nonzero Bragg peak at every $\log q$: nonlinear observables can recover frequencies absent from a particular linear count. Nor does pure point spectrum by itself prove (2.10.7). Estimating these low-frequency weights for the Euler critical-point process remains unresolved here.

Every nonempty projected process of positive intensity is nevertheless nonmixing. Otherwise its bounded local-count observables would have decaying correlations; their pure point covariance expansions force them all to be constant. That would make the random measure deterministic and invariant under every real translation, hence a multiple of Lebesgue measure, contrary to its nonzero locally finite atomic nature. Thus its law cannot equal a positive-intensity Poisson process or the sine-kernel determinantal process. For example their compact-test covariances tend to zero, by the Poisson covariance formula and the decay of the squared sine kernel, respectively. This is a fixed-strip, fixed-scale statement. It imposes no obstruction to a different limit under a changing strip or rescaling, and none to the GUE conjecture for critical-line zeta zeros.

2.10.4 A resolution of the generic-period modular-density question

Theorem 2.10.7 (Equidistribution of exponential-sum zeros modulo a length)**.** *Let*

$$f(s) = A_0 + \sum_{j=1}^{N} A_j e^{\omega_j s}, \qquad A_j \neq 0, \quad 0 < \omega_1 < \cdots < \omega_N,$$

where $\omega_1, \dots, \omega_N$ are linearly independent over $\mathbb{Q}$. Count zeros with multiplicity. If

$$\frac{2\pi}{L} \notin \operatorname{span}_{\mathbb{Q}}\{\omega_1, \dots, \omega_N\}, \tag{2.10.9}$$

then, for every interval $J \subset [0, L)$,

$$\frac{\#\{\rho : f(\rho) = 0,\ 0 < \Im\rho \le T, \Im\rho \bmod L \in J\}}{\#\{\rho : f(\rho) = 0,\ 0 < \Im\rho \le T\}} \longrightarrow \frac{|J|}{L}. \tag{2.10.10}$$

The denominator is $(\omega_N/(2\pi))T + o(T)$. Both conclusions are uniform when only the arguments of the A_j vary and their nonzero moduli are fixed. In particular, the ordinates are dense modulo L, and the assertion holds for all but countably many $L > 0$.

Proof. Multiplication by a nonzero constant does not change zeros, so write the phase family as $f_\theta(s) = a_0 + \sum_{j=1}^{N} a_j \theta_j e^{\omega_j s}$ with $a_j > 0$. The constant term dominates uniformly for sufficiently negative real part, and the final term dominates uniformly for sufficiently positive real part. Hence every member has all zeros in one common bounded real strip. None is identically zero. Lemma 2.10.1 applies, with flow $\theta_j \mapsto \theta_j e^{i\omega_j t}$, and gives a uniform bound on the number of zeros in every interval of unit height.

Choose $g \in C_c(\mathbb{R})$ equal to one throughout the common real zero strip, and $\eta \in C_c^\infty(\mathbb{R})$. The local observable

$$A_\eta(\theta) = \sum_{f_\theta(\rho)=0} g(\Re\rho)\eta(\Im\rho)$$

is continuous on $\mathbb{T}^N$. For any $\alpha \notin \Gamma := \sum_j \mathbb{Z}\omega_j$, character approximation gives, uniformly in θ,

$$\frac{1}{T}\int_0^T e^{i\alpha t} A_\eta(R_t\theta)\, dt \longrightarrow 0. \tag{2.10.11}$$

Indeed the integral of each approximating character has frequency $\alpha + \sum_j k_j\omega_j \neq 0$. Let H contain the support of η. By a change of variable for each zero, and the uniform zero bound in the two endpoint regions,

$$\int_0^T e^{i\alpha t} A_\eta(R_t\theta)\, dt = \widehat{\eta}(\alpha) \sum_{\substack{f_\theta(\rho)=0 \\ 0<\Im\rho\le T}} e^{i\alpha\Im\rho} + O_\eta(1). \tag{2.10.12}$$

The error is uniform in T and the initial phases. Choose η with $\widehat{\eta}(\alpha) \neq 0$, which is possible by taking a positive function supported in a sufficiently short interval. Equations (2.10.11)–(2.10.12) prove that the twisted zero sum is $o(T)$.

For completeness we compute the untwisted mean. Define

$$J(\sigma) = \int_{\mathbb{T}^N} \log |f_\theta(\sigma)| \, d\theta.$$

Circle Jensen integration and the positive constant term imply $\log a_0 \le J(\sigma) \le \log \sum_j a_j e^{\omega_j \sigma}$, where $\omega_0 = 0$. On compact real intervals these bounds also give integrability of the absolute value of the logarithm, since its positive part is uniformly bounded. Poincare–Lelong and Fubini show that the mean zero measure is $(2\pi)^{-1} J''(\sigma)\, d\sigma\, dt$, with J'' understood as a measure. Where the constant term dominates, the convergent logarithm series has no constant phase character, so $J = \log a_0$. Where the final term dominates, the same argument gives $J = \log a_N + \omega_N \sigma$. Therefore the total mean number per unit height is $\rho = \omega_N/(2\pi)$. Taking $\alpha = 0$ and $\int \eta = 1$ in (2.10.12), unique ergodicity gives $N(T) = \rho T + o(T)$ uniformly in the phases.

Condition (2.10.9) says exactly that $2\pi m/L \notin \Gamma$ for every nonzero integer m. The established twisted sums, divided by $N(T)$, are all the Weyl tests for uniform distribution modulo L. Trigonometric majorants and minorants of interval indicators prove (2.10.10), including uniformity in the initial phases. The excluded rational span is countable. □

There is also a short proof of the existence-and-density conclusion alone. A frozen phase choice has a simple real zero, obtained from $a_0 = \sum_{j \ge 1} a_j e^{\omega_j \sigma}$. Under (2.10.9), the joint flow of the N phases and t mod L is dense. Approximate the frozen phase and any desired residue simultaneously, then use Rouche's theorem near that real zero. This observation is sufficient for the existence question in [48]; the longer argument supplies uniform distribution and the exact density.

The nonresonance condition is sufficient, not asserted necessary. For instance $1 + e^{\omega s}$ has zeros on an arithmetic progression, so appropriate resonant lengths fail even density. In the Euler edge process there are explicit resonant obstructions: if $L_p = 2\pi/\log p$, (2.10.2) confines ordinates of zeros with $E - \epsilon < \Re\rho \le E$ modulo L_p to an arc of length $O_p(\sqrt{\epsilon})$, centered according to ω_p / z_p^*. Thus the exceptional periods are mathematically meaningful. Recovering the full classification of resonant L is outside Theorem 2.10.7.

2.10.5 Three atomic Gaussian frequencies: an explicit conditional law

The phase hull describes randomness generated by phases. Gaussian coefficients add invariant random amplitudes, and this distinction changes the macroscopic variance. We make the separation explicit. Let $0 < a < b$ with $a/b \notin \mathbb{Q}$ and

$$F(s) = \xi_0 + \xi_1 e^{-as} + \xi_2 e^{-bs},$$

where the ξ_j are independent standard circular complex Gaussians. Write $R_j = |\xi_j|$; their densities are $2re^{-r^2}\, dr$ on $(0, \infty)$, and their phases are independent uniform variables. Conditioning on $R = (R_0, R_1, R_2)$ and removing a common phase leaves a two-torus Kronecker system.

For fixed positive $r = (r_0, r_1, r_2)$ set

$$A = r_0, \qquad B = r_1 e^{-a\sigma}, \qquad C = r_2 e^{-b\sigma}, \qquad D_r(\sigma) = a^2B^2 + b^2C^2 + ab(A^2 - B^2 - C^2).$$

When A, B, C are the sides of a strict triangle, let

$$\mathcal{A}_r(\sigma) = \frac{1}{4}\sqrt{(A+B+C)(-A+B+C)(A-B+C)(A+B-C)}$$

be its area, and define

$$I_r(\sigma) = \frac{D_r(\sigma)}{4\pi^2 \mathcal{A}_r(\sigma)} \quad \text{in the strict triangle region,} \qquad I_r(\sigma) = 0 \quad \text{outside its closure.} \tag{2.10.13}$$

Values at its finitely many boundary points may be assigned arbitrarily.

Theorem 2.10.8 (Atomic conditional density and random limiting measure)**.** *For every fixed positive r, the mean zero measure conditional on $R = r$ has planar density $I_r(\sigma)$, independent of height. It has no mass on the boundary lines of the strict triangle region. For every finite real interval J and every choice of the coefficient phases,*

$$\frac{1}{T}\#\{\rho : F(\rho) = 0,\ \Re\rho \in J,\ 0 < \Im\rho \le T\} \longrightarrow M_r(J) := \int_J I_r(\sigma)\, d\sigma. \tag{2.10.14}$$

Consequently the Gaussian limiting real-coordinate measure has the explicit law $I_R(\sigma)\, d\sigma$. For a bounded interval of positive length J, writing $N_T(J)$ for the numerator above,

$$\lim_{T\to\infty} \frac{\operatorname{Var} N_T(J)}{T^2} = \operatorname{Var}_R M_R(J) > 0. \tag{2.10.15}$$

The expectation and variance on the right are ordinary integrals against $\prod_{j=0}^{2} 2r_j e^{-r_j^2}\, dr_j$.

Proof. At a fixed point $s = \sigma + it$, absorb t into the two uniform phases. The value of the function is $A + Be^{i\theta} + Ce^{i\phi}$. In the strict triangle region its zeros as a function of (θ, ϕ) are exactly two noncollinear configurations. At either configuration the absolute real Jacobian is

$$|\operatorname{Im}(Be^{i\theta}\overline{Ce^{i\phi}})| = 2\mathcal{A}_r(\sigma).$$

Also

$$|F'(s)|^2 = |-aBe^{i\theta} - bCe^{i\phi}|^2 = D_r(\sigma) > 0,$$

because $2\operatorname{Re}(Be^{i\theta}\overline{Ce^{i\phi}}) = A^2 - B^2 - C^2$. The change-of-variables formula for these two local phase branches therefore gives the zero intensity

$$\frac{1}{(2\pi)^2} \sum_{\text{two branches}} \frac{|F'(s)|^2}{2\mathcal{A}_r(\sigma)} = I_r(\sigma).$$

This is the two-dimensional Kac–Rice calculation, and can equivalently be obtained by applying the area formula on compact subsets where the two displayed Jacobians are nonzero. If a side is strictly larger than the sum of the others, cancellation is impossible.

The degenerate-triangle equations are real exponential polynomials with three distinct exponents. Each has finitely many real roots (at most two), by Rolle's theorem after division by an extreme exponential. At each such real coordinate there is only one phase configuration giving cancellation, modulo 2π. For bounded height, the set of initial phases which reaches this configuration is a smooth curve in the two-torus and has Haar measure zero. A countable union over height intervals still has measure zero. Thus the mean zero measure has no mass on these boundary lines, and (2.10.13) accounts for all of it.

For fixed amplitudes the divisor map is continuous on the compact two-torus and the flow is uniquely ergodic. Smooth real-coordinate test functions consequently have deterministic limiting densities given by the mean zero measure, using the smoothing and bounded-endpoint argument from Theorem 2.10.7. Since this mean measure is absolutely continuous, continuous majorants and minorants at the endpoints of J extend the limit to 1_J. This proves (2.10.14) for every initial phase, not just almost every one.

To pass to Gaussian variances, a uniform integrability argument is needed. Normalize any nonzero coefficient vector to have Euclidean norm one. The normalized vectors form a compact sphere; because the three exponentials are linearly independent, none gives an identically zero function. Lemma 2.10.1 gives a zero bound on $J + i[0, 1]$ uniform even when some coefficients vanish. Vertical translation preserves this sphere. Therefore $N_T(J) \le C_J(T + 1)$ for every coefficient vector, and dominated convergence proves (2.10.15) with a nonnegative constant. It is strictly positive: there is an open set of amplitudes on which the constant term dominates throughout J, so $M_R(J) = 0$; another open set gives a strict triangle at an interior point of J, and then $M_R(J) > 0$. Both open sets have positive Rayleigh probability. □

Feldheim already proves that an atomic spectral component gives a positive quadratic variance coefficient in this Gaussian setting [51, Theorem 1]. Equation (2.10.15) is not claimed as a new growth law; its contribution here is the explicit conditional coefficient supplied by (2.10.13). Conditional on the amplitudes, the limiting density is deterministic and the phase dynamics is pure point. Unconditionally, the random invariant amplitudes produce the quadratic term. One must therefore not transfer that quadratic term to the fixed-amplitude Euler phase hull.

These arguments give a complete answer to the new inverse question, an affirmative answer to the printed generic-period existence question with a stronger uniform-distribution theorem, and an explicit three-frequency instance of the atomic-density program. They leave the actual Euler Bragg weights, its fixed-strip hyperuniformity, the resonant-period classification, and general atomic Gaussian spectra undetermined.

2.11 Connections, limits, and the next mathematical bottlenecks

2.11.1 One graph gives two genuinely different invariants

Theorems 2.3.1 and 2.4.2 can be applied to the same covariance model, but they answer different questions. For an m-vertex covariance graph G, a compact parameter box of full dimension has

$$k = 2m + |E(G)|, \qquad q_* = \operatorname{gcr}(G) + 1, \qquad \mathcal{R}_B \asymp B^{-1/(k+2)}.$$

Here q_* is the exact number of shared-latent observations needed to identify an arbitrary mean/covariance law from one fixed group law. The dimension k governs the rate at which an arbitrary mixing law can be learned under a total budget with growing replication depths. This distinction is operational: infinitely many groups below q_* leave a positive risk floor, whereas unequal depths growing with B give consistency. A graph's likelihood-existence threshold is a third quantity. Existing computations for complete bipartite graphs show that it cannot replace q_*.

Thus the transfer produces an exact answer without computing every graph invariant. A substantive further advance would give new bounds on $\operatorname{gcr}(G)$, or prove a relation to the maximum likelihood threshold on a class where that relation is presently unknown. Transform uniqueness alone cannot supply those combinatorial bounds.

2.11.2 Shape constraints do not determine the information content

Strong log-concavity and arbitrarily high Gevrey regularity coexist with the nonidentifiability constructions in Theorems 2.3.1, 2.7.1, and 2.7.2. The three failures have different causes: a polynomial relation among transform probes; reflection of independent convolution factors; and a summable analytic zero set. These are explicit exact counterexamples, not examples of numerical instability.

On the other hand a quasianalytic restriction changes the heat problem completely: Proposition 2.7.3 identifies sources from any sequence of positive times tending to zero. Finite moments and very strong smoothness do not replace quasianalyticity. This identifies which qualitative prior changes the uniqueness argument, instead of merely improving constants in an estimate.

2.11.3 A probability class can be excluded by a mass budget

For the McKay classifications the decisive inverse object is a Thorin measure. If a candidate GGC law has an odd logarithmic derivative about a center, the representing measure must be symmetric about that center. Endpoint singularities and a central asymptotic then force specific atoms. Their sum cannot exceed the total mass. For the squared-Bessel family beyond the claimed range, this becomes the explicit contradiction

$$6\mu \le 4\mu + 1 \qquad (\mu > 1/2).$$

For the third family it becomes $\nu \le \mu + 1$. The positive half of each classification constructs a measure satisfying the same constraints through a classical hypergeometric ratio. This paired construction and obstruction is stronger than verifying a finite list of derivative signs.

The unresolved SD extension requires a different argument. A law can be self-decomposable without being GGC; a failed positive Thorin representation cannot by itself exclude self-decomposability. One must study the monotonicity of its Levy kernel, or find another valid SD criterion, on the remaining domain.

2.11.4 Full observations, coarse invariants, and the arithmetic gap

The exact radial law determines every metric weight, while its leading small-ball entropy retains only a coarse asymptotic scale. The near-edge critical-point divisor determines every Euler phase, while its projected count spectrum can have missing Bragg peaks. These inverse statements specify what is lost by each further compression of the observations.

The random recurrence theorem does not settle its deterministic prime-frequency analogue. Its proof uses independent anti-concentration to control a sum of inverse divisors. At one prescribed vector of prime logarithms that probabilistic step is unavailable. A theorem holding on a set of full measure cannot be substituted for an arithmetic estimate at that vector. Likewise, the exact criterion for hyperuniformity in (2.10.7) is not a proof that the Euler zero process satisfies it: the low-frequency Bragg weights still have to be estimated.

2.11.5 Four concrete continuation questions

The following problems isolate missing steps without claiming that they have been solved.

1. **The scalar endpoint boundary.** For arbitrary mixing of Uniform$[0,\theta]$, $\theta \in [1,2]$, is the total-budget W_1 rate $B^{-1/2}$, or is a logarithmic factor necessary? The present proved interval is
$$cB^{-1/2} \le \mathcal{R}_B^{\mathrm{ad}} \le \mathcal{R}_B^{\mathrm{fix}} \le CB^{-1/2}\log(eB).$$
The lower bound for this particular experiment must improve, or the critical multilevel summation must be bypassed.

2. **Sparse terminal lengths without covariance bounds.** For d-output terminal experiments with unbounded covariance laws, is harmonic divergence also necessary, or does multivariate coupling change the exact sparse criterion? The bounded-covariance case is settled, and harmonic divergence is sufficient without boundedness.

3. **Resonant modular lengths.** Theorem 2.10.7 settles all nonresonant lengths. Determine which resonant lengths still have density or equidistribution. The arithmetic-progression example and Euler edge phase confinement show why a universal assertion over resonant lengths would be false.

4. **Stability of phase recovery from perturbed zeros.** Theorem 2.10.3 gives an exact inverse and a weighted single-zero error bound. A finite-height statistical recovery theorem would also have to control how close to the edge one can get, how many zeros are available, and the sensitivity to uncertainty in ordinates. The deterministic arithmetic return problem is part of that difficulty.

2.11.6 Validation and interpretation of this draft

Each main branch has been read by a second reasoning agent in addition to its author; the assembled statements were also reviewed for matching assumptions, transform signs, endpoints, and scope. These are internal checks, not independent human peer review or a formal proof assistant verification. Numerical checks, where used, served only to detect normalization mistakes; no numerical scan replaces an analytic proof.

The source audit establishes connections to printed questions, not historical priority. That requires specialist review, including older exponential-family completeness theory for the Gaussian branch.

Part III

Binomial Mixture Likelihood and Exact Observation Thresholds

A Conjectured Range, Sparse Gaussian Sampling, and Resonant Zero Patterns

Source manuscript: 14 October 2026.

Source abstract. For an arbitrary mixing law on $[0,1]$, we prove that every nonparametric maximum likelihood estimator from N independent binomial counts with t trials has expected Wasserstein error $O_\varepsilon((t\log N)^{-1/2})$ throughout $\log N \le t \le N^{2/3-\varepsilon}$. This is the range conjectured by Vinayak, Kong, Valiant and Kakade in 2019. Two estimates enter: a contour integral removes a polynomial prefactor from the Chebyshev–Bernstein change of basis, and an aggregate ellipse bound gives finite likelihood brackets of logarithmic cardinality $O(\sqrt{t}\log^{3/2}(t/\delta))$. We derive consequences for replication caps, and develop separate exact results for sparse Gaussian sampling, self-decomposability on Bessel parameter families, and resonant periods of three-term exponential sums. Further results identify a deterministic small-divisor criterion for recurrence and prove limitations of finite continuous phase calibration and noisy analytic heat observations. The research register distinguishes published conjectures, new questions, partial classifications, and unresolved arithmetic obstacles. Earlier results are not counted as new resolutions. Historical priority of individual auxiliary lemmas is not asserted.

3.1 What is proved, and what the research programme did not close

This manuscript continues the 223-page integrated working paper [1], its 42-page sequel [29], and the 56-page sequel [53]. The new work starts at the unresolved boundaries of those drafts. None of their completed results is included in the present count of research targets. The most directly externally testable conclusion is Theorem 3.2.1: it has the same unrestricted mixing-law MLE, support interval, loss, and trial-count regime as the original 2019 conjecture. It does not replace the estimator or change the sampling design.

The improvement is a change in the range of a theorem, not a constant-factor gain. The originally proved exponent for the upper trial count was $2/9$; the later argument of Musco et al. reaches $1/4$; the theorem here reaches $2/3$, with an arbitrarily small fixed loss in each exponent. The target Wasserstein error stays $(t\log N)^{-1/2}$. The still larger near-linear range discussed in the 2026 author version of Musco et al. remains open here.

The other sections answer different questions about observation maps. They should not be represented as consequences of one universal theorem. The Gaussian section uses holomorphic functions on a positive matrix tube to construct a single joint invisible perturbation. The Bessel section uses polynomial root locations to make Levy kernels explicit. The zero section uses a lifted torus and triangle geometry to retain information lost by a nonresonant equidistribution argument. Topology and analytic approximation give negative results for two inverse problems with strong shape constraints.

3.1.1 Interpretation and validation

All the mathematical statements below are statements of this draft, not a claim of community certification. The main proofs have been checked by a second reasoning agent and by

the assembling reviewer; these checks are internal, not external peer review or formal proof verification. We give full arguments so that the results can be independently assessed. The literature comparison uses specific primary-source formulations available on 14 October 2026 and does not certify an exhaustive historical search.

The goal of an advance comparable to major work in analysis is a research direction, not a conclusion justified by ambition or by page count. In particular, no Riemann-hypothesis consequence, unrestricted McKay self-decomposability classification, or deterministic prime-flow recurrence asymptotic is asserted.

3.1.2 Research register

The register at the end records the precise scope of each attempted route and its outcome. A theorem for a new internal question is not counted as a solution of a previously published conjecture; two lemmas in one proof are not counted as two conjecture resolutions.

3.2 A wider validity range for the binomial-mixture maximum likelihood estimator

Consider N independent counts X_i obtained by drawing $\theta_i \sim G$ and then $X_i \mid \theta_i \sim \mathrm{Binomial}(t, \theta_i)$, where G is an arbitrary probability measure on $[0,1]$. Put

$$B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}, \qquad p_G(j) = \int B_j^t(x)\, dG(x), \quad 0 \le j \le t.$$

Let $\widehat{G}$ be any maximum likelihood solution over all probability measures on $[0,1]$. Such a solution exists: the attainable probability vectors form the compact convex hull of the continuous binomial curve, and the likelihood is upper semicontinuous. Nonuniqueness of the mixing measure is allowed throughout.

Vinayak, Kong, Valiant and Kakade [13, Theorem 3.2 and Remark 3.2] proved the medium-sample rate below for $t \le N^{2/9-\varepsilon}$ and conjectured the range $t \le N^{2/3-\varepsilon}$. Musco et al. [14, Theorem 20 and Section 6.2, author version] increased the proved exponent to $1/4$ and discussed further conditional improvements. The following argument reaches the originally conjectured range. It uses a bound with unspecified absolute constants for Chebyshev–Bernstein coefficients; it does not prove the sharper constant-one coefficient inequality discussed in those papers.

Theorem 3.2.1 (The conjectured medium-sample MLE range)**.** *For every $0 < \varepsilon < 2/3$ there are constants $C_\varepsilon, N_\varepsilon$ such that, whenever $N \ge N_\varepsilon$ and*

$$\log N \le t \le N^{2/3-\varepsilon}, \tag{3.2.1}$$

every mixing-measure MLE satisfies, uniformly over G,

$$\mathbb{P}_G\left\{ W_1(\widehat{G}, G) > \frac{C_\varepsilon}{\sqrt{t \log N}} \right\} \le N^{-2}, \qquad \mathbb{E}_G W_1(\widehat{G}, G) \le \frac{C_\varepsilon}{\sqrt{t \log N}}.$$

The same conclusion, with constants also depending on $a > 0$, holds if the lower limit in (3.2.1) is $a \log N$.

The matching lower bound in [13, Theorem 3.3] therefore makes the MLE minimax optimal throughout this range.

The proof separates two estimates that control different directions of the observation map: an aggregate analytic coefficient bound makes its image statistically small, whereas a contour bound controls polynomial inversion without a polynomial prefactor in t.

3.2.1 A contour estimate without a degree prefactor

Write $\widetilde{T}_m(x) = T_m(2x-1)$ and define the coefficients $C(t,m,j)$ by

$$\widetilde{T}_m(x) = \sum_{j=0}^{t} C(t,m,j) B_j^t(x), \qquad m \le t.$$

Lemma 3.2.2 (A uniform coefficient bound)**.** *There is an absolute C such that, for $0 \le m \le t/2$,*

$$\max_{0\le j\le t} |C(t,m,j)| \le C \exp(2m^2/t). \tag{3.2.2}$$

The bound includes $j = 0, t$; no restriction away from the parameter endpoints is imposed.

Proof. The assertion is immediate for $m = 0$ and for $j = 0, t$, where the coefficients have absolute value one. For $0 < j < t$, set

$$r = \frac{j}{t-j}, \qquad q = \frac{r}{(1+r)^2}, \qquad z = re^{i\phi}.$$

The polynomial

$$F(z) = (1+z)^t \widetilde{T}_m\left(\frac{z}{1+z}\right) = \sum_{j=0}^{t} \binom{t}{j} C(t,m,j) z^j$$

has the algebraic expression

$$F(z) = \frac{(1+z)^{t-m}}{2}\left\{(\sqrt{z}+i)^{2m} + (\sqrt{z}-i)^{2m}\right\}. \tag{3.2.3}$$

Although a square root is used in this identity, the sum is a polynomial in z and is independent of its choice. The identity follows from $T_m(w) = [(w+\sqrt{w^2-1})^m + (w-\sqrt{w^2-1})^m]/2$.

For $u = 2\sqrt{q}\,|\sin(\phi/2)| \in [0,1]$, one has

$$|1+z| = (1+r)\sqrt{1-u^2}, \qquad \max\{|\sqrt{z}+i|^2, |\sqrt{z}-i|^2\} = (1+r)(1+u).$$

Consequently, using $m \le t/2$,

$$\begin{aligned}\frac{|F(re^{i\phi})|}{(1+r)^t} &\le (1-u^2)^{(t-m)/2}(1+u)^m \\ &\le \exp(-tu^2/4 + mu) \le \exp(2m^2/t)\exp(-tu^2/8).\end{aligned}$$

The last step is the elementary maximum $\sup_{u\in\mathbb{R}}\{mu - tu^2/8\} = 2m^2/t$. Because $\sin(|\phi|/2) \ge |\phi|/\pi$ for $|\phi| \le \pi$,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \exp\{-tq\sin^2(\phi/2)/2\}\,d\phi \le \frac{C}{\sqrt{1+tq}}.$$

Cauchy's coefficient formula now yields

$$|C(t,m,j)| \le \frac{(1+r)^t}{\binom{t}{j} r^j} \frac{C\exp(2m^2/t)}{\sqrt{1+tq}}. \tag{3.2.4}$$

For completeness the binomial probability at its mean obeys

$$\binom{t}{j}(j/t)^j(1-j/t)^{t-j} \ge c\sqrt{\frac{t}{j(t-j)}} \ge \frac{c}{\sqrt{1+tq}}.$$

The first inequality follows directly by applying upper and lower Stirling bounds $c\sqrt{n}(n/e)^n \le n! \le C\sqrt{n}(n/e)^n$ to $t!, j!, (t-j)!$; here $j, t-j \ge 1$. This probability equals $\binom{t}{j} r^j/(1+r)^t$ and $tq = j(t-j)/t$. It cancels the square-root factor in (3.2.4), proving (3.2.2). $\square$

The weighted square-summability of the coefficients used next is explicitly developed in [14, Lemma 13]; we give the short periodic-function argument and the approximation step to make the dependence on constants self-contained.

Lemma 3.2.3 (Lipschitz approximation with bounded coefficient sum)**.** *For every* 1*-Lipschitz* $f : [0,1] \to \mathbb{R}$ *with* $f(0) = 0$ *and integer* $k \geq 2$*, there is* $Q_k = \sum_{m=0}^{k} a_m \widetilde{T}_m$ *such that*

$$\|Q_k - f\|_\infty \leq C/k, \qquad \sum_{m=0}^{k} |a_m| \leq C.$$

Proof. The even periodic function $g(\phi) = f((1+\cos\phi)/2)$ has $\|g\|_\infty \leq 1$ and almost-everywhere derivative bounded by $1/2$. If $g = \sum_{m\geq 0} c_m \cos(m\phi)$ in L^2, Parseval applied to its weak derivative gives $\sum_{m\geq 1} m^2 |c_m|^2 \leq C$. Hence $|c_0| + \sum_{m\geq 1} |c_m| \leq C$ by Cauchy–Schwarz.

Put $s = \lfloor k/2 \rfloor + 1$. Normalize the nonnegative even trigonometric polynomial

$$J_s(u) = c_s \left(\frac{\sin(su/2)}{\sin(u/2)} \right)^4$$

to integral one on $[-\pi, \pi]$. It has degree $2(s-1) \leq k$, and its first absolute moment is at most C/s. To check this, the unnormalized integral is comparable to s^3: on $|u| \leq 1/s$ the quotient is comparable to s, while everywhere it is at most $C \min(s, |u|^{-1})$. The same bound gives an unnormalized first moment at most Cs^2. Convolution $J_s * g$ therefore differs from g uniformly by C/k. It is even of degree at most k; each cosine coefficient is the corresponding c_m multiplied by a Fourier coefficient of a probability kernel, of absolute value at most one. Replacing $\cos(m\phi)$ by $\widetilde{T}_m(x)$ proves the assertion. □

For probability vectors put $h(p,q)^2 = \sum_j (\sqrt{p(j)} - \sqrt{q(j)})^2$.

Corollary 3.2.4 (An inverse modulus)**.** *For all mixing laws* G, H *and* $2 \leq k \leq t/2$*,*

$$W_1(G,H) \leq \frac{C}{k} + C \exp(2k^2/t) h(p_G, p_H). \tag{3.2.5}$$

Proof. Represent the polynomial from Lemma 3.2.3 in the degree-t Bernstein basis. Lemma 3.2.2 bounds every resulting coefficient by $C \exp(2k^2/t)$. Therefore

$$\left| \int f \, d(G - H) \right| \leq C/k + C \exp(2k^2/t) \|p_G - p_H\|_1.$$

Use $\|p - q\|_1 \leq 2h(p,q)$ and Kantorovich–Rubinstein duality. □

3.2.2 An aggregate ellipse bound and finite likelihood brackets

Expand the whole probability vector in shifted Chebyshev polynomials:

$$B_j^t(x) = \sum_{m=0}^{t} a_{jm} \widetilde{T}_m(x).$$

Lemma 3.2.5 (Aggregate Chebyshev decay)**.** *For* $m \geq 1$*,*

$$\sum_{j=0}^{t} |a_{jm}| \leq 2e^{-m^2/(2t)}; \qquad \sum_{j=0}^{t} |a_{j0}| = 1. \tag{3.2.6}$$

Proof. On the shifted Bernstein ellipse $z = (2 + w + w^{-1})/4$, $|w| = e^u$, its focal-distance identity is $|z| + |1 - z| = \cosh u$. Thus

$$\sum_{j=0}^{t} |B_j^t(z)| = (|z| + |1 - z|)^t = (\cosh u)^t \le e^{tu^2/2}.$$

The Laurent coefficient of w^m in $B_j^t((2 + w + w^{-1})/4)$ is $a_{jm}/2$ for $m \ge 1$. Cauchy's formula, followed by summation over j inside the integral, gives $\sum_j |a_{jm}| \le 2\exp(tu^2/2 - mu)$. Choose $u = m/t$. For $m = 0$, the real cosine integral expresses every a_{j0} as a nonnegative average of B_j^t; summing gives one. □

Lemma 3.2.6 (Small upper brackets for all binomial mixtures)**.** *For $0 < \delta < 1/2$, there is a finite collection $\mathcal{U}_\delta$ of nonnegative vectors such that every p_G is coordinatewise bounded above by some $u \in \mathcal{U}_\delta$ with*

$$1 \le \sum_j u(j) \le 1 + \delta,$$

and

$$\log |\mathcal{U}_\delta| \le C\sqrt{t}\,[\log(Ct/\delta)]^{3/2}. \tag{3.2.7}$$

Proof. Set

$$K = \min\left\{t, \left\lceil \sqrt{2t\log(16t/\delta)} \right\rceil\right\}.$$

Every Chebyshev moment $\mu_m(G) = \int \widetilde{T}_m\, dG$ lies in $[-1, 1]$, and $\mu_0 = 1$. Round $\mu_1, \dots, \mu_K$ onto a grid of mesh $\xi = \delta/(8K)$, giving values ν_m within ξ of the respective moments. For each possible grid vector define

$$v(j) = a_{j0} + \sum_{m=1}^{K} a_{jm}\nu_m, \qquad e(j) = \xi \sum_{m=1}^{K} |a_{jm}| + \sum_{m=K+1}^{t} |a_{jm}|.$$

Then $|v(j) - p_G(j)| \le e(j)$ for the law whose moments were rounded. Lemma 3.2.5 gives

$$\sum_j e(j) \le 2K\xi + 2te^{-(K+1)^2/(2t)} \le \delta/2,$$

where the tail is zero when $K = t$. Take $u(j) = \max\{v(j) + e(j), 0\}$. For each covered law, $u \ge p_G$ and $u - p_G \le 2e$ coordinatewise, so its total mass lies in $[1, 1 + \delta]$. Discard grid vectors whose upper vectors fail this mass condition. This does not discard the covering vector of any law. The number of grid vectors is at most $(CK/\delta)^K$. The definition of K implies (3.2.7) (also when $K = t$). □

Theorem 3.2.7 (Predictive Hellinger accuracy of the full MLE)**.** *There is an absolute C such that, for $0 < \eta < 1/2$,*

$$\mathbb{P}_G\left\{h(p_{\widehat{G}}, p_G)^2 > C\frac{\sqrt{t}[\log(CNt)]^{3/2} + \log(1/\eta)}{N}\right\} \le \eta. \tag{3.2.8}$$

This result holds for every $N, t \ge 1$, with a vacuous bound allowed when its right-hand threshold exceeds the maximal squared distance two.

Proof. Here is a direct finite-bracket likelihood argument. Fix a prospective error $0 < \rho \le \sqrt{2}$ and set $\delta = \rho^2/100$. For an upper bracket $u \ge p$, let $S = \sum u \in [1, 1 + \delta]$ and $q = u/S$. Then

$\|q-p\|_1 \le 2\delta$, hence $h(q,p) \le \sqrt{2\delta}$. If $h(p,p_G) \ge \rho$, it follows that $h(q,p_G) \ge 3\rho/4$. The affinity identity and $1-x \le e^{-x}$ imply

$$\sum_j \sqrt{q(j)p_G(j)} = 1 - \tfrac{1}{2}h(q,p_G)^2 \le e^{-9\rho^2/32}.$$

The MLE has likelihood at least that of the truth. If a bad MLE lies under u, then $q \ge p_{\widehat{G}}/(1+\delta)$ and therefore

$$\prod_{i=1}^{N} \frac{q(X_i)}{p_G(X_i)} \ge (1+\delta)^{-N}.$$

Zero true probabilities cause no difficulty: almost surely no such coordinate is observed, and the displayed expectation calculation is over the support of p_G. Markov's inequality applied to the square root likelihood ratio bounds the probability for this fixed bracket by

$$(1+\delta)^{N/2}\left(\sum_j \sqrt{q(j)p_G(j)}\right)^N \le e^{-N\rho^2/4}.$$

A union bound over Lemma 3.2.6 proves

$$\mathbb{P}_G\{h(p_{\widehat{G}},p_G) \ge \rho\} \le \exp\{C\sqrt{t}[\log(Ct/\rho^2)]^{3/2} - N\rho^2/4\}.$$

For ρ^2 a sufficiently large absolute multiple of the threshold in (3.2.8), one has $\rho^2 \ge 1/N$ and $\log(Ct/\rho^2) \le \log(CNt)$, proving the claim. If this choice exceeds two, the theorem is automatic. □

Proof of Theorem 3.2.1. Use Theorem 3.2.7 with $\eta = N^{-2}$. Throughout (3.2.1), it gives, except on an event of probability N^{-2},

$$h(p_{\widehat{G}},p_G) \le C\frac{t^{1/4}(\log N)^{3/4}}{\sqrt{N}}.$$

Choose $k = \lfloor \sqrt{c\varepsilon t \log N} \rfloor$ for a fixed sufficiently small absolute $c > 0$, for example $c = 1/100$. For all large N, $2 \le k \le t/2$. The inverse modulus gives

$$W_1(\widehat{G},G) \le \frac{C_\varepsilon}{\sqrt{t\log N}} + CN^{2c\varepsilon}\frac{t^{1/4}(\log N)^{3/4}}{\sqrt{N}}.$$

Multiplying the second term by $\sqrt{t\log N}$ bounds it by

$$CN^{2c\varepsilon-1/2}t^{3/4}(\log N)^{5/4} \le CN^{-(3/4-2c)\varepsilon}(\log N)^{5/4},$$

which is bounded as $N \to \infty$. This proves the probability bound. Since the diameter of $[0,1]$ is one, the exceptional event contributes at most N^{-2} to expected W_1, proving the expectation bound. For a lower limit $a\log N$, decrease c depending on a to ensure $k \le t/2$; the rest of the argument is unchanged. □

3.2.3 Scope of the resolution

Theorem 3.2.1 concerns the original unrestricted mixing-law MLE, not a moment estimator or a changed replication design. The parameter support includes both endpoints and arbitrary singular laws. It proves the range stated in Remark 3.2 of [13], provided the preceding self-contained argument withstands external review. The larger near-linear range discussed conditionally in [14, Section 6.2, author version] is not claimed. Nor do we prove the dimension-free $N^{-1/2}$ fingerprint-discrepancy inequality conjectured there: the predictive Hellinger bound here still contains the factor $t^{1/4}$, apart from logarithms. The generic contour, Jackson and likelihood-testing methods are classical; historical novelty of the individual coefficient or entropy lemmas is not asserted.

3.3 Adaptive replication caps: an exact low-cap phase

In the binomial experiment, now count each individual Bernoulli toss as one observation. A policy can start and revisit individuals adaptively, has at most B total tosses, and may toss each individual at most Q times. Write $R_{B,Q}$ for its minimax expected W_1 risk, allowing arbitrary randomized policies and arbitrary probability-valued estimators. The following concerns the full adaptive experiment, even though its upper bound uses equal groups.

Theorem 3.3.1 (Exact capped-budget rate in a polynomial range)**.** *For each $0 < \gamma < 2/5$, uniformly over integers $1 \le Q \le B^{2/5-\gamma}$ and all sufficiently large B,*

$$R_{B,Q} \asymp_\gamma \max\left\{\frac{1}{Q+1}, \frac{1}{\sqrt{Q\log(eB)}}\right\}. \tag{3.3.1}$$

The lower bound, with an absolute constant, holds for every $1 \le Q \le B$ without the polynomial-range restriction.

Lemma 3.3.2 (Explicit moment-matching laws)**.** *For every integer $K \ge 0$ and interval of length a, there are two probability laws on that interval whose moments through order K agree and whose W_1 distance is $a/(K+1)$.*

Proof. On $[0,1]$, split the signed measure

$$2^{-K}\sum_{j=0}^{K+1}(-1)^j\binom{K+1}{j}\delta_{j/(K+1)}$$

into its positive and negative parts. Each part has total mass one. The $(K+1)$st finite difference annihilates polynomials of degree at most K. Between consecutive support points the absolute cumulative signed mass is $2^{-K}\binom{K}{j}$, by the identity $\sum_{i=0}^{j}(-1)^i\binom{K+1}{i} = (-1)^j\binom{K}{j}$. Integration of the absolute cumulative mass gives $1/(K+1)$. An affine change of variables proves the general assertion. □

Adaptive lower bound in Theorem 3.3.1. Degree-Q moment-matching laws have exactly the same law of a full Q-toss vector from an individual. They are therefore indistinguishable under every capped adaptive policy: prepare independent full vectors when each individual is first queried, and reveal the requested prefixes. Lemma 3.3.2 and the triangle inequality give $R_{B,Q} \ge 1/[2(Q+1)]$.

For the second bound suppose first that $Q \ge K := \lceil 4\log(eB)\rceil$ and put

$$h^2 = \frac{K}{64eQ}.$$

Take the two laws from Lemma 3.3.2 on $[1/2-h, 1/2+h]$, with moments matched through order K. Under the fair-coin law for the full Q-toss vector, write its signs as independent Rademacher variables $Z_1,\dots,Z_Q$. For a bias $1/2+a$, its likelihood ratio is

$$\prod_{i=1}^{Q}(1+2aZ_i) = \sum_{S\subseteq\{1,\dots,Q\}}(2a)^{|S|}\prod_{i\in S}Z_i.$$

Walsh orthogonality and moment matching show that the squared L^2 distance between the two mixture likelihood ratios is at most

$$\begin{aligned}4\sum_{\ell>K}\binom{Q}{\ell}(4h^2)^\ell &\le 4\sum_{\ell>K}\left(\frac{4eQh^2}{\ell}\right)^\ell\\ &\le 4\sum_{\ell>K}16^{-\ell} \le C16^{-K}.\end{aligned}$$

Thus the total variation distance between their full-vector laws is at most $C4^{-K}$. An oracle revealing B independent full Q-toss vectors dominates every allowed policy: at most B individuals can ever be queried, and each one's observed history can be read from its own oracle vector. The total variation between the two oracle laws is at most $CB4^{-K} \le 1/4$ for all sufficiently large B. Their latent-law separation is $2h/(K+1)$, so the two-point testing inequality gives $R_{B,Q} \ge c/\sqrt{Q\log(eB)}$. If $Q < K$, the first bound already implies this second bound up to an absolute constant. This proves the lower bound for all caps. □

Upper bound in Theorem 3.3.1. Take $N = \lfloor B/Q \rfloor$ independent individuals, each observed $t = Q$ times, and use their MLE. For $Q \ge c_0 \log B$ with any fixed $c_0 > 0$, the assumptions of Theorem 3.2.1 hold for all large B, after changing its fixed ε and lower constant. In fact $\log N \asymp_\gamma \log B$ and $Q \le B^{2/5-\gamma}$ implies $Q \le N^{2/3-\gamma}$ for all sufficiently large B, since

$$(2/5-\gamma)(5/3-\gamma) < 2/3-\gamma.$$

It follows that the expected risk is at most $C_\gamma/\sqrt{Q\log B}$.

For $4 \le Q \le c_0 \log B$, use the inverse modulus with $k = \lfloor Q/2 \rfloor$. The Hellinger theorem with exceptional probability N^{-2} gives

$$\mathbb{E}W_1(\widehat{G}, G) \le C/Q + Ce^{Q/2}Q^{1/4}[\log(CNQ)]^{3/4}/\sqrt{N} + N^{-2}.$$

Choose, for example, $c_0 = 1/10$. Since $N \asymp B/Q$, the second term is $o(1/Q)$ uniformly in this range. The caps $Q \le 3$ are handled by the diameter bound one. The two ranges prove (3.3.1). □

The fully uncapped regular-channel rate proved earlier is $B^{-1/3}$. The new lower bound shows that a cap satisfying $Q \gtrsim B^{2/3}/\log(eB)$ is necessary to attain that rate within a fixed constant. The earlier multilevel allocation attains it with $Q \asymp B^{2/3}$. This leaves a logarithmic difference between a necessary and a sufficient depth. The intermediate region beyond the exact range of Theorem 3.3.1 is not declared solved.

3.4 The scalar endpoint critical case: a precise limitation of the current method

For $P_\theta = \mathrm{Uniform}[0,\theta]$, $\theta \in [1,2]$, the arbitrary-adaptive total-budget minimax bounds remain

$$cB^{-1/2} \le R_B \le CB^{-1/2}\log(eB).$$

We do not close this gap here. The following calculation establishes a specific obstruction to removing its logarithm merely by choosing a better allocation for the same independent-level CDF estimator.

Proposition 3.4.1 (A critical multilevel variance barrier)**.** *Fix a component endpoint $\theta \in [1,2]$. At level $\ell \ge 1$, let $r_\ell = 2^{\ell-1}$, couple the maxima M_{r_ℓ} and M_{2r_ℓ} by using the same initial observations, and put*

$$D_\ell(x) = \mathbf{1}_{\{M_{2r_\ell} \le x\}} - \mathbf{1}_{\{M_{r_\ell} \le x\}}, \qquad x \in [0,2].$$

Use n_ℓ independent copies at level ℓ, with independence across levels. For any independent base-level estimator, the sum of that base estimator and the level averages through L has integrated variance at least

$$\frac{cL^2}{B} \quad \text{if} \quad \sum_{\ell=1}^{L} 2r_\ell n_\ell \le B. \tag{3.4.1}$$

No choice of these allocation integers removes the squared logarithm in integrated variance when L is of order $\log B$.

Proof. For $0 \le x \le \theta$ write $z = x/\theta$. Monotonicity of the maxima gives

$$\mathbb{E}D_\ell(x) = z^{2r} - z^r, \qquad \mathbb{E}D_\ell(x)^2 = z^r - z^{2r}, \quad r = r_\ell.$$

Direct integration therefore gives

$$\int_0^2 \mathrm{Var}(D_\ell(x))\,dx = \theta\left\{\frac{1}{r+1} - \frac{2}{2r+1} + \frac{2}{3r+1} - \frac{1}{4r+1}\right\} \ge \frac{c}{r}.$$

For an elementary bound without cancellation, integrate only over $z \in [1 - 1/(4r), 1 - 1/(8r)]$. On this interval z^r stays in a fixed compact subinterval of $(0,1)$, so $\mathrm{Var}(D) = a(1-a)$ with $a = z^r - z^{2r}$ is bounded below by an absolute positive constant. The interval length is $1/(8r)$, proving the inequality. Independence gives integrated variance at least $c\sum_{\ell=1}^{L}(r_\ell n_\ell)^{-1}$. Cauchy–Schwarz bounds this below by $cL^2/\sum_\ell r_\ell n_\ell \ge cL^2/B$. □

This is an obstruction for a specified estimator architecture and for integrated squared CDF loss. It is not an unrestricted minimax lower bound in W_1: another estimator can exploit information not represented by these independent increments, and squared L^2 loss cannot simply be substituted for expected L^1 loss. In particular the proposition does not justify declaring the critical endpoint logarithm necessary. The familiar critical multilevel cost–variance phenomenon and its randomized counterparts have substantial prior literature [10, 54]; our contribution here is the explicit calculation locating that obstruction in this endpoint model.

3.5 An exact sparse terminal-design criterion without covariance moments

The previous manuscript [53, terminal-design theorem] left one direction open for multivariate Gaussian mixtures with unbounded covariance: harmonic divergence was sufficient for a sequence of d-output terminal designs, but necessity was proved only for finite design sets. We now prove necessity for every convergent harmonic subseries. The new ingredient is a Fourier–Laplace construction on the cone of positive semidefinite matrices. It produces one joint signed density, not a separate formal Laplace transform for each mean frequency.

Let (A, Σ) have an arbitrary Borel law on $\mathbb{R}^d \times \mathbb{S}_d^+$, and conditionally let X_j be iid $N_d(A, \Sigma)$. For $n \ge d$ put

$$\mathcal{O}_n = (X_1, \dots, X_{d-1}, S_n), \qquad S_n = X_1 + \cdots + X_n.$$

Different terminal lengths supply separate group laws. No joint law between distinct terminal-length experiments is assumed given.

Theorem 3.5.1 (Exact harmonic criterion for unrestricted covariance)**.** *For every $d \ge 1$ and every $\mathcal{N} \subset \{d, d+1, \dots\}$, the collection $\{\mathcal{L}(\mathcal{O}_n) : n \in \mathcal{N}\}$ identifies every Borel mixing law if and only if*

$$\sum_{n \in \mathcal{N}} \frac{1}{n} = \infty.$$

Whenever the sum converges, there are distinct absolutely continuous mixing laws with all polynomial moments finite and equal, with the same full A marginal and the same full Σ marginal, that produce the same observations for every selected n. Any prescribed lower covariance bound $\Sigma \succeq b_0 I$, $b_0 > 0$, can be imposed. No bounded upper covariance support is asserted.

We give the analytic realization in detail. On $\mathbb{S}_d$ use orthonormal coordinates for the Frobenius inner product $\langle V, S\rangle = \mathrm{tr}(VS)$, and write $p = d(d+1)/2$. All matrix Fourier integrals use the corresponding Lebesgue measure. For a complex symmetric matrix $T = X + iV$ with $X \succ 0$, all square roots are principal analytic matrix roots.

Lemma 3.5.2 (A matrix regularizer and a cone-supported inverse)**.** *Let F be a nonzero bounded holomorphic function on the right half-plane, satisfying $F(\overline{z}) = \overline{F(z)}$. Suppose F and all its holomorphic derivatives extend continuously and boundedly to the closed right half-plane. Suppose moreover that its boundary values $F(iy)$ are a Schwartz function of $y \in \mathbb{R}$, flat at zero. Choose a nonzero real even $h \in C_c^\infty(\mathbb{R}^d \setminus \{0\})$. Then*

$$\mathcal{H}(u,T) = h(u)F(u^T T^{-1}u)\exp\{-\operatorname{tr}(T^{1/2}) - \operatorname{tr}(T^{-1/2})\} \tag{3.5.1}$$

is the Fourier–Laplace transform

$$\mathcal{H}(u,T) = \int e^{iu^T a - \operatorname{tr}(T\Sigma)} g(a,\Sigma)\, da\, d\Sigma$$

of a nonzero real Schwartz function g on $\mathbb{R}^d \times \mathbb{S}_d$, supported where $\Sigma \succeq 0$. Both coordinate marginals of g vanish. Every joint polynomial moment of g is zero.

Proof. The inverse of an accretive matrix is accretive:

$$\operatorname{Re}(T^{-1}) = T^{-*} X T^{-1} \succ 0.$$

Thus $\operatorname{Re}(u^T T^{-1}u) > 0$ for every real $u \neq 0$, and the composition in (3.5.1) is well defined and holomorphic in T.

We first prove a uniform decay estimate on the tube. The principal square root of an accretive matrix is accretive. One direct justification uses

$$T^{1/2} = \frac{1}{\pi}\int_0^\infty t^{-1/2} T(tI+T)^{-1}\, dt.$$

Indeed $\|t(tI+T)^{-1}\| < 1$, so $\operatorname{Re}\{I - t(tI+T)^{-1}\} \succ 0$; the convergent integral has positive real part. The same reasoning applies to $T^{-1/2}$. Since T and its square root are symmetric, write $T^{1/2} = A + iB$ with real symmetric matrices A, B and $A \succ 0$. Squaring gives

$$A^2 - B^2 = X \succ 0, \qquad AB + BA = V.$$

Consequently $\|B\| \le \|A\|$ in operator norm and

$$\|V\| \le 2\|A\|\|B\| \le 2(\operatorname{tr} A)^2.$$

With $\Psi(T)$ denoting the exponential factor in (3.5.1), this proves the uniform bound

$$|\Psi(X+iV)| \le \exp\{-\sqrt{\|V\|/2}\} \qquad (X \succ 0). \tag{3.5.2}$$

In particular its right side is integrable in all matrix-frequency coordinates, independently of X.

Next consider the boundary $T = iV$ with V invertible and real symmetric. If its eigenvalues are $\lambda_1, \dots, \lambda_d$,

$$|\Psi(iV)| = \exp\left\{-\frac{1}{\sqrt{2}}\sum_{j=1}^{d}(|\lambda_j|^{1/2} + |\lambda_j|^{-1/2})\right\}. \tag{3.5.3}$$

All derivatives of the logarithm of $\Psi(iV)$ grow at most polynomially in $1+\|V\|$ and $\|V^{-1}\|$. Here is a functional-calculus justification that also handles repeated eigenvalues. Put $\delta = \min_j |\lambda_j|$. Take contours surrounding the spectrum of the normal matrix iV, at distance comparable to δ, grouped into unions of disks if necessary. They avoid zero and the negative real axis. The resolvent bound $\|(zI - iV)^{-1}\| \le \operatorname{dist}(z, i\operatorname{spec} V)^{-1}$, the Cauchy integral formula for matrix functions, and its differentiated formulas bound each derivative of $T^{\pm 1/2}$ by a polynomial in $1 + \|V\|$ and δ^{-1}.

On this boundary $u^T(iV)^{-1}u = -iu^TV^{-1}u$ is purely imaginary. The bounded boundary derivatives of F and the rational derivatives of V^{-1} give the same polynomial derivative bounds for its composition, uniformly for u in the compact support of h. At a zero of $u^TV^{-1}u$, use the flat extension of F at zero. Formula (3.5.3) dominates all these polynomial factors. It follows that the boundary function

$$H(u, V) = \mathcal{H}(u, iV)$$

extends by zero across $\det V = 0$ to a Schwartz function of (u, V), compactly supported in u and flat on the singular set. The same follows directly for each derivative from the displayed bounds, since distance to the singular set is comparable to δ.

Let

$$g_u(\Sigma) = (2\pi)^{-p} \int e^{i\,\mathrm{tr}(V\Sigma)} H(u, V)\, dV.$$

We prove its cone support, rather than merely invoking a tube Paley–Wiener theorem. For $X \succ 0$, define $g_{u,X}$ by the same inverse Fourier integral with $\mathcal{H}(u, X + iV)$ in place of H. Contour shifting inside the convex positive tube gives

$$g_{u,X}(\Sigma) = e^{-\,\mathrm{tr}(X\Sigma)} g_u(\Sigma). \tag{3.5.4}$$

For clarity, this identity can be proved without a multidimensional contour convention. Differentiate the inverse Fourier integral along a straight path of positive definite real parts. The Cauchy–Riemann equations and integration by parts in V give the differential equation $\partial_D g_{u,X} = -\,\mathrm{tr}(D\Sigma) g_{u,X}$. On a compact path inside the cone, Cauchy estimates and (3.5.2) justify differentiation and make all boundary terms vanish. Start at $X = \epsilon I$ and let $\epsilon \downarrow 0$; the same uniform bound gives dominated convergence to g_u for almost every invertible boundary V. The exceptional singular matrices, and the possible zero set of $u^TV^{-1}u$, have measure zero. At the latter set the prescribed flat limit agrees as well.

Boundedness of F, compact support of h, and (3.5.2) bound $|g_{u,X}(\Sigma)|$ by a constant independent of $X \succ 0$. If Σ is not positive semidefinite, choose $X_0 \succ 0$ with $\mathrm{tr}(X_0\Sigma) < 0$. Taking $X = RX_0$ in (3.5.4) and letting $R \to \infty$ proves $g_u(\Sigma) = 0$. Thus its support is in the required cone.

Finally set

$$g(a, \Sigma) = (2\pi)^{-d} \int e^{-iu^Ta} g_u(\Sigma)\, du.$$

It is Schwartz by the boundary estimates. The symmetry $H(-u, -V) = \overline{H(u, V)}$ makes it real. It is nonzero: the scalar argument takes an interval of positive real values when T varies through positive scalar matrices, and F is not identically zero. Fourier inversion of (3.5.4) proves the stated Fourier–Laplace identity, since g_u is Schwartz and supported on the cone.

The function g_u vanishes near $u = 0$. Fourier differentiation therefore gives $\int a^\gamma g(a, \Sigma)\, da = 0$ for every multi-index γ and every Σ. This proves the vanishing covariance marginal and every joint polynomial moment. For the other marginal, let $T = \epsilon I$. The exponential factor is $\exp\{-d(\sqrt{\epsilon} + \epsilon^{-1/2})\}$, while F is bounded. Thus $\mathcal{H}(u, \epsilon I) \to 0$ for every u. Dominated convergence against the integrable g, followed by Fourier uniqueness in a, proves the vanishing mean marginal. □

Lemma 3.5.3 (Scalar input with prescribed sparse zeros)**.** *If $\sum_{n\in\mathcal{N}} n^{-1} < \infty$, the function*

$$F(z) = \left\{ \prod_{n\in\mathcal{N}} \frac{2n - z}{2n + z} \right\} \exp\{-\sqrt{z} - z^{-1/2}\}, \qquad \mathrm{Re}\, z > 0,$$

satisfies all hypotheses of Lemma 3.5.2 and vanishes at every $2n$, $n \in \mathcal{N}$. Empty and finite sets are allowed.

Proof. The factors differ from one by $O_K(1/n)$ on each compact set of the right half-plane. The product is therefore nonzero holomorphic, bounded by one, and invariant under conjugation. These assertions hold continuously at every finite point of the closed half-plane. To control derivatives uniformly there, write $b_n(z) = (2n - z)/(2n + z)$. For $\operatorname{Re} z \geq 0$, $|b_n(z)| \leq 1$, and direct differentiation gives $|b_n^{(r)}(z)| \leq C_r n^{-r}$ for $r \geq 1$. The product rule for finite products, together with $\sum_n n^{-1} < \infty$, bounds every derivative uniformly in the number of factors and in the entire closed half-plane. The locally uniform differentiated products then give the same bounds for the infinite product, including its continuous boundary derivatives. There is no division by a factor that might vanish in this argument.

For $\operatorname{Re} z \geq 0$ and $z \neq 0$, the principal roots satisfy

$$\operatorname{Re} \sqrt{z} \geq |z|^{1/2}/\sqrt{2}, \qquad \operatorname{Re} z^{-1/2} \geq |z|^{-1/2}/\sqrt{2}.$$

Every derivative of the root exponential is that exponential times a finite sum of powers of $z^{1/2}$ and $z^{-1/2}$. It therefore decays faster than every power at zero and at infinity, uniformly on the closed half-plane. The product rule proves all the required bounded continuous derivative extensions, as well as boundary Schwartz decay and flatness at zero. □

Proof of Theorem 3.5.1. For completeness, recall the sufficient direction. For the signed difference of two candidate laws, put

$$H_u(T) = \int e^{iu^T a - \operatorname{tr}(T\Sigma)}\, d\rho.$$

The experiment is equivalent to independent conditional blocks of lengths $1, \dots, 1, n - d + 1$. Their Fourier arguments are exactly

$$T = \frac{uu^T}{2n} + C, \qquad C \succeq 0, \operatorname{rank} C \leq d - 1.$$

This follows by decomposing the weighted frequency matrix into its projection on the vector of square roots of block lengths and its orthogonal complement. Fix $u \neq 0$ and a rank-$d-1$ matrix $C \succeq 0$ with u not orthogonal to its kernel. Then $z \mapsto H_u(C + zuu^T)$ is bounded holomorphic on $\operatorname{Re} z > 0$ and vanishes at $1/(2n)$. Harmonic divergence violates the classical half-plane Blaschke condition for a nonzero bounded function [55], so this function is zero. Every $T \succ 0$ has the decomposition

$$T = C + \lambda uu^T, \quad \lambda = (u^T T^{-1} u)^{-1}, \quad C = T - \lambda uu^T \succeq 0, \quad \operatorname{rank} C = d - 1.$$

This gives $H_u(T) = 0$. Continuity supplies $u = 0$; analytic continuation in T followed by Fourier uniqueness after the tilt $e^{-\operatorname{tr}\Sigma}$ gives $\rho = 0$.

For necessity use the scalar input in Lemma 3.5.3 and the joint signed density in Lemma 3.5.2. At a positive definite argument of experiment n, the determinant identity gives

$$0 = \det(2nT - uu^T) = (2n)^d \det T \left(1 - \frac{u^T T^{-1} u}{2n}\right).$$

Thus $F(u^T T^{-1} u) = F(2n) = 0$, and the signed observation transform vanishes. Singular arguments follow by continuity: full-rank d-column frequency matrices are dense, and the signed density has finite total variation. At $u = 0$ it vanishes by the support of h.

Write $g = g_+ - g_-$ and $m = \int g_+ = \int g_- > 0$. Then $m^{-1} g_+$ and $m^{-1} g_-$ are distinct probability densities with the same laws of every selected experiment. Schwartz decay, vanishing marginals, and vanishing polynomial moments give the additional claims. Translating both covariance variables by $b_0 I$ multiplies the signed transform by $e^{-b_0 \operatorname{tr} T}$, preserving its zeros and all the asserted equalities. □

For bounded covariance support, any infinite terminal set was already sufficient because the scalar function in the positive proof becomes entire. For an infinite convergent harmonic subseries, the witnesses consequently require unbounded covariance support. The theorem closes the exact unrestricted sparse-design classification left open in the previous manuscript; it is not presented as a newly located published conjecture.

3.5.1 Two graph-theoretic routes that remain open

The analytic completion above does not settle the separate relation between generic completion rank and maximum likelihood threshold. Blekherman and Sinn ask whether some function f satisfies

$$\operatorname{mlt}(G) \le k \quad \Longrightarrow \quad \operatorname{gcr}(G) \le f(k) \qquad \text{for every finite graph } G$$

[34, Introduction, final question]. Their complete bipartite examples have generic completion rank of order m and likelihood threshold of order $\sqrt{m}$ for $K_{m,m}$. They rule out a linear comparison, but do not supply bounded-threshold, unbounded-rank examples. The signed polynomial annihilators underlying the earlier observation-rank criterion do not provide such examples either: the likelihood threshold asks for positive semidefinite certificates, whereas a nonzero annihilator can be indefinite. Thus an algebraic rank calculation alone does not decide this question.

A second published target is equality of the two parameters with high probability for $G(n, c/n)$ at every fixed $c > 0$ [35, Conjecture 2.4]. Peled's August 2026 revision establishes the generic-rigidity phase transition and sharp asymptotic rank information. Its concluding discussion still identifies a globally rigid large core as a missing step toward the likelihood-threshold conjecture [56, Section 7, item 3, p. 30]. Our transform construction adds neither that global rigidity nor the necessary positivity. Both graph targets therefore remain open in this work. This is an explicit boundary of the method, rather than a claimed consequence of the harmonic classification.

The literature check for the new terminal-design theorem was bounded: the primary sources above and grouped-sample identifiability sources were inspected, and no exact earlier statement for this sparse terminal experiment was located. This does not certify historical priority. In particular, neither the scalar Blaschke condition nor the general Fourier–Laplace principle is claimed as new.

3.6 Self-decomposability beyond the McKay Thorin boundaries

Two different questions in the published source motivate this section. Baricz, Prabhu, Singh, and Vijesh [3], Section 2.3, ask for the largest infinite-divisibility domain of the generalized McKay distribution. Their Section 4.3, pp. 282–283, poses a multipart problem asking, among other things, for the self-decomposability domains of the third and fourth McKay families. The two families are cases of that one multipart problem, not two separately posed conjectures.

The full GGC classifications established in the preceding working manuscript [53] were

$$x^{\nu-1}e^{-bx}I_\mu(ax): \quad \nu \le \mu + 1, \qquad x^{2\mu}e^{-bx}I_\mu(ax)^2: \quad \mu \le \tfrac{1}{2}.$$

Those classifications are not new results of this section. We now enter their complementary parameter regions. We prove infinite divisibility and determine the exact self-decomposability boundary on infinitely many parameter rays. For the fourth family at $\mu = 3/2$, the boundary is specified by a unique critical point, and an explicit rational choice of parameters gives an infinitely divisible law that is not self-decomposable. The remaining noninteger parameter regions are explicitly left open.

All densities below are on $(0,\infty)$ and are normalized by a positive constant. Write ID for infinite divisibility, SD for self-decomposability, and GGC for generalized gamma convolution. The positive scale parameter a is always strictly positive.

3.6.1 An exact criterion for exponential tilts

Lemma 3.6.1 (The tilt boundary from a positive spectral kernel)**.** *Let $d>0$, and let K be a positive real C^1 function on $[0,\infty)$ such that K and K' are $O(e^x)$. Suppose a probability law with Laplace transform L_b satisfies*

$$-\frac{d}{ds}\log L_b(s)=\int_0^\infty e^{-st}k_b(t)\,dt,\qquad k_b(t)=e^{-bt}K(dt),\qquad b>d. \tag{3.6.1}$$

Then the law is ID, with Lévy density $k_b(t)/t$. It is SD if and only if

$$\frac{b}{d}\ge\sup_{x\ge0}\frac{K'(x)}{K(x)}. \tag{3.6.2}$$

Proof. The assumptions imply $\int_0^\infty(1\wedge t)k_b(t)\,dt/t<\infty$. Integrating (3.6.1) from 0 to s, and using $L_b(0)=1$, gives

$$h(s):=-\log L_b(s)=\int_0^\infty(1-e^{-st})\frac{k_b(t)}{t}\,dt.$$

This is the Lévy–Khintchine representation of a positive ID law.

For completeness, in this smooth, exponentially decaying setting the SD criterion is exactly $k_b'\le0$. If k_b is nonincreasing, then for each $0<c<1$,

$$h(s)-h(cs)=\int_0^\infty(1-e^{-st})\frac{k_b(t)-k_b(t/c)}{t}\,dt$$

is a Lévy exponent. Therefore $L_b(s)/L_b(cs)$ is a Laplace transform, which is the defining SD property.

Conversely, suppose all the ratios $R_c(s)=L_b(s)/L_b(cs)$ are Laplace transforms. Each function $-R_c'(s)/(1-c)$ is completely monotone. Differentiating the explicit smooth expression for R_c shows that these functions and all their derivatives converge locally on $(0,\infty)$, as $c\uparrow1$, to the corresponding derivatives of $(sh'(s))'$. Hence this last function is completely monotone. Integration by parts gives

$$(sh'(s))'=-\int_0^\infty e^{-st}tk_b'(t)\,dt.$$

The signed measure on the right has finite total variation. Bernstein's representation theorem and uniqueness of Laplace transforms therefore imply $-tk_b'(t)\ge0$. Continuity yields $k_b'\le0$ everywhere. Finally, $k_b'(t)=e^{-bt}\{dK'(dt)-bK(dt)\}$, proving (3.6.2). □

Lemma 3.6.2 (Simultaneous returns of finitely many phases)**.** *For real numbers $\theta_1,\dots,\theta_m$, there are arbitrarily large positive x for which all $e^{i\theta_j x}$ are arbitrarily close to 1.*

Proof. Partition the m-torus into N^m equal boxes and apply the pigeonhole principle to the first N^m+1 integer multiples of $(\theta_1,\dots,\theta_m)$ modulo 2π. There is a positive integer $q_N\le N^m$ for which all phases are within $2\pi/N$ of 1. If these integers have an unbounded subsequence, the assertion follows. Otherwise a fixed positive integer occurs along a sequence with $N\to\infty$, is an exact common period, and its multiples give the assertion. □

3.6.2 All integer power biases of the third family

Theorem 3.6.3 (ID and an exact SD threshold on every integer-bias ray)**.** *Let $\mu > -1/2$, $n \geq 2$ be an integer, and $\nu = \mu + 1 + n$. Put $\alpha = \mu + 1/2 > 0$. For every $b > a > 0$, the law with density*

$$f_{\mu,n,a,b}(x) = Cx^{\mu+n}e^{-bx}I_\mu(ax) \tag{3.6.3}$$

is ID. Define polynomials by

$$P_0(z) = 1, \qquad P_{j+1}(z) = 2(\alpha + j)zP_j(z) - (1 + z^2)P_j'(z). \tag{3.6.4}$$

The n roots $\omega_1, \ldots, \omega_n$ of P_n are real and simple. Set

$$K_{\alpha,n}(x) = 2(\alpha + n)\cosh x - \sum_{j=1}^{n} e^{i\omega_j x}. \tag{3.6.5}$$

This is a positive real even function. The constant

$$\beta_{\alpha,n} := \max_{x\geq 0} \frac{K_{\alpha,n}'(x)}{K_{\alpha,n}(x)} \tag{3.6.6}$$

exists, is attained at a finite positive point, and satisfies

$$1 < \beta_{\alpha,n} \leq \frac{2(\alpha + n) + n\sqrt{(n-1)/(2\alpha+1)}}{2\alpha + n}. \tag{3.6.7}$$

The exact classification on this ray is

$$f_{\mu,n,a,b} \text{ is SD} \quad \Longleftrightarrow \quad b/a \geq \beta_{\alpha,n}. \tag{3.6.8}$$

In particular, each ray contains both ID laws that are not SD and SD laws outside the GGC domain.

Proof. The elementary Bessel Laplace integral gives, with a positive constant C_0 independent of u,

$$J_0(u) := \int_0^\infty x^\mu e^{-ux} I_\mu(ax)\, dx = C_0(u^2 - a^2)^{-\alpha}, \qquad u > a.$$

For example, this follows directly by integrating the positive power series of I_μ and using the gamma duplication formula. All differentiations under the integral are justified by exponential decay. Consequently

$$J_n(u) := (-1)^n J_0^{(n)}(u) = C_0 \frac{Q_n(u)}{(u^2 - a^2)^{\alpha+n}}, \tag{3.6.9}$$

where

$$Q_0(u) = 1, \qquad Q_{j+1}(u) = 2(\alpha + j)uQ_j(u) - (u^2 - a^2)Q_j'(u).$$

The leading coefficient of Q_j is $(2\alpha)_j > 0$. The substitution $Q_j(iaz) = i^j a^j P_j(z)$ gives (3.6.4).

We verify the root assertion rather than assume a special-function zero theorem. The polynomial $P_1(z) = 2\alpha z$ has a simple real root and positive leading coefficient. Suppose P_j has ordered simple real roots $r_1 < \cdots < r_j$ and positive leading coefficient. At each root,

$$P_{j+1}(r_\ell) = -(1 + r_\ell^2)P_j'(r_\ell).$$

These signs alternate. There is a root in each of the $j - 1$ intervening intervals. The signs at the two outer roots, together with the positive leading coefficient $(2\alpha)_{j+1}$, give one root on

each outer interval. These $j+1$ distinct roots exhaust the degree. Induction proves simplicity, reality, and interlacing. Parity in (3.6.4) also gives invariance of the roots under $\omega \mapsto -\omega$.

The Laplace transform of (3.6.3) is $L_b(s) = J_n(s+b)/J_n(b)$. Its negative logarithmic derivative, with $u = s + b$, is

$$-\frac{d}{ds}\log L_b(s) = \frac{\alpha+n}{u-a} + \frac{\alpha+n}{u+a} - \sum_{j=1}^{n} \frac{1}{u - ia\omega_j}. \tag{3.6.10}$$

Taking elementary Laplace transforms yields the kernel $k_b(t) = e^{-bt}K_{\alpha,n}(at)$ in (3.6.5). Root symmetry makes it real and even, and

$$K_{\alpha,n}(x) \geq 2(\alpha+n)\cosh x - n \geq 2\alpha + n > 0. \tag{3.6.11}$$

Lemma 3.6.1 proves ID and the asserted SD equivalence, once the statements about the maximum are verified.

The logarithmic derivative K'/K is continuous, equals 0 at 0, and tends to 1 at infinity. Moreover,

$$K'(x) - K(x) = -2(\alpha+n)e^{-x} + \sum_{j=1}^{n}(1 - i\omega_j)e^{i\omega_j x}. \tag{3.6.12}$$

At simultaneous returns of the phases, the sum on the right is arbitrarily close to n, since $\sum_j \omega_j = 0$. Lemma 3.6.2 therefore gives arbitrarily large x for which $K'(x) > K(x)$. It follows that the supremum is strictly larger than 1 and is attained in a compact subinterval of $(0, \infty)$.

For the quantitative upper bound, put $S = \sum_j |\omega_j|$ and $A = 2(\alpha+n)$. Then

$$\frac{K'(x)}{K(x)} \leq \frac{A\sinh x + S}{A\cosh x - n} \leq \frac{A+S}{A-n}.$$

Comparison of the two leading coefficients in the recurrence for Q_n gives

$$Q_n(u) = (2\alpha)_n \left(u^n + \frac{n(n-1)a^2}{2(2\alpha+1)} u^{n-2} + \text{lower powers} \right).$$

This coefficient identity follows inductively: if the second-to-first coefficient ratio is r_j, then $r_{j+1} = \{(2\alpha+j+2)r_j + j\}/(2\alpha+j)$, starting at $r_1 = 0$. Vieta's formulas now give $\sum_j \omega_j^2 = n(n-1)/(2\alpha+1)$. Cauchy–Schwarz yields the bound for S in (3.6.7).

Finally, $\nu = \mu + 1 + n > \mu + 1$ puts the entire ray strictly beyond the previously established GGC boundary. The intervals $1 < b/a < \beta_{\alpha,n}$ and $b/a \geq \beta_{\alpha,n}$ supply the two asserted subclasses. □

The first ray covered here, $n = 2$, already has the explicit kernel

$$K_{\alpha,2}(x) = 2(\alpha+2)\cosh x - 2\cos\left(\frac{x}{\sqrt{2\alpha+1}}\right). \tag{3.6.13}$$

The case $n = 1$ is the second McKay family, whose SD classification was obtained in the earlier research round; it is deliberately excluded from the new-result count. Theorem 3.6.3 also supplies new ID parameter rays for the separate maximal-domain question in Section 2.3 of the source. It does not determine that maximal domain.

The maximization in (3.6.6) has a certified finite search region. If $K'(x_0)/K(x_0) = 1 + \delta$ with $\delta > 0$, then (3.6.12) gives

$$\frac{K'(x)}{K(x)} - 1 \leq \frac{n+S}{2(\alpha+n)\cosh x - n}.$$

Once the right side falls below δ, no subsequent point can maximize the ratio. This is an exact tail estimate, not an appeal to numerical truncation.

3.6.3 All half-integer orders of the fourth family

Theorem 3.6.4 (An ID extension and exact SD thresholds for squared Bessel laws)**.** *Let $n \geq 1$ be an integer, $\mu = n + 1/2$, $d = 2a$, and $b > d$. The law with density*

$$g_{n,a,b}(x) = Cx^{2n+1}e^{-bx}I_{n+1/2}(ax)^2 \tag{3.6.14}$$

is ID. Define

$$q_n(z) = \sum_{k=0}^{n}(-1)^k a_k z^{2(n-k)}, \qquad a_k = \binom{n}{k}\frac{(1/2)_k}{(n+3/2)_k}, \tag{3.6.15}$$

and let $\lambda_1, \ldots, \lambda_{2n}$ denote all its roots, counted with multiplicity. Put

$$\rho_n^2 = \max_{1\leq k\leq n}\frac{(n-k+1)(k-1/2)}{k(n+k+1/2)} < 1, \tag{3.6.16}$$

$$K_n(x) = (2n+1)(1+2\cosh x) - \sum_{j=1}^{2n} e^{\lambda_j x}. \tag{3.6.17}$$

All $|\lambda_j| \leq \rho_n$, and K_n is a positive real even function. Its exact SD threshold is

$$\beta_n = \max_{x\geq 0}\frac{K_n'(x)}{K_n(x)}, \qquad g_{n,a,b} \text{ is SD} \iff \frac{b}{2a} \geq \beta_n. \tag{3.6.18}$$

The maximum is attained at a finite positive point and

$$1 < \beta_n \leq \frac{4n+2+2n\rho_n}{2n+2} < 3. \tag{3.6.19}$$

Thus every such order has a nonempty ID but non-SD interval, and $b \geq 6a$ is a uniform sufficient condition for SD at all these orders. All the laws in this theorem lie outside the GGC domain.

Proof. The Laplace integral in the source, equation (2-4), gives

$$J(u) = C_1 u^{-(4n+3)}{}_2F_1(n+1, 2n+3/2; n+3/2; d^2/u^2), \qquad u > d,$$

where $C_1 > 0$ does not depend on u. Euler's hypergeometric transformation [6] gives

$${}_2F_1(n+1, 2n+3/2; n+3/2; z) = (1-z)^{-(2n+1)}{}_2F_1(-n, 1/2; n+3/2; z).$$

Consequently, with $Q_n(u) = d^{2n}q_n(u/d)$,

$$J(u) = C_1\frac{Q_n(u)}{u^{2n+1}(u^2-d^2)^{2n+1}}. \tag{3.6.20}$$

Here is a direct root bound. A polynomial $p(z) = \sum_{j=0}^{m} c_j z^j$ with $0 < c_0 \leq \cdots \leq c_m$ has no root with $|z| > 1$. Indeed,

$$(1-z)p(z) = c_0 + \sum_{j=1}^{m}(c_j - c_{j-1})z^j - c_m z^{m+1},$$

and at such a root the absolute value of the last term strictly exceeds the sum of the preceding absolute-value bounds. Rescaling shows that a polynomial with arbitrary positive coefficients has all roots in $|z| \leq \max_j c_j/c_{j+1}$. Apply this fact to $p(v) = \sum_{k=0}^{n} a_k v^{n-k}$, whose roots are the negatives of the squared roots of q_n. Since

$$\frac{a_k}{a_{k-1}} = \frac{(n-k+1)(k-1/2)}{k(n+k+1/2)} < 1,$$

the claimed bound $|\lambda_j| \le \rho_n < 1$ follows.

Set $B = 2n + 1$. Logarithmic differentiation of $L_b(s) = J(s + b)/J(b)$ gives

$$-\frac{d}{ds}\log L_b(s) = \frac{B}{u} + \frac{B}{u - d} + \frac{B}{u + d} - \sum_{j=1}^{2n} \frac{1}{u - d\lambda_j}, \qquad u = s + b. \tag{3.6.21}$$

The roots are invariant under conjugation and negation, so (3.6.17) is real and even. Pairing opposite roots and using $|\cosh z| \le \cosh |z|$ gives

$$K_n(x) \ge B + 2B\cosh x - 2n\cosh(\rho_n x) \ge B + (2n + 2)\cosh x > 0. \tag{3.6.22}$$

Equation (3.6.21) is precisely the Laplace transform of $e^{-bt}K_n(dt)$. Lemma 3.6.1 proves ID and the threshold equivalence.

To prove the strict lower bound for β_n, first observe that some root has positive real part. Indeed, Vieta's formulas give

$$\sum_{j=1}^{2n} \lambda_j = 0, \qquad \sum_{j=1}^{2n} \lambda_j^2 = 2a_1 > 0.$$

If every root had real part zero, the last sum would be nonpositive; negation symmetry excludes the possibility that all real parts are nonpositive with one strictly negative. Thus $r := \max_j \operatorname{Re}\lambda_j$ satisfies $0 < r < 1$. Let $\mathcal{J}$ be the multiset of roots with real part r and $m = |\mathcal{J}|$. The real trigonometric polynomial

$$T(x) = \sum_{\lambda \in \mathcal{J}} (1 - \lambda) e^{i \operatorname{Im}\lambda\, x}$$

satisfies $T(0) = m(1 - r) > 0$. Simultaneous returns from Lemma 3.6.2 give arbitrarily large x with $T(x) > T(0)/2$. At these points,

$$K_n'(x) - K_n(x) = -B(1 + 2e^{-x}) + e^{rx}T(x) + O(e^{r'x}) > 0,$$

where $r' < r$ bounds the real parts of the remaining roots; an absent remainder is simply omitted. As K_n'/K_n equals 0 at the origin and tends to 1 at infinity, its maximum is greater than 1 and is attained at a finite positive point.

Opposite-root pairing also gives

$$\left| \sum_j \lambda_j e^{\lambda_j x} \right| \le 2n\rho_n \sinh(\rho_n x) \le 2n\rho_n \sinh x.$$

Together with (3.6.22), this proves

$$\frac{K_n'(x)}{K_n(x)} \le \frac{(2B + 2n\rho_n)\sinh x}{B + (2B - 2n)\cosh x} \le \frac{2B + 2n\rho_n}{2B - 2n} < 3.$$

The final GGC statement uses the earlier boundary $\mu \le 1/2$, since $\mu = n + 1/2 > 1/2$ here. □

The exact thresholds again admit finite tail certification. From (3.6.22), $K_n(x) \ge e^x$, and hence

$$\left| \frac{K_n'(x)}{K_n(x)} - 1 \right| \le \{3(2n + 1) + 2n(1 + \rho_n)\} e^{-(1-\rho_n)x}. \tag{3.6.23}$$

A single point with ratio $1 + \delta > 1$ therefore supplies an explicit finite interval containing every global maximizer.

3.6.4 A uniquely determined boundary at order three halves

Theorem 3.6.5 (The order-3/2 SD transition and an exact counterexample)**.** *Let $a > 0$, $b > 2a$, and let X have density*

$$f(x) = Cx^3 e^{-bx} I_{3/2}(ax)^2.$$

Put $c = 1/\sqrt{5}$ and

$$K(x) = 3 + 6\cosh x - 2\cosh(cx), \tag{3.6.24}$$

$$D(x) = 36 + 4c^2 + 18\cosh x - 6c^2\cosh(cx) - 6(1-c)^2\cosh((1+c)x) - 6(1+c)^2\cosh((1-c)x). \tag{3.6.25}$$

There is exactly one positive zero τ of D. With

$$\beta = \frac{6\sinh\tau - 2c\sinh(c\tau)}{3 + 6\cosh\tau - 2\cosh(c\tau)}, \tag{3.6.26}$$

the law of X is ID for every $b > 2a$, and is SD exactly when $b/(2a) \geq \beta$. In particular:

1. *$b = 1001a/500$ gives an ID law that is not SD;*
2. *$b = 4a$ gives an SD law that is not GGC.*

The boundary is larger than $1001/1000$.

Proof. For $n = 1$, (3.6.15) is $q_1(z) = z^2 - 1/5$, which gives (3.6.24). Direct differentiation and the addition identities for hyperbolic functions show that $K''K - (K')^2 = D$ as displayed in (3.6.25). Thus the sign of D is the sign of $(K'/K)'$.

Write $D(x) = d_0 + \sum_{m\geq 1} d_m x^{2m}/(2m)!$. Then $d_0 = 196/5$, and

$$d_m = 6\left[3 - c^{2m+2} - (1-c^2)^2\{(1+c)^{2m-2} + (1-c)^{2m-2}\}\right], \qquad m \geq 1. \tag{3.6.27}$$

For $m = 1, 2, 3, 4$ the bracketed quantities are respectively

$$\frac{42}{25}, \qquad \frac{182}{125}, \qquad \frac{82}{625}, \qquad -\frac{9058}{3125}.$$

They strictly decrease with m. To see this, set $S_j = (1+c)^j + (1-c)^j$. Expansion by the binomial theorem gives $S_{2m} - S_{2m-2} \geq 2c^2$ for every $m \geq 1$. Therefore the increase in the subtracted quantity in (3.6.27) is at least

$$(1-c^2)^2(2c^2) - (1-c^2)c^{2m+2} \geq \frac{16}{25}\frac{2}{5} - \frac{4}{5}\frac{1}{25} = \frac{28}{125} > 0.$$

It follows that $d_1, d_2, d_3 > 0$ and $d_m < 0$ for all $m \geq 4$.

On $(0, \infty)$, the function

$$\frac{D(x)}{x^6} = \frac{d_0}{x^6} + \frac{d_1}{2!x^4} + \frac{d_2}{4!x^2} + \frac{d_3}{6!} + \sum_{m\geq 4} \frac{d_m x^{2m-6}}{(2m)!}$$

is strictly decreasing. Termwise differentiation is justified on each compact subinterval by convergence of the entire power series, and every nonconstant summand has negative derivative. The limits are $+\infty$ at zero and $-\infty$ at infinity, the latter also following from the negative dominant term $-6(1-c)^2\cosh((1+c)x)$ in (3.6.25). Thus D has exactly one positive zero τ. The ratio K'/K strictly increases before τ and strictly decreases afterwards. Theorem 3.6.4 now proves the exact threshold formula.

We prove the non-SD example without decimal approximations. Let $r = 1001/1000$ and take $x = \log 81 = 4\log 3$. The inequalities $4/9 < c < 9/20$ imply

$$e^{cx} = 3^{4c} > 3^{16/9} > 7,$$

where the last inequality is the integer comparison $3^{16} = 43046721 > 40353607 = 7^9$. Hence $\cosh(cx) > 7/2$ and $\sinh(cx) < \cosh(cx)$. Using $e^x = 81$, we obtain

$$\begin{aligned} K'(x) - rK(x) &= 6\sinh x - 6r\cosh x + 2r\cosh(cx) - 2c\sinh(cx) - 3r \\ &> -\frac{6}{81} - \frac{6}{1000}\frac{81 + 1/81}{2} + 7\left(r - \frac{9}{20}\right) - 3r = \frac{604}{1125} > 0. \end{aligned}$$

Therefore $\beta > r$, proving the first example. For the second, (3.6.19) gives

$$\beta \le \frac{6 + 2/\sqrt{5}}{4} < 2.$$

Thus $b = 4a$, for which $b/(2a) = 2$, is SD. The preceding GGC classification excludes GGC at $\mu = 3/2$. □

Numerical root finding, provided only as an illustration of the uniquely specified constant, gives

$$\tau \approx 4.988970295384644, \qquad \beta \approx 1.005210740181144.$$

No assertion above depends on these decimal values. In particular, the existence of the non-SD interval and the rational counterexample were proved by exact inequalities.

3.6.5 What remains unresolved

These results resolve the SD question on the rays $\nu - \mu - 1 \in \{2, 3, \ldots\}$ with $\mu > -1/2$, and on the orders $\mu \in \{3/2, 5/2, \ldots\}$ of the squared-Bessel family. They do not classify the full real parameter domains in the source. For the third family, the regions with noninteger $\nu - \mu - 1 > 0$, and the admissible negative orders $-1 < \mu \le -1/2$, are not settled here. For the fourth family, non-half-integer orders $\mu > 1/2$ remain outside the argument.

The specific obstruction is mathematical: the proofs above turn the Laplace integrals into a power of a quadratic times a finite polynomial. Interlacing or coefficient ratios then control every zero and produce a positive, finite exponential expression for the Lévy spectral kernel. At noninteger parameters the terminating polynomials become nonterminating hypergeometric functions. Their logarithmic derivatives can have additional branch contributions, and the present root bounds provide no sign control for the full inverse Laplace transform. Exponential tilting alone does not remove this gap. No full solution of either source question, and no priority claim for the entire literature, is inferred from the partial classifications proved here.

3.7 All modular periods for a three-term exponential sum

This section advances the modular-ordinate problem rather than the previous edge-phase reconstruction argument. Podesta [48, Section 1, paragraphs after the mean vertical zero density] asks whether the imaginary parts of zeros of a rationally independent exponential sum can be dense modulo a suitable length, and explicitly defers the solution. The nonresonant periods follow from a compact-torus argument. Here we classify *every* period for a three-term sum, including the resonant periods. The proof uses triangle geometry and a three-dimensional orbit closure. These ingredients are classical; the claimed contribution is the explicit complete

classification below. A targeted primary-source search on 14 October 2026 located the printed question but no subsequent resolution of this exact classification. This limited search is not a claim of historical priority.

Fix

$$f(s) = -1 + Be^{as+i\phi} + Ce^{bs+i\psi}, \qquad B, C > 0, \quad 0 < a < b, \quad a/b \notin \mathbb{Q}. \tag{3.7.1}$$

For $L > 0$, put $q = 2\pi/L$ and let

$$\mathcal{R}_L = \overline{\{q \operatorname{Im}\rho \bmod 2\pi : f(\rho) = 0,\ \operatorname{Im}\rho > 0\}} \subset \mathbb{R}/(2\pi\mathbb{Z}).$$

Thus $\mathcal{R}_L$ is the circle precisely when the positive ordinates are dense modulo L. Neither multiplicity nor finitely many low zeros affects this question.

Theorem 3.7.1 (Complete trinomial classification)**.** *Let $r = a/b$ and $\kappa = BC^{-r}$. There is a unique $x_0 > 0$ satisfying*

$$x_0^{2-2r} - x_0^{-2r} = \kappa^2(2r-1). \tag{3.7.2}$$

Define

$$u(x) = \frac{x + x^{-1} - \kappa^2 x^{2r-1}}{2}, \qquad \Theta = \arccos\bigl(\max\{-1, u(x_0)\}\bigr) \in (0, \pi]. \tag{3.7.3}$$

Then the following statements hold.

1. *If $q \notin \operatorname{span}_{\mathbb{Q}}\{a, b\}$, then $\mathcal{R}_L = \mathbb{R}/(2\pi\mathbb{Z})$.*
2. *Otherwise write, uniquely,*

$$q = (ma + nb)/d, \qquad m, n \in \mathbb{Z}, \quad d \geq 1, \quad \gcd(m, n, d) = 1.$$

   If $m \neq 0$, then $\mathcal{R}_L = \mathbb{R}/(2\pi\mathbb{Z})$.

3. *If $m = 0$, necessarily $n > 0$ and $\gcd(n, d) = 1$, and*

$$\mathcal{R}_L = \{z \bmod 2\pi :\ dz + n\psi \bmod 2\pi \in [-n\Theta, n\Theta] \bmod 2\pi\}. \tag{3.7.4}$$

   In particular, this set is the circle if and only if $n\Theta \geq \pi$.

Equivalently, the complete set of nondense periods is

$$\left\{ \frac{2\pi d}{nb} :\ n, d \geq 1,\ \gcd(n, d) = 1,\ n\Theta < \pi \right\}. \tag{3.7.5}$$

This criterion does not depend on ϕ or ψ, although the location of the missing arcs depends on ψ. The assertion concerns density, not equidistribution at resonant periods.

Lemma 3.7.2 (Triangle interval and endpoint phases)**.** *Put $x = Ce^{b\sigma}$ and $U = \kappa x^r$. The numbers $1, U, x$ are the side lengths of a possibly degenerate triangle exactly for*

$$K = \{x > 0 : -1 \leq u(x) \leq 1\}.$$

The set K is either one compact interval or two compact intervals, and its nondegenerate part is dense in it. On K define the continuous angles

$$\theta(x) = \arccos\frac{1 + U^2 - x^2}{2U}, \qquad \eta(x) = -\arccos\frac{1 + x^2 - U^2}{2x} = -\arccos u(x). \tag{3.7.6}$$

All solutions of $1 = Ue^{i\theta'} + xe^{i\eta'}$ are, modulo 2π, $(\theta', \eta') = (\theta, \eta)$ or $(-\theta, -\eta)$. At the leftmost endpoint of K these angles are $(0, 0)$, and at the rightmost endpoint they are $(\pi, 0)$. If K has two components, the two endpoints facing its gap both have angles $(0, -\pi)$. Moreover,

$$\max_{x \in K} |\eta(x)| = \Theta.$$

Proof. The triangle inequalities are $|1-x| \le U \le 1+x$, which are equivalent to $-1 \le u(x) \le 1$. Differentiation gives

$$2x^{2-2r}u'(x) = x^{2-2r} - x^{-2r} - \kappa^2(2r-1).$$

The function $x^{2-2r} - x^{-2r}$ is strictly increasing from $-\infty$ to $+\infty$. Consequently u decreases up to the unique x_0 in (3.7.2) and increases after it. Also $u(x) \to +\infty$ at both ends of $(0,\infty)$ and $u(1) = 1 - \kappa^2/2 < 1$. This proves the claimed structure of K, including the case $\min u = -1$, when the strict triangle set has two components meeting at a degenerate limiting point. If $\min u < -1$, the interval where $u < -1$ is the only gap.

The cosine rule gives (3.7.6); the signs are opposite because the sum is positive real. In the strict triangle case these are exactly the two conjugate configurations. Continuity gives the same assertion at degeneracy, with the configurations possibly equal. The two outer endpoints solve $U = |1-x|$. The left one is below 1, so $U + x = 1$ and its angles are $(0,0)$. The right one is above 1, so $x = 1 + U$ and its angles are $(\pi, 0)$. The endpoints of a gap solve $U = 1 + x$, yielding $(0, -\pi)$. Finally, the smallest value of u attained on K is $\max\{-1, u(x_0)\}$, proving the formula for the maximum angle. □

Lemma 3.7.3 (Orbit realization of every compatible triangle)**.** *For x in the strict triangle set, take either angle pair $(\theta', \eta') = (\theta(x), \eta(x))$ or its negative. If $q \notin \operatorname{span}_{\mathbb{Q}}\{a, b\}$, every $z \in \mathbb{R}/(2\pi\mathbb{Z})$ is a limit of residues of positive ordinates of zeros with real parts tending to $b^{-1}\log(x/C)$. In the resonant case $q = (ma+nb)/d$, the same conclusion holds exactly for the compatible residues*

$$dz = m(\theta' - \phi) + n(\eta' - \psi) \pmod{2\pi}. \tag{3.7.7}$$

The closure of all zero residues is obtained by allowing all $x \in K$, both angle pairs, and all compatible residues.

Proof. Consider the orbit

$$t \longmapsto (at, bt, qt) \pmod{2\pi}$$

in the three-dimensional torus. In the nonresonant case its coordinates are rationally independent, so the orbit is dense in the entire torus. In the resonant case its integer relation lattice is exactly $\mathbb{Z}(m, n, -d)$. Indeed, substituting $q = (ma+nb)/d$ into any relation and using the rational independence of a, b proves this claim; primitivity of (m, n, d) makes the multiplier an integer. The orbit closure is therefore the subtorus

$$H = \{(v, w, z) : mv + nw - dz = 0 \pmod{2\pi}\}.$$

This is the elementary character form of Kronecker's theorem. Both claims also follow by averaging characters, then approximating continuous functions by trigonometric polynomials. In particular, each point of the orbit closure can be approached along times $t_j \to +\infty$, since discarding a bounded initial segment leaves the same orbit closure.

For a compatible target, choose these t_j so that $at_j \to \theta' - \phi$, $bt_j \to \eta' - \psi$, and $qt_j \to z$, all modulo 2π. Locally uniformly in s,

$$f(s + it_j) \longrightarrow f_*(s) = -1 + Be^{as + i\theta'} + Ce^{bs + i\eta'}.$$

At $\sigma = b^{-1}\log(x/C)$ the function f_* vanishes. This zero is simple: if both $Ue^{i\theta'} + xe^{i\eta'} = 1$ and $aUe^{i\theta'} + bxe^{i\eta'} = 0$, solving the two linear equations forces both summands to be real, contrary to a strict triangle. Rouche's theorem on a sufficiently small disk gives zeros $s_j \to \sigma$ of $f(s + it_j)$. Thus $\rho_j = s_j + it_j$ are zeros of f, their ordinates tend to $+\infty$, and $q \operatorname{Im}\rho_j \to z$ modulo 2π.

Conversely, every zero gives a triangle, and its angles satisfy (3.7.7) in the resonant case. The possible real parts form the compact set corresponding to K. Taking convergent subsequences therefore proves containment of the residue closure in the stated compact set. Strict triangle

points are dense in K, so the preceding realization argument supplies all its endpoint residues as well. □

Proof of Theorem 3.7.1. Lemma 3.7.3 proves the nonresonant assertion. For a resonance define the real-valued function $g(x) = m\theta(x) + n\eta(x)$ on K. Its image is an interval containing 0 and $m\pi$. To see connectedness when K has two components, observe that the image of each component is an interval, and both images contain the value $-n\pi$ at the two inner endpoints. Consequently, with $M = \max_{x\in K} |g(x)|$,

$$g(K) \cup [-g(K)] = [-M, M].$$

The orbit realization lemma now gives the exact general formula

$$\mathcal{R}_L = \{z \bmod 2\pi : \ dz + m\phi + n\psi \bmod 2\pi \in [-M, M] \bmod 2\pi\}. \tag{3.7.8}$$

Multiplication by the positive integer d is surjective on the circle, so the set in (3.7.8) is the whole circle if and only if $M \geq \pi$. If $m \neq 0$, the endpoint value $m\pi$ proves this inequality. If $m = 0$, positivity of q gives $n > 0$, and Lemma 3.7.2 gives $M = n\Theta$. This proves (3.7.4) and the complete list (3.7.5). □

Corollary 3.7.4 (A finite numerator obstruction)**.** *Only finitely many positive numerators n can appear in the exceptional list (3.7.5), namely $n < \pi/\Theta$. If $\Theta = \pi$, every period $L > 0$ has dense ordinates. If $\Theta < \pi$, the periods $2\pi d/b$, $d = 1, 2, \ldots$, all fail density. At such a period the closure consists of d closed arcs whose total relative circle length is Θ/π.*

Proof. The first three statements are direct substitutions in (3.7.5). In the last case $n = 1$ and the arc $[-\Theta, \Theta]$ has length $2\Theta < 2\pi$; its inverse image under multiplication by d has d components, each of length $2\Theta/d$. □

Corollary 3.7.5 (Closed coefficient thresholds)**.** *For $n \geq 1$, set $\tau_n = \pi/n$, $c_n = \cos\tau_n$ and*

$$x_n = \frac{(1-2r)c_n + \sqrt{(1-2r)^2c_n^2 + 4r(1-r)}}{2(1-r)}, \qquad K_n(r) = x_n^{-r}\sqrt{1 + x_n^2 - 2c_nx_n}.$$

Then the resonant periods $L = 2\pi d/(nb)$ in lowest terms have dense ordinates if and only if $\kappa \geq K_n(r)$. In particular, every positive period has dense ordinates if and only if

$$BC^{-a/b} \geq \frac{1}{r^r(1-r)^{1-r}}, \qquad r = a/b. \tag{3.7.9}$$

Proof. The inequality $\Theta \geq \tau_n$ is equivalent to the existence of $x > 0$ such that $u(x) \leq c_n$, hence to

$$\kappa^2 \geq \min_{x>0} x^{-2r}(1 + x^2 - 2c_nx).$$

The function minimized tends to infinity at both endpoints. Its derivative has the sign of $(1-r)x^2 - (1-2r)c_nx - r$, which has exactly one positive root, the displayed x_n. This proves the formula. When $n = 1$, $c_1 = -1$, $x_1 = r/(1-r)$, and $K_1(r) = r^{-r}(1-r)^{-(1-r)}$. Since the length $2\pi/b$ is dense exactly when $\Theta = \pi$, the last assertion follows. □

Remark 3.7.6 (What the classification does and does not close). The theorem answers the printed existence question in the three-term case and additionally decides every resonant length. The nonresonant existence argument was already present in the preceding work and is not counted as a new result. The all-resonance formula and the finite numerator obstruction are the increment. General sums with four or more terms have higher-dimensional polygon configuration spaces; the present proof does not classify their resonant lengths. The proof also gives no equidistribution assertion at a resonant length.

3.8 Two attempted extensions: a precise obstruction and an unresolved estimate

3.8.1 Unknown frequencies cannot be recovered in an unrestricted Dirichlet class

One must specify the model before asking whether rightmost zeros recover unknown frequencies. The following elementary obstruction survives both a fixed constant coefficient and a fixed first frequency and coefficient. It is recorded as a counterexample, not as a new general inverse theorem and not as an Euler-product counterexample.

Proposition 3.8.1 (Normalized right-divisor nonidentifiability)**.** *Fix $c < R$, $a > 0$, and $F(s) = 1 - e^{aR}e^{-as}$. For every $\lambda > a$ with λ/a irrational, choose a nonzero ϵ such that $|\epsilon|e^{-\lambda c} < 1$, and put*

$$G_{\lambda,\epsilon}(s) = F(s)(1 + \epsilon e^{-\lambda s}) = 1 - e^{aR}e^{-as} + \epsilon e^{-\lambda s} - \epsilon e^{aR}e^{-(a+\lambda)s}.$$

Then all these finite Dirichlet sums have constant coefficient 1, the same smallest positive frequency a, and the same coefficient $-e^{aR}$ at that frequency. Their complete zero divisors in $\{\Re s > c\}$ agree, including multiplicities, and their rightmost real zero coordinate is the common value R. Nevertheless their remaining frequencies and coefficients vary.

Proof. The factor $1 + \epsilon e^{-\lambda s}$ has no zero in $\Re s > c$ by the displayed strict modulus bound. It thus preserves the entire zero divisor there. Its own zeros all have real part $\lambda^{-1}\log|\epsilon| < c$, whereas the zeros of F are $R + 2\pi ik/a$, $k \in \mathbb{Z}$, and are simple. The frequency and coefficient statements follow from the explicit expansion and $\lambda > a$. □

This example shows that even the entire right divisor, stronger than an edge-approaching subset, does not identify a normalized general Dirichlet sum. It does not settle the narrower problem for logarithmic derivatives of Euler products with prescribed local factor form, nor for sums whose full list of nonconstant frequencies is rationally independent. The product construction introduces an additive frequency relation. Those restrictions must be retained in any proposed inverse theorem.

3.8.2 The actual Euler hyperuniformity problem remains open here

For independent Haar phases ω_p, consider the actual Euler hull

$$Q_\omega(s) = \sum_p \frac{(\log p)\omega_p p^{-s}}{1 - \omega_p p^{-s}}, \qquad \Re s > 1,$$

and count zeros whose real parts lie in a fixed compact interval $I \subset (1, \infty)$. Write N_T for the count with ordinates in $[0, T]$. The previously established pure point expansion gives

$$\operatorname{Var} N_T = \sum_{k \neq 0} |c_k|^2 \frac{4\sin^2(\lambda_k T/2)}{\lambda_k^2}, \qquad \lambda_k = \sum_p k_p \log p. \tag{3.8.1}$$

The c_k are the Bragg amplitudes of this particular zero-count observable. The exact missing estimate is

$$\sum_{0<|\lambda_k|\le r} |c_k|^2 = o(r) \qquad (r \downarrow 0), \tag{3.8.2}$$

which is equivalent to $\operatorname{Var} N_T = o(T)$. No estimate establishing (3.8.2) for the actual Euler amplitudes is proved in this work.

Three attempted shortcuts fail for specific reasons. First, pure point spectrum alone controls the *location* of the spectral mass, not its accumulation at zero. Second, an argument-principle contour can place its right boundary in a zero-free region, but a left boundary through the zero-bearing strip has a nontrivial winding count; one cannot replace it by a globally bounded continuous logarithm. Third, a finite prime cutoff permits Diophantine lower bounds for nonzero frequencies, but those constants depend on the cutoff and the Fourier degrees. Neither their passage to infinitely many primes nor the necessary regularity of the sharp zero-count observable is established.

For clarity, the first failure can be demonstrated on the *same* prime torus, although with a random density measure instead of a zero point process. This is only a test of the proposed spectral argument.

Proposition 3.8.2 (Pure point spectrum is insufficient even on the prime torus)**.** *There exists a positive continuous function A on $\Omega = \prod_p \mathbb{T}$ such that the stationary random measure $A(R_t\omega)\,dt$, where $R_t\omega = (\omega_p p^{-it})_p$, is a factor of the prime Kronecker flow and has pure point spectrum, but its mass M_T in $[0,T]$ satisfies*

$$\limsup_{T\to\infty} \operatorname{Var}(M_T)/T = \infty.$$

Proof. Choose distinct rationals $r_j > 1$ with $\lambda_j = \log r_j < 2^{-4j}$. Unique factorization gives distinct characters χ_{k_j} with frequency λ_j. Set

$$A(\omega) = 2 + \sum_{j\ge 1} 2^{-j} \operatorname{Re} \chi_{k_j}(\omega).$$

The series converges uniformly, so A is continuous and $1 \le A \le 3$. Orthogonality of the distinct positive and negative frequency characters gives

$$\operatorname{Var} M_T = \sum_{j\ge 1} \frac{2^{-2j}}{2} \frac{4\sin^2(\lambda_j T/2)}{\lambda_j^2}.$$

At $T_j = 1/\lambda_j$, its jth summand implies

$$\frac{\operatorname{Var} M_{T_j}}{T_j} \ge 2\sin^2(1/2)\frac{2^{-2j}}{\lambda_j} > 2\sin^2(1/2)\,2^{2j}.$$

Stationarity follows from Haar invariance. Every square-integrable observable of a factor of this Kronecker flow lies in a pure point unitary representation, establishing the spectral claim. □

This counterexample concerns a continuous random density, not Euler zeros. It prevents an invalid inference from the previously proved pure point theorem but does not decide either direction of (3.8.2). Work on Gaussian Weyl–Heisenberg zeros [57] exhibits both hyperuniform and non-hyperuniform behaviour in other models; its hypotheses do not apply to the Euler series. The actual Euler count-variance question is a new formulation in this project, not a published conjecture claimed to be solved.

3.9 A deterministic resonance criterion and a computable prime-like flow

The random-frequency theorem in the preceding manuscript [53] identifies uniform covering time with small-ball entropy for rapidly decreasing weights. Its almost-sure conclusion does not identify any specified frequency sequence. We now give a necessary and sufficient arithmetic

condition for that conclusion and a deterministic, computable sequence satisfying it. The sequence approximates every prime logarithm with any prescribed fixed polynomial accuracy. It is not the sequence of prime logarithms itself.

The underlying finite-dimensional transference principle is classical. Bounemoura [44] treats ergodization by geometry of numbers, and the Euclidean covering-radius inequality used below is due to Banaszczyk [58]; see the formulation and improvement in Aggarwal–Stephens-Davidowitz [59]. The conclusions specific to this section are the weighted infinite-dimensional formulation, its exact-entropy criterion, and an effective deterministic realization. Their historical priority requires further investigation.

3.9.1 A weighted dual profile controls actual covering

Let $w_j > 0$, $\sum_j w_j < \infty$, and $\lambda_j \in \mathbb{R}$. On the product torus with angles of period 2π, put

$$E_w(\theta) = \sum_j w_j(1 - \cos\theta_j), \qquad \Phi_t\theta = \theta - t\lambda.$$

Let $R_{w,\lambda}(\varepsilon)$ be the infimum of the times T for which every starting point meets $\{E_w < \varepsilon\}$ during $[0, T]$; it is infinite if no such time exists. For finitely supported integer vectors set

$$Q_w(k)^2 = \sum_j \frac{k_j^2}{w_j}, \qquad \Psi_{w,\lambda}(u) = \max_{0<Q_w(k)\le u^{-1/2}} |k\cdot\lambda|^{-1}. \tag{3.9.1}$$

Exact resonances contribute infinity and the empty maximum is zero. The character set is finite: a nonzero k_j requires $w_j \ge u$. Let $n(\varepsilon)$ be the least positive integer with $\sum_{j>n} w_j \le \varepsilon/8$.

Theorem 3.9.1 (Weighted resonance transference)**.** *There is an absolute constant $K \ge 1$ such that*

$$\frac{\pi}{2}\Psi_{w,\lambda}(\varepsilon) \le R_{w,\lambda}(\varepsilon/8), \qquad R_{w,\lambda}(\varepsilon) \le 2Kn(\varepsilon)\Psi_{w,\lambda}\left(\frac{\varepsilon}{K^2 n(\varepsilon)^2}\right), \tag{3.9.2}$$

provided the character set in the upper bound is nonempty. The bound concerns actual covering, without any product-kernel restriction.

Proof. For principal angles, $1 - \cos\theta \ge 2\theta^2/\pi^2$, whence

$$|k\cdot\theta| \le (\pi/\sqrt{2})Q_w(k)\sqrt{E_w(\theta)}.$$

For $Q_w(k) \le \varepsilon^{-1/2}$ the character is within angular distance $\pi/4$ of 1 throughout $\{E_w < \varepsilon/8\}$. Starting the character at -1 keeps its angular distance from 1 at least $\pi/2$ for $0 \le t \le \pi/(2|k\cdot\lambda|)$. This proves the lower bound, including the infinite bound for an exact resonance.

Classical Euclidean transference supplies an absolute C_0 with

$$\mu(\Lambda)\lambda_1(\Lambda^*) \le C_0 n \tag{3.9.3}$$

for every full-rank lattice in $\mathbb{R}^n$. Here μ is covering radius and λ_1 the shortest nonzero Euclidean vector. This weaker constant form follows from the cited Banaszczyk theorem; its sharp constant is not needed. Fix $K \ge \max(1, 2\pi C_0)$, write $n = n(\varepsilon)$ and $P = \Psi_{w,\lambda}(\varepsilon/(K^2n^2))$, and put $T = KnP$. We may assume $P < \infty$. Set $D = \operatorname{diag}(\sqrt{\varepsilon/w_1}, \dots, \sqrt{\varepsilon/w_n})$. The full-dimensional ellipsoid

$$\mathcal{E} = \{tT\lambda_{\le n} + Du : t^2 + \|u\|_2^2 \le 1\}$$

has support function $h_{\mathcal{E}}(k)^2 = T^2(k\cdot\lambda)^2 + \varepsilon Q_w(k)^2$. For every nonzero integer k, either $Q_w(k) \ge Kn/\sqrt{\varepsilon}$ or $|k\cdot\lambda| \ge 1/P$. In either case $h_{\mathcal{E}}(k) \ge Kn$.

Write $\mathcal{E} = AB_2^n$ with A symmetric positive definite. The dual lattice of $A^{-1}(2\pi\mathbb{Z}^n)$ is $A\mathbb{Z}^n/(2\pi)$ and has shortest vector at least $Kn/(2\pi)$. Equation (3.9.3) therefore gives covering radius at most $2\pi C_0/K \le 1$. It follows that $\mathcal{E}+2\pi\mathbb{Z}^n = \mathbb{R}^n$. Every head target is approximated at a signed time in $[-T,T]$ with error Du, $\|u\|_2 \le 1$, whose energy is at most $\frac{1}{2}\sum_{j\le n} w_j(Du)_j^2 \le \varepsilon/2$. The arbitrary tail has energy at most $2\sum_{j>n} w_j \le \varepsilon/4$. Thus total energy is strictly less than ε. Translating the signed segment gives the same covering conclusion for $[0,2T]$. □

3.9.2 The exact deterministic entropy criterion

Assume

$$\log w_j = -aj^b + o(j^b), \quad a,b>0, \qquad |\lambda_j| \le C(1+j)^d, \tag{3.9.4}$$

and write

$$\ell = \log(1/\varepsilon), \qquad H(\varepsilon) = \frac{b}{2(b+1)} a^{-1/b}\ell^{1+1/b}.$$

Nonmonotone weights are allowed. The elementary spectral estimates are

$$n(\varepsilon) \sim (\ell/a)^{1/b}, \qquad \frac{1}{2}\sum_j (L+\log w_j)_+ \sim \frac{b}{2(b+1)} a^{-1/b}L^{1+1/b}. \tag{3.9.5}$$

To verify them, sandwich late weights between $e^{-(a+\delta)j^b}$ and $e^{-(a-\delta)j^b}$. Tail comparison gives $\log\sum_{j>n} w_j = -an^b + o(n^b)$; summing $L - cj^b$ through $(L/c)^{1/b}$ and then sending δ to zero gives the second estimate.

For context, the established small-ball scale is

$$-\log \mu\{E_w < \varepsilon\} \sim H(\varepsilon). \tag{3.9.6}$$

It is not a new small-deviation theorem. One direct verification uses $\mathbb{E}e^{-v(1-\cos\Theta)} \le C(1+v)^{-1/2}$ for a uniform angle: Markov's inequality at parameter $1/\varepsilon$, applied to indices with $w_j \ge \varepsilon$, gives the lower bound for the negative log probability with error $O(\ell^{1/b})$. For the other bound, constrain the first $n = n(\varepsilon)$ angles by $|\theta_j| \le \min(1, \sqrt{\varepsilon/(2nw_j)})$ and leave the tail free. The resulting box lies in the ball. Its negative log probability is $H(\varepsilon)(1+o(1))$ by (3.9.5), because $n\log n = o(H(\varepsilon))$.

Lemma 3.9.2 (Universal dual entropy obstruction)**.** *Under* (3.9.4),

$$\liminf_{\varepsilon\downarrow 0} \log \Psi_{w,\lambda}(\varepsilon)/H(\varepsilon) \ge 1,$$

with infinity allowed.

Proof. Set $n = n(\varepsilon)$ and $D_j = \lfloor\sqrt{w_j/(n\varepsilon)}\rfloor$, $j \le n$. There are $N = \prod_{j\le n}(D_j+1)$ integer vectors with $0 \le k_j \le D_j$. Equation (3.9.5) shifted by $\log n = o(\ell)$ gives $\log N \sim H(\varepsilon)$. Their linear forms lie in an interval of length at most

$$M = \sum_{j\le n} D_j|\lambda_j| \le (n\varepsilon)^{-1/2}\sum_j \sqrt{w_j}|\lambda_j|.$$

The last sum is finite. Coincident forms give an exact resonance. Otherwise some two consecutive values differ by at most $M/(N-1)$. Their difference vector has $Q_w(k)^2 \le \sum_{j\le n} D_j^2/w_j \le 1/\varepsilon$. Hence $\log\Psi \ge \log(N-1) - \log M \ge (1-o(1))H$, since $\ell = o(H)$. □

Theorem 3.9.3 (Necessary and sufficient arithmetic condition)**.** *Under* (3.9.4)*, the following conditions are equivalent:*

$$\log R_{w,\lambda}(\varepsilon) \sim H(\varepsilon), \tag{3.9.7}$$

$$\log \Psi_{w,\lambda}(\varepsilon) \sim H(\varepsilon), \tag{3.9.8}$$

$$\limsup_{\varepsilon \downarrow 0} \frac{\log \Psi_{w,\lambda}(\varepsilon)}{H(\varepsilon)} \le 1. \tag{3.9.9}$$

Proof. Lemma 3.9.2 makes the last two equivalent. The lower bound in (3.9.2) shows that the first implies the last, since $H(\varepsilon/8) \sim H(\varepsilon)$. Conversely, $\log(K^2 n(\varepsilon)^2/\varepsilon) = \ell + O(\log \ell)$, so the rescaling in the upper bound of (3.9.2) leaves H unchanged to leading order. That bound proves $\limsup \log R/H \le 1$. The lower bound and Lemma 3.9.2, with 8ε in place of ε, prove the matching lower estimate. □

3.9.3 A computable sequence near the prime logarithms

Here computable means that a finite algorithm, given j, q, outputs a rational number within 2^{-q} of λ_j. No feasible running-time bound is asserted.

Theorem 3.9.4 (Effective realization without random choices)**.** *Let $0 < \rho \le 1$ be rational and $A \ge 1$ an integer. There is a computable sequence with $|\lambda_j - \log p_j| \le \rho p_j^{-A}$ for all j, having no nonzero finitely supported integer resonance. With $w_j = e^{-j}$,*

$$\log R_{w,\lambda}(\varepsilon) = \tfrac{1}{4}\ell^2 + O(\ell \log \ell), \qquad \ell = \log(1/\varepsilon). \tag{3.9.10}$$

The implicit constant may depend on ρ, A.

Proof. On $X = [-1,1]^{\mathbb{N}}$ with product uniform probability put $r_j = \rho p_j^{-A}$ and $\lambda_j(x) = \log p_j + r_j x_j$. For $m \ge 1$ define

$$\mathcal{K}_m = \{k \ne 0 : k_j = 0 \ (j > m),\ |k_j| \le D_{mj} \ (j \le m)\}, \quad D_{mj} = \lceil u_{mj} \rceil,$$

where u_{mj} is a specified rational upper approximation to $e^{(m-j)/2}$ with error at most $1/4$. Such approximations are obtained by summing the Taylor series and using its geometric tail bound; fix the first truncation which certifies the stated error. In particular, $e^{(m-j)/2} \le D_{mj} \le e^{(m-j)/2} + 5/4$. Set $N_m = |\mathcal{K}_m|$, $L_m = p_m^A/(2\rho)$ and

$$\delta_m = \frac{2^{-m-4}}{L_m(N_m+1)}, \qquad B_m = \bigcup_{k \in \mathcal{K}_m} \{x : |k \cdot \lambda(x)| < \delta_m\}.$$

These are computable data, with only ceilings of rational numbers used. Conditioning on any coordinate with $k_j \ne 0$ bounds the density of the form by L_m. Consequently

$$\mathbb{P}(B_m) \le 2 L_m N_m \delta_m \le 2^{-m-3}, \qquad \mathbb{P}\left(\bigcup_{m>M} B_m\right) \le 2^{-M-3}. \tag{3.9.11}$$

The closed set $G = X \setminus \bigcup_{m \ge 1} B_m$ has probability at least $7/8$.

To obtain a computable point, positive measure alone is insufficient; we use the effective tail bound. For every closed rational cylinder C, the number $\mathbb{P}(G \cap C)$ is computable. Truncate the union at $m = M$; its omitted probability is bounded by (3.9.11). The remaining finite union of linear slabs intersected with a rational box has computable volume. Indeed approximate the computable slab offsets by rationals and compute inner and outer rational-polytope volumes. Their difference tends effectively to zero: every hyperplane boundary strip of thickness η in the

linear form has probability at most $2L_M\eta$, and there are only finitely many boundaries. Rational box boundaries have measure zero. This supplies certified upper and lower approximations to $\mathbb{P}(G \cap C)$ to any requested absolute error.

Start with $C_0 = X$. At step s, partition C_{s-1} into finitely many closed dyadic cylinders whose first s coordinates have lengths at most 2^{-s}. Boundaries have probability zero, so some child has positive G-intersection measure. Dovetail the computable volume approximations until a strictly positive lower bound is certified for one child, using a fixed enumeration to resolve ties, and call it C_s. This finite search terminates. The nested cylinders specify a unique point x^* coordinate by coordinate with an effective error bound. The compact sets $G \cap C_s$ are nonempty; compactness gives $x^* \in G$. Thus $\lambda_j(x^*)$ is computable and has the stated approximation property.

For $w_j = e^{-j}$, every integer vector with $Q_w(k) \le e^{m/2}$ belongs to $\mathcal{K}_m$, so $\Psi_{w,\lambda}(e^{-m}) \le \delta_m^{-1}$. Furthermore,

$$\log(N_m + 1) = \sum_{j=1}^{m} \log(2D_{mj} + 1) = m(m-1)/4 + O(m).$$

Bertrand's postulate gives $p_m \le 2^m$, hence $\log L_m = O(m)$ and $\log \delta_m^{-1} = m^2/4 + O(m)$. Every fixed nonzero finite character belongs to some $\mathcal{K}_m$, proving nonresonance. Monotonicity gives

$$\log \Psi_{w,\lambda}(\varepsilon) \le \ell^2/4 + O(\ell). \tag{3.9.12}$$

For these exact weights the pigeonhole proof of Lemma 3.9.2 also gives $\log \Psi \ge \ell^2/4 - O(\ell \log \ell)$: here $n = \ell + O(1)$, and the spectral sum with $L = \ell - \log n$ differs from $\ell^2/4$ by $O(\ell \log \ell)$. Theorem 3.9.1 changes ℓ by only $O(\log \ell)$ and proves (3.9.10). □

The proof specifies a deterministic terminating rule at every requested precision. Its potentially enormous computational cost is a separate issue. Closeness to prime logarithms does not justify a limit passage to the exact prime flow: the construction controls one fixed sequence, and its constants and time scales need not be uniform as the perturbations tend to zero.

3.9.4 The unresolved arithmetic obstacle at exact primes

For the actual prime flow with $w_p = p^{-\sigma}$, $\sigma > 3$, put $S_\sigma = \sum_p p^{-\sigma}(\log p)^2 < \infty$. Unique factorization associates to each nonzero finite integer vector distinct positive integers N, D with $k \cdot \lambda = \log(N/D)$. Cauchy–Schwarz gives

$$\log \max(N, D) \le \sum_p |k_p| \log p \le S_\sigma^{1/2} Q_w(k).$$

The mean-value theorem for log implies

$$|k \cdot \lambda| \ge e^{-S_\sigma^{1/2} Q_w(k)}, \qquad \log \Psi_{w,\log p}(\varepsilon) \le S_\sigma^{1/2} \varepsilon^{-1/2}. \tag{3.9.13}$$

This is the elementary height barrier, not an improvement. The preceding manuscripts give entropy exponent $1/(\sigma - 1)$ for the lower covering bound but upper exponent $1/2$. The new transference theorem does not close that gap: the head dimension now grows as a power of $1/\varepsilon$, so rescaling by $n(\varepsilon)^2$ is no longer negligible. Substitution of (3.9.13) into (3.9.2) is actually weaker than the earlier optimized positive-kernel upper bound.

The exact-prime attempt therefore remains open. The proved new conclusions within this project are the deterministic fast-weight criterion and its computable prime-like realization. Neither proves a new zero-free region for the Riemann zeta function or a theorem about its critical-line zeros.

3.10 Two limits of supplementary information in analytic inverse problems

The results here address two precisely stated inverse problems. Neither is advertised as a resolution of a previously named open problem. The first combines convolution ambiguity with a topological obstruction to arbitrary finite continuous side information. The second separates analytic uniqueness from statistical recovery when the analytic class has no common derivative envelope. The principal imported ingredients are classical: the Borsuk–Ulam theorem and localized smooth annihilators for a summable Müntz sequence.

3.10.1 Complete diffraction and arbitrary finite continuous side data

For $f \in L^1(\mathbb{R})$ put $\widehat{f}(\xi) = \int e^{-i\xi x} f(x)\,dx$ and $\mathcal{I}(f) = |\widehat{f}|^2$, with the whole real line observed. A density is κ-strongly log-concave when $e^{\kappa x^2/2} f(x)$ is log-concave. Gevrey order σ means $\|f^{(r)}\|_\infty \le AB^r (r!)^\sigma$ for some A, B.

Theorem 3.10.1 (A uniform obstruction to finite continuous calibration)**.** *Fix $N \ge 1$, $L, \kappa > 0$, $\sigma > 1$, and $K \ge 1$. There exist a compact family $\mathcal{F} \subset C^\infty(\mathbb{R})$ and $d_N > 0$, chosen independently of the observation map below, with the following properties. Every member of $\mathcal{F}$ is a mean-zero probability density supported exactly on $[-L, L]$, positive inside, κ-strongly log-concave, and Gevrey of order σ with common constants. All members have the same moments through order K. For every continuous map $A : \mathcal{F} \to \mathbb{R}^N$ there exist $f, g \in \mathcal{F}$ such that*

$$\mathcal{I}(f) = \mathcal{I}(g), \qquad A(f) = A(g), \qquad \|f - g\|_1 \ge d_N.$$

These two densities are not translates or reflected translates of one another. Continuity of A is required for the entire observation map, with $\mathcal{F}$ carrying its C^∞ topology.

Proof. Set $d = N + 1$, and use $d + 1$ convolution factors, indexed by $0, \dots, d$. Let $a = L/(d+1)$ and $c = (d+1)\kappa$. Choose strictly increasing odd integers $K < m_0 < m_1 < \cdots < m_d$. Put $p = (\sigma - 1)^{-1}$ and let

$$b(x) = C \exp(-cx^2 - (a - x)^{-p} - (a + x)^{-p}) \mathbf{1}_{|x|<a}$$

have integral one. This is an even Gevrey density with $(\log b)'' \le -2c$ in its support. For each j choose an even, nonnegative, nonzero Gevrey bump ψ_j supported in $(-a/2, a/2)$, and put $h_j = \psi_j^{(m_j)}$. Integration by parts gives

$$\int x^r h_j(x)\,dx = 0 \quad (r < m_j), \qquad \int x^{m_j} h_j(x)\,dx = -m_j! \int \psi_j \ne 0.$$

All h_j are odd. Choose positive ε_j sufficiently small that $b + u\varepsilon_j h_j$ is positive and c-strongly log-concave for every $|u| \le 1$. This follows from a uniform C^2 perturbation on the compact support of h_j, with the strict curvature reserve provided by b. Mass and mean remain one and zero.

Fix $q_0 = b + \varepsilon_0 h_0$. For $t = (t_1, \dots, t_d) \in S^{d-1} = S^N$ define

$$q_{j,t_j} = b + t_j \varepsilon_j h_j \quad (1 \le j \le d), \qquad f_t = q_0 * q_{1,t_1} * \cdots * q_{d,t_d}.$$

Convolution of densities of curvatures $c_0, \dots, c_d$ has curvature at least $(\sum_j c_j^{-1})^{-1}$: completing the square and applying the log-concavity theorem for marginals proves this rule. Thus f_t has curvature at least $c/(d+1) = \kappa$. Its support and interior positivity follow by convolution of interval-supported positive densities. Differentiation of the fixed factor gives $\|f_t^{(r)}\|_\infty \le \|q_0^{(r)}\|_\infty$,

which proves common Gevrey bounds. The map $t \mapsto f_t$ is continuous in C^∞ (indeed polynomial in the finitely many coordinates), so its image $\mathcal{F}$ is compact. The moment cancellations give common moments through K.

Since $q_{j,-t_j}(x) = q_{j,t_j}(-x)$, reflection conjugates each varying factor's Fourier transform. Consequently $\mathcal{I}(f_t) = \mathcal{I}(f_{-t})$ for every t. It remains to check that this antipodal pair is uniformly nontrivial. The first nonzero odd cumulant of q_0 has order m_0 and is nonzero. For $j \geq 1$, all odd cumulants of q_{j,t_j} below m_j vanish and its m_jth cumulant is $t_j\beta_j$ for a fixed $\beta_j \neq 0$. This follows from the moment–cumulant identities and the displayed moment cancellations. Cumulants add under convolution. If j is the first index with $t_j \neq 0$, the m_jth cumulants of f_t and f_{-t} differ by $2t_j\beta_j \neq 0$. Their common nonzero m_0th odd cumulant rules out reflection, while their common mean rules out translation. In particular,

$$d_N := \min_{t \in S^N} \|f_t - f_{-t}\|_1 > 0$$

by compactness and continuity. Finally the continuous map $t \mapsto A(f_t)$ from S^N to $\mathbb{R}^N$ has an antipodal coincidence by the Borsuk–Ulam theorem. That pair proves the claim. □

Corollary 3.10.2 (Finite sensor values and a decision-theoretic floor)**.** *The theorem applies to any fixed finite list of complex Fourier values, heat values at arbitrary sensor locations and positive times, moments, or continuous nonlinear summaries, counting each real component as one observation. Any reconstruction from $(\mathcal{I}(f), A(f))$ has worst-case L^1 error at least $d_N/2$. The same lower bound holds for worst-case expected error in any statistical experiment whose law depends only on these data.*

Proof. The listed evaluations are continuous on $\mathcal{F}$. The two densities in the theorem have identical data, hence identical data laws. The triangle inequality, integrated under that common law, gives the risk bound. □

Scope and prior-art boundary. Convolution-factor reflection and Borsuk–Ulam information lower bounds are classical. The additional construction realizes a uniformly separated antipodal obstruction under simultaneous support, curvature, smoothness, and moment constraints while the entire Fourier intensity is supplied for free. Priority for this combination is unverified. It strengthens the earlier fixed-moment ambiguity construction by allowing arbitrary finite continuous side data, but supplies no rate for d_N as N grows. It is not a claim about arbitrary adaptive or randomized sensing. In particular, Krieg–Novak–Ullrich [60, Theorem 1 and Section 3] show that individually continuous adaptive measurements can have a discontinuous joint transcript and escape naive dimension bounds. Fixed sensors, and feedback designs whose complete transcript is continuous, are covered here; arbitrary switching is not. Nor does the result cover a second complete diffraction pattern, which is an infinite-dimensional observation.

3.10.2 Analytic uniqueness without any uniform statistical modulus

Fix $0 < a < b$ and distinct $t_j \downarrow 0$ with $\sum_j t_j < \infty$. Write

$$(\mathcal{H}f)_j = (4\pi t_j)^{-1/2} \int_a^b e^{-x^2/(4t_j)} f(x)\, dx.$$

Analyticity below is a property on a neighborhood of the closed source interval. The density is set to zero outside $[a, b]$ and is not asserted to be globally analytic across its support endpoints.

Theorem 3.10.3 (A statistical discontinuity inside the analytic class)**.** *Fix $\kappa > 0$ and an integer $R \geq 2$. There exist $m, M, d > 0$ such that the class $\mathcal{A}$ of probability densities on $[a, b]$ which extend to entire functions and satisfy*

$$f \geq m, \qquad \|f\|_{C^R([a,b])} \leq M, \qquad (\log f)'' \leq -\kappa$$

has the following properties:

1. *$\mathcal{H}$ is injective on $\mathcal{A}$.*
2. *There are $f_n^+, f_n^- \in \mathcal{A}$ with*

$$\|f_n^+ - f_n^-\|_1 \geq d, \qquad \|\mathcal{H}f_n^+ - \mathcal{H}f_n^-\|_{\ell^2} + \|\mathcal{H}f_n^+ - \mathcal{H}f_n^-\|_{\ell^\infty} \longrightarrow 0.$$

3. *With adversarial coordinatewise noise bounded by any $\delta > 0$, the minimax L^1 reconstruction error is at least $d/2$, even if the whole infinite sequence is observed. With independent Gaussian noise of any standard deviation $\delta > 0$ at every sampled time, the minimax expected L^1 error is at least $d/4$.*

The constants do not depend on the noise level or the number of observed coordinates. No common complex-neighborhood norm or quasianalytic derivative envelope is imposed.

Proof. For injectivity let $h = f - g$ be nonzero and analytic near $[a, b]$. There is a least $r \geq 0$ with $h^{(r)}(a) \neq 0$. Substituting $x = a + u/s$ and using Taylor expansion with exponential domination gives

$$\int_a^b e^{-sx^2} h(x)\, dx \sim \frac{h^{(r)}(a) e^{-sa^2}}{(2a)^{r+1} s^{r+1}} \qquad (s \to \infty).$$

For completeness, on a fixed endpoint neighborhood the leading term is $h^{(r)}(a)u^r/(r!s^r)$ and the remaining exponential tends to e^{-2au}; the integral of $u^r e^{-2au}$ is $r!/(2a)^{r+1}$. The part away from a is exponentially smaller. The transform therefore has no sufficiently large positive zeros. Since $s_j = (4t_j)^{-1} \to \infty$, equality of all observations is impossible.

The smooth annihilator construction for summable Müntz sequences provides a nonzero real $h \in C_c^\infty((a, b))$ with

$$\int h = 0, \qquad \int_a^b e^{-x^2/(4t_j)} h(x)\, dx = 0 \quad (j \geq 1).$$

One may obtain this directly from the localized Mellin annihilator with zeros s_j, by the substitution $y = e^{-x^2}$; its existence uses $\sum_j 1/s_j < \infty$. This is the classical existence input of Luxemburg–Korevaar, in the explicit form restated by Wellershoff [19, Theorem 21].

Let $f_0 = Ce^{-\kappa x^2}$ on $[a, b]$, normalized to have mass one. Its logarithmic curvature is -2κ. Choose $\varepsilon > 0$ small enough that $f_0 \pm \varepsilon h$ are positive and have logarithmic curvature at most $-3\kappa/2$. There are polynomials p_n with $p_n \to h$ in $C^R([a, b])$ and $\int p_n = 0$. Indeed, approximate $h^{(R)}$ uniformly by polynomials, integrate R times with the prescribed endpoint derivatives, and subtract the vanishing constant mass correction. Put $f_n^\pm = f_0 \pm \varepsilon p_n$. These functions are entire, have mass one, and for all sufficiently large n satisfy the asserted bounds with fixed m, M and curvature at most $-\kappa$. Their L^1 separation tends to $2\varepsilon\|h\|_1 > 0$; reduce $d > 0$ and discard a finite prefix.

Let $k_j = (4\pi t_j)^{-1/2} e^{-a^2/(4t_j)}$. Then $k \in \ell^2 \cap \ell^\infty$: for small t, $t^{-1}e^{-a^2/(2t)} \leq C_a t$, and $\sum_j t_j < \infty$. Consequently

$$\|\mathcal{H}(f_n^+ - f_n^-)\|_{\ell^q} \leq 2\varepsilon \|k\|_{\ell^q} \|p_n - h\|_1 \longrightarrow 0, \qquad q = 2, \infty.$$

The midpoint of two data sequences whose sup distance is at most 2δ is admissible under both hypotheses. The triangle inequality proves the adversarial-noise lower bound.

For Gaussian noise choose n so large that the ℓ^2 distance of the two mean sequences is at most δ. Their product Gaussian laws have relative entropy at most $1/2$: the finite-dimensional formula is $\|\Delta\|_2^2/(2\delta^2)$, and the same formula for the infinite product follows by convergence of the Gaussian likelihood ratios when $\Delta \in \ell^2$. Pinsker's inequality gives total variation at most $1/2$. Integration against the common part of the two laws gives worst-case expected metric loss at least $d(1 - \mathrm{TV})/2 \geq d/4$. □

What this does and does not settle. The theorem gives a precise obstruction to seeking a noise rate from analytic or quasianalytic membership alone. It persists with any fixed finite number of uniform derivative bounds. A fixed full quasianalytic envelope, by contrast, can yield a compact class and a uniformly continuous inverse; its sharp modulus is a different problem and is not computed here. The mechanism is density of analytic functions in a smooth nonidentifiable class, so this is a rigorous barrier, not a new solution of a general inverse-heat stability problem.

3.11 Seventeen research routes: exact outcomes and remaining barriers

This register counts investigations, not independent historically open conjectures. "Complete" means complete for the explicitly stated question in that row. "Partial" records a proved subdomain of a larger question. A supporting lemma, a numerical illustration, or a corollary is not an additional investigation. The seven complete rows, three partial rows, and seven unresolved rows below are therefore not a claim to have solved seven or seventeen famous conjectures.

ID	Exact target	Outcome and scope
1	Published 2019 binomial-mixture MLE range conjecture	**Complete proof in this draft.** Theorem 3.2.1 reaches $t \le N^{2/3-\varepsilon}$ for the same unrestricted MLE and Wasserstein loss. This is the clearest direct external conjecture resolution claimed by the manuscript.
2	Near-linear MLE range discussed in the 2026 author version	**Unresolved.** The obtained predictive Hellinger rate still has a $t^{1/4}$ factor. Neither the exact constant-one coefficient inequality nor the proposed dimension-free fingerprint bound is proved.
3	Adaptive Bernoulli recovery under a per-individual cap Q	**Partial full-phase answer.** Theorem 3.3.1 gives matching risk throughout $Q \le B^{2/5-\gamma}$ and lower bounds for every cap. The remaining intermediate-cap range is open.
4	Scalar uniform-endpoint critical logarithm	**Unresolved.** Proposition 3.4.1 proves only a limitation of the specified independent-level CDF estimator, not a Wasserstein minimax lower bound.
5	Sparse Gaussian terminal designs with unrestricted covariance	**Complete.** Theorem 3.5.1 establishes harmonic divergence as necessary and sufficient, with joint invisible laws in every dimension when it fails. This closes a question left in the previous draft.
6	Does bounded maximum-likelihood threshold force bounded generic completion rank?	**Unresolved.** The Gaussian identification transfer computes neither invariant in the required generality. The original graph question is identified in the Gaussian section.
7	Sparse random-graph likelihood/completion-rank conjectures	**Unresolved.** The latest rigidity result located in the source audit still leaves the global-rigidity step needed by the likelihood conjecture. No graph-conjecture closure is inferred.
8	Full ID/SD domain of the third McKay family	**Partial.** Theorem 3.6.3 proves ID and an exact SD threshold on every integer-bias ray $n \ge 2$, $\mu > -1/2$. Noninteger parameters and further negative orders remain open.
9	Full SD domain of the fourth McKay family	**Partial.** Theorem 3.6.4 treats all half-integer orders above $1/2$. Theorem 3.6.5 uniquely specifies the $3/2$ threshold and proves an exact rational non-SD example. These are parts of the same published multipart problem as row 8.
10	Every modular period for irrational-frequency trinomials	**Complete for three terms.** Theorem 3.7.1 determines density and missing arcs for every resonance. General four-or-more-term classification and resonant equidistribution are not included.
11	Unknown Euler frequencies from a right zero divisor	**Unresolved in the Euler class.** Proposition 3.8.1 disproves the broader normalized Dirichlet-sum formulation. The counterexample does not preserve the required Euler-factor structure.
12	Actual Euler critical-zero hyperuniformity	**Unresolved.** The required low-frequency Bragg-weight estimate is displayed in (3.8.2). The accompanying counterexample invalidates a spectral shortcut, not the conjectured Euler conclusion itself.

ID	Exact target	Outcome and scope
13	A deterministic exact-entropy criterion for fast weights	**Complete.** Theorem 3.9.3 gives a necessary and sufficient dual-resonance condition for actual uniform covering. The finite-dimensional transference input is classical and explicitly attributed.
14	A computable deterministic entropy-optimal frequency sequence	**Complete.** Theorem 3.9.4 constructs one with prescribed polynomial closeness to every prime logarithm. It is not the sequence of prime logarithms; feasible runtime is not claimed.
15	The exact-prime recurrence exponent gap for $\sigma > 3$	**Unresolved.** The height bound in (3.9.13) and the dimension loss in transference do not improve the previous optimized upper bound.
16	Complete Fourier intensity plus finite continuous side data	**Complete negative answer in the stated class.** Theorem 3.10.1 constructs uniformly separated strongly log-concave Gevrey ambiguities for every jointly continuous finite-dimensional side map. Arbitrary switching is not covered.
17	A uniform noise modulus from analytic heat-source membership	**Complete negative answer without a common full derivative envelope.** Theorem 3.10.3 gives a fixed minimax floor despite uniqueness and common finite-order derivative bounds. A sharp modulus under a fixed full envelope remains a different, unresolved question.

3.11.1 Which changes are substantial, and which are supporting tools?

The binomial theorem is the strongest link to a previously printed conjecture. It combines a forward complexity estimate with an inverse coefficient estimate. Controlling only one side would not give the stated range: the generic simplex complexity retains a larger trial factor, while a polynomial prefactor in the inverse map consumes the gain from the smaller observation class. Their simultaneous removal to the stated orders is the relevant proof mechanism.

The sparse Gaussian theorem has a different source of difficulty. For each fixed mean frequency, constructing a scalar vanishing function is not enough: such functions might fail to arise from any single joint law. The matrix-tube realization solves this consistency problem and gives a Schwartz signed density supported on the covariance cone. Its vanishing marginals show that the hidden information can lie entirely in dependence between mean and covariance, even after every joint polynomial moment is also supplied.

The three-term zero theorem classifies all exceptional periods instead of merely proving existence of good periods. Its geometry is elementary once the appropriate lifted torus is used. This makes it a concise complete classification; it does not establish a new general method for the critical strip of the Riemann zeta function.

The Bessel results are genuine extensions beyond the previous GGC boundary, but they are partial classifications of the full published questions. The exact threshold expressed by a positive finite spectral kernel is meaningful because all its zeros and its positivity have been proved. The unique order-3/2 maximum and the rational counterexample give an independently checkable instance. Counting every integer order as a separate solved conjecture would misstate the contribution.

The deterministic recurrence and inverse-problem results are new formulations within this programme. They specify when a transfer is valid and when extra information remains insufficient. Their classical ingredients receive credit. They should be judged by the precise conclusions and their usefulness, not by a nominal count of solved problems.

3.11.2 The next bottlenecks are not removed by aggregation

For the binomial MLE, reaching a near-linear number of trials needs more than the global Hellinger estimate proved here. In the present argument the comparison to the target loss has the factor $t^{3/4}/\sqrt{N}$, up to logarithms and a small adjustable power of N. This locates the 2/3 boundary of this particular proof. Functional estimates sharper than global Hellinger control, or a different inverse argument, would be needed to go beyond it.

For the McKay laws, nonterminating hypergeometric logarithmic derivatives replace the finite root sums used here. Their additional contributions must be controlled before a full real-parameter SD classification is justified. For exact prime frequencies, an almost-sure theorem or a computable nearby sequence supplies no estimate at the prescribed arithmetic point. For Euler variance, pure point spectrum supplies no bound on the mass accumulating at arbitrarily small nonzero frequencies.

These are concrete missing estimates. Combining the completed sections does not by itself prove them, and no Annals-level or Fields-level assessment is inferred from combining the papers. The immediate review priority is the complete binomial MLE argument, followed by the matrix-tube construction and the resonant-period classification.

Part IV

Transport, Spectral Positivity, and Sharp Observation Boundaries

Near-linear Mixture Likelihood, Full McKay Domains, and Critical Sampling Budgets

Source manuscript: 14 October 2026.

Source abstract. We develop six continuations of a program on observation and recovery. For unrestricted binomial mixtures we prove that every maximum likelihood estimator attains the minimax Wasserstein order $(t \log N)^{-1/2}$ whenever $a \log N \le t \le N^{1-\varepsilon}$, for fixed $a, \varepsilon > 0$. The proof transfers kernel transportation inequalities through the likelihood score equations and then uses derivative-stable polynomial inversion. Under a total observation budget we determine the full predetermined Bernoulli replication-cap curve and prove that the rate $(\log B)/\sqrt{B}$ is unavoidable for uniform-endpoint mixtures even under arbitrary adaptive sampling. For two modified-Bessel McKay families we establish infinite divisibility throughout their admissible parameter domains; the first answers a maximal-domain question posed in the published literature. Signed hypergeometric spectral measures have positive inverse Laplace kernels despite their changing signs. Further results locate a logarithmic tail threshold for sparse Gaussian identification, characterize random-clock Lévy snapshot identification, classify all modular zero periods of independent exponential sums, and recover a hidden finite spectrum from its density-transition curve. Finally, a fixed analytic envelope gives exponential exact-data recovery but no sampling-sequence-uniform noise rate. Full arguments and an eighteen-target register distinguish established statements, partial advances, and unresolved transfers. No Riemann-hypothesis or general prime-statistics conclusion is claimed.

4.1 What changes from the previous program

The 223-page manuscript *Observation and Recovery in Arithmetic and Probability* and its three sequels [1, 29, 53, 61] developed inverse questions in probability, special functions, and exponential-sum dynamics. The present paper contains new continuations, rather than another concatenation of those earlier texts. Definitions and the arguments for the new claims are given here; the earlier papers are used to identify which boundaries were previously unresolved. Their results are not assumed to be externally certified.

The most substantial advances come from changing which intermediate quantity is controlled. In the likelihood problem, total variation of the count fingerprint is unnecessarily strong. The transport geometry of that fingerprint retains exactly the information needed by a Lipschitz inverse test. In the special-function problem, positivity of each spectral contribution is unnecessarily restrictive. A signed representation can still have a positive inverse Laplace kernel, once its total mass and pole locations are controlled. In adaptive sampling, a bound on the information supplied by an entire individual can obscure the decisive issue: the conditional information gained by one additional observation. We obtain an exact posterior identity for that increment. These are three distinct mechanisms; the paper does not assert a single universal theorem subsuming all six branches.

4.1.1 The principal quantitative advances

Let N be the number of independent binomial groups and t their common size. For their latent mixing law G, Theorem 4.2.1 proves, for every unrestricted MLE $\widehat{G}$,

$$\sup_G \mathbb{E}_G W_1(\widehat{G}, G) \le \frac{C_{\varepsilon,a}}{\sqrt{t \log N}}, \qquad a \log N \le t \le N^{1-\varepsilon}.$$

This improves the exponent $2/3$ in our immediate predecessor to every exponent below one. It reaches the near-linear range discussed conditionally by Musco et al. [14, Section 6.2], with a matching lower bound from Vinayak et al. [13]. It does not prove the separate constant-one coefficient conjecture. The all-N, t interpolation estimate is an upper bound, and its entire transition is not asserted to be minimax sharp.

For a total raw-observation budget B and per-individual cap Q, our predetermined Bernoulli design has the exact risk order

$$\max\left\{B^{-1/3},\ Q^{-1},\ [Q \log(e + B/Q^{3/2})]^{-1/2}\right\}.$$

The proof optimizes a frequency-dependent allocation; it is not a claim about the equal-group MLE. For adaptive designs the same analysis gives an exact order through every $Q \le B^{2/3-\gamma}$, extending the earlier $2/5$ exponent, but leaves a critical window open. For Uniform$[0, \theta]$ individuals with $\theta \in [1, 2]$, Theorem 4.3.6 closes the logarithmic gap completely:

$$U_B^{\mathrm{ad}} \asymp \frac{\log B}{\sqrt{B}}.$$

The lower bound holds for every adaptive policy, even when the latent log-parameter has a positive uniformly Lipschitz density.

Theorems 4.4.3 and 4.4.6 establish ID for the normalized densities

$$\begin{aligned} &Cx^{\nu-1}e^{-bx}I_\mu(ax), \quad \mu > -1, \quad \mu + \nu > 0, \quad b > a > 0, \\ &Cx^{2\mu}e^{-bx}I_\mu(ax)^2, \quad \mu > -\tfrac{1}{4}, \quad b > 2a > 0. \end{aligned}$$

The first gives the whole domain requested in [3, Section 2.3]. The second extends that paper's Theorem 4 to every admissible real order. Explicit Lévy kernels also give exact spectral SD tests. They do not give an elementary expression for every threshold, nor settle every strictness question.

4.1.2 Identification, zero geometry, and stability

The Gaussian threshold uses $\omega_a(v) = \sqrt{v}/\log^a(e + v)$. An exponential moment in $\omega_a(\operatorname{tr}\Sigma)$ with $a \le 1$ makes every infinite set of terminal lengths identifying. For $a > 1$, a scalar construction with dyadic terminal lengths defeats identification while preserving both marginals and every joint polynomial moment. The sharpness is scalar; additional anchors obstruct a direct counterexample in a fixed matrix dimension greater than one. A parallel Fourier–Laplace argument gives the exact sparse-time criterion for dependent random locations and Lévy clocks.

For finite rationally independent exponential sums, every period is classified, including its missing arcs. Only the largest frequency can create an exception. Removing the largest frequency in a normally convergent infinite sum removes all exceptions. An amplitude-dependent transition then determines all magnitudes and relative frequencies of an unknown finite spectrum. Its observation is an entire exact transition curve, and several phases remain invisible. It is not a reconstruction of primes from a single rightmost zero.

The analytic inverse results separate exact information from noisy information on one fixed class. Any n distinct positive heat parameters give error $\exp(-\Theta(n))$, uniformly over their locations. Yet no noise-convergence modulus works uniformly over all sampling sequences, even under that same analytic envelope. For every fixed sequence the risk still tends to zero. Thus the negative claim is not a positive asymptotic error floor on a compact analytic class.

4.1.3 How to read the claims

Section 4.10 records eighteen targets: twelve completed statements at their specified scope, three partial advances, and three unresolved transfers. These are research targets, not eighteen named conjectures. Among the completed statements, the third-McKay domain problem is a directly located published open question, and the MLE result reaches a published conditional target. Other rows are full extensions or internally formulated inverse and minimax problems. Consequences of the same theorem are not counted as additional famous problem resolutions.

Primary-source comparisons were checked through 14 October 2026, including the May 2026 version of [14], the published PJM version of [3], and the recent distinct smooth-density problem in [16]. This is a bounded literature comparison, not a certification of priority. The proofs have undergone cross-checks within this AI-assisted drafting process. External expert review, and therefore any claim of final acceptance or exceptional historical importance, remain outstanding.

4.2 Near-linear optimality of the unrestricted binomial-mixture MLE

The earlier manuscript established optimality through the replication range $t \le N^{2/3-\varepsilon}$ by controlling the entire predictive distribution in Hellinger distance. Here we change the statistical geometry: the relevant fingerprint error is transportation distance on the normalized count grid, not total variation. Likelihood self-consistency transfers a transportation inequality for each binomial kernel to the fitted-versus-empirical discrepancy. Together with derivative-stable polynomial inversion, this gives the near-linear range discussed in [14, Section 6.2].

We use natural logarithms. For a probability measure G on $[0,1]$, write

$$B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}, \qquad p_G(j) = \int B_j^t(x)\, dG(x), \quad 0 \le j \le t.$$

The observations $X_1, \dots, X_N$ are independent with common law p_G. Their empirical probability vector is $q_j = N^{-1}\#\{i : X_i = j\}$. Let $\widehat{G}$ be any maximizer of $\sum_j q_j \log p_H(j)$ over all probability measures H on $[0,1]$, and write $\widehat{p} = p_{\widehat{G}}$. Such maximizers exist by compactness of the binomial moment body. Every maximizer assigns positive probability to each observed count. Write $W_{1,t}$ for Wasserstein distance on $\{0, \dots, t\}$ with cost $|j-l|/t$, and D for relative entropy, with the usual extended-value conventions.

Theorem 4.2.1 (Near-linear MLE optimality and a uniform interpolation bound)**.** *There are absolute constants C, A such that for all integers $N, t \ge 1$, all mixing laws G, and every unrestricted mixing-measure MLE,*

$$\mathbb{E}_G W_1(\widehat{G}, G) \le C \left\{ \frac{1}{t} + \frac{1}{\sqrt{t \log(1 + N/t)}} \right\}. \tag{4.2.1}$$

More specifically, for $2 \le k \le t/2$,

$$\mathbb{E}_G W_1(\widehat{G}, G) \le \frac{C}{k} + \frac{C \exp(Ak^2/t)}{\sqrt{N}}. \tag{4.2.2}$$

For every fixed $\varepsilon \in (0,1)$ and $a > 0$, there are constants $C_{\varepsilon,a}, N_{\varepsilon,a}$ such that, whenever

$$N \geq N_{\varepsilon,a}, \qquad a \log N \leq t \leq N^{1-\varepsilon},$$

$$\mathbb{P}_G \left\{ W_1(\widehat{G}, G) > \frac{C_{\varepsilon,a}}{\sqrt{t \log N}} \right\} \leq N^{-2}, \qquad \mathbb{E}_G W_1(\widehat{G}, G) \leq \frac{C_{\varepsilon,a}}{\sqrt{t \log N}}. \tag{4.2.3}$$

The probability statement holds simultaneously for all MLE solutions.

The theorem makes no smoothness, separation, finite-support, or interiority assumption on G. Its estimator is the original unrestricted MLE. The bound in (4.2.3) has the matching minimax order from [13, Theorem 3.3]. We do not claim a matching lower bound for the entire interpolation formula (4.2.1); in particular, its transition when t is close to N may be suboptimal. The exact constant-one Chebyshev–Bernstein inequality appearing in [14, Section 6.2] is not needed or proved here. The recent smooth-density analysis of Lee and Kennedy [16] concerns pointwise density loss under Hölder regularity and a regularized series estimator. The unrestricted-law, Wasserstein-loss MLE question addressed here is different.

4.2.1 An entropy–transport principle for mixture likelihood

The next statement isolates the structural step. The likelihood directional condition is classical mixture geometry [62]; the exponential-moment/transport duality used below is classical [63]. We include their elementary finite-space arguments. The transfer does not assume that the mixture itself has the same concentration as its individual kernels.

Lemma 4.2.2 (Likelihood transfers kernel transportation inequalities)**.** *Let $(\mathcal{X}, d)$ be a finite metric space and K_θ a family of probability vectors. Suppose that, for every θ and every probability vector r,*

$$W_{1,d}(r, K_\theta)^2 \leq 2cD(r\|K_\theta). \tag{4.2.4}$$

For a probability vector q, let $\widehat{H}$ maximize $\sum_x q(x) \log p_H(x)$ over mixtures $p_H = \int K_\theta \, dH(\theta)$, and suppose this maximum is finite. Then, with $\widehat{p} = p_{\widehat{H}}$,

$$W_{1,d}(q, \widehat{p})^2 \leq 2cD(q\|\widehat{p}). \tag{4.2.5}$$

Consequently $W_{1,d}(q, \widehat{p})^2 \leq 2cD(q\|p_H)$ for every comparison mixture p_H.

Proof. The first directional derivative of likelihood in the direction $\delta_\theta - \widehat{H}$ gives

$$S(\theta) := \sum_{x:q(x)>0} \frac{q(x)K_\theta(x)}{\widehat{p}(x)} \leq 1.$$

All denominators on the displayed support are positive. Since $\int S \, d\widehat{H} = \sum_x q(x) = 1$, one has $S = 1$ almost surely under $\widehat{H}$. Thus

$$R_\theta(x) = \frac{q(x)K_\theta(x)}{\widehat{p}(x)} \quad (q(x) > 0), \qquad R_\theta(x) = 0 \quad (q(x) = 0)$$

is a probability vector for almost every θ, and $\int R_\theta \, d\widehat{H}(\theta) = q$. The entropy chain rule reduces here to the finite-sum identity

$$\begin{aligned} \int D(R_\theta\|K_\theta) \, d\widehat{H}(\theta) &= \sum_x \int R_\theta(x) \log \frac{q(x)}{\widehat{p}(x)} \, d\widehat{H}(\theta) \\ &= D(q\|\widehat{p}). \end{aligned}$$

Terms with $K_\theta(x) = R_\theta(x) = 0$ are zero. By the dual formula for Wasserstein distance, followed by (4.2.4) and Cauchy–Schwarz,

$$W_{1,d}(q, \widehat{p}) \le \int W_{1,d}(R_\theta, K_\theta)\, d\widehat{H}(\theta) \le \sqrt{2c \int D(R_\theta \| K_\theta)\, d\widehat{H}(\theta)}.$$

Finally, maximizing likelihood is equivalent to minimizing $D(q\|p_H)$, because $\sum_x q(x) \log q(x)$ is independent of H. □

Lemma 4.2.3 (Binomial transportation)**.** *For every probability vector r on $\{0, \dots, t\}$ and $\theta \in [0, 1]$,*

$$W_{1,t}(r, \mathrm{Binomial}(t, \theta))^2 \le \frac{1}{2t} D(r \| \mathrm{Binomial}(t, \theta)). \tag{4.2.6}$$

Hence every binomial-mixture MLE obeys the deterministic inequality

$$W_{1,t}(q, \widehat{p}) \le \sqrt{\frac{D(q\|\widehat{p})}{2t}} \le \sqrt{\frac{D(q\|p_G)}{2t}}. \tag{4.2.7}$$

Proof. Let $Z_1, \dots, Z_t$ be independent Bernoulli variables of parameter θ. For any function u that is 1-Lipschitz on the normalized count grid, changing one Z_i changes $u(\sum_i Z_i)$ by at most $1/t$. The Doob martingale which reveals these variables one at a time has conditional increment range at most $1/t$. Hoeffding's elementary bounded-variable inequality, applied conditionally and iterated, gives

$$\log \mathbb{E} \exp\{\lambda(u - \mathbb{E}u)\} \le \lambda^2/(8t), \qquad \lambda \in \mathbb{R}.$$

For clarity, the bounded-variable inequality says that a centered variable with range length b has logarithmic moment generating function at most $\lambda^2 b^2/8$; it follows by convexity from the two-point distribution on the range endpoints and two integrations of its variance bound $b^2/4$. For r absolutely continuous with respect to the binomial law K, nonnegativity of relative entropy with respect to the tilted law $e^{\lambda u} K / \mathbb{E}_K e^{\lambda u}$ gives

$$\mathbb{E}_r u - \mathbb{E}_K u \le D(r\|K)/\lambda + \lambda/(8t), \qquad \lambda > 0.$$

Minimization over λ and the Wasserstein dual formula prove (4.2.6). If entropy is infinite the assertion is automatic; the degenerate kernels at $\theta = 0, 1$ cause no exception. Lemma 4.2.2, with $c = 1/(4t)$, proves (4.2.7). □

Lemma 4.2.4 (Dimension-free predictive transportation error)**.** *Uniformly in G, N, t,*

$$\mathbb{E}_G W_{1,t}(q, \widehat{p}) \le \frac{1}{\sqrt{2N}}, \qquad \mathbb{E}_G W_{1,t}(q, p_G) \le \frac{1}{2\sqrt{N}}. \tag{4.2.8}$$

Moreover, for $0 < \eta < 1/2$, except on an event of probability η, simultaneously for all MLEs,

$$W_{1,t}(q, \widehat{p}) + W_{1,t}(q, p_G) \le C \sqrt{\frac{\log(N+1) + \log(2/\eta)}{N}}. \tag{4.2.9}$$

Proof. If p_G has support size $s \le t + 1$, then $\log x \le x - 1$ implies

$$D(q\|p_G) \le \sum_{j : p_G(j) > 0} \frac{(q_j - p_G(j))^2}{p_G(j)}, \qquad \mathbb{E}_G D(q\|p_G) \le \frac{s-1}{N} \le \frac{t}{N}.$$

Apply (4.2.7) and Jensen's inequality. For the second expectation bound, identify the counts with $j/t \in [0,1]$. Their empirical and population distribution functions satisfy

$$\mathbb{E}|F_N(x) - F(x)| \le \sqrt{F(x)(1-F(x))/N} \le 1/(2\sqrt{N}).$$

Integrating the one-dimensional formula $W_1(q,p_G) = \int_0^1 |F_N - F|$ gives the result.

For the probability assertion, a multinomial type v has probability at most $\exp\{-ND(v\|p_G)\}$. Indeed, its multinomial coefficient is at most $\exp\{NH(v)\}$, as follows by taking its term in the multinomial expansion of $(\sum_j v_j)^N = 1$. There are at most $(N+1)^s$ types. Consequently, with probability at least $1-\eta/2$,

$$D(q\|p_G) \le \frac{(t+1)\log(N+1) + \log(2/\eta)}{N}.$$

In addition, changing one observation changes $W_{1,t}(q,p_G)$ by at most $1/N$. The same conditional bounded-variable argument used above, followed by Chernoff's inequality, bounds it by its expectation plus $\sqrt{\log(2/\eta)/(2N)}$ except on an event of probability $\eta/2$. Combine these estimates with (4.2.7) and $(t+1)/t \le 2$. □

4.2.2 Polynomial inversion that preserves a derivative bound

The following coefficient estimate applies to every bounded polynomial, not just to individual Chebyshev polynomials. Its contour mechanism is the one used in the preceding manuscript, with the elementary Bernstein–Walsh principle replacing a sum over Chebyshev modes.

Lemma 4.2.5 (Bernstein coefficients of an arbitrary bounded polynomial)**.** *There is an absolute constant C with the following property. If P is a real polynomial of degree at most $m \le n/2$ and*

$$P(x) = \sum_{j=0}^{n} b_j B_j^n(x),$$

then

$$\max_j |b_j| \le C e^{2m^2/n} \|P\|_{L^\infty[0,1]}. \tag{4.2.10}$$

Proof. Write $M = \|P\|_\infty$. For a complex number w with $|w| \ge 1$, the maximum principle applied to the polynomial $v^m P((2+v+v^{-1})/4)$ on $|v| \le 1$ gives

$$\left| P\left(\frac{2+w+w^{-1}}{4}\right)\right| \le M|w|^m.$$

For $0 < j < n$, set $r = j/(n-j)$, $q_* = r/(1+r)^2$, $z = re^{i\phi}$, and $u = 2\sqrt{q_*}|\sin(\phi/2)|$. The two inverse Joukowski choices corresponding to $z/(1+z)$ have reciprocal moduli, with the larger equal to

$$\frac{\max\{|\sqrt{z}+i|^2, |\sqrt{z}-i|^2\}}{|1+z|} = \frac{1+u}{\sqrt{1-u^2}}.$$

Therefore the polynomial $F(z) = (1+z)^n P(z/(1+z))$ satisfies

$$\begin{aligned}\frac{|F(re^{i\phi})|}{M(1+r)^n} &\le (1-u^2)^{(n-m)/2}(1+u)^m \\ &\le e^{-nu^2/4+mu} \le e^{2m^2/n} e^{-nu^2/8}.\end{aligned}$$

Values where a displayed rational expression is undefined follow by continuity of F. Since $\sin(|\phi|/2) \ge |\phi|/\pi$ for $|\phi| \le \pi$, the average of the last Gaussian factor over $[-\pi,\pi]$ is at most $C/\sqrt{1+nq_*}$. Cauchy's coefficient formula and $F(z)=\sum_j \binom{n}{j} b_j z^j$ now yield

$$|b_j| \le M e^{2m^2/n} \frac{(1+r)^n}{\binom{n}{j} r^j} \frac{C}{\sqrt{1+nq_*}}.$$

Stirling's upper and lower factorial bounds give

$$\binom{n}{j}(j/n)^j(1-j/n)^{n-j} \ge \frac{c}{\sqrt{1+nq_*}},$$

which cancels the displayed square-root factor. At the two endpoints, $b_0 = P(0)$ and $b_n = P(1)$; the constant case is also immediate. □

Lemma 4.2.6 (A derivative-stable algebraic Jackson approximation)**.** *For every 1-Lipschitz function $f:[0,1]\to\mathbb{R}$ and integer $k \ge 2$, there is a real polynomial P_k of degree at most k such that*

$$\|P_k - f\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le 2. \tag{4.2.11}$$

Proof. Subtract $f(0)$, so that $|f| \le 1$. Reflect f from $[0,1]$ to $[1,2]$ by $F(x) = f(2-x)$ and extend with period two. This is a globally 1-Lipschitz function with $\|F\|_\infty \le 1$. Convolve it with the nonnegative normalized, period-two Jackson kernel

$$J_s(u) = c_s \left(\frac{\sin(s\pi u/2)}{\sin(\pi u/2)}\right)^4, \qquad -1 \le u \le 1.$$

The quotient is interpreted continuously at zero. The resulting real trigonometric polynomial Q_s has frequencies $|l| \le 2(s-1)$, satisfies $\|Q_s'\|_\infty \le 1$ by positivity and translation invariance of convolution, and has error $\|Q_s - F\|_\infty \le C/s$. Here the kernel has first absolute moment $O(1/s)$: its unnormalized integral is comparable to s^3, while its unnormalized first absolute moment is at most Cs^2, using $|\sin(s\pi u/2)/\sin(\pi u/2)| \le C\min(s, |u|^{-1})$.

Write $Q_s(x) = \sum_{|l|\le 2s} a_l e^{i\pi l(x-1/2)}$. Each $|a_l| \le 1$, because it is a Fourier coefficient of F multiplied by a Fourier coefficient of a probability kernel. Replace each exponential by its Taylor polynomial at $x = 1/2$ of degree $M = 100s$, and call the resulting algebraic polynomial P. Conjugate symmetry makes P real. On $[0,1]$, $|\pi l(x-1/2)| \le \pi s$, so Taylor's remainder gives

$$\|P - Q_s\|_\infty \le (4s+1) e^{\pi s} \frac{(\pi s)^{M+1}}{(M+1)!},$$
$$\|P' - Q_s'\|_\infty \le (4s+1) 2\pi s e^{\pi s} \frac{(\pi s)^M}{M!}.$$

The bound $M! \ge (M/e)^M$ shows that both expressions are at most e^{-100s} for $s \ge 1$. Thus $\|P'\|_\infty \le 2$ and $\|P - f\|_\infty \le C/s$. For $k \ge 100$, take $s = \lfloor k/100 \rfloor$, which has $100s \le k$ and $s \ge k/200$. For $2 \le k < 100$, use the constant polynomial $f(0)$ and enlarge the absolute approximation constant. Finally add back $f(0)$. □

Corollary 4.2.7 (Lipschitz Bernstein inversion)**.** *For a 1-Lipschitz f and $2 \le k \le t/2$, its approximating polynomial can be written $P_k = \sum_{j=0}^{t} b_j B_j^t$ with*

$$\max_{0\le j<t} t|b_{j+1} - b_j| \le C e^{4k^2/t}. \tag{4.2.12}$$

Consequently, for any mixing laws G, H,

$$W_1(G,H) \le C/k + C e^{4k^2/t} W_{1,t}(p_G, p_H). \tag{4.2.13}$$

Proof. Differentiate the Bernstein expansion:

$$P_k'(x) = t\sum_{j=0}^{t-1}(b_{j+1}-b_j)B_j^{t-1}(x).$$

Its degree is at most $k-1 \le (t-1)/2$ and its supremum norm is at most two. Lemma 4.2.5 applies, with $2(k-1)^2/(t-1) \le 4k^2/t$, proving (4.2.12). The coefficient vector $j \mapsto b_j$ is therefore Lipschitz on the normalized count grid with the displayed constant. The approximation error contributes at most $2\|P_k - f\|_\infty$, and the remaining integral is $\sum_j b_j(p_G(j) - p_H(j))$. Apply Wasserstein duality on both spaces. □

4.2.3 Proof of the rate statements

Proof of Theorem 4.2.1. Insert $H = \widehat{G}$ into (4.2.13) and use

$$W_{1,t}(p_G, \widehat{p}) \le W_{1,t}(p_G, q) + W_{1,t}(q, \widehat{p}).$$

Lemma 4.2.4 proves (4.2.2) with $A = 4$. It also proves, with probability at least $1-\eta$,

$$W_1(\widehat{G}, G) \le \frac{C}{k} + Ce^{4k^2/t}\sqrt{\frac{\log(N+1) + \log(2/\eta)}{N}}, \tag{4.2.14}$$

simultaneously for all MLEs.

For the near-linear range, set $\eta = N^{-2}$ and $k = \lfloor \sqrt{c\varepsilon t \log N} \rfloor$, where $c > 0$ is sufficiently small, depending only on the stated lower-limit constant a when necessary. For large N, this is in $[2, t/2]$. The second term of (4.2.14), multiplied by $\sqrt{t \log N}$, is at most

$$CN^{4c\varepsilon - 1/2}\sqrt{t}\log N \le CN^{-(1/2-4c)\varepsilon}\log N.$$

Choose $c < 1/8$. This expression is bounded for large N, proving the probability claim. The expectation claim follows directly from (4.2.2), or from integration since the loss is at most one and the exceptional probability is N^{-2}.

For completeness, the uniform interpolation bound can be obtained without asymptotic restrictions. Put $x = N/t$ and $L = \log(1+x)$, and fix $0 < c < 1/32$. For all parameter choices for which $z = \min\{t/2, \sqrt{ctL}\} \ge 4$, choose $k = \lfloor z \rfloor$. Then $k \ge z/2$, so

$$1/k \le C\{1/t + 1/\sqrt{tL}\}.$$

If $z = \sqrt{ctL}$, the exponential term in (4.2.2) satisfies

$$\frac{e^{4k^2/t}}{\sqrt{N}}\sqrt{tL} \le (1+x)^{4c}\sqrt{\frac{\log(1+x)}{x}} \le C.$$

The last expression is bounded on $(0, \infty)$ because $4c < 1/2$. If $z = t/2$, then $L \ge t/(4c)$ and

$$\frac{e^{4k^2/t}}{\sqrt{N}} \le \frac{e^t}{\sqrt{t(e^{t/(4c)} - 1)}} \le C/t.$$

If $z < 4$, at least one of $1/t$ and $1/\sqrt{tL}$ is bounded below by an absolute positive constant, and the trivial loss bound one suffices. This proves (4.2.1) for every $N, t \ge 1$. □

4.2.4 Why a uniform bounded-fingerprint shortcut is false

Section 6.2 of [14] discusses a dimension-free bound for the fingerprint term tested against Bernstein coefficients. Our result proves the needed uniform control in a Lipschitz test norm after constructing a suitable approximation. It must not be interpreted as a uniform $N^{-1/2}$ total-variation bound over all bounded coefficient vectors. The latter strengthening is false, even if the fitted mixture is chosen by an oracle rather than maximum likelihood.

Proposition 4.2.8 (An obstruction for all bounded fingerprint tests)**.** *Let the true mixing law be uniform on* $[0,1]$*, so that* $p_G(j) = 1/(t+1)$*. There are absolute constants* $c, C > 0$ *such that, whenever* $N \geq t+1$ *and* $t \geq 8$*,*

$$\mathbb{E}_G \inf_H \|q - p_H\|_1 \geq c\sqrt{t/N} - C/\sqrt{t}. \tag{4.2.15}$$

In particular, along $t = \lfloor N^{3/4} \rfloor$ *the left side is at least* $cN^{-1/8}$ *for all sufficiently large* N*. Consequently there is no uniform* $O(N^{-1/2})$ *expectation bound for* $\sup_{\|b\|_\infty \leq 1} |\sum_j b_j(q_j - p_H(j))|$*, for any fitting rule whose output is a binomial mixture.*

Proof. Set $L = \lceil t/4 \rceil$, $U = \lfloor 3t/4 \rfloor$, and $V(v) = \sum_{j=L}^{U-1} |v_{j+1} - v_j|$. Each binomial probability vector is unimodal. Its variation on this central interval is at most twice its largest value there, which is at most $C/\sqrt{t}$. Indeed, for each central j, $B_j^t(\theta)$ is largest at $\theta = j/t$, and Stirling's factorial bounds give $B_j^t(j/t) \leq C/\sqrt{t}$. Integrating over H therefore gives

$$V(p_H) \leq C/\sqrt{t} \tag{4.2.16}$$

for every mixture.

For adjacent bins under the uniform count law, let $Z = N(q_{j+1} - q_j) = \sum_{i=1}^N Y_i$, where the independent variables Y_i take values $1, -1$ each with probability $1/(t+1)$ and zero otherwise. They are centered, and $v = \mathbb{E}Z^2 = 2N/(t+1) \geq 2$. Their fourth moment satisfies

$$\mathbb{E}Z^4 = v + 3N(N-1)\left(\frac{2}{t+1}\right)^2 \leq v + 3v^2 \leq \frac{7}{2}v^2.$$

For a nonnegative random variable Y, Cauchy–Schwarz gives $\mathbb{P}\{Y \geq \frac{1}{2}\mathbb{E}Y\} \geq (\mathbb{E}Y)^2/(4\mathbb{E}Y^2)$: split its first moment over that event and its complement. Apply this with $Y = Z^2$ to obtain $\mathbb{E}|Z| \geq c\sqrt{v}$. There are at least ct adjacent central pairs, and hence $\mathbb{E}V(q) \geq c\sqrt{t/N}$. Finally, the triangle inequality yields

$$V(q) \leq V(p_H) + 2\|q - p_H\|_1.$$

Use (4.2.16), take the infimum and expectation, and absorb absolute constants. The final assertion follows from the dual identity between the ℓ_1 norm and bounded coefficient vectors. □

This proposition concerns a supremum over all bounded tests, including tests selected after seeing the data. It does not disprove a conjecture with one fixed deterministic coefficient vector, or a conjecture restricted to a carefully chosen low-degree approximation class. The distinction is essential: the transportation norm used in our proof exploits cancellation between nearby count bins, whereas unrestricted bounded tests can select the signs of the local sampling fluctuations.

4.3 Exact replication caps and the critical endpoint experiment

This section advances two total-budget questions that were left open in our preceding observation papers. For Bernoulli populations we determine the complete cap-dependent minimax curve for

predetermined, possibly unequal, replication counts, and extend the exact adaptive curve to every polynomially subcritical cap. For scalar uniform-endpoint populations we prove that the logarithm at the critical component-estimation exponent is necessary even under fully adaptive sampling. These are classification problems for the experiments specified below; we do not describe them as previously named external conjectures.

Let an individual draw an independent latent parameter from an unknown probability law G. A raw observation costs one unit, and repeated observations of that individual are conditionally independent. A policy may start individuals, revisit them, and randomize using all previous observations. Its total cost is at most a deterministic integer B. A cap Q means that no individual is observed more than Q times. The superscripts ad and fix respectively allow all such policies and restrict the replication counts to a possibly randomized schedule chosen independently of the data. Estimators are arbitrary probability-valued measurable procedures; the loss is expected W_1.

4.3.1 The Bernoulli cap curve

For $P_\theta = \mathrm{Bernoulli}(\theta)$, $0 \le \theta \le 1$, write $R^{\mathrm{ad}}_{B,Q}$ and $R^{\mathrm{fix}}_{B,Q}$ for these risks. Set

$$\Psi(B,Q) = \max\left\{ B^{-1/3}, \frac{1}{Q}, \frac{1}{\sqrt{Q\log(e + B/Q^{3/2})}} \right\}. \tag{4.3.1}$$

Theorem 4.3.1 (The complete predetermined cap curve)**.** *There are absolute constants $0 < c < C < \infty$ such that, for all integers $B \ge 2$ and $1 \le Q \le B$,*

$$c\Psi(B,Q) \le R^{\mathrm{fix}}_{B,Q} \le C\Psi(B,Q).$$

The upper bound uses a predetermined collection of binomial observations at geometrically varying replication depths, followed by a weighted Chebyshev-moment projection. It does not assert a property of the MLE. In particular the optimal uncapped order $B^{-1/3}$ is attained within a fixed constant precisely when $Q \gtrsim B^{2/3}$, with constants allowed to depend on that fixed risk constant.

Theorem 4.3.2 (Adaptive caps below the critical power)**.** *For every fixed $0 < \gamma < 2/3$, uniformly for $1 \le Q \le B^{2/3-\gamma}$ as $B \to \infty$,*

$$R^{\mathrm{ad}}_{B,Q} \asymp_\gamma \max\left\{ \frac{1}{Q}, \frac{1}{\sqrt{Q\log(eB)}} \right\}.$$

For arbitrary $1 \le Q \le B$ the following valid sandwich remains:

$$c\max\left\{ B^{-1/3}, Q^{-1}, [Q\log(eB)]^{-1/2} \right\} \le R^{\mathrm{ad}}_{B,Q} \le C\Psi(B,Q).$$

In particular the fully adaptive cap curve in a shrinking neighborhood of $B^{2/3}$ is not claimed to be settled.

The preceding cap theorem only reached $Q \le B^{2/5-\gamma}$, by substituting an equal-group MLE bound. The new procedure estimates different frequency bands at different costs. The weighted moment inequality below is an established Chebyshev-moment principle, explicitly developed by Musco, Musco, Rosenblatt and Singh [14]. Trial-profile-dependent spectral estimation also appears in Lee and Kennedy [16]; their target is pointwise estimation of a Hölder mixing density, whereas ours is arbitrary-law W_1 recovery under an optimized observation budget and a cap. Neither statement is presented as a special case of the other.

Lemma 4.3.3 (Moment recovery and binomial statistics)**.** *Write* $\widetilde{T}_m(x) = T_m(2x-1)$ *and* $\mu_m(G) = \int \widetilde{T}_m \, dG$*. For* $K \geq 2$*,*

$$W_1(G,H) \leq \frac{C}{K} + C\left\{\sum_{m=1}^{K} \frac{|\mu_m(G) - \mu_m(H)|^2}{m^2}\right\}^{1/2}. \tag{4.3.2}$$

For $m \leq r/2$ *there is a statistic* $A_{r,m}$ *of* $X \sim \mathrm{Binomial}(r,\theta)$ *satisfying*

$$\mathbb{E}_\theta A_{r,m}(X) = \widetilde{T}_m(\theta), \qquad \max_{0\leq j\leq r} |A_{r,m}(j)| \leq C\exp(2m^2/r).$$

Proof. For a 1-Lipschitz f the periodic function $g(u) = f((1+\cos u)/2)$ has bounded weak derivative. Its cosine coefficients satisfy $\sum_{m\geq 1} m^2|c_m|^2 \leq C$ by Parseval. Convolution with the normalized nonnegative Jackson kernel $[\sin(su/2)/\sin(u/2)]^4$, $s = \lfloor K/2 \rfloor + 1$, has degree at most K, changes g uniformly by at most C/K, and multiplies its coefficients by factors of absolute value at most one. The first-moment estimate for this kernel follows by integrating $\min(s^4, |u|^{-4})$ and $|u|\min(s^4,|u|^{-4})$. Cauchy–Schwarz applied to the resulting polynomial proves (4.3.2) by Kantorovich duality.

Take $A_{r,m}(j)$ to be the degree-r Bernstein coefficients of $\widetilde{T}_m$. We include the coefficient estimate to make the cost calculation self-contained. For $0 < j < r$ put $z = te^{iu}$, $t = j/(r-j)$, $q = t/(1+t)^2$. The generating polynomial

$$(1+z)^r \widetilde{T}_m\!\left(\frac{z}{1+z}\right) = \frac{(1+z)^{r-m}}{2}\{(\sqrt{z}+i)^{2m} + (\sqrt{z}-i)^{2m}\}$$

has coefficient $\binom{r}{j}A_{r,m}(j)$ at z^j. With $v = 2\sqrt{q}\,|\sin(u/2)|$, its modulus divided by $(1+t)^r$ is at most

$$(1-v^2)^{(r-m)/2}(1+v)^m \leq e^{2m^2/r}e^{-rv^2/8}.$$

The Cauchy integral supplies a factor $C(1+rq)^{-1/2}$. The binomial probability $\binom{r}{j}t^j/(1+t)^r$ is at least $c/\sqrt{1+rq}$ by Stirling's inequalities. Its reciprocal cancels the preceding factor. The endpoint coefficients have absolute value one. The Bernstein identity proves unbiasedness. □

Lemma 4.3.4 (A spectral allocation bound)**.** *There are absolute constants such that, whenever* $K \geq 2$ *is a power of two,* $K \leq Q/2$*, and* $K^2 \leq cB$*, a predetermined capped procedure satisfies*

$$\sup_G \mathbb{E}_G W_1(\widehat{G}, G) \leq C\left\{K^{-1} + \sqrt{K/B}\, e^{2K^2/Q}\right\}. \tag{4.3.3}$$

Proof. Partition $1 \leq m \leq K$ into dyadic bands with upper endpoints $k = 1, 2, 4, \ldots, K$. At band k use $r_k = \min\{Q, \max(2, k^2)\}$ observations per fresh individual. Then $r_k \geq 2m$ throughout that band. Estimate every moment in it by the average of $A_{r_k,m}$ over its n_k individuals. The expected squared weighted moment error in the band is at most a_k/n_k, where

$$a_k = Ck^{-1}e^{4k^2/r_k}.$$

Let $S = \sum_k \sqrt{r_k a_k}$ and take $n_k = \lceil B\sqrt{a_k/r_k}/(2S) \rceil$. The total cost is at most $B/2 + \sum_k r_k \leq B$ after reducing the absolute constant in $K^2 \leq cB$. Moreover

$$\sum_k \frac{a_k}{n_k} \leq \frac{2S^2}{B}, \qquad S \leq C\sum_k \sqrt{k}\, e^{2+2K^2/Q} \leq C\sqrt{K}e^{2K^2/Q}.$$

Project the estimated moment vector onto the compact moment set of probability laws on $[0,1]$, in the norm $(\sum_{m=1}^K |z_m|^2/m^2)^{1/2}$. Its distance to the true moment vector is at most twice the estimation error. A measurable approximate projection over a countable dense set of finitely supported rational laws suffices and adds an arbitrarily small error. Apply (4.3.2) and Jensen's inequality. □

Upper bound in Theorem 4.3.1. Put $L = \log(e + B/Q^{3/2})$ and $H = \min\{B^{1/3}, Q, \sqrt{QL}\}$. Choose the largest power of two $K \le c_0 H$, where $c_0 > 0$ is a sufficiently small absolute constant. If there is no $K \ge 2$, the diameter bound gives the assertion by increasing C. Otherwise Lemma 4.3.4 applies, with finitely many small budgets absorbed into the constant. If $Q \ge B^{2/3}$, then $K^2/Q \le c_0^2$ and $K^3 \le c_0^3 B$. If $Q < B^{2/3}$, write $x = B/Q^{3/2} > 1$. Then

$$\frac{K^{3/2}}{\sqrt{B}} e^{2K^2/Q} \le c_0^{3/2} \frac{\log(e+x)^{3/4}}{\sqrt{x}} (e+x)^{2c_0^2} \le C,$$

provided, for example, $2c_0^2 < 1/4$. Thus both terms in (4.3.3) are at most C/H, and $H^{-1} = \Psi(B, Q)$. □

The projection can be implemented by a finite convex quadratic program. Indeed $|\widetilde{T}'_m(x)| \le 2m^2$ on $[0,1]$, so rounding a probability law to a grid of mesh K^{-3} changes its weighted moment vector by at most $CK^{-3}(\sum_{m \le K} m^2)^{1/2} \le C/K$. Minimizing the squared weighted discrepancy over the simplex on this grid, to additive objective error $O(K^{-2})$, therefore preserves the rate. The grid has $O(K^3)$ points; its entries and the Bernstein statistics are rational and have polynomial encoding length. Thus the construction admits a polynomial-time implementation in the usual rational-input model; no efficient evaluation of a continuum argmin is being assumed.

Lemma 4.3.5 (Localized moment matching with an information charge)**.** *There are absolute constants $c_0, C_0 > 0$ and $0 < \rho < 1$ with the following property. Let $K \ge 1$, $Q \ge K$, and $h^2 = c_0 K/Q$, with $h \le 1/32$. For any center $u \in [1/4, 3/4]$ there are laws ν_+, ν_- supported on $[u-h, u+h]$ such that*

$$W_1(\nu_+, \nu_-) = \frac{h}{K+1}, \qquad \mathrm{KL}(P_{r,+} \| P_{r,-}) \le C_0 r h^2 \rho^K \quad (1 \le r \le Q),$$

where $P_{r,\pm}$ is the full r-toss mixture law.

Proof. Split the signed measure $2^{-K} \sum_{j=0}^{K+1} (-1)^j \binom{K+1}{j} \delta_{j/(K+1)}$ into two probability laws, and transport them to $[u-h, u+h]$. They match moments through order K and have W_1 distance $2h/(K+1)$: integrate their cumulative signed mass, using $\sum_{i=0}^{j} (-1)^i \binom{K+1}{i} = (-1)^j \binom{K}{j}$. Mix each law with weight $1/2$ with the common atom δ_u. This gives the displayed separation and bounds each observation-law likelihood ratio below by $1/2$ relative to r independent Bernoulli(u) observations.

Set $v = u(1-u) \ge 3/16$ and $Z_i = (X_i - u)/\sqrt{v}$ under that reference law. Its product likelihood for bias $u + a$ is $\prod_i (1 + aZ_i/\sqrt{v})$. Orthogonality of the products $\prod_{i \in S} Z_i$ and moment matching imply

$$\mathrm{KL}(P_{r,+} \| P_{r,-}) \le \chi^2(P_{r,+}, P_{r,-}) \le C \sum_{j > K} \binom{r}{j} (h^2/v)^j \le C r h^2 \left(\frac{CQh^2}{K} \right)^K.$$

The last inequality follows from $\binom{r}{j} \le (er/j)^j$; when $r \le K$ the sum is zero. Choose c_0 small enough. □

Lower bound in Theorem 4.3.1. Moment-matching laws through degree Q have identical full Q-toss laws and separation $1/(Q+1)$, proving the Q^{-1} term even for adaptive policies.

We recall a short fully adaptive argument for the $B^{-1/3}$ term. Place $M \asymp h^{-1}$ separated cells in $[1/4, 3/4]$, each with two possible point biases separated by h, and give every cell mass $1/M$. Strengthen the experiment by allowing the statistician to select a known cell for each toss. This oracle simulates an original policy by drawing a uniform cell label whenever an individual starts. Two parameter vectors differing at cell j have transcript relative entropy at most $Ch^2 \mathbb{E} N_j$,

where N_j is its number of oracle queries. Sum this inequality over hypercube edges and use $\sum_j N_j \le B$; Pinsker and Cauchy–Schwarz bound the mean edge variation by $C\sqrt{Bh^2/M}$. For $h = cB^{-1/3}$ this is below $1/2$. The separated-cell W_1 identity and nearest-hypercube decoding then give risk at least ch. This proof allows arbitrary adaptation.

For the remaining term put $L = \log(e + B/Q^{3/2})$ and choose $K = \lceil AL \rceil$ with a sufficiently large fixed A. If $Q < C_1 L$, the desired term is already at most $\sqrt{C_1}/Q$. Otherwise take $h^2 = c_0 K/Q$ as in Lemma 4.3.5, increasing C_1 so $h \le 1/32$, and pack $M \asymp h^{-1}$ disjoint intervals of radius h inside $[1/4, 3/4]$. At each interval choose either law from that lemma and assign it mass $1/M$. Reveal the interval label of every individual for free. For any predetermined group sizes $r_i \le Q$ with $\sum_i r_i \le B$, neighboring hypercube laws in this stronger experiment have relative entropy at most

$$\frac{1}{M}\sum_i C_0 r_i h^2 \rho^K \le CBh^3\rho^K \le C\frac{B}{Q^{3/2}}K^{3/2}\rho^K \le \frac{1}{8}.$$

The last inequality holds by increasing A, since $K \asymp L$. It is uniform over randomized schedules after including their independent seed in the transcript. For two hypercube vertices the cumulative mass difference returns to zero at each cell boundary, and hence their W_1 distance is exactly $h/[M(K+1)]$ times their Hamming distance. Pinsker, binary testing, and nearest-hypercube decoding give minimax risk at least $ch/(K+1) \asymp [QL]^{-1/2}$. This completes the proof. □

Theorem 4.3.2. Besides the preceding adaptive lower bounds, take $K = \lceil 4\log(eB) \rceil$ and $h^2 = K/(64eQ)$ when $Q \ge K$. Two moment-matching laws through degree K supported on $[1/2 - h, 1/2 + h]$ have W_1 separation $2h/(K+1)$. The Walsh expansion used above bounds the squared L^2 difference of their full Q-toss likelihood ratios, relative to the fair coin, by $C16^{-K}$. Their total variation is at most $C4^{-K}$. An oracle of B independent full Q-toss vectors simulates every adaptive capped policy by revealing requested prefixes. Its total variation is at most $CB4^{-K} < 1/4$, proving the $[Q\log(eB)]^{-1/2}$ lower bound. If $Q < K$, the moment floor already gives it up to a constant. For $Q \le B^{2/3-\gamma}$,

$$\log(e + B/Q^{3/2}) \asymp_\gamma \log(eB),$$

and $B^{-1/3}$ is bounded by the other rate terms for all sufficiently large B. The predetermined upper bound therefore matches. □

The narrow remaining adaptive cap region is consequential. At $Q = B^{2/3}/\log B$, predetermined designs have risk of order $B^{-1/3}\sqrt{\log B/\log\log B}$, whereas the present adaptive lower bound is only $cB^{-1/3}$. We neither claim equality of these risks nor infer an adaptive advantage from the gap.

4.3.2 The scalar uniform-endpoint critical rate

Now let $P_\theta = \mathrm{Uniform}[0, \theta]$ with $1 \le \theta \le 2$, without a replication cap, and write U_B for the fully adaptive minimax risk. The previously known multilevel upper bound and one-scale lower bound left a logarithmic gap.

Theorem 4.3.6 (The critical logarithm is unavoidable)**.** *For absolute constants $0 < c < C < \infty$ and all sufficiently large B,*

$$c\frac{\log B}{\sqrt{B}} \le U_B \le C\frac{\log B}{\sqrt{B}}.$$

The lower bound allows arbitrary adaptive randomized allocation of the B observations. It already holds on a class for which the density of $\log\theta/\log 2$ is bounded above and below by

positive constants and has a common Lipschitz constant. A predetermined multilevel design attains the upper bound.

The main new step is an information bound for the entire adaptive experiment, not merely an integrated-variance calculation for one multilevel architecture. Bayesian information inequalities and multilevel distribution recovery are established methods [64, 9]. We supply the required absolute-loss inequality and the adaptive information calculation in full.

Lemma 4.3.7 (An absolute-loss Bayesian information inequality)**.** *Let (β, Y) have a joint density $p(\beta, y)$ on $\mathbb{R}^D \times \mathcal{Y}$ with integrable weak derivatives in β and positive definite finite Fisher matrix*

$$J = \int (\nabla_\beta \log p)(\nabla_\beta \log p)^T p.$$

For every $v \in \mathbb{R}^D$ and estimator $T(Y)$,

$$\mathbb{E}|T(Y) - v^T\beta| \geq \frac{1}{4}\sqrt{v^T J^{-1} v}.$$

Proof. For $p_h(\beta, y) = p(\beta + h, y)$, integration along translations and Cauchy–Schwarz give

$$\mathrm{TV}(p, p_h) \leq \tfrac{1}{2}\sqrt{h^T J h}.$$

The triangle inequality, integrated against $\min(p, p_h)$, gives

$$2\mathbb{E}|T(Y) - v^T\beta| \geq |v^T h|\{1 - \mathrm{TV}(p, p_h)\}.$$

Take $h = J^{-1}v/\sqrt{v^T J^{-1} v}$; the case $v = 0$ is immediate. □

Lemma 4.3.8 (A Fourier trace inequality)**.** *For $k = 1, \ldots, K$ let*

$$v_{k,c}(x) = \frac{\sqrt{2}\sin(2\pi k x)}{2\pi k}, \qquad v_{k,s}(x) = -\frac{\sqrt{2}\cos(2\pi k x)}{2\pi k},$$

and concatenate them into $v(x) \in \mathbb{R}^{2K}$. For every positive definite J,

$$\int_0^1 \sqrt{v(x)^T J^{-1} v(x)}\, dx \geq \frac{c}{\sqrt{\operatorname{tr} J}} \sum_{k=1}^{K} \frac{1}{k}.$$

Proof. The map $J \mapsto \sqrt{v^T J^{-1} v}$ is convex, since it is the supremum over $z \neq 0$ of $|v^T z|/(z^T J z)^{1/2}$, each of which is convex in J. The integral is invariant under conjugation by the block rotations corresponding to $x \mapsto x + t$. Average these conjugates over $0 \leq t \leq 1$ and apply Jensen. The average matrix has blocks $j_k I_2$, where $2\sum_k j_k = \operatorname{tr} J$. Its integrand is the constant $(\sum_k 1/(2\pi^2 k^2 j_k))^{1/2}$. Cauchy–Schwarz gives $\sum_k 1/(k^2 j_k) \geq (\sum_k 1/k)^2 / \sum_k j_k$. □

Lemma 4.3.9 (A constant information charge for an adaptive endpoint query)**.** *Let $a = \log 2$, $U = \log\theta/a \in [0, 1]$, and let its density be*

$$w_\beta(u) = 1 + \sum_{k=1}^{K} \{\beta_{k,c}\sqrt{2}\cos(2\pi k u) + \beta_{k,s}\sqrt{2}\sin(2\pi k u)\}.$$

Suppose $1/2 \leq w_\beta \leq 3/2$ and $\mathrm{Lip}(w_\beta) \leq L_0$ throughout the parameter set. For every adaptive policy of at most B raw observations, the trace of its Fisher information matrix in β is at most $C(1 + L_0^2)B$, uniformly in K and β.

Proof. After r observations of an individual, its posterior under the baseline density $w_0 = 1$ is a truncated exponential

$$q(u) = \frac{\lambda e^{-\lambda(u-b)}}{1 - e^{-\lambda d}} \mathbf{1}_{[b,1]}(u), \qquad \lambda = ar, \quad d = 1 - b,$$

where b is the maximum of zero and the scaled logarithms of that individual's observations. The uniform case is the limit $\lambda = 0$. This remains its baseline posterior after conditioning on the entire adaptive history: the action probabilities contribute no parameter factor once the observed history is fixed. A fresh individual has $q = \mathbf{1}_{[0,1]}$.

A new scaled log-observation is $Y = U - E$ with E exponential of rate a. Let q_Y be the updated baseline posterior. Direct integration shows the exact conditional identity

$$\mathbb{E}_0 \|q_Y - q\|_2^2 = \frac{a}{2}. \tag{4.3.4}$$

Here is the calculation, after translating b to zero. Put $Z = 1 - e^{-\lambda d}$. The current squared norm is $\lambda(1 + e^{-\lambda d})/(2Z)$. The event $Y \le 0$ has probability $\lambda(1 - e^{-(\lambda+a)d})/[(\lambda + a)Z]$, and its posterior norm is $(\lambda + a)\coth((\lambda + a)d/2)/2$. For $0 < y < d$ the predictive density is

$$\frac{a\lambda}{(\lambda + a)Z} e^{-\lambda y}(1 - e^{-(\lambda+a)(d-y)}).$$

Multiply it by $(\lambda + a)\coth((\lambda + a)(d - y)/2)/2$ and integrate. The sum with the $Y \le 0$ contribution is $a/2 + \|q\|_2^2$. The martingale identity $\mathbb{E}_0 q_Y = q$ proves (4.3.4). The limit $\lambda = 0$ follows by continuity.

Uniformly over these posteriors,

$$\|q\|_2^2 \operatorname{Var}_q(U) \le C. \tag{4.3.5}$$

If $\lambda d \le 1$, the density is at most C/d and the variance at most d^2. If $\lambda d > 1$, the squared norm is at most $C\lambda$ and the variance at most C/λ^2; since $\lambda > 1/d \ge 1$, their product is again bounded.

Set $z = \int w_\beta q$ and $z_Y = \int w_\beta q_Y$; both lie in $[1/2, 3/2]$. The predictive density under β, relative to its baseline counterpart, is z_Y/z. The score increment for this observation is the vector of Fourier coefficients of $q_Y/z_Y - q/z$, in the $2K$ functions occurring in w_β. Bessel's inequality therefore bounds its squared length by

$$\|q_Y/z_Y - q/z\|_2^2 \le C\|q_Y - q\|_2^2 + C\|q\|_2^2 |z_Y - z|^2.$$

Kantorovich duality and a product coupling give

$$\mathbb{E}_0 |z_Y - z|^2 \le L_0^2 \mathbb{E}_0 W_1(q_Y, q)^2 \le 2L_0^2 \operatorname{Var}_q(U).$$

For the last step, couple conditionally independent draws from q_Y and q, and then average over Y; the first draw has marginal q by the posterior martingale property. Equations (4.3.4)–(4.3.5), and the bounded predictive likelihood ratio, prove the conditional trace bound $C(1 + L_0^2)$ under the actual parameter β. Score increments are martingale differences. Conditional independence of action randomization from the parameter, and the deterministic budget bound, allow their squared lengths to be summed through all B adaptive queries. This proves the assertion. □

Lower bound in Theorem 4.3.6. Use the $2K$-parameter family in Lemma 4.3.9. Independently put on every coefficient the density

$$\pi_b(t) = b^{-1} \cos^2(\pi t/(2b)) \mathbf{1}_{[-b,b]}(t), \qquad b = c_1 K^{-2}.$$

Choose the absolute c_1 small enough. Then every density in the prior support lies between $3/4$ and $5/4$, and $\mathrm{Lip}(w_\beta) \le Cb\sum_{k\le K} k \le C$. Each prior coordinate has Fisher information π^2/b^2, so the prior information trace is at most CK^5. The Bayesian joint information matrix, with the policy's independent random seed included in its data, consequently satisfies

$$\operatorname{tr} J \le C(B + K^5).$$

The likelihood and prior score cross term vanishes. Compact support causes no boundary term because π_b and its first derivative vanish at the endpoints, and its Fisher information is finite.

For a law on $[0,1]$, write $F^\circ(x) = F(x) - \int_0^1 F(t)dt$. For two laws,

$$\|\widehat{F}^\circ - F^\circ\|_1 \le 2\|\widehat{F} - F\|_1 = 2W_1(\widehat{G}, G).$$

The centered CDF of w_β is $x - 1/2 + v(x)^T\beta$, with the vector from Lemma 4.3.8; centering removes the otherwise non-rotating primitive constants. Apply Lemma 4.3.7 at every x to the centered estimated CDF minus $x - 1/2$, then integrate and use Lemma 4.3.8. Every policy and estimator have Bayes, and hence worst-case, W_1 risk in the U parameter at least

$$c\frac{\sum_{k=1}^K k^{-1}}{\sqrt{B + K^5}}.$$

Taking $K = \lfloor B^{1/5} \rfloor$ gives $c \log B/\sqrt{B}$. Finally $|e^{au} - e^{av}| \ge a|u - v|$ transfers the same lower order to $\theta \in [1,2]$. □

Upper bound in Theorem 4.3.6. Let $T_r = \max(1, M_r)$, where M_r is an individual's sample maximum after r observations. Uniformly in $\theta \in [1,2]$, $\mathbb{E}_\theta|T_r - \theta| \le 2/(r+1)$. Take dyadic depths $r_\ell = 2^\ell$ up to $R = 2^L \asymp \sqrt{B}$. Estimate the CDF of T_1 at the base level; at level $\ell \ge 1$ average independent copies of

$$D_\ell(x) = \mathbf{1}_{\{T_{r_\ell} \le x\}} - \mathbf{1}_{\{T_{r_{\ell-1}} \le x\}}, \qquad 1 \le x \le 2,$$

using prefixes of the same individual's observations within a copy, and fresh individuals across copies and levels. Its integrated variance is at most $\mathbb{E}|T_{r_\ell} - T_{r_{\ell-1}}| \le C/r_\ell$. Use $n_\ell = \lceil cB/[(L+1)r_\ell] \rceil$ copies, with a small fixed $c > 0$. The total cost is at most B for all large B, because the rounding overhead is $O(R) = O(\sqrt{B})$. The sum of the level means has integrated variance at most $C(L+1)^2/B$ and mean equal to the CDF of T_R. Jensen and Cauchy–Schwarz give expected integrated absolute error $C(L+1)/\sqrt{B}$; its bias to the latent CDF is at most $2/(R+1)$. Project onto probability CDFs in $L^1([1,2])$, or use a measurable approximate projection over a countable dense set, losing at most a factor two and an arbitrarily small error. The asserted upper bound follows. □

The result closes the scalar endpoint critical case for this observation model. It does not claim a universal critical-rate theorem for every component family with a $1/r$ estimator: the posterior trace identity and its stable adaptive extension use the uniform-endpoint likelihood. The capped Bernoulli proof is independent of the near-linear MLE result in Section 4.2. A change of estimator and observation design is essential to the cap theorem.

4.4 The full ID domains of two McKay families

The parameter-domain question in Section 2.3 of Baricz, Prabhu K, Singh, and Vijesh [3] asks how far infinite divisibility extends for

$$f_{\mu,\nu,a,b}(x) = Cx^{\nu-1}e^{-bx}I_\mu(ax), \qquad x > 0.$$

The answer proved below is the entire domain where the source defines this probability density. This replaces the integer-bias restriction in the preceding manuscript. The proof uses the classical zero-count of Runckel, Jacobi Sturm oscillation, and a signed Cauchy representation. The new step is that the sign-changing spectral representation has a positive inverse Laplace transform throughout the admissible domain. A second application reaches the entire parameter domain specified in the same source for the squared-Bessel McKay density as well. Throughout this section, "admissible domain" means the domain stated in that source; exceptional negative-integer Bessel orders outside its convention are not included in a maximality claim.

Write $F(a,b;c;z) = {}_2F_1(a,b;c;z)$. ID means infinitely divisible and SD means self-decomposable, in each case for probability laws on $[0,\infty)$.

4.4.1 Negative zeros and a signed spectral measure

Lemma 4.4.1 (A special case of the classical zero count)**.** *Suppose*

$$\alpha > -\tfrac{1}{2}, \qquad \lambda > 0, \qquad \alpha + \lambda > 0, \qquad c = \alpha + \tfrac{1}{2}.$$

If 2λ is not an integer, the function

$$V(z) = F(-\lambda, \tfrac{1}{2} - \lambda; c; z)$$

has exactly $N = \lceil \lambda - \frac{1}{2} \rceil$ zeros in the cut plane $\mathbb{C} \setminus [1,\infty)$. All are simple and negative real. There are no zeros on either bank of $(1,\infty)$.

Proof. Runckel's zero-count theorem, in the form of [65, Theorem 3], gives

$$N = \lfloor \lambda \rfloor + \frac{1 + \operatorname{sgn}\{\Gamma(-\lambda)\Gamma(\frac{1}{2} - \lambda)\}}{2}.$$

Indeed, the other two gamma arguments in that theorem are $c + \lambda > 0$ and $\alpha + \lambda > 0$; its smallest argument is $-\lambda$, and its largest is positive. The duplication formula reduces the sign in the display to that of $\Gamma(-2\lambda)$. The resulting integer is $\lceil \lambda - \frac{1}{2} \rceil$. The nonvanishing on the cut follows from the nondegenerate case of Runckel's theorem, as recorded in [65, Lemma 3].

It remains to locate every counted zero. Pfaff's transformation [6, Eq. 15.8.1] gives, for $z < 0$ and $y = z/(z-1) \in (0,1)$,

$$V(z) = (1-z)^{\lambda} v_\lambda(y), \qquad v_\lambda(y) = F(-\lambda, \alpha + \lambda; c; y).$$

Put $\mu = \alpha - 1/2 > -1$. The latter function is the solution regular at zero of the Jacobi Sturm equation

$$-(pv')' = E_\lambda w v, \qquad p(y) = y^{\mu+1}(1-y)^{1/2}, \quad w(y) = y^{\mu}(1-y)^{-1/2}, \quad E_\lambda = \lambda(\lambda + \alpha), \tag{4.4.1}$$

normalized by $v(0) = 1$. The left boundary condition is $\lim_{y \downarrow 0} p(y)v'(y) = 0$. At the right endpoint take the Dirichlet condition $v(1) = 0$. These are the usual separated Jacobi boundary conditions; the positive quadratic form is $\int_0^1 p|v'|^2$ in $L^2(w\,dy)$. Gauss's evaluation gives

$$v_\lambda(1) = \frac{\Gamma(c)\Gamma(1/2)}{\Gamma(c+\lambda)\Gamma(1/2-\lambda)}. \tag{4.4.2}$$

The complete eigenfunctions can also be written explicitly:

$$v_{n+1/2}(y) = (1-y)^{1/2} F(-n, n + \mu + 3/2; \mu + 1; y), \qquad n \geq 0.$$

They are positive scalar multiples of $(1-y)^{1/2}P_n^{(\mu,1/2)}(1-2y)$. Multiplication by $(1-y)^{1/2}$ is a unitary map from the Jacobi space with weight $y^\mu(1-y)^{1/2}$ onto $L^2(w\,dy)$. Jacobi completeness therefore supplies the entire spectrum, with eigenvalues $E_{n+1/2}$, $n=0,1,2,\dots$. The zero-flux condition at zero excludes the other local solution $y^{-\mu}$, or the logarithmic solution when $\mu=0$. There are no nonpositive eigenvalues: the quadratic form is nonnegative, and its only zero-energy solution satisfying the left condition is constant, which cannot satisfy the right condition. Also E_λ is strictly increasing on $\lambda>\max(0,-\alpha)$, the parameter range under consideration.

Sturm oscillation therefore says that v_λ has as many zeros in $(0,1)$ as there are eigenvalues $E_{n+1/2}<E_\lambda$. This is precisely N. The statement applies to the possibly singular left endpoint by the stated zero-flux boundary condition; equivalently one can exhaust by regular separated problems and take the regular solution limit. All interior zeros are simple by uniqueness for the second-order equation. Their images under $z=y/(y-1)$ are N simple negative zeros of V, exhausting Runckel's total count. □

Lemma 4.4.2 (The signed Cauchy measure)**.** *Under the noninteger hypotheses of Lemma 4.4.1, put*

$$q=2\lambda,\qquad A=\alpha+q,\qquad R(w)=\frac{\lambda}{w}-\frac{V'(1/w)}{w^2V(1/w)}.$$

Write the negative zeros of $V(1/w)$ as $-\tau_1,\dots,-\tau_N$, where each $\tau_j>0$. There is a finite real measure ρ on $(0,1)$ such that

$$R(w)=\sum_{j=1}^{N}\frac{1}{w+\tau_j}+\int_0^1\frac{\rho(dt)}{w-t}. \tag{4.4.3}$$

It has constant sign, total mass $\lambda-N$, and density

$$\frac{d\rho}{dt}=-\frac{\Gamma(c)^2}{\Gamma(-\lambda)\Gamma(1/2-\lambda)\Gamma(c+\lambda)\Gamma(\alpha+\lambda)}\frac{t^{-q-1/2}(1-t)^{A-1}}{|V(1/t+i0)|^2},\qquad 0<t<1. \tag{4.4.4}$$

In particular ρ has no endpoint atoms.

Proof. Although $w^\lambda V(1/w)$ itself need not be single-valued around the cut, its logarithmic derivative R is a well-defined real meromorphic function on $\mathbb{C}\setminus[0,1]$. Its only poles there are the simple zeros identified in Lemma 4.4.1, each with residue one, and $R(w)=\lambda/w+O(w^{-2})$ at infinity.

For clarity we derive the jump. Write $V(x+i0)=U(x)+iW(x)$ for $x>1$. The hypergeometric equation implies

$$(UW'-U'W)(x)=D\,x^{-c}(x-1)^{A-1}.$$

The connection formula at one gives

$$D=\frac{\pi\Gamma(c)^2}{\Gamma(-\lambda)\Gamma(1/2-\lambda)\Gamma(c+\lambda)\Gamma(\alpha+\lambda)}.$$

For example, for noninteger A its two leading connection coefficients are $P=\Gamma(c)\Gamma(A)/[\Gamma(c+\lambda)\Gamma(\alpha+\lambda)]$ and $Q=\Gamma(c)\Gamma(-A)/[\Gamma(-\lambda)\Gamma(1/2-\lambda)]$. On the upper bank the second term has phase $e^{-i\pi A}$, and $D=-PQA\sin(\pi A)$ gives the displayed value by reflection. Integer $A>0$ follows by continuity. Thus $\operatorname{Im}(V'/V)(x+i0)=Dx^{-c}(x-1)^{A-1}/|V(x+i0)|^2$. The map $w\mapsto 1/w$ swaps the banks, so $-\operatorname{Im}R(t+i0)/\pi$ is exactly (4.4.4). Its sign is constant.

The connection formula at infinity, whose exponent difference is $1/2$, gives a nonzero constant term followed by a square-root term in $w^\lambda V(1/w)$ near zero. The constant is nonzero

because 2λ is not an integer and $c+\lambda, \alpha+\lambda > 0$. Hence $R(w) = O(|w|^{-1/2})$ there. At one, Gauss's evaluation gives a finite nonzero value since $A > 0$. The derivative has at worst $O(1 + |1-w|^{A-1}(1 + |\log(1-w)|))$ growth. These estimates make both small-circle integrals vanish and the jump absolutely integrable. Cauchy's formula on a keyhole contour around $[0,1]$, after subtracting the residue-one poles, proves (4.4.3). Comparison of the coefficients of $1/w$ at infinity gives $\rho((0,1)) = \lambda - N$. The endpoint estimates also exclude endpoint atoms. □

4.4.2 The maximal ID domain

Theorem 4.4.3 (Every admissible generalized McKay law is ID)**.** *For every*

$$\mu > -1, \qquad \mu + \nu > 0, \qquad b > a > 0, \tag{4.4.5}$$

the normalized density

$$f_{\mu,\nu,a,b}(x) = C x^{\nu-1} e^{-bx} I_\mu(ax), \qquad x > 0,$$

is infinitely divisible. Thus the maximal parameter domain asked for in [3, Section 2.3], within the density family and normalization convention of that source, is the whole domain (4.4.5)*.*

More explicitly, if $q = \nu - \mu - 1 > 0$, let $\alpha = \mu + 1/2$, $M = \mu + \nu = 2\alpha + q$, and $A = \alpha + q$. For noninteger q, use the zeros and signed measure in Lemma 4.4.2 and define

$$K_{\alpha,q}(x) = 2A\cosh x - 2\sum_{j=1}^{N} \cos(\sqrt{\tau_j}\,x) - 2\int_0^1 \cosh(\sqrt{t}\,x)\,\rho(dt). \tag{4.4.6}$$

Then the Lévy density is $e^{-bt}K_{\alpha,q}(at)/t$, and

$$K_{\alpha,q}(x) \geq M\cosh x > 0 \qquad (x \geq 0). \tag{4.4.7}$$

Proof. For $q \leq 0$, the source's Theorem 3 proves ID on precisely this part of the admissible domain. Consider $q > 0$. Scaling permits $a = 1$ in the transform calculation. The positive Bessel series, integrated termwise, gives

$$J(u) := \int_0^\infty x^{\nu-1} e^{-ux} I_\mu(x)\,dx = C_0 u^{-M} F(M/2, (M+1)/2; \mu+1; u^{-2}), \qquad u > 1,$$

where $C_0 > 0$. Euler's transformation yields

$$J(u) = C_0 \frac{u^q V(u^{-2})}{(u^2-1)^A}. \tag{4.4.8}$$

Notice that $\alpha > -1/2$ and $\alpha + q/2 = M/2 > 0$, exactly the hypotheses needed in the two lemmas. For noninteger q, logarithmic differentiation gives

$$-\frac{J'(u)}{J(u)} = \frac{A}{u-1} + \frac{A}{u+1} - 2\sum_{j=1}^{N} \frac{u}{u^2+\tau_j} - 2\int_0^1 \frac{u}{u^2-t}\,\rho(dt).$$

Each term is the Laplace transform of its corresponding term in (4.4.6). The signed integral is legitimate because ρ has finite total variation and $u > 1$.

The essential positivity argument uses only the total mass and sign of ρ. If $d = \lambda - N \geq 0$, then

$$K_{\alpha,q}(x) \geq 2A\cosh x - 2N - 2d\cosh x = M\cosh x + 2N(\cosh x - 1) \geq M\cosh x.$$

If $d \le 0$, then

$$K_{\alpha,q}(x) \ge 2A\cosh x - 2N + 2(N-\lambda) = 2A\cosh x - q \ge M\cosh x.$$

Thus the signed spectral measure cannot produce a negative Lévy kernel. Since $K = O(e^x)$ and $K(0) = M$, the measure $e^{-bt}K(at)\,dt/t$ satisfies the Lévy integrability condition for $b > a$. Integrating the logarithmic derivative and using $L(0) = 1$ gives

$$-\log L(s) = \int_0^\infty (1 - e^{-st})e^{-bt}\frac{K_{\alpha,q}(at)}{t}\,dt.$$

This proves ID.

For integer $q \ge 1$, one may take a sequence of admissible noninteger parameters tending to q: the normalized densities converge in L^1 by the usual Bessel bounds at zero and infinity, and the ID class is closed under weak convergence. There is also a direct formula. If $q = 2m$, V is a degree-m polynomial with simple negative zeros, $N = m$, and $\rho = 0$. If $q = 2m+1$, V is a degree-m polynomial with simple negative zeros and the logarithmic derivative of $u^q V(u^{-2})$ has one additional $1/u$ term. The negative-zero assertion follows from the same Jacobi Sturm argument, including the right-endpoint eigenvalue case. Hence the integer kernel is

$$K_{\alpha,q}(x) = 2A\cosh x - 2\sum_{j=1}^{\lfloor q/2\rfloor}\cos(\sqrt{\tau_j}\,x) - \mathbf{1}_{\{q\ \text{odd}\}},$$

and it too satisfies $K \ge M\cosh x$. □

Corollary 4.4.4 (A spectral SD boundary on the full excess-power domain)**.** *For every admissible (α, q) with $q > 0$ define the explicit kernel of Theorem 4.4.3, using its integer convention when necessary, and put*

$$\beta_{\alpha,q} = \sup_{x\ge 0}\frac{K'_{\alpha,q}(x)}{K_{\alpha,q}(x)}.$$

Then $1 \le \beta_{\alpha,q} < \infty$, and the generalized McKay law is SD if and only if $b/a \ge \beta_{\alpha,q}$. Moreover $\beta_{\alpha,q} > 1$ whenever q is a positive integer or $q \in (2m, 2m+1)$ for an integer $m \ge 0$; in these cases the supremum is attained at a finite positive point.

Proof. The kernel is positive and smooth. Its cosine terms are bounded, and ρ has no atom at one, so dominated convergence gives

$$e^{-x}K_{\alpha,q}(x) \longrightarrow A, \qquad e^{-x}K'_{\alpha,q}(x) \longrightarrow A.$$

Thus $K'/K \to 1$, is bounded, and its supremum is at least one. The SD criterion for a positive ID law with smooth spectral function $k(t)$ is $k'(t) \le 0$. Applied to $k(t) = e^{-bt}K(at)$ this is exactly the stated inequality.

For completeness this criterion follows directly from the defining SD ratios. A nonincreasing k makes $-\log L(s) + \log L(cs)$ a Bernstein exponent for each $0 < c < 1$, with Lévy density $[k(t) - k(t/c)]/t$. Conversely, if all $L(s)/L(cs)$ are Laplace transforms, their negative derivatives divided by $1 - c$ converge, with derivatives locally, to $(s[-\log L(s)]')'$. This is completely monotone, while integration by parts identifies it as the Laplace transform of $-tk'(t)$. Uniqueness of the Laplace transform forces $k' \le 0$.

For noninteger $q \in (2m, 2m+1)$ the measure ρ is strictly positive on $(0, 1)$. In the difference $K' - K$, its contribution is

$$2\int_0^1 [\cosh(\sqrt{t}\,x) - \sqrt{t}\sinh(\sqrt{t}\,x)]\,\rho(dt).$$

Any compact subinterval of $(0,1)$ has positive mass, so this quantity grows exponentially. The cosine contribution is bounded, and the remaining contribution is $-2Ae^{-x}$. It follows that $K'-K>0$ for all sufficiently large x. For positive integer q, simultaneous returns of the finitely many cosine phases make the corresponding bounded contribution to $K'-K$ arbitrarily close to $q>0$ at arbitrarily large times. This gives $\beta>1$ in both cases. Since $K'/K\to 1$, the larger maximum is attained on a compact interval. □

The supremum criterion is a spectral classification, not a claim that its maximizing point has a closed elementary expression. For $q\in(2m+1,2m+2)$ the preceding theorem proves ID and the same exact SD test, but does not assert that $\beta_{\alpha,q}>1$ for every α. That strictness question is kept separate.

4.4.3 The full ID domain of the squared-Bessel family

Lemma 4.4.5 (A zero-free disk for the transformed squared-Bessel factor)**.** *For every $r>0$, the function*

$$F_r(z)=F(-r,1/2;r+3/2;z)$$

has no zero in the closed unit disk. If r is not an integer, its total number of zeros in $\mathbb{C}\setminus[1,\infty)$ is

$$N_r=2\left\lfloor\frac{r+1}{2}\right\rfloor,$$

and it has no zero on either bank of $(1,\infty)$.

Proof. The classical quadratic transformation [6, Eq. 15.8.16] gives

$$F_r(z)=(1-z)^{-1/2}F\left(1/4,r+3/4;r+3/2;\frac{-4z}{(1-z)^2}\right). \tag{4.4.9}$$

The map $z\mapsto -4z/(1-z)^2$ maps the open unit disk onto $\mathbb{C}\setminus[1,\infty)$. This follows either from the usual slit-plane map or by solving its quadratic inverse; on $z=e^{i\theta}\neq 1$ its value is $1/\sin^2(\theta/2)$. Euler's beta integral for the last hypergeometric function is an average of $(1-ZT)^{-1/4}$, with $T\sim\mathrm{Beta}(r+3/4,3/4)$. For nonreal Z, these integrands lie in a common sector of width at most $\pi/4$, so their average is nonzero. For real $Z<1$ they are positive. This proves nonvanishing in the open disk.

For finite $Z>1$, the upper and lower boundary integrals also exist: the interior singularity has exponent $-1/4$ and is integrable. All their integrands have arguments either zero or $\pm\pi/4$, so their real parts are positive. At $Z=1$ the beta weight gives an integrable endpoint singularity and the integral is positive. Formula (4.4.9) therefore excludes zeros on the unit circle except possibly at $z=1$. At that point, Gauss's formula gives the positive value

$$F_r(1)=\frac{\Gamma(r+3/2)\Gamma(2r+1)}{\Gamma(2r+3/2)\Gamma(r+1)}>0.$$

Finally Runckel's theorem gives, for noninteger r,

$$N_r=\lfloor r\rfloor+\frac{1+\operatorname{sgn}\Gamma(-r)}{2}=2\left\lfloor\frac{r+1}{2}\right\rfloor.$$

Its nondegenerate-cut assertion excludes all zeros on the banks. All off-cut zeros are simple: a simultaneous zero of F_r and F_r', away from the regular singular points, would contradict uniqueness for the hypergeometric differential equation. The values at zero and one are nonzero. □

Theorem 4.4.6 (Every admissible squared-Bessel McKay law is ID)**.** *For every $\mu > -1/4$ and $b > 2a > 0$, the normalized density*

$$g_{\mu,a,b}(x) = Cx^{2\mu}e^{-bx}I_\mu(ax)^2, \qquad x > 0, \tag{4.4.10}$$

is infinitely divisible.

For an explicit formula when $r = \mu - 1/2 > 0$ is not an integer, put $B = 2r+1$, $c = r+3/2$, and let $w_1, \dots, w_N$ be the reciprocals of all zeros of F_r, with $N = N_r$ as in Lemma 4.4.5. In particular $|w_j| < 1$. Define the real signed measure

$$\frac{d\sigma_r}{dt} = -\frac{\Gamma(c)^2}{\Gamma(-r)\Gamma(1/2)\Gamma(2r+3/2)\Gamma(r+1)} \frac{t^{-r-1/2}(1-t)^{2r}}{|F_r(1/t+i0)|^2}, \qquad 0 < t < 1. \tag{4.4.11}$$

It has constant sign and total mass $r - N$. The Lévy spectral function of (4.4.10) is $e^{-bt}K_r(2at)$, where

$$K_r(x) = B(1 + 2\cosh x) - 2\sum_{j=1}^{N} \cosh(\sqrt{w_j}\, x) - 2\int_0^1 \cosh(\sqrt{t}\, x)\, \sigma_r(dt). \tag{4.4.12}$$

The choice of either square root gives the same summand. The kernel is real, positive, and even.

Proof. For $-1/4 < \mu \le 1/2$, ID follows from the source's Theorem 4 (and the GGC representation in the earlier manuscript). Let $r = \mu - 1/2 > 0$. The Bessel-square Laplace integral, with scale $2a$ set to one, and Euler's transformation give

$$J(u) = C_1 u^{-4r-3} F(r+1, 2r+3/2; r+3/2; u^{-2}) = C_1 \frac{u^{2r} F_r(u^{-2})}{u^{2r+1}(u^2-1)^{2r+1}}. \tag{4.4.13}$$

For noninteger r, apply the same logarithmic-derivative keyhole argument as in Lemma 4.4.2 to $H_r(w) = w^r F_r(1/w)$. It gives

$$\frac{H_r'(w)}{H_r(w)} = \sum_{j=1}^{N} \frac{1}{w - w_j} + \int_0^1 \frac{\sigma_r(dt)}{w - t}.$$

The Wronskian computation with parameters $(-r, 1/2; c)$ gives exactly (4.4.11). Near zero the leading term of H_r is a nonzero constant; its next nonanalytic exponent is $r + 1/2 > 0$, with a possible logarithm when this is an integer. Thus there is no pole of its logarithmic derivative at zero, and the jump is integrable there. At one $H_r(1) = F_r(1) > 0$ and the singular exponent is $2r + 1 > 1$. There is no endpoint atom. Comparison at infinity gives the mass $r - N$. Taking Laplace transforms in (4.4.13) now gives (4.4.12).

Complex conjugation pairs the w_j, making the kernel real. Let $\varrho = \max_j \sqrt{|w_j|} < 1$ if there are any zeros; otherwise omit all zero terms. The bound $|\cosh z| \le \cosh |z|$ gives

$$\left| \sum_{j=1}^{N} \cosh(\sqrt{w_j}\, x) \right| \le N \cosh(\varrho x) \le N \cosh x.$$

If $r - N \ge 0$, use the positive measure's mass to obtain

$$K_r(x) \ge B + 2(B - r)\cosh x = B + (2r+2)\cosh x > 0. \tag{4.4.14}$$

If $r - N \le 0$, instead use $\cosh(\sqrt{t}\, x) \ge 1$ for its positive contribution, giving

$$K_r(x) \ge B + 2(B - N)\cosh x + 2(N - r) = 2N + 1 + 2(B - N)\cosh x > 0. \tag{4.4.15}$$

Here $N < r+1$ for noninteger r, so $B - N > r > 0$. The kernel has growth $O(e^x)$ and $K_r(0) = 4r+3$. The positive Lévy representation, with $b > 2a$, follows exactly as in Theorem 4.4.3.

For integer $r = n \geq 1$, F_r is a polynomial of degree n. Lemma 4.4.5 still puts all reciprocal roots strictly inside the unit disk. Use these n roots and set $\sigma_r = 0$ in (4.4.12). Its logarithmic derivative is a finite partial-fraction sum, and $K_r(x) \geq B + 2(B-n)\cosh x > 0$ proves ID directly. □

Corollary 4.4.7 (The squared-Bessel SD boundary above order one half)**.** *For every $r > 0$, use the explicit kernel in Theorem 4.4.6, with its polynomial convention at integers, and set*

$$\beta_r = \sup_{x \geq 0} \frac{K_r'(x)}{K_r(x)}.$$

Then $1 \leq \beta_r < \infty$, and

$$g_{r+1/2,a,b} \text{ is SD} \quad \Longleftrightarrow \quad b/(2a) \geq \beta_r.$$

For $0 < r < 1$ one has the quantitative strict bound

$$1 < \beta_r \leq \frac{2r+1}{r+1} < \frac{3}{2},$$

and the maximum is attained at a finite positive point. More generally $\beta_r > 1$ for $r \in (2m, 2m+1)$, $m \geq 0$.

Proof. All zeros have square roots of modulus less than one, and the finite measure σ_r has no atom at one. Therefore $e^{-x}K_r(x) \to B$ and $e^{-x}K_r'(x) \to B$. The positivity bounds in the theorem show that K'/K is continuous and bounded; its limit is one. The SD criterion proved in Corollary 4.4.4 gives the equivalence.

If $0 < r < 1$, there are no zero terms and σ_r is a positive measure of mass r. Thus $K_r' \leq 2B\sinh x$, and (4.4.14) gives the stated upper bound. In general, on $r \in (2m, 2m+1)$ the density of σ_r is strictly positive throughout $(0,1)$. The positive integral contribution to $K_r' - K_r$ contains exponential growth of every rate less than one. Choose such a rate strictly larger than $\max_j \sqrt{|w_j|}$. It dominates all zero contributions and the negative term $-B(1+2e^{-x})$. Hence $K_r'/K_r > 1$ at some finite point and the supremum is attained. □

4.4.4 A quantitative tail for the spectral threshold

Proposition 4.4.8 (Polynomially certified tail control)**.** *For fixed admissible parameters with noninteger $q > 0$ in the third family or noninteger $r > 0$ in the fourth family, there exist finite positive constants C, c_0 such that, for $x \geq 1$,*

$$\left| \frac{K'(x)}{K(x)} - 1 \right| \leq C\{e^{-c_0 x} + x^{-\gamma}\}, \qquad \gamma = \begin{cases} \alpha + q, & \text{third family,} \\ 2r+1, & \text{fourth family.} \end{cases}$$

The constants can be bounded from the displayed cut densities and the finite zero sets. In particular, a certified point with $K'(x)/K(x) > 1 + \delta$ reduces the search for the exact maximum to a finite interval.

Proof. In both cases the explicit density $h(t)$ of the cut measure obeys $|h(t)| \leq C_1(1-t)^{\gamma-1}$ on $[1/2, 1)$: the denominator has a positive continuous modulus, including its nonzero limit at one. On $(0, 1/2)$ its absolute mass is finite. The kernel is bounded below by $c_1 \cosh x$ from (4.4.7),

(4.4.14), or (4.4.15). After division by this bound, the absolute cut contribution to $K' - K$ is at most a constant times

$$\int_0^1 e^{-(1-\sqrt{t})x}|h(t)|\,dt \le C_2 e^{-(1-1/\sqrt{2})x} + C_1 \int_0^1 e^{-(1-t)x/2}(1-t)^{\gamma-1}\,dt$$
$$\le C_2 e^{-(1-1/\sqrt{2})x} + C_1\Gamma(\gamma)(2/x)^{\gamma}.$$

The finite zero terms are bounded in the third family, and grow at a rate strictly smaller than one in the fourth family. The remaining elementary terms decay exponentially after the same division. These estimates prove the claim. □

Scope and priority. The full third-family ID statement closes the maximal-domain question in the stated density domain, subject to independent verification of this proof. The fourth-family theorem also reaches its entire admissible real-order domain. Runckel's zero count, Jacobi oscillation, and the hypergeometric transformation identities are classical inputs and are not new results. The SD answers are explicit spectral threshold classifications; they do not provide elementary closed forms for every threshold, and universal strictness above one in the negative-cut intervals has not been established. These distinctions are necessary when counting externally posed problems rather than individual consequences.

4.5 A sharp logarithmic tail boundary for sparse Gaussian observation

The unrestricted terminal-design theorem in the preceding manuscript requires divergence of a harmonic subseries. Bounded covariance support, in contrast, makes every infinite set of terminal lengths sufficient. The results below locate a sharp boundary between these two regimes. The boundary is finer than the square-root exponential moment: it has a logarithmic correction. The uniqueness argument uses classical Stieltjes moment determinacy. The converse requires more than a pair of moment-equivalent measures: it constructs one joint mixing density whose whole observation transforms vanish at the selected designs.

Write

$$\omega_a(v) = \frac{\sqrt{v}}{\{\log(e+v)\}^a}, \qquad v \ge 0, \quad a \ge 0.$$

Let (A, Σ) be a random element of $\mathbb{R}^d \times \mathbb{S}_d^+$ and, conditionally on it, let $X_1, X_2, \dots$ be iid $N_d(A, \Sigma)$. As before the separate experiment of terminal length $n \ge d$ observes

$$\mathcal{O}_n = (X_1, \dots, X_{d-1}, S_n), \qquad S_n = \sum_{j=1}^{n} X_j.$$

Separate experiments supply their separate population laws; their joint cross-design distribution is not assumed available.

Theorem 4.5.1 (Logarithmic tail threshold)**.** *1. For every $d \ge 1$, every infinite set $\mathcal{N} \subset \{d, d+1, \dots\}$, and every $0 \le a \le 1$, the laws $\{\mathcal{L}(\mathcal{O}_n) : n \in \mathcal{N}\}$ identify the mixing law within the class*

$$\mathbb{E}\exp\{c\omega_a(\operatorname{tr}\Sigma)\} < \infty \quad \text{for some } c > 0. \tag{4.5.1}$$

No moment or support assumption on A is needed.

2. For $d = 1$ and $\mathcal{N} = \{2^j : j \ge 0\}$, the assertion fails for every $a > 1$. There are distinct absolutely continuous mixing laws with identical A marginals, identical variance marginals,

and identical finite joint polynomial moments, producing the same law of S_n for every $n \in \mathcal{N}$. Both laws satisfy

$$\mathbb{E}\exp\{c\omega_a(\Sigma)\} < \infty \qquad \text{for every } c > 0. \tag{4.5.2}$$

Any fixed lower bound $\Sigma \geq v_0 > 0$ may be imposed.

Thus $a = 1$ is the exact boundary for the assertion that this tail condition makes *every* infinite terminal set identifying. Sharpness for the full matrix-covariance experiment in each fixed dimension $d > 1$ is not asserted. A scalar counterexample cannot simply be placed in one matrix coordinate: the additional anchors would then provide extra scalar observations.

4.5.1 Boundary zeros, moments, and the positive result

We use the classical Stieltjes–Carleman criterion: a finite positive measure on $[0, \infty)$ with moments m_k is determined by its moments if $\sum_{k\geq 1} m_k^{-1/(2k)} = \infty$. Hardy's square-root exponential condition and its relation to this criterion are classical [66, 67].

Lemma 4.5.2 (A logarithmic moment bound)**.** *Suppose $V \geq 0$ and $\mathbb{E}e^{c\omega_a(V)} < \infty$ for some $c > 0$ and $a \geq 0$. Then a finite constant C satisfies*

$$(\mathbb{E}V^k)^{1/(2k)} \leq Ck\{\log(e + k)\}^a, \qquad k \geq 1.$$

Consequently the Stieltjes–Carleman condition holds when $a \leq 1$. The same conclusion holds for any finite positive measure dominated by a finite sum of laws satisfying the displayed exponential bound, possibly with different positive constants c.

Proof. Put $Y = \sqrt{V}$. Since $\log(e+Y^2)$ and $\log(e+Y)$ are comparable, $\mathbb{E}\exp\{c_1 Y/\log^a(e+Y)\} = M < \infty$ for some $c_1 > 0$. Set $b_k = k\log^a(e + k)$. For $y = b_k w$ with $w \geq 1$,

$$\log(e + b_k w) \leq C_a \log(e + k)(1 + \log w), \qquad \frac{y}{\log^a(e + y)} \geq c_a \frac{kw}{(1 + \log w)^a}.$$

It follows that

$$\sup_{y\geq 0} y^{2k} \exp\{-c_1 y/\log^a(e + y)\} \leq (C_1 b_k)^{2k},$$

because $w^2 \exp\{-c_1 c_a w/(1 + \log w)^a\}$ is bounded on $[1, \infty)$; for $w < 1$ the claim is immediate. Hence $\mathbb{E}Y^{2k} \leq M(C_1 b_k)^{2k}$. Absorb $\max(1, \sqrt{M})$ into C. The series $\sum[k\log^a(e + k)]^{-1}$ diverges precisely for $a \leq 1$. A finite sum only changes the constants. □

Lemma 4.5.3 (Accumulating positive zeros under a moment envelope)**.** *Let ν be a finite complex measure on $[0, \infty)$, dominated in total variation by a finite positive measure μ with all moments finite and satisfying the Stieltjes–Carleman condition. If*

$$\int e^{-t_j v}\, d\nu(v) = 0 \quad \text{for distinct } t_j > 0 \text{ tending to } 0,$$

then $\nu = 0$.

Proof. The Laplace transform $f(t)$ is C^∞ on $[0, \infty)$, with $f^{(k)}(0) = (-1)^k \int v^k\, d\nu(v)$, by domination. Continuity and the zeros first give $f(0) = 0$. Inductively, if the derivatives of orders below k vanish at zero, Taylor's formula at t_j gives $0 = f^{(k)}(0)t_j^k/k! + o(t_j^k)$, so the kth derivative also vanishes. This argument applies directly to a complex function. The nonnegative measures $\mu \pm \operatorname{Re}\nu$ have the same moments as μ, so moment determinacy gives $\operatorname{Re}\nu = 0$. The same argument gives $\operatorname{Im}\nu = 0$. □

Proof of the positive assertion of Theorem 4.5.1. Let ρ be the signed difference of two candidate mixing laws. The blocks

$$X_1, \dots, X_{d-1}, S_n - X_1 - \cdots - X_{d-1}$$

are conditionally independent with respective lengths $m_1 = \cdots = m_{d-1} = 1$, $m_d = n - d + 1$. Their Fourier arguments for the mixing transform are exactly

$$u \in \mathbb{R}^d, \qquad T = \frac{uu^T}{2n} + C, \quad C \succeq 0, \quad \operatorname{rank} C \le d - 1. \tag{4.5.3}$$

Indeed, let the frequency columns be $\sqrt{m_i} w_i$. Project their matrix onto the unit vector

$$n^{-1/2}(\sqrt{m_1}, \dots, \sqrt{m_d}).$$

The parallel part gives $uu^T/(2n)$ and the orthogonal part gives C. Conversely every such C is realized by a matrix on that orthogonal complement.

Fix $u \neq 0$ and a rank-$d-1$ positive semidefinite matrix C with u not orthogonal to its kernel. Define a complex measure ν by pushing forward

$$e^{iu^T A - \operatorname{tr}(C\Sigma)}\, d\rho(A, \Sigma) \quad \text{under } (A, \Sigma) \longmapsto v = u^T \Sigma u.$$

It is dominated by the pushforward of the sum of the two candidate laws. Since $v \le \|u\|^2 \operatorname{tr} \Sigma$, the latter measure satisfies the moment bound of Lemma 4.5.2. Equality of observations and (4.5.3) give zeros at $t = 1/(2n)$ for every selected n. Therefore Lemma 4.5.3 gives

$$\int e^{iu^T A - \operatorname{tr}((C + zuu^T)\Sigma)}\, d\rho = 0, \qquad z > 0.$$

For any $T \succ 0$ the choice

$$z = (u^T T^{-1} u)^{-1}, \qquad C = T - zuu^T$$

has precisely the required properties. Thus the full transform vanishes for all real $u \neq 0$ and $T \succ 0$. Continuity gives $u = 0$. Holomorphy in the positive matrix tube, followed by Fourier uniqueness after multiplication by $e^{-\operatorname{tr} \Sigma}$, gives $\rho = 0$. □

The proof also gives a more general assertion: a fixed covariance moment envelope $\mathbb{E}(\operatorname{tr} \Sigma)^k \le M_k$ with $\sum M_k^{-1/(2k)} = \infty$ suffices. Specifying a common envelope avoids an unwarranted closure assertion about arbitrary unions of moment-determinate classes.

4.5.2 A sector construction at the logarithmic endpoint

The following scalar construction is also used for the transfer to non-Gaussian noise. Throughout, inverse Laplace transforms are ordinary integrable functions, not distributions supported at the boundary.

Lemma 4.5.4 (Lacunary zeros with nearly square-root decay)**.** *Let $b_j = b_0 2^{-j}$, $j \ge 0$, with $b_0 > 0$. For every $q > 1$ there exists a nonzero real Schwartz function f on $\mathbb{R}$, supported in $[0, \infty)$, such that*

$$F(z) := \int_0^\infty e^{-zv} f(v)\, dv, \qquad \operatorname{Re} z > 0, \tag{4.5.4}$$

$$F(b_j) = 0 \quad (j \ge 0), \qquad \int_0^\infty v^k f(v)\, dv = 0 \quad (k \ge 0), \tag{4.5.5}$$

$$|f^{(m)}(v)| \le C_m \exp\{-c_m \omega_q(v)\}, \qquad v \ge 1, \quad m = 0, 1, 2, \dots. \tag{4.5.6}$$

Proof. Put $p = 2(q-1) > 0$, choose $K > 1$ with $\log K \geq \max(2p, 2)$, and use principal branches. In the cut plane $\mathbb{C} \setminus (-\infty, 0]$ set

$$B(z) = \prod_{j \geq 0} \frac{z - b_j}{z + b_j}, \qquad Q(w) = \frac{w}{\{\log(K + w)\}^p}, \qquad F(z) = B(z) \exp\{-z^{1/2} - Q(1/z)\}. \tag{4.5.7}$$

The product converges locally uniformly away from zero and its negative poles. The other functions in (4.5.7) are holomorphic in the cut plane: $K + w$ never lies on the logarithm's cut there, and $\log(K + w)$ can lie on the nonpositive real axis only when w is negative real.

We record a uniform bound that is special to geometric zeros. For every fixed $\theta_0 < \pi$,

$$|B(z)| \leq C_{\theta_0}, \qquad 0 < |z| < \infty, \quad |\arg z| \leq \theta_0. \tag{4.5.8}$$

To prove it, split the factors into $b_j \leq |z|/2$, $b_j \geq 2|z|$, and the remaining at most three indices. For the first group the positive part of the logarithm of the factor's modulus is at most $C_{\theta_0} b_j / |z|$; for the second it is at most $C_{\theta_0} |z| / b_j$. Both geometric sums are bounded. The middle factors are bounded because z stays a fixed angular distance from their poles. This proves (4.5.8). On the right half-plane the sharper bound $|B| \leq 1$ holds.

For $\operatorname{Re} w \geq 0$ one has $\operatorname{Re} Q(w) \geq 0$. For example, if $\arg w = \vartheta \in [0, \pi/2]$, write $\log(K + w) = a + i\varphi$. Then $a \geq \log K$ and $0 \leq \varphi \leq \vartheta$, so

$$0 \leq p \arg(a + i\varphi) \leq (p / \log K)\vartheta \leq \vartheta.$$

Hence $0 \leq \arg Q(w) \leq \vartheta$; the lower half-plane follows by conjugation. In particular F is bounded on the right half-plane and is conjugation invariant.

There is stronger decay near zero, including in a narrow region to the left of the imaginary axis. Put $r = e^{-L}$ and

$$\delta(r) = \frac{p\pi}{8L}.$$

For all sufficiently large L, uniformly for $|\arg z| \leq \pi/2 + \delta(r)$ and $|z| = r$,

$$\operatorname{Re} Q(1/z) \geq \frac{c_p}{r L^{p+1}}. \tag{4.5.9}$$

Here is a direct check of the endpoint angular estimate. Write $1/z = e^{L + i\vartheta}$, $|\vartheta| \leq \pi/2 + \delta(r)$. Uniformly over this range,

$$\log(K + e^{L+i\vartheta}) = L + i\vartheta + O(e^{-L}), \qquad \arg Q(e^{L+i\vartheta}) = \vartheta - p \arctan(\vartheta / L) + O(e^{-L}/L).$$

The expression before the error is increasing in ϑ for large L; its endpoint is $\pi/2 - 3p\pi/(8L) + O(L^{-2})$. The modulus of Q is comparable to e^L / L^p. Its real part therefore has the lower bound (4.5.9).

On the imaginary axis all derivatives of B near zero have at most polynomial growth in $1/r$: this follows either by Cauchy's inequality in disks of radius proportional to r inside a fixed sector and (4.5.8), or by differentiating the product. Derivatives of $Q(1/z)$ have the same property after allowing powers of $\log(1/r)$. Estimate (4.5.9) dominates all of them. At infinity, $\exp(-\sqrt{z})$ and its derivatives provide rapid decay, while $Q(1/z)$ and B remain bounded with polynomially bounded derivatives. Thus $F(iy)$ extends across $y = 0$ to a Schwartz function, flat at zero. Define

$$f(v) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{iyv} F(iy)\, dy.$$

It is real, Schwartz, and nonzero. It is supported in $[0, \infty)$: for $x > 0$, Fourier inversion on the line $x + i\mathbb{R}$ gives $f_x(v) = e^{-xv} f(v)$ by contour shifting, and

$$|F(x + iy)| \leq e^{-\operatorname{Re}\sqrt{x+iy}} \leq e^{-\sqrt{|y|/2}}$$

gives a bound on $f_x(v)$ independent of x. If $v < 0$, letting $x \to \infty$ forces $f(v) = 0$. The same line-shift identity gives (4.5.4). Flatness at $z = 0$ and dominated differentiation give (4.5.5).

It remains to prove the quantitative tail; Schwartz decay alone would not suffice. Deform the inverse contour near zero onto

$$z_\pm(r) = r \exp\{\pm i(\pi/2 + \delta(r))\}, \qquad 0 < r \le r_0,$$

where r_0 is sufficiently small. The line element is bounded by a constant times dr. Continue both curves to infinity along a fixed angle in $(\pi/2, \pi)$, joined in a compact region entirely within the left half-plane. No negative pole or branch cut is crossed. The connecting compact pieces contribute $O(e^{-cv})$. The large-radius pieces are controlled by $e^{-cvr - c\sqrt{r}}$. Small arcs around zero disappear by (4.5.9); large closing arcs disappear by $e^{-\sqrt{z}}$ and $\operatorname{Re} z \le 0$. These bounds justify the deformation for the inverse of $z^m F(z)$ as well, for every fixed $m \ge 0$.

For $0 < r \le r_0$, writing $L = \log(1/r)$, the integrand is bounded by a constant times

$$r^m \exp\left\{-c\left(\frac{vr}{L} + \frac{1}{rL^{p+1}}\right)\right\}. \tag{4.5.10}$$

For all sufficiently large v and all $L \ge L_0$,

$$\frac{ve^{-L}}{L} + \frac{e^L}{L^{p+1}} \ge c_p \frac{\sqrt{v}}{(\log v)^{1+p/2}}. \tag{4.5.11}$$

Indeed, on $\frac{1}{4}\log v \le L \le \log v$ the arithmetic– geometric mean inequality gives the bound. Below that interval the first term is at least a constant times $v^{3/4}/\log v$. Above it the second term, eventually increasing in L, is at least $v/(\log v)^{p+1}$. Reserve half the exponent in (4.5.10) and use the other half with (4.5.11); the remaining r integral is bounded uniformly. The same estimates apply to every m, proving (4.5.6) because $q = 1 + p/2$. □

Lemma 4.5.5 (Joint realization with equal marginals)**.** *Let f and F be as in Lemma 4.5.4, and let $\psi : \mathbb{R}^D \setminus \{0\} \to (0, \infty)$ be even and C^∞. Choose a nonzero real even $h \in C_c^\infty(\mathbb{R}^D \setminus \{0\})$. Then*

$$g(a, v) = (2\pi)^{-D} \int_{\mathbb{R}^D} e^{-iu^T a} h(u)\psi(u) f(\psi(u)v)\, du \tag{4.5.12}$$

is a nonzero real Schwartz function supported in $v \ge 0$, with Fourier–Laplace transform

$$\int e^{iu^T a - zv} g(a, v)\, da\, dv = h(u) F(z/\psi(u)). \tag{4.5.13}$$

The right side is taken to be zero wherever h vanishes. Both coordinate marginals and all joint polynomial moments of g vanish. Moreover, for every $M \ge 0$ there are $C_M, c_M > 0$ such that

$$|g(a, v)| \le C_M (1 + \|a\|)^{-M} e^{-c_M \omega_q(v)}, \qquad v \ge 1. \tag{4.5.14}$$

Proof. On the compact support of h, ψ is bounded above and bounded away from zero, and all of its derivatives are bounded. Every fixed number of u derivatives of $h(u)\psi(u)f(\psi(u)v)$ is therefore bounded by a polynomial in v times $e^{-c\omega_q(v)}$. Scaling the argument by a fixed positive constant preserves this estimate. Polynomial factors can be absorbed by decreasing c. Repeated integration by parts in u gives (4.5.14); applying the same argument to all a, v derivatives proves the Schwartz assertion. Reality follows from evenness, and nonzeroness from Fourier uniqueness. Fubini and the change of variable $w = \psi(u)v$ give (4.5.13).

The transform in a vanishes in a neighborhood of $u = 0$, so its derivatives there give $\int a^\alpha g(a, v)\, da = 0$ for every multi-index α. This proves the vanishing v marginal and all joint polynomial moments. The a marginal vanishes because $F(0) = \int f = 0$; equivalently integrate the Fourier transform in v. All these differentiations and integrations are justified by (4.5.14) and the Schwartz bounds. □

Proof of the negative assertion of Theorem 4.5.1. Fix $a > 1$ and choose $1 < q < a$. Apply Lemma 4.5.4 with $b_j = 2^{-j-1}$ and then Lemma 4.5.5 with $D = 1$ and $\psi(u) = u^2$. Since g is real, nonzero, and of total integral zero, its positive and negative parts have the same strictly positive mass m. They define probability densities g_+/m and g_-/m for (A, V). The characteristic function of S_n/n, conditionally on (A, V), is $e^{iuA-u^2V/(2n)}$. The difference of the two unconditional characteristic functions is

$$m^{-1}h(u)F(1/(2n)) = 0 \qquad (n = 2^j).$$

The equal marginals and moments follow from Lemma 4.5.5. Integrating (4.5.14) in a shows that both variance marginals have density bounded by $Ce^{-c_0\omega_q(v)}$ for $v \geq 1$. Since $\omega_a(v)/\omega_q(v) \to 0$, this implies (4.5.2). Replacing V by $V+v_0$ multiplies the signed transform by $e^{-v_0 z}$ and preserves all observed zeros and equalities. The tail condition is unchanged by a fixed translation. □

As a simpler consequence, every tail condition $\mathbb{E}e^{c\Sigma^\beta} < \infty$ with $\beta < 1/2$ fails to ensure identification from an arbitrary infinite terminal set. The preceding witnesses satisfy this condition for every $c > 0$. At $\beta = 1/2$, identification holds; it even holds after weakening the exponent to $\sqrt{\Sigma}/\log(e + \Sigma)$.

4.6 Transfer to random clocks of symmetric Lévy noise

The joint realization is not specific to Gaussian kernels. It gives an exact sparse-time observation criterion for a class of random-clock models. Let $(L_t)_{t\geq 0}$ be a known symmetric Lévy process in $\mathbb{R}^D$ with

$$\mathbb{E}e^{iu^T L_t} = e^{-t\psi(u)},$$

where ψ is continuous, even, strictly positive away from zero, and C^∞ away from zero. These assumptions include Brownian motion and the isotropic symmetric stable symbols $\psi(u) = \|u\|^\alpha$, $0 < \alpha \leq 2$. The latent pair (A, V) has an arbitrary Borel law on $\mathbb{R}^D \times [0, \infty)$, independent of the driving process. Only the separate marginal laws of

$$Y_t = A + L_{tV}$$

are observed. Dependence between the latent position A and clock rate V is unrestricted. The given data are not joint trajectory laws.

Theorem 4.6.1 (An exact sparse-time criterion and its tail refinement)**.** *Let $t_j > 0$ be distinct and decrease to zero.*

1. *The collection $\{\mathcal{L}(Y_{t_j}) : j \geq 1\}$ identifies every Borel law of (A, V) if and only if*

$$\sum_j t_j = \infty.$$

   When the sum converges, distinct absolutely continuous mixing laws with identical full A and V marginals and identical finite joint polynomial moments produce the same observations.

2. *For every sequence $t_j \downarrow 0$, identification holds within each class $\mathbb{E}e^{c\omega_a(V)} < \infty$ for some $c > 0$, with $0 \leq a \leq 1$.*

3. *For the geometric times $t_j = 2^{-j}$ and every $a > 1$, the counterexamples in part 1 can satisfy $\mathbb{E}e^{c\omega_a(V)} < \infty$ for every $c > 0$. This statement holds in every spatial dimension D and for every symbol ψ just specified.*

Proof. For the difference ρ of candidate laws and fixed $u \neq 0$, set

$$H_u(z) = \int e^{iu^T a - zv}\, d\rho(a, v), \qquad \operatorname{Re} z > 0.$$

It is bounded and holomorphic, with zeros $z = t_j\psi(u)$ if the observations agree. If $\sum_j t_j = \infty$, these zeros violate the right-half-plane Blaschke condition for a nonzero bounded holomorphic function [55]. Hence H_u vanishes. Continuity gives $u = 0$; evaluating at $z = 1 + i\tau$ and using Fourier uniqueness for $e^{-v}\rho$ gives $\rho = 0$. Under the tail condition, apply Lemmas 4.5.2 and 4.5.3 to the pushforward $\int e^{iu^T a}\, d\rho(a, v)$ instead; this proves part 2.

For completeness, necessity in part 1 needs a scalar realization for every summable sequence, not only geometric sequences. Suppose $\sum_j t_j < \infty$ and set

$$F_0(z) = \left\{ \prod_j \frac{z - t_j}{z + t_j} \right\} \exp\{-\sqrt{z} - z^{-1/2}\}. \tag{4.6.1}$$

The product is nonzero, locally uniformly convergent in the right half-plane, bounded by one, and conjugation invariant. Its boundary derivatives near $iy = 0$ have at most polynomial growth in $1/|y|$. Indeed, for each factor $b_j(z) = (z - t_j)/(z + t_j)$,

$$|b_j(iy)| = 1, \qquad |b_j^{(k)}(iy)| \leq 2k! t_j |y|^{-k-1}, \quad k \geq 1.$$

The product rule and $\sum_j t_j < \infty$ bound each derivative by a finite sum of powers of $1/|y|$. The same estimates, or local uniform convergence away from zero, justify differentiated limits. Thus the exponential in (4.6.1) makes its boundary values Schwartz and flat at zero. Uniform integrable right-tube bounds follow from $|F_0(x + iy)| \leq e^{-\sqrt{|y|/2}}$. The inverse Fourier and line-shift argument used in Lemma 4.5.4 produces a nonzero real Schwartz inverse Laplace transform f_0, supported on $[0, \infty)$, with every moment zero. No exponential tail is claimed for this arbitrary summable design.

Use (4.5.12) with f_0 and the present symbol ψ. The proof of Lemma 4.5.5, without the quantitative tail estimate, gives a real nonzero joint Schwartz density with transform $h(u)F_0(z/\psi(u))$ and vanishing marginals and moments. Its positive and negative parts, normalized to probability laws, have the same observed characteristic functions, because at $z = t_j\psi(u)$ this transform is zero. This proves necessity and the supplementary assertions in part 1.

For part 3 choose $1 < q < a$, use the sector construction with zeros $b_j = 2^{-j}$, and repeat the same joint realization. Estimate (4.5.14) gives the claimed exponential integrability exactly as in the proof of Theorem 4.5.1. □

For Brownian noise this identifies a precise limitation of recovering a population's joint starting-position and diffusivity distribution from separate spatial snapshots. All coordinate marginals and all polynomial mixed moments can agree while every selected sparse snapshot agrees as well. The logarithmic tail boundary gives a condition under which this ambiguity disappears. For isotropic stable noise the same conclusion applies to a fractional diffusion clock. These are exact-data identifiability statements; no finite-sample estimation rate or stable numerical inversion is claimed.

4.6.1 Priority and unresolved transfers

The classical moment-determinacy criteria, the half-plane Blaschke condition, and Fourier–Laplace uniqueness are credited as inputs. The new claims here are their exact observation consequences together with the joint-density converse, especially its critical logarithmic tail

estimate. The bounded literature check included the primary moment-problem sources and grouped-mixture identification [31]. It did not locate the sparse random-clock criterion above, but does not certify historical priority. These results settle questions formulated within the present program; they are not described as resolutions of a located, previously published conjecture.

The graph-theoretic comparison between maximum likelihood threshold and generic completion rank remains outside these arguments. The question of a graph-independent bound on generic completion rank in terms of maximum likelihood threshold was posed by Blekherman and Sinn [34]. Neither a moment-determinacy argument nor an annihilating transform supplies the positive semidefinite stress certificates needed there. We do not count this graph problem as solved. For the matrix experiment, a converse with the precise logarithmic tail bound in each fixed dimension $d > 1$, and sharp statistical stability at the tail boundary, also remain open here.

4.7 Every modular period of a rationally independent exponential sum

The three-term restriction in the preceding manuscript can be removed. The key observation is that it is unnecessary to describe the entire polygon configuration space: configurations in which the summands form only two aligned groups already force every nonexceptional character to cover the circle. Only the largest frequency can produce a missing arc. This answers the existence question in Podesta [48, Introduction] for arbitrary finite sums satisfying its rational-independence hypothesis, and gives the stronger classification of every period. The existence part was already obtained by the nonresonant torus argument in our preceding work; the new assertion is the complete classification for any number of terms. Polygon criteria and Kronecker–Rouche arguments for real projections are classical; see Sepulcre [68, Theorems 5–6]. Historical priority of the modular classification beyond these checked sources is not asserted.

Let $m \geq 2$, and consider

$$f(s) = -1 + \sum_{j=1}^{m} B_j e^{\lambda_j s + i\phi_j}, \qquad B_j > 0, \quad 0 < \lambda_1 < \cdots < \lambda_m, \tag{4.7.1}$$

where $\lambda_1, \dots, \lambda_m$ are linearly independent over $\mathbb{Q}$. Write $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and

$$\mathcal{R}_L(f) = \overline{\{(2\pi/L)\operatorname{Im}\rho \bmod 2\pi : f(\rho) = 0,\ \operatorname{Im}\rho > 0\}} \subset \mathbb{T}.$$

Set $r_j = \lambda_j/\lambda_m$, $\kappa_j = B_j B_m^{-r_j}$ for $j < m$, and define

$$S(x) = \sum_{j<m} \kappa_j x^{r_j}, \qquad u(x) = \frac{1 + x^2 - S(x)^2}{2x}, \qquad \Theta = \arccos\Big(\max\{-1, \min_{x>0} u(x)\}\Big). \tag{4.7.2}$$

The minimum exists, and $0 < \Theta \leq \pi$.

Theorem 4.7.1 (Only the largest frequency can obstruct modular density)**.** *For* (4.7.1)*, every* $L > 0$ *has* $\mathcal{R}_L(f) = \mathbb{T}$*, except possibly*

$$L = \frac{2\pi d}{n\lambda_m}, \qquad n, d \in \mathbb{N}, \quad \gcd(n, d) = 1.$$

At such a period the exact residue closure is

$$\mathcal{R}_L(f) = \{z \in \mathbb{T} : dz + n\phi_m \in [-n\Theta, n\Theta] \pmod{2\pi}\}. \tag{4.7.3}$$

Consequently the complete exceptional set is

$$\left\{ \frac{2\pi d}{n\lambda_m} : n, d \in \mathbb{N},\ \gcd(n,d) = 1,\ n\Theta < \pi \right\}. \tag{4.7.4}$$

In particular, only finitely many integers n can occur. The classification and every residue closure are independent of $\phi_1, \dots, \phi_{m-1}$. If $n\Theta < \pi$, the closure in (4.7.3) consists of d closed arcs, whose total relative circle length is $n\Theta/\pi$.

Lemma 4.7.2 (Realization of compatible polygon configurations)**.** *Put*

$$\mathcal{Z} = \{(\sigma, \theta) \in \mathbb{R} \times \mathbb{T}^m : \textstyle\sum_j B_j e^{\lambda_j \sigma + i\theta_j} = 1\}.$$

This is a nonempty compact set. For $q = 2\pi/L$, if $q \notin \operatorname{span}_{\mathbb{Q}}\{\lambda_j\}$, then $\mathcal{R}_L(f) = \mathbb{T}$. Otherwise write

$$q = (k \cdot \lambda)/d, \qquad k \in \mathbb{Z}^m, \quad d \geq 1, \quad \gcd(k_1, \dots, k_m, d) = 1.$$

Then

$$\mathcal{R}_L(f) = \{z \in \mathbb{T} : dz + k \cdot \phi \in \{k \cdot \theta : (\sigma, \theta) \in \mathcal{Z}\}\}. \tag{4.7.5}$$

Every residue on the right is approached by ordinates tending to $+\infty$. Multiple limiting zeros are allowed.

Proof. For sufficiently negative σ, the sum of all lengths is less than 1; for sufficiently positive σ, the last length exceeds 1 plus all the others. Neither region contains a configuration. This proves compactness. The sum of the lengths is strictly increasing from zero to infinity, so there is a unique σ_0 where its value is 1; the all-zero angle vector is then a configuration.

The orbit $t \mapsto (t\lambda, qt)$ has closure $\mathbb{T}^{m+1}$ in the nonresonant case. In the resonant case its character relations are exactly the multiples of $(k, -d)$, so its closure is the subtorus $\{(v, z) : k \cdot v - dz = 0\}$. This is the character form of Kronecker's theorem. Removing any bounded initial part of the orbit does not change its closure. Given a configuration and a compatible residue, choose $t_\nu \to +\infty$ such that $t_\nu \lambda \to \theta - \phi$ and $qt_\nu \to z$ on the tori. Locally uniformly,

$$f(s + it_\nu) \longrightarrow -1 + \sum_j B_j e^{\lambda_j s + i\theta_j} =: f_*(s).$$

The limit has a zero at σ and is not identically zero, since its constant term is -1 and all its exponents are positive. Rouche's theorem, on successively smaller circles around this isolated zero, therefore supplies zeros $s_\nu \to \sigma$ along a subsequence. The zeros $s_\nu + it_\nu$ have the required residues. Conversely, every actual zero defines a configuration satisfying the character relation. Compactness proves the converse containment after taking closures. □

Lemma 4.7.3 (Two aligned groups exhaust every nonextreme character)**.** *If $k \in \mathbb{Z}^m \setminus \{0\}$ is not supported solely at the index m, then*

$$\{k \cdot \theta : (\sigma, \theta) \in \mathcal{Z}\} = \mathbb{T}.$$

Proof. For a nonempty proper subset $A \subset \{1, \dots, m\}$ set

$$U(\sigma) = \sum_{j \in A} B_j e^{\lambda_j \sigma}, \quad V(\sigma) = \sum_{j \notin A} B_j e^{\lambda_j \sigma}, \quad K_A = \sum_{j \in A} k_j, \quad K_{A^c} = \sum_{j \notin A} k_j.$$

Whenever $|U - V| \leq 1 \leq U + V$, the aligned-group configurations are obtained from

$$\alpha = \arccos \frac{1 + U^2 - V^2}{2U}, \qquad \beta = -\arccos \frac{1 + V^2 - U^2}{2V},$$

by setting $\theta_j = \alpha$ on A and $\theta_j = \beta$ on A^c, or by conjugating all angles. Their character is $K_A\alpha + K_{A^c}\beta$.

Suppose first that k has at least two nonzero entries. There is an index j for which $k_j \neq 0$ and $k_j \neq \sum_i k_i$. Indeed, if the total is zero every nonzero entry works; if the total is nonzero and every nonzero entry equals it, there can only be one nonzero entry. Take $A = \{j\}$. Both K_A and K_{A^c} are nonzero. Starting at the unique σ_0 with $U + V = 1$, let $\sigma_1 > \sigma_0$ be the first point where $|U - V| = 1$. It exists because the group containing the largest exponent eventually dominates. On $[\sigma_0, \sigma_1]$ the angles are continuous, start at $(0, 0)$, and end at either $(\pi, 0)$ or $(0, -\pi)$. The character therefore joins zero to a nonzero integral multiple of π. Its image and the conjugate image together cover the circle.

It remains to consider k supported at one $j < m$. Again take $A = \{j\}$; now $V - U \to +\infty$. Let $\sigma_1 > \sigma_0$ be the first point where $V - U = 1$. If there is a point in $[\sigma_0, \sigma_1)$ where $U - V = 1$, let σ_2 be the last such point; otherwise put $\sigma_2 = \sigma_0$. The triangle inequalities hold on $[\sigma_2, \sigma_1]$, and α runs continuously from 0 to π. The character $k_j\alpha$, together with its negative, again covers the circle. This proves the lemma. □

Lemma 4.7.4 (The exact range of the largest angle)**.** *The projection of $\mathcal{Z}$ onto θ_m is exactly $[-\Theta, \Theta] \pmod{2\pi}$, with (4.7.2).*

Proof. Put $x = B_m e^{\lambda_m \sigma}$. The other lengths sum to $S(x)$. A necessary condition for a prescribed angle $\eta = \theta_m$ is

$$|1 - xe^{i\eta}| \le S(x) \quad \text{for some } x > 0. \tag{4.7.6}$$

It is also sufficient after allowing the real part to vary. The continuous difference $|1 - xe^{i\eta}| - S(x)$ tends to 1 as $x \to 0$ and to $+\infty$ as $x \to \infty$, since $0 < r_j < 1$. If it is nonpositive anywhere, it vanishes somewhere. At such a point all the smaller summands can be aligned with the nonzero vector $1 - xe^{i\eta}$, giving the required configuration. Thus no lower polygon-radius condition has been discarded.

Squaring (4.7.6) gives $\cos\eta \ge u(x)$. The function u tends to $+\infty$ at both ends and $u(1) = 1 - S(1)^2/2 < 1$. The stated arc and $0 < \Theta \le \pi$ follow. □

Proof of Theorem 4.7.1. Apply Lemma 4.7.2. A resonant character not supported at m covers the circle by Lemma 4.7.3. A character supported at m has $k_m = n > 0$, since $q > 0$, and $\gcd(n, d) = 1$. Lemma 4.7.4 then gives (4.7.3). The interval $[-n\Theta, n\Theta]$ covers the circle exactly when $n\Theta \ge \pi$. If it is shorter, its inverse image under multiplication by d has d components of equal length $2n\Theta/d$. All assertions follow. □

Corollary 4.7.5 (Each resonant decision is a strictly concave scalar problem)**.** *Let $n \ge 1$, $c_n = \cos(\pi/n)$ and*

$$R_n(x) = \sqrt{1 + x^2 - 2c_n x}, \qquad D_n(x) = S(x) - R_n(x).$$

The function D_n is strictly concave on $(0, \infty)$ and has a unique maximizer x_n, characterized by

$$\sum_{j<m} r_j \kappa_j x_n^{r_j - 1} = \frac{x_n - c_n}{\sqrt{1 + x_n^2 - 2c_n x_n}}. \tag{4.7.7}$$

Every period $2\pi d/(n\lambda_m)$ in lowest terms is dense if and only if $D_n(x_n) \ge 0$. In particular, all periods are dense if and only if

$$\max_{x>0} \{S(x) - x\} \ge 1. \tag{4.7.8}$$

The unique maximizer in (4.7.8) solves $S'(x) = 1$.

Proof. The criterion $n\Theta \geq \pi$ is equivalent to $S(x) \geq R_n(x)$ for some x. Because $S'' < 0$ and R_n is convex, $D_n'' < 0$; for $n = 1$, $R_1(x) = 1 + x$ and the same conclusion holds. Moreover $D_n'(x) \to +\infty$ as $x \to 0$ and $D_n'(x) \to -1$ as $x \to \infty$. Hence its unique critical point is the unique maximum, giving (4.7.7). The case $n = 1$ proves (4.7.8) and the last assertion. □

4.7.1 An infinite spectrum without a largest frequency has no exceptional period

Theorem 4.7.6 (Universal modular density for normally convergent infinite sums)**.** *Let* $\{\lambda_j : j \geq 1\} \subset (0, \infty)$ *be linearly independent over* $\mathbb{Q}$ *and have no largest member. Suppose* $B_j > 0$ *and*

$$\sum_{j \geq 1} B_j e^{R\lambda_j} < \infty \qquad \text{for every } R > 0. \tag{4.7.9}$$

For arbitrary real phases ϕ_j*, the entire function*

$$f(s) = -1 + \sum_{j \geq 1} B_j e^{\lambda_j s + i\phi_j}$$

has positive zero ordinates dense modulo every $L > 0$*. In particular, this holds for the smoothed prime sum*

$$-1 + \sum_{p \text{ prime}} e^{-p} p^s. \tag{4.7.10}$$

Proof. Condition (4.7.9) gives locally uniform convergence, uniformly over all choices of coefficient phases. The positive length sum $M(\sigma) = \sum_j B_j e^{\lambda_j \sigma}$ is continuous and strictly increasing, tends to zero at $-\infty$ by dominated convergence, and tends to infinity at $+\infty$. Hence there is again a unique σ_0 with $M(\sigma_0) = 1$.

The continuous-time orbit on the countable phase torus, augmented by qt for $q = 2\pi/L$, has the same character description as in Lemma 4.7.2: its continuous characters involve only finitely many coordinates. It is either the entire product torus, or the subtorus with the single primitive relation $k \cdot v - dz = 0$, where k has finite support and $q = k \cdot \lambda/d$. This follows by applying the finite-dimensional Kronecker theorem to each finite collection of coordinates and then using the product topology. Every point can be approached at times tending to $+\infty$. Coordinatewise phase convergence implies local uniform convergence of the entire functions, by the uniform tail bound in (4.7.9). The same Rouche argument therefore realizes each compatible polygon configuration. The limiting function is not identically zero, since it tends to -1 on the negative real axis.

The nonresonant case follows at once from the configuration at σ_0. In the resonant case the proof of Lemma 4.7.3 still applies. When k has at least two nonzero entries choose its singleton group exactly as there. When it has only one, choose that singleton group. Its length U is one exponential, whereas the complementary group V contains a term with a strictly larger frequency, so $V - U \to +\infty$. The two continuous triangle-angle paths used in that lemma are therefore available in both cases, and every nonzero finite character covers the circle. This proves density for every period. Finally, $\{\log p\}$ is rationally independent by unique factorization, has no largest element, and $\sum_p e^{-p} p^R < \infty$ for every R. Thus (4.7.10) satisfies the hypotheses. □

This theorem removes the finite-sum exceptional direction by removing the largest frequency itself. Its normal-convergence assumption is essential to the argument; it does not cover an arbitrary Dirichlet series converging only on a half-plane. In particular the smoothed prime example is not the Riemann zeta function or an Euler product.

4.8 Recovering an unknown spectrum from density transitions

The modular classification permits a different inverse problem from phase recovery at an Euler edge. Here the relative frequencies are unknown, while an external amplitude can be varied. The observation is only the transition between dense and nondense modular ordinates. The following problem is formulated here; it is not presented as a second previously published conjecture.

Fix $b > 0$ and distinct unknown numbers $0 < r_1 < \cdots < r_{m-1} < 1$ such that $br_1, \dots, br_{m-1}, b$ are rationally independent. For $A, C > 0$ consider

$$f_{A,C}(s) = -1 + A \sum_{j<m} B_j e^{br_j s + i\phi_j} + Ce^{bs + i\phi_m}, \qquad B_j > 0. \tag{4.8.1}$$

The normalizations of A, C and the constant coefficient are part of the model. Observe, for each C, the smallest common amplitude A at which every period has dense ordinates:

$$T(C) = \inf\{A > 0 : \mathcal{R}_L(f_{A,C}) = \mathbb{T} \text{ for every } L > 0\}.$$

Let $S_0(x) = \sum_{j<m} B_j x^{r_j}$.

Theorem 4.8.1 (Exact spectral recovery from a binary transition curve)**.** *The transition is attained, and*

$$T(C) = \min_{x>0} \frac{1 + Cx}{S_0(x)}, \qquad \mathcal{R}_L(f_{A,C}) = \mathbb{T} \text{ for every } L > 0 \iff A \geq T(C). \tag{4.8.2}$$

Its entire curve determines the positive power sum by the exact inverse

$$S_0(x) = \inf_{C>0} \frac{1 + Cx}{T(C)}. \tag{4.8.3}$$

It therefore determines the number of smaller frequencies, all ratios r_j, and all magnitudes B_j, up to relabeling. The curve is smooth, strictly increasing and strictly concave. If $x(C)$ is the unique minimizer in (4.8.2), then

$$x(C) = \frac{T'(C)}{T(C) - CT'(C)}, \qquad S_0(x(C)) = \frac{1}{T(C) - CT'(C)}. \tag{4.8.4}$$

As C ranges over $(0, \infty)$, $x(C)$ ranges bijectively over $(0, \infty)$ in the opposite direction.

Proof. The all-period criterion says that $AS_0(x) \geq 1 + Cx$ for some x. The quotient in (4.8.2) tends to infinity at both ends, so its infimum is attained. This proves the transition formula. For every $C, x > 0$ it immediately gives

$$S_0(x) \leq \frac{1 + Cx}{T(C)}.$$

To obtain equality at a prescribed x, put

$$H(x) = S_0(x) - xS_0'(x) = \sum_{j<m} (1 - r_j) B_j x^{r_j} > 0, \qquad C_x = \frac{S_0'(x)}{H(x)}.$$

Strict concavity gives, for every $y > 0$,

$$S_0(y) \leq S_0(x) + S_0'(x)(y - x) = H(x)(1 + C_x y),$$

with equality only at $y = x$. It follows that $T(C_x) = 1/H(x)$ and that x is its unique minimizing point. This proves (4.8.3). Direct differentiation yields

$$C'_x = \frac{S_0''(x)S_0(x)}{H(x)^2} < 0.$$

The smallest and largest power terms show respectively that $C_x \sim r_1/((1-r_1)x)$ as $x \to 0$ and $C_x \sim r_{m-1}/((1-r_{m-1})x)$ as $x \to \infty$. Thus $x \mapsto C_x$ is a smooth decreasing bijection. The envelope identity gives

$$T'(C) = \frac{x(C)}{S_0(x(C))} > 0, \qquad T(C) - CT'(C) = \frac{1}{S_0(x(C))} > 0.$$

Also the derivative of $x/S_0(x)$ with respect to x is $H(x)/S_0(x)^2 > 0$, while $x'(C) < 0$, proving $T''(C) < 0$ and (4.8.4).

Finally, a finite positive power sum has a unique representation with distinct exponents. Explicitly,

$$r_1 = \lim_{x \downarrow 0} \frac{\log S_0(x)}{\log x}, \qquad B_1 = \lim_{x \downarrow 0} x^{-r_1} S_0(x).$$

Subtract $B_1 x^{r_1}$ and repeat until the remainder is zero. This establishes uniqueness of the term count, exponents and coefficients; it makes no finite-data numerical claim about detecting the last term. □

The inverse is a supporting-line reconstruction of a strictly concave function. This convex-duality step is classical; the specific content here is that the observable $T(C)$ is exactly the modular-zero density transition, so it determines an unknown finite spectrum.

Proposition 4.8.2 (Relative stability of the recovered power-sum profile)**.** *Suppose $0 < \epsilon < 1$ and a positive observed curve satisfies $(1-\epsilon)T(C) \le \widetilde{T}(C) \le (1+\epsilon)T(C)$ for all $C > 0$. Define $\widetilde{S}(x) = \inf_{C>0}(1+Cx)/\widetilde{T}(C)$. Then*

$$\frac{S_0(x)}{1+\epsilon} \le \widetilde{S}(x) \le \frac{S_0(x)}{1-\epsilon} \qquad (x > 0).$$

If the curve is observed only on $[C_-, C_+]$, the same bounds hold for the infimum restricted to that interval at every x with $C_x \in [C_-, C_+]$.

Proof. Apply the pointwise multiplicative inequalities to $(1+Cx)/\widetilde{T}(C)$ and take infima. On a restricted interval the true infimum is still $S_0(x)$ whenever its tangent value C_x belongs to the observed interval. □

Proposition 4.8.3 (An exact phase blind spot)**.** *Even knowing all closures $\mathcal{R}_L(f_{A,C})$, for every $A, C, L > 0$, cannot identify any of the phases $\phi_1, \ldots, \phi_{m-1}$ when B_j, r_j, b, ϕ_m are fixed.*

Proof. Theorem 4.7.1 expresses every such closure in terms of the magnitudes, frequencies and ϕ_m alone. Arbitrarily changing the remaining phases leaves all these data unchanged. □

This is a statement about closures, not equality of the zero divisors, the empirical residue distributions, or zero correlations. The stable reconstruction above concerns S_0; extracting closely spaced exponents or very small coefficients from noisy restricted data is a separate inverse problem. Neither theorem applies directly to an infinite Euler product: it has no largest frequency, and its prime-power frequencies are not rationally independent. The actual Euler hyperuniformity estimate and the exact-prime recurrence exponent remain unproved in this branch. The two completed targets here are the finite-sum all-period classification and the newly formulated transition-curve inverse problem; the scalar test and phase blind spot are consequences rather than additional conjecture closures.

4.9 A common analytic envelope: exact information versus noisy information

This section addresses two problems left separate by the earlier analytic instability construction. Does a *fixed* complex-neighborhood bound yield a quantitative finite-sample recovery rate? Does that same bound yield a noise rate uniform over all sequences of sampling times? The answers are respectively yes and no. Exact-data recovery has uniform exponential order, while noise convergence can be arbitrarily slow along one fixed sampling sequence, without changing the analytic class. These are internally formulated inverse problems. No previously named open conjecture is claimed to be resolved here.

Fix $I = [a, b]$ with $0 < a < b$, put $\ell = b - a$, and fix $\rho > 1$. Let E_ρ be the open Bernstein ellipse with foci a, b and parameter ρ: its boundary is the image of $|w| = \rho$ under

$$w \longmapsto \frac{a+b}{2} + \frac{\ell}{4}(w + w^{-1}).$$

Write $\|f\|_{E_\rho} = \sup_{z \in E_\rho} |f(z)|$. For positive constants m, M, κ let $\mathcal{A}_\rho$ be the class of functions holomorphic in E_ρ, real on I, satisfying

$$\int_I f = 1, \qquad f \geq m, \qquad (\log f)'' \leq -\kappa \quad \text{on } I, \qquad \|f\|_{E_\rho} \leq M. \tag{4.9.1}$$

The functions are probability densities on I; no analyticity across the endpoints is claimed for their zero extensions. The common bound in (4.9.1), unlike finitely many derivative bounds, controls *every* derivative uniformly on I. Use inverse-time coordinates

$$H_s f = \sqrt{s/\pi} \int_I e^{-sx^2} f(x)\, dx, \qquad s > 0, \tag{4.9.2}$$

so that $H_s f$ is the heat observation at location zero and time $t = 1/(4s)$.

4.9.1 Exponential exact-data recovery, uniformly over the times

For distinct $s_1, \ldots, s_n > 0$, let $e_n(s_1, \ldots, s_n)$ be the smallest worst-case $L^1(I)$ error over all reconstructions from the n exact real numbers $H_{s_j} f$. No computational restriction is imposed on a reconstruction.

Theorem 4.9.1 (Uniform exponential order)**.** *For every a, b, ρ, κ as above, there exist $m, M, c > 0$ and $0 < \vartheta < 1$, depending only on these four parameters, such that*

$$\frac{c\rho^{-(n+1)}}{(n+2)^2} \;\leq\; e_n(s_1, \ldots, s_n) \;\leq\; 2M\ell\,\vartheta^{n+1} \tag{4.9.3}$$

for every $n \geq 1$ and every collection of distinct positive sampling parameters. In particular $e_n = \exp(-\Theta(n))$, with constants independent of the sampling parameters.

The lower bound also holds for every deterministic procedure using at most n adaptively selected real linear measurements on the unknown density, even if those measurements are not heat observations and their selection depends discontinuously on previous exact answers.

Proof. We give the upper and lower arguments separately.

Sign changes from the data. Suppose that $f, g \in \mathcal{A}_\rho$ have the same n observations, and write $h = f - g$. Then h is orthogonal in the ordinary integral pairing to

$$1, e^{-s_1 x^2}, \ldots, e^{-s_n x^2}.$$

This is an extended Chebyshev system on I. To see this directly, make the strictly increasing change of variable $y = x^2$ and order the distinct exponents $0, -s_1, \dots, -s_n$. Any nonzero combination of $r+1$ exponentials with distinct real exponents has at most r zeros, counted with multiplicity: divide by one exponential, differentiate, and apply Rolle's theorem inductively.

If h is nonzero and has at most n sign changes, say r changes at $x_1 < \cdots < x_r$, choose a nonzero combination v of any $r+1$ of the listed functions vanishing at those r points. The zero bound shows that these zeros are simple and are its only zeros on I. After changing its overall sign, $hv \geq 0$ everywhere and $hv > 0$ on an open subinterval. This contradicts $\int_I hv = 0$. Consequently a nonzero h has at least $n+1$ distinct real zeros in I.

Zeros in a common analytic neighborhood. Let $\phi : E_\rho \to \mathbb{D}$ be a conformal map and set

$$\vartheta = \max_{x,y \in I} \left| \frac{\phi(x) - \phi(y)}{1 - \overline{\phi(y)}\phi(x)} \right|.$$

Compact containment gives $0 < \vartheta < 1$. The bounded holomorphic function $h/(2M)$, transferred to the disk, has the $n+1$ zeros just found. Dividing by their finite Blaschke product leaves a function bounded by one. Equivalently, repeated Schwarz factorization gives

$$|h(x)| \leq 2M \prod_{k=1}^{n+1} \left| \frac{\phi(x) - \phi(x_k)}{1 - \overline{\phi(x_k)}\phi(x)} \right| \leq 2M\vartheta^{n+1}, \qquad x \in I.$$

Integration bounds the diameter of every exact-data fiber. Selecting any member of a nonempty fiber proves the asserted upper bound.

An analytic perturbation ball. Normalize $f_0(x) = C\exp(-\kappa x^2)$ to have integral one on I. Then $(\log f_0)'' = -2\kappa$. Cauchy estimates on disks of a fixed radius around I show that there is $\epsilon > 0$ such that every real holomorphic h satisfying

$$\|h\|_{E_\rho} \leq \epsilon, \qquad \int_I h = 0$$

has $f_0 \pm h \geq m$ and $(\log(f_0 \pm h))'' \leq -\kappa$, where $m > 0$ is fixed. Indeed the $C^2(I)$ norm of h is at most a fixed multiple of ϵ, and the strict curvature and positivity reserves of f_0 persist under sufficiently small C^2 perturbations. Choose $M > \|f_0\|_{E_\rho} + \epsilon$. Thus every such perturbation belongs to the same class (4.9.1).

A polynomial null direction. For fixed measurements, choose a nonzero real polynomial p of degree at most $d = n+1$ annihilated by all n measurements and by integration. There are at most $n+1$ homogeneous linear constraints on a space of dimension $n+2$. Normalize $\|p\|_{I,\infty} = 1$. The polynomial growth inequality

$$\|p\|_{E_\rho} \leq \rho^d \tag{4.9.4}$$

follows from the maximum principle applied outside the unit disk to $w^{-d}p((a+b)/2 + \ell(w + w^{-1})/4)$. Also

$$\|p\|_{L^1(I)} \geq \frac{\ell}{(d+1)^2}. \tag{4.9.5}$$

For a short proof, expand in the orthonormal shifted Legendre polynomials. Their reproducing kernel has diagonal at most $(d+1)^2/\ell$, since $|P_k(x)| \leq 1$ on $[-1,1]$. Hence $1 \leq (d+1)\|p\|_2/\sqrt{\ell}$; combining this with $\|p\|_2^2 \leq \|p\|_\infty \|p\|_1 = \|p\|_1$ proves the claim.

Set $h = \epsilon\rho^{-d}p$. The pair $f_0 \pm h$ is admissible, has identical data, and has L^1 distance at least $2\epsilon\ell\rho^{-(n+1)}/(n+2)^2$. The triangle inequality proves the lower bound with $c = \epsilon\ell$.

For adaptive linear measurements, run the procedure at f_0 first and record the at most n linear functionals selected along that transcript. Choose p annihilating those functionals and integration. An induction over the query index shows that $f_0 + h$, $f_0 - h$, and f_0 generate exactly the same transcript and therefore the same measurement choices. This proof makes no continuity assumption on the selection rule. □

Interpretation and boundary. The exponent constants in (4.9.3) are not matched; "exponential order" means upper and lower bounds $\exp(-c_1 n)$ and $\exp(-c_2 n)$, not a sharp exponential constant. The upper bound is uniform even when observations are taken at extraordinarily small or nearly identical positive times. It uses exact real numbers. It is not a numerically stable reconstruction algorithm. The lower bound addresses linear adaptive information; it does not settle the problem of arbitrary adaptive analytic functionals posed by Krieg–Novak–Ullrich [60, Section 3].

4.9.2 A quantitative noise obstruction inside the same analytic class

For an increasing sequence $\mathbf{s} = (s_j)_{j\ge 1}$ tending to infinity, define the deterministic minimax risk

$$R_{\mathbf{s}}^{\infty}(\delta) = \inf_{\widehat{f}} \sup_{f\in\mathcal{A}_\rho} \sup_{\|e\|_{\ell^\infty}\le\delta} \|\widehat{f}((H_{s_j} f + e_j)_{j\ge 1}) - f\|_1.$$

Define $R_{\mathbf{s}}^{G}(\delta)$ similarly using expected loss and independent $N(0,\delta^2)$ errors in all the coordinates. The estimator receives the entire infinite noisy sequence in both definitions.

Theorem 4.9.2 (Tail-controlled minimax lower bounds)**.** *Use the fixed class from Theorem 4.9.1. Suppose $s_1 \ge 1$ and $s_{j+1} \ge s_j + 1$. There is a constant $C_a > 0$ such that, for*

$$\delta_n = C_a \sqrt{s_{n+1}+1}\, e^{-a^2 s_{n+1}}, \tag{4.9.6}$$

and all sufficiently large n,

$$\min\{R_{\mathbf{s}}^{\infty}(\delta_n), R_{\mathbf{s}}^{G}(\delta_n)\} \ge \frac{c\rho^{-(n+1)}}{2(n+2)^2}. \tag{4.9.7}$$

The constant C_a can be chosen independently of the sequence and n; c is the constant of Theorem 4.9.1.

Proof. Use the polynomial construction in the preceding proof to obtain $f_n^{\pm} = f_0 \pm h_n$ with identical first n observations and separation

$$D_n := \|f_n^+ - f_n^-\|_1 \ge \frac{2c\rho^{-(n+1)}}{(n+2)^2}.$$

Both are probability densities, so $D_n \le 2$. For $j > n$,

$$|H_{s_j}(f_n^+ - f_n^-)| \le 2\sqrt{s_j/\pi}\, e^{-a^2 s_j}.$$

Writing $S = s_{n+1}$, unit spacing and eventual monotonicity of $ue^{-2a^2 u}$ give

$$\sum_{j>n} s_j e^{-2a^2 s_j} \le \sum_{k=0}^{\infty} (S+k) e^{-2a^2(S+k)} \le C_a'(S+1)e^{-2a^2 S}.$$

Increase C_a in (4.9.6) so that the entire mean-vector difference has ℓ^2 norm at most δ_n, and hence supremum norm at most δ_n. For adversarial noise the midpoint of the two mean sequences is compatible with both laws, giving risk at least $D_n/2$. For Gaussian noise the two product Gaussian measures have relative entropy at most $1/2$; their mean difference is in ℓ^2, so the usual finite-dimensional identity passes to the product by the convergent Gaussian likelihood ratio. Pinsker's inequality gives total variation at most $1/2$. Integration against the common part of the measures gives risk at least $D_n/4$, which is the displayed lower bound. □

Corollary 4.9.3 (Geometric sampling rules out a Hölder noise modulus)**.** *If $s_j = q^j$ with $q > 1$, then for all sufficiently small δ,*

$$\min\{R_{\mathbf{s}}^{\infty}(\delta), R_{\mathbf{s}}^{G}(\delta)\} \geq c_q \frac{[\log(1/\delta)]^{-\log \rho/\log q}}{[\log\log(1/\delta)]^2}.$$

Consequently no positive power of δ bounds either minimax risk on this class. The displayed logarithmic exponent is a lower-bound exponent; matching upper bounds are not asserted.

Proof. A finite initial segment may be discarded so that the spacing condition holds. The proof of Theorem 4.9.2 itself is unchanged by that initial segment, which remains among the annihilated coordinates. For $L = \log(1/\delta)$ choose the smallest n such that $q^{n+1} \geq 2L/a^2$. Then $\delta_n \leq \delta$ for large L and $n + 1 \leq \log_q L + C_{a,q}$. Apply the two-point argument at noise level δ; its mean difference is at most δ in ℓ^2. Substitution in (4.9.7) proves the result. □

Theorem 4.9.4 (Arbitrarily slow consistency with a fixed analytic envelope)**.** *Fix the class $\mathcal{A}_\rho$ of Theorem 4.9.1 once and for all. For every function $\omega : (0, 1) \to (0, \infty)$ with $\omega(\delta) \to 0$ as $\delta \downarrow 0$, there exists one sequence $s_j \uparrow \infty$, with $s_{j+1} \geq 2s_j$, such that*

$$\limsup_{\delta\downarrow 0} \frac{R_{\mathbf{s}}^{\infty}(\delta)}{\omega(\delta)} = \limsup_{\delta\downarrow 0} \frac{R_{\mathbf{s}}^{G}(\delta)}{\omega(\delta)} = \infty.$$

Nevertheless, for this same sequence,

$$R_{\mathbf{s}}^{\infty}(\delta) \longrightarrow 0, \qquad R_{\mathbf{s}}^{G}(\delta) \longrightarrow 0.$$

In particular $t_j = 1/(4s_j)$ is summable, the noiseless observation map is injective, and there is no noise modulus uniform over all such sampling sequences, even under a common full analytic envelope and strong log-concavity.

Proof. Recursively choose $s_{n+1} \geq 2s_n$ so large that

$$\omega\left(C_a\sqrt{s_{n+1}+1}e^{-a^2 s_{n+1}}\right) \leq \frac{c\rho^{-(n+1)}}{2n(n+2)^2}.$$

This is possible because the argument tends to zero. Theorem 4.9.2 gives a risk ratio at least n along the corresponding sequence $\delta_n \downarrow 0$, for both observation models.

It remains to prove the consistency assertions rather than infer them from membership in an analytic class. The class $\mathcal{A}_\rho$ is compact in $C^2(I)$ by normal-family compactness and Cauchy estimates. The constraints in (4.9.1) are closed under that convergence; a limiting holomorphic function on E_ρ still has bound M. In particular the class is compact in $L^1(I)$. The complete map $f \mapsto (H_{s_j} f)_{j\geq 1}$ is injective, because Theorem 4.9.1 bounds the diameter of every first-n fiber by a quantity tending to zero.

For any $\eta > 0$, compactness and this injectivity give a finite J and $r > 0$ such that

$$\max_{1\leq j\leq J} |H_{s_j} f - H_{s_j} g| \leq r \quad \Longrightarrow \quad \|f - g\|_1 < \eta. \tag{4.9.8}$$

Otherwise pairs separated by at least η and converging in their first J data coordinates for every growing J would have convergent subsequences with different limits and identical full data, a contradiction. For deterministic noise, choose any class member fitting the first J observations within δ; if $2\delta \leq r$, its error is less than η. For Gaussian noise, minimize the maximum residual over the first J coordinates on the compact class (a measurable approximate minimizer also suffices). On the event $\max_{j\leq J} |Z_j| \leq r/(3\delta)$, its data discrepancy from the truth is at most r if the optimization error is at most $r/3$. The event's complement has probability tending to zero as $\delta \downarrow 0$. The loss is at most two, since both outputs and truths are densities, so the expected risk tends to at most η. Let $\eta \downarrow 0$. □

What has changed from the earlier barrier. Previously a positive noise-risk floor was obtained by allowing the full analytic envelope to vary. Here the entire class, the complex neighborhood, its norm bound, the density lower bound, and its logarithmic curvature are fixed. Every fixed sampling sequence is consistent, yet a sequence can force convergence slower than any prescribed candidate rate. At the same time *every* sequence has exponential exact-data recovery in the number of distinct samples. The distinction is between exact real-number information and accuracy of that information, rather than a failure of analyticity or identifiability.

Unresolved directions and prior-art boundary. The arguments combine classical Chebyshev-system oscillation, Schwarz–Blaschke factorization, polynomial growth, and two-point statistical lower bounds. Priority for this particular combined statement has not been established. The sharp noise rate even for $s_j = q^j$ remains open in this work, as does the optimal exponential constant in Theorem 4.9.1. A second whole exponentially weighted diffraction curve is not counted as a new discovery: parallel-line phase-retrieval theory already supplies closely related uniqueness mechanisms; see Wellershoff [69]. The adaptive analytic-functional question of [60, Section 3] remains unresolved: our adaptive lower bound uses linearity essentially, and does not extend by merely replacing "linear" with "analytic".

4.10 An eighteen-target register and the remaining barriers

Here "complete" means that the precisely scoped statement has a proof in this manuscript. It does not mean that every broader problem bearing the same name has been resolved. "Partial" records a proved range or one-sided bound. "Open" records a transfer for which the attempted mechanism does not supply a proof. Supporting lemmas are not separate targets.

No.	Target	Status	Exact outcome and provenance
1	Near-linear binomial-mixture MLE	Complete	Theorem 4.2.1; all mixing laws and all MLE solutions, for every fixed exponent below one. Reaches the published conditional range in [14, Section 6.2].
2	Simultaneous bounded-test fingerprint control	Complete, negative	Proposition 4.2.8; the uniform-over-tests strengthening fails. This is not a counterexample to the original fixed deterministic test conjecture.
3	Predetermined Bernoulli replication caps	Complete	Theorem 4.3.1; matching upper and lower bounds for all $1 \le Q \le B$. Internal minimax problem, arbitrary latent laws.
4	Fully adaptive Bernoulli caps	Partial	Theorem 4.3.2; exact for $Q \le B^{2/3-\gamma}$, but not throughout the shrinking critical window near $B^{2/3}$.
5	Uniform-endpoint critical budget rate	Complete	Theorem 4.3.6; the logarithm is necessary even with fully adaptive allocation. Closes an internal critical-rate gap.
6	Maximal ID domain of the third McKay family	Complete	Theorem 4.4.3; the entire admissible domain. Directly addresses the question in [3, Section 2.3].
7	Full real-order fourth McKay ID domain	Complete	Theorem 4.4.6; every $\mu > -1/4$. An extension of the published family and theorem, not separately labeled as a named conjecture.
8	Universal strictness of the SD thresholds	Partial	Exact spectral criteria and tail bounds are proved in Corollaries 4.4.4, 4.4.7 and Proposition 4.4.8. Strictness above one in all negative-cut intervals is not established.
9	Logarithmic covariance-tail boundary	Complete	Theorem 4.5.1; sufficiency in all dimensions, sharpness for the scalar dyadic experiment, exactly within the stated log-power scale. Internal observation problem.

No.	Target	Status	Exact outcome and provenance
10	Sharp converse with matrix covariance in fixed $d > 1$	Open	The scalar construction does not preserve the extra anchored observations. No matrix-dimensional converse is claimed.
11	Sparse random-clock Lévy snapshots	Complete	Theorem 4.6.1; the sum-of-times criterion, and the logarithmic tail boundary for the stated symbols and scalar clocks. Internal identification problem.
12	All modular periods of exponential sums	Complete	Theorems 4.7.1 and 4.7.6; complete finite-sum classification and a normally convergent infinite extension. The earlier existence answer to [48] is not counted again.
13	Spectrum from a modular-density transition	Complete	Theorem 4.8.1; exact continuum data recover the finite positive power sum. A newly formulated inverse problem; lower-frequency phases remain invisible.
14	Transfer to Euler-product fluctuation and prime recurrence laws	Open	Prime-power relations and half-plane convergence invalidate the independent-frequency transfer. Neither Euler hyperuniformity nor an exact-prime recurrence exponent follows.
15	Exact heat information under one analytic envelope	Complete	Theorem 4.9.1; uniform exponential order, with a lower bound also for deterministic adaptive linear measurements. The two exponential constants need not match.
16	A sampling-uniform analytic noise modulus	Complete, negative	Theorem 4.9.4; one fixed analytic class admits no common rate over all sampling sequences, while each fixed sequence remains consistent. Internal minimax problem.
17	Sharp noisy rate for geometric heat times	Partial	Corollary 4.9.3; a logarithmic lower bound rules out Hölder stability. A matching sharp upper bound is not proved.
18	Nonlinear continuous-measurement transfer	Open	The adaptive linear annihilator does not establish the distinct continuous nonlinear measurement claims studied in [60]. No theorem in that paper is strengthened here by assertion.

4.10.1 Three concrete next obstructions

The fully adaptive cap window. At $Q = B^{2/3}/\log B$, the predetermined rate is $B^{-1/3}\sqrt{\log B/\log\log B}$, while the available adaptive lower bound is only $cB^{-1/3}$. The fixed-design lower proof charges whole groups using local moment matching. A fully adaptive policy may revisit individuals selected by earlier outcomes. An exact extension would require a conditional information charge stable under this selection, or an explicit adaptive construction beating the predetermined rate. The endpoint experiment supplies such a charge for a different likelihood; it does not automatically supply it for Bernoulli populations.

Matrix tail sharpness. The scalar logarithmic counterexample annihilates a family of one-dimensional Laplace arguments. Anchored matrix observations probe additional positive-semidefinite directions. The scalar embedding therefore fails as a proof, even though the sufficiency argument extends to arbitrary dimension. A sharp converse must annihilate the whole anchored transform set while preserving its quantitative tail.

Arithmetic transfer. In the finite-sum theorem, a primitive character relation and a largest frequency produce the complete list of missing arcs. In a genuine Euler product, prime-power frequencies have exact multiplicative relations and the relevant series need not converge normally on the whole plane. Removing the largest frequency from an absolutely convergent toy model does not solve this arithmetic obstacle. No Riemann-hypothesis implication or recovery of all

prime data from one right-hand zero is established by this manuscript.

4.10.2 Other distinctions that survive the new results

The MLE proof does not establish the exact constant-one Chebyshev–Bernstein coefficient estimate, and the negative simultaneous-test result leaves the original fixed-test fingerprint question intact. The positive mixture transport transfer is the route around those assertions, not a proof of them.

An exact SD threshold written as $\sup K'/K$ is more informative once the kernel and a certified tail are explicit, but it is not an elementary formula for every maximizing point. Likewise, relative stability of an entire recovered power-sum profile does not imply stable extraction of colliding frequencies. Exponential recovery from exact real numbers does not imply a comparable noisy rate. These separations are part of the mathematical conclusions.

Verification and use of computation. The asserted theorems are analytic arguments. Numerical checks were not substituted for positivity, uniqueness, lower bounds, or quantifiers over adaptive policies. The drafting process included separate reviews of the likelihood transport and approximation steps, the adaptive endpoint information calculation, the signed hypergeometric representations, the logarithmic inverse-Laplace construction, and the infinite-torus limit. Such reviews can miss errors and do not replace external refereeing. This document is a standalone single-source LaTeX manuscript; it requires no external data files, figures, or bibliography database to compile.

Part V

Adaptive Analytic Information and Sharp Recovery Boundaries

Polynomial Measurements, Complete Adaptive Budgets, and Nonlinear Spectral Reconstruction

Source manuscript: 14 October 2026.

Source abstract. We continue a program on observation and recovery by changing the admissible measurement geometry. Every vector in $\mathbb{R}^m$ can be recovered to any prescribed accuracy using at most $\min\{m, \lceil \log_2 m \rceil + 2\}$ adaptive measurements, each a globally convex polynomial. The same count is possible with globally convex, real-analytic, one-Lipschitz measurements. This gives a positive complexity answer to the analytic and convex information directions raised by Krieg, Novak, and Ullrich; it does not determine the optimal number of queries. For populations of Bernoulli parameters and uniform endpoints, we determine complete minimax replication-cap curves, including arbitrary adaptive policies. A global background in the Bernoulli hard family controls information after every possible observation history. Further results give the sharp deterministic binomial inverse modulus, certified approximate likelihood at statistical precision, the exact view threshold for random low-rank Gaussian covariance, and all finite iterated-log refinements of a scalar tail boundary. For McKay laws we classify which powers restore hyperbolic complete monotonicity, with a first-Bessel-zero asymptotic at the critical power. Any $2m$ prescribed transition thresholds recover an m-atomic positive spectrum, with sharp collision-stable noise exponent $1/(2m-1)$. Finally, lacunary heat observations obey an exact logarithmic-risk information-depth law. A target register distinguishes complete internal classifications, positive answers to external questions, partial advances, and unresolved arithmetic transfers.

5.1 The new questions and the change of viewpoint

The 223-page manuscript *Observation and Recovery in Arithmetic and Probability*, its three intermediate sequels, and the preceding *Transport, Spectral Positivity, and Sharp Observation Boundaries* form the starting point of this paper [1, 29, 53, 61, 76]. The present text is a new continuation. It does not count results already proved in those manuscripts as new closures, and it does not treat unreviewed earlier claims as established literature. Its principal statements are supplied with proofs here. The signed spectral kernel in the final SD extension is an explicitly identified dependency on the preceding paper.

There are three useful changes of viewpoint. First, a measurement rule may admit a robust discrete implementation even when its original exact membership test cannot be analytic. This allows approximation of the measurement functions themselves. Second, in an adaptive statistical experiment the necessary lower-bound object is conditional information at the next query, including rare histories. An unconditional comparison of whole individuals is insufficient. Third, nonlinear threshold data can sometimes be converted into separated samples by contact geometry, then into moments by a compact differential-equation argument. These mechanisms transfer between problems, but no single theorem is claimed to encompass all the branches below.

5.1.1 A direct answer in the analytic-information direction

Krieg, Novak, and Ullrich ask for complexity results when continuous information is restricted to analytic, convex, or homogeneous mappings [60, Section 3]. We prove an $O(\log m)$ upper

bound with polynomial queries, and also with globally convex polynomial queries, for arbitrary prescribed reconstruction accuracy. The convex-polynomial bound on all of $\mathbb{R}^m$ is

$$\min\{m, \lceil \log_2 m \rceil + 2\}.$$

For example, it is twelve queries in dimension one thousand. The statement counts real-valued oracle queries. The degree, coefficients, precision cost, and computational work of such a query are not bounded by this count. A separate construction permits every query to be globally convex, real analytic, and Euclidean one-Lipschitz, at the same query count. The optimal minimum remains open; the result establishes that analytic regularity and global convexity alone do not remove the exponential separation from linear information.

The crucial proof step is to retain proximity to a union of admissible cells rather than exact membership. A polynomial can then replace each distance query with a controlled error. A final polynomial distinguishes finitely many separated expanded cells. An initial norm measurement provides a compact sphere, whose lower-dimensional colored cover improves the count, and the radial value needed to subtract a convexifying polynomial. Thus the polynomial result is obtained by a robustification of the known colored-cover construction, not by claiming that exact membership indicators are analytic.

5.1.2 Complete adaptive budget curves

For a total budget B and a per-individual cap Q, the Bernoulli population risk is, uniformly over $1 \le Q \le B$,

$$R^{\mathrm{ad}}_{B,Q} \asymp R^{\mathrm{fix}}_{B,Q} \asymp \max\left\{B^{-1/3}, Q^{-1}, [Q\log(e + B/Q^{3/2})]^{-1/2}\right\}.$$

The preceding paper had left a critical adaptive window open. The new lower bound uses localized moment cancellation together with a positive global background. It bounds the next-toss Fisher information for every possible success/failure history, closing that window. At $Q \asymp B^{2/3}/\log B$, the remaining cost is $B^{-1/3}\sqrt{\log B/\log\log B}$, even with adaptation.

For an arbitrary joint endpoint law on $[1,2]^d$, one observation is a conditionally independent coordinate vector from the corresponding uniform intervals. The exact capped risk is

$$U^{\mathrm{ad}}_{B,Q,d} \asymp U^{\mathrm{fix}}_{B,Q,d} \asymp \begin{cases} \max\{\log(eB)/\sqrt{B}, (BQ)^{-1/3}\}, & d = 1, \\ \max\{B^{-1/(d+1)}, (BQ^{2d-1})^{-1/(3d)}\}, & d \ge 2. \end{cases}$$

Dependence among the latent endpoints is unrestricted. The lower bound uses a stronger Poisson record experiment and the full vector of posterior scores, including all face terms. Thus the scalar critical logarithm and its disappearance in higher dimensions are consequences of the same allocation and information calculations.

5.1.3 Finite observations and sharp inverse boundaries

The deterministic binomial inverse modulus is determined at every replication size and error level, rather than just along a statistical asymptotic regime. Its lower bound identifies a precise limitation of the global count-Wasserstein approach to MLE. A separate posterior repair argument shows that normalized likelihood deficit $O(N^{-1})$ is enough to retain the preceding statistical guarantees; a rational finite-grid algorithm certifies this deficit.

For Gaussian location-covariance mixtures whose covariance rank is at most $r < d$, $r + 1$ independent conditional blocks identify an arbitrary mixing law without moment conditions. Every experiment with at most r blocks, even all block lengths together, can fail on compact

exact-rank models. The obstruction is a determinant differential operator that survives the orthogonal covariance-factor quotient. The positive theorem uses the covariance range revealed by repeated Gaussian contrasts.

For a positive m-atomic spectrum with exponents in $(0, 1)$, an entire transition curve can be replaced by any $2m$ distinct prescribed positive settings. When the spectrum is a probability law in a fixed compact subinterval, the sharp noisy recovery rate is $\delta^{1/(2m-1)}$ without atom separation. This transfers the classical collision exponent from moment data to nonlinear optimized thresholds. These thresholds are population observables; the theorem does not assert recovery from a finite-height list of Riemann-zeta zeros or recover prime phases.

5.1.4 Scope of the claimed advances

The target register at the end supplies individual statuses and theorem references. Some entries are newly formulated inverse problems, some close gaps left in this research program, and the analytic/convex query results address directions explicitly asked in an external paper. A logarithmic-risk equivalence is distinguished from a sharp minimax exponent, and a partial parameter strip from a full classification. Neither a count of targets nor the length of the manuscript establishes novelty, correctness, or publication level. Internal independent proof reviews and a targeted primary-source comparison were performed; external specialist review and a broader priority check remain necessary.

5.2 Adaptive recovery by convex polynomial measurements

Krieg–Novak–Ullrich [60, Section 3] ask for complexity results when adaptive continuous information is restricted to homogeneous, convex, or analytic functionals. Their continuous construction uses distance oracles and exact membership decisions. We prove that a robust version of the color search permits polynomial oracles instead. An initial energy measurement then converts every subsequent polynomial oracle into a globally convex polynomial without increasing the number of subsequent measurements.

The information model is the exact-real-number model of that question. At each step an algorithm selects a scalar functional on $\mathbb{R}^m$ using only earlier answers and obtains its exact value. The selected functional must belong to the indicated class, but its degree, coefficients, and the selection rule have no complexity bound. The final reconstruction may be discontinuous. These conventions matter: neither finite-precision stability nor efficient evaluation of the polynomials is asserted.

Theorem 5.2.1 (Logarithmic complexity with convex polynomials)**.** *Let $m \geq 1$, let $\|\cdot\|$ be any norm on $\mathbb{R}^m$, and let $\varepsilon > 0$. There is an adaptive procedure recovering every $x \in \mathbb{R}^m$ with*

$$\|x - \widehat{x}\| \leq \varepsilon$$

using at most

$$\min\{m, \lceil \log_2 m \rceil + 2\} \tag{5.2.1}$$

scalar measurements, each of which is a globally convex real polynomial on $\mathbb{R}^m$.

In particular, an exponential reduction in the number of exact measurements remains possible under the simultaneous restrictions "convex" and "real analytic". The theorem gives an upper bound; it does not determine the optimal number of such measurements.

The following compact-space statement is the central step.

Lemma 5.2.2 (Polynomial measurements from a finite colored cover)**.** *Let $K \subset \mathbb{R}^m$ be compact. Suppose finitely many nonempty compact sets $D_i \subset \mathbb{R}^m$ cover K, have diameter at most $\varepsilon/2$ in the chosen norm, and have colors in $\{1, \dots, M\}$. Suppose distinct sets of the same color are at distance at least $\sigma > 0$. Then K can be recovered to error at most ε by at most*

$$\lceil \log_2 M \rceil + 1$$

adaptive real polynomial measurements on $\mathbb{R}^m$. No convexity of these intermediate polynomials is required.

Proof. Discard colors not used by any set and decrease σ, if necessary, so that $\sigma \le \varepsilon$. Let $r = \lceil \log_2 M \rceil$. If $M = 1$, only the final coding measurement below is needed. Otherwise put $\eta = \sigma/(8r)$. For a nonempty set J of colors define

$$E_J = \bigcup_{i:\, \mathrm{color}(D_i) \in J} D_i, \qquad d_J(x) = \mathrm{dist}(x, E_J).$$

For every such J, select a real polynomial p_J with

$$\sup_{x \in K} |p_J(x) - d_J(x)| \le \eta. \tag{5.2.2}$$

These approximations exist by multivariate polynomial approximation on a box containing K, since d_J is continuous on the whole box. For clarity, no approximation of a discontinuous membership indicator is used.

Start with all colors retained and error allowance $e_0 = 0$. After j queries maintain the invariant

$$\mathrm{dist}(x, E_J) \le e_j, \qquad e_j = 2j\eta. \tag{5.2.3}$$

If more than one color remains, divide J into two nonempty sets J_1, J_2 with sizes differing by at most one and query $p_{J_1}(x)$. Retain J_1 if this answer is at most $e_j + \eta$, and otherwise retain J_2. In the first case (5.2.2) gives $d_{J_1}(x) \le e_j + 2\eta$. In the second case it gives $d_{J_1}(x) > e_j$; since $\min(d_{J_1}(x), d_{J_2}(x)) \le e_j$, necessarily $d_{J_2}(x) \le e_j$. Thus the invariant holds with allowance e_{j+1} in both cases. After at most r queries there is one retained color c and

$$\mathrm{dist}(x, E_{\{c\}}) \le \sigma/4.$$

For the finitely many sets of color c, write

$$F_i = \{y : \mathrm{dist}(y, D_i) \le \sigma/3\}.$$

They are compact and pairwise separated by at least $\sigma/3$. There is a continuous real function g_c on $\mathbb{R}^m$ which equals the distinct integer i on F_i. For an explicit choice, put

$$g_c(y) = \sum_{i:\, \mathrm{color}(D_i) = c} i \max\{0, 1 - 6\,\mathrm{dist}(y, F_i)/\sigma\}.$$

Its supports do not intersect at points where both summands are positive. Approximate g_c on a box containing K by a polynomial q_c with error less than $1/4$, and query $q_c(x)$. The point x belongs to one of the F_i, so rounding the answer to the nearest integer identifies that i. Output a fixed $x_i \in D_i$. Then

$$\|x - x_i\| \le \mathrm{dist}(x, D_i) + \mathrm{diam}(D_i) \le \sigma/4 + \varepsilon/2 < \varepsilon.$$

If $M = 1$, x already belongs to some D_i and the same coding query applies directly.

The polynomial approximation invoked twice above can be obtained explicitly from multivariate Bernstein polynomials after rescaling the box to $[0, 1]^m$. Their value at a point is the expectation of the target function at independent binomial grid coordinates. Uniform continuity and the uniform bound on their coordinate variances prove uniform convergence. Thus no regularity beyond continuity of the two distance-based functions is needed. □

Lemma 5.2.3 (A small colored cover of a sphere)**.** *For $m \geq 2$, $R > 0$, and $d > 0$, the Euclidean sphere $S_R^{m-1} \subset \mathbb{R}^m$ has a finite cover by nonempty compact sets of diameter at most d in any prescribed norm, colored with m colors, such that distinct sets of the same color are a positive distance apart.*

Proof. Take a finite triangulation of the sphere, refine it so that each closed vertex star after barycentric subdivision has diameter at most d, and use that subdivision. Such a triangulation can be obtained from the boundary of a cross-polytope by radial projection; successive subdivision and uniform continuity give arbitrarily small stars. Vertices in the barycentric subdivision correspond to faces in the preceding triangulation. Color each vertex by the dimension of that face, one of $0, \ldots, m-1$. Vertices of the same color never belong to one simplex, since a simplex of the subdivision corresponds to a strictly nested chain of faces.

Let φ_v be the continuous piecewise affine barycentric coordinate associated with a vertex v, transferred to the sphere. The sets

$$D_v = \{x : \varphi_v(x) \geq 1/m\}$$

cover the sphere: at any point at most m barycentric coordinates are nonzero and their sum is one. Each D_v is compact and contained in its small closed star. For distinct vertices of one color, $\varphi_v \varphi_w = 0$, so D_v and D_w are disjoint. There are finitely many such pairs; compactness makes their minimum distance strictly positive. Unused or empty sets may be discarded. □

Lemma 5.2.4 (Convexification using known energy)**.** *For every real polynomial P on $\mathbb{R}^m$ there are an integer $D \geq 1$ and a constant $C \geq 0$ such that*

$$Q(x) = P(x) + C(1 + \|x\|_2^2)^D$$

is a globally convex polynomial. If $E = \|x\|_2^2$ is already known, one evaluation of $Q(x)$ therefore determines $P(x)$ exactly.

Proof. If d is the degree of P, its Hessian satisfies

$$\|\nabla^2 P(x)\|_{2\to 2} \leq C_P(1 + \|x\|_2^2)^{\max\{d-2,0\}/2}.$$

Choose an integer D with $D - 1 \geq \max\{d-2, 0\}/2$. Direct differentiation gives

$$\nabla^2 (1 + \|x\|_2^2)^D \succeq 2D(1 + \|x\|_2^2)^{D-1} I_m.$$

A sufficiently large C makes $\nabla^2 Q$ positive semidefinite everywhere, which proves convexity. Subtract $C(1+E)^D$ from the measured value to recover $P(x)$. □

Proof of Theorem 5.2.1. The coordinate functionals provide exact recovery in m queries; they are convex polynomials. For the other bound, first query the convex polynomial $E(x) = \|x\|_2^2$. If the answer is zero, output zero. Otherwise the input is now known to belong to the compact sphere of radius $R = \sqrt{E}$. Apply Lemma 5.2.3 with $d = \varepsilon/2$, and then Lemma 5.2.2. This requires at most $\lceil \log_2 m \rceil + 1$ further polynomial measurements. Each selected polynomial is converted into a convex polynomial by Lemma 5.2.4; the initial energy allows its original value to be recovered exactly from the same single answer. All later branch decisions therefore agree with the robust polynomial algorithm. Including the energy query gives (5.2.1). The case $m = 1$ is already covered by the coordinate procedure. □

Corollary 5.2.5 (Convex analytic measurements with unit Lipschitz bound)**.** *The bound (5.2.1) also holds with every measurement globally convex, globally real analytic, and Lipschitz with constant at most one in the Euclidean norm. These measurements need not be polynomials.*

Proof. First query $A(x) = \sqrt{1 + \|x\|_2^2}$, which has all three properties and determines $E = \|x\|_2^2$ exactly. If $E = 0$, output zero and stop. Run the same polynomial algorithm on the resulting sphere. To simulate a selected polynomial P, put $F(y) = P(y)e^{-\|y\|_2^2}$. The function F and all its derivatives are polynomials times a Gaussian. In particular

$$M = \sup_y (1 + \|y\|_2^2)^{3/2} \|\nabla^2 F(y)\|_{2\to 2} < \infty, \qquad L = \sup_y \|\nabla F(y)\|_2 + M < \infty.$$

Since $\nabla^2 A(y) \succeq (1 + \|y\|_2^2)^{-3/2} I_m$, the function $Q = F + MA$ is convex. It is globally real analytic and has Lipschitz constant at most L. Query $Q/\max\{1, L\}$. Multiply by its known scaling factor, subtract the already known $MA(x)$, and multiply by e^E. This recovers $P(x)$ exactly from one measurement, with no new query. The coordinate alternative is obtained by scaling each coordinate functional if needed. □

5.2.1 Positively homogeneous convex information

There is a related conclusion for homogeneity, with its convention specified explicitly. A function F is positively one-homogeneous when $F(tx) = tF(x)$ for $t \geq 0$. We do not require $F(-x) = -F(x)$.

Theorem 5.2.6 (Convex positively homogeneous measurements)**.** *The recovery and measurement-count bound of Theorem 5.2.1 also hold when every measurement is required instead to be globally convex and positively one-homogeneous. The measurements can be continuous everywhere and real analytic on $\mathbb{R}^m \setminus \{0\}$, with Euclidean Lipschitz constant at most one after a known rescaling. They are not asserted to be real analytic at the origin.*

Proof. First query $R = \|x\|_2$ and output zero if $R = 0$. On the known sphere, use the same finite polynomial algorithm. For any selected polynomial P, put

$$H(y) = \frac{\|y\|_2}{R} P\left(\frac{Ry}{\|y\|_2}\right) \quad (y \neq 0), \qquad H(0) = 0.$$

This function is continuous, positively one-homogeneous and real analytic away from zero, and $H(x) = P(x)$ for the unknown input. It can be convexified by adding $C\|y\|_2$. Indeed Euler's identity gives $\nabla^2 H(y)y = 0$, and the Hessian is homogeneous of degree -1. Smoothness on the unit sphere therefore gives a constant C_1 with

$$\nabla^2 H(y) \succeq -\frac{C_1}{\|y\|_2}\left(I_m - \frac{yy^{\mathsf{T}}}{\|y\|_2^2}\right).$$

The matrix on the right, without its negative coefficient, is the Hessian of $\|y\|_2$. Thus $Q(y) = H(y) + C\|y\|_2$ has positive semidefinite Hessian away from zero when $C \geq C_1$. Increase C so that $C \geq \sup_{\|v\|_2=1} |H(v)|$. On a line passing through zero the one-sided slopes of $Q(tv)$ then have the correct order, because $Q(v) + Q(-v) \geq 0$. On every other line convexity follows from the Hessian inequality. Hence Q is globally convex. It is positively one-homogeneous and querying it determines $P(x) = Q(x) - CR$, since R is known. The number of queries is unchanged. Each constructed Q is continuous, convex, and positively one-homogeneous, hence subadditive. Its bounded values on the unit sphere imply a finite global Lipschitz constant. Dividing by a known upper bound and undoing that scaling in the scalar answer makes the constant at most one. □

Corollary 5.2.7 (Homogeneous polynomials of unrestricted degree)**.** *If "homogeneous polynomial" permits a different nonnegative integer degree for each measurement, then recovery on $\mathbb{R}^m$ to any prescribed error requires at most*

$$\min\{m, 2\lceil \log_2 m \rceil + 3\}$$

adaptive homogeneous polynomial measurements. The homogeneity identity holds for every real scalar. Convexity of these polynomials is not asserted.

Proof. Query $E = \|x\|_2^2$ first and stop if it is zero. Run the polynomial sphere algorithm. For each polynomial $P(y) = \sum_\alpha c_\alpha y^\alpha$, split its monomials into their even and odd total-degree parts P_0, P_1. For $e \in \{0,1\}$ with nonzero part, choose an integer D_e of parity e at least as large as the degree of P_e and define

$$H_e(y) = \sum_{|\alpha| \equiv e \,(\mathrm{mod}\, 2)} c_\alpha y^\alpha \left(\frac{\|y\|_2^2}{E} \right)^{(D_e - |\alpha|)/2}.$$

This is a real homogeneous polynomial of degree D_e. At the actual input, $H_e(x) = P_e(x)$. At most two queries therefore simulate each of the $\lceil \log_2 m \rceil + 1$ polynomial queries after the energy. The coordinate alternative gives the minimum with m. □

The resolved direction and the remaining question. In the exact information model, restricting adaptive measurements to globally convex polynomials does not restore the m-measurement requirement of linear information. Logarithmically many queries still suffice. The same is true for convex positively homogeneous information. Thus these theorems answer the exponential-speedup direction of the cited complexity questions. They do not prove an optimal count, nor do they provide a bounded-degree or stable finite-precision implementation.

The global domain in Theorem 5.2.1 is handled by an exact preliminary energy query, rather than by assuming a uniform polynomial approximation on all of $\mathbb{R}^m$. Finally, a globally real-analytic, positively one-homogeneous function is linear: differentiability at zero gives $F(x) = \lim_{t \downarrow 0} F(tx)/t = DF(0)x$. Accordingly the two theorems impose different simultaneous restrictions; they do not claim logarithmic recovery using functionals that are both globally analytic and one-homogeneous.

5.3 The sharp binomial inverse modulus and certified approximate likelihood

The preceding manuscript proved that the unrestricted binomial mixture maximum likelihood estimator is rate optimal throughout every fixed polynomially sublinear replication range. Two distinct questions remain: whether that estimator is optimal at the statistical transition, and how accurately its likelihood must be optimized. We resolve a deterministic inverse problem underlying the first question, and give a quantitative, computable answer to the second. The first result also proves that one particular strategy cannot settle the remaining statistical transition. We do not claim to settle that transition itself.

For a probability measure G on $[0,1]$, let

$$B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}, \qquad p_G(j) = \int B_j^t(x)\, dG(x), \qquad 0 \le j \le t.$$

Write $W_{1,t}$ for Wasserstein distance on the count grid with cost $|j-l|/t$. All logarithms are natural. Constants below are absolute unless specified otherwise.

5.3.1 A sharp modulus, uniformly in replication and measurement error

Define the deterministic inverse modulus

$$\omega_t(\delta) = \sup\{W_1(G,H) : W_{1,t}(p_G, p_H) \le \delta\}, \qquad 0 \le \delta \le 1.$$

Theorem 5.3.1 (Sharp binomial inverse modulus)**.** *For every integer $t \geq 1$ and $0 \leq \delta \leq 1$,*

$$c\Phi_t(\delta) \leq \omega_t(\delta) \leq C\Phi_t(\delta), \qquad \Phi_t(\delta) = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(1+1/(t\delta^2))}}\right\}. \tag{5.3.1}$$

At $\delta = 0$ the second term is defined to be zero. In particular, the order of the entire interpolation expression, including its transition, is necessary for deterministic inversion from count Wasserstein error alone.

The spectral calculation used below is the classical beta–binomial Gibbs diagonalization of Diaconis, Khare and Saloff-Coste [70, Section 4.1]. Its use here is to construct sharp Wasserstein inverse witnesses. We provide the calculation to make the result independent of normalization conventions for Jacobi and Hahn polynomials. Neither those eigenvalues nor their diagonalization are claimed as new.

Lemma 5.3.2 (Oscillatory witnesses)**.** *For every integer $k \geq 1$, put*

$$dG_k^{\pm}(x) = \{1 \pm T_k(2x-1)\}\frac{dx}{\pi\sqrt{x(1-x)}}.$$

These are probability measures, and

$$\frac{c}{k} \leq W_1(G_k^+, G_k^-) = \frac{1}{\pi k}\int_0^\pi |\sin(k\theta)|\sin\theta\, d\theta \leq \frac{C}{k}. \tag{5.3.2}$$

For $1 \leq k \leq t$,

$$W_{1,t}(p_{G_k^+}, p_{G_k^-}) \leq \frac{C}{k}\left\{\prod_{i=0}^{k-1}\frac{t-i}{t+1+i}\right\}^{1/2} \leq \frac{C}{k}e^{-k^2/(4t)}. \tag{5.3.3}$$

For $k > t$ the two count laws are identical.

Proof. Let π be the arcsine law on $[0,1]$, and let $h_k(x) = T_k(2x-1)$. Under $x = (1+\cos\theta)/2$, π is uniform on $[0,\pi]$ and $h_k(x) = \cos(k\theta)$. Thus $\int h_k\, d\pi = 0$, $|h_k| \leq 1$, and the signed cumulative distribution function is $-2\sin(k\theta)/(\pi k)$. Integration gives (5.3.2). Its lower bound follows, for example, by restricting to $[\pi/4, 3\pi/4]$, where $\sin\theta \geq 1/\sqrt{2}$; the integral of $|\sin(k\theta)|$ on this interval is bounded below uniformly in the integer $k \geq 1$.

Draw $X \sim \pi$ and $J \mid X \sim \mathrm{Binomial}(t, X)$. Let $Qf(j) = \mathbb{E}[f(X) \mid J = j]$ and $Q^*v(x) = \mathbb{E}[v(J) \mid X = x]$. The operator $A = Q^*Q$ is self-adjoint on $L^2(\pi)$ and preserves polynomials of degree at most $k \leq t$. Indeed, the posterior law is $\mathrm{Beta}(j+1/2, t-j+1/2)$, so

$$\mathbb{E}[X^k \mid J = j] = \frac{(j+1/2)(j+3/2)\cdots(j+k-1/2)}{(t+1)(t+2)\cdots(t+k)}.$$

The leading coefficient of Ax^k is therefore

$$\lambda_{t,k} = \frac{t(t-1)\cdots(t-k+1)}{(t+1)(t+2)\cdots(t+k)}.$$

Since h_k is orthogonal to every lower-degree polynomial, self-adjointness and degree preservation give $Ah_k = \lambda_{t,k}h_k$. Consequently

$$\mathbb{E}[Qh_k(J)^2] = \langle h_k, Ah_k\rangle_{L^2(\pi)} = \lambda_{t,k}/2.$$

If p_π denotes the beta–binomial count law, the signed difference of the two count laws is $\Delta(j) = 2p_\pi(j)Qh_k(j)$. Hence $\sum_j |\Delta(j)| \leq \sqrt{2\lambda_{t,k}}$. Moreover, $\sum_j P(j/t)\Delta(j) = 0$ whenever P

has degree less than k: condition on X and use polynomial orthogonality. Approximate an arbitrary 1-Lipschitz function on $[0,1]$ by a polynomial of degree at most $k-1$ with uniform error C/k. Such a Jackson approximation is constructed in Lemma 5.3.3 below; the case $k=1$ uses a constant. Thus its integral against Δ is at most $C\sqrt{\lambda_{t,k}}/k$. Wasserstein duality proves the first inequality in (5.3.3). For the second, use

$$\log \lambda_{t,k} = \sum_{i=0}^{k-1} \log\left(1 - \frac{1+2i}{t+1+i}\right) \le -\frac{k^2}{t+k} \le -\frac{k^2}{2t}.$$

Finally, for $k > t$, each B_j^t has degree below k and is orthogonal to h_k, proving exact equality of the count laws. □

Lemma 5.3.3 (A derivative-preserving inverse inequality)**.** *For any two mixing laws and any integer $2 \le k \le t/2$,*

$$W_1(G,H) \le C/k + Ce^{4k^2/t} W_{1,t}(p_G, p_H). \tag{5.3.4}$$

Proof. We recall the quantitative ingredients, with proofs. A 1-Lipschitz function f on $[0,1]$, after subtraction of $f(0)$, extends by reflection to a period-two 1-Lipschitz function of supremum norm at most one. Convolution with the positive, normalized Jackson kernel

$$J_s(u) = c_s \left\{ \frac{\sin(s\pi u/2)}{\sin(\pi u/2)} \right\}^4$$

gives a trigonometric polynomial Q_s, of frequencies $|l| \le 2s$, with $\|Q_s - f\|_\infty \le C/s$ and $\|Q_s'\|_\infty \le 1$. The approximation estimate follows from the kernel's first absolute moment $O(1/s)$: the unnormalized integral is comparable to s^3 and the first moment is $O(s^2)$, using the bound $C\min(s, |u|^{-1})$ for the sine quotient. Every Fourier coefficient of Q_s has modulus at most one. Replacing each $e^{i\pi l(x-1/2)}$ by its degree-$100s$ Taylor polynomial changes both the function and its derivative by at most e^{-100s}, by the factorial bound $m! \ge (m/e)^m$. Choosing $s = \lfloor k/100 \rfloor$ for $k \ge 100$, and a constant for smaller k, gives a polynomial P_k with

$$\deg P_k \le k, \qquad \|P_k - f\|_\infty \le C/k, \qquad \|P_k'\|_\infty \le 2.$$

We also need the following Bernstein coefficient estimate: if $P = \sum_{j=0}^n b_j B_j^n$ has degree $m \le n/2$, then

$$\max_j |b_j| \le Ce^{2m^2/n} \|P\|_\infty. \tag{5.3.5}$$

Here is a short contour proof. The maximum principle applied to $v^m P((2+v+v^{-1})/4)$ gives $|P((2+w+w^{-1})/4)| \le \|P\|_\infty |w|^m$ for $|w| \ge 1$. For $0 < j < n$ set $r = j/(n-j)$, $q = r/(1+r)^2$, and $u = 2\sqrt{q}\,|\sin(\phi/2)|$. The larger inverse Joukowski modulus corresponding to $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Thus, for $F(z) = (1+z)^n P(z/(1+z))$,

$$\frac{|F(re^{i\phi})|}{\|P\|_\infty (1+r)^n} \le (1-u^2)^{(n-m)/2} (1+u)^m \le e^{2m^2/n} e^{-nu^2/8}.$$

The average of the last Gaussian factor is at most $C/\sqrt{1+nq}$. Cauchy's coefficient formula and the elementary Stirling lower bound $\binom{n}{j}(j/n)^j(1-j/n)^{n-j} \ge c/\sqrt{1+nq}$ prove (5.3.5). Endpoint coefficients are just $P(0), P(1)$.

Write $P_k = \sum_j b_j B_j^t$. Since $P_k' = t\sum_{j=0}^{t-1}(b_{j+1}-b_j)B_j^{t-1}$, (5.3.5) gives $\max_j t|b_{j+1}-b_j| \le Ce^{4k^2/t}$. Apply Wasserstein duality to this Lipschitz coefficient vector; the two polynomial approximation errors contribute C/k. □

Proof of Theorem 5.3.1. For the upper bound let $x = 1/(t\delta^2)$ and $L = \log(1+x)$. Choose a small absolute $a > 0$, and $k = \lfloor \min(t/2, \sqrt{atL}) \rfloor$ whenever that minimum is at least four. The inverse inequality then yields $C/t + C/\sqrt{tL}$. In fact, in the uncapped case its exponential term, multiplied by $\sqrt{tL}$, is at most

$$C(1+x)^{4a}\sqrt{L/x},$$

which is bounded when $4a < 1/2$. In the capped case $L \geq t/(4a)$, and $\delta e^t \leq C/t$. If the minimum is below four, the trivial bound one gives the result. At $\delta = 0$, use k of order t and handle bounded t trivially.

For the lower bound, Lemma 5.3.2 with $k = t+1$ always gives $\omega_t(\delta) \geq c/t$. Also, endpoint mixing laws differing by mass δ give $\omega_t(\delta) \geq \delta$. If $x = 1/(t\delta^2) \leq 1$, then $L \geq x/2$ and $1/\sqrt{tL} \leq \sqrt{2}\delta$, so these two witnesses suffice. Suppose $x > 1$. Let A be a sufficiently large absolute constant and $k = \lceil A\sqrt{tL} \rceil$. If $k \leq t$, then

$$\frac{Ck^{-1}e^{-k^2/(4t)}}{\delta} \leq \frac{C\sqrt{x}}{A\sqrt{L}}(1+x)^{-A^2/4} \leq 1.$$

The final inequality holds for all $x > 1$ after increasing A. The same witness has latent distance at least $c/k \geq c'/\sqrt{tL}$. If $k > t$, then, apart from bounded t, the inequality $A\sqrt{tL} > t - 1$ implies $1/\sqrt{tL} \leq 2A/t$, so the exact moment-matching witness already suffices. Combining the lower bounds, and clipping at one, proves the result. □

Corollary 5.3.4 (A limitation of global transport inversion)**.** *At $t = \lfloor N/\log N \rfloor$,*

$$\omega_t(N^{-1/2}) \asymp \sqrt{\frac{\log N}{N \log\log N}}.$$

Consequently a deterministic argument using only a count-Wasserstein error bound of order $N^{-1/2}$ cannot guarantee latent error of order $N^{-1/2}$ at this transition.

This does not lower-bound the statistical risk of the MLE: the witness pairs need not represent its fitted-versus-true pairs on likely samples. Closing the statistical transition would require additional likelihood structure or a more informative, localized observation norm. The open coefficient and fixed-test questions in [14, Section 6.2] are not answered by this deterministic obstruction.

5.3.2 Entropy–transport with an optimization certificate

Let $(\mathcal{X}, d)$ be a finite metric space of diameter D_0, and let K_θ be probability kernels. For a probability vector q write $\ell(H) = \sum_x q(x) \log p_H(x)$ and $\ell^* = \sup_H \ell(H)$. The following statement extends the exact likelihood transport identity; it does not assume that mixtures inherit the concentration of their components.

Theorem 5.3.5 (Posterior repair of approximate mixture likelihood)**.** *Suppose each kernel satisfies $W_{1,d}(r, K_\theta)^2 \leq 2cD(r\|K_\theta)$ for all r. Let H have $p_H > 0$ on the support of q, and put $e = \ell^* - \ell(H)$. Define*

$$S_H(\theta) = \sum_{x:q(x)>0} \frac{q(x)K_\theta(x)}{p_H(x)}, \qquad dH^+(\theta) = S_H(\theta)\, dH(\theta), \qquad d_H = D(H^+\|H).$$

Then H^+ is a probability measure and

$$W_{1,d}(q, p_H) \leq \sqrt{2c\{D(q\|p_H) - d_H\}} + D_0\|H^+ - H\|_{\mathrm{TV}}, \qquad d_H \leq \ell(H^+) - \ell(H) \leq e. \quad (5.3.6)$$

Here total variation is one half the L^1 norm. In particular,

$$W_{1,d}(q, p_H) \le \sqrt{2cD(q\|p_H)} + D_0\sqrt{e/2}. \tag{5.3.7}$$

Alternatively, if $\sigma = \sup_\theta S_H(\theta) \le 1 + \eta$, then

$$e \le \log(1+\eta), \qquad W_{1,d}(q, p_H) \le \sqrt{2cD(q\|p_H)} + D_0\frac{\eta}{1+\eta}. \tag{5.3.8}$$

Proof. The identity $\int S_H\, dH = 1$ is immediate. For $S_H(\theta) > 0$ let

$$R_\theta(x) = \frac{q(x)K_\theta(x)}{p_H(x)S_H(\theta)}.$$

This is a probability vector, and $\int R_\theta\, dH^+ = q$. Finite-sum calculation gives the exact entropy decomposition

$$D(q\|p_H) = D(H^+\|H) + \int D(R_\theta\|K_\theta)\, dH^+(\theta). \tag{5.3.9}$$

Terms on $S_H = 0$ are absent. Convexity of transportation cost, the kernel inequality and Cauchy–Schwarz bound $W_{1,d}(q, p_{H^+})$ by the square root in (5.3.6). Coupling the common part of H, H^+ gives $W_{1,d}(p_{H^+}, p_H) \le D_0\|H^+ - H\|_{\mathrm{TV}}$.

The posterior of θ at an observed x under H is $K_\theta(x)dH(\theta)/p_H(x)$. Jensen's inequality yields

$$\begin{aligned}\ell(H^+) - \ell(H) &= \sum_x q(x) \log \mathbb{E}_H[S_H(\theta) \mid x] \\ &\ge \sum_x q(x)\mathbb{E}_H[\log S_H(\theta) \mid x] = \int S_H \log S_H\, dH = d_H.\end{aligned}$$

This is the familiar EM monotonicity calculation, here retaining the specific divergence needed for transportation. Pinsker's inequality gives (5.3.7). For any comparison mixing law F, a second application of Jensen gives

$$\ell(F) - \ell(H) \le \log \sum_x q(x)\frac{p_F(x)}{p_H(x)} = \log \int S_H\, dF \le \log \sigma.$$

Finally, when $0 \le S_H \le 1 + \eta$ and $\int S_H\, dH = 1$,

$$\|H^+ - H\|_{\mathrm{TV}} = \int (S_H - 1)_+\, dH \le \frac{\eta}{1+\eta}\int S_H\, dH.$$

These prove the score statements. □

Theorem 5.3.6 (Statistically sufficient likelihood precision)**.** *Let q be the empirical count law from N independent observations from p_G. Suppose a data-dependent mixing law $\widetilde{G}$ satisfies $\ell^* - \ell(\widetilde{G}) \le e_0$ for every sample, where the likelihood is normalized by N. Then, for $2 \le k \le t/2$,*

$$\mathbb{E}_G W_1(\widetilde{G}, G) \le \frac{C}{k} + Ce^{4k^2/t}\{N^{-1/2} + \sqrt{e_0}\}. \tag{5.3.10}$$

In particular $e_0 \le C_0/N$ preserves the preceding manuscript's uniform interpolation bound and its near-linear minimax rate. More explicitly, for every fixed $a, \varepsilon, C_0 > 0$ and $a\log N \le t \le N^{1-\varepsilon}$,

$$\mathbb{E}_G W_1(\widetilde{G}, G) \le \frac{C_{a,\varepsilon,C_0}}{\sqrt{t\log N}}.$$

For all $t \ge N$, the same optimization tolerance gives risk $O_{C_0}(N^{-1/2})$.

Proof. For a 1-Lipschitz function on the normalized binomial grid, changing one Bernoulli trial changes its value by at most $1/t$. Conditional Hoeffding bounds give $\log \mathbb{E}\exp\{\lambda(f-\mathbb{E}f)\} \le \lambda^2/(8t)$. The entropy variational inequality, optimized over $\lambda > 0$, then gives $W_{1,t}(r, K_\theta)^2 \le D(r\|K_\theta)/(2t)$. Theorem 5.3.5 applies with $c = 1/(4t)$ and $D_0 = 1$. Also

$$D(q\|p_{\widetilde{G}}) \le D(q\|p_G) + e_0, \qquad \mathbb{E}D(q\|p_G) \le t/N.$$

The last assertion follows from $D \le \chi^2$ and the elementary multinomial variance identity. Moreover $\mathbb{E}W_{1,t}(q, p_G) \le 1/(2\sqrt{N})$ by integrating $\mathbb{E}|F_N - F| \le 1/(2\sqrt{N})$. Thus

$$\mathbb{E}W_{1,t}(p_{\widetilde{G}}, p_G) \le C\{N^{-1/2} + \sqrt{e_0}\}.$$

Use Lemma 5.3.3. For the stated near-linear range choose k a sufficiently small fixed multiple of $\sqrt{\varepsilon t \log N}$; the same optimization as in the upper bound of Theorem 5.3.1 gives the all-range interpolation bound. When $t \ge N$, that bound is $O(N^{-1/2})$. □

Proposition 5.3.7 (Sharpness of root-sample likelihood precision)**.** *There are constants $c, C, e_* > 0$ such that, for all sufficiently large N, all $t \ge 1$, and $C/N \le e_0 \le e_*$, there exists an e_0-approximate likelihood maximizer whose risk at $G = (\delta_0 + \delta_1)/2$ is at least $c\sqrt{e_0}$. Consequently likelihood deficit alone cannot guarantee root-N risk with a worst-case optimization tolerance larger in order than N^{-1}.*

Proof. At this G all observed counts are endpoints. Let r be the empirical mass at t. The endpoint law $H_r = (1-r)\delta_0 + r\delta_1$ attains the global maximum likelihood: for any mixture $p(0)+p(t) \le 1$, and the endpoint masses $1-r, r$ maximize the resulting two-coordinate likelihood. Put $h = \sqrt{e_0}/4$. On the event $r \in [1/4, 3/4]$, output H_{r+h}; otherwise output H_r. For e_* small enough, $r + h \in [1/4, 7/8]$, and

$$\ell(H_r) - \ell(H_{r+h}) = D(\text{Bernoulli}(r)\|\,\text{Bernoulli}(r+h)) \le \frac{h^2}{(r+h)(1-r-h)} \le e_0.$$

Outside the endpoint-data set define the estimator to be any exact MLE. On the stated event its loss is $|r + h - 1/2| \ge h - |r - 1/2|$. Hoeffding's bound and $\mathbb{E}|r - 1/2| \le 1/(2\sqrt{N})$ give expected loss at least $h(1 - 2e^{-N/8}) - 1/(2\sqrt{N})$, which is at least $c\sqrt{e_0}$ when C is large enough. □

5.3.3 A polynomial-time, fully certified construction

Approximate mixture likelihood by convex optimization is an established computational approach [71]. The next result specifies an elementary global certificate at the precision needed above. Its purpose is a polynomial complexity guarantee, not a competitive runtime claim.

Theorem 5.3.8 (Certified finite-grid likelihood optimization)**.** *For rational q and rational $0 < e_0 \le 1/4$, an e_0-approximate unrestricted binomial-mixture MLE can be computed with a number of rational arithmetic operations polynomial in t, e_0^{-1} and the input bit length. Its support lies on a grid of $M + 1 = O(t^2/e_0)$ points. The algorithm terminates with a computable upper bound on its unrestricted likelihood deficit.*

Proof. Let $M \ge \max(2t, 4t^2/e_0)$ be an integer and set $\alpha = (M)_t/M^t$, where $(M)_t = M(M-1)\cdots(M-t+1)$. For any $x \in [0,1]$, draw $Z \sim \text{Binomial}(M, x)$. Sampling t indices with replacement from these M Bernoulli trials shows, coordinatewise,

$$\mathbb{E}B_j^t(Z/M) \ge \alpha B_j^t(x). \tag{5.3.11}$$

Indeed, the indices are distinct with probability α, and on that event the sampled values are independent Bernoulli(x) variables; the remaining event contributes a nonnegative probability.

Integrating over any mixing law gives a grid-supported law with likelihood at least $\ell(H) + \log \alpha$. Furthermore $-\log \alpha \le t(t-1)/M \le e_0/4$, using $-\log(1-u) \le 2u$ for $u \le 1/2$.

Let u denote uniform weights on the grid $\{0, 1/M, \dots, 1\}$ and put $\gamma = e_0/4$. The grid law generated by applying the above randomization to uniform x has exactly these weights, because $\int_0^1 B_m^M(x)\,dx = 1/(M+1)$. Its predictive probabilities are therefore at least $\alpha/(t+1) \ge 1/(2(t+1))$. Restrict weights to $w = (1-\gamma)v + \gamma u$, with v in the simplex. Replacing any grid weights by this mixture loses at most $-\log(1-\gamma) \le e_0/3$ in normalized likelihood. Every feasible predictive coordinate is at least $\gamma/(2(t+1))$.

Write $L(v) = \ell((1-\gamma)v + \gamma u)$. At a current v compute the usual simplex directional gap

$$g(v) = \max_m \partial_m L(v) - \sum_m v_m \partial_m L(v).$$

Concavity gives $\max L - L(v) \le g(v)$. Hence the explicitly computable quantity

$$-\log \alpha - \log(1-\gamma) + g(v) \tag{5.3.12}$$

upper-bounds the unrestricted likelihood deficit. Stop when $g(v) \le e_0/4$; the total bound is then below e_0.

For completeness a simple fixed-step conditional-gradient method finds such a v in polynomial time. Along a segment from any feasible v to a simplex vertex, $|L''| \le C_* = 4(t+1)^2/\gamma^2$, because every predictive coordinate has the preceding lower bound and its directional change is at most one. Start at $v = u$. If $g(v) > e_0/4$, move toward a maximizing vertex by the fixed rational step $s = e_0/(4C_*)$. Taylor's theorem gives an increase at least $e_0^2/(32C_*)$. The initial likelihood is at least $-\log(2(t+1))$, and all likelihoods are at most zero. Therefore the number of iterations is at most

$$1 + 32C_* \log(2(t+1))/e_0^2 = O(t^2 e_0^{-4} \log(2t)).$$

All kernel probabilities are rational numbers of bit length $O(t \log M)$, and every update uses the same rational step. After polynomially many iterations all weights have polynomial bit length. The derivatives and the stopping comparison are rational, so no logarithm evaluation is needed by the algorithm. The displayed certificate may equivalently be replaced by its rational upper bound $t(t-1)/M + \gamma/(1-\gamma) + g(v)$. □

Status and scope. The sharp deterministic inverse modulus and the certified approximate-MLE extension are completed targets in the present program. They do not constitute two newly resolved published conjectures. The published near-linear likelihood question was addressed in the preceding manuscript; its extension to the full statistical transition remains unproved here. The inverse theorem proves a sharp limitation of one proof strategy, while the optimization theorem removes the need for an exact optimizer at statistically relevant precision. Classical beta–binomial spectral theory, Jackson approximation, EM monotonicity and conditional-gradient optimization are explicitly credited as ingredients.

5.4 Complete adaptive replication thresholds for endpoint populations

The scalar uniform-endpoint experiment in the preceding paper has the critical uncapped rate $(\log B)/\sqrt{B}$. We now determine its entire replication-cap curve, and obtain a different, noncritical phase diagram in every higher dimension. The mixing law may have arbitrary dependence between its coordinates. The lower bound permits sequential decisions within a strictly stronger continuous-time observation experiment.

Fix an integer $d \geq 1$. An individual has a latent vector $\theta \in [1,2]^d$ drawn independently from an unknown probability law G. A raw observation is a vector whose coordinates, conditionally on θ, are independent Uniform$[0,\theta_j]$ variables. Raw vectors from the same individual are conditionally independent. A policy spends at most B raw observations and observes any individual at most Q times, where $1 \leq Q \leq B$. Let $U^{\mathrm{ad}}_{B,Q,d}$ be the minimax expected Euclidean W_1 loss over arbitrary adaptive policies and probability-valued estimators; $U^{\mathrm{fix}}_{B,Q,d}$ restricts the replication counts to a schedule chosen independently of the data. Constants below may depend on the fixed dimension.

Theorem 5.4.1 (The complete endpoint cap curve)**.** *Uniformly for integers $B \geq 2$ and $1 \leq Q \leq B$,*

$$U^{\mathrm{ad}}_{B,Q,1} \asymp U^{\mathrm{fix}}_{B,Q,1} \asymp \max\left\{\frac{\log(eB)}{\sqrt{B}}, (BQ)^{-1/3}\right\}.$$

For every fixed $d \geq 2$,

$$U^{\mathrm{ad}}_{B,Q,d} \asymp U^{\mathrm{fix}}_{B,Q,d} \asymp \max\left\{B^{-1/(d+1)}, (BQ^{2d-1})^{-1/(3d)}\right\}.$$

Consequently, the least cap that attains the uncapped minimax order, within constants, is

$$Q \asymp \frac{\sqrt{B}}{\log^3(eB)} \quad (d=1), \qquad Q \asymp B^{1/(d+1)} \quad (d \geq 2).$$

Adaptive sampling changes none of these orders.

This theorem concerns a new classification problem for the specified experiment, rather than a named conjecture from the literature. Its $Q = 1$ exponent $B^{-1/(3d)}$ is consistent with ordinary-smooth deconvolution: after taking logarithms, each coordinate is contaminated by a one-sided exponential error. Classical Wasserstein deconvolution rates and inversion methods are developed in [72, 73]. Those results do not optimize replication counts under a total budget or permit selection of the next individual from the observed history. The new ingredient here is a uniform information charge for every adaptive history. Recent multivariate CDF deconvolution work [74] distinguishes integrated CDF loss from W_1 for dependent laws. Accordingly, our multivariate proof uses Lipschitz dual tests, and never identifies these two losses.

5.4.1 Frequency-dependent replication gives the upper bound

Put $a = \log 2$ and $U_j = \log \theta_j / a$. The maps between $[0,1]^d$ and $[1,2]^d$ are bi-Lipschitz, so it suffices to estimate the law of U. After r observations of an individual, the coordinate log-maxima form a vector

$$Y_r = U - (E_{r,1}, \ldots, E_{r,d}), \qquad E_{r,j} \overset{\mathrm{iid}}{\sim} \mathrm{Exp}(ar),$$

with errors independent of U.

We use a torus of side length 4, containing $[0,1]^d$ without a metric wrap-around, and write $\omega = \pi/2$. For a probability law H on the cube define $m_k(H) = \int e^{i\omega k\cdot u}\, dH(u)$. For every integer $K \geq 2$,

$$W_1(H,H') \leq \frac{C_d}{K} + C_d \left\{ \sum_{0<\|k\|_\infty \leq K} \frac{|m_k(H) - m_k(H')|^2}{\|k\|_2^2} \right\}^{1/2}. \tag{5.4.1}$$

Indeed, a 1-Lipschitz function on the cube extends to a 1-Lipschitz function on this torus by the McShane formula. Convolve it with the product of positive one-dimensional Jackson kernels of degree at most K. The uniform approximation error is C_d/K, while Parseval bounds the

weighted square sum of its Fourier coefficients by its bounded Dirichlet integral. Cauchy–Schwarz and Kantorovich duality give (5.4.1). The use of the larger torus is essential: identifying the endpoints of $[0,1]$ itself would destroy the claimed Euclidean transportation bound.

For each k an unbiased statistic of Y_r is

$$A_{r,k}(Y_r) = e^{i\omega k\cdot Y_r}\prod_{j=1}^{d}\left(1+\frac{i\omega k_j}{ar}\right), \qquad |A_{r,k}| \le C_d(1+\|k\|_\infty/r)^d.$$

Partition frequencies into dyadic bands of upper radius k, and use $r_k = \min(Q,k)$ observations per individual in that band. With n_k fresh individuals, its contribution to the expected squared norm in (5.4.1) is at most a_k/n_k, where

$$a_k = C_d k^{d-2}(1+k/r_k)^{2d}.$$

Let $S = \sum_{k\le K}\sqrt{r_k a_k}$ and take $n_k = \lceil B\sqrt{a_k/r_k}/(2S)\rceil$. The cost is at most $B/2 + \sum r_k \le B/2 + CK$, and the total squared error at most $2S^2/B$. Rounding is harmless for the choices of K below, since $K \le \sqrt{B}$; adjust constants for bounded B. Project the estimated moment vector onto the compact set of moment vectors of probability laws on the cube, in the displayed weighted Euclidean norm. A measurable approximate minimizer over a countable dense set gives a probability-valued estimator and loses only a factor two and an arbitrarily small additive error.

The geometric sums give

$$S \le C\{\log(eK) + \sqrt{K/Q}\}, \qquad d = 1, \tag{5.4.2}$$

$$S \le C_d\{K^{(d-1)/2} + K^{(3d-2)/2}Q^{-(2d-1)/2}\}, \qquad d \ge 2. \tag{5.4.3}$$

For $d = 1$ choose a dyadic K within constants of

$$\min\left\{\frac{\sqrt{B}}{\log(eB)}, (BQ)^{1/3}\right\}.$$

Then $K^{-1} + S/\sqrt{B}$ is at most a constant times the claimed maximum. For $d \ge 2$ take K within constants of

$$\min\{B^{1/(d+1)}, (BQ^{2d-1})^{1/(3d)}\}.$$

Both inequalities $K^{d+1} \le B$ and $K^{3d} \le BQ^{2d-1}$ hold, and (5.4.3) gives error at most C_d/K. This proves every upper bound in Theorem 5.4.1.

5.4.2 A Poisson experiment that dominates every capped policy

For one coordinate of one individual, expose a Poisson point process on $[0,\lambda]\times[0,U_j]$ with unit intensity. The second coordinate of its highest point, with value zero when there is no point, has the same law as $\max(0, U_j - E)$, $E \sim \mathrm{Exp}(\lambda)$. An exposure block of length a therefore simulates one scaled log-uniform observation: when no point occurs, attach independently a negative $\mathrm{Exp}(a)$ value, and otherwise use the block maximum. This reconstructed observation is $U_j - \mathrm{Exp}(a)$. The full point configuration supplies at least as much information.

Allow the statistician to allocate exposure continuously and separately to coordinates and individuals, with total exposure at most daB and exposure of every coordinate of any individual at most aQ. This stronger experiment simulates the original one, even with its adaptation. It is enough to prove a lower bound in the stronger experiment. Conditionally on a fixed adaptive history, an individual's likelihood for $u \in [0,1]^d$ is proportional to

$$e^{-\lambda\cdot u}\mathbf{1}_{\{u\ge b\}}, \qquad 0 \le \lambda_j \le aQ, \tag{5.4.4}$$

where b_j is the largest observed mark in its jth coordinate. The observed choice probabilities have no parameter factor once their history is fixed. Consequently (5.4.4) remains valid under arbitrary adaptive allocations.

Lemma 5.4.2 (Localized information charge)**.** *Fix a nonzero smooth function ψ supported in $(0,1)$ with $\int \psi = 0$ and $\|\psi\|_\infty \le 1$. Pack $M \asymp_d h^{-d}$ disjoint cubes of side h inside $[1/4, 3/4]^d$, and set*

$$\psi_j(u) = \prod_{\ell=1}^{d} \psi((u_\ell - c_{j,\ell})/h), \qquad w_\beta(u) = 1 + \varepsilon \sum_{j=1}^{M} \beta_j \psi_j(u), \quad |\beta_j| \le 1,$$

where $\varepsilon \in (0, 1/4)$ is fixed. For every policy in the stronger Poisson experiment, the trace of its Fisher information in β is at most

$$C_d B Q^{2d-1} h^{2d}. \tag{5.4.5}$$

Proof. Write $\Psi(t) = \int_{-\infty}^{t} \psi(s)\, ds$. The full tensor antiderivative of ψ_j is $\Phi_j(u) = h^d \prod_\ell \Psi((u_\ell - c_{j,\ell})/h)$. It is supported in its cube, and disjointness gives $\|(\Phi_j(u))_{j=1}^M\|_2 \le C_d h^d$ at every u. For a posterior state (λ, b) set

$$Z(b) = \int_{u \ge b} w_\beta(u) e^{-\lambda \cdot u}\, du, \qquad H_j(b) = \varepsilon \int_{u \ge b} \psi_j(u) e^{-\lambda \cdot u}\, du, \qquad s(b) = H(b)/Z(b).$$

Integration by parts in each coordinate, followed by the triangle inequality in ℓ_2, gives

$$\|H(b)\|_2 \le C_d h^d e^{-\lambda \cdot b}.$$

There are at most 2^d terms: each consists of a boundary evaluation or an integral with factor λ_ℓ, and $\lambda_\ell \int_{b_\ell}^1 e^{-\lambda_\ell u}\, du \le e^{-\lambda_\ell b_\ell}$. If every $b_\ell \le 3/4$, then

$$Z(b) \ge c_d e^{-\lambda \cdot b} \prod_\ell (1 + \lambda_\ell)^{-1}, \qquad \|s(b)\|_2 \le C_d h^d \prod_\ell (1 + \lambda_\ell). \tag{$*$}$$

If any $b_\ell > 3/4$, all the $H_j(b)$ vanish, so $s(b) = 0$.

While exposing coordinate i, the conditional intensity of a new mark $y > b_i$ is $T_i(y) = Z(b^{i,y})/Z(b)$, where $b^{i,y}$ replaces b_i by y. Its score vector is $s(b^{i,y}) - s(b)$. Marks below b_i have intensity one and zero score. The conditional Fisher trace per unit exposure is therefore

$$\int_{b_i}^{1} T_i(y) \|s(b^{i,y}) - s(b)\|_2^2\, dy. \tag{5.4.6}$$

For completeness, on a time interval dt the probability of a mark in dy is $T_i(y) dy\, dt + o(dt)$; the squared score of the no-mark outcome contributes $o(dt)$. Summing these conditional score variances, or taking the corresponding compensated counting-process martingale limit, gives (5.4.6). Scores and intensities here are bounded on each fixed finite-dimensional family, so truncation and dominated convergence justify this calculation.

The posterior density is between fixed constant multiples of the product of truncated exponential densities. Hence

$$\int_{b_i}^{1} T_i(y)\, dy = \mathbb{E}_\beta(U_i - b_i \mid \text{history}) \le \frac{C}{1 + \lambda_i}.$$

Using $(*)$, with zero score beyond the support when necessary, bounds (5.4.6) by

$$C_d h^{2d} (1 + \lambda_i) \prod_{\ell \ne i} (1 + \lambda_\ell)^2 \le C_d h^{2d} Q^{2d-1}.$$

This is a conditional bound for every history, rather than an average over predetermined sample sizes. Integrating over total exposure at most daB and using orthogonality of score increments proves (5.4.5). Independent policy randomization adds no information and may be included in the observed transcript. □

5.4.3 Turning trace information into transportation loss

We record the elementary absolute-loss information bound used below. If a joint density $p(\beta, Y)$ has finite positive definite Fisher matrix J and integrable weak parameter derivatives, then every estimator $T(Y)$ and vector v satisfy

$$\mathbb{E}|T(Y) - v^T\beta| \geq \frac{1}{4}\sqrt{v^T J^{-1} v}. \tag{5.4.7}$$

Indeed translation by z changes total variation by at most $\sqrt{z^T J z}/2$. Integrating the triangle inequality against the minimum of the original and translated densities gives $2\mathbb{E}|T - v^T\beta| \geq |v^T z|(1 - \mathrm{TV})$. Take $z = J^{-1}v/\sqrt{v^T J^{-1} v}$.

Give each β_j independently the density $\cos^2(\pi\beta_j/2)\mathbf{1}_{[-1,1]}(\beta_j)$. Its Fisher information is π^2, so the joint prior-and-data matrix satisfies

$$\operatorname{tr} J \leq C_d\{M + BQ^{2d-1}h^{2d}\}.$$

Let f_j be a smooth function supported in the jth cube, with Lipschitz constant bounded independently of h, such that $\int f_j\psi_j = c_\psi h^{d+1}$ for a nonzero fixed c_ψ. For example, take $f_j = h\psi_j$, so $c_\psi = (\int_0^1 \psi(t)^2 dt)^d > 0$. Disjoint support and zero boundary values ensure that every sum $\sum_j \sigma_j f_j$, $|\sigma_j| \leq 1$, is C_d-Lipschitz. Thus every probability-valued estimator obeys

$$W_1(\widehat{G}, G_\beta) \geq c_d h^{d+1} \sum_{j=1}^{M} |T_j(\widehat{G}) - \beta_j|, \qquad T_j(H) = \frac{\int f_j\, dH - \int f_j(u)\, du}{\varepsilon c_\psi h^{d+1}}.$$

Apply (5.4.7) to each coordinate. Since $(J^{-1})_{jj} \geq 1/J_{jj}$ and $\sum_j J_{jj}^{-1/2} \geq M^{3/2}/\sqrt{\operatorname{tr} J}$, the Bayes, and hence minimax, risk is at least

$$\frac{c_d h^{d+1} M^{3/2}}{\sqrt{M + BQ^{2d-1}h^{2d}}}.$$

Choose h within fixed small constants of $(BQ^{2d-1})^{-1/(3d)}$, and use $M \asymp_d h^{-d}$. The denominator is $O_d(\sqrt{M})$, proving

$$U^{\mathrm{ad}}_{B,Q,d} \geq c_d(BQ^{2d-1})^{-1/(3d)}. \tag{5.4.8}$$

Every density in this lower-bound family is bounded above and below by fixed positive constants. No density regularity is assumed in the minimax problem; the derivatives of these localized bumps increase as h decreases.

5.4.4 The uncapped obstruction and the scalar critical dimension

For $d \geq 2$ the uncapped lower bound $B^{-1/(d+1)}$ has a short fully adaptive proof. Pack $M \asymp_d h^{-d}$ separated cells in the interior of the latent cube. In each cell choose one of two point parameters separated by h in the first coordinate, and give every cell mass $1/M$. Strengthen the experiment by allowing the statistician to select the cell for every raw observation. A new individual in the original experiment is simulated by an independent uniform cell label; revisits retain the label. The oracle is therefore more informative.

The total variation between the two component observation laws at a cell is at most $C_d h$. Couple two adaptive transcripts, differing only at cell j, until their first different answer. Conditional on agreement, each query to cell j disagrees with probability at most $C_d h$. Consequently their transcript total variation is bounded by $C_d h\, \mathbb{E}_\omega N_j$, with expectation under either specified starting vertex ω. Averaging this bound over all directed hypercube edges and using $\sum_j N_j \leq B$ gives mean edge variation at most $C_d Bh/M$. On the other hand localized Lipschitz functions

show that the W_1 separation between two vertices is at least $c_d h/M$ times their Hamming distance. Binary testing and nearest-vertex decoding give risk at least $c_d h$ when Bh/M is a sufficiently small constant. Taking $h = c_d B^{-1/(d+1)}$ proves the claimed term. This argument also gives $B^{-1/2}$ in dimension one but misses its critical logarithm.

The necessary scalar logarithm was proved in the preceding paper. We include its information argument to specify the exact input. For $U \in [0,1]$ take

$$w_\beta(u) = 1 + \sum_{k=1}^{K} \sqrt{2}\{\beta_{k,c}\cos(2\pi k u) + \beta_{k,s}\sin(2\pi k u)\}, \quad |\beta_{k,*}| \le cK^{-2}.$$

These densities and their Lipschitz constants are uniformly bounded, and the densities are bounded away from zero. Under baseline density one, a posterior after r raw observations is a truncated exponential $q(u) \propto e^{-aru}\mathbf{1}_{[b,1]}(u)$. If q_Y is the posterior after the next observation, direct integration gives

$$\mathbb{E}_0\|q_Y - q\|_2^2 = a/2, \qquad \|q\|_2^2 \operatorname{Var}_q(U) \le C.$$

For clarity, writing $\lambda = ar$, $D = 1-b$, the current squared norm is $\lambda\coth(\lambda D/2)/2$; the expectation of the next one is this quantity plus $a/2$. The identity then follows from $\mathbb{E}_0 q_Y = q$, including the limit $\lambda = 0$. If $z = \int w_\beta q$ and $z_Y = \int w_\beta q_Y$, the one-step score is the vector of Fourier coefficients of $q_Y/z_Y - q/z$. Bessel's inequality, $z, z_Y \asymp 1$, and

$$\mathbb{E}_0|z_Y - z|^2 \le \operatorname{Lip}(w_\beta)^2 \mathbb{E}_0 W_1(q_Y, q)^2 \le 2\operatorname{Lip}(w_\beta)^2 \operatorname{Var}_q(U)$$

bound its information trace by an absolute constant for every history. Thus any adaptive B-query experiment has trace at most CB.

A product cosine-squared prior of width cK^{-2} contributes CK^5. For the centered CDF, the coefficient vector is

$$v_{k,c}(x) = \frac{\sqrt{2}\sin(2\pi k x)}{2\pi k}, \qquad v_{k,s}(x) = -\frac{\sqrt{2}\cos(2\pi k x)}{2\pi k}.$$

The integrated absolute error of the centered CDF is at most twice W_1. Applying (5.4.7) at every x gives a lower bound by a constant times $\int_0^1 \sqrt{v(x)^T J^{-1} v(x)}dx$. This functional of J is convex: for fixed v it is $\sup_{z\neq 0} |v^T z|/\sqrt{z^T J z}$. Average J over the block rotations induced by translating x. The averaged matrix has blocks $j_k I_2$, and Cauchy–Schwarz yields

$$\int_0^1 \sqrt{v(x)^T J^{-1} v(x)}dx \ge \frac{c\sum_{k=1}^K k^{-1}}{\sqrt{\operatorname{tr} J}} \ge \frac{c\log(eK)}{\sqrt{B + K^5}}.$$

Taking $K \asymp B^{1/5}$ proves $c\log(eB)/\sqrt{B}$ even without a cap, and hence for every cap. Together with (5.4.8) and the preceding upper bounds this completes Theorem 5.4.1.

5.5 The complete adaptive Bernoulli replication-cap curve

The preceding paper determined the Bernoulli cap curve for predetermined designs and for adaptive designs at polynomially subcritical caps. Its remaining transition window is closed by a different least-favourable family: localized moment-matching perturbations are immersed in a global uniform background. The background controls posterior normalizers even after unusually favourable sequences of tosses. This supplies the conditional information estimate that a fixed-prefix divergence bound alone cannot imply.

Let $R^{\mathrm{ad}}_{B,Q}$ and $R^{\mathrm{fix}}_{B,Q}$ denote the minimax expected W_1 risk for estimating an arbitrary law of $\theta \in [0,1]$ from conditionally independent Bernoulli(θ) observations, with total budget B and per-individual cap Q.

Theorem 5.5.1 (No adaptive improvement at any Bernoulli cap)**.** *For absolute constants, uniformly over integers $B \geq 2$, $1 \leq Q \leq B$,*

$$R^{\mathrm{ad}}_{B,Q} \asymp R^{\mathrm{fix}}_{B,Q} \asymp \Psi(B,Q) := \max\left\{B^{-1/3}, Q^{-1}, [Q\log(e+B/Q^{3/2})]^{-1/2}\right\}.$$

In particular, at $Q \asymp B^{2/3}/\log B$ the adaptive minimax risk has order $B^{-1/3}\sqrt{\log B/\log\log B}$. The uncapped order $B^{-1/3}$ is attainable within constants if and only if $Q \gtrsim B^{2/3}$; the implicit cap constant may depend on the allowed risk constant.

We give the new adaptive lower bound first. All constants in its proof are absolute and can be chosen in the stated order.

Lemma 5.5.2 (A background-stabilized conditional information bound)**.** *There are constants $c_0, C > 0$ and $\rho \in (0,1)$ such that the following holds. Let $1 \leq K < Q$, put $h^2 = c_0K/Q$, and suppose $h \leq 1/100$. Choose $M \asymp h^{-1}$ intervals $I_j = [u_j - h, u_j + h]$ with centers in $[3/8, 5/8]$ separated by at least $4h$. There exist probability laws $\nu_{j,+}, \nu_{j,-}$ on I_j, matching moments through degree K, with $W_1(\nu_{j,+}, \nu_{j,-}) = 2h/(K+1)$. Set $\delta_j = \nu_{j,+} - \nu_{j,-}$, $\mu_j = (\nu_{j,+} + \nu_{j,-})/2$, and*

$$G_\beta = \tfrac{1}{2}\,\mathrm{Uniform}[1/4, 3/4] + \frac{1}{2M}\sum_{j=1}^{M}(\mu_j + \tfrac{1}{2}\beta_j\delta_j), \qquad |\beta_j| \leq 1. \tag{5.5.1}$$

For any adaptive policy with total budget B and cap Q, the trace of its Fisher information matrix in β is at most

$$CBh^2\rho^K. \tag{5.5.2}$$

Proof. Split $2^{-K}\sum_{i=0}^{K+1}(-1)^i\binom{K+1}{i}\delta_{i/(K+1)}$ into positive and negative probability parts and rescale to I_j. Finite differences prove moment matching. Integrating the absolute cumulative signed mass, and using $\sum_{i=0}^{l}(-1)^i\binom{K+1}{i} = (-1)^l\binom{K}{l}$, gives the stated W_1 distance. Every law in (5.5.1) obeys, for every interval I,

$$G_\beta(I) \leq C(|I| + h), \qquad G_\beta \geq \tfrac{1}{2}\,\mathrm{Uniform}[1/4, 3/4]. \tag{5.5.3}$$

The first bound follows by counting the $4h$-separated cells meeting I; each cell has total mass $1/(2M) \asymp h$.

Consider an individual with s successes in r previous tosses, where $0 \leq r \leq Q-1$. Write

$$L(x) = x^s(1-x)^{r-s}, \quad m = \int L\,dG_\beta, \quad p = \frac{\int xL(x)\,dG_\beta(x)}{m}.$$

Its next outcome is Bernoulli(p), with $p \in [1/4, 3/4]$. Conditioning on the entire adaptive history adds no further factor to this individual's posterior once β is fixed. Therefore its conditional Fisher trace is at most

$$C\sum_{j=1}^{M}|\partial_{\beta_j}p|^2, \qquad \partial_{\beta_j}p = \frac{1}{4Mm}\int(x-p)L(x)\,d\delta_j(x). \tag{5.5.4}$$

For $r < K$ these derivatives vanish by moment matching.

Suppose $r \geq K$, set $v = r+1$, and fix one center $u = u_j$. Let $A = (\log L)'(u) = (s - ru)/(u(1-u))$. Uniformly in all the possible histories,

$$m \geq c\,v^{-1/2}L(u)\exp(cA^2/v), \tag{5.5.5}$$

$$|p - u| \leq C\{|A|/v + h + v^{-1/2}\}. \tag{5.5.6}$$

Here are details that in particular cover extreme success counts. On $[1/4, 3/4]$, $(\log L)'' \ge -Cr$ and $|A| \le Cr$. For a sufficiently large absolute C_1, integrate the lower Taylor bound $\log L(u+t) \ge \log L(u) + At - C_1 v t^2$ over an interval of length $c/\sqrt{v}$ centered at $t = A/(2C_1 v)$. Taking C_1 large keeps this interval inside $[1/4 - u, 3/4 - u]$. Completing the square and using the uniform background proves (5.5.5).

If $z = s/r \in [1/4, 3/4]$, then $L(x) \le L(z) e^{-cr(x-z)^2}$ on that interval, while the background gives $m \ge cL(z)/\sqrt{v}$ by one-sided integration when needed. The interval mass bound (5.5.3), summed over annuli of width $v^{-1/2}$ around z, gives

$$\int |x - z| L(x)\, dG_\beta(x) \le CL(z)(v^{-1} + hv^{-1/2}).$$

Thus $|p - z| \le C(v^{-1/2} + h)$, and $|u - z| \le C|A|/v$. If z lies outside $[1/4, 3/4]$, its distance from $u \in [3/8, 5/8]$ is at least $1/8$, so $|A|/v$ is bounded below by a positive absolute constant. Then (5.5.6) follows from $p, u \in [1/4, 3/4]$ directly.

Fix a small absolute $\kappa > 0$ and let $R = \kappa\sqrt{(K+1)/v} \le \kappa$. For complex z with $|z| \le R$, Taylor expansion of the analytic logarithms around u gives

$$|L(u+z)| \le L(u) \exp\{|A|R + CrR^2\}.$$

Choose c_0 sufficiently small, after κ, so that $h/R < 1/4$. Cauchy's estimate for the Taylor remainder of the polynomial $F(x) = (x-p)L(x)$, and annihilation of its terms through degree K, give

$$\left| \int F\, d\delta_j \right| \le CL(u)(|u - p| + R) e^{|A|R + CrR^2} (h/R)^{K+1}.$$

Combining this with (5.5.5)–(5.5.6), $M^{-1} \asymp h$, and $h\sqrt{v} \le \sqrt{c_0 K}$ gives

$$|\partial_{\beta_j} p| \le Ch\sqrt{K+1}\, e^{-cA^2/v} \{\eta \sqrt{v/Q}\}^{K+1}, \tag{$**$}$$

where η can be made any prescribed positive small constant by reducing c_0. To check this last absorption, set $z_0 = |A|/\sqrt{v}$. Before absorption the extra factor is bounded by $(z_0 + \sqrt{K+1}) e^{-cz_0^2 + \kappa\sqrt{K+1} z_0 + C\kappa^2(K+1)}$. Young's inequality bounds it by $C\sqrt{K+1} e^{-c' z_0^2} e^{C'(K+1)}$; the last exponential is absorbed into $(h/R)^{K+1} \le \{\sqrt{c_0}\sqrt{v/Q}/\kappa\}^{K+1}$. All constants precede the final choice of c_0.

The centers are $4h$-separated, and $A_j = (s - ru_j)/(u_j(1-u_j))$. A Gaussian sum therefore gives

$$\sum_j e^{-cA_j^2/v} \le C(1 + (h\sqrt{v})^{-1}).$$

Indeed $|A_j|/\sqrt{v} \ge c\sqrt{v}\,|u_j - s/r|$ since $r \ge 1$. Let $x = v/Q \le 1$. Squaring $(**)$ and summing shows that (5.5.4) is at most

$$Ch^2(K+1)\eta^{2(K+1)} x^{K+1} \{1 + (c_0 K x)^{-1/2}\} \le Ch^2 \rho^K,$$

for some fixed $\rho < 1$, after taking $\eta < 1/4$ and absorbing the polynomial factors in K. This bound holds for every adaptive history and every β in the parameter cube. Orthogonality of conditional score increments through at most B tosses proves (5.5.2). Policy randomization has zero parameter score. □

Adaptive lower bound in Theorem 5.5.1. Use the family in Lemma 5.5.2, and independent cosine-squared priors on the coordinates $\beta_j \in [-1, 1]$. The joint Fisher matrix has

$$\operatorname{tr} J \le C\{M + Bh^2 \rho^K\}.$$

For each cell choose a 1-Lipschitz Kantorovich potential for δ_j. Subtract its linear interpolant between the two cell endpoints. Its integral against δ_j is unchanged because the zeroth and first moments vanish; its Lipschitz constant is at most two and its endpoint values are zero. Extend by zero outside the cell and call the resulting function f_j. Then $\int f_j \, d\delta_j = 2h/(K+1)$ and all signed sums of the f_j are 2-Lipschitz. Defining the affine estimator coordinate T_j from $\int f_j \, d\widehat{G}$ as in the endpoint argument gives

$$W_1(\widehat{G}, G_\beta) \geq \frac{ch}{M(K+1)} \sum_{j=1}^{M} |T_j - \beta_j|.$$

The absolute-loss inequality (5.4.7) and the trace bound thus give risk at least

$$\frac{ch}{K+1} \left(1 + Bh^3 \rho^K\right)^{-1/2}. \tag{5.5.7}$$

Set $L = \log(e + B/Q^{3/2})$ and $K = \lceil AL \rceil$, where A is a sufficiently large absolute constant. When $Q \geq C_2 L$, choose C_2 large enough for $K < Q$ and $h \leq 1/100$. Since $h^2 = c_0 K/Q$,

$$Bh^3 \rho^K = C c_0^{3/2} (B/Q^{3/2}) K^{3/2} \rho^K \leq C,$$

uniformly, by increasing A. Equation (5.5.7) therefore gives $c[QL]^{-1/2}$. When $Q < C_2 L$, this term is dominated, within a constant, by Q^{-1}.

The remaining two lower bounds also allow adaptation. Two probability laws matching moments through degree Q and separated in W_1 by $1/(Q+1)$ have identical full Q-toss laws. An oracle of independent full Q-toss vectors simulates every capped policy, proving the Q^{-1} floor. For $B^{-1/3}$, pack $M \asymp h^{-1}$ cells with two point biases at distance h in each, inside $[1/4, 3/4]$, and reveal or even allow selection of cell labels. Neighbouring hypercube transcripts have KL at most $Ch^2 \mathbb{E} N_j$. Averaging over edges, using $\sum_j N_j \leq B$, and applying Pinsker gives mean edge variation at most $C\sqrt{Bh^2/M}$. With $h = cB^{-1/3}$ it is below $1/2$, and the separated-cell transportation loss yields risk at least ch. Combining the three bounds proves the adaptive lower bound. □

Predetermined upper bound in Theorem 5.5.1. We recall the frequency allocation argument to make the relation to the lower bound explicit. Let $\widetilde{T}_m(x) = T_m(2x-1)$. The Chebyshev-moment inequality

$$W_1(G, H) \leq C/K + C \left\{ \sum_{m=1}^{K} m^{-2} \left| \int \widetilde{T}_m \, d(G - H) \right|^2 \right\}^{1/2}$$

follows by applying the positive Jackson kernel and Parseval to $f((1+\cos u)/2)$; see also [14]. For $m \leq r/2$, the degree-r Bernstein coefficients of $\widetilde{T}_m$ are bounded by $Ce^{2m^2/r}$, and hence define an unbiased statistic of a binomial(r, θ) observation with that bound. For clarity, on the coefficient contour $z = te^{iu}$, $t = j/(r-j)$, put $v = 2\sqrt{t}\,|\sin(u/2)|/(1+t)$. The generating polynomial is

$$(1+z)^r \widetilde{T}_m\left(\frac{z}{1+z}\right) = \tfrac{1}{2}(1+z)^{r-m}\{(\sqrt{z}+i)^{2m} + (\sqrt{z}-i)^{2m}\}.$$

Its normalized modulus is bounded by $(1-v^2)^{(r-m)/2}(1+v)^m \leq e^{2m^2/r} e^{-rv^2/8}$. The Gaussian contour integral contributes $C(1+rt/(1+t)^2)^{-1/2}$, canceled by the reciprocal binomial probability in Stirling's formula. The endpoint coefficients are one in absolute value. This proves the required coefficient bound.

For the dyadic band ending at $k \leq K \leq Q/2$, use $r_k = \min(Q, \max(2, k^2))$ trials per fresh individual. Its weighted moment variance is at most a_k/n_k, where $a_k = Ck^{-1} e^{4k^2/r_k}$. The

allocation used earlier in this section gives $\sum \sqrt{r_k a_k} \le C\sqrt{K} e^{2K^2/Q}$ and total cost at most B when $K^2 \le cB$. Moment projection therefore gives

$$\sup_G \mathbb{E} W_1(\widehat{G}, G) \le C/K + C\sqrt{K/B}\, e^{2K^2/Q}.$$

Choose a dyadic K within a sufficiently small absolute factor of $\min\{B^{1/3}, Q, \sqrt{QL}\}$, with L as above. If $Q \ge B^{2/3}$ the exponential is bounded. Otherwise put $x = B/Q^{3/2} > 1$; the stochastic term divided by $1/K$ is at most $C \log(e+x)^{3/4}(e+x)^{\eta}/\sqrt{x}$ with any prescribed small $\eta > 0$, by making the factor defining K small enough. It is bounded for $\eta < 1/4$. This proves the upper bound, with bounded small parameters covered by the diameter bound. □

The new Bernoulli and endpoint classifications use different observation kernels but share a principle: the lower bound must control the information of the next adaptively selected query conditionally on its entire history. In the endpoint model tensor antiderivatives provide this control. In the Bernoulli model it is the uniform background, the complex likelihood remainder, and the Gaussian sum over localized cells. A fixed-prefix divergence estimate without this conditional step would not justify either conclusion about adaptive designs.

5.6 Power restoration of hyperbolic complete monotonicity

The preceding manuscripts classified the GGC and HCM properties of the original McKay families and then proved their full ID domains. Those results are not counted again here. The new question is whether a power transformation can restore HCM after the untransformed density has failed it. This section proves an exact exponent boundary for restoration under exponential tilting, gives a constructive sufficient bound for a fixed law, and extends the strict SD threshold into part of the previously unresolved negative-cut range.

A positive function f is HCM if, for each $u > 0$, the function $f(uv)f(u/v)$ is completely monotone in $v + v^{-1}$. Multiplication by any real monomial preserves HCM. We use the classical facts that an HCM function with value one at zero is the Laplace transform of a GGC, and hence has a Stieltjes logarithmic derivative, and that every HCM density is GGC. Equivalently, the normalized HCM functions considered below have the representation

$$g(x) = \exp\left\{ -d_0 x - \int_0^\infty \log(1 + x/t)\, U(dt) \right\}, \qquad d_0 \ge 0, \quad U \ge 0. \tag{5.6.1}$$

The constructed functions with $p < 1/2$ below have $d_0 = 0$. See [75, Chapter 5, especially Theorem 5.4.1] and [4]. Sufficiency in our result can also be checked directly from finite products $(1 + x/t)^{-c}$ and passage to their pointwise limits.

For $\mu > -1$ write

$$\mathcal{I}_\mu(z) = 2^\mu \Gamma(\mu+1) z^{-\mu} I_\mu(z) = {}_0F_1(;\mu+1; z^2/4), \qquad R_\mu(z) = I_{\mu+1}(z)/I_\mu(z).$$

The first expression is understood by its entire even continuation. Thus $\mathcal{I}_\mu(0) = 1$. Let $j_{\mu,n}$ be the positive zeros of J_μ. The classical product and its logarithmic derivative give

$$\mathcal{I}_\mu(z) = \prod_{n \ge 1} \left(1 + z^2/j_{\mu,n}^2\right), \qquad R_\mu(z) = \sum_{n \ge 1} \frac{2z}{z^2 + j_{\mu,n}^2}. \tag{5.6.2}$$

The series is locally uniform away from its poles. For $\mu > -1$ all nonzero zeros of J_μ are real and simple; see [6, Sections 10.21(i), 10.21(iii)].

5.6.1 A positive spectral measure after changing the power

Lemma 5.6.1 (A ray estimate from a tilted beta law)**.** *For $\mu \geq 1/2$, $y > 0$, and $0 < \theta < \pi/2$,*

$$0 < \frac{\operatorname{Im}\{e^{i\theta} R_\mu(ye^{i\theta})\}}{\sin\theta} \leq \sec\theta. \tag{5.6.3}$$

For every fixed $\mu > -1$ and θ in the same range the middle quantity is positive, continuous in y, bounded, tends to zero at zero, and tends to one at infinity.

Proof. The product formula gives, term by term,

$$\begin{aligned}\frac{\operatorname{Im}\{e^{i\theta} R_\mu(ye^{i\theta})\}}{\sin\theta} &= 4y\cos\theta \sum_{n\geq 1} \frac{j_{\mu,n}^2}{j_{\mu,n}^4 + 2y^2 j_{\mu,n}^2 \cos(2\theta) + y^4} \\ &\leq \frac{1}{\cos\theta}\{R_\mu(y) + yR'_\mu(y)\}.\end{aligned}$$

Indeed the denominator in each summand is at least $\cos^2\theta (j_{\mu,n}^2 + y^2)^2$.

We give an elementary proof that $R_\mu(y) + yR'_\mu(y) \leq 1$ for $\mu \geq 1/2$. Put $c = \mu - 1/2 \geq 0$. The Bessel beta integral represents $R_\mu(y)$ as the expectation of T under the probability density proportional to $e^{yt}(1-t^2)^c$ on $(-1,1)$. Consequently $y\{1 - R_\mu(y)\}$ is the mean of $Z = y(1-T)$, whose density is proportional to

$$z^c (2 - z/y)^c e^{-z}\, \mathbf{1}_{(0,2y)}(z).$$

If $y_2 > y_1$, the ratio of these densities on their common support, apart from a positive constant, is $\{(2 - z/y_2)/(2 - z/y_1)\}^c$. It is nondecreasing in z, and the second density has the larger right endpoint. The elementary single-crossing criterion for likelihood-ratio order therefore shows that Z is stochastically nondecreasing in y. Its mean is nondecreasing. Differentiating that mean gives the desired inequality. The case $c = 0$ is the same truncated-exponential ordering.

For arbitrary $\mu > -1$, positivity follows from the displayed sum. There are no poles on the fixed ray. The expansion at zero gives a limit of zero, and the standard Bessel asymptotic in every closed subsector of $|\arg z| < \pi/2$ gives $R_\mu(z) = 1 + O(1/z)$ at infinity. Thus the stated limit is one, and continuity proves boundedness. □

Theorem 5.6.2 (An exact exponent boundary for HCM restoration)**.** *Let $\mu > -1$, $a > 0$, $p > 0$, and $\eta > 0$. Consider the normalized density*

$$f_{p,b}(x) = Cx^{\eta-1} e^{-bx^p} \mathcal{I}_\mu(ax^p), \qquad x > 0, \qquad b > a. \tag{5.6.4}$$

There exists a $b > a$ for which this density is HCM if and only if

$$0 < p < \tfrac{1}{2}. \tag{5.6.5}$$

For $0 < p < 1/2$, put $\theta = \pi p$ and

$$c_\mu(p) = \sup_{y>0} \frac{\operatorname{Im}\{e^{i\theta} R_\mu(ye^{i\theta})\}}{\sin\theta}. \tag{5.6.6}$$

Then $1 \leq c_\mu(p) < \infty$, and for every $b > a$ the exact test is

$$f_{p,b} \text{ is HCM} \quad \Longleftrightarrow \quad b/a \geq c_\mu(p).$$

In particular, for $\mu \geq 1/2$ the elementary sufficient bound is

$$b/a \geq \sec(\pi p). \tag{5.6.7}$$

The upper bound in (5.6.7) is not asserted to be the optimal tilt threshold.

Proof. The density is integrable: at zero its non-power factor tends to one, and at infinity its exponential decay is $e^{-(b-a)x^p}$ times a power. Define $g(x) = e^{-bx^p}\mathcal{I}_\mu(ax^p)$, so $g(0) = 1$. Since powers preserve HCM, the density is HCM exactly when g is.

Assume first $0 < p < 1/2$. On the slit plane $D = \mathbb{C} \setminus (-\infty, 0]$, with principal powers, g is holomorphic and nonzero by (5.6.2). Its negative logarithmic derivative is

$$h(z) = -\frac{g'(z)}{g(z)} = pz^{p-1}\{b - aR_\mu(az^p)\}. \tag{5.6.8}$$

The upper-bank boundary value yields the real density

$$u_{p,b}(t) = -\frac{1}{\pi}\operatorname{Im} h(-t + i0) = \frac{pt^{p-1}}{\pi}\left[b\sin\theta - a\operatorname{Im}\{e^{i\theta}R_\mu(at^pe^{i\theta})\}\right]. \tag{5.6.9}$$

A keyhole application of Cauchy's formula gives the signed identity

$$h(z) = \int_0^\infty \frac{u_{p,b}(t)}{z+t}\,dt. \tag{5.6.10}$$

Here no residue is omitted. At zero, $h(z) = O(|z|^{p-1})$; at infinity, uniformly up to both banks, $h(z) = p(b-a)z^{p-1} + O(1/z)$. The circle terms in the Cauchy formula vanish respectively as $O(\varepsilon^p)$ and $O(R^{p-1})$. The density is $O(t^{p-1})$ at both endpoints, so the signed integral is absolutely convergent for $z \in D$.

The measure $U(dt) = u_{p,b}(t)\,dt$ is nonnegative exactly when $b/a \geq c_\mu(p)$. In that event integration of (5.6.10) from zero to x, justified by the just stated endpoint bounds, proves (5.6.1). This gives HCM, either by its classical representation or by positive finite-product approximation. Conversely, if g is HCM, its normalization at zero makes it a GGC Laplace transform. Its logarithmic derivative is a Stieltjes function. Stieltjes inversion in (5.6.10) forces $u_{p,b} \geq 0$; its continuity rules out a negative value even at a single interior point. This proves the exact test. Lemma 5.6.1 proves finiteness, the lower bound one, and (5.6.7). Choosing $b/a > c_\mu(p)$ supplies the claimed existence for every $\mu > -1$.

At the boundary $p = 1/2$, the same product formula gives

$$h(z) = \frac{b}{2\sqrt{z}} - \sum_{n\geq 1}\frac{a^2}{a^2z + j_{\mu,n}^2}. \tag{5.6.11}$$

At each $z = -j_{\mu,n}^2/a^2$ the residue of the second term is -1, while $b/(2\sqrt{z})$ has a finite one-sided boundary value there. A Stieltjes transform of a nonnegative measure can only have nonnegative residues at negative-axis poles. More explicitly, $\lim_{\varepsilon\downarrow 0} i\varepsilon h(-j_{\mu,n}^2/a^2 + i\varepsilon) = -1$, whereas that limit must be the nonnegative mass of the corresponding atom. Thus g cannot be HCM.

For $p > 1/2$, the point

$$z_n = (j_{\mu,n}/a)^{1/p}\exp\{i\pi/(2p)\}$$

lies strictly inside D and is a zero of the analytic continuation of g. A normalized HCM function is a GGC Laplace transform and has a zero-free continuation to D, contradicting this zero. This excludes every b and completes the equivalence. □

Corollary 5.6.3 (A constructive power for a fixed McKay law)**.** *Suppose X has density*

$$Cx^{\nu-1}e^{-bx}I_\mu(ax), \qquad x > 0,$$

with $\mu > -1$, $\mu + \nu > 0$, and $b > a > 0$. No positive power X^d with $d \leq 2$ has an HCM density. If $\mu \geq 1/2$, every power satisfying

$$d \geq \frac{\pi}{\arccos(a/b)} \tag{5.6.12}$$

has an HCM density, and consequently is GGC, SD, and ID. The assertion for a negative power d is identical with that for $|d|$.

Proof. The change of variables has the form (5.6.4) with $p = 1/d$ and $\eta = p(\mu + \nu) > 0$. The explicit sufficient bound follows from Theorem 5.6.2. For negative powers, the reciprocal-density transformation multiplies $f(1/x)$ by x^{-2}; inversion and monomials preserve HCM. □

5.6.2 Which fixed laws can be repaired by any power?

Theorem 5.6.4 (The fixed-tilt restoration criterion and its order boundary)**.** *For $\mu > -1$ set*

$$M_\mu = \sup_{y>0}\{R_\mu(y) + yR'_\mu(y)\}.$$

Then $1 \le M_\mu < \infty$, and

$$M_\mu = 1 \quad \Longleftrightarrow \quad \mu \ge 1/2. \tag{5.6.13}$$

If $-1 < \mu < 1/2$, the strictly larger maximum is attained. For a generalized McKay random variable X with fixed admissible parameters, the following are equivalent:

1. *Some positive power X^d has an HCM density.*
2. *Every sufficiently large positive power X^d has an HCM density.*
3. *$b/a > M_\mu$.*

A constructive sufficient power, valid for every order, is

$$d \ge \frac{\pi}{\arccos(aM_\mu/b)} \qquad \text{when } b/a > M_\mu. \tag{5.6.14}$$

Consequently $\mu \ge 1/2$ is exactly the set of orders for which every *admissible exponential tilt can be restored by a power.*

Proof. Let $D_\mu(y) = R_\mu(y) + yR'_\mu(y)$. Formula (5.6.2) shows $D_\mu(y) > 0$ and $D_\mu(y) \to 0$ at zero. The Bessel asymptotic, differentiated in a real subsector, gives

$$D_\mu(y) = 1 - \frac{4\mu^2 - 1}{8y^2} + O(y^{-3}). \tag{5.6.15}$$

Thus M_μ is finite and at least one. Lemma 5.6.1's real estimate gives $M_\mu = 1$ when $\mu \ge 1/2$. If $-1/2 < \mu < 1/2$, (5.6.15) implies $D_\mu(y) > 1$ at large y. If $-1 < \mu \le -1/2$, the function $H(y) = y(1 - R_\mu(y))$ is positive at sufficiently small positive y and tends to $\mu + 1/2 \le 0$ at infinity. Its derivative must be negative somewhere; since $H' = 1 - D_\mu$, again $M_\mu > 1$. The endpoint limits of D_μ imply attainment when $M_\mu > 1$.

The ray expression in Lemma 5.6.1 satisfies the two-sided pointwise estimate

$$\cos\theta\, D_\mu(y) \le \frac{\operatorname{Im}\{e^{i\theta}R_\mu(ye^{i\theta})\}}{\sin\theta} \le \sec\theta\, D_\mu(y).$$

The lower estimate uses the opposite elementary bound $j^4 + 2j^2y^2\cos(2\theta) + y^4 \le (j^2 + y^2)^2$. In particular,

$$\cos(\pi p)M_\mu \le c_\mu(p) \le \sec(\pi p)M_\mu, \qquad \lim_{p\downarrow 0} c_\mu(p) = M_\mu. \tag{5.6.16}$$

If $b/a > M_\mu$, choosing $p = 1/d$ as in (5.6.14) proves HCM by Theorem 5.6.2, for every sufficiently large d.

For necessity suppose a power is HCM. Its normalized non-power factor $g(x) = e^{-bx^p}\mathcal{I}_\mu(ax^p)$ is then a nonconstant HCM function with $g(0) = 1$. The classical GGC representation, allowing a nonnegative drift d_0, gives

$$\frac{d}{dx}\left(x\frac{g'(x)}{g(x)}\right) = -d_0 - \int_0^\infty \frac{t}{(x+t)^2}\,U(dt) < 0.$$

Strictness holds because g is nonconstant and its normalization at zero excludes an atom of the Thorin measure at zero. Direct differentiation, with $y = ax^p$, makes the same derivative

$$\frac{p^2 y}{x}\{-b/a + D_\mu(y)\}.$$

Therefore $b/a > D_\mu(y)$ for every $y > 0$. If $\mu < 1/2$, the maximum is attained, forcing $b/a > M_\mu$. If $\mu \geq 1/2$, admissibility already gives $b/a > 1 = M_\mu$. This proves all equivalences including their equality cases. □

For example $R_{-1/2}(y) = \tanh y$. The derivative of $D_{-1/2}(y) = \tanh y + y \operatorname{sech}^2 y$ is $2 \operatorname{sech}^2 y(1 - y \tanh y)$. Since $y \tanh y$ increases strictly from zero to infinity, there is a unique $y_* > 0$ satisfying $y_* \tanh y_* = 1$, and at that point $D_{-1/2}(y_*) = y_*$. Thus $M_{-1/2} = y_*$, approximately 1.19968. For this order, no power restores HCM when $1 < b/a \leq y_*$, whereas some power does when $b/a > y_*$. This is an explicit one-equation transition for a family whose original density never has the HCM property.

Corollary 5.6.5 (The cost of approaching the square-power barrier)**.** *For fixed $\mu > -1$ and $0 < p < 1/2$,*

$$\frac{1}{j_{\mu,1} \cos(\pi p)} \leq c_\mu(p) \leq \frac{M_\mu}{\cos(\pi p)}.$$

More precisely,

$$\lim_{p \uparrow 1/2} \cos(\pi p) c_\mu(p) = \frac{1}{j_{\mu,1}}. \tag{5.6.17}$$

If $r = b/a > M_\mu$, let $d_*(r)$ be the infimum of the powers $d > 0$ for which the corresponding McKay law has an HCM density. For all sufficiently large r,

$$\frac{\pi}{\arccos(1/(r j_{\mu,1}))} \leq d_*(r) \leq \frac{\pi}{\arccos(M_\mu/r)}.$$

In particular the first Bessel zero determines the leading cost:

$$d_*(r) = 2 + \frac{4}{\pi j_{\mu,1} r} + o(r^{-1}), \qquad r \to \infty. \tag{5.6.18}$$

Proof. In the positive series of Lemma 5.6.1, set $y = j_{\mu,1}$ and retain the first term. It equals $1/(j_{\mu,1} \cos\theta)$ exactly. The upper bound is (5.6.16). These bounds give the displayed arccos bounds for $d_*(r)$.

We supply the uniform estimate needed for the sharper constant. Write $\varepsilon = \cos\theta$ and abbreviate $j_n = j_{\mu,n}$. The exact ray formula becomes

$$\varepsilon \frac{\operatorname{Im}\{e^{i\theta} R_\mu(y e^{i\theta})\}}{\sin\theta} = \frac{1}{y} \sum_{n \geq 1} \frac{4y^2 j_n^2 \varepsilon^2}{(j_n^2 - y^2)^2 + 4y^2 j_n^2 \varepsilon^2}. \tag{5.6.19}$$

The Bessel zero asymptotic $j_n = \pi(n + \mu/2 - 1/4) + O(n^{-1})$ ensures a positive minimum gap δ_μ and the counting bound $\#\{n : j_n \leq t\} \leq C_\mu(1+t)$. For $y \geq 1$, split the sum at $y/2$ and $2y$. The ranges $j_n < y/2$ and $j_n > 2y$, after the factor $1/y$, contribute at most $C_\mu \varepsilon^2$: use respectively $C\varepsilon^2 j_n^2/y^2$ and $C\varepsilon^2 y^2/j_n^2$ to bound the summand. In the middle range, retain one closest zero at cost at most one. The other terms have successive distances at least a constant times $\delta_\mu k$, and their sum is bounded by

$$C \sum_{k \geq 1} \frac{1}{1 + (\delta_\mu k/(Cy\varepsilon))^2} \leq C_\mu y \varepsilon.$$

Consequently the right side of (5.6.19) is at most $1/y + C_\mu \varepsilon$ for $y \geq 1$. On every bounded y interval, the summands tend to zero uniformly away from the finitely many zeros. Near $y = j_n$

the corresponding summand is at most one and the other summands vanish uniformly; the prefactor tends to $1/j_n$. The same uniform convergence to zero near $y = 0$ follows directly from $\sum_n j_n^{-2} < \infty$. Thus the supremum on a fixed bounded interval containing j_1 tends to $1/j_1$. The just proved tail bound, followed by taking its starting point to infinity, proves (5.6.17).

Finally, as $d \downarrow 2$, $\cos(\pi/d) \sim \pi(d-2)/4$. The exact HCM test $r \geq c_\mu(1/d)$ and (5.6.17) imply $c_\mu(1/d) \sim 4/[\pi j_1(d-2)]$. For every fixed $\epsilon > 0$, the lower estimate $c_\mu(1/d) \geq (4/(\pi j_1) - \epsilon)/(d-2)$ holds for every sufficiently small $d - 2 > 0$, excluding all smaller powers at once. The corresponding upper estimate supplies a feasible power $d = 2 + (4/(\pi j_1) + 2\epsilon)/r$. Together with the earlier upper bound, which already gives $d_*(r) \downarrow 2$, these estimates prove (5.6.18) without a monotonicity assumption. □

Corollary 5.6.6 (Positive products and arbitrary real Bessel powers)**.** *Let $m \geq 1$, $\mu_j \geq 1/2$, $a_j > 0$, $\kappa_j > 0$, $\eta > 0$, and $0 < p < 1/2$. If*

$$b \geq \sec(\pi p) \sum_{j=1}^{m} \kappa_j a_j,$$

then the normalized density

$$C x^{\eta-1} e^{-bx^p} \prod_{j=1}^{m} \mathcal{I}_{\mu_j}(a_j x^p)^{\kappa_j}$$

is HCM, hence GGC, SD, and ID.

Proof. The negative logarithmic derivative of the non-power factor is the sum of the corresponding functions (5.6.8), with boundary measure

$$\frac{p t^{p-1}}{\pi} \left[b \sin(\pi p) - \sum_j \kappa_j a_j \operatorname{Im}\{e^{i\pi p} R_{\mu_j}(a_j t^p e^{i\pi p})\} \right] dt.$$

Lemma 5.6.1 makes this nonnegative. Integrating its Stieltjes representation proves (5.6.1) again. Integrability follows from $\eta > 0$ and $b > \sum_j \kappa_j a_j$, which the stated condition implies. □

5.6.3 A strict SD boundary in the negative-cut intervals

Write $\alpha = \mu + 1/2$, $q = \nu - \mu - 1 > 0$, $A = \alpha + q$, and $M = 2\alpha + q$. Admissibility means $\alpha > -1/2$ and $M > 0$; in particular $A > 0$. The preceding manuscript's signed spectral construction supplies a positive smooth even kernel K for which

$$-J'(u)/J(u) = \int_0^\infty e^{-ux} K(x)\, dx, \quad K(0) = M, \quad e^{-x} K(x) \longrightarrow A, \qquad u > 1, \tag{5.6.20}$$

where

$$J(u) = C \frac{u^q V(u^{-2})}{(u^2 - 1)^A}, \qquad V(z) = {}_2F_1(-q/2, (1-q)/2; \alpha + 1/2; z).$$

Moreover $K'/K \to 1$ and $\beta_{\alpha,q} = \sup_{x \geq 0} K'(x)/K(x) < \infty$ is the exact SD boundary: the law is SD precisely when $b/a \geq \beta_{\alpha,q}$. These background kernel identities, established in [76, Section 4], are not new claims of this section. The old strictness proof covered positive-cut intervals and integer q. The following new argument also works in the negative-cut intervals, without having to locate a trigonometric oscillation.

Proposition 5.6.7 (A regular-part obstruction to threshold one)**.** *For every admissible $q > 0$ and $-1/2 < \alpha \leq 1$,*

$$\beta_{\alpha,q} > 1.$$

The supremum is attained at a finite positive point. Thus, for every $-1 < \mu \le 1/2$ and every admissible excess power $q > 0$, there is a nonempty interval of admissible tilts $b > a$ for which the law is ID but not SD.

Proof. Suppose instead $\beta_{\alpha,q} = 1$. Then $h(x) = e^{-x}K(x)$ is nonincreasing and tends to A, so $K(x) \ge Ae^x$ for every $x \ge 0$. If $\alpha < 0$, the inequality at zero contradicts $K(0) - A = \alpha$. If $\alpha = 0$, a nonincreasing function with $h(0) = h(\infty) = A$ must be constant; but $h'(0) = K'(0) - K(0) = -M < 0$.

Let $0 < \alpha \le 1$ and first suppose $A > 1$. Gauss's value and the derivative identity give

$$\frac{V'(1)}{V(1)} = \frac{q(q-1)}{4(A-1)}.$$

Consequently, direct logarithmic differentiation yields

$$\begin{aligned}\lim_{u\downarrow 1}\left\{-\frac{J'(u)}{J(u)} - \frac{A}{u-1}\right\} &= \frac{A}{2} - q + \frac{q(q-1)}{2(A-1)} \\ &= \frac{\alpha(\alpha-1)}{2(A-1)} \le 0. \end{aligned} \tag{5.6.21}$$

On the other hand the expression in braces is $\int_0^\infty e^{-ux}\{K(x) - Ae^x\}\,dx$. Its integrand would be nonnegative and strictly positive near zero, because $K(0) - A = \alpha > 0$. Monotone convergence as $u \downarrow 1$ therefore makes its limit strictly positive, a contradiction, including at $\alpha = 1$.

The only uncovered case has $0 < \alpha < 1$ and $A \le 1$, hence $0 < q < 1$. Here there are no negative spectral zeros and the signed measure is positive. For completeness, the background kernel is

$$K(x) = 2A\cosh x - 2\int_0^1 \cosh(\sqrt{t}\,x)\,\rho(dt),$$

where ρ has positive mass on every compact subinterval of $(0,1)$. In $K' - K$, the integral contributes $2\int(\cosh(\sqrt{t}\,x) - \sqrt{t}\sinh(\sqrt{t}\,x))\rho(dt)$; this grows exponentially along any such subinterval, whereas the remaining contribution is $-2Ae^{-x}$. Thus $K' - K > 0$ for large x, again a contradiction. This last case reproduces the previously known positive-cut argument only to cover the statement's endpoints. Finally, K'/K tends to one and is continuous; any value strictly above one ensures that its supremum is attained on a compact interval. □

Scope of the progress. The power-restoration problem is formulated here as a new transfer problem. Its existence boundary $p < 1/2$ and fixed-tilt criterion $b/a > M_\mu$ are complete, including the failure at equality. The explicit sufficient exponent need not be the smallest possible exponent for a fixed law. The SD proposition is a partial advance on the external classification problem in [3, Section 4.3]; it does not close strictness for all $\mu > 1/2$ in the negative-cut intervals, nor give an elementary formula for the full SD threshold. The earlier full ID and GGC classifications are prerequisites, not additional closures in this round.

Power transformations as a tool for HCM are classical, including Bosch's work on stable quotients [77]; the contribution here is the Bessel-specific phase boundary and fixed-tilt classification. Sjödin's recent preprint [78] concerns preservation of GGC under powers larger than one. That conclusion is distinct from HCM restoration, which is stronger and here also applies to original McKay laws lying outside the GGC domain.

5.7 Random covariance ranges and the exact number of Gaussian views

Low covariance rank reduces the required number of repeated observations, but uncertainty about the covariance range has an irreducible cost. We work throughout with arbitrary Borel mixing laws: the number of mixture components is not assumed finite. Conditional on (A, Σ), let X_i be independent $N_d(A, \Sigma)$ vectors, allowing singular covariance matrices. A conditional block of length $m > 0$ means a vector with law $N_d(mA, m\Sigma)$; disjoint blocks are conditionally independent. Integer block lengths are the usual partial-sum experiments.

Theorem 5.7.1 (Exact rank-adaptive view threshold)**.** *Fix $1 \le r < d$ and restrict the mixing law to* $\operatorname{rank}\Sigma \le r$*.*

1. *One experiment consisting of $r+1$ conditionally independent Gaussian blocks of any known positive lengths identifies the full joint law of (A, Σ). No moment assumptions are required. In particular $(X_1, \dots, X_r, S_n)$ identifies it for every $n \ge r+1$.*
2. *Even the population laws of* all *experiments with at most r blocks, allowing arbitrary positive lengths, do not identify the mixing law. Two distinct witnesses can have compact support, covariance rank exactly r, equal mean marginals, equal covariance marginals, and equal joint polynomial moments through any prescribed finite degree. Their nonzero covariance eigenvalues can be bounded above and bounded away from zero.*

Consequently r anchors are necessary and sufficient in this class. The failure with $r-1$ anchors persists even when every terminal length is available and the covariance support is bounded.

Here anchors are the separately observed X_i preceding the terminal sum. Thus the theorem counts $r+1$ independent blocks, not $r+1$ independent draws of the mixing parameter. Its negative assertion is stronger than failure at one terminal length.

Lemma 5.7.2 (Covariance mixtures of bounded rank)**.** *For finite complex measures supported on $\{\Sigma \succeq 0 : \operatorname{rank}\Sigma \le r\}$, the kernel*

$$\Sigma \longmapsto \mathcal{L}(Y_1, \dots, Y_r \mid \Sigma), \qquad Y_j \mid \Sigma \overset{\text{iid}}{\sim} N_d(0, \Sigma),$$

is injective.

Proof. Write $Y = [Y_1 \ \cdots \ Y_r]$. Conditional on a covariance of rank $k \le r$, one has $\operatorname{range} Y = \operatorname{range}\Sigma$ almost surely. Indeed in orthonormal coordinates on that range, Y is an invertible square root of the covariance times a $k \times r$ standard Gaussian matrix, whose row rank is k almost surely.

This observable range takes values in the finite disjoint union of Grassmannians. Disintegrate the total variation of a candidate signed or complex measure over that range, and disintegrate the measure itself using its bounded Radon–Nikodym derivative. If its mixture kernel is zero, the corresponding conditional mixture on almost every range E is zero. This follows by testing against products of an arbitrary bounded measurable function of the observable range and a member of a countable determining class of functions of Y.

Choose a measurable orthonormal frame on each Grassmannian chart. On a fixed k-dimensional range the covariance becomes a positive definite $k \times k$ matrix B. The characteristic function of its r Gaussian vectors is

$$\int \exp\{-\tfrac{1}{2}\operatorname{tr}(WW^T B)\}\, d\nu_E(B), \qquad W \in \mathbb{R}^{k\times r}.$$

As $r \ge k$, the matrices $WW^T/2$ fill the positive semidefinite cone. Uniqueness of the matrix Laplace transform gives $\nu_E = 0$. For completeness, zero on the open positive cone implies zero in its complex tube by successive one-variable identity theorems; evaluation at $I + iV$ is then Fourier uniqueness for the finite measure $e^{-\operatorname{tr} B}\nu_E$. The zero-dimensional range is immediate. Integrating the zero conditional measures proves the lemma. □

Proof of the positive assertion of Theorem 5.7.1. Let the block lengths be $m_1, \dots, m_{r+1}$ and set $n = \sum m_j$. Divide block j by $\sqrt{m_j}$, and apply an orthogonal change of coordinates in the block index with first row $(\sqrt{m_1}, \dots, \sqrt{m_{r+1}})/\sqrt{n}$. The resulting vectors are conditionally independent, with

$$Z \mid (A, \Sigma) \sim N_d(\sqrt{n}A, \Sigma), \qquad Y_1, \dots, Y_r \mid (A, \Sigma) \overset{\text{iid}}{\sim} N_d(0, \Sigma).$$

For the difference ρ of two mixing laws giving the same experiment, Fourier transformation in $Z/\sqrt{n}$ gives, for every $u \in \mathbb{R}^d$, the zero covariance mixture associated with

$$\nu_u(d\Sigma) = \int e^{iu^T a - u^T \Sigma u/(2n)}\, d\rho(a, \Sigma).$$

Lemma 5.7.2 implies $\nu_u = 0$. Multiplying this zero measure by $e^{u^T \Sigma u/(2n)}$, first on $\operatorname{tr}\Sigma \le M$ and then passing to the limit, shows that $\int e^{iu^T a}\, d\rho(a, d\Sigma) = 0$. Fourier uniqueness in a, applied after restriction to each covariance Borel set, now gives $\rho = 0$. One may first use rational u and a countable generating class of covariance sets, then use continuity, to avoid any issue with uncountably many null sets. □

5.7.1 An exterior differential annihilator

The obstruction is easiest to see before quotienting the covariance factorization. The general implication from a parameter differential annihilator to nonidentifiability is established by Le, Bariletto, Rinaldo, and Ho [8]; the preceding integrated manuscript also used differential annihilators. The new points here are a single exterior operator valid for every allowed block design, its nonzero descent through the covariance quotient, and the matching positive rank threshold. Write $\Sigma = VV^T$ with $V = [v_1 \ \cdots \ v_r] \in \mathbb{R}^{d \times r}$ of full column rank. Choose $r + 1$ distinct row indices $I = (i_0, \dots, i_r)$, and form the constant-coefficient differential operator

$$D_I = \det\big[\partial_{a_i} \ \partial_{(v_1)_i} \ \cdots \ \partial_{(v_r)_i}\big]_{i \in I}. \tag{5.7.1}$$

The determinant is an ordinary polynomial in commuting derivatives. For

$$K_{u,T}(a, V) = e^{iu^T a - \operatorname{tr}(V^T T V)}, \qquad T \succeq 0,$$

direct differentiation gives

$$D_I K_{u,T} = i(-2)^r \det[u_I \ (Tv_1)_I \ \cdots \ (Tv_r)_I] K_{u,T}. \tag{5.7.2}$$

There are no mixed Gaussian derivative terms: each column derivative acts on a different v_j.

Proof of the negative assertion of Theorem 5.7.1. For $q \le r$ blocks with lengths $m_j > 0$ and Fourier arguments w_j, the parameter transform is $K_{u,T}$ with

$$u = \sum_{j=1}^{q} m_j w_j, \qquad T = \tfrac{1}{2}\sum_{j=1}^{q} m_j w_j w_j^T.$$

Thus $\operatorname{rank} T \le q \le r$ and $u \in \operatorname{range} T$. All $r + 1$ columns in the determinant in (5.7.2) lie in a subspace of dimension at most r, so $D_I K_{u,T} = 0$ for every such observation.

Choose a nonzero real C_c^∞ function $\chi(a, V)$ supported on full-rank V and invariant under $V \mapsto VO$, $O \in O(r)$. For instance, average a nonnegative nonzero compact bump over that compact group. Its support can be chosen inside $cI \prec V^T V \prec CI$ for any fixed $0 < c < C$. Let

$$g = D_I^{2M} \chi, \qquad M \geq 1.$$

The polynomial symbol of D_I^{2M} is not zero. A nonzero compactly supported smooth function cannot be annihilated by a nonzero constant-coefficient differential operator: its Fourier transform would vanish on the open set where the polynomial symbol is nonzero, and hence everywhere by real analyticity. Therefore $g \neq 0$.

Under right multiplication by O, D_I transforms by $\det O$. Its even power consequently preserves $O(r)$ invariance, so $g(a, V)$ is constant on the fibers of $\pi(a, V) = (a, VV^T)$. Those fibers are exactly the right $O(r)$ orbits for full-rank V. It follows that the signed pushforward $\rho = \pi_*(g\, da\, dV)$ is nonzero: its sign is a measurable function of (a, VV^T), so integrating that sign against ρ gives $\int |g| > 0$. This step prevents a nonzero factor-space perturbation from disappearing on the covariance quotient.

Integration by parts and $D_I K_{u,T} = 0$ show that every selected observation transform of ρ is zero. Its total mass is zero. Write its positive and negative parts as ρ_+ and ρ_-; their common positive mass is m, and $\rho_+/m, \rho_-/m$ are the required distinct mixing laws. Alternatively push forward g_+/m and g_-/m; invariance makes these the same Jordan decomposition. They arise from smooth compact densities in the covariance factor coordinates and are supported on the smooth rank-r covariance stratum.

Every term of D_I^{2M} contains $2M$ derivatives in a and $2M$ derivatives in each column v_j. Hence integration over all a gives zero identically in V, and integration over all V gives zero identically in a. The covariance and mean marginals therefore agree. If $2M > L$, integration against a polynomial in (a, VV^T) of total degree at most L is also zero, because its degree in a is smaller than $2M$. This gives the asserted finite moment matching. Compactness and the chosen bounds on $V^T V$ give all remaining support assertions. □

Remark 5.7.3 (Why the full-rank endpoint is different). The determinant (5.7.1) requires $r + 1 \leq d$. For full rank $r = d$, the preceding manuscript showed that $d - 1$ anchors can suffice when infinitely many terminal lengths are used; without a covariance tail restriction the exact condition is $\sum_{n \in \mathcal{N}} n^{-1} = \infty$. The current theorem proves that substituting the intrinsic rank r for d in that result is false when the covariance range is itself random. If the range is a fixed known r-plane, reduction to that plane is valid instead. Thus unknown orientation, not simply the number of nonzero eigenvalues, accounts for the extra anchor.

5.7.2 A hierarchy beyond the first logarithmic boundary

The previous logarithmic tail theorem has a sharp scalar converse. Its endpoint is not the weakest possible sufficient tail envelope: one can repeatedly insert another iterated logarithm. We give the complete finite hierarchy, including joint observation counterexamples. Fix an integer $m \geq 1$. Choose E so large that every logarithm below is at least one, and put

$$L_1(x) = \log(E + x), \qquad L_{j+1}(x) = \log L_j(x).$$

Define the envelope

$$W_{m,a}(v) = \frac{\sqrt{v}}{L_1(v) \cdots L_{m-1}(v) L_m(v)^a}, \quad a \geq 0.$$

The empty product is one. Changing E affects none of the assertions.

Theorem 5.7.4 (Every finite iterated-log boundary)**.** *Consider the separate population laws of* $(X_1, \dots, X_{d-1}, S_n)$ *for an infinite terminal set* $\mathcal{N} \subset \{d, d+1, \dots\}$.

1. *For any* $d \geq 1$, *any* $m \geq 1$, *and* $0 \leq a \leq 1$, *these laws identify the joint mixing law within the class*
$$\mathbb{E} \exp\{cW_{m,a}(\operatorname{tr} \Sigma)\} < \infty \quad \text{for some } c > 0.$$
There is no moment condition on A.

2. *For* $d = 1$, $\mathcal{N} = \{2^j : j \geq 0\}$, *and each* $a > 1$, *identification fails. Distinct absolutely continuous witnesses can have equal coordinate marginals and all equal joint polynomial moments, while each satisfies the displayed exponential integrability for* every $c > 0$. *A fixed positive lower bound on the variance can be imposed.*

The moment-determinacy input is classical; general quasi-analytic weight criteria already occur in de Jeu [79]. The assertion specific to the present experiment is the simultaneous construction of equal-marginal joint mixing laws annihilating the entire prescribed sequence of observation transforms.

Lemma 5.7.5 (Iterated-log moment estimate)**.** *If* $V \geq 0$ *and* $\mathbb{E}e^{cW_{m,a}(V)} < \infty$ *for some* $c > 0$, *then*
$$(\mathbb{E}V^k)^{1/(2k)} \leq CkL_1(k) \cdots L_{m-1}(k)L_m(k)^a, \qquad k \geq 1.$$
In particular the Stieltjes–Carleman series diverges for $a \leq 1$.

Proof. Set $D(x) = L_1(x) \cdots L_{m-1}(x)L_m(x)^a$ and $b_k = kD(k)$. Iterated logarithms of y^2 and of y differ by bounded multiplicative factors, so the exponential moment implies $\mathbb{E}e^{c_1\sqrt{V}/D(\sqrt{V})} < \infty$. For all $w \geq 1$, elementary repeated logarithm inequalities give
$$D(b_k w) \leq CD(k)(1 + \log w)^{m-1+a}.$$
For clarity, $L_j(b_k) \asymp L_j(k)$ at every fixed depth, since $\log b_k = \log k + O(\log\log k)$; and $L_j(b_k w) \leq CL_j(b_k)(1 + \log w)$ follows successively from $\log(xw) \leq \log x + \log w$. Therefore
$$\sup_{y \geq 0} y^{2k} e^{-c_1 y/D(y)} \leq (Cb_k)^{2k},$$
by substituting $y = b_k w$ and bounding $w^2 \exp\{-c_2 w/(1 + \log w)^{m-1+a}\}$. The moment assertion follows. The integral test, with successive logarithmic substitutions, gives divergence of $\sum[kL_1(k) \cdots L_{m-1}(k)L_m(k)^a]^{-1}$ exactly for $a \leq 1$. □

Proof of the positive part of Theorem 5.7.4. Let ρ be the difference of two candidate laws. The Fourier arguments of the independent blocks of lengths $1, \dots, 1, n - d + 1$ have the form
$$(u, T), \qquad T = C + \frac{uu^T}{2n}, \quad C \succeq 0, \quad \operatorname{rank} C \leq d - 1.$$
This follows by projecting their d frequency columns onto the vector of square roots of the block lengths. Fix $u \neq 0$ and a rank-$d-1$ matrix C with $u \not\perp \ker C$. Push forward $e^{iu^T a - \operatorname{tr}(C\Sigma)}\rho$ under $(a, \Sigma) \mapsto v = u^T \Sigma u$. Its total variation is dominated by the pushforward of the sum of the candidate probability laws. The preceding lemma, unchanged by fixed positive rescaling of v, gives a Stieltjes–Carleman moment envelope.

Its Laplace transform vanishes at $1/(2n)$ for the selected n. These zeros accumulate at zero. The transform is C^∞ there; successive Taylor expansions show that all its derivatives vanish. Thus the complex measure has zero moments. If μ is its positive dominating measure, $\mu \pm \operatorname{Re}\nu$ and $\mu \pm \operatorname{Im}\nu$ have the same moments as μ; Stieltjes–Carleman uniqueness forces $\nu = 0$.

It follows that the full mixing transform vanishes at $T = C + zuu^T$ for all $z > 0$. Every $T \succ 0$ has this form, with $z = (u^T T^{-1} u)^{-1}$ and $C = T - zuu^T$. Continuity includes $u = 0$, and matrix Laplace–Fourier uniqueness concludes the proof. □

Lemma 5.7.6 (Lacunary zeros at any finite iterated-log depth)**.** *Let $b_j = b_0 2^{-j}$, $b_0 > 0$. For every $m \geq 1$ and $q > 1$ there is a nonzero real Schwartz function f, supported on $[0, \infty)$, whose Laplace transform F satisfies $F(b_j) = 0$ for all j, whose moments all vanish, and such that for each fixed $\ell \geq 0$*

$$|f^{(\ell)}(v)| \leq C_\ell e^{-c_\ell W_{m,q}(v)}, \qquad v \geq 1.$$

Proof. Put $p = 2(q-1) > 0$. For a sufficiently large fixed K, define nested holomorphic logarithms in the cut plane by

$$J_1(w) = \log(K + w), \qquad J_{j+1}(w) = \log(K + J_j(w)), \qquad Q(w) = \frac{w}{J_m(w)^p}.$$

All branches are principal. In the upper cut half-plane each J_j has positive imaginary part, and conjugation handles the lower half; there are no unintended zeros or cuts. On the positive axis all J_j are positive. Increasing K if necessary gives $0 \leq p \arg J_m(w) \leq \arg w$ when $0 \leq \arg w \leq \pi/2$: each logarithm has real part at least $\log K$ on that half-plane, and its argument is at most its input argument divided by $\log K$. Hence $\operatorname{Re} Q(w) \geq 0$ on the right half-plane.

Define

$$B(z) = \prod_{j \geq 0} \frac{z - b_j}{z + b_j}, \qquad F(z) = B(z) \exp\{-\sqrt{z} - Q(1/z)\}.$$

The geometric product is bounded uniformly on every fixed sector $|\arg z| \leq \theta_0 < \pi$, and by one on the right half-plane. Indeed split the factors at $b_j \leq |z|/2$, $b_j \geq 2|z|$, and the bounded number of intervening terms. The two tail sums of the positive logarithms of the moduli are bounded by geometric sums of $b_j/|z|$ and $|z|/b_j$.

We give the quantitative estimate needed for inversion. Write $r = e^{-L}$ and, for large L, put

$$\lambda_1 = L, \quad \lambda_{j+1} = \log \lambda_j, \quad P_m(L) = \lambda_1 \cdots \lambda_m, \quad \delta(r) = \frac{p\pi}{8 P_m(L)}.$$

Uniformly for bounded real θ, repeated expansion of the nested logarithms gives

$$|J_m(e^{L+i\theta})| \sim \lambda_m, \qquad \arg J_m(e^{L+i\theta}) = \frac{\theta}{P_m(L)}(1 + o(1)).$$

The second relation and its first θ derivative hold uniformly on bounded intervals. This can also be checked inductively: the imaginary part of the next logarithm is the previous imaginary part divided by its real part, with a relative error tending to zero; the fixed additive K changes only lower-order terms. Consequently for $|\arg z| \leq \pi/2 + \delta(r)$ and $|z| = r$ small enough,

$$\operatorname{Re} Q(1/z) \geq \frac{c}{r P_m(L) \lambda_m^p}. \tag{5.7.3}$$

At the angular endpoints the argument of Q is at most $\pi/2 - 3p\pi/(8P_m(L)) + o(P_m(L)^{-1})$ in modulus; inside them the argument is monotone for large L. The modulus of Q is comparable to $1/(r\lambda_m^p)$, proving the bound.

The estimate and Cauchy bounds in fixed sectors show that $F(iy)$ extends to a Schwartz function flat at $y = 0$; every fixed derivative is controlled because the exponential in (5.7.3) dominates all inverse powers of r and all logarithmic factors. At infinity $e^{-\sqrt{z}}$ gives rapid decay. Define f by inverse Fourier transformation of $F(iy)$. It is real and Schwartz. For $x > 0$,

$$|F(x + iy)| \leq e^{-\sqrt{|y|/2}},$$

so vertical contour shifts give $f_x(v) = e^{-xv} f(v)$ with f_x uniformly bounded. Letting $x \to \infty$ proves $f(v) = 0$ for $v < 0$ and proves the Laplace identity. Flatness at zero gives all zero moments.

For the tail, bend the inverse contour near zero onto $z = r \exp\{\pm i(\pi/2 + \delta(r))\}$, then continue it to infinity on fixed rays with arguments between $\pi/2$ and π in modulus. No negative pole or branch cut is crossed. Small arcs disappear by (5.7.3); large arcs disappear by the square root exponential. Compact joining portions are chosen strictly in the left half-plane and give $O(e^{-cv})$. The small-radius integrand for the inverse of $z^\ell F(z)$ is bounded by a constant times

$$r^\ell \exp\left\{-c\left(\frac{vr}{P_m(L)} + \frac{1}{rP_m(L)\lambda_m^p}\right)\right\} dr.$$

On $\frac{1}{4}\log v \le L \le \log v$, arithmetic–geometric mean gives a lower bound of a constant times

$$\frac{\sqrt{v}}{\left(\prod_{j=1}^{m-1} \log_j v\right)(\log_m v)^{1+p/2}},$$

with the evident interpretation for $m = 1$. Below this range the first exponential term is at least $v^{3/4}$ divided by a fixed product of logarithms; above it the second term is eventually increasing and is at least v divided by such a product. Both dominate the same lower bound. Reserving half the exponent for an integrable majorant proves the stated estimate, with $q = 1 + p/2$, for every fixed ℓ. Nonzeroness follows from that of F. □

Proof of the negative part of Theorem 5.7.4. Choose $1 < q < a$, use the preceding lemma with $b_j = 2^{-j-1}$, and let $h \in C_c^\infty(\mathbb{R} \setminus \{0\})$ be real, even, and nonzero. Define

$$g(a, v) = \frac{1}{2\pi} \int e^{-iua} h(u) u^2 f(u^2 v)\, du.$$

Its Fourier–Laplace transform is $h(u)F(z/u^2)$. At the observation argument $z = nu_0^2/2$ and $u = nu_0$, this is zero because $z/u^2 = 1/(2n)$. The case $u = 0$ follows from the support of h. Thus every selected sum has zero signed observation law.

Integration by parts in u and the derivative bounds for f give

$$|g(a, v)| \le C_N (1 + |a|)^{-N} e^{-c_N W_{m,q}(v)} \quad (v \ge 1)$$

for every fixed N. The function is real, nonzero, and Schwartz. Its v marginal and all joint polynomial moments vanish because its Fourier transform in a vanishes near $u = 0$; its a marginal vanishes because $\int f = 0$. The normalized positive and negative parts therefore give distinct probability densities with all the required equalities. Since $W_{m,q}(v)/W_{m,a}(v) = L_m(v)^{a-q} \to \infty$, their tails satisfy every asserted exponential moment. A common translation $v \mapsto v + v_0$ preserves all transform zeros and all equalities. □

5.7.3 Scope, comparison, and an unresolved matrix converse

Theorem 5.7.1 concerns arbitrary mixing measures with bounded component covariance rank. It is distinct from finite-component grouped-mixture identifiability, where the number of components controls the required group size; see Vandermeulen and Scott [31]. The present sufficiency uses the observed random covariance range, and the necessity uses an exterior differential operator that survives the orthogonal factorization quotient. The literature search did not locate this precise classification, but is not an exhaustive priority certificate. Both targets here are internally formulated questions, not newly located named published conjectures.

The converse question about absence of finite differential barriers in [8] was already answered negatively in the 223-page integrated manuscript, including by a jointly entire positive kernel. It is not counted again as a new closure in this section.

The fixed-dimensional matrix sharp-tail converse remains open here: for $d > 1$ and $d - 1$ anchors we do not construct nonidentifiable laws at every supercritical $W_{m,a}$ envelope. The old matrix-tube annihilator composes a scalar zero product with $u^T T^{-1} u$. Near indefinite imaginary matrix arguments, cancellation in that quadratic form prevents the scalar bent-contour estimate from transferring directly. Schwartz decay of the matrix inverse does not imply the required nearly square-root exponential tail. The compact low-rank witness above uses fewer than $d - 1$ anchors and therefore does not resolve or contradict the matrix sufficiency theorem.

5.8 Finite spectral recovery from modular-density thresholds

The preceding manuscript recovered a finite positive spectrum from an entire transition curve. Here we replace that curve by finitely many scalar measurements, prove the exact sample count for unnormalized spectra, and determine the optimal noise exponent without any separation between the unknown frequencies. These are two newly formulated inverse problems, not two previously published conjectures. The exponent in the noise theorem is the familiar finite-moment exponent from mixture estimation; see Wu and Yang [80]. The additional issue resolved here is that the observations are nonlinear minima with unknown minimizing points, rather than moments or values of an exponential sum.

Let

$$S_\mu(x) = \int_{(0,1)} x^r \, d\mu(r), \qquad T_\mu(C) = \min_{x>0} \frac{1 + Cx}{S_\mu(x)}, \quad C > 0, \tag{5.8.1}$$

where μ is a nonzero positive measure with finite support in $(0, 1)$. The normalization of μ will be specified separately for each theorem. If $\mu = \sum_{j=1}^{k} B_j \delta_{r_j}$ and $1, r_1, \dots, r_k$ are rationally independent, the preceding modular classification gives the following interpretation. For any $b > 0$ and any phases, the function

$$-1 + A \sum_{j=1}^{k} B_j e^{b r_j s + i\phi_j} + C e^{bs + i\phi_*}$$

has positive zero ordinates dense modulo every period if and only if $A \geq T_\mu(C)$. Thus $T_\mu(C)$ is a scalar transition threshold. Our inverse statements concern measurements of this threshold. They do not assert that a threshold can be certified by observing zeros up to a fixed finite height.

5.8.1 The exact finite sample count

Theorem 5.8.1 (Finite determination and the necessary sample count)**.** *Fix an integer $m \geq 1$ and any $2m$ distinct positive numbers $C_1, \dots, C_{2m}$. The vector*

$$(T_\mu(C_1), \dots, T_\mu(C_{2m}))$$

determines every positive measure μ on $(0, 1)$ with at most m atoms. It determines its actual number of atoms, locations, and masses. More generally, two measures with at most m_1 and m_2 atoms are equal if their thresholds agree at $m_1 + m_2$ distinct positive settings.

On the open parameter space of exactly m distinct atoms with positive, unnormalized masses, $K < 2m$ fixed distinct settings never give local identifiability: through every parameter point there is a real-analytic fiber of dimension $2m - K$ with the same data. At $K = 2m$ the parameter

map has a nonsingular Jacobian everywhere. Consequently its inverse is locally Lipschitz at every spectrum with distinct atoms and positive masses.

Lemma 5.8.2 (Contact points and common supporting lines)**.** *Write $S = S_\mu$ and $H(x) = S(x) - xS'(x)$. Then $S' > 0$, $S'' < 0$, $H > 0$, and*

$$C(x) = \frac{S'(x)}{H(x)}, \qquad C'(x) = \frac{S''(x)S(x)}{H(x)^2} < 0$$

is a bijection from $(0,\infty)$ onto $(0,\infty)$. The minimum defining $T(C)$ has the unique contact point $x(C)$, the inverse of this bijection, and

$$T(C) = \frac{1}{H(x(C))}, \quad T'(C) = \frac{x(C)}{S(x(C))}, \quad T(C) - CT'(C) = \frac{1}{S(x(C))}. \tag{5.8.2}$$

The line

$$L_C(x) = \frac{1 + Cx}{T(C)}$$

majorizes S and touches it only at $x(C)$.

If two such profiles have the same threshold at N increasing settings C_i, their difference has at least N distinct zeros on $(0,\infty)$.

Proof. All derivative signs follow by differentiating x^r, with $0 < r < 1$. Moreover $C(x)x$ is the ratio of the r-weighted and $(1-r)$-weighted positive sums, so it stays between two positive constants. Together with $C' < 0$ this proves the bijection. The tangent inequality for a strictly concave function gives $S(y) \le H(x)(1 + C(x)y)$, with equality only at $y = x$. It proves all minimum assertions; differentiation proves (5.8.2).

For the last assertion, let T_i be the common value and put $L_i(x) = (1 + C_i x)/T_i = a_i + b_i x$. The slopes strictly increase with C, since

$$\frac{d}{dC}\frac{C}{T(C)} = \frac{T(C) - CT'(C)}{T(C)^2} > 0.$$

For $i < N$, let z_i be the intersection of L_i and L_{i+1}. For either profile its contact with L_i lies strictly to the right of z_i, and its contact with L_{i+1} lies strictly to the left. Indeed the profile is below both lines, and equality with both at one point would give two different tangent slopes to a differentiable function. Put $z_0 = +\infty$ and $z_N = 0$. Both contacts of L_i therefore belong to the interval (z_i, z_{i-1}); in particular these intervals are nonempty and disjoint.

At the contact for the first profile its difference from the second is nonnegative; at the contact for the second it is nonpositive. There is therefore a zero between the two contacts, including their common point if they coincide. These zeros belong to the N disjoint intervals. □

Lemma 5.8.3 (Real exponential interpolation, including collisions)**.** *A nonzero solution of*

$$\prod_{j=1}^{n} (D - r_j)h = 0, \qquad r_j \in \mathbb{R},$$

has at most $n-1$ distinct real zeros. Repeated roots are allowed. Fix n, a compact interval of allowed roots, a compact interval of sampling points, and a positive minimum separation between n sampling points. On every fixed compact interval J there is a uniform constant A such that all these solutions satisfy

$$\sup_{u \in J} |h(u)| \le A \max_{1 \le i \le n} |h(u_i)|. \tag{5.8.3}$$

Proof. For the zero count, conjugate by $e^{r_1 u}$ and differentiate. Rolle's theorem reduces the asserted count to the equation of order $n-1$ with real roots $r_j - r_1$, $j \ge 2$. If the derivative is identically zero, the original function is a nonzero multiple of $e^{r_1 u}$ and has no zeros. Induction proves the result.

Write the order-n equation as a first-order system with companion matrix. Its coefficients depend continuously on the roots, including coincident roots. The matrix taking the initial jet $(h(0), \dots, h^{(n-1)}(0))$ to $(h(u_i))_{i=1}^{n}$ is nonsingular by the zero count. The allowed roots and separated sampling tuples form a compact set, so the inverses of these matrices are uniformly bounded. The fundamental matrices on J are uniformly bounded as well. This proves (5.8.3) without a frequency-separation assumption. □

Proof of Theorem 5.8.1. Under $x = e^u$, the difference of two profiles is a sum of at most $m_1 + m_2$ real exponentials. Unless it vanishes identically, it has at most $m_1 + m_2 - 1$ zeros by Lemma 5.8.3. The last assertion of Lemma 5.8.2 contradicts that bound. Linear independence of distinct real exponentials now identifies all atoms and masses.

For local necessity, order the unknown locations and use coordinates $(B_1, r_1, \dots, B_m, r_m)$. Write $x_i = x(C_i)$ and $u_i = \log x_i$. The envelope derivative is

$$\frac{\partial T(C_i)}{\partial B_j} = -\frac{T(C_i)}{S(x_i)} e^{r_j u_i}, \qquad \frac{\partial T(C_i)}{\partial r_j} = -\frac{T(C_i)}{S(x_i)} B_j u_i e^{r_j u_i}.$$

The u_i are distinct. After nonzero row and column rescalings this is the evaluation matrix of $e^{r_j u}, u e^{r_j u}$, $j = 1, \dots, m$. Any initial set of K functions in this order is the full solution space of an order-K differential equation with real roots, repeated as needed. Lemma 5.8.3 makes its square evaluation matrix invertible. Thus the full Jacobian has rank K for $K \le 2m$. The analytic implicit-function theorem gives the fibers for $K < 2m$; the inverse-function theorem gives the remaining assertion.

The local nonidentifiability also persists if the initial frequencies $1, r_1, \dots, r_m$ are rationally independent and this restriction is imposed on alternatives. Choose a nonconstant analytic curve in a fiber through the initial point. Every rational affine relation in the frequencies either has no zeros on the curve or has isolated zeros; it cannot vanish identically since it does not vanish at the initial point. Removing the countable union of these discrete zero sets still leaves distinct admissible alternatives arbitrarily nearby. □

The necessity statement uses unnormalized masses. It does not assert that $2m$ is the smallest global sample count under every additional normalization. For example, even a single probability atom can fail to be identified from one threshold: for $\mu = \delta_r$,

$$T_\mu(C) = \frac{C^r}{r^r (1-r)^{1-r}},$$

and this function of r need not be injective.

5.8.2 A sharp noise exponent from the same finite experiment

Fix $0 < a < b < 1$, an integer $m \ge 1$, and $2m$ distinct positive settings $\mathcal{C} = (C_1, \dots, C_{2m})$. Let $\mathcal{P}_m[a, b]$ denote probability measures on $[a, b]$ with at most m atoms, and define

$$d_{\mathcal{C}}(\mu, \nu) = \max_i |T_\mu(C_i) - T_\nu(C_i)|.$$

All constants below may depend on $m, a, b, \mathcal{C}$, but not on frequency separations or individual atom masses.

Theorem 5.8.4 (Optimal resolution without frequency separation)**.** *For all $\mu, \nu \in \mathcal{P}_m[a,b]$,*

$$W_1(\mu,\nu) \le K\, d_{\mathcal{C}}(\mu,\nu)^{1/(2m-1)}. \tag{5.8.4}$$

The exponent cannot be increased. More strongly, if I is any fixed compact interval in $(0,\infty)$, there are measures μ_h, ν_h, each with m atoms in $[a,b]$, such that, as $h \downarrow 0$,

$$W_1(\mu_h,\nu_h) \ge ch, \qquad \sup_{C\in I} |T_{\mu_h}(C) - T_{\nu_h}(C)| \le K_I h^{2m-1}. \tag{5.8.5}$$

Both spectra can be required to be rationally independent together with 1.

Consequently, for adversarial error at most δ in each of the $2m$ thresholds, the minimax Wasserstein reconstruction error is

$$\asymp \delta^{1/(2m-1)} \qquad (\delta \downarrow 0). \tag{5.8.6}$$

The same order remains a lower bound if the entire threshold curve on a fixed compact interval is observed with uniform error δ.

Lemma 5.8.5 (Separated samples from noisy tangents)**.** *There are constants $\delta_0, c_0, K_0 > 0$ and a compact interval $X \subset (0,\infty)$ such that, whenever $d_{\mathcal{C}}(\mu,\nu) \le \delta \le \delta_0$, there are $2m$ points $y_i \in X$ with pairwise distances at least c_0 and*

$$|S_\mu(y_i) - S_\nu(y_i)| \le K_0\delta.$$

These constants are uniform over $\mathcal{P}_m[a,b]$.

Proof. Arrange $C_1 < \cdots < C_n$, $n = 2m$. Since

$$Cx = \frac{\int r x^r \, d\mu}{\int (1-r)x^r \, d\mu},$$

every contact lies in the fixed interval

$$X = \left[\frac{a}{(1-a)C_n}, \frac{b}{(1-b)C_1}\right].$$

On X, the functions S, H are uniformly bounded above and away from zero, and $-S''$ is uniformly bounded above and away from zero. The formula for $C'(x)$ in Lemma 5.8.2 therefore bounds $|dx/dC|$ above and away from zero. Consecutive contacts for any one profile are separated by a fixed $d_0 > 0$.

Write $L_i^\mu = (1 + C_i x)/T_\mu(C_i)$ and similarly for ν. Their coefficients differ by $O(\delta)$, uniformly, since all thresholds are uniformly bounded above and away from zero. Let z_i^μ be the intersection of L_i^μ and L_{i+1}^μ. The consecutive slope differences are uniformly bounded above and away from zero, by (5.8.2). It follows that $|z_i^\mu - z_i^\nu| \le K_1\delta$.

There is also a uniform positive margin between each intersection and its two neighboring contacts. To see this, if the contacts are $v > w$ and $-S'' \ge \kappa$ on X, the tangent at w exceeds $S(v)$ by at least $\kappa(v-w)^2/2 \ge \kappa d_0^2/2$. Divide by the bounded difference of tangent slopes to bound $v - z_i$ away from zero. The same argument gives a positive lower bound for $z_i - w$. Let a common lower bound be $\eta > 0$.

For δ_0 small enough, the contacts of *both* profiles at C_i lie in the interval between z_i^μ and z_{i-1}^μ, with distance at least $\eta/2$ from each finite endpoint. Here the outer endpoints are interpreted as 0 and $+\infty$. These intervals are ordered and separated by the specified margins.

Put $D = S_\mu - S_\nu$. At the contact of S_μ with L_i^μ, the inequality $S_\nu \le L_i^\nu$ gives $D \ge -K_2\delta$. At the contact of S_ν, the reversed argument gives $D \le K_2\delta$. By continuity there is a point between the two contacts with $|D| \le K_2\delta$: if either endpoint already has this property use it, and otherwise the signs are opposite. The resulting points for successive i are separated by at least η. □

Lemma 5.8.6 (A finite-support moment bound)**.** *Fix a compact real interval J and $m \geq 1$. If probability measures α, β on J each have at most m atoms, and*

$$\left| \int z^k \, d(\alpha - \beta)(z) \right| \leq \epsilon, \qquad 1 \leq k \leq 2m-1,$$

then $W_1(\alpha, \beta) \leq A_{J,m} \epsilon^{1/(2m-1)}$.

Proof. For $0 < \epsilon \leq 1$, let F be the difference of the two cumulative distribution functions and $E = \int_J |F| = W_1(\alpha, \beta)$. There are at most $2m-1$ intervals on which F is nonzero and constant, and hence at most $d \leq 2m-2$ sign changes, after zero intervals are ignored. Choose a root $t_j \in J$ at each sign change, anywhere in an intervening zero interval when necessary. With a suitable sign, $P(z) = \pm \prod_{j=1}^{d} (z - t_j)$ satisfies $P(z)F(z) \geq 0$. For $d = 0$ take $P = 1$ or -1 as appropriate.

If $d \geq 1$, the union of the intervals of radius $E/(4d)$ about the roots has length at most $E/2$. Since $|F| \leq 1$, at least $E/2$ of the $|F|$ integral remains outside this union. Therefore

$$\int_J |P(z)F(z)| \, dz \geq \frac{E}{2} \left(\frac{E}{4d} \right)^d.$$

For $d = 0$ the integral equals E. If $Q' = P$, integration by parts gives $\int PF = -\int Q \, d(\alpha - \beta)$. The degree of Q is $d + 1 \leq 2m - 1$, and its coefficients are uniformly bounded because all roots belong to J; its constant term is irrelevant since the signed measure has mass zero. Thus $E^{d+1} \leq A\epsilon$. For $\epsilon \leq 1$ this implies the asserted exponent, and for larger ϵ the diameter bound is sufficient. □

Proof of Theorem 5.8.4. First suppose $\delta = d_{\mathcal{C}}(\mu, \nu)$ is smaller than the constant in Lemma 5.8.5. The function

$$h(u) = S_\mu(e^u) - S_\nu(e^u)$$

is a sum of at most $2m$ real exponentials with exponents in $[a, b]$. It solves an order-$2m$ equation as in Lemma 5.8.3: if necessary, pad its annihilating polynomial with further roots in $[a, b]$. The points $\log y_i$ lie in a fixed compact interval and are uniformly separated. Lemmas 5.8.5 and 5.8.3, with the target interval $[0, 2m-1]$, give

$$|h(k)| \leq K_3 \delta, \qquad k = 1, \ldots, 2m-1.$$

Push both measures forward by $r \mapsto z = e^r$. These are exactly their moment differences of orders $1, \ldots, 2m-1$. Lemma 5.8.6 proves the required Wasserstein bound for the pushed-forward measures. The map $\log z$ is Lipschitz on $[e^a, e^b]$, so the same bound holds for μ, ν. For $\delta \geq \delta_0$, increase the constant using $W_1(\mu, \nu) \leq b - a$. This proves (5.8.4).

For sharpness, let $q = 2m - 1$ and choose r_0 in the interior of $[a, b]$. For sufficiently small $h > 0$, define probability measures by placing masses $2^{-(q-1)} \binom{q}{j}$ at $r_0 + jh$, assigning even j to μ_h and odd j to ν_h. Each measure has m atoms. Their Laplace-transform difference is exactly

$$S_{\mu_h}(e^u) - S_{\nu_h}(e^u) = 2^{-(q-1)} e^{r_0 u} (1 - e^{hu})^q. \tag{5.8.7}$$

On any fixed compact u interval this is $O(h^q)$. For C in a fixed compact interval, all minimizing x belong to one fixed compact subinterval of $(0, \infty)$, by the contact bound in Lemma 5.8.5. Both denominators S are bounded away from zero there. Taking minima in (5.8.1) thus gives the second bound of (5.8.5) uniformly in C.

The elementary identity $\sum_{j=0}^{k} (-1)^j \binom{q}{j} = (-1)^k \binom{q-1}{k}$ shows, by integrating the absolute difference of cumulative distribution functions over the q gaps, that $W_1(\mu_h, \nu_h) = h$ exactly. The arithmetic progression is not rationally independent. To enforce that requirement, perturb

every location by less than h^{q+1}, choosing locations outside the countable union of rational affine hyperplanes for each spectrum. This preserves the two masses and the number of atoms. It changes the Wasserstein distance by at most $2h^{q+1}$ and each transform on a fixed compact window by $O(h^{q+1})$. The stated lower construction therefore remains valid in the class interpreted by the modular-zero theorem.

Finally the space $\mathcal{P}_m[a,b]$ is compact and the finite threshold map is continuous. A minimum-distance fit to data within δ of the truth has threshold discrepancy at most 2δ from the truth, proving the upper minimax bound. For the lower bound, take h a fixed small multiple of $\delta^{1/q}$. The midpoint of the two data vectors is within δ of both. Every estimator must make error at least half their Wasserstein distance on one of them. The same midpoint argument works for the two entire curves on I. □

5.8.3 What this transfer does and does not settle

The preceding exact inverse used infinitely many amplitude settings. Theorem 5.8.1 reduces them to $2m$ and identifies the local obstruction to using fewer when masses are unnormalized. Theorem 5.8.4 proves that this same finite experiment already attains the best possible collision exponent, even compared with the entire noisy curve on a bounded interval. There is no frequency-separation hypothesis in that theorem; its deterioration with m is unavoidable. Neither theorem estimates the original complex phases, which remain invisible to these threshold data.

The interpretation of a scalar threshold uses an infinite-height zero closure. A finite-height experiment requires an additional uniform recurrence theorem. In particular, these results supply no improved small-divisor estimate for the exact prime vector $(\log p)_p$. The fixed-strip Euler-zero variance and the entropy-sharp maximal-gap exponent at the exact prime vector remain open in this round. The almost-sure perturbed-frequency theorem and finite-spectrum interpolation cannot remove that arithmetic gap. Thus this section closes two internal inverse problems and does not claim a new theorem about the Riemann hypothesis, prime hyperuniformity, or actual prime recurrence exponents.

5.9 Precision complexity of sparse heat observations

The previous manuscript proved exponential recovery from exact heat observations under a common analytic envelope, but only lower bounds for geometrically sparse noisy observations. We now obtain a two-sided *information-depth law* for every Hadamard-lacunary sampling sequence. It determines the order of the logarithm of the minimax risk, although not its sharp leading constant. A second result shows that choosing arithmetic inverse times changes logarithmic precision dependence into power dependence, and that a power obstruction persists for every adaptive heat experiment. These are internally formulated problems. The separate polynomial-information theorem 5.2.1 addresses the logarithmic upper-bound direction of the analytic-information question of Krieg–Novak–Ullrich [60]; optimal measurement counts remain open.

5.9.1 A fixed analytic class and the two experiments

Fix $I=[a,b]$ with $0<a<b$, $\ell=b-a$, and $\rho>1$. The open Bernstein ellipse E_ρ has foci a,b and boundary

$$z=\frac{a+b}{2}+\frac{\ell}{4}(w+w^{-1}),\qquad |w|=\rho.$$

Fix $\kappa > 0$. We use constants $m, M > 0$ for which the class

$$\mathcal{A} = \left\{ f \in H^\infty(E_\rho) : f|_I \text{ real}, \quad \int_I f = 1, \quad f \geq m, \quad (\log f)'' \leq -\kappa, \quad \|f\|_{E_\rho} \leq M \right\} \tag{5.9.1}$$

contains a fixed analytic perturbation ball. More precisely, normalize $f_0(x) = c_0 e^{-\kappa x^2}$ on I and choose $\epsilon > 0$ so small that

$$\int_I h = 0, \quad h|_I \text{ real}, \quad \|h\|_{E_\rho} \leq \epsilon \quad \Longrightarrow \quad f_0 \pm h \in \mathcal{A}. \tag{5.9.2}$$

Such m, M, ϵ exist: Cauchy estimates control the $C^2(I)$ norm of h, whereas f_0 has a positive lower bound and $(\log f_0)'' = -2\kappa$. All constants defining this class remain fixed throughout the section.

The observation at inverse time $s > 0$ is

$$H_s f = \sqrt{s/\pi} \int_a^b e^{-sx^2} f(x)\, dx. \tag{5.9.3}$$

For an increasing sequence $\mathbf{s} = (s_j)$, let $R^\infty_{\mathbf{s}}(\delta)$ be the minimax $L^1(I)$ error from the entire sequence $(H_{s_j} f + e_j)_{j \geq 1}$ with $\sup_j |e_j| \leq \delta$. Let $R^G_{\mathbf{s}}(\delta)$ be the minimax expected L^1 error when e_j are independent $N(0, \delta^2)$ variables. Reconstructions may be nonlinear; all upper bounds below can be achieved by measurable approximate minimum-residual estimators taking their values in $\mathcal{A}$. Normal-family compactness makes $\mathcal{A}$ compact in $C^2(I)$, so existence and measurability can also be obtained from a countable dense subset.

5.9.2 The information-depth law

Suppose

$$s_{j+1} \geq q s_j \quad (j \geq 1), \qquad q > 1, \qquad N(S) = \#\{j : s_j \leq S\}. \tag{5.9.4}$$

The count N is taken in inverse time, not in physical heat time.

Theorem 5.9.1 (Noise risk and usable observation depth)**.** *For every sequence satisfying (5.9.4), there are constants $c, C > 0$ and $\vartheta \in (0,1)$ such that, for all sufficiently small δ and $L = \log(1/\delta)$,*

$$\frac{c\rho^{-N(L)}}{(N(L)+2)^2} \;\leq\; \min\{R^\infty_{\mathbf{s}}(\delta), R^G_{\mathbf{s}}(\delta)\} \;\leq\; \max\{R^\infty_{\mathbf{s}}(\delta), R^G_{\mathbf{s}}(\delta)\} \;\leq\; C\vartheta^{N(L)}. \tag{5.9.5}$$

Consequently, for either noise model,

$$-\log R_{\mathbf{s}}(\delta) = \Theta(N(\log(1/\delta)))\,. \tag{5.9.6}$$

Constants may depend on the analytic class, q, and s_1, but not on the later arrangement of the sampling sequence. The theorem matches the order of the logarithm of the risk. It does not identify a sharp exponent or assert that the two endpoints in (5.9.5) are equivalent up to multiplicative constants.

We give a self-contained proof. The elementary matrix estimate below is the Hilbert-space instance of the classical lacunary Müntz/Riesz-sequence mechanism; see Fricain–Lefèvre [81, Corollary 3.2] and Gaillard–Lefèvre [82].

Lemma 5.9.2 (Uniform conditioning on a finite interval)**.** *For $d > 0$ and $q > 1$ there are $S_0, c_q, C_q > 0$ such that, whenever $S_0 \leq t_1 < \cdots < t_n$ and $t_{j+1} \geq q t_j$,*

$$c_q \sum_{j=1}^n |v_j|^2 \leq \int_0^d \left| \sum_{j=1}^n v_j \sqrt{2t_j}\, e^{-t_j y} \right|^2 dy \leq C_q \sum_{j=1}^n |v_j|^2.$$

The constants are independent of n and the chosen t_j.

Proof. The Gram matrix on the half-line has entries

$$G_{ij} = \frac{2\sqrt{t_i t_j}}{t_i + t_j}.$$

For $i \neq j$, $G_{ij} \le 2q^{-|i-j|/2}$, so its row sums are bounded by $K_q = 1 + 4\sum_{r\ge 1} q^{-r/2}$. The Cauchy inverse formula gives

$$(G^{-1})_{ij} = G_{ij} A_i A_j, \qquad A_i = \prod_{k\neq i} \frac{t_i + t_k}{t_i - t_k}.$$

This formula also follows by partial fractions: the rational function with poles at $-t_k$ and prescribed values at the t_j gives the columns of the inverse. Separation yields

$$|A_i| \le A_q := \prod_{r=1}^{\infty} \left(\frac{1+q^{-r}}{1-q^{-r}}\right)^2 < \infty.$$

Thus $\|G^{-1}\|_{2\to 2} \le A_q^2 K_q$ by the row-sum bound, and $\lambda_{\min}(G) \ge (A_q^2 K_q)^{-1}$. The Gram matrix on $[0, d]$ is $G - DGD$, where $D = \operatorname{diag}(e^{-dt_j})$. Since $\|DGD\| \le K_q e^{-2dt_1}$, choosing S_0 sufficiently large proves the result. □

Lemma 5.9.3 (Correcting finitely many noisy observations)**.** *There are $S_0, K, C > 0$, depending only on I, E_ρ, q, with the following property. If $S_0 \le t_1 < \cdots < t_n$ satisfy the same ratio separation, then, for arbitrary real $d_1, \dots, d_n$, there is a real entire function p with*

$$H_{t_j} p = d_j \quad (1 \le j \le n), \qquad \|p\|_{E_\rho} \le C e^{Kt_n} \max_j |d_j|. \tag{5.9.7}$$

Proof. Put $u = x^2 - a^2 \in [0, b^2 - a^2]$ and $\phi_j(x) = \sqrt{2t_j} e^{-t_j(x^2-a^2)}$. The matrix $Q_{ij} = \int_I \phi_i \phi_j$ satisfies $Q \ge cI_n$: use Lemma 5.9.2 and $dx = du/(2\sqrt{a^2+u}) \ge du/(2b)$. For $p = \sum c_j \phi_j$, the interpolation equations become

$$Qc = b, \qquad b_i = \sqrt{2\pi}\, e^{a^2 t_i} d_i.$$

In particular $\|c\|_2 \le C\sqrt{n}\, e^{a^2 t_n} \max |d_i|$. For $D = \max\{0, \sup_{z\in E_\rho} \operatorname{Re}(a^2 - z^2)\}$,

$$\|p\|_{E_\rho} \le Cn\sqrt{t_n}\, e^{(a^2+D)t_n} \max |d_i|.$$

Since $n \le 1 + \log_q(t_n/S_0)$, increasing K, C absorbs the polynomial factor and gives (5.9.7). □

Lemma 5.9.4 (An analytic fiber diameter estimate)**.** *There is $\vartheta \in (0,1)$, depending only on I, E_ρ, such that a real holomorphic function h bounded by B on E_ρ and orthogonal on I to n exponentials $e^{-t_1 x^2}, \dots, e^{-t_n x^2}$ with distinct positive t_j satisfies*

$$\|h\|_{L^1(I)} \le B\ell\vartheta^n.$$

Consequently, for $f, g \in \mathcal{A}$ and separated $t_j \ge S_0$,

$$\|f - g\|_1 \le C\vartheta^n + Ce^{Kt_n} \max_{j\le n} |H_{t_j}(f-g)|. \tag{5.9.8}$$

Proof. Exponentials with distinct real exponents form a Chebyshev system: after $y = x^2$, dividing by one exponential, differentiating, and using Rolle's theorem proves that a nonzero combination of $r+1$ of them has at most r real zeros, counted with multiplicity. If a nonzero h had fewer than n sign changes, a combination of as many exponentials as needed could be chosen with simple zeros exactly at those sign-change points and the same sign as h. Their

integral pairing would be strictly positive, a contradiction. Thus h has at least n distinct zeros in I.

Let $\phi : E_\rho \to \mathbb{D}$ be conformal and set

$$\vartheta = \max_{x,y \in I} \left| \frac{\phi(x) - \phi(y)}{1 - \overline{\phi(y)}\phi(x)} \right| < 1.$$

Repeated Schwarz factorization at the n zeros bounds $|h(x)|$ by $B\vartheta^n$ on I. For the last assertion apply Lemma 5.9.3 to the data of $f - g$. The corrected function $h = f - g - p$ has zero data and analytic norm at most $2M + \|p\|_{E_\rho}$. Adding back p proves (5.9.8). □

Proof of Theorem 5.9.1. For the upper bound, discard the finitely many parameters below S_0 and use all the remaining parameters at most $c_0 L$, with $c_0 > 0$ chosen so small that $K c_0 < 1/2$. Write n for their number. With deterministic noise, a class member fitting those observations to error at most δ has discrepancy from the truth at most 2δ. Formula (5.9.8) gives error at most $C\vartheta^n + C\delta^{1/2}$.

With Gaussian noise, minimize the maximum residual on the same coordinates. On the event $\max_{j \le n} |Z_j| \le A\sqrt{n}$, its discrepancy is at most $2A\delta\sqrt{n}$, up to an arbitrarily small optimization error. Choose fixed A large enough that $2ne^{-A^2 n/2} \le C\vartheta^n$. Outside this event the loss is at most two. Since $n = O(\log L)$, the resulting expected risk is at most $C\vartheta^n + C\sqrt{n}\,\delta^{1/2}$, hence at most $C\vartheta^n$ for small δ. Every multiplicative interval $[cL, CL]$ contains at most $1 + \log_q(C/c)$ sample points. Therefore $n = N(L) + O(1)$, and the stated upper bound follows.

For completeness we reconstruct the lower bound in the same fixed analytic class. Put $n = N(C_0 L)$, where $C_0 > 2/a^2$. There is a nonzero real polynomial P of degree at most $n + 1$ annihilated by integration and the first n observations. Normalize $\|P\|_{I,\infty} = 1$. The Bernstein–Walsh inequality and the Legendre reproducing-kernel bound give

$$\|P\|_{E_\rho} \le \rho^{n+1}, \qquad \|P\|_1 \ge \frac{\ell}{(n+2)^2}.$$

The first inequality follows directly by the maximum principle after the Joukowski parametrization of the ellipse. For the second, the degree-$(n+1)$ Legendre kernel has diagonal at most $(n+2)^2/\ell$, so $1 \le (n+2)\|P\|_2/\sqrt{\ell}$ and $\|P\|_2^2 \le \|P\|_1$. Set $h = \epsilon\rho^{-(n+1)}P$. By (5.9.2), $f_\pm = f_0 \pm h$ are in $\mathcal{A}$, agree in the first n observations, and have L^1 separation at least $c\rho^{-n}/(n+2)^2$.

For every $j > n$, their difference is bounded by

$$|H_{s_j}(f_+ - f_-)| \le 2\sqrt{s_j/\pi}\, e^{-a^2 s_j}.$$

After another finite initial segment, the sequence has spacing at least one. Eventual monotonicity of $se^{-2a^2 s}$ then gives, with $S = s_{n+1} > C_0 L$,

$$\sum_{j>n} |H_{s_j}(f_+ - f_-)|^2 \le C_a(S+1)e^{-2a^2 S} \le \delta^2.$$

For deterministic errors, the midpoint mean sequence is compatible with both densities, giving half their separation as a lower bound. For Gaussian errors, the two product laws have relative entropy at most $1/2$. Pinsker's inequality and the common part of the two laws give one quarter of their separation. Finally $N(C_0 L) = N(L) + O(1)$ proves (5.9.5). Since $N(L) \to \infty$ and $\log(N(L) + 2) = o(N(L))$, (5.9.6) follows. □

Corollary 5.9.5 (Three different precision scales)**.** *For either noise model, with the analytic class unchanged:*

$$\begin{aligned} s_j = q^j \ (q>1) \quad &\Longrightarrow \quad R_{\mathbf{s}}(\delta) = [\log(1/\delta)]^{-\Theta(1)}, \\ s_j = e^{j^p} \ (p>1) \quad &\Longrightarrow \quad R_{\mathbf{s}}(\delta) = \exp\{-\Theta((\log\log(1/\delta))^{1/p})\}, \\ s_j = e^{q^j} \ (q>1) \quad &\Longrightarrow \quad R_{\mathbf{s}}(\delta) = [\log\log(1/\delta)]^{-\Theta(1)}. \end{aligned}$$

Here $A^{-\Theta(1)}$ means two bounds with possibly different positive exponents. In particular it is not a claim about a single sharp exponent.

Proof. Substitute the respective counting functions into (5.9.6). □

What the depth law closes. The earlier construction established arbitrarily slow consistency by making the sampling sequence increasingly sparse. Theorem 5.9.1 now computes that slowdown, at the level of $-\log R$, for *every* sequence with a fixed multiplicative gap. For geometric sampling it supplies the missing logarithmic-type upper bound. The optimal exponent in that logarithmic rate is still open.

5.9.3 Designing the times: power accuracy and a universal obstruction

The next result separates a limitation of an excessively sparse design from the limitation of the full heat-observation family. For adversarial errors let $R_{\mathrm{all}}^{\infty}(\delta)$ be the infimum of worst-case L^1 errors over all adaptive procedures allowed to query $H_s f + e$ at any $s > 0$, with $|e| \le \delta$ per query and no upper bound on the number of queries. Randomization does not invalidate the lower bound below if expected loss is used. Let $R_B^{\mathrm{ad},G}$ denote the corresponding minimax expected loss with at most B adaptive queries, each with fresh independent unit-variance Gaussian noise.

Choose $\tau > 1$ so close to one that

$$\mathrm{Re}(z^2) \ge c_* > 0 \qquad (z \in E_\tau), \qquad \beta = \frac{\log\rho}{\log\rho + \log\tau} \in (0,1). \tag{5.9.9}$$

The ellipse here has the same foci a, b as before; such a choice exists because $a > 0$.

Theorem 5.9.6 (Power-type accuracy under designed sampling)**.** *For fixed $h > 0$ there are constants $c, C, \alpha > 0$ such that*

$$\frac{c\delta^\beta}{1+\log(1/\delta)} \le R_{\mathrm{all}}^{\infty}(\delta) \le C\delta^\alpha \tag{5.9.10}$$

for small δ. The upper bound uses only $O(\log(1/\delta))$ nonadaptive observations at $s_j = jh$, $1 \le j \le n$. It also holds in expected loss for Gaussian errors of standard deviation δ on those observations, after possibly reducing α.

For the unit-variance Gaussian budget experiment there are $\alpha_0, c, C > 0$ such that

$$\frac{cB^{-\beta/2}}{1+\log B} \le R_B^{\mathrm{ad},G} \le CB^{-\alpha_0} \qquad (B \ge 2). \tag{5.9.11}$$

The upper bound is achieved by a fixed design with repeated arithmetic inverse times. Thus $-\log R_B^{\mathrm{ad},G} = \Theta(\log B)$. Neither statement matches the exact power exponent.

Lemma 5.9.7 (Arithmetic data correction)**.** *For any fixed $h > 0$, there are $A > 1, C > 0$ such that*

$$\|f - g\|_1 \le C\vartheta^{n+1} + CA^n \max_{1\le j\le n} |H_{jh}(f-g)|, \qquad f, g \in \mathcal{A}. \tag{5.9.12}$$

Proof. Put $z = e^{-hx^2} \in J = [e^{-hb^2}, e^{-ha^2}]$. The measure dx becomes $w(z)\,dz$, with w continuous and bounded above and below by positive constants on J. For $v(z) = \sum_{j=0}^{n} c_j z^j$, the monomial Gram matrix

$$Q_{ij} = \int_J z^{i+j} w(z)\,dz, \qquad 0 \le i, j \le n,$$

satisfies $\|Q^{-1}\|_{2\to 2} \le C_1^{n+1}$ for some $C_1 > 1$. Here is an elementary justification of the exponential bound. Expand v in the shifted orthonormal Legendre polynomials on J. The Euclidean norm of the monomial coefficient vector of the degree-k polynomial is at most C_2^{k+1}: this follows by induction from the three-term Legendre recurrence and the fixed affine change of variable. Cauchy–Schwarz then bounds $\|c\|_2 \le C_3^{n+1}\|v\|_{L^2(J)}$. The positive lower bound on w proves the asserted inverse bound, after enlarging the constant.

Apply this Gram inverse to find $p(x) = \sum_{j=0}^{n} c_j e^{-jhx^2}$ such that $\int_I p = 0$ and $H_{jh}p = H_{jh}(f-g)$ for $1 \le j \le n$. The right-hand side of the moment equations is zero at $j = 0$ and $\sqrt{\pi/(jh)}H_{jh}(f-g)$ at $j \ge 1$. The Gram estimate, followed by the bound $|e^{-jhz^2}| \le e^{jh\sup_{E_\rho}(-\operatorname{Re} z^2)}$, gives

$$\|p\|_{E_\rho} \le CA^n \max_{j\le n} |H_{jh}(f-g)|$$

after increasing A to absorb polynomial factors. The corrected function $f - g - p$ is orthogonal to the $n+1$ functions $1, e^{-hx^2}, \dots, e^{-nhx^2}$. The argument of Lemma 5.9.4, which also permits the exponent zero, proves (5.9.12). □

Proof of Theorem 5.9.6. For the deterministic upper bound, take $n = \lfloor c_1 \log(1/\delta) \rfloor$ with $c_1 > 0$ small enough that $c_1 \log A < 1/2$. A minimum-residual estimate in the class has data discrepancy at most 2δ. Lemma 5.9.7 bounds its error by $C\vartheta^n + C\delta A^n \le C\delta^\alpha$ for some $\alpha > 0$. Under Gaussian noise, use the event $\max_{j\le n} |Z_j| \le A_0\sqrt{n}$ exactly as in the depth theorem. Its complement has exponentially small probability in n, and its extra factor $\sqrt{n}$ is absorbed by decreasing α.

For the lower bounds, let P_n be the shifted Legendre polynomial of degree $n \ge 1$, normalized by $\|P_n\|_{I,\infty} = 1$. It is orthogonal to every polynomial of degree less than n, including the constant. Moreover,

$$\|P_n\|_{E_\rho} \le \rho^n, \qquad \|P_n\|_1 \ge \|P_n\|_2^2 = \frac{\ell}{2n+1}.$$

Thus $f_\pm = f_0 \pm \epsilon\rho^{-n}P_n$ belong to the fixed class and have separation at least $c\rho^{-n}/(n+1)$.

The heat kernels $k_s(z) = \sqrt{s/\pi}e^{-sz^2}$ are uniformly bounded on E_τ for all $s > 0$, by (5.9.9) and $\sup_{s>0}\sqrt{s}\,e^{-c_* s} < \infty$. Their degree-$(n-1)$ Chebyshev truncations consequently approximate them on I with error at most $C\tau^{-n}$, uniformly in s. To verify this standard estimate directly, substitute the Joukowski map into k_s and use the Laurent coefficients on circles $|w| < \tau$; Cauchy's bound gives geometric coefficients and summing the tail gives the estimate. Orthogonality of P_n therefore gives the simultaneous bound

$$\sup_{s>0} |H_s(f_+ - f_-)| \le C(\rho\tau)^{-n}. \tag{5.9.13}$$

Choose n as the smallest integer with the right side at most 2δ. An adversarial oracle can always return $H_s f_0$. This response is consistent with both f_+ and f_- at every query, including queries chosen adaptively from all earlier responses. Thus unlimited adaptive querying cannot distinguish the pair. Their separation proves $R^{\infty}_{\mathrm{all}}(\delta) \ge c\delta^\beta/(1+\log(1/\delta))$. The same common-response argument works after conditioning on the random seed of a randomized procedure.

For the Gaussian budget lower bound, the chain rule for relative entropy applies also to an adaptive experiment: conditional on the past, the algorithm chooses the same query rule under

both models, and the next conditional divergence is half the squared difference of their two means. Hence (5.9.13) gives

$$D(P_+^{(B)} \| P_-^{(B)}) \le CB(\rho\tau)^{-2n}.$$

Choose $n = \lceil (\log B + C_4)/(2\log(\rho\tau)) \rceil$ with C_4 large enough that the divergence is at most $1/2$. The two-point inequality proves the lower bound in (5.9.11).

For its upper bound, take $n = \lfloor c_2 \log B \rfloor$ arithmetic locations and repeat each at least $\lfloor B/n \rfloor$ times. The averaged noise standard deviation is at most $C\sqrt{n/B}$. Except on an event of probability $Ce^{-c_3 n}$, the maximum averaged noise is at most $Cn/\sqrt{B}$. Lemma 5.9.7 then bounds the expected loss by

$$C\vartheta^n + CA^n n/\sqrt{B} + Ce^{-c_3 n}.$$

Choose c_2 sufficiently small and absorb the factor n into a slightly smaller positive power of B. This proves the upper bound. Bounded small values of B are covered by adjusting C. □

Scope and unresolved boundaries. The new result is a sampling-dependent minimax classification under a single analytic envelope: all multiplicatively separated sequences obey the depth law, whereas a designed arithmetic sequence achieves power dependence on precision. Classical Riesz-sequence conditioning, analytic approximation, and two-point lower bounds are credited as ingredients, not new theorems. General inverse-Laplace regularization has a substantial literature; for example Maréchal–Triki–Simo Tao Lee [83] study a different global logarithmic-stability problem. Priority for the specific fixed-envelope, infinite-noisy-sequence classification here has not been established by the literature search.

The following remain open in this work: the exact exponent for $s_j = q^j$; the optimal power exponent and constant for designed heat sampling; and the optimal number of arbitrary adaptive analytic measurements in the sense of [60, Section 3], beyond the logarithmic upper bound proved in Theorem 5.2.1. In particular, the adaptive lower bound in Theorem 5.9.6 is for the concrete family of heat functionals. It does not extend to all analytic nonlinear information merely because each heat functional is analytic.

5.10 A target register and the unresolved transfers

The entries below record distinct questions rather than count individual lemmas as separate discoveries. “Complete” refers to the mathematical scope stated in that row, subject to the internal-review status of this manuscript. The externally posed analytic and convex directions overlap: the same convex-polynomial theorem addresses both, so they are not represented as independent proof breakthroughs. The homogeneous variant is grouped with them. Similarly, the several Bessel power criteria are grouped into one restoration problem.

No.	Target and origin	Result, status, and precise limit
1	Analytic and convex information; externally posed complexity directions	Positive logarithmic-query answer, Theorem 5.2.1. At most $\min\{m, \lceil \log_2 m \rceil + 2\}$ globally convex polynomial queries suffice on all of $\mathbb{R}^m$. A convex analytic one-Lipschitz variant has the same count. Positively homogeneous variants are also supplied. Optimal query count, bounded-degree complexity and finite-precision costs remain separate questions.

No.	Target and origin	Result, status, and precise limit
2	Full adaptive Bernoulli replication cap; inherited gap	Complete rate classification for every $1 \le Q \le B$, Theorem 5.5.1. The previous critical adaptive window is closed. No improvement in minimax order from adaptation in this model.
3	Scalar endpoint cap; new internal extension	Complete curve $\max\{\log(eB)/\sqrt{B}, (BQ)^{-1/3}\}$, Theorem 5.4.1. The uncapped logarithm was already proved; the capped classification is new here.
4	Dependent multivariate endpoint cap; new internal problem	Complete curve for every fixed $d \ge 2$, Theorem 5.4.1, without independence of the latent coordinates. Constants may depend on d.
5	Deterministic binomial inverse modulus; new internal problem	Complete two-parameter modulus, Theorem 5.3.1. It is a channel-inversion result and is not the statistical minimax risk of the MLE.
6	Certified approximate MLE; new internal problem	Likelihood deficit $O(N^{-1})$ preserves the stated statistical rates; this order is necessary for a worst-case root-N guarantee based on deficit alone. Theorems 5.3.6 and 5.3.8 give a polynomial-time certified construction.
7	Power restoration of HCM; new internal transfer	Complete existence and fixed-tilt classification, Theorems 5.6.2 and 5.6.4. The first Bessel zero controls the critical power cost, Corollary 5.6.5. The smallest exponent at an arbitrary fixed tilt is specified by a supremum test, not a general elementary formula.
8	Random covariance rank versus repeated views; new internal problem	Complete threshold for rank at most $r < d$, Theorem 5.7.1. $r+1$ blocks suffice; every collection of designs with at most r blocks can fail on compact exact-rank laws.
9	Finite iterated-log tail hierarchy; new internal extension	Complete at every finite depth with an all-dimensional sufficiency and a sharp scalar dyadic converse, Theorem 5.7.4. A fixed-$d > 1$ converse is not claimed.
10	Finite transition-threshold identification; new internal problem	Any $2m$ distinct settings identify a positive at-most-m-atom spectrum, Theorem 5.8.1. Fewer are locally insufficient for unnormalized exactly-m-atom spectra. No minimal count for normalized masses is asserted.
11	Noisy finite-threshold recovery; new internal problem	Complete minimax error exponent $1/(2m-1)$ without separation, Theorem 5.8.4. The classical moment exponent is credited; the new transfer starts from nonlinear optimized thresholds.
12	Noisy lacunary heat information depth; inherited inverse gap	Complete order of $-\log R$ for every Hadamard-lacunary sequence, Theorem 5.9.1. Sharp risk exponents are not matched.
13	Designed versus sparse heat sampling; new internal transfer	Complete distinction between power-type and logarithmic-type precision dependence, Theorem 5.9.6, including adaptive lower bounds. The optimal power exponent remains open.
14	McKay SD strictness; external classification direction	Partial extension to every admissible $q > 0$ and $-1 < \mu \le 1/2$, Proposition 5.6.7. The entire negative-cut, high-order classification is not closed.
15	Full statistical MLE transition; inherited boundary	Open near $t \asymp N/\log N$. The new deterministic modulus proves a limitation of the global count-Wasserstein argument, rather than a statistical lower bound ruling out an optimal MLE.
16	Fixed-dimensional matrix tail converse; inherited transfer	Open for each $d > 1$ at the supercritical iterated-log envelopes. Cancellation near indefinite Fourier matrices prevents the scalar contour estimate from transferring directly.
17	Exact-prime recurrence exponent; inherited arithmetic target	Open. Generic frequency perturbations and continuum inverse theorems do not supply quantitative recurrence for the actual prime-log vector.

No.	Target and origin	Result, status, and precise limit
18	Euler-product zero variance or hyperuniformity; inherited target	Open. The needed quantitative Fourier/small-divisor control is absent. No Riemann-hypothesis or Speiser-type conclusion follows from the spectral threshold results.
19	Exact optimal query count in analytic information; external refinement	Open. The logarithmic upper bound answers the speedup direction, while sharp lower bounds beyond the standard continuous-information obstruction are not established.
20	Sharp heat inverse exponents; inherited precision refinement	Partial. Both sparse and designed regimes now have matching logarithmic-risk orders, but their leading exponents need not coincide.

5.10.1 What is actually gained over the preceding paper

The adaptive Bernoulli boundary is now complete, while the previous paper only had a complete predetermined curve and polynomially subcritical adaptive regimes. The endpoint result now includes every cap and arbitrary fixed dimension. The continuum spectral inverse has become a finite-data inverse with an optimal noise exponent. The Gaussian result now distinguishes intrinsic covariance rank from the ambient dimension through unknown orientation. The heat section computes how arbitrary lacunarity changes the usable information depth, rather than giving a geometric lower bound alone. The Bessel section changes the variable by a power and obtains HCM restoration even where the original density lacks it.

Most significantly for the externally posed questions, the analytic information restriction does not block the colored-cover method after its decisions are made tolerant to approximation. Convexification using a known radial measurement transfers this construction into still more restricted query classes. This is an upper-complexity theorem in an exact oracle model. It should be assessed on that scope, not advertised as a practical scheme for recording an arbitrary high-dimensional object in a few finite-bit numbers.

5.10.2 Why the arithmetic targets do not follow

An exact inverse theorem answers which latent object is determined by an observable. A quantitative arithmetic theorem also needs to show that the observable is reached, or estimated, at a controlled height for the specific arithmetic frequencies. The latter step is not provided by identifiability, continuity, generic rational independence, or a worst-case statistical inverse modulus. In particular, the transition thresholds used here belong to a finite exponential model. Replacing them by a list of right-half-plane zeta zeros is a different observation problem and is not justified by any result of this paper.

The prime recurrence and zero-variance entries are therefore retained as open. A credible next advance there would require a new quantitative arithmetic estimate, not another relabeling of an abstract uniqueness statement. Similarly, the full statistical MLE transition requires more than a global count-transport estimate, and the matrix tail converse requires more than a scalar flat transform.

5.10.3 Review record and provenance

All main sections received an independent internal proof review in addition to author self-review. Checks focused on conditional versus unconditional information, zero-mass normalization, singular covariance strata, quotient cancellation, atom collisions, boundary residues, and the distinction between exact and noisy information. The combined manuscript was compiled and

its rendered pages inspected. These are internal checks, not a formal proof certificate or external peer review.

The external comparisons use the cited primary papers. A targeted search can find direct overlaps and verify the statement of an open question; it cannot certify exhaustive priority. The colored-cover idea, classical quasi-analytic criteria, beta–binomial eigenvalues, HCM/GGC representations, finite-moment collision exponent, and lacunary exponential conditioning are attributed to their respective sources. The contributions claimed here are the stated transfers and complete boundaries built from those ingredients. No journal-level or prize-level evaluation is implied.

Part VI

Boundary Laws for Observation and Recovery

Prime-Zeta Counting, Global Likelihood Optimality, and Sharp Inverse Thresholds

Source manuscript: 15 October 2026.

Source abstract. We continue an observation-and-recovery program through several distinct changes of analytic representation. For the actual prime zeta function, we prove that the number of zeros in $\Re s > 1$ up to height T is $\Theta(T)$, with a multiplicity upper bound and a distinct-zero lower bound. This supplies the endpoint upper bound explicitly proposed in recent literature and extends to every fixed derivative. For arbitrary binomial mixing laws we prove that every unrestricted maximum-likelihood estimator attains the complete minimax order $\min\{1, t^{-1} + (t\log(eN))^{-1/2} + N^{-1/2}\}$ for all sample sizes N and replication counts t. A low-frequency transportation estimate and a small-amplitude high-frequency Hellinger estimate close the previous transition gap. Further results give exact and degree-dependent adversarial-noise floors for restricted adaptive information, sharp iterated-log covariance-tail boundaries in every fixed Gaussian dimension, positivity certificates for nonlinear spectral thresholds, and three critical regimes for Bessel power restoration. Fixed analytic population classes exhibit double- and triple-logarithmic replication penalties. Sixteen precisely scoped targets receive proofs; dependent consequences, classical ingredients, one partial SD advance, and remaining questions are identified explicitly. The statements are research claims supported by proofs and independent automated readings, not externally certified priority or solutions of sixteen named classical conjectures.

6.1 The new step beyond the preceding manuscripts

The starting point is the 223-page manuscript *Observation and Recovery in Arithmetic and Probability* and its sequels [1, 29, 53, 61, 76, 84]. The present paper contains new continuations rather than a consolidation of their already claimed results. Earlier manuscripts are unpublished research sources, not substitutes for external validation. New arguments are supplied here; explicit older dependencies are stated where they are used, particularly for the Bessel signed spectral kernel.

The principal new statement in arithmetic concerns an actual prime Dirichlet series, without independent random phases. The earlier fixed-$\gamma > 1$ zero counts do not control accumulation toward the line $\Re s = 1$. We remove the prime-power tail from the Euler logarithm, exponentiate, and count zeros of a single-valued function defined in a larger half-plane. Classical second moments then suffice for an averaged Jensen argument. The result addresses precisely the endpoint question recorded in [85, p. 16]; it does not concern the Riemann hypothesis or a zero-counting formula on continuation branches to the left of one.

The statistical advance is a different kind of boundary result. The preceding paper obtained the optimal MLE order in polynomially sublinear replication ranges, but its deterministic count-transport modulus lost a logarithmic factor around $t = N/\log N$. That modulus was a real obstruction for a proof using only that norm. A fitted likelihood also supplies Hellinger information. Splitting the inverse test polynomial at degree about $\sqrt{t}$ makes the high-frequency remainder small in supremum norm, so this second observation norm closes the transition. This extends the statistical range questions in [13, 14], without resolving their separate constant-one coefficient conjecture.

6.1.1 Four transfers with different mathematical content

The common discipline is to retain the observation model while changing the object on which an available theorem acts. The actual mechanisms are different and no universal theorem subsuming them is asserted.

1. A logarithmic arithmetic series is lifted to a meromorphic function; positive logarithmic means then control zeros at a boundary.
2. An inverse statistical test is split between two norms; the amplitude of the difficult component determines which norm is useful.
3. A meromorphic zero product is replaced by an entire canonical product. Its slower growth can be absorbed by a matrix regularizer, allowing contour deformation through all matrix signatures.
4. Nonlinear transition values yield signed inequalities at contact points. Positive exponential interpolation then turns those inequalities into certificates against arbitrary positive competitors.

The regularity and Bessel results additionally show why a boundary must be analyzed before a supremum or minimization is taken. Analytic population smoothness leaves an increasing number of active frequency bands. At Bessel order $1/2$, exponentially small terms determine a critical power that cannot be detected by a finite algebraic expansion.

6.1.2 Sixteen-target register

In the table, **E** means an externally stated target in its exact scope, **F** a sharpened or restricted continuation of an external question, **I** a problem formulated in this research program, and **D** a dependent transfer or consequence. "Proved" means a complete argument is presented in this manuscript, subject to external mathematical review. The rows do not count parameter values, derivative orders, or elementary lemmas as separate conjectures.

No.	Type	Target	Result and location
1	E	Prime-zeta count at the boundary $\Re s > 1$	$M_1(T) = \Theta(T)$; Theorem 6.2.1.
2	D	Derivative and fixed-level arithmetic transfer	Every fixed derivative has $\Theta(T)$ zeros; each level has an $O(T)$ upper bound; Theorem 6.2.1.
3	F	Complete unrestricted binomial-MLE curve	All N, t, including $t \asymp N/\log N$; Theorem 6.3.1.
4	D	Equal-replication Bernoulli budget	Exact order $(B\log(eB))^{-1/4}$; Theorem 6.3.8.
5	D	Analytic component-model budget transfer	Sufficient hypotheses and a verified factorial Legendre example; Theorem 6.3.9.
6	F	Dimension-three restricted exact information	Three queries are necessary and sufficient for polynomial or convex analytic information; Theorem 6.4.1.
7	I	Range-normalized convex noisy recovery	Exact infinite-budget floor $\delta/(2-\delta)$, including convex analytic and convex polynomial queries; Theorem 6.4.2.
8	I	Degree-limited polynomial noisy recovery	Sharp order δ/D with an adaptive construction; Theorem 6.4.3.
9	I	Matrix Gaussian tail converse at every finite logarithmic depth	Sharp threshold $a = 1$ for polynomially sparse terminal sets in each fixed dimension; Theorem 6.5.1.

No.	Type	Target	Result and location
10	I	Denser Gaussian terminal designs	The first-log converse holds for $n_j \geq cj \log^{\beta}(e+j)$, $\beta > 1$; Theorem 6.5.2.
11	I	Normalized finite-spectrum threshold count	Exactly $2m-1$ suitably placed settings for at most m atoms; Theorem 6.6.1.
12	I	Blind positivity certification	$2m$ normalized settings certify an m-atomic truth against every positive competitor; Theorem 6.6.1.
13	I	Blind spectral consistency under noise	Sharp uniform exponent $1/(2m)$; Theorem 6.6.2.
14	I	Bessel critical power and order endpoints	Inverse-square-root, logarithmic, and bounded regimes; Theorems 6.7.2 and 6.7.3.
15	I	Stretched-exponential population regularity	Adaptive and fixed-design budget risk $\asymp \log\log B/\sqrt{B}$; Theorem 6.8.1.
16	I	Double-exponential population regularity	Adaptive and fixed-design budget risk $\asymp \log\log\log B/\sqrt{B}$; Theorem 6.8.2.

The approximate-MLE extension and the unknown-atom-count stopping rule are useful further corollaries, not additional independent entries. The SD moment hierarchy in Proposition 6.7.4 is a *partial* advance, and is not included among the sixteen completed targets. The linear positive-measure recovery principle and generic moment exponents already occur in prior work [90, 80]; the spectral contribution here is the nonlinear threshold transfer with its precise hypotheses.

6.1.3 Notation and verification status

W_1 denotes Wasserstein distance with the specified ground metric. The symbols $\asymp$, O, and Θ allow positive constants depending only on the fixed parameters named in each statement. Dimension is fixed in the matrix-tail results. Exact population thresholds, exact real-valued queries, and finite noisy samples are distinct observation models and are never silently interchanged.

All branches received independent automated proof readings, with separate checks of adaptive conditional information, zero-cell likelihood envelopes, singular matrix contours, confluent interpolation, and parameter-uniform Bessel limits. These checks can detect errors but are neither formal verification nor external peer review. Literature comparisons were checked against primary sources available on 15 October 2026; absence of a matching result in that search does not establish exhaustive priority. Journal-level significance must be judged after independent expert evaluation of the proofs and their novelty.

6.2 The prime-zeta counting problem at the boundary of absolute convergence

Let

$$P(s) = \sum_{p\ mprime} p^{-s} \qquad (\Re s > 1).$$

The question of Belovas, Čepaitytė, and Sabaliauskas [21, Section 3] includes the endpoint upper bound

$$M_1(T) := \#\{s : P(s) = 0,\ \Re s > 1,\ 0 < \Im s < T\} = O(T).$$

It is also recorded explicitly in [85, p. 16, Conjecture 2.1.4 and the sentence following it]. The preceding manuscripts proved uniform counting statements in each fixed half-plane $\Re s > \gamma > 1$, but explicitly left this endpoint open. Those statements alone do not justify sending γ down to

one. Here a different observation, which retains the exact prime frequencies, gives the missing estimate. The proof uses classical mean-square estimates for the Riemann zeta function; it is an application of those estimates, not a new zeta mean-value theorem or a claim about the Riemann hypothesis.

Theorem 6.2.1 (Endpoint counting and the derivative tower)**.** *For every fixed integer $m \geq 0$, let $N_m(T)$ count the zeros of $P^{(m)}$ in $\Re s > 1$, $0 < \Im s < T$, with multiplicity, and let $N_m^{\mathrm{dist}}(T)$ count them without multiplicity. There are constants $0 < c_m < C_m < \infty$ such that, for all sufficiently large T,*

$$c_m T \leq N_m^{\mathrm{dist}}(T) \leq N_m(T) \leq C_m T. \tag{6.2.1}$$

In particular the proposed endpoint estimate $M_1(T) = O(T)$ holds. For every fixed $a \in \mathbb{C}$, the number of solutions of $P(s) = a$ in the same half-plane and height interval, counted with multiplicity, is $O_a(T)$.

The case $m = 0$ addresses the displayed external endpoint question. The derivative tower and level-set upper bounds are transfers of the same argument. They are not counted as infinitely many separately solved published conjectures. No limiting coefficient $N_m(T)/T$ at the endpoint, short-interval bound uniform in the initial height, or zero-variance asymptotic is asserted.

6.2.1 An averaged Jensen lemma

We write $\log^+ u = \max\{0, \log u\}$ for $u > 0$, and set $\log^+ 0 = 0$.

Lemma 6.2.2 (Vertical logarithmic means control endpoint zero counts)**.** *Let $\alpha < \beta$, and let G be a nonzero function holomorphic on $\Re s > \alpha$. Suppose that there are $R > \beta$ and $c > 0$ such that G has no zeros in $\Re s \geq R$ and $|G(R + it)| \geq c$ for every real t. Suppose also that for every compact interval $I \subset (\alpha, \infty)$,*

$$\sup_{\sigma \in I} \int_{-T}^{T} \log^+ |G(\sigma + it)|\, dt = O_I(T + 1). \tag{6.2.2}$$

Then the number of zeros of G in $\Re s > \beta$, $0 < \Im s < T$, with multiplicity, is $O(T + 1)$.

Proof. Choose fixed radii $R - \beta < r < \rho < R - \alpha$, and put $h = (r^2 - (R - \beta)^2)^{1/2} > 0$. Let $n_r(t)$ count the zeros in the closed disk of radius r centered at $R + it$. Jensen's formula gives

$$n_r(t) \log(\rho/r) \leq \frac{1}{2\pi} \int_0^{2\pi} \log^+ |G(R + it + \rho e^{i\theta})|\, d\theta - \log c.$$

One may replace $-\log c$ by $|\log c|$ and use limiting radii at the exceptional circles containing zeros. Integrating over $-h < t < T + h$, applying Tonelli to the nonnegative integrand, and using (6.2.2) for $\sigma \in [R - \rho, R + \rho]$ shows $\int_{-h}^{T+h} n_r(t)\, dt = O(T + 1)$. Every zero $x + iy$ with $\beta < x < R$, $0 < y < T$, belongs to these disks for a set of center ordinates of length $2(r^2 - (R - x)^2)^{1/2} \geq 2h$. Thus its multiplicity contributes at least $2h$ to that integral. There are no zeros farther right, which proves the lemma. □

6.2.2 Removal of the prime powers

The function

$$H(s) = \sum_p \sum_{k=2}^{\infty} \frac{p^{-ks}}{k} \tag{6.2.3}$$

is holomorphic on $\Re s > 1/2$. For each $j \geq 0$ and each fixed $\sigma_0 > 1/2$, its derivative $H^{(j)}$ is bounded uniformly on $\Re s \geq \sigma_0$. Indeed its absolute derivative series is majorized by

$$\sum_p (\log p)^j \sum_{k\geq 2} k^{j-1} p^{-k\sigma_0} \leq C_{j,\sigma_0} \sum_p (\log p)^j p^{-2\sigma_0} < \infty.$$

The convention $k^{j-1} = k^{-1}$ applies when $j = 0$. On the original half-plane of absolute convergence,

$$\log \zeta(s) = P(s) + H(s), \qquad \zeta(s) = \exp(P(s) + H(s)). \tag{6.2.4}$$

Here the logarithm is the Euler-product branch on $\Re s > 1$. We never continue that branch across a zero of ζ.

We use the classical mean-square estimate, locally uniform in $\sigma > 1/2$ away from the pole at height zero,

$$\sup_{\sigma\in I} \int_1^T |\zeta^{(j)}(\sigma + it)|^2 \, dt = O_{I,j}(T + 1), \qquad I \Subset (1/2, \infty), \quad j \geq 0. \tag{6.2.5}$$

The $j = 0$ input is the standard off-critical-line mean-square theorem [86, Chapter VII]; see also [87, Lecture II, Section 3] for its basic form. For clarity, the derivative version needs no additional moment conjecture. Choose a fixed Cauchy radius smaller than both $1/2$ and the distance of I from $1/2$, apply Cauchy's formula and Cauchy–Schwarz on the circle, and then integrate in t. The circle's real parts remain in a fixed larger compact interval, and its imaginary parts differ from t by a bounded amount. The $j = 0$ bound therefore gives (6.2.5). A fixed initial interval is harmless, and the corresponding negative-height estimate follows by conjugation.

6.2.3 The endpoint upper bound

For $a \in \mathbb{C}$, define the single-valued meromorphic function

$$F_a(s) = \zeta(s) - \exp(a + H(s)), \qquad \Re s > 1/2.$$

Its only possible pole in this half-plane is the simple pole at one. On $\Re s > 1$, identity (6.2.4) gives

$$F_a(s) = e^{H(s)}(e^{P(s)} - e^a). \tag{6.2.6}$$

Consequently a solution of $P(s) = a$ is a zero of F_a with exactly the same multiplicity. The converse need not hold: $F_a = 0$ also allows $P(s) = a + 2\pi ik$. Overcounting these additional solutions is legitimate for an upper bound.

If $e^a \neq 1$, then $F_a(s) \to 1 - e^a \neq 0$ uniformly as $\Re s \to \infty$. If $e^a = 1$, the absolutely convergent expansions give instead

$$2^s F_a(s) = 1 + o(1) \quad (\Re s \to \infty),$$

uniformly in the ordinate: the leading term of P is 2^{-s}, and $\zeta(s)(1 - e^{-P(s)}) = P(s) + O(2^{-2\Re s})$. Set $b_a = 0$ in the first case and $b_a = \log 2$ in the second, and put

$$G_a(s) = e^{b_a s} \frac{s - 1}{s + 1} F_a(s).$$

The pole at one is removable, so G_a is holomorphic on $\Re s > 1/2$. Its far-right limit is nonzero uniformly in the ordinate: the extra rational factor tends to one uniformly there. Thus for sufficiently large R it is zero-free in $\Re s \geq R$ and bounded away from zero on $R + i\mathbb{R}$. For real parts in a fixed compact interval and $|t| \geq 1$,

$$\log^+ |G_a(\sigma + it)| \leq C + \log(1 + |\zeta(\sigma + it)|) \leq C + |\zeta(\sigma + it)|^2.$$

The bounded factors include e^H and $(s-1)/(s+1)$; the latter has modulus at most one on $\Re s > 0$. On the remaining compact region, G_a itself is bounded, including at the removed pole. Equation (6.2.5) verifies (6.2.2). Lemma 6.2.2, with $\alpha = 1/2$ and $\beta = 1$, gives the asserted $O_a(T)$ upper bound. For $a = 0$ this proves $N_0(T) = O(T)$ without any interchange of limits.

6.2.4 All fixed derivatives, using only second moments

For $m \ge 1$ define differential polynomials in ζ recursively by

$$A_1 = \zeta', \qquad A_{m+1} = \zeta A_m' - m\zeta' A_m. \tag{6.2.7}$$

Where $\zeta \neq 0$, induction gives $A_m = \zeta^m (\log \zeta)^{(m)}$. In particular,

$$F_m := A_m - \zeta^m H^{(m)} = \zeta^m P^{(m)} \qquad (\Re s > 1). \tag{6.2.8}$$

The definition on the left is single-valued and meromorphic on $\Re s > 1/2$, even at zeros of ζ; no division by ζ is used there. Its only possible pole is at one, of order at most $2m$. This follows either from the recursion, or from the fact that every monomial of A_m has m factors and total derivative order m. Since ζ does not vanish for $\Re s > 1$, the zeros of F_m and $P^{(m)}$ in that half-plane agree, including multiplicities.

Normalize by

$$G_m(s) = \frac{2^s}{(-\log 2)^m} \left(\frac{s-1}{s+1}\right)^{2m} F_m(s).$$

This is holomorphic on $\Re s > 1/2$, and tends to one uniformly as $\Re s \to \infty$. Indeed $P^{(m)}(s) = (-\log 2)^m 2^{-s}(1 + o(1))$ uniformly there and $\zeta(s) \to 1$. The finite polynomial expression for F_m, boundedness of $H^{(m)}$, and boundedness of the remaining factors on compact real-part intervals give, for $|t| \ge 1$,

$$\log^+ |G_m(\sigma + it)| \le C_m + C_m \sum_{j=0}^{m} \log(1 + |\zeta^{(j)}(\sigma + it)|). \tag{6.2.9}$$

Integration and the mean-square bound give the required logarithmic mean estimate. The compact region around the removed pole contributes a bounded amount. Lemma 6.2.2 now proves $N_m(T) = O_m(T)$. The logarithm in (6.2.9) is essential to the economy of the argument: high powers in a differential polynomial do not require high moments of ζ for this counting conclusion.

6.2.5 The matching lower order from the exact prime flow

For a fixed $m \ge 0$, put $a_p = (\log p)^m$ and let $S_m > 1$ be the unique solution of

$$\sum_{p \ge 3} \frac{a_p}{a_2} (2/p)^{S_m} = 1.$$

Existence follows because this sum decreases continuously from infinity to zero as the real argument increases from one to infinity. For $m = 0$, divergence at one is Euler's divergence of the sum of reciprocal primes; the other cases follow by comparison. Consider

$$Q_m(s) = a_2 2^{-s} - \sum_{p \ge 3} a_p p^{-s}.$$

It has a simple zero at S_m, because the derivative of its normalization by $a_2 2^{-s}$ at that point is

$$\sum_{p \ge 3} \frac{a_p}{a_2} (2/p)^{S_m} \log(p/2) > 0.$$

Choose a small closed disk D about S_m, wholly inside $\Re s > 1$, containing exactly this one zero, and with no boundary zeros. A sufficiently accurate approximation to the coefficient phases $(1, -1, -1, \ldots)$ makes $\sum_p a_p \omega_p p^{-s}$ uniformly close to Q_m on ∂D. This requires only a finite open cylinder of phase conditions: first make the normally convergent tail uniformly small and then constrain the finitely many remaining phases. Rouché's theorem then gives exactly one zero in D.

The exact prime flow $\omega_p(t) = e^{-it \log p}$ visits this cylinder relatively densely. To see the precise input, finite nonzero integer vectors satisfy $\sum_p k_p \log p \neq 0$ by unique factorization. Finite-dimensional Kronecker density, followed by compactness of the finite torus, says that the visits to any nonempty open cylinder have bounded gaps. More explicitly, translates of the cylinder by the forward flow cover that compact torus; a finite subcover bounds the waiting time from every starting point. Thus some fixed L_m ensures that every sufficiently long height interval contains a zero of $\sum_p a_p p^{-s} = (-1)^m P^{(m)}(s)$ in the fixed strip containing D. Allowing for the disk's bounded imaginary radius only changes L_m. Disjoint intervals of that length give at least $c_m T$ distinct zeros up to height T. This proves the lower bound in (6.2.1) and completes the theorem.

6.2.6 What this resolves, and what the arithmetic still does not give

The external endpoint $M_1(T) = O(T)$ is obtained for the actual prime zeta function, and the two-sided linear order extends to every fixed derivative. This is stronger than a fixed $\gamma > 1$ statement, but it is not a claim that its proof supplies a new technique comparable to a major zero-density breakthrough. The new step within this program is to lift the logarithmic prime series to a single-valued meromorphic function and then control its positive logarithmic mean.

The growing-dimensional recurrence problem uses a different input: small values of $\sum_p k_p \log p$ with the support and coefficients growing as the required phase error shrinks. The present argument uses only qualitative finite-dimensional Kronecker density for the lower bound, so it gives no entropy-sharp bound for those return times. Likewise, a first-moment count of zeros provides no variance or hyperuniformity theorem. Both the exact-prime recurrence exponent gap and fixed-strip Euler-zero variance remain open here. No result is claimed about zeros on different branches of P to the left of one.

6.3 The complete binomial-mixture likelihood transition

The near-linear MLE theorem in the preceding manuscript left open the transition $t \asymp N/\log N$. Its sharp deterministic inverse modulus also showed that global count-Wasserstein error alone cannot close that transition. We now use two different statistical norms on two different parts of a test polynomial. The low-degree part is controlled by likelihood transportation. The remaining polynomial has small amplitude, and is controlled by a predictive Hellinger bound. This yields the full replication curve, including its transition and its parametric plateau.

For a probability measure G on $[0, 1]$, put

$$K_x(j) = B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}, \qquad p_G(j) = \int B_j^t(x)\, dG(x), \quad 0 \le j \le t.$$

We observe N independent counts from p_G. Their empirical law is q. The unrestricted mixing-measure MLE is any maximizer $\widehat{G}$ of $\ell(H) = \sum_j q_j \log p_H(j)$. Write $\widehat{p} = p_{\widehat{G}}$, $L_N = \log(eN)$, and

$$\mathcal{R}_{N,t} = \inf_{\widetilde{G}} \sup_{G} \mathbb{E}_G W_1(\widetilde{G}, G), \qquad r_{N,t} = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{tL_N}} + \frac{1}{\sqrt{N}}\right\}.$$

The infimum allows randomized estimators. Wasserstein distance on the count space uses the normalized cost $|j-l|/t$ and is denoted $W_{1,t}$. Our Hellinger convention is $h(p,v)^2 = \sum_j(\sqrt{p_j}-\sqrt{v_j})^2$, also for nonnormalized nonnegative vectors v.

Theorem 6.3.1 (All-replication optimality of the unrestricted MLE)**.** *There are absolute constants $0<c<C<\infty$ such that, for every pair of integers $N,t\ge 1$,*

$$cr_{N,t} \le \mathcal{R}_{N,t} \le \sup_{G} \mathbb{E}_G W_1(\widehat{G},G) \le Cr_{N,t}. \tag{6.3.1}$$

The upper bound holds for every measurable choice of an unrestricted MLE. In particular, at $t \asymp N/\log(eN)$ its risk is $O(N^{-1/2})$, and the same order holds for every larger t.

The statistical question extends the replication-range conjectures of Vinayak, Kong, Valiant and Kakade [13] and the discussion in Section 6.2 of Musco, Musco, Rosenblatt and Singh [14]. The preceding near-linear result already answered the fixed-polynomial-range question in that discussion. The new assertion is the complete transition, not a second claim of novelty for the old near-linear range. We do not prove the separate constant-one Chebyshev–Bernstein coefficient conjecture from those papers.

6.3.1 An elementary finite-envelope bound for predictive likelihood

The following entropy argument is stated with its details because it must apply even when the true distribution has zero cells. Neither an interior mixing density nor a regular parametric likelihood is assumed.

Lemma 6.3.2 (Small covers of the binomial moment body)**.** *For $0<\beta<1$ the set $\{p_G : G\in \mathcal{P}([0,1])\}$ has a finite ℓ_1 cover of radius β, with centers in that same set, whose cardinality M satisfies*

$$\log M \le m\log\frac{C(m+1)}{\beta}, \qquad m=\min\left\{t,\left\lceil\sqrt{2t\log(16t/\beta)}\right\rceil\right\}. \tag{6.3.2}$$

Proof. View $K_x=(B_0^t(x),\dots,B_t^t(x))$ as a vector-valued polynomial. Its shifted Chebyshev expansion is

$$K_x=\sum_{k=0}^{t} v_k T_k(2x-1).$$

We claim $\|v_k\|_1 \le 2\exp\{-k^2/(2t)\}$ for $k\ge 1$. Indeed, the even periodic function $F(z)=K_{(1+\cos z)/2}$ has Fourier coefficient $v_k/2$. For $z=\theta+iu$, the ellipse identity gives

$$\left|\frac{1+\cos z}{2}\right|+\left|\frac{1-\cos z}{2}\right|=\cosh u.$$

The binomial formula therefore yields $\|F(\theta+iu)\|_1=(\cosh u)^t\le e^{tu^2/2}$. Shift the Fourier contour in the appropriate direction and take $u=k/t$. This proves the claim. In particular, truncating after degree $m<t$ has vector ℓ_1 error at most $2te^{-(m+1)^2/(2t)}\le\beta/8$, uniformly in x. For $m=t$ the error is zero.

Each mixing law contributes its first m shifted Chebyshev moments in $[-1,1]^m$. Partition that cube into cells of coordinate diameter at most $\beta/[8(m+1)]$, and choose one mixing-law representative from every nonempty cell. If G,H lie in one cell, their truncated transforms differ in ℓ_1 norm by at most $2m\beta/[8(m+1)]\le\beta/4$. The two tails contribute at most $\beta/4$. Thus their predictive laws have distance at most β. The number of cells is at most $[C(m+1)/\beta]^m$, as asserted. □

Lemma 6.3.3 (Likelihood envelopes, including zero cells)**.** *Let $\varepsilon \ge 0$ and $\eta > 0$. Let p be the true law on a finite alphabet with d cells, and suppose a model admits an ℓ_1 cover of radius η/d, with M probability centers in the model. Let $\widetilde{p}$ be any fitted model law satisfying*

$$\sum_{i=1}^{N} \log \widetilde{p}(X_i) \ge \sum_{i=1}^{N} \log p(X_i) - N\varepsilon.$$

For every $x \ge 0$, simultaneously for all such fitted laws, except on an event of probability at most e^{-x},

$$h(p, \widetilde{p}) \le \sqrt{\eta} + \sqrt{\varepsilon + \eta + 2(\log M + x)/N}. \tag{6.3.3}$$

Consequently

$$\mathbb{E}h(p, \widetilde{p})^2 \le C\left\{\varepsilon + \eta + \frac{\log M + 1}{N}\right\}. \tag{6.3.4}$$

Proof. For each center p_a, form the nonnegative vector $u_a(j) = p_a(j) + \eta/d$. Every model law is bounded coordinatewise by at least one u_a, and $\sum_j u_a(j) = 1 + \eta$. If $\widetilde{p} \le u_a$, then

$$h(u_a, \widetilde{p})^2 \le \sum_j (u_a(j) - \widetilde{p}(j)) = \eta.$$

Thus $h(p, \widetilde{p}) > \sqrt{\eta} + r$ implies $h(p, u_a) > r$. Moreover the likelihood comparison implies $\prod_i u_a(X_i)/p(X_i) \ge e^{-N\varepsilon}$. The ratios need only be defined on the support of p, where all sampled observations lie almost surely. Markov's inequality with exponent $1/2$ gives

$$\begin{aligned}
\Pr\left\{\prod_i \frac{u_a(X_i)}{p(X_i)} \ge e^{-N\varepsilon}\right\} &\le e^{N\varepsilon/2}\left(\sum_j \sqrt{p(j)u_a(j)}\right)^N \\
&= e^{N\varepsilon/2}\left(1 + \frac{\eta}{2} - \frac{h(p, u_a)^2}{2}\right)^N \\
&\le \exp\{N(\varepsilon + \eta - r^2)/2\}
\end{aligned}$$

when $h(p, u_a) > r$. Taking the union over the M envelopes proves (6.3.3). Integrating that tail, and using $(a+b)^2 \le 2a^2 + 2b^2$, proves (6.3.4). □

Corollary 6.3.4 (A sufficient predictive Hellinger rate)**.** *For $1 \le t \le N$, every exact MLE satisfies*

$$\mathbb{E}_G h(p_G, \widehat{p}) \le \frac{Ct^{1/4}L_N^{3/4}}{\sqrt{N}}. \tag{6.3.5}$$

More generally, if $\ell(\widetilde{G}) \ge \ell^* - \varepsilon$, the right side can be increased by $C\sqrt{\varepsilon}$.

Proof. Set $\eta = (eN)^{-2}$ and $\beta = \eta/(t+1)$ in the preceding lemmas. For $t \le N$, (6.3.2) gives $m \le C\sqrt{tL_N}$ and $\log M \le C\sqrt{t}\, L_N^{3/2}$. The MLE likelihood is at least that of the true law. The approximate likelihood condition loses at most $N\varepsilon$. Apply (6.3.4) and Jensen's inequality. □

The logarithmic factors in (6.3.5) need not be sharp. The argument below only needs the strict improvement from effective dimension t to a quantity of order $\sqrt{t}$ up to logarithms.

6.3.2 The two-scale inversion inequality

We record the two approximation ingredients used in the preceding manuscripts. They are included with proofs to specify the uniformity needed here.

Lemma 6.3.5 (Stable approximation and bounded-polynomial inversion)**.** *There is an absolute C with the following properties. For every 1-Lipschitz $f : [0,1] \to \mathbb{R}$ and integer $k \geq 1$, there is a polynomial P_k of degree at most k such that*

$$\|P_k - f\|_\infty \leq C/k, \qquad \|P_k'\|_\infty \leq 2.$$

If $P = \sum_{j=0}^{t} b_j B_j^t$ has degree $k \leq t/2$, then

$$\max_j |b_j| \leq C e^{2k^2/t} \|P\|_\infty. \tag{6.3.6}$$

For the derivative-stable P_k, its coefficient vector also satisfies

$$\max_{j<t} t|b_{j+1} - b_j| \leq C e^{4k^2/t}. \tag{6.3.7}$$

Proof. Subtract $f(0)$, reflect f on $[0,1]$ to a period-two 1-Lipschitz function, and convolve with the normalized positive Jackson kernel $c_s[\sin(s\pi u/2)/\sin(\pi u/2)]^4$. Its first absolute moment is $O(1/s)$, so the resulting trigonometric polynomial Q_s has error $O(1/s)$ and derivative norm at most one. Its frequencies have magnitude at most $2s$, and its Fourier coefficients have modulus at most one. Replacing each exponential by its Taylor polynomial at $1/2$ of degree $100s$ changes the function and derivative by at most e^{-100s}. Indeed $|\pi l(x - 1/2)| \leq \pi s$ and the factorial inequality $M! \geq (M/e)^M$ bounds both remainders, including the $4s+1$ terms and the extra derivative factor. Take $s = \lfloor k/100 \rfloor$ for $k \geq 100$; use a constant polynomial and enlarge C for smaller k.

For (6.3.6), put $M = \|P\|_\infty$ and $F(z) = (1+z)^t P(z/(1+z))$. The maximum principle in the inverse Joukowski variable gives $|P((2+w+w^{-1})/4)| \leq M|w|^k$ for $|w| \geq 1$. For $0 < j < t$, set $r = j/(t-j)$, $v = r/(1+r)^2$, and $u = 2\sqrt{v}|\sin(\phi/2)|$. The larger inverse Joukowski modulus for $re^{i\phi}/(1+re^{i\phi})$ is $(1+u)/\sqrt{1-u^2}$. Hence

$$\frac{|F(re^{i\phi})|}{M(1+r)^t} \leq (1-u^2)^{(t-k)/2}(1+u)^k \leq e^{2k^2/t} e^{-tu^2/8}.$$

The last inequality follows from $k \leq t/2$ and completing the square. The average of the final Gaussian factor is at most $C/\sqrt{1+tv}$. Cauchy's coefficient formula, together with $\binom{t}{j}(j/t)^j(1 - j/t)^{t-j} \geq c/\sqrt{1+tv}$, proves (6.3.6). The last lower bound follows directly from Stirling's factorial bounds; endpoint coefficients are $P(0)$ and $P(1)$. Finally differentiate the Bernstein expansion: $P_k' = t\sum_{j<t}(b_{j+1} - b_j)B_j^{t-1}$. Apply (6.3.6) at degree $k-1 \leq (t-1)/2$ and use $2(k-1)^2/(t-1) \leq 4k^2/t$. □

Lemma 6.3.6 (Likelihood transport, recalled)**.** *For every binomial-mixture MLE,*

$$W_{1,t}(q, \widehat{p})^2 \leq \frac{D(q\|\widehat{p})}{2t} \leq \frac{D(q\|p_G)}{2t}. \tag{6.3.8}$$

Consequently

$$\mathbb{E}_G W_{1,t}(\widehat{p}, p_G) \leq C/\sqrt{N}. \tag{6.3.9}$$

Proof. In score sums, terms with $q_j = 0$ are set to zero; every observed cell has positive fitted probability. The likelihood score condition gives $S(x) = \sum_j q_j K_x(j)/\widehat{p}(j) \leq 1$, and its integral

against $\widehat{G}$ is one. Thus $S = 1$ almost surely under $\widehat{G}$. The posterior vectors $R_x(j) = q_j K_x(j)/\widehat{p}(j)$ are therefore probability laws, with mixture q, and

$$\int D(R_x \| K_x)\, d\widehat{G}(x) = D(q\|\widehat{p}).$$

Every normalized-count binomial kernel satisfies $W_{1,t}(R, K_x)^2 \le D(R\|K_x)/(2t)$: a 1-Lipschitz test of the normalized sum of t Bernoulli variables has centered logarithmic moment generating function at most $\lambda^2/(8t)$ by bounded differences; entropy duality and optimization over λ give the inequality. Mix the couplings, use Cauchy–Schwarz, and then use likelihood optimality to prove (6.3.8). Also $\mathbb{E}D(q\|p_G) \le t/N$, by $D(q\|p_G) \le \sum_{j:p_G(j)>0}(q_j - p_G(j))^2/p_G(j)$. Finally $\mathbb{E}W_{1,t}(q, p_G) \le 1/(2\sqrt{N})$ by integrating the standard deviation of the empirical distribution function. The triangle inequality proves (6.3.9). □

Proposition 6.3.7 (Two-scale deterministic inversion)**.** *For integers $1 \le s \le k \le t/2$ and any two mixing laws G, H,*

$$W_1(G, H) \le \frac{C}{k} + Ce^{4s^2/t} W_{1,t}(p_G, p_H) + \frac{Ce^{2k^2/t}}{s} h(p_G, p_H). \tag{6.3.10}$$

Proof. For any 1-Lipschitz test f, take the polynomials P_k, P_s from Lemma 6.3.5. The approximation error in integrating P_k instead of f against $G - H$ is at most C/k. The Bernstein coefficients of P_s have normalized-grid Lipschitz constant at most $Ce^{4s^2/t}$. On the other hand,

$$\|P_k - P_s\|_\infty \le C/k + C/s \le C/s.$$

Thus the Bernstein coefficients of $P_k - P_s$ have supremum norm at most $Ce^{2k^2/t}/s$. Test P_s by count transportation and $P_k - P_s$ by $\|p_G - p_H\|_1 \le 2h(p_G, p_H)$. All constants are uniform over f, so taking the Wasserstein dual supremum proves the result. □

The small amplitude $1/s$ is essential. Applying only the Hellinger bound to the entire polynomial loses the low-frequency empirical $N^{-1/2}$ scale; applying only transportation loses the critical transition. Proposition 6.3.7 retains both.

Upper bound in Theorem 6.3.1. It suffices first to treat $t \le N$. Bounded N or bounded t are handled by the loss bound one. Fix a sufficiently small absolute $a > 0$, for example $a = 1/64$, and take

$$k = \left\lfloor \min\{t/2, \sqrt{atL_N}\} \right\rfloor, \qquad s = \lfloor \sqrt{t}/4 \rfloor.$$

For sufficiently large N and t, these satisfy $1 \le s \le k \le t/2$, $s \asymp \sqrt{t}$, and

$$1/k \le C\{1/t + (tL_N)^{-1/2}\}, \qquad e^{4s^2/t} \le C.$$

Use Proposition 6.3.7, Lemma 6.3.6, and Corollary 6.3.4. This gives

$$\mathbb{E}_G W_1(\widehat{G}, G) \le \frac{C}{k} + \frac{C}{\sqrt{N}} + \frac{Ce^{2k^2/t} L_N^{3/4}}{\sqrt{N}\, t^{1/4}}. \tag{6.3.11}$$

The final term, multiplied by $\sqrt{tL_N}$, is at most

$$C(eN)^{2a} N^{-1/2} t^{1/4} L_N^{5/4} \le C(eN)^{2a} N^{-1/4} L_N^{5/4} \le C.$$

The last bound is uniform because $2a < 1/4$. This proves the desired upper bound when $t \le N$.

For $t \ge N$, couple every mixing parameter x to its normalized binomial count. The expected absolute difference is at most $\sqrt{x(1-x)/t} \le 1/(2\sqrt{t})$. Hence, after identifying count laws with measures on $[0, 1]$,

$$W_1(\widehat{G}, G) \le 1/\sqrt{t} + W_{1,t}(\widehat{p}, p_G).$$

Take expectations and apply (6.3.9). This is $O(N^{-1/2})$ for $t \ge N$, and completes the proof of the upper bound. □

6.3.3 Matching lower bounds and the equal-replication budget

Lower bound in Theorem 6.3.1. The $N^{-1/2}$ term follows from mixing laws supported on $\{0,1\}$ with weights $1/2 \pm c/\sqrt{N}$. Each count then reveals exactly one endpoint label, regardless of t. Their N-sample total variation stays bounded away from one for sufficiently small absolute c.

For the other terms let $\pi(dx) = dx/[\pi\sqrt{x(1-x)}]$ and $h_k(x) = T_k(2x-1)$. Define $G_k^{\pm} = (1 \pm h_k/2)\pi$. Integration of their signed cumulative function gives

$$W_1(G_k^+, G_k^-) = \frac{1}{2\pi k}\int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \asymp 1/k.$$

If $k = t+1$, the count laws are identical, proving the $1/t$ lower bound. For $k \le t$, draw $X \sim \pi$, $J \mid X \sim \mathrm{Binomial}(t, X)$ and put $Qh(j) = \mathbb{E}[h(X) \mid J = j]$. The posterior is $\mathrm{Beta}(j+1/2, t-j+1/2)$. The self-adjoint operator Q^*Q preserves polynomial degrees and has leading degree-k coefficient

$$\lambda_{t,k} = \prod_{i=0}^{k-1} \frac{t-i}{t+1+i}.$$

Orthogonality of the Chebyshev polynomials therefore gives $Q^*Qh_k = \lambda_{t,k}h_k$ and $\mathbb{E}[Qh_k(J)^2] = \lambda_{t,k}/2$. This is the classical beta–binomial diagonalization [70, Section 4.1]. The count laws are $p_\pi(1 \pm Qh_k/2)$, so

$$\chi^2(p_{G_k^+} \| p_{G_k^-}) \le 2\mathbb{E}[Qh_k(J)^2] = \lambda_{t,k} \le e^{-k^2/(2t)}.$$

Take $k = \lceil\sqrt{8t\log(8N)}\rceil$ if this is at most t. The chi-squared divergence of the N-fold products is $(1+\chi^2)^N - 1 \le e^{N(8N)^{-4}} - 1$, so their total variation is bounded by an absolute constant smaller than one. The two-point testing bound yields risk at least $c/k \ge c'/\sqrt{tL_N}$. If this choice has $k > t$, then $t \le CL_N$ apart from bounded t, and $1/\sqrt{tL_N} \le C'/t$; the exact moment-matching lower bound already suffices. The maximum of the three lower bounds is comparable to their sum clipped at one, proving (6.3.1). □

Theorem 6.3.8 (Exact equal-replication budget order)**.** *Suppose a total of at most B Bernoulli observations may be used, with one common replication count t for every sampled individual. Then*

$$\inf_{N,t \ge 1:\, Nt \le B} \mathcal{R}_{N,t} \asymp \{B\log(eB)\}^{-1/4}, \qquad B \ge 1. \tag{6.3.12}$$

One order-optimal choice is

$$N \asymp \sqrt{B\log(eB)}, \qquad t \asymp \sqrt{B/\log(eB)}.$$

The unrestricted MLE attains this order. This closes the logarithmic gap in the equal-group Bernoulli allocation problem left in the previous manuscripts; it does not change the previously proved, smaller risk available with unequal group sizes.

Proof. For any admissible N, t, the lower bound in Theorem 6.3.1 contains both $cN^{-1/2}$ and $c(t\log(eN))^{-1/2}$. Their geometric mean is at least $c\{Nt\log(eN)\}^{-1/4} \ge c\{B\log(eB)\}^{-1/4}$. The clipped case only alters the absolute constant. The displayed choices of N, t, rounded down while preserving $Nt \le B$, give the matching upper bound by Theorem 6.3.1. Bounded budgets are immediate. □

Theorem 6.3.9 (A matching budget transfer to analytic component models)**.** *Let a component model P_θ, $0 \le \theta \le 1$, admit a statistic Y based on b_0 independent component observations such that $|Y| \le M$ and $\mathbb{E}_\theta Y = \theta$, where b_0, M are fixed. Then its equal-replication population problem has the upper bound*

$$\mathcal{R}_B^{\mathrm{eq}} \le C_{M,b_0}\{B\log(eB)\}^{-1/4}.$$

Suppose in addition that, for some interior θ_0 and $R > 0$, the likelihood ratios $f_z = dP_{\theta_0+z}/dP_{\theta_0}$ admit a holomorphic extension to $|z| \le R$ in $L^2(P_{\theta_0};\mathbb{C})$ satisfying

$$\int f_z \, dP_{\theta_0} = 1, \qquad \|f_z - 1\|_2 \le C_0|z|. \tag{6.3.13}$$

Then $\mathcal{R}_B^{\mathrm{eq}} \asymp \{B\log(eB)\}^{-1/4}$. Constants may depend on the displayed model parameters. The analytic condition is a sufficient lower-bound condition, not a consequence of existence of an unbiased statistic.

Proof. With an independent uniform U, the variable $Z = \mathbf{1}_{\{U \le (Y+M)/(2M)\}}$ is Bernoulli with parameter $(\theta + M)/(2M)$. Disjoint observation blocks give independent such variables conditionally on θ. Apply Theorem 6.3.8 with budget $\lfloor B/b_0 \rfloor$ and push the estimated mixing law through the clipped inverse $z \mapsto \min(1, \max(0, 2Mz - M))$, which is $2M$-Lipschitz. This proves the upper bound.

We include the lower-bound argument, already used in the preceding manuscript, to specify the scope of this transfer. Let q be the common number of component observations per individual. An oracle revealing all N latent parameters gives the lower bound $cN^{-1/2}$ by varying the weights of a two-point mixing law. For the other lower bound, write

$$F_z = f_z^{\otimes q} = \sum_{j \ge 0} T_j z^j \quad \text{in } L^2(P_{\theta_0}^{\otimes q};\mathbb{C}).$$

The zero integral of $f_z - 1$ cancels the cross term, so $\|F_z\|_2 \le e^{C_0^2 q|z|^2/2}$. For $0 < h \le R/2$, Cauchy's estimate on $|z| = 2h$ gives $\|T_j\|_2 \le e^{2C_0^2 qh^2}(2h)^{-j}$. Let $n = m + 1$ and set $\theta_j = \theta_0 - h + 2hj/n$. The even and odd binomial-weight laws

$$G = 2^{1-n} \sum_{j \text{ even}} \binom{n}{j} \delta_{\theta_j}, \qquad H = 2^{1-n} \sum_{j \text{ odd}} \binom{n}{j} \delta_{\theta_j}$$

match moments through degree m and have $W_1(G, H) = 2h/(m + 1)$. The latter identity follows from the signed cumulative values $2^{1-n}(-1)^j \binom{n-1}{j}$ between successive grid points. If $Q_q(G) = \int P_\theta^{\otimes q} \, dG(\theta)$, then

$$\left\| \frac{dQ_q(G) - dQ_q(H)}{dP_{\theta_0}^{\otimes q}} \right\|_2 \le 2e^{2C_0^2 qh^2} \sum_{j > m} 2^{-j} = 2e^{2C_0^2 qh^2} 2^{-m}.$$

Take

$$m = \left\lceil \frac{4}{3\log 2} \log(4N) \right\rceil, \qquad h = \min\left\{ R/2, \sqrt{\frac{m \log 2}{8 \max(1, C_0^2) q}} \right\},$$

after decreasing R to keep all parameters in $[0, 1]$. One-group total variation is at most $1/(4N)$; the N-group distance is at most $1/4$. Therefore

$$\mathcal{R}_{N,q} \ge c \max\left\{ N^{-1/2}, \min\{L_N^{-1}, (qL_N)^{-1/2}\} \right\}.$$

If $q \ge L_N$, the geometric mean of the first and last terms is $(NqL_N)^{-1/4} \ge \{B\log(eB)\}^{-1/4}$. If $q < L_N$, the lower bound $c/L_N \ge c/\log(eB)$ is larger than the required budget rate for all sufficiently large B. Bounded budgets are absorbed in the constant. □

Corollary 6.3.10 (The factorial Legendre equal-group gap is closed)**.** *Let L_j denote shifted Legendre polynomials and consider*

$$k_*(x \mid \theta) = 1 + \sum_{j \ge 2} \frac{L_j(x) L_j(\theta)}{j!}, \qquad x, \theta \in [0, 1].$$

For this model the optimal expected population W_1 risk under common replication and total budget B is $\Theta(\{B\log(eB)\}^{-1/4})$.

Proof. The kernel is bounded below by $3 - e > 0$. The factorial series and its parameter derivatives converge uniformly on compact complex parameter sets, so (6.3.13) holds at $\theta_0 = 1/2$. For $j \geq 2$ define $A_j(x) = j!(2j+1)L_j(x)$; orthogonality gives $\mathbb{E}_\theta A_j(X) = L_j(\theta)$. The Legendre product identity

$$L_2 L_3 = \frac{10}{21} L_5 + \frac{4}{15} L_3 + \frac{9}{35} L_1$$

then shows that the two-observation statistic

$$Y(x,y) = \frac{1}{2} + \frac{35}{18}\left\{A_2(x)A_3(y) - \frac{10}{21}A_5(x) - \frac{4}{15}A_3(x)\right\}$$

has expectation θ and absolute value less than 2100. Apply Theorem 6.3.9. The unbiased statistic and the lower bound were established previously; the new ingredient completing the matching upper bound is Theorem 6.3.1. □

6.3.4 Approximate likelihood and the remaining coefficient question

Corollary 6.3.11 (The complete rate survives N^{-1} optimization error)**.** *If a fitted mixing law satisfies $\ell^* - \ell(\widetilde{G}) \leq 1/N$, then*

$$\sup_G \mathbb{E}_G W_1(\widetilde{G}, G) \leq C r_{N,t}$$

for all $N, t \geq 1$. The likelihood gap is measured per observation.

Proof. We give the needed approximate transport argument. As above, all score sums are over observed cells, with zero terms elsewhere. Set $H = \widetilde{G}$, $p_H = p$, and $S(x) = \sum_j q_j K_x(j)/p(j)$. The measure $H^+ = SH$ is a probability measure. For $S(x) > 0$ set $R_x(j) = q_j K_x(j)/(p(j)S(x))$. Then

$$\int D(R_x \| K_x)\, dH^+(x) = D(q\|p) - D(H^+\|H).$$

The same component transport inequality as before implies $W_{1,t}(q, p_{H^+}) \leq \sqrt{D(q\|p)/(2t)}$. Furthermore Jensen's inequality applied to the posterior mixing law at each observed cell gives $D(H^+\|H) \leq \ell(H^+) - \ell(H) \leq 1/N$. Pinsker's inequality therefore gives $W_{1,t}(p_{H^+}, p_H) \leq \|H^+ - H\|_{\mathrm{TV}} \leq 1/\sqrt{2N}$. Finally $D(q\|p_H) \leq D(q\|p_G) + 1/N$. Thus $\mathbb{E}W_{1,t}(p_H, p_G) \leq C/\sqrt{N}$ still holds. Corollary 6.3.4 adds at most $C/\sqrt{N}$ to its original right side. Repeating (6.3.11) absorbs this addition because $t \geq 1$ and $L_N \geq 1$. The high-replication kernel-coupling argument is unchanged. □

The constant-one inequality $|C(t,m,j)| \leq e^{m^2/t}$ for the Bernstein coefficients of a shifted Chebyshev polynomial remains open in this manuscript. The full statistical curve does not require it. Nor do we obtain optimal leading constants, sharp predictive Hellinger logarithms, or a finite-sample confidence band for the entire mixing law. Those are separate questions from the order-optimal unrestricted-MLE result proved above.

6.4 Restricted adaptive information: exact lower counts and sharp noise floors

Krieg–Novak–Ullrich ask how adaptive recovery changes when the admissible scalar measurements are restricted to convex, homogeneous, or analytic functions, and ask for lower bounds already in dimension three [60, Section 3]. Their question concerns exact real answers. The earlier manuscript established logarithmic upper bounds using convex polynomial measurements. We first give a lower bound which is sharp in dimension three, and then show that

range-normalized convex information has a very different behavior under adversarial noise. For the latter model the relevant comparison is Krieg–Novak–Plaskota–Ullrich [88]: unrestricted continuous information can correct bounded deterministic noise to arbitrarily small reconstruction error. Convex information cannot do so.

The results below concern deterministic adaptive algorithms. A measurement chosen after a transcript must have the stated properties as a function of the unknown input, for that fixed transcript. The selection rule and decoder need not be continuous. The exact-real and noisy models are kept separate throughout.

6.4.1 Two exact queries cannot suffice with a polynomial or quasiconvex first query

A function is called quasiconvex if all its sublevel sets are convex. No differentiability is part of this definition.

Theorem 6.4.1 (A two-query obstruction)**.** *Let $m \geq 3$. Consider an algorithm on $\mathbb{R}^m$ which first observes $f(x)$ exactly, then observes $g_{f(x)}(x)$ exactly, where every $g_t : \mathbb{R}^m \to \mathbb{R}$ is continuous. Suppose that f is either a real polynomial or a continuous quasiconvex function. For every decoder $\Phi : \mathbb{R}^2 \to \mathbb{R}^m$,*

$$\sup_{x \in \mathbb{R}^m} \left\| x - \Phi\big(f(x), g_{f(x)}(x)\big) \right\|_2 = \infty.$$

The same conclusion holds with any norm on $\mathbb{R}^m$. In particular, the minimum number of adaptive exact measurements needed for uniformly bounded-error recovery on $\mathbb{R}^3$ is exactly three in each of the following classes: real polynomials; globally convex polynomials; and globally convex real-analytic functions. In the last class the measurements may additionally be required to have Lipschitz constant at most one.

Proof. We use an elementary consequence of connectedness. Suppose that for arbitrarily large R some level of f contains a continuous graph

$$\gamma(z) = z + \tau(z)v, \qquad z \in H, \quad \|z - z_0\|_2 \leq R, \qquad H = v^{\perp},$$

where v is a unit vector. On the sphere $\{z \in H : \|z - z_0\|_2 = R\}$ the function

$$g_t(\gamma(z_0 + w)) - g_t(\gamma(z_0 - w))$$

is continuous and odd in w. This sphere has dimension $m - 2 \geq 1$ and is connected, so the function vanishes somewhere. The two corresponding inputs have identical transcripts and are at distance at least $2R$, because their orthogonal projections onto H are antipodal. At least one reconstruction error is at least R.

Suppose first that f is a nonconstant polynomial of degree d. Choose a direction v on which its highest homogeneous part is nonzero, and negate f if necessary. In the coordinates $z + sv$, with $z \in H$, we have

$$f(z + sv) = as^d + \sum_{j<d} a_j(z)s^j, \qquad a > 0.$$

On any fixed transverse ball of radius $2R$ all coefficients and their derivatives are bounded. Thus there is a $T > 0$ such that $\partial_s f(z + sv) > 0$ for all $s \geq T$ and all z in that ball, and $f(z + sv) \to \infty$ uniformly there as $s \to \infty$. Choose $t > \max_{\|z\|_2 \leq 2R} f(z + Tv)$. The equation $f(z + sv) = t$ has a unique solution $s = \tau(z) > T$ on the ball. The implicit function theorem makes τ continuous, and the preceding graph argument applies.

Now suppose that f is continuous and quasiconvex. Write $s = \sup_{\mathbb{R}^m} f \in \mathbb{R} \cup \{\infty\}$ and $U = \{x : f(x) < s\}$, interpreting $U = \mathbb{R}^m$ when $s = \infty$. If U is a proper subset, it is an open

convex set. Unless it is empty, separation of a proper open convex set gives a halfspace disjoint from U. On that halfspace $f = s$, and it contains balls of arbitrarily large radius. A single continuous second query on the sphere of such a ball identifies an antipodal pair. The case $U = \varnothing$, including constant f, is immediate in the same way.

It remains to consider $U = \mathbb{R}^m$. For any $R > 0$, choose

$$\max_{\|x\|_2 \le 2R} f(x) < t < s.$$

Then $K = \{x : f(x) \le t\}$ is a proper closed convex set containing the radius-$2R$ ball in its interior. Separation gives a unit vector v and a finite b such that $\langle v, x\rangle \le b$ on K. For $z \in v^{\perp}$ with $\|z\|_2 < 2R$, set

$$\tau(z) = \sup\{a : z + av \in K\}.$$

This is finite, positive and concave on that ball. A finite concave function is continuous on the interior of its domain. Its graph lies in the boundary of K, hence in $\{f = t\}$ by continuity of f. The graph argument again applies.

Equivalence of norms proves the assertion for every norm. Three coordinate measurements recover $\mathbb{R}^3$ exactly and belong to all the stated classes, including the unit-Lipschitz class. □

Scope of the lower bound. The theorem resolves the three-dimensional count for the indicated restricted classes. It does not settle whether two arbitrary continuous measurements can approximate every vector in $\mathbb{R}^3$ with prescribed uniform accuracy, the unrestricted question in [60]. Nor does it determine the optimal count in higher dimensions. The unbounded input domain is essential to the argument; it is not a claim that every two-query method on a fixed compact ball has infinite error.

6.4.2 The exact noise floor for bounded convex measurements

Let $C_m = [-1, 1]^m$. For a class Λ of real functions on C_m with values in $[-1, 1]$, let $e_n(\Lambda, \delta)$ denote the infimum of the worst-case ℓ_∞ recovery error over algorithms using at most n adaptive observations

$$y_j = \lambda_j(x; y_1, \dots, y_{j-1}) + \xi_j, \qquad |\xi_j| \le \delta. \tag{6.4.1}$$

The supremum is over $x \in C_m$ and all admissible noise sequences. The range bound applies to every selected query on the whole cube, not merely on the currently feasible subset. Without this normalization, multiplying a query by an arbitrary constant would make the noise restriction vacuous.

Write Λ_{cvx} for restrictions of globally convex functions to C_m, subject to this range bound, and $\Lambda_{\mathrm{ca}} \subset \Lambda_{\mathrm{cvx}}$ for those which are also globally real analytic. Let Λ_{cp} be the subclass consisting of globally convex polynomials, with no common degree restriction.

Theorem 6.4.2 (Sharp convex noise floor)**.** *For every $m \ge 1$ and $0 < \delta < 1$, put*

$$r_\delta = \frac{\delta}{2 - \delta}.$$

For every $n \ge 0$,

$$e_n(\Lambda_{\mathrm{cvx}}, \delta) \ge r_\delta. \tag{6.4.2}$$

For every integer $k \ge 0$,

$$e_{mk}(\Lambda_{\mathrm{cp}}, \delta) \le r_\delta + \left(\frac{\delta}{2}\right)^k (1 - r_\delta). \tag{6.4.3}$$

Consequently,

$$\inf_{n\ge 0} e_n(\Lambda_{\mathrm{cvx}},\delta)=\inf_{n\ge 0} e_n(\Lambda_{\mathrm{ca}},\delta)=\inf_{n\ge 0} e_n(\Lambda_{\mathrm{cp}},\delta)=\frac{\delta}{2-\delta}. \tag{6.4.4}$$

The lower bound remains true if the admissible functions are only required to be convex on the cube.

Lower bound. For a convex $h:[-1,1]\to[-1,1]$ and $0\le t<1$, convexity on the interval $[-t,1]$ gives

$$h(t)-h(-t)\le \frac{2t}{1+t}\{h(1)-h(-t)\}\le \frac{4t}{1+t}.$$

Applying the same argument with the orientation reversed shows

$$|h(t)-h(-t)|\le \frac{4t}{1+t}. \tag{6.4.5}$$

Set $x_\pm=\pm r_\delta e_1$. Every admissible query, restricted to the central first-coordinate line, therefore satisfies

$$|\lambda(x_+)-\lambda(x_-)|\le 2\delta.$$

At every step return the midpoint of these two query values. The answer is within δ of the correct answer under both inputs. Inductively the algorithm has the same transcript under x_+ and x_-. Any output has error at least r_δ for one of them. This works for every finite number of adaptive queries. □

Upper bound for convex queries. It suffices to recover one coordinate $z\in[-1,1]$. Maintain a closed feasible interval $I=[l,u]$, with center c and radius r. If I is a singleton, no further query is needed. Otherwise choose whichever of the following two convex functions has the smaller denominator:

$$g_+(z)=-1+\frac{2(z-l)_+}{1-l},$$
$$g_-(z)=-1+\frac{2(u-z)_+}{1+u}.$$

Both are globally convex and lie in $[-1,1]$ on $[-1,1]$. On I the chosen function is affine, with slope magnitude $2/\min(1-l,1+u)$. Intersect I with the interval compatible with its noisy answer. The new feasible interval has radius at most

$$\frac{\delta}{2}\min(1-l,1+u)=\frac{\delta}{2}(1+r-|c|)\le\frac{\delta}{2}(1+r).$$

Starting from $r_0=1$, iteration yields

$$r_k\le r_\delta+(\delta/2)^k(1-r_\delta).$$

Run this construction for each coordinate and output the midpoints. It uses mk queries and proves the stated upper bound for convex measurements. □

Real-analytic and polynomial convex queries. The hinges can be approximated without abandoning global convexity or the range normalization. Set

$$H_\tau(s)=\tau\log(1+e^{s/\tau}),\qquad \tau>0.$$

This function is globally real analytic and convex, with $0\le H_\tau(s)-s_+\le\tau\log 2$. Given $\eta>0$, replace g_+ by the raw function

$$\widetilde{g}(z)=-1+\frac{2H_\tau(z-l)}{1-l},\qquad \tau=\frac{\eta(1-l)}{2\log 2},$$

and then by

$$h(z) = \frac{\widetilde{g}(z) - \eta/2}{1 + \eta/2}.$$

On $[-1, 1]$ this function lies in $[-1, 1]$ and satisfies $|h - g_+| \le \eta$. The analogous construction applies to g_-. Every resulting query is globally convex and real analytic. An answer to h with noise bound δ is an answer to the hinge with error bound $\delta + \eta$. Use the interval procedure with that enlarged bound, taking $\eta < 1 - \delta$. To obtain globally convex polynomials as well, fix one of these smooth convex functions h on $[-1, 1]$. Approximate the continuous function $\sqrt{h''}$ uniformly there by a real polynomial q. Define the polynomial P by

$$P'' = q^2, \qquad P(0) = h(0), \qquad P'(0) = h'(0).$$

It is convex on all of $\mathbb{R}$, and $\|P - h\|_{[-1,1]} \le \frac{1}{2}\|q^2 - h''\|_{[-1,1]}$. This error can be made arbitrarily small. If it is at most ζ, replace P by $P/(1 + \zeta)$ to restore its range to $[-1, 1]$; the additional uniform error is at most ζ. Thus either globally convex analytic functions or globally convex polynomials approximate each hinge with arbitrarily small uniform error while preserving normalization. For fixed k the enlarged noise bound tends to δ as these approximation errors tend to zero. Since e_{mk} is an infimum over algorithms, (6.4.3) follows for the polynomial subclass. Finally let $k \to \infty$ and combine with the lower bound. □

The limit in (6.4.4) is an infimum over finite-query algorithms, not an assumption that an infinite transcript can be read. Formula (6.4.3) supplies a finite algorithm for every desired excess error above the floor. The analytic queries have no common imposed Lipschitz constant, and the polynomial queries have no common degree bound; either would be an additional restriction.

A geometric identity behind the obstruction. The sharp lower bound is a special case of a useful exact dual formula. Let K be a compact convex body, let $x \ne y$ lie in its interior, and let z be the intersection of the ray $x + \mathbb{R}_+(y - x)$ with ∂K. Put

$$q_K(x, y) = \frac{\|y - x\|_2}{\|z - x\|_2}.$$

Then

$$\sup_{\substack{f \text{ convex on } K \\ -1 \le f \le 1}} \{f(y) - f(x)\} = 2q_K(x, y). \tag{6.4.6}$$

Indeed, $y = (1 - q)x + qz$ proves the upper bound. For equality choose a supporting linear functional a at z, put $b = \langle a, z\rangle$, and use

$$f(w) = \max\left\{-1, -1 + 2\frac{\langle a, w - x\rangle}{b - \langle a, x\rangle}\right\}.$$

The denominator is positive because x is interior; this convex function lies between -1 and 1 on K and attains the desired difference. Thus the symmetric discrepancy induced by all bounded convex measurements is $2\max\{q_K(x, y), q_K(y, x)\}$. This identity is included as the geometric explanation of the noise floor, not as a separately counted resolution of an external open problem.

6.4.3 Degree is a precision resource for nonconvex polynomial information

Let Λ_D consist of real polynomials on $\mathbb{R}^m$ of total degree at most the integer $D \ge 1$, with $|P(x)| \le 1$ on C_m. The polynomial is allowed to depend on earlier noisy answers, but its degree and range bounds are uniform. Write $e_{n,D}(\delta) = e_n(\Lambda_D, \delta)$.

Theorem 6.4.3 (Sharp order of the bounded-degree noise floor)**.** *For every $m, D \geq 1$, every $n \geq 0$, and $0 < \delta < 1$,*

$$e_{n,D}(\delta) \geq \frac{\delta}{\sqrt{D^2 + \delta^2}}. \tag{6.4.7}$$

If $0 < \delta \leq 1/72$, set $a = 18\delta \leq 1/4$ and

$$k = 2 + \left\lceil \frac{\log D}{\log(1/a)} \right\rceil.$$

Then

$$e_{mk,D}(\delta) \leq \frac{18\delta}{D}. \tag{6.4.8}$$

In particular, uniformly in m, D and $0 < \delta \leq 1/72$,

$$\inf_{n \geq 0} e_{n,D}(\delta) \asymp \frac{\delta}{D}.$$

The constants in the two inequalities are not claimed optimal.

Lower bound. The classical Bernstein inequality for algebraic polynomials states that, if p has degree at most D and $|p| \leq 1$ on $[-1, 1]$, then

$$|p'(s)| \leq \frac{D}{\sqrt{1 - s^2}}, \qquad -1 < s < 1;$$

see, for example, the treatments in [89]. For $0 < t < 1$ this implies

$$|p(t) - p(-t)| \leq \frac{2Dt}{\sqrt{1 - t^2}}.$$

Take $t = \delta/\sqrt{D^2 + \delta^2}$ and restrict each selected multivariate query to the line se_1. Its values at $\pm te_1$ differ by at most 2δ. The common-midpoint transcript argument from Theorem 6.4.2 gives error at least t. □

Upper bound. Again work coordinate by coordinate. One linear query gives a feasible interval $I = [c - r, c + r] \subset [-1, 1]$ with $r \leq \delta$. We show that whenever $0 < r \leq 1/2$ one further degree-D query reduces its radius to at most

$$a \max\{r, D^{-1}\}. \tag{6.4.9}$$

Let

$$K = \min\{D, (2r)^{-1}\},$$

and choose the largest positive odd integer $j \leq K$. Since $K \geq 1$, we have $j \geq K/3$. The function

$$P_{j,c}(z) = \sin\left(j \arcsin \frac{z - c}{2}\right)$$

is a real polynomial of degree j: for odd j it is, up to sign, a Chebyshev polynomial. Since $c, z \in [-1, 1]$, it is bounded by one on the whole input interval. For $z \in I$ we have

$$\left| j \arcsin \frac{z - c}{2} \right| \leq jr \leq \frac{1}{2},$$

where $\arcsin(r/2) \leq r$ for $r \leq 1/2$. Therefore

$$P'_{j,c}(z) = \frac{j \cos(j \arcsin((z - c)/2))}{2\sqrt{1 - (z - c)^2/4}} \geq \frac{j}{3}.$$

The polynomial is strictly increasing on the current feasible interval. Intersecting with the preimage of the answer interval of width 2δ leaves a feasible interval of radius at most

$$\frac{3\delta}{j} \le \frac{9\delta}{K} \le 18\delta \max\{r, D^{-1}\},$$

which proves (6.4.9).

Write r_q for the radius after q queries in one coordinate. From $r_1 \le \delta$ and (6.4.9), induction gives

$$r_q \le \max\{a^{q-1}\delta, a/D\} \qquad (q \ge 1).$$

The induction uses $a \le 1$; all radii stay below $1/2$. With the stated k we have $a^{k-2}D \le 1$, and consequently $a^{k-1}\delta \le a/D$. Thus $r_k \le 18\delta/D$. Applying the procedure to all m coordinates proves (6.4.8). □

What the degree theorem adds. Bernstein's derivative bound is classical, and the entropy bounds for unrestricted noisy information in [88] are not new claims here. The result is the matching recovery construction within a fixed polynomial degree: degree D supplies precisely an order-D improvement in the irreducible precision scale. Its upper bound uses nonconvex polynomials. No corresponding improvement is possible within the convex class, even if the degree is unbounded and real analyticity is allowed.

6.4.4 Audit and unresolved extensions

The prior exact-real convexification argument passes a direct audit: an exact energy answer permits subtracting a known radial convex term; Gaussian damping permits globally analytic, globally Lipschitz convex queries. The new noise result explains why this does not automatically yield a stable noise-correcting construction. Range normalization and errors in the compensating answer are substantive costs, and the sharp floor above prohibits eliminating them by an arbitrary number of bounded convex queries.

Three limitations are material. First, the two-query theorem does not cover a general nonconvex real-analytic first measurement. Second, the full finite-n minimax curve for bounded convex queries is not determined here; the exact result is its infinite-budget infimum and the explicit convergent upper bound. Third, the degree theorem identifies the uniform order δ/D, not its optimal leading constant or the smallest query count achieving it. These are research targets, not corollaries already proved.

6.5 Sharp covariance-tail boundaries in every fixed dimension

The preceding manuscripts established logarithmic tail sufficiency in every dimension, but their sharp counterexamples were scalar. Embedding a scalar witness in one coordinate does not preserve invisibility to the additional Gaussian anchors. We resolve that matrix obstruction by replacing a meromorphic zero product with an entire one. This replacement permits a contour deformation across the imaginary matrix boundary even when the matrix is indefinite and its inverse quadratic form vanishes.

Conditional on $(A, \Sigma) \in \mathbb{R}^d \times \mathbb{S}_d^+$, let X_i be independent $N_d(A, \Sigma)$ vectors, and write

$$\mathcal{O}_n = (X_1, \dots, X_{d-1}, S_n), \qquad S_n = \sum_{i=1}^{n} X_i, \quad n \ge d.$$

Only the separate population laws of these experiments are supplied. There is no cross-design coupling information. Fix a sufficiently large E, put $L_1(v) = \log(E + v)$, $L_{j+1}(v) = \log L_j(v)$, and set

$$W_{m,a}(v) = \frac{\sqrt{v}}{L_1(v) \cdots L_{m-1}(v) L_m(v)^a}, \qquad m \geq 1, \quad a \geq 0.$$

Theorem 6.5.1 (The fixed-dimension tail boundary)**.** *Fix any dimension $d \geq 1$ and any finite logarithmic depth $m \geq 1$.*

1. *If $0 \leq a \leq 1$, every infinite terminal set identifies the joint mixing law in the class*

$$\mathbb{E} \exp\{c W_{m,a}(\operatorname{tr} \Sigma)\} < \infty \quad \text{for some } c > 0. \tag{6.5.1}$$

   No restriction on the mean marginal is needed.

2. *Suppose $n_1 < n_2 < \cdots$ and $n_j \geq c_0 j^s$ for some $s > 1$. For every $a > 1$, these terminal experiments fail to identify the mixing law in (6.5.1). The two distinct witnesses can be absolutely continuous, have the same complete mean and covariance marginals and the same joint polynomial moments of every degree, and satisfy (6.5.1) for* every *$c > 0$. One can impose any fixed lower bound $\Sigma \succeq b_0 I$ with $b_0 > 0$.*

The sufficiency assertion is recalled to state a sharp boundary; the new assertion is its converse in each fixed $d > 1$, together with the enlarged class of terminal designs. The proof also gives a substantially denser design theorem at the first logarithmic depth.

Theorem 6.5.2 (Nearly harmonic terminal designs)**.** *Fix $d \geq 1$ and $\beta > 1$. If*

$$n_j \geq c_0 j \{\log(e + j)\}^\beta, \qquad j \geq 1,$$

then for every $a > 1$ there are two witnesses with all the properties in the negative part of Theorem 6.5.1, with $m = 1$. Thus the logarithmic covariance-tail threshold remains exactly $a = 1$ for each such fixed design sequence, even though it can be much denser than a geometric sequence.

These are population nonidentifiability results for arbitrary mixing laws. They do not contradict finite Gaussian-mixture identifiability. The general differential-annihilator approach to infinite-mixture nonidentifiability is developed by Le, Bariletto, Rinaldo, and Ho [8]. The construction here simultaneously cancels infinitely many matrix observation surfaces and imposes a quantitative covariance tail; it is not a consequence of a single finite-order annihilator.

6.5.1 Entire products and quantitative matrix regularization

For large positive R, abbreviate

$$P_m(R) = \prod_{j=1}^{m} \log_j R, \qquad D_{m,p}(R) = P_m(R)(\log_m R)^p, \qquad p > 0.$$

All uses below have R large; on bounded intervals these functions may be extended smoothly and positively. Choose K sufficiently large and define, on the cut plane, the principal holomorphic functions

$$J_1(w) = \log(K + w), \qquad J_{j+1}(w) = \log(K + J_j(w)), \qquad Q(w) = \frac{w}{J_m(w)^p}. \tag{6.5.2}$$

On the upper cut half-plane all nested logarithms have positive imaginary part, and on the positive axis they are positive; conjugation handles the lower half-plane. Thus these formulas introduce no additional cuts or zeros.

Lemma 6.5.3 (Scalar sector bounds)**.** *For suitable constants $c, C, R_0 > 0$ the following hold.*

1. $\operatorname{Re} Q(w) \geq 0$ *on the right half-plane, and*

$$\operatorname{Re} Q(w) \geq c\frac{|w|}{D_{m,p}(|w|)} \quad (\operatorname{Re} w \geq 0,\ |w| \geq R_0).$$

2. *The latter estimate remains valid when $|w| \geq R_0$ and*

$$|\arg w| \leq \frac{\pi}{2} + \frac{c}{P_m(|w|)}.$$

3. *There is a smooth positive function $\eta(r)$, $r > 0$, with bounded derivative, bounded above by an arbitrarily small fixed constant, and*

$$\eta(r) \asymp \frac{r}{P_m(1/r)} \quad (r \downarrow 0), \tag{6.5.3}$$

such that, for every real $\lambda \neq 0$ and $0 < r \leq |\lambda|$,

$$\operatorname{Re} Q\big((-t\eta(r) + i\lambda)^{-1}\big) \geq 0, \qquad 0 \leq t \leq 1.$$

For $|\lambda|$ small the last expression is at least $c/(|\lambda| D_{m,p}(1/|\lambda|))$.

Proof. If $0 \leq \arg w \leq \pi/2$, each nested logarithm has real part at least $\log K$. Successive argument estimates give $0 \leq p \arg J_m(w) \leq \arg w$ once K is large enough. This proves the first nonnegativity assertion. Uniformly for bounded real θ, repeated logarithmic expansion gives

$$|J_m(Re^{i\theta})| \sim \log_m R, \qquad \arg J_m(Re^{i\theta}) = \frac{\theta}{P_m(R)}(1 + o(1)).$$

The second relation and its first angular derivative are uniform. They follow inductively because the imaginary part after each logarithm is the previous imaginary part divided by its real part, up to a relative error tending to zero. Fixed additive constants only affect the lower-order terms. Near $\theta = \pi/2$, the argument of Q is consequently at most $\pi/2 - c/P_m(R)$, even if θ exceeds $\pi/2$ by a sufficiently small multiple of $1/P_m(R)$. Away from those endpoints its real part is larger. This proves the two large-modulus assertions.

For the last part, take a sufficiently small multiple of $r/P_m(1/r)$ near zero and extend it smoothly to a small positive constant for large r, with bounded derivative. For small $|\lambda|$, the angular overshoot of $(-t\eta(r) + i\lambda)^{-1}$ is at most $\eta(r)/|\lambda| \leq c/P_m(1/|\lambda|)$; here the function $r/P_m(1/r)$ is increasing near zero. The sector estimate applies. On compact intervals of nonzero λ, strict positivity on the imaginary axis and continuity allow a common small upper bound on η. Finally, at zero the Taylor expansion is $Q(w) = b_1 w - b_2 w^2 + O(w^3)$ with $b_1, b_2 > 0$. For large $|\lambda|$ it yields

$$\operatorname{Re} Q((-t\eta + i\lambda)^{-1}) = \frac{b_2 - b_1 t\eta}{\lambda^2} + O(|\lambda|^{-3}),$$

uniformly for bounded η. Reducing its upper bound once more completes the proof. The expansion's positivity follows directly from $J_m(0) > 0$ and $J_m'(0) > 0$. □

Let $a_j = 2n_j$. Since $\sum_j a_j^{-1} < \infty$ in both theorems, the canonical product

$$E_0(z) = \prod_{j \geq 1} (1 - z/a_j) \tag{6.5.4}$$

is a nonzero entire function, real on the real axis, with the required positive zeros. The absence of negative poles is essential.

Lemma 6.5.4 (Growth of the design product). *1. If $a_j \ge cj^s$, $s > 1$, then $\log|E_0(z)| \le C(1+|z|^{1/s})$ throughout the plane.*

2. If $a_j \ge cj\log^\beta(e+j)$, $\beta > 1$, put

$$J(R) = \frac{1}{2}\sum_j \log(1+R^2/a_j^2), \qquad S(R) = \sum_j \frac{a_j}{a_j^2+R^2}.$$

For large R,

$$J(R) \le \frac{CR}{\log^\beta R}, \qquad S(R) \le \frac{C}{\log^{\beta-1} R}.$$

Moreover, for every complex z,

$$\log|E_0(z)| \le J(|z|) + (-\operatorname{Re} z)_+ S(|z|). \tag{6.5.5}$$

Consequently, uniformly when $|z| \le C_0 R$ and $(-\operatorname{Re} z)_+ \le C_0 R/\log R$,

$$\log|E_0(z)| \le \frac{CR}{\log^\beta R}. \tag{6.5.6}$$

The same bounds, with altered constants, apply to each fixed derivative of E_0 on the indicated sets.

Proof. In the first case, $\sum_j \log(1+|z|/(cj^s)) \le C(1+|z|^{1/s})$, by splitting at $j \asymp |z|^{1/s}$ or comparing with the corresponding integral. In the second case, the counting function satisfies $N(t) = \#\{j : a_j \le t\} \le Ct/\log^\beta(e+t)$. Integration by parts gives

$$J(R) = \int_0^\infty \frac{R^2 N(t)}{t(t^2+R^2)}\,dt.$$

Split at $\sqrt{R}$ and R. The first portion is $O(\sqrt{R})$; on the middle portion use $\log t \asymp \log R$; on the last portion use $R^2/(t^2+R^2) \le R^2/t^2$. This proves the stated bound for J. For S, the terms $a_j \le R$ sum to at most $N(R)/R$. Partial summation bounds the remaining terms by $C\int_R^\infty dt/(t\log^\beta t)$, proving the assertion.

For $x = (-\operatorname{Re} z)_+$ and $r = |z|$,

$$|1 - z/a|^2 \le (1 + r^2/a^2)\exp\{2xa/(a^2+r^2)\}.$$

Summing logarithms proves (6.5.5). If $r \ge \sqrt{R}$, its right side is bounded by $CR/\log^\beta R$ under the extra hypotheses. If $r < \sqrt{R}$, use instead $\log|E_0(z)| \le r\sum a_j^{-1} = O(\sqrt{R})$. Fixed derivatives follow from Cauchy's formula on circles of radius one; enlarging a set by that radius preserves all the displayed bounds after a harmless change in constants. □

On $\mathbb{S}_d$ use orthonormal coordinates for $\langle V, W\rangle = \operatorname{tr}(VW)$, and set $D = d(d+1)/2$. Choose a nonzero real even $h \in C_c^\infty(\mathbb{R}^d \setminus \{0\})$ and define, for complex symmetric T with $\operatorname{Re} T \succ 0$,

$$\mathcal{H}(u,T) = h(u)E_0(u^T T^{-1}u)\exp\{-\operatorname{tr} T^{1/2} - \operatorname{tr} Q(T^{-1})\}. \tag{6.5.7}$$

Matrix functions here are principal holomorphic functions. In the polynomially sparse case $m \ge 1$ and $p > 0$ are arbitrary. In the nearly harmonic case take $m = 1$ and $0 < p < \beta - 1$.

Lemma 6.5.5 (A quantitatively decaying invisible matrix density). *Under either set of hypotheses just specified, $\mathcal{H}$ is the Fourier–Laplace transform of a nonzero real Schwartz function g on $\mathbb{R}^d \times \mathbb{S}_d$ supported where $\Sigma \succeq 0$. Both coordinate marginals and all joint polynomial moments of g vanish. For every integer $M \ge 0$ there are $c_M, C_M > 0$ such that*

$$|g(a,\Sigma)| \le C_M(1+|a|)^{-M}\exp\{-c_M W_{m,1+p/2}(\operatorname{tr}\Sigma)\}, \qquad \Sigma \succeq 0. \tag{6.5.8}$$

Proof. We prove the tube estimates, the support assertion, and the quantitative contour estimate separately.

Tube estimates. If $T = X + iV$ with $X \succ 0$, then $R = T^{-1}$ is accretive because $\operatorname{Re} R = T^{-*}XT^{-1} \succ 0$. For any accretive $d \times d$ matrix,

$$\|R\| \le C_d \sum_{j=1}^{d} |\lambda_j(R)|. \tag{6.5.9}$$

Indeed, put R in unitary Schur form. Positivity of its Hermitian part gives $|R_{ij}| \le 2\sqrt{\operatorname{Re}\lambda_i \operatorname{Re}\lambda_j}$ for each strictly upper triangular entry. The Frobenius norm is therefore bounded by a constant times $\sum |\lambda_i|$. This also handles repeated eigenvalues and non-normal matrices. Lemma 6.5.3, the trace identity for holomorphic matrix functions, and (6.5.9) imply

$$\operatorname{Re}\operatorname{tr} Q(R) \ge c\frac{\|R\|}{D_{m,p}(\|R\|)} - C \tag{6.5.10}$$

when $\|R\|$ is large. To see this directly, at least one eigenvalue has modulus at least $c_d\|R\|$, all other real contributions are nonnegative, and fixed rescaling preserves the displayed asymptotic order.

In the first design class, the product growth is $O(\|R\|^{1/s})$; in the second, $u^T Ru$ lies in the right half-plane and Lemma 6.5.4 bounds its logarithmic growth by $C\|R\|/\log^{\beta}(e + \|R\|)$. These quantities are $o(\|R\|/D_{m,p}(\|R\|))$ in their respective parameter ranges. Thus the negative exponent in (6.5.10) absorbs the entire product, uniformly for u in the support of h.

There is also a uniform bound at large matrix frequencies:

$$|\mathcal{H}(u, X + iV)| \le C\exp\{-c\sqrt{\|V\|}\}, \qquad X \succ 0. \tag{6.5.11}$$

For completeness, write $T^{1/2} = B + iC$ with real symmetric B, C. The principal square root of an accretive matrix is accretive (for example, use its convergent resolvent integral), so $B \succ 0$. The equations $B^2 - C^2 = X \succ 0$ and $BC + CB = V$ give $\|C\| \le \|B\|$ and $\|V\| \le 2(\operatorname{tr} B)^2$. This proves (6.5.11) after the preceding absorption.

On the boundary $T = iV$ with V real symmetric and invertible, functional calculus is diagonal in an orthonormal real eigenbasis. Put

$$r(V) = \{\operatorname{tr}(V^{-2})\}^{-1/2}. \tag{6.5.12}$$

It is smooth on the invertible locus, $\min|\lambda_j(V)|/\sqrt{d} \le r(V) \le \min|\lambda_j(V)|$, and $|\nabla r| \le 1$. The last assertion follows from $\nabla r = r^3 V^{-3}$ and $\operatorname{tr} V^{-6} \le (\operatorname{tr} V^{-2})^3$. The boundary value $H(u, V) = \mathcal{H}(u, iV)$ decays faster than every power at both $\|V\| = \infty$ and $r(V) = 0$. The same is true of every derivative. Here are the details needed when a Cauchy neighborhood leaves the positive tube. Around iV take complex symmetric matrix perturbations of operator norm at most $cr/P_m(1/r)$, with c sufficiently small. Since iV is normal, every perturbed eigenvalue is within this distance of its real-normal reference spectrum $i\operatorname{spec} V$. Its inverse therefore lies in the enlarged scalar sector of Lemma 6.5.3, and the inverse matrix norm is $O(1/r)$ by the Neumann-series bound. Grouping overlapping spectral disks preserves the eigenvalue count of each cluster along the perturbation. The cluster containing an eigenvalue of smallest modulus has diameter at most $2d$ times the perturbation radius. At least one inverse eigenvalue therefore has modulus comparable to $1/r$; all the real Q contributions remain nonnegative. Thus the trace regularization still gives $c/(rD_{m,p}(1/r))$ damping. For the nearly harmonic product one also needs its real argument control. For any invertible T,

$$\operatorname{Re}(T^{-1}) = T^{-*}(\operatorname{Re} T)T^{-1}.$$

In these neighborhoods $\|\operatorname{Re} T\| = O(r/P_m(1/r))$, so the negative part of the quadratic form $u^T T^{-1} u$ is at most $C/(rP_m(1/r))$. When $m = 1$ this is exactly the range of (6.5.6); the polynomially sparse product needs only the inverse norm bound. Cauchy's formula in these neighborhoods therefore bounds each fixed matrix derivative by the same exponential damping times powers of $P_m(1/r)/r$ and of $1 + \|V\|$. At infinity, neighborhoods can be reduced if needed; the principal square-root trace retains its $\sqrt{\|V\|}$ damping, up to a constant. For u derivatives apply the scalar Cauchy formula for E_0 on circles of radius one and the chain rule; their extra factors are polynomial in $\|V^{-1}\|$. The exponential bounds absorb all these factors. Consequently H extends by zero across $\det V = 0$ to a Schwartz function, compactly supported in u.

Fourier inversion and cone support. Set

$$g_u(\Sigma) = (2\pi)^{-D} \int_{\mathbb{S}_d} e^{i \operatorname{tr}(V\Sigma)} H(u, V)\, dV, \qquad g(a, \Sigma) = (2\pi)^{-d} \int e^{-iu^T a} g_u(\Sigma)\, du.$$

Both are Schwartz. For $X \succ 0$ let $g_{u,X}$ be the analogous inverse Fourier integral of $\mathcal{H}(u, X + iV)$. Cauchy–Riemann equations and integration by parts along a compact path in the positive tube give

$$g_{u,X}(\Sigma) = e^{-\operatorname{tr}(X\Sigma)} g_u(\Sigma).$$

One first starts at $X = \epsilon I$ and then sends ϵ to zero; (6.5.11) justifies dominated convergence. The same bound shows that $g_{u,X}$ is uniformly bounded for all $X \succ 0$. If Σ has a negative eigenvalue, choose $X_0 \succ 0$ with $\operatorname{tr}(X_0\Sigma) < 0$ and send $X = tX_0$ to infinity. Thus $g_u(\Sigma) = 0$ outside the positive semidefinite cone. Fourier inversion now proves the asserted Fourier–Laplace identity.

The variable matrix contour. For invertible real symmetric V deform the contour from iV to

$$T(V) = -\eta(r(V))I + iV. \tag{6.5.13}$$

All matrices along the homotopy $-t\eta(r(V))I + iV$, $0 \le t \le 1$, are normal and have nonreal eigenvalues; their spectra and the spectra of their inverses avoid the negative real axis. Thus (6.5.7) is holomorphic in a neighborhood of every point of the homotopy. The entire function E_0 introduces no further pole hypersurfaces.

At least one eigenvalue of V has modulus comparable to $r(V)$. Lemma 6.5.3 therefore gives, uniformly along this homotopy when $r = r(V)$ is small,

$$\operatorname{Re} \operatorname{tr} Q(T^{-1}) \ge \frac{c}{rD_{m,p}(1/r)}.$$

All other summands are nonnegative. Also $\operatorname{Re} \operatorname{tr} T^{1/2} \ge c\sqrt{\|V\|}$ outside a fixed bounded set: the real shift is bounded, so each large imaginary eigenvalue has a principal square root with real part comparable to its square-root modulus. In the first design class the product growth is $O(r^{-1/s})$ and is absorbed by the first displayed damping term.

In the nearly harmonic case put $R = 1/r$. Since $m = 1$, $\eta(r) \le Cr/\log(1/r)$. In a real eigenbasis of V, for $z = u^T T^{-1} u$ one has

$$|z| \le CR, \qquad (-\operatorname{Re} z)_+ = t\eta(r) \sum_j \frac{u_j^2}{t^2\eta(r)^2 + \lambda_j(V)^2} \le \frac{CR}{\log R}.$$

Equation (6.5.6) bounds its logarithmic product growth by $CR/\log^{\beta} R$. This is absorbed because $p + 1 < \beta$. Every fixed derivative in u obeys the same estimate, up to polynomial factors in R, by Cauchy's formula for the scalar entire function and the chain rule. We have therefore proved, for each fixed multi-index α,

$$|\partial_u^\alpha \mathcal{H}(u, T(V))| \le C_\alpha \exp\left\{ -c_\alpha \sqrt{\|V\|} - \frac{c_\alpha}{rD_{m,p}(1/r)} \right\} \tag{6.5.14}$$

for small r; on r bounded away from zero only the first exponential is needed.

Here the contour substitution has bounded Jacobian. In orthonormal matrix coordinates it is a rank-one change of iI_D, since its real displacement is always a scalar multiple of the fixed matrix I. Its determinant has modulus at most $1 + \sqrt{d}\,\|\nabla\eta(r(V))\|$, which is bounded by $1 + \sqrt{d}\sup|\eta'|$. The deformation can be justified using only one complex variable, which avoids any assumption about the geometry of the singular matrix hypersurface. Write

$$V = W + \frac{t}{\sqrt{d}}I, \qquad \operatorname{tr} W = 0,$$

using orthonormal coordinates, and first fix W. The singular real t values are the finitely many numbers $t_j = -\sqrt{d}\,\lambda_j(W)$. On each interval between successive distinct t_j, deform the real t contour to

$$z = t + i\sqrt{d}\,\eta(r(W + tI/\sqrt{d})).$$

The corresponding matrix is $iW + izI/\sqrt{d}$, exactly (6.5.13). Every eigenvalue has imaginary part of a fixed nonzero sign in the interior of this vertical strip. Its spectrum and that of its inverse therefore avoid all branch cuts; the integrand is holomorphic throughout the region swept out by the contour. Near a finite endpoint the deformation height tends to zero, and the short joining segment has a vanishing integral by (6.5.14), because r is comparable to the endpoint distance up to constants depending only on d. The unbounded intervals have vanishing joining integrals at infinity by the square-root damping. Cauchy's theorem proves the identity for each fixed W, with line element $(1 + i\sqrt{d}\,\partial_t\eta)\,dt$. Its modulus is bounded uniformly, as already noted. The global bounds above, and $|e^{\operatorname{tr}(T\Sigma)}| \le 1$ for $\Sigma \succeq 0$ along the homotopy, make both original and deformed integrals absolutely integrable in (t, W). Fubini's theorem therefore proves the full matrix contour identity. The same reasoning applies after any fixed number of u derivatives. Repeated eigenvalues of W merely repeat an endpoint and cause no change in this argument. No contour passes through a singular matrix.

Write $v = \operatorname{tr}\Sigma$. The modulus of the deformed inverse integrand, for small r, is bounded by

$$C\exp\left\{-c\sqrt{\|V\|} - c\left(\frac{vr}{P_m(1/r)} + \frac{1}{rP_m(1/r)(\log_m(1/r))^p}\right)\right\}.$$

For v large, the expression in parentheses is at least $cW_{m,1+p/2}(v)$ uniformly in small r. Indeed, on $\frac{1}{4}\log v \le \log(1/r) \le \log v$ all the fixed-depth logarithms are comparable to the corresponding logarithms of v, and the arithmetic–geometric mean inequality gives the assertion. Below that interval the first term is at least $v^{3/4}$ divided by a fixed product of logarithms; above it the second is at least v divided by such a product. Both are larger than the required bound. On r bounded away from zero, $\eta(r)$ has a positive lower bound and the estimate is $O(e^{-cv})$ instead. Reserve half of the exponents in (6.5.14) for integrability in V. We conclude

$$|\partial_u^\alpha g_u(\Sigma)| \le C_\alpha e^{-c_\alpha W_{m,1+p/2}(v)}.$$

Integration by parts in the compactly supported u integral now proves (6.5.8).

Finally g is real because $\mathcal{H}(-u, \overline{T}) = \overline{\mathcal{H}(u, T)}$. It is nonzero since, at a point where $h(u) \neq 0$, varying $T = tI$ avoids the discrete zeros of E_0. The function g_u vanishes near $u = 0$, so $\int a^\gamma g(a, \Sigma)\,da = 0$ for every multi-index γ. This gives the covariance marginal and all joint polynomial moments. For the other marginal send $T = \epsilon I$ to zero in (6.5.7). Its Q damping dominates the product growth and forces a zero limit. Dominated convergence and Fourier uniqueness in a show $\int g(a, \Sigma)\,d\Sigma = 0$. □

6.5.2 Identification, counterexamples, and the exact scope

Proof of the two main theorems. We first recall the sufficient argument, including the classical moment input. If $V \ge 0$ satisfies $\mathbb{E}e^{cW_{m,a}(V)} < \infty$, then

$$(\mathbb{E}V^k)^{1/(2k)} \le CkL_1(k)\cdots L_{m-1}(k)L_m(k)^a.$$

To verify this estimate, put $Y = \sqrt{V}$, $D(x) = L_1(x)\cdots L_{m-1}(x)L_m(x)^a$, and $b_k = kD(k)$. For $w \ge 1$, iterated logarithm inequalities give $D(b_k w) \le CD(k)(1+\log w)^{m-1+a}$. Substitution $Y = b_k w$ in $\sup_y y^{2k}e^{-cy/D(y)}$ proves the displayed moment bound. The Stieltjes–Carleman series consequently diverges when $a \le 1$. This moment-determinacy mechanism, and much more general quasi-analytic weights, are classical; see de Jeu [79].

The Fourier arguments of the d independent blocks with lengths $1,\dots,1,n-d+1$ are exactly

$$(u,T), \qquad T = C + \frac{uu^T}{2n}, \quad C \succeq 0, \quad \operatorname{rank} C \le d-1. \tag{6.5.15}$$

This follows by projecting the matrix of weighted block frequencies onto the vector of square roots of the lengths and its orthogonal complement. Let ρ be the difference of two mixing laws with the same observations. Fix $u \ne 0$ and a rank-$d-1$ matrix C with u not orthogonal to its kernel. The pushforward of $e^{iu^Ta-\operatorname{tr}(C\Sigma)}\rho$ under $(a,\Sigma) \mapsto u^T\Sigma u$ is a finite complex measure whose Laplace transform vanishes at $1/(2n)$ for every selected n. The sum of the candidate laws dominates it and supplies the common Stieltjes–Carleman moment bound. Since the zeros accumulate at zero, successive Taylor expansions there force all moments to vanish. Moment determinacy applied to the positive dominating measure plus and minus the real and imaginary parts makes the complex measure zero. Every $T \succ 0$ can be written as $C + zuu^T$ of the above type: take $z = (u^TT^{-1}u)^{-1}$. Thus the full mixing transform vanishes on $u \in \mathbb{R}^d$, $T \succ 0$. Matrix Laplace–Fourier uniqueness proves identification.

For the negative assertion of Theorem 6.5.1, choose $0 < p < 2(a-1)$ and use Lemma 6.5.5. For Theorem 6.5.2 choose $0 < p < \min\{\beta - 1, 2(a-1)\}$ and set $m = 1$. At each positive definite observation argument in (6.5.15), the matrix determinant lemma gives $u^TT^{-1}u = 2n$. Hence $\mathcal{H}(u,T) = 0$ by (6.5.4). Singular observation arguments are limits of positive definite ones on the same observation surface: perturb the square $d \times d$ weighted frequency matrix to an invertible one. Continuity of the finite-measure transform therefore includes them as well. The signed observation law of g is zero at every chosen terminal length.

Write $g = g_+ - g_-$. Its nonzero positive and negative parts have the same positive total mass b; use g_+/b and g_-/b as the two mixing densities. Their coordinate marginals and all joint moments agree by the preceding lemma. Since $1 + p/2 < a$, the ratio $W_{m,1+p/2}(v)/W_{m,a}(v)$ tends to infinity. Equation (6.5.8), integrated first in a and then on the finite-dimensional positive cone, proves the required exponential integrability for every $c > 0$. Translating both covariances by $b_0 I$ multiplies the transform by $e^{-b_0 \operatorname{tr} T}$; all zeros, equalities, and tail properties persist. This imposes the positive lower bound and completes both proofs. □

Remark 6.5.6 (What has and has not been closed). The fixed-$d > 1$ logarithmic converse and its finite iterated-log extensions were explicit gaps in the preceding manuscripts and are closed here. The nearly harmonic design theorem is an additional extension, not a restatement of the scalar lacunary example. The exact joint classification for *every* convergent harmonic subseries and *every* covariance-tail envelope is not asserted. In particular, the present estimates do not settle highly irregular design sequences or supply sharp finite-sample stability at the quasi-analytic boundary. No named external conjecture is claimed as resolved by these results, and priority requires a broader external literature review.

6.6 Blind spectral certification from nonlinear transition thresholds

The earlier threshold theorem assumed an a priori bound on the number of atoms in both competing spectra. Here we remove that assumption from the competitor. One additional, appropriately designed scalar measurement certifies an atomic spectrum against every positive measure, including continuous spectra. The corresponding collision exponent changes from $1/(2m-1)$ to $1/(2m)$. These are inverse problems for the nonlinear threshold observable below. They are not asserted to be previously published open problems. In particular, positivity certificates and the $2m+1$ sample principle for *linear* Chebyshev measurements are established results of Eftekhari–Tanner–Thompson–Toader–Tyagi [90]; moment comparison by interpolation is developed in Wu–Yang [80]. The new step here is the transfer from unobserved contacts of nonlinear minima to those certificates, with uniform constants through collisions.

Fix $0 < a < b < 1$. For a nonzero finite positive Borel measure on $[a, b]$, put

$$S_\mu(x) = \int x^r \, d\mu(r), \qquad T_\mu(C) = \min_{x>0} \frac{1 + Cx}{S_\mu(x)}. \tag{6.6.1}$$

Write $\mathcal{P}[a, b]$ for all probability measures and $\mathcal{P}_m[a, b]$ for those with at most m atoms. A list of distinct settings is called one-sided here if

$$0 < C_i < \frac{a}{1-a} \quad \text{for every } i. \tag{6.6.2}$$

This is a design condition, independent of the unknown measure. It ensures that every contact in (6.6.1) is strictly larger than 1. Settings all larger than $b/(1-b)$ give the symmetric version with contacts below 1.

Theorem 6.6.1 (Exact normalized count and blind certification)**.** *The following assertions hold.*

1. *Any $2m-1$ one-sided settings identify every member of $\mathcal{P}_m[a, b]$ among members of that same class. Fewer fixed settings cannot identify every exactly m-atomic probability measure with distinct interior atoms and positive masses.*
2. *Any $2m$ one-sided settings identify each member of $\mathcal{P}_m[a, b]$ among* all *of $\mathcal{P}[a, b]$. This count is necessary: for every exactly m-atomic probability measure with distinct interior atoms and every list of at most $2m-1$ fixed positive settings, there is a different, positive $(m+1)$-atomic probability measure giving exactly the same data.*
3. *Without a mass normalization, any $2m+1$ distinct positive settings identify an at most m-atomic positive measure among all finite positive measures on $[a, b]$. At every spectrum with exactly m distinct interior atoms, $2m$ settings admit positive $(m+1)$-atomic alternatives.*

The first count is optimal for a known sparsity bound; the second and third are optimal for certifying that bound from the data themselves.

Theorem 6.6.2 (Sharp blind stability and the cost of certification)**.** *For a fixed one-sided list $\mathcal{C}$ of $2m$ settings there is a constant K such that*

$$W_1(\mu, \nu) \le K \left(\max_{C \in \mathcal{C}} |T_\mu(C) - T_\nu(C)| \right)^{1/(2m)} \qquad (\mu \in \mathcal{P}_m[a, b], \ \nu \in \mathcal{P}[a, b]). \tag{6.6.3}$$

No atom separation or lower mass bound is needed. The exponent is optimal, even if the entire threshold curve on a fixed compact interval of settings is observed. In contrast, for two members of $\mathcal{P}_m[a, b]$, already $2m-1$ one-sided settings give the sharp exponent $1/(2m-1)$.

Equivalently, the largest Wasserstein diameter between an m-atomic truth and an arbitrary positive probability measure consistent with threshold errors of size δ is $\asymp \delta^{1/(2m)}$. This is a blind consistency radius*; it is not the minimax estimation rate when the estimator is told that the truth has at most m atoms.*

6.6.1 Contacts, separated near-zeros, and positivity

We include the needed geometric facts so that the extension to non-atomic competitors is explicit. Put $H = S - xS'$. Differentiation under the integral gives

$$C(x) = \frac{S'(x)}{H(x)}, \qquad C'(x) = \frac{S''(x)S(x)}{H(x)^2} < 0, \qquad T(C(x)) = \frac{1}{H(x)}. \tag{6.6.4}$$

Moreover

$$\frac{a}{1-a} \le C(x)x \le \frac{b}{1-b}.$$

Thus $C(x)$ is a decreasing bijection and $L_C(x) = (1 + Cx)/T(C)$ is the unique supporting line at its contact. These assertions hold for every finite positive measure in the stated class, not just atomic ones.

Lemma 6.6.3 (Geometric conversion of threshold data)**.** *If two positive profiles have equal thresholds at n distinct settings, their difference has a zero in each of n disjoint intervals of contact points. Under (6.6.2), these intervals lie strictly above 1.*

For probability measures on $[a, b]$ and a fixed list of settings, there are constants $\delta_0, c, K > 0$ and a compact interval J such that threshold discrepancy at most $\delta \le \delta_0$ gives points $u_1, \dots, u_n \in J$, pairwise separated by at least c, with

$$\left| \int e^{u_i r} \, d(\mu - \nu)(r) \right| \le K\delta. \tag{6.6.5}$$

For one-sided settings they are also uniformly separated from 0.

Proof. Order the settings increasingly. The slopes of their supporting lines increase strictly, because

$$\frac{d}{dC}\frac{C}{T(C)} = \frac{T(C) - CT'(C)}{T(C)^2} > 0, \qquad T(C) - CT'(C) = \frac{1}{S(x(C))}.$$

The intersection of two consecutive supporting lines lies strictly between their two contacts. For common lines the contacts for both profiles therefore belong to the same pairwise disjoint intervals. At the first profile's contact its difference from the second is nonnegative; at the second profile's contact it is nonpositive. The intermediate value theorem proves the exact assertion.

All contacts for probability measures lie in the fixed interval

$$X = \left[\frac{a}{(1-a)\max C_i}, \frac{b}{(1-b)\min C_i} \right].$$

On X, $S, H, -S''$ are bounded above and away from zero uniformly. Equation (6.6.4) therefore separates consecutive contacts uniformly. If two contacts are $v > w$, the tangent at w exceeds $S(v)$ by at least $\kappa(v - w)^2/2$, where $\kappa = \inf_X(-S'') > 0$. Division by the bounded difference of the tangent slopes gives a uniform positive distance between their intersection and either contact.

Threshold errors of size δ change the supporting-line coefficients by $O(\delta)$; their consecutive slope differences are bounded away from zero, so the intersections change by $O(\delta)$ as well. For

small δ, both sets of contacts remain within uniformly separated versions of the same intervals. At the first contact the profile difference is at least $-K\delta$, and at the second it is at most $K\delta$. Some intermediate point consequently has difference of absolute value at most $K\delta$. Taking its logarithm proves (6.6.5). The bound on X under (6.6.2) also separates these logarithms from zero. □

The following uniform certificate is the step that allows an arbitrary positive competitor. A finite-dimensional differential equation for the competitor's transform would be unavailable.

Lemma 6.6.4 (Uniform one-sided exponential interpolation)**.** *Fix compact intervals I, J, an integer $m \ge 1$, and $c > 0$. Let $v_1 < \cdots < v_{2m+1}$ belong to J with successive gaps at least c. Suppose μ is positive with at most m atoms in I, and ν is any finite positive measure on I. Then, for every $\varphi \in C^{2m}(I)$,*

$$\left| \int \varphi \, d(\nu - \mu) \right| \le K \|\varphi\|_{C^{2m}(I)} \max_{1 \le i \le 2m+1} \left| \int e^{v_i r} \, d(\nu - \mu)(r) \right|. \tag{6.6.6}$$

Here K is uniform over collisions of atoms and over their masses.

Proof. Pad the atom list to m nodes $t_1, \dots, t_m \in I$ if necessary; repetitions mean higher multiplicity in what follows. Put $n = 2m$ and $P(r) = \prod_{j=1}^{m} (r - t_j)^2$. Let $R\varphi$ be the Hermite interpolant of φ at the node list $t_1, t_1, \dots, t_m, t_m$ in $E = \operatorname{span}\{e^{v_1 r}, \dots, e^{v_n r}\}$.

For completeness, this interpolation operator has uniformly bounded coefficients even when the nodes coalesce. Write the interpolation equations in successive divided differences at the ordered node list. Their matrix and right side extend continuously when nodes coincide. The limiting matrix is always nonsingular: a nonzero member of E cannot have n zeros counted with multiplicity. This zero count follows by repeatedly dividing by one exponential, differentiating, and applying Rolle's theorem with multiplicities. Compactness of the node and exponent tuples now bounds the inverse matrices. Divided differences of φ are bounded by its derivatives. Thus the coefficients of $R\varphi$, and its C^n norm, are at most $K\|\varphi\|_{C^n}$.

Define $q = e^{v_{n+1} r} - R(e^{v_{n+1} r})$. It vanishes at the specified nodes with the prescribed even multiplicities. The $(n+1)$-dimensional exponential system has at most n zeros, counted with multiplicity; hence q/P never vanishes on I, including at coincident nodes where the quotient is interpreted by continuity. Change its sign if necessary. Uniformly in all parameters,

$$q(r) \ge c_1 P(r) \quad (r \in I), \tag{6.6.7}$$

and the coefficients of q in the exponential basis are bounded. To justify uniformity precisely, the quotient is the ordinary divided difference of q at its n zeros and the additional point r. It extends continuously to every collision. At a limiting tuple it is nonzero by the same multiplicity zero count. Its absolute value therefore has a positive minimum on the compact parameter set.

The ordinary Hermite remainder estimate applied to $\varphi - R\varphi$ yields

$$|\varphi(r) - R\varphi(r)| \le K \|\varphi\|_{C^n} P(r) \le K' \|\varphi\|_{C^n} q(r).$$

Consequently $R\varphi \pm K' \|\varphi\|_{C^n} q$ are exponential majorants and minorants of φ, agreeing with it on the support of μ. Integrate these inequalities against ν, subtract the equal integrals against μ, and use the bounded exponential coefficients. This proves (6.6.6). □

6.6.2 Proof of the exact counts

Proof of Theorem 6.6.1. If $\mu, \nu \in \mathcal{P}_m[a, b]$, the function $h(u) = \int e^{ur} \, d(\mu - \nu)(r)$ is a sum of at most $2m$ real exponentials and satisfies $h(0) = 0$. Equal thresholds at $2m - 1$ one-sided settings

provide $2m-1$ further distinct zeros, all on one side of 0, by Lemma 6.6.3. The exponential zero count forces $h=0$, hence $\mu=\nu$.

For an m-atomic probability μ and arbitrary ν, $2m$ one-sided settings provide $2m$ transform zeros, which, together with $h(0)=0$, give $2m+1$ distinct samples. Apply Lemma 6.6.4 with zero right side to obtain equality of integrals against every smooth function and hence equality of measures. Without normalization, use $2m+1$ settings directly.

We prove the claimed lower bounds locally, including settings for which a contact happens to equal 1. Write an interior atomic measure as $\sum_{j=1}^{m} B_j\delta_{r_j}$ and use its $2m$ mass and location coordinates. If u_i is the logarithm of the contact for C_i, the envelope derivatives of $T(C_i)$ are, up to a nonzero row factor,

$$(e^{r_j u_i},\ B_j u_i e^{r_j u_i})_{j=1}^{m}.$$

Thus $K\le 2m$ distinct threshold rows have rank K. This is again the multiplicity zero count for the system $e^{r_j u}, ue^{r_j u}$, or its initial subsystems.

For normalized measures also impose the mass row, which is evaluation at $u=0$. If no $u_i=0$, these $K+1\le 2m$ rows are independent. If one $u_i=0$, replace that threshold constraint by the first moment constraint. This adds the derivative row at 0; together with the mass row and the other $K-1$ contact rows it again has rank $K+1$. Preserving mass and first moment indeed preserves the omitted threshold: for probability measures a contact at 1 occurs exactly when $\int r\,d\mu = C_i/(1+C_i)$, and its threshold is then $1+C_i$.

For $K<2m-1$ the analytic implicit-function theorem therefore gives nonconstant normalized m-atomic curves with the same constraints, proving necessity in part (i). For the stronger lower bound, add an atom of small positive mass ϵ at a fixed new interior point. The same full row rank lets us adjust the original masses and locations analytically in ϵ to preserve all K thresholds and the normalization, whenever $K\le 2m-1$. The original masses remain positive and the atoms remain distinct, so the result has exactly $m+1$ atoms. In the case of a contact at 1, use the replaced first-moment constraint as above. Without normalization, the $K\le 2m$ threshold rows alone allow the identical construction. □

Corollary 6.6.5 (Exact discovery of an unknown finite spectrum)**.** *Choose any infinite list of distinct settings satisfying (6.6.2). An exact threshold oracle permits the following information-theoretic procedure with no prior bound on atom count. After the first $2k$ observations, test whether a member of $\mathcal{P}_k[a,b]$ fits them exactly. Stop at the first successful test and return that measure. If the truth has exactly M atoms, the procedure stops at $k=M$, after $2M$ scalar observations, and returns the truth uniquely.*

Proof. A successful fit with $k<M$ would contradict the blind uniqueness in Theorem 6.6.1. At $k=M$ the truth is an admissible fit, and that same uniqueness theorem identifies it. □

This is an exact-data existence procedure. No polynomial-time claim is made for its nonlinear feasibility tests. Nor can exact atom count be uniformly recovered with positive noise: adding an arbitrarily small extra atom makes the threshold data arbitrarily close. The next result quantifies the meaningful distributional guarantee that remains.

6.6.3 Proof of the sharp blind noise exponent

Lemma 6.6.6 (One atomic side in a moment comparison)**.** *On a fixed compact interval, if α is an at most m-atomic probability and β any probability, and their moments of orders $1,\dots,2m$ differ by at most ϵ, then*

$$W_1(\alpha,\beta)\le K\epsilon^{1/(2m)}. \tag{6.6.8}$$

If both have at most m atoms, orders $1,\dots,2m-1$ suffice with exponent $1/(2m-1)$.

Proof. Let $F = F_\alpha - F_\beta$ and $E = \int |F| = W_1(\alpha, \beta)$. Between consecutive atoms of α, F is nonincreasing. Before its first atom $F \le 0$ and after its last atom $F \ge 0$. There are therefore at most $2m-1$ changes of sign, after zero sets are ignored: upward changes can occur only at its m atoms, and the initial and final signs allow at most one fewer downward change. Choose a point t_j at each sign change, including a point of an intervening zero interval when necessary. For $d \le 2m-1$ such points, a sign choice gives $P(x) = \pm \prod_{j=1}^{d}(x - t_j)$ with $P(x)F(x) \ge 0$ almost everywhere.

For $d \ge 1$, remove intervals of radius $E/(4d)$ about these roots. They remove at most $E/2$ from the integral of $|F|$, since $|F| \le 1$. On their complement $|P| \ge (E/(4d))^d$. Thus

$$\int PF \ge \frac{E}{2}\left(\frac{E}{4d}\right)^d.$$

For $d = 0$ the integral is E. An antiderivative of P has degree at most $2m$, with uniformly bounded coefficients. Integration by parts and the moment assumptions bound $\int PF$ by $K\epsilon$. This proves (6.6.8) for $\epsilon \le 1$; a diameter bound handles larger errors. If both measures are at most m-atomic, their CDF difference has at most $2m-1$ nonzero constant intervals and hence at most $2m-2$ sign changes. The same proof gives the final assertion. □

Proof of Theorem 6.6.2. Let δ be the threshold discrepancy. For small δ, Lemma 6.6.3 supplies $2m$ uniformly separated near-zeros of the transform difference. Adding its exact zero at 0 gives the $2m+1$ uniformly separated samples needed by Lemma 6.6.4. Apply that lemma to $\varphi(r) = e^{kr}$, $k = 1, \dots, 2m$. Thus the first $2m$ moment differences of the pushforwards under $r \mapsto e^r$ are $O(\delta)$. Lemma 6.6.6 and the Lipschitz property of log on $[e^a, e^b]$ prove (6.6.3). Larger δ are handled by increasing the constant.

For two at most m-atomic measures, $2m-1$ near-zeros and the exact zero at 0 are enough. Their transform difference solves an order-$2m$ constant-coefficient differential equation whose real roots are in $[a, b]$, padding the root list if needed. Evaluation at $2m$ separated points uniformly controls its initial jet: the evaluation matrix in the initial-jet basis is continuous in the roots, including collisions, and invertible by the real-exponential zero count. Compactness uniformly bounds its inverse. Its values at $1, \dots, 2m-1$ are consequently $O(\delta)$, and the two-atomic-side part of Lemma 6.6.6 proves the stated exponent.

For sharpness in the blind case, put $q = 2m$ and take sufficiently small $h > 0$ with $r_0, r_0 + qh$ in the interior of $[a, b]$. Split the binomial masses

$$2^{-(q-1)}\binom{q}{j}, \qquad 0 \le j \le q,$$

between their even and odd locations $r_0 + jh$. Let μ_h be the odd part, with m atoms, and ν_h the even part, with $m+1$ atoms. Both have total mass one, and

$$\int e^{ur}\, d(\nu_h - \mu_h)(r) = 2^{-(q-1)} e^{r_0 u}(1 - e^{hu})^q.$$

This is $O(h^{2m})$ on each fixed compact u interval. The contact bound in Lemma 6.6.3 then gives, on every fixed compact setting interval $I \subset (0, \infty)$,

$$\sup_{C \in I} |T_{\mu_h}(C) - T_{\nu_h}(C)| \le K_I h^{2m}.$$

The identity $\sum_{j=0}^{k}(-1)^j\binom{q}{j} = (-1)^k\binom{q-1}{k}$ gives $W_1(\mu_h, \nu_h) = h$ by integrating their CDF difference over the q gaps. This proves the optimal blind exponent. Taking $q = 2m-1$ instead gives m atoms on both sides and proves the other sharp exponent. If rational independence with 1 is needed for the original modular-zero interpretation, perturb the locations by $o(h^q)$ outside the countably many rational affine hyperplanes. The bounds and atom counts persist. □

6.6.4 The precise scope of this transfer

For normalized compact spectra the distinction is now exact:

comparison class	designed threshold count	sharp collision exponent
both at most m atoms	$2m-1$	$1/(2m-1)$
one at most m atoms, other arbitrary	$2m$	$1/(2m)$

The data are still the ideal scalar transitions $T(C)$, defined through infinite-height modular-zero closure in their motivating exponential-sum model. These theorems do not turn finitely many observed zeros into exact thresholds, do not recover the phases invisible to those thresholds, and do not give a new recurrence estimate for the exact prime frequencies. A quantitative finite-height threshold estimator is a separate remaining problem.

6.7 Three critical regimes for Bessel power restoration

This section develops the power-restoration criterion of the preceding manuscript, rather than counting that criterion or the full McKay ID domain a second time. The new conclusions are a sharp three-regime transition in the necessary power, uniqueness of the critical real maximizer, and the endpoint behavior of the obstruction. A separate moment test advances, but does not finish, the remaining SD problem. The published questions about the McKay families are in [3, Sections 4.3–4.4]. Our power-transition question is a newly formulated transfer problem; we do not identify it as a previously named conjecture.

Write

$$R_\mu(y)=\frac{I_{\mu+1}(y)}{I_\mu(y)},\qquad D_\mu(y)=R_\mu(y)+yR_\mu'(y),\qquad M_\mu=\sup_{y>0}D_\mu(y),\qquad \mu>-1.$$

For a generalized McKay variable with density

$$Cx^{\nu-1}e^{-bx}I_\mu(ax),\qquad \mu+\nu>0,\quad b>a>0,$$

put $r=b/a$. Its least admissible HCM power is

$$d_\mu^*(r)=\inf\{d>0:X^d \text{ has an HCM density}\},\qquad \inf\varnothing=+\infty.$$

Monomial factors do not change HCM, so this definition does not depend on ν. Negative powers give the same criterion with $|d|$.

We recall the exact criterion from [84, Theorems 6.2 and 6.4], in a form that also fixes all normalizations. For $0<\theta<\pi/2$, define

$$B_\mu(y,\theta)=\frac{\operatorname{Im}\{e^{i\theta}R_\mu(ye^{i\theta})\}}{\sin\theta},\qquad C_\mu(\theta)=\sup_{y>0}B_\mu(y,\theta). \tag{6.7.1}$$

The HCM test is

$$X^d \text{ is HCM}\quad\Longleftrightarrow\quad d>2 \text{ and } r\ge C_\mu(\pi/d). \tag{6.7.2}$$

Moreover, $M_\mu=1$ for $\mu\ge 1/2$, $M_\mu>1$ for $\mu<1/2$, and some power exists exactly when $r>M_\mu$. For reference, the product over the positive Bessel zeros gives

$$R_\mu(z)=\sum_{n\ge1}\frac{2z}{z^2+j_{\mu,n}^2},$$

$$B_\mu(y,\theta)=4y\cos\theta\sum_{n\ge1}\frac{j_{\mu,n}^2}{(j_{\mu,n}^2-y^2)^2+4y^2j_{\mu,n}^2\cos^2\theta}. \tag{6.7.3}$$

In particular $B_\mu > 0$ and

$$\cos\theta D_\mu(y) \le B_\mu(y,\theta) \le \sec\theta D_\mu(y). \tag{6.7.4}$$

These are classical product identities followed by the previous Stieltjes boundary test, not new assertions of this section. The underlying Bessel formulas can be found in [6]; the HCM representation and its relation to GGC are in [75].

Here is a concise verification of the criterion used below. Put $p = 1/d$ and $\mathcal{I}_\mu(z) = 2^\mu\Gamma(\mu+1)z^{-\mu}I_\mu(z)$. Up to a monomial, the powered density is $g(x) = e^{-bx^p}\mathcal{I}_\mu(ax^p)$, with $g(0) = 1$. For $0 < p < 1/2$, its negative logarithmic derivative on the principal slit plane is

$$h(z) = pz^{p-1}\{b - aR_\mu(az^p)\}.$$

There are no poles in that plane. A keyhole contour gives

$$h(z) = \int_0^\infty \frac{u(t)}{z+t}\,dt, \qquad u(t) = \frac{pa\sin(\pi p)}{\pi}t^{p-1}\{r - B_\mu(at^p, \pi p)\}.$$

The small and large contour terms vanish by the bounds $h(z) = O(|z|^{p-1})$ at zero and $h(z) = p(b-a)z^{p-1} + O(z^{-1})$ at infinity. The signed integral converges absolutely. The normalized HCM/GGC representation and Stieltjes inversion say exactly that g is HCM if and only if $u \ge 0$, proving the test for these powers. At $p = 1/2$, h has residue -1 at every $-j_{\mu,n}^2/a^2$, incompatible with a positive Stieltjes measure. For $p > 1/2$, g has a zero at $(j_{\mu,n}/a)^{1/p}e^{i\pi/(2p)}$ inside the slit plane, whereas a normalized HCM function is zero-free there. Finally the HCM representation implies $(xg'/g)' < 0$ for this nonconstant g; direct differentiation gives the necessary strict inequalities $r > D_\mu(y)$ for every $y > 0$. When $M_\mu > 1$ is attained they imply $r > M_\mu$; when $M_\mu = 1$ the same strict inequality is already imposed by $b > a$. Conversely (6.7.4) makes every $r > M_\mu$ feasible at a sufficiently small angle. This verifies the background facts without counting them among the new conclusions.

6.7.1 A unique maximizer and its exact curvature

Theorem 6.7.1 (The shape of the real obstruction)**.** *If $\mu \ge 1/2$, D_μ is strictly increasing from zero to one. If $-1 < \mu < 1/2$, it has a unique global maximum, at a point $y_\mu > 0$, and*

$$D_\mu(y_\mu) = M_\mu > 1, \qquad D_\mu''(y_\mu) = -\frac{2(M_\mu^2 - 1)}{y_\mu} < 0. \tag{6.7.5}$$

For $-1/2 \le \mu < 1/2$ this is its only stationary point. For $-1 < \mu < -1/2$ it has exactly one further stationary point, a minimum below one, after which it increases to one.

Proof. Set $\alpha = \mu + 1/2$, $v(y) = R_\mu(y) + \alpha/y$, and $D = D_\mu$. The Bessel recurrence yields

$$R' = 1 - 2\alpha R/y - R^2, \qquad v' = 1 - v^2 + \frac{\alpha(\alpha-1)}{y^2} = \frac{D - v}{y}, \qquad D' = 2(1 - vD). \tag{6.7.6}$$

At a stationary point of v,

$$v'' = -2\alpha(\alpha-1)/y^3.$$

At a stationary point of D, positivity of D, supplied by the product formula, gives $v = 1/D > 0$ and hence

$$D'' = -2Dv' = -2(D^2 - 1)/y. \tag{6.7.7}$$

No stationary point has $D = 1$. Such a point would have $v = 1$ and $v' = 0$, forcing $\alpha(\alpha-1) = 0$ in the Riccati equation; the two exceptional solutions are $v = \tanh y$ for $\alpha = 0$ and $v = \coth y$ for $\alpha = 1$, neither of which equals one at a finite positive point.

The endpoint expansions are

$$D(y) = \frac{y}{\mu+1} + O(y^3) \quad (y \downarrow 0),$$
$$v(y) = 1 + \frac{\alpha(\alpha-1)}{2y^2} + O(y^{-3}),$$
$$D(y) = 1 - \frac{\alpha(\alpha-1)}{2y^2} + O(y^{-3}) \quad (y \to \infty),$$

with differentiated expansions valid as well. If $0 < \alpha < 1$, v' is initially negative and eventually positive, and every zero of v' is a strict upward crossing. Thus v has exactly one minimum. Before it, $D < v$, so every possible stationary point of D would be a minimum below one. There can be none because $D' > 0$ initially. After it, $D > v$, so every stationary point of D is a maximum above one. Since D' is eventually negative, there is exactly one such maximum.

If $\alpha < 0$, v' is initially positive and eventually negative, and every zero is a strict downward crossing. Thus v has one maximum. Before it every stationary point of D is a maximum above one; after it every stationary point is a minimum below one. There is a value $D > 1$: otherwise $[y(1 - R_\mu(y))]' = 1 - D \geq 0$, whereas this expression starts at zero and tends to $\alpha < 0$. Since D is eventually below one and increasing, the stated maximum and minimum both exist and are unique. For $\alpha = 0$, the explicit solution $R = \tanh y$ gives the unique maximum from $y \tanh y = 1$.

If $\alpha > 1$, v' is negative initially, and every possible zero is a downward crossing; hence there is no zero and $v' < 0$ throughout. All possible stationary points of D would be minima below one, which cannot be the first zero of D' because $D' > 0$ initially. Consequently $D' > 0$ throughout. The same conclusion follows from $v = \coth y$ when $\alpha = 1$. The endpoint limits complete all cases. Equation (6.7.7) proves the asserted curvature. □

6.7.2 The critical power has three different asymptotic regimes

Theorem 6.7.2 (The three-regime critical law)**.** *For fixed $\mu > -1$, the following statements hold.*

1. *If $-1 < \mu < 1/2$, then, as $r \downarrow M_\mu$,*

$$d^*_\mu(r) \sim \pi \sqrt{\frac{y_\mu(M_\mu^2 - 1)}{3(r - M_\mu)}}. \tag{6.7.8}$$

2. *At $\mu = 1/2$, as $r \downarrow 1$,*

$$d^*_{1/2}(r) = \log \frac{1}{r-1} + \log 2 - 2 + o(1). \tag{6.7.9}$$

3. *If $\mu > 1/2$, there is a finite $d_0(\mu) > 2$ such that every $d \geq d_0(\mu)$ works simultaneously for every $r > 1$. In particular*

$$\sup_{r>1} d^*_\mu(r) < \infty. \tag{6.7.10}$$

Thus the order $1/2$ separates an inverse-square-root divergence, a logarithmic divergence, and a uniformly bounded power cost.

Proof. For the first case, analytic expansion at $\theta = 0$ on each compact positive y interval gives

$$B_\mu(y, \theta) = D_\mu(y) - \frac{\theta^2}{6}\{3yD'_\mu(y) + y^2 D''_\mu(y)\} + O(\theta^4). \tag{6.7.11}$$

One can obtain this by applying the operator $1 + y\partial_y$ to R_μ three times in the numerator of (6.7.1) and expanding $\sin\theta$. The error and its first two y derivatives are uniform on such compact sets.

When $M_\mu > 1$, all maximizers for sufficiently small θ lie in a fixed compact interval. Indeed, D_μ tends to zero at zero and to one at infinity; (6.7.4) excludes both ends. The unique maximum in Theorem 6.7.1 and its negative second derivative then give, by the implicit function theorem, a unique maximizing point $y(\theta) = y_\mu + O(\theta^2)$. Substitution in (6.7.11) gives

$$C_\mu(\theta) = M_\mu + \frac{y_\mu(M_\mu^2 - 1)}{3}\theta^2 + O(\theta^4). \tag{6.7.12}$$

The coefficient is strictly positive. To justify inversion at the least feasible power, note also that $C_\mu(\theta) > M_\mu$ for each $\theta > 0$: equality would make the normalized non-power factor HCM at tilt $r = M_\mu$, contrary to the strict fixed-tilt obstruction at the attained maximum of D_μ. Continuity of C_μ on compact subintervals of $(0, \pi/2)$ follows from its endpoint limits in y and the locally uniform Bessel asymptotics. Finally (6.7.3), evaluated at $y = j_{\mu,1}$, gives $C_\mu(\theta) \geq 1/(j_{\mu,1}\cos\theta)$. Thus feasible powers must tend to infinity as $r \downarrow M_\mu$; (6.7.2) and (6.7.12) give (6.7.8).

At $\mu = 1/2$, $R_\mu(z) = \coth z - 1/z$. Put $k = \cot\theta$ and $t = 2y\sin\theta$. Direct algebra gives

$$B_{1/2}(y,\theta) - 1 = \frac{\cos t - k\sin t - e^{-kt}}{\cosh(kt) - \cos t}. \tag{6.7.13}$$

For $0 < t \leq \pi$, the numerator is nonpositive: the derivative of $e^{kt}(\cos t - k\sin t)$ is $-(1 + k^2)e^{kt}\sin t \leq 0$, and its value at zero is one. For $t \geq \pi$ the denominator is $\frac{1}{2}e^{kt}(1 + O(e^{-\pi k}))$, uniformly as $k \to \infty$. Consequently the supremum of (6.7.13) equals

$$2\sup_{t\geq\pi} e^{-kt}(\cos t - k\sin t) + O((1+k)e^{-2\pi k}).$$

The first positive maximum of the last expression occurs at $t = \pi + 2\arctan(1/k)$ and has trigonometric factor exactly one. Subsequent positive maxima are smaller by successive factors $e^{-2\pi k}$. It follows that

$$C_{1/2}(\theta) - 1 \sim 2\exp\{-\pi\cot\theta - 2\} \qquad (\theta \downarrow 0). \tag{6.7.14}$$

For every fixed $\theta > 0$ the right side of (6.7.13) is positive somewhere: use sufficiently large t in a positive trigonometric interval. Thus $C_{1/2}(\theta) > 1$. Compactness and the same resonance lower bound as above show that feasible powers tend to infinity when $r \downarrow 1$. Since $\pi\cot(\pi/d) = d + o(1)$, inversion of (6.7.14) proves (6.7.9).

For $\mu > 1/2$, put $a_2 = (4\mu^2 - 1)/8 > 0$. The fixed-order Bessel expansion, uniform in a closed sector $|\arg z| \leq \theta_1 < \pi/2$, gives

$$B_\mu(y,\theta) = 1 - a_2/y^2 + O(y^{-3}) \quad (y \to \infty), \qquad 0 \leq \theta \leq \theta_1. \tag{6.7.15}$$

The value at zero angle is the continuous extension D_μ. Uniformity after division by $\sin\theta$ follows by applying Cauchy's estimates in a slightly wider sector to the remainder; its imaginary part vanishes on the positive real axis and is $O(\theta y^{-3})$ after multiplication by $e^{i\theta}$. Choose Y large enough that the expression is strictly below one for $y \geq Y$ and all these angles. At the other end the power series at zero makes $B_\mu < 1/2$ uniformly for small y and small angles. On the intervening compact interval, Theorem 6.7.1 gives $D_\mu < 1$; continuity therefore gives $B_\mu < 1$ there for all sufficiently small positive θ. We have proved $B_\mu(y,\theta) < 1$ for every finite y at all sufficiently small angles. Its limit as $y \to \infty$ is one, so $C_\mu(\theta) = 1$ at all those angles. The exact test now proves (6.7.10). □

The last part strengthens the previous statement that every fixed admissible tilt can be repaired. Above order one half, the power can be chosen before the tilt is specified. At and below one half such a uniform choice is impossible near the critical tilt.

6.7.3 The two order endpoints of the obstruction

Theorem 6.7.3 (Endpoint asymptotics of the critical tilt)**.** *As $\varepsilon \downarrow 0$,*

$$M_{-1+\varepsilon} \sim \frac{3\sqrt{3}}{8\sqrt{\varepsilon}}, \qquad y_{-1+\varepsilon} \sim \frac{2\sqrt{\varepsilon}}{\sqrt{3}}, \tag{6.7.16}$$

$$M_{1/2-\varepsilon} - 1 \sim \frac{2\varepsilon}{\log^2(1/\varepsilon)}, \qquad y_{1/2-\varepsilon} \sim \frac{1}{2}\log(1/\varepsilon). \tag{6.7.17}$$

Proof. The positive product series gives

$$D_\mu(y) = \sum_{n\geq 1} \frac{4y j_{\mu,n}^2}{(y^2 + j_{\mu,n}^2)^2}.$$

When $\mu = -1+\varepsilon$, the first zero satisfies $j_{\mu,1}^2 = 4\varepsilon + O(\varepsilon^2)$. For completeness, multiply the power series ${}_0F_1(;\varepsilon;-w/4)$ by ε; near $(w,\varepsilon) = (0,0)$ it is $\varepsilon - w/4 + O(w^2)$, so the analytic implicit function theorem gives this small zero. The remaining zeros stay uniformly away from zero and have uniformly linear growth in their index; they converge to the positive zeros of J_1, since $J_{-1} = -J_1$. These facts also follow directly from the standard zero-continuity and large-index formulas in [6, Section 10.21]. Splitting the remaining sum at $j_n = y$ shows that its contribution is $O(1)$ uniformly in $y > 0$ and in small ε. The first summand, with $j = j_{\mu,1}$ and $y = ju$, is

$$\frac{1}{j}\frac{4u}{(1+u^2)^2}.$$

Its unique maximum is $3\sqrt{3}/(4j)$, at $u = 1/\sqrt{3}$. Uniform boundedness of the other terms and the separated strict maximum of this function yield both assertions in (6.7.16).

For the second endpoint, write $L = \log(1/\varepsilon)$. We use the parameter-uniform estimate

$$D_{1/2-\varepsilon}(y) - D_{1/2}(y) = \frac{\varepsilon(1-\varepsilon)}{2y^2} + O(\varepsilon y^{-3}), \qquad y \geq Y, \tag{6.7.18}$$

where Y and the constant are independent of small ε. This follows from the fixed-order expansion for D_μ, locally uniform and analytic in the order near $\mu = 1/2$. More explicitly, after subtracting $1 - (4\mu^2-1)/(8y^2)$ the remainder is $O(y^{-3})$ uniformly on a small complex disk in μ; Cauchy's derivative estimate and the mean value formula give an $O(\varepsilon y^{-3})$ difference. Thus an error independent of ε is not being substituted into this singular limit.

The exact half-order formula gives

$$D_{1/2}(y) - 1 = \coth y - y\,\mathrm{csch}^2 y - 1 = -(4y-2)e^{-2y} + O(ye^{-4y}).$$

At $y = L/2 + 2\log L$, these formulas give

$$D_{1/2-\varepsilon}(y) - 1 \geq (2 - o(1))\varepsilon/L^2.$$

On every fixed compact positive y interval, $D_{1/2} < 1$, so the maximizer tends to infinity. For any fixed $0 < \eta < 1$, the negative exponential term dominates the positive perturbation uniformly when $Y \leq y \leq (1-\eta)L/2$, once Y is sufficiently large and ε sufficiently small. In particular the maximizing point lies beyond $(1-\eta)L/2$. For larger y, (6.7.18) and $D_{1/2} < 1$ imply

$$D_{1/2-\varepsilon}(y) - 1 \leq \frac{\varepsilon}{2y^2}(1 + O(1/y)).$$

This bounds the maximum by $(2+o(1))\varepsilon/[(1-\eta)^2L^2]$. Letting $\eta \downarrow 0$ proves the first assertion of (6.7.17). The same bound excludes $y \geq (1+\eta)L/2$ from being a maximizing point, because it is strictly smaller than the already established lower bound there. The two exclusions prove the location asymptotic. □

6.7.4 A finite moment obstruction for the remaining SD problem

Write $\alpha = \mu + 1/2$, $q = \nu - \mu - 1 > 0$, $A = \alpha + q$, and $M = 2\alpha + q$. Admissibility means $\alpha > -1/2$ and $M > 0$. The signed-spectral construction of [76, Theorem 4.3 and Corollary 4.4] gives a positive even kernel K with

$$-J'(u)/J(u) = \int_0^\infty e^{-ux} K(x)\,dx, \quad J(u) = C\frac{u^q\,{}_2F_1(-q/2, (1-q)/2; \alpha + 1/2; u^{-2})}{(u^2-1)^A},$$

and $e^{-x}K(x) \to A$, $K(0) = M$. The exact SD boundary is $\beta_{\alpha,q} = \sup_{x\geq 0} K'(x)/K(x) \geq 1$. Positive integers and positive-cut intervals $q \in (2m, 2m+1)$ were already proved to have $\beta > 1$. The previous regular-part argument also covered $\alpha \leq 1$. The following criterion is new progress in the remaining negative-cut intervals.

Proposition 6.7.4 (An explicit extension and a hierarchy of certificates)**.** *In a negative-cut interval $q \in (2m+1, 2m+2)$,*

$$1 < \alpha \leq \frac{3 + \sqrt{1+8q}}{2} \quad \Longrightarrow \quad \beta_{\alpha,q} > 1. \tag{6.7.19}$$

More generally, for $\alpha > 1$ in such an interval, define

$$\begin{aligned} h_0 &= \frac{\alpha(\alpha-1)}{2(A-1)}, \\ h_n &= \frac{(n+1)h_{n-1} - 2\sum_{i+j=n-1} h_i h_j - \sum_{i+j=n-2} h_i h_j}{2(A-n-1)}, \qquad n < A-1, \end{aligned} \tag{6.7.20}$$

where an empty sum is zero. If any defined value satisfies $(-1)^n h_n \leq 0$, then $\beta_{\alpha,q} > 1$. More generally, if a Hankel matrix built from $m_n = (-1)^n n! h_n$ within this index range is not positive semidefinite, the same conclusion follows.

Proof. Suppose $\beta = 1$. Then $k(x) = e^{-x}K(x)$ is nonincreasing to A, and $k - A \geq 0$, strictly positive near zero since $k(0) - A = \alpha > 0$. Let

$$H(s) = -J'(1+s)/J(1+s) - A/s = \int_0^\infty e^{-sx}(k(x) - A)\,dx.$$

In a negative-cut interval the signed spectral formula shows $k(x) - A = O(x^{-A})$ at infinity; the finite zero terms and the elementary terms decay exponentially. Hence moments through every integer $n < A - 1$ are finite, and $h_n = H^{(n)}(0)/n!$ satisfy $(-1)^n h_n > 0$. All moment Hankel matrices, both unshifted and shifted when their indices are available, must be positive semidefinite.

The Bessel differential equation, integrated against the defining Laplace density, gives

$$(u^2-1)J'' + (2\nu+1)uJ' + (\nu^2 - \mu^2)J = 0.$$

Putting $-J'/J = A/s + H(s)$ with $u = 1 + s$ yields the identity

$$s(s+2)(H^2 - H') + (2A - 2 - 2s)H + \alpha(1-\alpha) = 0.$$

Expansion through any available finite derivative order gives exactly (6.7.20). A failed sign or moment-matrix test therefore contradicts $\beta = 1$.

In particular $A > 2$ in the stated range, and

$$h_1 = \frac{h_0(1-h_0)}{A-2}.$$

If $h_0 \leq 1$, this is nonnegative, contradicting the strict negative derivative of the Laplace transform of $k - A$ at zero. The inequality $h_0 \leq 1$ is precisely $\alpha(\alpha-1) \leq 2(\alpha+q-1)$, which gives (6.7.19), including equality. □

The hierarchy can go beyond its first-derivative corollary. For example $(\alpha, q) = (5, 3/2)$ is outside (6.7.19), but the exact recurrence gives $h_0 = 20/11$, $h_1 = -40/121$, and $h_2 = -360/1331$. The negative second moment certifies $\beta_{5,3/2} > 1$. This illustrates the same test, not an additional problem closure.

Adversarial scope check. The hierarchy is a sufficient obstruction, not a proved terminating decision procedure. We have not proved that it succeeds for every $\alpha > 1$ and every negative-cut q, and we therefore do not claim universal SD strictness. The three-regime HCM theorem and its endpoint asymptotics are complete within their stated model. Their proof uses uniform estimates before taking a supremum, accounts for the unattained maximum $M_\mu = 1$, and treats $\mu = 1/2$ separately; replacing that case by a power-series approximation would miss the exponential transition.

The new preprint [39] reports a proof of the Ismail–Miller Bessel-ratio ID conjecture. That is a different problem and is not claimed as a closure here. No published priority for our three-regime statement was identified in the primary sources checked; this is a limited literature check, not a guarantee of novelty.

6.8 Analytic populations and iterated critical budget penalties

The preceding endpoint theorem showed that the unrestricted scalar risk is $\log(B)/\sqrt{B}$, even under adaptive replication. We now ask whether analytic regularity of the population removes this critical penalty. It reduces it, but does not in general remove it. A spectral smoothness class leaves a smaller, precisely calculable number of resolution levels. The following results concern new internally formulated minimax problems.

An individual has latent $U \in [0, 1]$, with density w, and observations conditionally uniform on $[0, 2^U]$. A policy may revisit individuals and start new ones using at most B scalar observations. There is no additional replication cap. Loss is expected W_1 for the law of U; by a bi-Lipschitz change of variable the same rates hold for 2^U. Let $\mathcal{W}_\phi(L)$ consist of real, one-periodic densities with

$$w \geq 1/2, \qquad \int_0^1 w = 1, \qquad |\widehat{w}(k)| \leq L e^{-\phi(|k|)} \quad (k \in \mathbb{Z} \setminus \{0\}),$$

where $\widehat{w}(k) = \int_0^1 e^{-2\pi i k u} w(u)\, du$. The periodic restriction is part of the class, not an identification of the two endpoints in the transportation metric. All constants L, τ, ν below are fixed positive numbers, independent of B.

Theorem 6.8.1 (A double logarithm for every fixed stretched-exponential spectrum)**.** *For $\phi(k) = \tau k^\nu$, $\nu > 0$, the adaptive and predetermined minimax risks on $\mathcal{W}_\phi(L)$ satisfy*

$$R_B^{\mathrm{ad}} \asymp R_B^{\mathrm{fix}} \asymp \frac{\log\log B}{\sqrt{B}}, \qquad B \longrightarrow \infty.$$

For $\nu = 1$ this is an analytic class with a fixed strip-width envelope. For $\nu > 1$ all its densities extend to entire periodic functions. Despite that stronger regularity, the same double logarithm remains.

Theorem 6.8.2 (A triple logarithm for a double-exponential spectrum)**.** *For $\phi(k) = \tau e^{\nu k}$ the corresponding risks satisfy*

$$R_B^{\mathrm{ad}} \asymp R_B^{\mathrm{fix}} \asymp \frac{\log\log\log B}{\sqrt{B}}, \qquad B \longrightarrow \infty.$$

The equivalences are asymptotic; one may replace the iterated logarithms by positive shifted versions to state bounds for all $B \geq 2$.

These statements count the distinct regularity classes as two targets, not every possible value of ν as a separate solved problem. Classical scale mixtures of uniforms and fixed-noise deconvolution are studied in [37, 72]; the experiment here instead optimizes repeated observations under a total budget and allows arbitrary adaptive selection of the next individual.

6.8.1 A mean correction avoids circular transportation

Lemma 6.8.3 (A spectral upper bound on an interval)**.** *For any $w, v \in \mathcal{W}_\phi(L)$ and $K \ge 1$,*

$$W_1(w(u)\,du, v(u)\,du) \le |\mu(w) - \mu(v)| + C\left(\sum_{1\le|k|\le K} \frac{|\widehat{w}(k) - \widehat{v}(k)|^2}{k^2}\right)^{1/2} + CLe^{-\phi(K)},$$

where $\mu(w) = \int_0^1 uw(u)\,du$ and ϕ is increasing.

Proof. Put $F(x) = \int_0^x (w - v)$ and $\bar{F} = \int_0^1 F$. The ordinary interval Wasserstein identity is $W_1 = \int_0^1 |F|$. Integration by parts gives $\bar{F} = \mu(v) - \mu(w)$. For $k \ne 0$, the Fourier coefficient of F is $(\widehat{w}(k) - \widehat{v}(k))/(2\pi ik)$, because $F(0) = F(1) = 0$. Hence $W_1 \le |\bar{F}| + \|F - \bar{F}\|_2$, and Parseval gives the low-frequency term. The squared remaining tail is bounded by $CL^2 e^{-2\phi(K)} \sum_{k>K} k^{-2}$, which proves the assertion. In particular a periodic Fourier metric alone is not substituted for Euclidean transportation; the separately estimated mean is essential. □

Write $a = \log 2$. With r observations from an individual, the log maximum has distribution $Y_r = U - \mathrm{Exp}(ar)$. Thus

$$A_{r,k}(Y_r) = e^{-2\pi ikY_r}\left(1 - \frac{2\pi ik}{ar}\right)$$

is unbiased for $\widehat{w}(k)$, and is uniformly bounded when $r \asymp k$. The statistic $Y_1 + 1/a$ is unbiased for $\mu(w)$, with uniformly bounded variance. Spend one quarter of the budget on this mean statistic. Partition the frequencies up to a dyadic K into $J = O(\log(eK))$ bands. For the band ending at k, use $r = k$ observations on each of $n_k = \lceil cB/(Jk)\rceil$ fresh individuals. Choose the fixed $c > 0$ small enough. The total band cost is at most $B/2 + CK$, and

$$\sum_{1\le|l|\le K} \frac{\mathbb{E}|\widehat{m}_l - \widehat{w}(l)|^2}{l^2} \le C \sum_{k \text{ dyadic}\le K} \frac{1}{kn_k} \le CJ^2/B.$$

The extra rounding cost is harmless for the polylogarithmic cutoffs used below. Project the empirical mean and Fourier vector onto the set supplied by $\mathcal{W}_\phi(L)$, using the weighted Euclidean norm appearing in Lemma 6.8.3. Fourier summability makes this density class compact in the uniform norm. A measurable approximate minimizer, or a countable dense approximation, therefore supplies a probability estimator. The projection increases its discrepancy from the truth by at most a factor two. We have proved the upper bound

$$R_B^{\mathrm{fix}} \le C\left\{e^{-\phi(K)} + \frac{\log(eK)}{\sqrt{B}}\right\}, \qquad K \le c'B. \tag{6.8.1}$$

6.8.2 A uniform information bound after every history

For completeness we specify the adaptive lower-bound input, which also underlies the unrestricted critical theorem in the preceding manuscript [84]. For $K \ge 1$ put

$$w_\beta(u) = 1 + \sqrt{2}\sum_{k=1}^{K}\{\beta_{k,c}\cos(2\pi ku) + \beta_{k,s}\sin(2\pi ku)\}, \qquad |\beta_{k,*}| \le b_K := cK^{-2}e^{-\phi(K)}.$$

Choose c depending only on L and small enough. These densities belong to $\mathcal{W}_\phi(L)$, are between $1/2$ and $3/2$, and have a uniformly bounded Lipschitz constant. These bounds hold for every K.

Lemma 6.8.4 (Adaptive trace bound for Fourier perturbations)**.** *For every adaptive B-observation policy, the Fisher information trace in the $2K$ parameters β is at most CB, uniformly in K and β in the displayed cube.*

Proof. Under the baseline density one, the posterior of an individual after r observations is $q(u) \propto e^{-aru}\mathbf{1}_{[b,1]}(u)$ for some $0 \le b < 1$. This remains the posterior conditional on the full adaptive history, since the known selection rule contributes no parameter-dependent factor beyond the observed values. Denote by q_Y the posterior after the next observation. Two elementary truncated-exponential identities are

$$\mathbb{E}_0\|q_Y - q\|_2^2 = a/2, \qquad \|q\|_2^2 \operatorname{Var}_q(U) \le C.$$

To check the first, put $\lambda = ar$, $D = 1-b$, $\mu = \lambda + a$, and $Z_\lambda = (1-e^{-\lambda D})/\lambda$, with $Z_0 = D$. The current squared norm is $(1+e^{-\lambda D})/(2Z_\lambda)$. The next observation has a case corresponding to no increase in b, plus a density for a new record $b+v$, $0 < v < D$. Their respective contributions to the expected next squared norm give

$$\begin{aligned}\mathbb{E}_0\|q_Y\|_2^2 &= \frac{1+e^{-\mu D}}{2Z_\lambda} + \frac{a}{2Z_\lambda}\int_0^D e^{-\lambda v}\{1+e^{-\mu(D-v)}\}\,dv \\ &= \frac{a}{2} + \frac{1+e^{-\lambda D}}{2Z_\lambda}.\end{aligned}$$

Now $\mathbb{E}_0 q_Y = q$ proves the identity. The formulas extend by continuity to $\lambda = 0$. For the second identity, the squared norm is $O(\lambda + D^{-1})$ and the variance is $O(\min(D^2, \lambda^{-2}))$, with the evident convention at zero; use $D \le 1$.

Put $z = \int w_\beta q$ and $z_Y = \int w_\beta q_Y$. Both are uniformly bounded above and away from zero. The parameter score of the conditional next-observation density has coordinates

$$\int \varphi_j(u)\,\{q_Y(u)/z_Y - q(u)/z\}\,du,$$

where φ_j runs over the orthonormal sine and cosine functions. The true conditional law has density z_Y/z relative to the baseline conditional law. Bessel's inequality and boundedness of these ratios reduce the information trace to

$$C\mathbb{E}_0\|q_Y - q\|_2^2 + C\|q\|_2^2\mathbb{E}_0|z_Y - z|^2.$$

The second term is bounded because

$$\mathbb{E}_0|z_Y - z|^2 \le C\mathbb{E}_0 W_1(q_Y, q)^2 \le 2C\operatorname{Var}_q(U).$$

For the last inequality couple independently a draw from q_Y with a draw from q, and use $\mathbb{E}_0 q_Y = q$. Every next-query information trace is therefore bounded by a constant. Conditional score increments are orthogonal martingale differences, so their traces add through at most B queries. Independent policy randomization has zero score. This proves the lemma. □

Use independent cosine-squared priors on $[-b_K, b_K]$. Their total information is CKb_K^{-2}, so the joint prior-and-data Fisher matrix satisfies

$$\operatorname{tr} J \le C\{B + K^5 e^{2\phi(K)}\}.$$

We recall explicitly the absolute-loss information inequality. For any fixed vector v and any data estimator T,

$$\mathbb{E}|T - v^T\beta| \geq \tfrac{1}{4}\sqrt{v^T J^{-1} v}. \tag{6.8.2}$$

Indeed, let $p(\beta, y)$ be the joint parameter-data density and put $p_h(\beta, y) = p(\beta - hz, y)$. The total variation between these two distributions is at most $|h|\sqrt{z^T J z}/2$, by the fundamental theorem of calculus and Cauchy–Schwarz. Translation is valid after extending the density by zero; the prior vanishes quadratically at its boundary. The target under p is $v^T\beta$, and the translated target under p_h is $v^T(\beta - hz)$. The risks for these respective targets are equal by change of variables. On the common part of the two densities, the targets differ by $hv^T z$, so the two losses sum to at least that absolute difference. Choosing $|h| = (z^T J z)^{-1/2}$ gives the bound with $|v^T z|$ in place of the quadratic expression. Optimizing z proves (6.8.2).

For the centered CDF the vector multiplying β is

$$v_{k,c}(x) = \frac{\sqrt{2}\sin(2\pi k x)}{2\pi k}, \qquad v_{k,s}(x) = -\frac{\sqrt{2}\cos(2\pi k x)}{2\pi k}.$$

Centering a CDF error increases its integrated absolute value by at most a factor two. Applying (6.8.2) at each x therefore gives a risk lower bound by $c\int_0^1 \sqrt{v(x)^T J^{-1} v(x)}\,dx$. For fixed v, this is convex in J, since

$$\sqrt{v^T J^{-1} v} = \sup_{z \neq 0} \frac{|v^T z|}{\sqrt{z^T J z}}.$$

Average J over the simultaneous frequency rotations corresponding to translation of x. This kills cross-frequency terms and gives blocks $j_k I_2$; the integral is unchanged by each rotation and can only decrease under averaging. The resulting integrand is independent of x, and Cauchy–Schwarz gives

$$\sum_{k=1}^{K} \frac{1}{k^2 j_k} \geq \frac{(\sum_{k=1}^{K} k^{-1})^2}{\sum_{k=1}^{K} j_k}.$$

Consequently every adaptive procedure has risk at least

$$R_B^{\mathrm{ad}} \geq \frac{c\log(eK)}{\sqrt{B + K^5 e^{2\phi(K)}}}. \tag{6.8.3}$$

Proof of Theorems 6.8.1 and 6.8.2. For $\phi(k) = \tau k^\nu$, choose an upper-bound cutoff within a fixed factor of $(\log B/\tau)^{1/\nu}$, large enough that $\phi(K) \geq \log B$. Equation (6.8.1) is then $O(\log\log B/\sqrt{B})$. For the lower bound choose $K = \lfloor(\log B/(4\tau))^{1/\nu}\rfloor$. Then $K^5 e^{2\phi(K)} \leq B$ for all sufficiently large B, and $\log K \asymp \log\log B$. Equation (6.8.3) matches the upper bound. Taking a nearby dyadic integer for the upper bound changes only constants depending on τ, ν.

For $\phi(k) = \tau e^{\nu k}$ choose an upper cutoff at least $\nu^{-1}\log(\log B/\tau)$, rounding upward. The bias is at most B^{-1} and $\log K \asymp \log\log\log B$. For the lower bound round down $K = \nu^{-1}\log(\log B/(4\tau))$. Again $K^5 e^{2\phi(K)} \leq B$ eventually, and the same third iterated logarithm appears in (6.8.3). Dyadic rounding in the upper bound preserves $K \asymp \log\log B$ and only improves its bias. □

Interpretation and scope. The penalty is the number of statistically relevant dyadic resolution bands, divided by $\sqrt{B}$. Finitely many derivatives leave a polynomial frequency cutoff and the old $\log B$ penalty. A stretched-exponential spectral envelope leaves a polylogarithmic cutoff and a $\log\log B$ penalty. A double-exponential envelope leaves a logarithmic-logarithmic cutoff and a $\log\log\log B$ penalty. Only the last two assertions are new theorems of this section; the first is an explanation of the earlier critical mechanism, not an additional claimed closure.

The results use a known, fixed periodic regularity envelope and do not prove adaptation to unknown smoothness, a capped version, or a uniform statement over all analytic densities without a common envelope. They also do not imply that every individual analytic density incurs the minimax penalty. The lower bound ranges over a fixed infinite-dimensional class whose least-favourable elements depend on the budget.

6.9 What the new results leave open

The preceding proofs close the sixteen scoped targets in the register, not sixteen independent established conjectures. Several consequences share one mechanism. This distinction is especially important for the derivative tower and the equal-replication budget: a family of parameter values is not an infinite collection of separate breakthroughs.

1. **Arithmetic precision.** The endpoint count does not give an asymptotic density constant, a variance law, or an entropy-sharp quantitative return time for growing collections of exact prime phases. Finite-dimensional Kronecker density suffices for the present lower bound and supplies none of those quantitative statements.

2. **The coefficient conjecture.** The full statistical MLE curve bypasses the constant-one Chebyshev–Bernstein bound. It does not prove that bound. Optimal leading risk constants and sharp predictive Hellinger logarithms are also not determined here.

3. **Unrestricted exact information.** Three-query optimality is proved for polynomial or convex analytic classes on all of $\mathbb{R}^3$. Two arbitrary continuous queries, or two nonconvex real-analytic queries, remain outside that lower-bound argument. The global unbounded domain is part of the theorem.

4. **The full design-tail classification.** The matrix converse includes polynomially sparse and nearly harmonic design sets, but does not classify every irregular sequence with $\sum_j n_j^{-1} < \infty$ against every possible covariance-tail envelope. Population nonidentifiability is also distinct from a sharp finite-sample rate.

5. **Finite-height spectral observations.** The spectral branch uses ideal nonlinear transition values. It gives no theorem turning finitely many arithmetic zeros below a prescribed height into those values with a controlled error. Its blind consistency exponent is not the restricted m-atomic minimax exponent.

6. **Complete SD strictness.** The finite moment hierarchy provides new sufficient certificates in the remaining negative-cut region. We have not proved that it must terminate successfully for every admissible pair of parameters.

These are different missing inputs. In particular the arithmetic and spectral results cannot be composed into a finite-height prime recovery theorem without a new quantitative link between the observation models. The matrix-tail construction cannot establish a positive identification theorem in a class where it already furnishes indistinguishable laws. The MLE result is a statistical order theorem, not an assertion about unlimited computational accuracy or every possible optimization routine.

6.9.1 Research claims suitable for separate external scrutiny

The arithmetic endpoint proof and the complete MLE proof can be read independently of the longer research program. The first uses an analytic lift and classical zeta second moments; the second includes its approximation, entropy, transport, and matching lower-bound arguments.

These are the first two branches for which outside experts could assess both correctness and novelty without accepting earlier unpublished theorems. The matrix converse supplies a separate construction whose crucial contour step is reduced to one complex trace coordinate.

The remaining branches supply exact model classifications and sharpened boundaries. Their potential value depends on the observation model and on how they compare with the full literature, not on the number of theorem labels. No result in this manuscript proves the Riemann hypothesis, a general Euler-phase recovery claim from a finite zero set, or universal SD strictness. The appropriate next mathematical step is independent expert review of the strongest complete arguments and their specific prior-art claims.

Part VII

Exact Recovery Laws and a Coefficient Conjecture

Endpoint Zero Densities, Robust Populations, and Structural Obstructions

Source manuscript: 16 October 2026.

Source abstract. We continue a program on observation and recovery with sixteen precisely scoped targets. A Hahn-polynomial inequality, an optimized complex contour, and an exact rational finite certificate prove the constant-one Chebyshev–Bernstein coefficient conjecture posed by Vinayak, Kong, Valiant, and Kakade. For binomial populations we determine the full total-variation inverse modulus and the minimax curve under an unknown Huber contamination proportion. For the actual prime zeta function and each fixed derivative, we strengthen linear-order counting in $\Re s > 1$ to $c_m T + o(T)$, control limiting boundary mass, and identify every fixed-window counting moment. A definable-game argument proves that fixed-complexity adaptive information cannot compress intrinsic dimension; a separate geometric formula identifies convex-query noise floors through the Jung constant and gives an exact Euclidean formula at every noise level. Further classifications concern arbitrary sparse scalar Gaussian designs, fixed-shape stable mixtures, nonlinear spectral order detection, finite-height zero observations, gamma deconvolution and McKay self-decomposability, a critical HCM power cost, and complete replication-cap curves for analytic populations. Dependencies on earlier unpublished manuscripts and classical results are explicit. The sixteen targets are not sixteen named historical conjectures: they include a published coefficient conjecture, internal gap closures, and newly formulated transfer problems. Proofs and an executable exact certificate support the research claims; external refereeing and a complete priority assessment remain outstanding.

7.1 Scope, target ledger, and the change from the preceding work

The starting sources are the 223-page integrated manuscript *Observation and Recovery in Arithmetic and Probability* and its subsequent papers, most recently the 48-page *Boundary Laws for Observation and Recovery* [1, 91]. This paper supplies new continuations. Earlier linear-order prime-zeta counting, clean binomial MLE optimality, uncapped analytic budget rates, and signed Bessel kernel identities are not counted again as new achievements. These earlier manuscripts are unpublished inputs, not external confirmations of their claims.

The most directly traceable external target here is

$$T_m(2x-1) = \sum_{j=0}^{t} C(t,m,j)\binom{t}{j} x^j (1-x)^{t-j}, \qquad |C(t,m,j)| \le e^{m^2/t}.$$

The exact inequality is explicitly conjectured in the 2019 source [13, Supplement, Remark 4.1, equation (23)] and is still discussed as a conjecture in the May 2026 version of [14, Section 6.2]. We prove it by joining a classical Hahn bound to a sharper contour exponent. An analytic overlap covers $t \ge 512$; a short verifier proves all remaining cases using rational arithmetic. It is included verbatim in the single source file. The historical Hahn bound is credited to Dette, not presented as our new inequality. The searches performed for this draft do not constitute an exhaustive priority determination.

The other main changes have distinct meanings. Exact prime-zeta density replaces a pair of order bounds by a limiting law and local moment convergence. Robust statistics asks what

the inverse experiment can retain when the reported counts are contaminated, even when the contamination rate is unknown. The definable-information theorem shows that the complexity of allowed queries is essential to dimension compression. The McKay argument identifies an explicit signed gamma remainder, whose forced negative interval closes a strictness gap left by finite-moment tests. These are different conclusions linked by changes of representation, not consequences of one universal theorem.

7.1.1 Sixteen targets, with dependent conclusions grouped

In the table, "published" refers to an explicitly located conjecture; "gap" means a question left by our earlier manuscripts; and "transfer" means a new problem formulated here. These labels are not claims of publication priority. All entries are research theorems with proofs below, subject to their stated hypotheses and inherited inputs.

No.	Status of target	Result and exact scope	Location
1	Published	Constant-one Chebyshev–Bernstein coefficient bound for every $0 \le m \le t$ and every coefficient.	Thm. 7.2.1
2	Transfer	Complete TV inverse modulus for arbitrary binomial mixing laws.	Thm. 7.2.3
3	Transfer	Full Huber minimax curve, attained without knowing the contamination proportion.	Thm. 7.2.5
4	Gap / strengthening	Exact endpoint density $N_m(T) = c_m T + o(T)$ for actual prime zeta and its fixed derivatives; linear limiting boundary-strip control.	Thms. 7.3.1, 7.3.2
5	Transfer	All positive moments of fixed-window arithmetic zero counts converge to random-prime moments.	Thm. 7.3.1
6	Transfer	Fixed definable query families cannot recover below intrinsic dimension; fixed-degree polynomials need exactly m queries on an m-ball.	Thm. 7.4.1
7	Transfer	Exact convex-body noise-radius formula, Jung coefficient, and complete finite-noise Euclidean formula.	Thms. 7.4.6, 7.4.8
8	Gap	Scalar Gaussian identification classified jointly by every terminal design and finite iterated-log tail class.	Thm. 7.5.1
9	Gap	Fixed non-Gaussian stable shape $\alpha \neq 1$: necessity as well as sufficiency of $\sum n^{-\lvert 1-\alpha\rvert} = \infty$.	Thm. 7.6.1
10	Gap	Fixed skew index-one stable shape: exact criterion $\sum (1+\log^2 n)^{-1} = \infty$.	Thm. 7.6.1
11	Transfer	Sharp weak-mass/separation law for nonlinear spectral model-order detection.	Thm. 7.7.1
12	Gap / transfer	Quantitative finite-height zero-to-threshold bridge under an explicit Diophantine recurrence condition.	Thm. 7.7.4
13	Gap / transfer	Universal strict SD boundary for excess-power McKay laws, via an exact gamma-factor obstruction.	Thms. 7.8.2, 7.8.3
14	Gap	Exact logarithmic critical cost of a tilt-uniform HCM power above Bessel order 1/2.	Thm. 7.8.4
15	Transfer	Full adaptive replication-cap curve for stretched-exponential Fourier envelopes.	Thm. 7.9.1
16	Transfer	Full adaptive replication-cap curve for double-exponential Fourier envelopes.	Thm. 7.9.1

Items 4–5 use one common endpoint method; items 9–10 concern different stable sampling geometries; items 15–16 specialize one cap theorem to two regularity classes. We do not

count every derivative, every parameter value, the robust budget corollary, algebraicity of a fixed-degree error, the Jung constant, or gamma extraction as extra independent closures.

7.1.2 The six principal proof transfers

1. A coefficient becomes a discrete orthogonal polynomial. Two estimates that are individually insufficient cover complementary degree ranges, leaving a finite exact certificate.
2. A single inverse norm becomes two predictive norms: inexpensive transportation controls low frequencies, while total variation controls a small-amplitude high-frequency remainder.
3. Arithmetic vertical translates become random analytic functions only after a joint weak-convergence argument; logarithmic Jensen bounds supply the uniform integrability needed for zero moments.
4. An unrestricted adaptive rule becomes a definable finite game. Definable choice exposes a dimension obstruction that is invisible if the decision rules are considered one at a time.
5. A putative positive convolution factor becomes an explicit signed inverse Laplace transform. A differential identity forces a negative interval, excluding every nonnegative remainder.
6. A replication cap becomes a conditional information constraint in a stronger Poisson experiment. Boundary-vanishing perturbations keep that constraint uniform even after exceptional histories.

Conventions and verification status. Each section defines its own experiment and symbols; no unstated identification of observation models is intended. Constants in $\asymp$ may depend on explicitly fixed class parameters. The finite-certificate code was executed using exact rational arithmetic. Separate automated proof readings examined each section, including the endpoint uniform-integrability argument, the definable game, the signed gamma inverse, and the adaptive information bound. Such readings are useful error checks, not formal proof verification or independent human refereeing. No claim about acceptance by a journal, the Riemann hypothesis, or a general solution of famous conjectures follows from this manuscript.

7.2 The constant-one coefficient conjecture and robust population recovery

This section has two independent contributions. First, a classical Hahn polynomial inequality, combined with a contour estimate and a small exact certificate, proves the coefficient conjecture in Remark 4.1 of the supplement to Vinayak–Kong–Valiant–Kakade [13]. Second, the two-scale inverse method from the preceding manuscript yields the complete contamination curve for unrestricted binomial populations, with an estimator that does not know the contamination proportion. The all-replication optimality of the uncontaminated MLE is a prior result [91], not a new result of this section.

7.2.1 A constant-one Chebyshev–Bernstein bound

Write $B_j^t(x) = \binom{t}{j} x^j (1-x)^{t-j}$ and define $C(t,m,j)$ by

$$T_m(2x-1) = \sum_{j=0}^{t} C(t,m,j) B_j^t(x), \qquad 0 \le m \le t.$$

Theorem 7.2.1 (The coefficient conjecture)**.** *For all integers $t \geq 1$, $0 \leq m \leq t$, and $0 \leq j \leq t$,*

$$|C(t, m, j)| \leq \exp(m^2/t). \tag{7.2.1}$$

The proof below is a finite-certificate proof: its only computational part uses exact rational arithmetic for $t < 512$. The complete verifier is included in the text.

This is the precise constant-one assertion posed in [13, Supplement, Remark 4.1, equation (23)], rather than a bound with an unspecified multiplicative constant. The distinction was retained in the subsequent discussion [14, Section 6.2] and in our preceding manuscript. No optimality claim is made for the exponent constant one.

Lemma 7.2.2 (Three complementary coefficient bounds)**.** *For $1 \leq m \leq t$,*

$$|C(t, m, j)| \leq 2^m, \tag{7.2.2}$$

$$|C(t, m, j)| \leq (t+1)\exp\{tH(m/t)\} \leq (t+1)\exp\{(\log 2)m^2/t\}, \tag{7.2.3}$$

where $H(a) = \frac{1}{2}[(1+a)\log(1+a) + (1-a)\log(1-a)]$ and $0 \log 0 = 0$. For $m < t$ there is also the bound

$$|C(t, m, j)| \leq D(t, m) := \prod_{l=1}^{m-1} \frac{t+l}{t-l}. \tag{7.2.4}$$

In particular, $D(t, m) \leq e^{m^2/t}$ whenever

$$(m+6)(m-1)^2 \leq 6t^2. \tag{7.2.5}$$

Proof. At degree m the coefficients are $(-1)^{m-i}\binom{2m}{2i}/\binom{m}{i}$. The successive absolute-value ratio is $(2m-2i-1)/(2i+1)$, so their maximum occurs at a middle index. Vandermonde's identity bounds the numerator at that index by $2^m \max_i \binom{m}{i}$, proving (7.2.2) at degree m. Degree elevation replaces coefficients by convex combinations and proves the bound at every $t \geq m$.

For (7.2.3), put

$$F(z) = (1+z)^t T_m\left(\frac{z-1}{z+1}\right) = \sum_{j=0}^{t} \binom{t}{j} C(t, m, j) z^j.$$

Endpoint coefficients have absolute value one. For $0 < j < t$, set $r = j/(t-j)$ and, on $z = re^{i\theta}$, set $u = 2\sqrt{r}\,|\sin(\theta/2)|/(1+r)$. The larger inverse Joukowski modulus of $(z-1)/(z+1)$ is $\sqrt{(1+u)/(1-u)}$. Consequently

$$\frac{|F(re^{i\theta})|}{(1+r)^t} \leq (1-u^2)^{t/2}\left(\frac{1+u}{1-u}\right)^{m/2} \leq \exp\{tH(m/t)\}.$$

The last maximum is attained at $u = m/t$; limiting values handle $u = 1$. Cauchy's formula and $\binom{t}{j} r^j/(1+r)^t \geq 1/(t+1)$ give the first inequality in (7.2.3). This probability is a mode of a Binomial$(t, j/t)$ law, which proves the stated lower bound. Finally

$$H(a) = \sum_{k \geq 1} \frac{a^{2k}}{2k(2k-1)} \leq a^2 \sum_{k \geq 1} \frac{1}{2k(2k-1)} = a^2 \log 2.$$

To apply the older bound (7.2.4), combine the hypergeometric formula with the identity

$$x^r = \sum_j (j)_{\underline{r}} B_j^t(x)/(t)_{\underline{r}}.$$

This gives

$$C(t,m,j)=(-1)^m\,{}_3F_2\left(\begin{matrix}-m,m,-j\\1/2,-t\end{matrix};1\right)=(-1)^m Q_m(j;-1/2,-1/2,t).$$

Dette's Hahn bound [92, Corollary 3.6(b)] is exactly (7.2.4). Its stipulated range $m<t$ is respected here. Thus this inequality itself is prior art. For the final assertion, using the positive series for arctanh gives

$$\begin{aligned}\log D(t,m)&=2\sum_{l=1}^{m-1}\operatorname{arctanh}(l/t)\\&\le\frac{m(m-1)}{t}+\frac{m^2(m-1)^2}{6t^3\{1-(m-1)^2/t^2\}}.\end{aligned}$$

The second term is at most m/t precisely under (7.2.5). □

Proof of Theorem 7.2.1. The case $m=0$ is immediate. Suppose first $t\ge 512$. If $m^2/t\ge\frac{10}{3}\log(t+1)$, the contour bound proves the assertion because $\log 2<7/10$. Otherwise,

$$m^2<\frac{10}{3}t\log(t+1)\le\frac{25}{9}t^{4/3},\qquad m<\frac{5}{3}t^{2/3}.$$

Here $\log(t+1)\le(5/6)t^{1/3}$ for $t\ge512$: at 512 it follows from $513<729$, $\log 3<11/10$, and $33/5<20/3$; the difference $(5/6)t^{1/3}-\log(t+1)$ has positive derivative thereafter. In particular $m<t$, and

$$(m+6)(m-1)^2\le m^3+6m^2\le\left(\frac{125}{27}+\frac{25}{96}\right)t^2<6t^2.$$

The last estimate uses $t^{2/3}\ge64$. The Hahn bound therefore applies. The elementary logarithmic inequalities just used can, for example, be verified by positive Taylor lower bounds for $e^{7/10}$ and $e^{11/10}$.

For $1\le t<512$, the following verifier covers every pair (t,m). Its first branch uses (7.2.5); its second uses $2^m\le e^{7m/10}\le e^{m^2/t}$; its third uses the contour bound with $\log 2<7/10$. Every other pair is checked with the exact identity

$$C(t,m,j)=\frac{1}{\binom{t}{j}}\sum_{i=\max(0,j-t+m)}^{\min(m,j)}(-1)^{m-i}\binom{2m}{2i}\binom{t-m}{j-i}.\tag{7.2.6}$$

The function `exp_lower` returns a rational number no larger than e^x for $x\ge0$, so each successful assertion is a rigorous inequality in the required direction. There are no floating-point comparisons. The complete output is

$$(34784,39518,56329,185),\qquad 12734\text{ direct coefficient comparisons},\quad \max t_{\mathrm{direct}}=145.$$

The four counts sum to $511\cdot512/2$, as required. □

```
from fractions import Fraction
from math import comb

def exp_lower(x, degree=32):
    term = total = Fraction(1)
    for k in range(1, degree + 1):
        term *= x / k
        total += term
    return total

counts = [0, 0, 0, 0]
comparisons = largest_direct = 0
for t in range(1, 512):
    for m in range(1, t + 1):
        if m < t and (m+6)*(m-1)**2 <= 6*t*t:
            counts[0] += 1
            continue
        if 10*m >= 7*t:
            counts[1] += 1
            continue
        if exp_lower(Fraction(3*m*m, 10*t)) >= t+1:
            counts[2] += 1
            continue
        lower = exp_lower(Fraction(m*m, t))
        for j in range(t + 1):
            numerator = sum(
                (-1)**(m-i)*comb(2*m, 2*i)*comb(t-m, j-i)
                for i in range(max(0,j-t+m), min(m,j)+1)
            )
            assert Fraction(abs(numerator), comb(t,j)) <= lower
            comparisons += 1
        counts[3] += 1
        largest_direct = max(largest_direct, t)
print(counts, comparisons, largest_direct)
assert sum(counts) == 511*512//2
```

7.2.2 A sharp total-variation inverse modulus

For $G \in \mathcal{P}([0,1])$, write $p_G(j) = \int B_j^t \, dG$. Total variation is $\|p-q\|_{\mathrm{TV}} = \frac{1}{2}\sum_j |p_j - q_j|$; $W_{1,t}$ uses cost $|j-k|/t$ on the count alphabet. Define

$$\omega_t^{\mathrm{TV}}(\eta) = \sup\{W_1(G,H) : \|p_G - p_H\|_{\mathrm{TV}} \le \eta\}.$$

Throughout, $\log(e/0) = +\infty$ and $1/\sqrt{+\infty} = 0$.

Theorem 7.2.3 (The complete TV inverse curve)**.** *Uniformly for integers $t \ge 1$ and $0 \le \eta \le 1/4$,*

$$\omega_t^{\mathrm{TV}}(\eta) \asymp \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(e/\eta)}} + \eta\right\}. \tag{7.2.7}$$

All constants are absolute. In particular, infinite sample size does not remove an arbitrary predictive contamination of size η: its inverse cost is generally larger than η.

We use the following prior approximation ingredients, recording their proofs to specify the uniform constants. They do not require the constant-one theorem above.

Lemma 7.2.4 (Two-scale inversion in two predictive metrics)**.** *For $1 \le s \le k \le t/2$,*

$$W_1(G,H) \le \frac{C}{k} + Ce^{4s^2/t} W_{1,t}(p_G, p_H) + \frac{Ce^{2k^2/t}}{s}\|p_G - p_H\|_{\mathrm{TV}}. \tag{7.2.8}$$

Proof. For every 1-Lipschitz function f, there is a degree-k polynomial P_k such that $\|P_k - f\|_\infty \le C/k$ and $\|P_k'\|_\infty \le 2$. One elementary construction reflects f to a period-two Lipschitz function, convolves it with a positive Jackson kernel of order a, and replaces its $O(a)$ Fourier exponentials by their Taylor polynomials of degree $100a$. The Jackson first-moment bound is $O(1/a)$, convolution does not increase the Lipschitz constant, and the Taylor error in both the function and its derivative is exponentially small. Taking $a = \lfloor k/100 \rfloor$ and treating bounded k by a constant polynomial gives the assertion.

We also need a bound for arbitrary bounded polynomials, not merely a single Chebyshev polynomial. If $P = \sum b_j B_j^t$ has degree at most $k \le t/2$, then

$$\max_j |b_j| \le C e^{2k^2/t} \|P\|_\infty. \tag{7.2.9}$$

For completeness, Bernstein's complex polynomial bound gives $|P((2+w+w^{-1})/4)| \le \|P\|_\infty |w|^k$ for $|w| \ge 1$. Use the same coefficient contour as above, with $v = r/(1+r)^2$ and $u = 2\sqrt{v}\,|\sin(\theta/2)|$. Its normalized integrand is at most

$$(1-u^2)^{(t-k)/2}(1+u)^k \le e^{2k^2/t} e^{-tu^2/8}.$$

The average of the last factor is $O((1+tv)^{-1/2})$, whereas Stirling's factorial inequalities give $\binom{t}{j}(j/t)^j(1-j/t)^{t-j} \ge c(1+tv)^{-1/2}$. This proves (7.2.9), including endpoint coefficients by direct evaluation. Differentiating the Bernstein representation shows that the coefficient vector of P_s has normalized-grid Lipschitz constant at most $Ce^{4s^2/t}$. The polynomial $P_k - P_s$ has supremum norm $O(1/s)$, so its coefficients are bounded by $Ce^{2k^2/t}/s$. Test the first part by $W_{1,t}$, the second by TV, and take the Kantorovich dual supremum over f. □

Proof of Theorem 7.2.3. For the upper bound, $W_{1,t} \le \|\cdot\|_{\mathrm{TV}}$ because the normalized alphabet has diameter one. For sufficiently large t, take

$$k = \left\lfloor \min\{t/2, \sqrt{at\log(e/\eta)}\} \right\rfloor, \qquad s = \lfloor \sqrt{t}/16 \rfloor,$$

where $a = 1/64$; at $\eta = 0$ take $k = \lfloor t/2 \rfloor$. Bounded t are absorbed by the loss bound one. These choices have $1 \le s \le k$, $s \asymp \sqrt{t}$, and $e^{4s^2/t} \le C$. For $\eta > 0$ the last term of (7.2.8) is at most

$$C\eta e^{2a\log(e/\eta)}/\sqrt{t} \le C/\sqrt{t\log(e/\eta)}.$$

The approximation term is bounded by the other two inverse terms in (7.2.7). At $\eta = 0$ both predictive terms vanish.

For the lower bounds, endpoint mixing laws differing by mass η have both population W_1 and predictive TV equal to η. For the inverse term let

$$\pi(dx) = \frac{dx}{\pi\sqrt{x(1-x)}}, \qquad h_k(x) = T_k(2x-1), \qquad G_k^{\pm} = (1 \pm h_k/2)\pi.$$

Direct integration of their signed distribution function gives $W_1(G_k^+, G_k^-) \asymp 1/k$, uniformly in $k \ge 1$. For $k = t+1$ the predictive laws coincide, giving the $1/t$ term. For $k \le t$, the beta–binomial conditional-expectation operator $Qh(j) = \mathbb{E}_\pi[h(X) \mid J = j]$ satisfies

$$Q^*Qh_k = \lambda_{t,k} h_k, \qquad \lambda_{t,k} = \prod_{l=0}^{k-1} \frac{t-l}{t+1+l} \le e^{-k^2/(2t)}.$$

Indeed it is self-adjoint, preserves polynomial degree, and has the displayed leading coefficient; orthogonality then proves the identity. This classical diagonalization is discussed in [70]. Consequently

$$\|p_{G_k^+} - p_{G_k^-}\|_{\mathrm{TV}} = \tfrac{1}{2}\mathbb{E}|Qh_k(J)| \le \tfrac{1}{2\sqrt{2}}\sqrt{\lambda_{t,k}} \le \tfrac{1}{2\sqrt{2}} e^{-k^2/(4t)}.$$

Take $k = \lceil \sqrt{8t \log(e/\eta)} \rceil$ if it is at most t. This gives the required inverse lower bound. If that degree exceeds t, then $1/\sqrt{t \log(e/\eta)} \le C/t$, and the exactly indistinguishable pair suffices. A maximum of the three lower bounds is comparable to their clipped sum. □

7.2.3 Unknown Huber contamination: the full minimax curve

We observe N independent counts with common law

$$R = (1-\varepsilon)p_G + \varepsilon Q, \qquad G \in \mathcal{P}([0,1]), \quad Q \in \mathcal{P}(\{0,\dots,t\}), \qquad 0 \le \varepsilon \le 1/4.$$

Contamination occurs at the level of whole individuals' reported counts, not independently within each individual's t trials. Write

$$\mathcal{R}^{\mathrm{Hub}}_{N,t}(\varepsilon) = \inf_{\widehat{G}} \sup_{G,Q} \mathbb{E}_R W_1(\widehat{G}, G).$$

Theorem 7.2.5 (Adaptive robust population estimation)**.** *There are absolute positive constants such that, for every $N, t \ge 1$ and $0 \le \varepsilon \le 1/4$,*

$$\mathcal{R}^{\mathrm{Hub}}_{N,t}(\varepsilon) \asymp \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(eN)}} + \frac{1}{\sqrt{N}} + \varepsilon + \frac{1}{\sqrt{t\log(e/\varepsilon)}}\right\}. \tag{7.2.10}$$

One sequence of estimators, depending on N, t but not on ε, attains the upper bound simultaneously for all ε in this interval. The estimator is a finite-model minimum-distance estimator; no robustness claim for the ordinary MLE is made.

General contamination theory identifies TV moduli as the relevant lower bound mechanism [93]. What requires a separate argument here is attaining both the sampling transition and the contamination transition without allowing the predictive TV entropy cost to spoil the parametric low-frequency term.

Lemma 7.2.6 (Simultaneous predictive control without knowing contamination)**.** *Let a family of laws on an ordered alphabet of diameter one have a finite TV cover $p_1, \dots, p_M$ of radius β, with representatives in the family. Let q be the empirical law of N independent draws from R, and suppose $\|R - p\|_{\mathrm{TV}} \le \varepsilon$ for a member p of the family. There is an estimator $\widehat{p}$ in the cover, not using ε, such that*

$$\mathbb{E} W_1(\widehat{p}, p) \le C(\varepsilon + \beta + N^{-1/2}), \tag{7.2.11}$$

$$\mathbb{E}\|\widehat{p} - p\|_{\mathrm{TV}} \le C\left(\varepsilon + \beta + \sqrt{\log(2M)/N}\right). \tag{7.2.12}$$

With probability at least $1 - 3e^{-x}$, the same two bounds hold with $N^{-1/2}$ replaced by $\sqrt{(1+x)/N}$ and $\sqrt{\log(2M)/N}$ by $\sqrt{(\log(2M)+x)/N}$, respectively.

Proof. Let $\mathcal{A} = \{\{j : p_b(j) > p_c(j)\} : 1 \le b, c \le M\}$, $D_a = \max_{A \in \mathcal{A}} |p_a(A) - q(A)|$, and

$$b_N = \sqrt{\frac{\log(2M^2)}{2N}}, \qquad F_a = W_1(p_a, q) + (D_a - b_N)_+.$$

Choose the smallest index minimizing F_a. If $S = \max_{A \in \mathcal{A}} |q(A) - R(A)|$, the union bound and Hoeffding's inequality imply

$$\Pr\{S > b_N + u\} \le e^{-2Nu^2}, \qquad \mathbb{E}(S - b_N)_+ \le C/\sqrt{N}.$$

Choose a true nearby center p_{a_*}. Then $D_{a_*} \le \beta + \varepsilon + S$ and $W_1(p_{a_*}, q) \le \beta + \varepsilon + W_1(R, q)$. Integration of empirical CDF standard deviations gives $\mathbb{E}W_1(R, q) \le 1/(2\sqrt{N})$. Thus $\mathbb{E}F_{a_*} \le 2(\beta + \varepsilon) + C/\sqrt{N}$. The minimizing index has $W_1(\widehat{p}, q) \le F_{a_*}$ and $D_{\widehat{a}} \le b_N + F_{a_*}$. The triangle inequality proves (7.2.11). Since the set $\{\widehat{p} > p_{a_*}\}$ belongs to $\mathcal{A}$,

$$\|\widehat{p} - p_{a_*}\|_{\mathrm{TV}} \le D_{\widehat{a}} + D_{a_*}.$$

Together with $\mathbb{E}S \le b_N + C/\sqrt{N}$, this proves (7.2.12). The probability assertion follows from the same inequalities, the displayed tail bound for S, and the empirical-CDF bound $\Pr\{W_1(R, q) > \sqrt{x/(2N)}\} \le 2e^{-x}$. □

Lemma 7.2.7 (Effective dimension of the binomial moment body)**.** *For $t \le N$ the set $\{p_G\}$ has a TV cover of radius $\beta = (eN)^{-2}$, with centers in that set, satisfying*

$$\log(2M) \le C\sqrt{t}\,\log(eN)^{3/2}.$$

Proof. The vector polynomial $K_x = (B_0^t(x), \dots, B_t^t(x))$ has shifted Chebyshev coefficients v_l satisfying $\|v_l\|_1 \le 2e^{-l^2/(2t)}$ for $l \ge 1$. To see this, shift the Fourier contour of $K_{(1+\cos z)/2}$ by $u = l/t$ and use

$$\|K_{(1+\cos(\theta+iu))/2}\|_1 = (\cosh u)^t \le e^{tu^2/2}.$$

Truncation at $m = \min(t, \lceil\sqrt{2t\log(16t/\beta)}\rceil)$ has uniform ℓ_1 error $O(\beta)$. Partition the first m Chebyshev moments, each in $[-1, 1]$, into coordinate intervals of length $c\beta/(m+1)$ and retain one mixing representative per occupied cell. The resulting TV radius is at most β after reducing the absolute constant c, and $\log M \le m\log(C(m+1)/\beta)$. The asserted estimate follows. □

Upper bound in Theorem 7.2.5. For $t \le N$ use the cover of Lemma 7.2.7 and the estimator of Lemma 7.2.6; return its representative mixing law. Writing $L_N = \log(eN)$ and $L_\varepsilon = \log(e/\varepsilon)$ gives

$$\mathbb{E}W_{1,t}(\widehat{p}, p_G) \le C(\varepsilon + N^{-1/2}), \quad \mathbb{E}\|\widehat{p} - p_G\|_{\mathrm{TV}} \le C\left(\varepsilon + \frac{t^{1/4}L_N^{3/4}}{\sqrt{N}}\right).$$

In Lemma 7.2.4, choose

$$k = \left\lfloor \min\{t/2, \sqrt{at\min(L_N, L_\varepsilon)}\}\right\rfloor, \qquad s = \lfloor\sqrt{t}/16\rfloor, \qquad a = 1/64.$$

As above, bounded t, N are harmless. The approximation term has the required three inverse contributions. The low-frequency term costs $C(\varepsilon + N^{-1/2})$. The contamination part of the last term is at most $C/\sqrt{tL_\varepsilon}$. Its sampling part is bounded by

$$\frac{Ce^{2aL_N}L_N^{3/4}}{\sqrt{N}\,t^{1/4}} \le \frac{C}{\sqrt{tL_N}},$$

because $t \le N$ and $(eN)^{2a}N^{-1/4}L_N^{5/4}$ is uniformly bounded. The parameter ε appears only in the proof's choice of test degree, not in the estimator.

For $t > N$, return the empirical law of the normalized counts J_i/t as the estimated mixing distribution. Coupling a binomial count to its parameter gives $W_1(G, p_G) \le 1/(2\sqrt{t})$, and hence its expected loss is at most $\varepsilon + 1/(2\sqrt{N}) + 1/(2\sqrt{t})$. This is $O(\varepsilon + N^{-1/2})$, as required in this range. □

Lower bound in Theorem 7.2.5. Taking $Q = p_G$ recovers the uncontaminated experiment. Its elementary lower bound is $c\min\{1, t^{-1} + (tL_N)^{-1/2} + N^{-1/2}\}$: the exactly moment-matched pair above gives t^{-1}; the same beta–binomial pair with $k = \lceil\sqrt{8t\log(8N)}\rceil \le t$ has bounded

N-fold total variation and gives $(tL_N)^{-1/2}$; endpoint mixtures with weights $1/2 \pm c/\sqrt{N}$ give $N^{-1/2}$. If the stated k exceeds t, its contribution is absorbed by t^{-1}.

For the contamination lower bound, the neighborhoods $\{(1-\varepsilon)p+\varepsilon Q\}$ and $\{(1-\varepsilon)p'+\varepsilon Q'\}$ intersect whenever $\|p-p'\|_{\mathrm{TV}} \le \varepsilon/(1-\varepsilon)$. Indeed $(1-\varepsilon)\max(p,p')$ has total mass at most one; add any nonnegative remaining mass to produce a common probability law. Its differences from $(1-\varepsilon)p$ and $(1-\varepsilon)p'$ are nonnegative and each has mass ε. Every estimator then has maximum expected population loss at least $W_1(G,G')/2$, regardless of N. Apply Theorem 7.2.3 with predictive threshold $\eta = \varepsilon$, which is no larger than $\varepsilon/(1-\varepsilon)$. Combining the two lower bounds proves the theorem. At $\varepsilon = 0$ only the clean lower bound is needed. □

Corollary 7.2.8 (Equal-replication budgets remain robust at their optimum)**.** *For common replication and total trial budget $Nt \le B$, the optimal Huber-contaminated population risk is*

$$\inf_{Nt\le B} \mathcal{R}^{\mathrm{Hub}}_{N,t}(\varepsilon) \asymp \min\{1, (B\log(eB))^{-1/4} + \varepsilon\}.$$

The uncontaminated order-optimal allocation $N \asymp \sqrt{B\log(eB)}$, $t \asymp \sqrt{B/\log(eB)}$ continues to attain this order with the adaptive robust estimator.

Proof. The clean budget lower bound and the endpoint contamination lower bound give the assertion from below. Put $r = (B\log(eB))^{-1/4}$ and use the displayed allocation. Its clean terms are $O(r)$. If $\varepsilon \le r$, then $\log(e/\varepsilon) \ge c\log(eB)$, so the remaining inverse term is $O(r)$. If $r < \varepsilon \le 1/4$, the function $x^2\log(e/x)$ is increasing in this interval. Thus $t\varepsilon^2\log(e/\varepsilon) \ge tr^2\log(e/r) \ge c$, and the inverse term is $O(\varepsilon)$. Integer rounding and bounded budgets only change absolute constants. □

The coefficient conjecture is a published target. The TV curve and the adaptive contamination curve are newly formulated and resolved transfer problems in this manuscript; they are not presented as previously named conjectures. The budget statement is a corollary, not an additional independent conjecture closure. The exact best exponent constant in (7.2.1), computationally efficient robust fitting, and sharp confidence-radius constants remain separate questions.

7.3 Exact endpoint densities and local moments for prime-zeta zeros

The endpoint theorem in the preceding manuscript established $N_m(T) \asymp T$ for each fixed derivative of the prime zeta function. It deliberately did not identify a limiting coefficient. We now prove that the coefficient exists, that no positive density of zeros can escape into a shrinking strip adjacent to $\Re s = 1$, and that every fixed-height local counting moment has a random-prime limit. These are endpoint extensions of the earlier result, not repetitions of its linear upper bound. The published questions motivating this direction are those of Belovas, Čepaitytė, and Sabaliauskas [21], also discussed in [85]. We do not attribute the stronger moment statement below to a published conjecture.

Write $P(s) = \sum_p p^{-s}$ for $\Re s > 1$. Let $(\omega_p)_p$ be independent random variables uniform on the unit circle, and put

$$P_\omega(s) = \sum_p \omega_p p^{-s}, \qquad Q_{m,\omega}(s) = P_\omega^{(m)}(s). \tag{7.3.1}$$

These random series converge locally uniformly almost surely on $\Re s > 1/2$; a proof is included below. Zeros are counted with multiplicity unless a superscript dist is displayed. For $h > 0$

define

$$X_{m,h}(t) = \#\{s : P^{(m)}(s) = 0,\ \Re s > 1,\ t < \Im s < t + h\},$$
$$X^{\omega}_{m,h} = \#\{s : Q_{m,\omega}(s) = 0,\ \Re s > 1,\ 0 < \Im s < h\}.$$

Changing an open horizontal edge to a closed edge affects only a null set of starting heights in any integral below.

Theorem 7.3.1 (Endpoint density and every fixed-window moment)**.** *For every fixed integer $m \geq 0$ there is a number $0 < c_m < \infty$ such that*

$$N_m(T) = c_m T + o(T), \qquad N_m^{\mathrm{dist}}(T) = c_m T + o(T). \tag{7.3.2}$$

Here both counts include all zeros in $\Re s > 1$, $0 < \Im s < T$. For every fixed $h > 0$, as t is sampled uniformly from $[0, T]$,

$$X_{m,h}(t) \implies X^{\omega}_{m,h}. \tag{7.3.3}$$

Moreover, for every real $q > 0$,

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T X_{m,h}(t)^q \, dt = \mathbb{E}[(X^{\omega}_{m,h})^q] < \infty. \tag{7.3.4}$$

The same statements hold for distinct-zero counts. In particular, $c_m = \mathbb{E}X^{\omega}_{m,1}$ and the limiting mean and variance in a fixed window are $c_m h$ and $\mathrm{Var}(X^{\omega}_{m,h})$, respectively.

Theorem 7.3.2 (Quantitative absence of endpoint mass)**.** *Fix $m \geq 0$. There are $\epsilon_0 > 0$ and $C_m < \infty$ such that for $0 < \epsilon < \epsilon_0$,*

$$\limsup_{T\to\infty} \frac{1}{T} \#\{s : P^{(m)}(s) = 0,\ 1 < \Re s \leq 1 + \epsilon,\ 0 < \Im s < T\} \leq C_m \epsilon. \tag{7.3.5}$$

If $c_m(\gamma)$ denotes the limiting density in $\Re s > \gamma > 1$, then

$$0 \leq c_m - c_m(1 + \epsilon) \leq C_m \epsilon.$$

Thus the endpoint coefficient is a controlled one-sided limit of interior coefficients. No rate in T is asserted in (7.3.5).

7.3.1 The random analytic limit of the single-valued lifts

Set

$$H(s) = \sum_p \sum_{k\geq 2} \frac{p^{-ks}}{k}, \qquad H_\omega(s) = \sum_p \sum_{k\geq 2} \frac{\omega_p^k p^{-ks}}{k}.$$

Both are holomorphic and locally bounded, with every derivative, on $\Re s > 1/2$; the bounds for H_ω are uniform in ω. Let

$$Z_\omega(s) = \exp(P_\omega(s) + H_\omega(s)).$$

It is almost surely holomorphic and nowhere zero on that half-plane.

Lemma 7.3.3 (Joint weak convergence, with its approximation input)**.** *On every fixed bounded open rectangle whose closure lies in $\Re s > 1/2$, the joint laws of*

$$(\zeta(s + it), H(s + it))$$

for uniform $t \in [t_0, T]$ converge, in the topology of locally uniform convergence, to $(Z_\omega(s), H_\omega(s))$. Here t_0 is chosen so that the moving pole of ζ lies outside the rectangle. The same assertion holds jointly with any fixed number of derivatives.

Proof. We give the approximation argument, rather than assuming that a random model automatically represents arithmetic averages. The classical off-critical-line mean-square theorem and its fixed-coefficient Fourier formula imply, for $\sigma > 1/2$,

$$\lim_{T\to\infty} \frac{1}{T}\int_{t_0}^{T} \left|\zeta(\sigma+it) - \sum_{n\le M} n^{-\sigma-it}\right|^2 dt = \sum_{n>M} n^{-2\sigma}. \tag{7.3.6}$$

The estimates needed to integrate this identity over a compact real-part interval are locally uniform. These are standard consequences of the zeta mean-square formula [86, Chapter VII]. To specify the Fourier input, $T^{-1}\int_{t_0}^{T} \zeta(\sigma+it)n^{it}dt \to n^{-\sigma}$ for each fixed n; expanding the square together with $T^{-1}\int |\zeta(\sigma+it)|^2 dt \to \zeta(2\sigma)$ gives (7.3.6). One can obtain that fixed Fourier formula by moving the vertical integral to a line $\Re s > 1$, where the Dirichlet series is absolute; the horizontal connecting integrals are $o(T)$ by the classical polynomial growth bound for ζ on a fixed strip in $\Re s > 1/2$.

For compact rectangles $K \Subset U$ the submean inequality for holomorphic functions gives

$$\sup_K |f|^2 \le C_{K,U} \int_U |f(s)|^2 \, dA(s).$$

Apply this to the difference in (7.3.6), then average over t, integrate over U, and let $M \to \infty$. It follows that the Dirichlet polynomials approximate the translated zeta functions in mean square for the compact-uniform topology. The finitely many initial heights have no effect.

For clarity, the random analytic objects require no conjectural Euler product convergence. On a bounded rectangle in $\Re s > 1/2$,

$$\sum_p \mathbb{E}\|\omega_p p^{-s}\|^2_{L^2(U)} < \infty.$$

The independent centered Hilbert-space series therefore converges almost surely and in mean square in $L^2(U)$. The submean inequality on smaller rectangles promotes it to locally uniform convergence. A countable exhaustion proves the assertion about P_ω. Also, writing $\omega(n) = \prod_{p^a\|n} \omega_p^a$, orthogonality $\mathbb{E}\omega(n)\overline{\omega(k)} = \mathbf{1}_{n=k}$ gives

$$\mathbb{E}\left\| \sum_{M<n\le L} \omega(n) n^{-s} \right\|^2_{L^2(U)} = \int_U \sum_{M<n\le L} n^{-2\Re s}\, dA(s).$$

Thus the random Dirichlet polynomials converge in mean square in the compact-uniform topology. Their limit agrees with Z_ω on $\Re s > 1$, by the absolutely convergent Euler product, and hence on the whole connected half-plane by the identity theorem.

For each fixed truncation, the vertical phases $(p^{-it})_{p\le M}$ are equidistributed on their finite torus: no nonzero integer combination of distinct prime logarithms is zero. Consequently the finite Dirichlet polynomials and finite-prime truncations of H converge jointly in law to their random versions. The remaining tail of H tends uniformly to zero on compact sets, uniformly in the phases. The deterministic and random mean-square approximations above now prove the joint weak convergence, first for bounded Lipschitz functionals and then for the compact-uniform metric. Derivatives are continuous maps on holomorphic function spaces by Cauchy's formula. This proves the lemma. The argument is the classical probabilistic limit-theorem method of [94, 95], specialized here to the joint prime-power correction that is needed for zero counts. □

Define $A_1 = \zeta'$ and $A_{m+1} = \zeta A_m' - m\zeta' A_m$. Set

$$F_0 = \zeta - e^H, \qquad F_m = A_m - \zeta^m H^{(m)} \quad (m \ge 1), \qquad r(s) = \frac{s-1}{s+1},$$

and remove the sole possible pole by

$$L_0 = rF_0, \qquad L_m = r^{2m}F_m \quad (m \geq 1).$$

Every L_m is holomorphic on $\Re s > 1/2$. On $\Re s > 1$,

$$F_0 = e^H(e^P - 1), \qquad F_m = \zeta^m P^{(m)} \quad (m \geq 1). \tag{7.3.7}$$

Since $r(s+it) \to 1$ uniformly on compact sets as $t \to \infty$, Lemma 7.3.3 and the differential-polynomial identities give

$$L_m(s+it) \Longrightarrow V_{m,\omega}(s), \qquad V_{0,\omega} = e^{H_\omega}(e^{P_\omega} - 1), \qquad V_{m,\omega} = Z_\omega^m Q_{m,\omega} \ (m \geq 1). \tag{7.3.8}$$

For $m = 0$ this limit includes all levels $P_\omega = 2\pi ik$, not merely the level zero. We will use it only as an upper bound near the boundary. This avoids any unproved interchange of logarithm branches across $\Re s = 1$.

7.3.2 Uniform integrability of arithmetic local zero counts

Lemma 7.3.4 (Every logarithmic counting moment is bounded)**.** *Let K be any fixed compact rectangle in $\Re s > 1/2$, and let $Y_m(t)$ be the number of zeros of $L_m(s+it)$ in K. For every fixed real $q > 0$,*

$$\sup_{T \geq T_0} \frac{1}{T} \int_{t_0}^{T} Y_m(t)^q \, dt < \infty. \tag{7.3.9}$$

The corresponding random zero counts of $V_{m,\omega}$ have all finite moments as well.

Proof. Choose $\alpha > 1/2$ to the left of K. For sufficiently large fixed R, the functions $L_m(R+it)$ are uniformly bounded away from zero for all real t. Indeed, the leading term of F_0 is 2^{-s} and that of F_m is $(-\log 2)^m 2^{-s}$, uniformly as $\Re s \to \infty$, while r is bounded away from zero on $\Re s = R > 1$. Increasing R if necessary, a disk centered at $R + iy_0$, with fixed radii $r_1 < r_2 < R - \alpha$, contains K in its smaller disk. This is possible for any fixed height of K because increasing R makes the available vertical aperture arbitrarily large.

Jensen's formula, applied after the vertical shift, bounds

$$Y_m(t) \leq C + C \int_0^{2\pi} \log^+ |L_m(R + i(t+y_0) + r_2 e^{i\theta})| \, d\theta.$$

For $|\Im s| \geq 1$ and real parts in the fixed disk's range,

$$\log^+ |L_m(s)| \leq C_m + C_m \sum_{j=0}^{m} \log(1 + |\zeta^{(j)}(s)|).$$

All H derivatives are uniformly bounded there. Jensen's inequality for powers at least one, and the elementary inequality $\log^q(1+x) \leq C_q(1+x^2)$, reduce the average of $Y_m(t)^q$ to the off-critical-line mean-square bounds for finitely many zeta derivatives. The derivative bounds follow from the ordinary mean-square theorem by Cauchy's formula on fixed smaller circles. The remaining bounded-height region is compact and contributes only a finite constant after pole removal. Powers $0 < q < 1$ follow from the bound for $q = 1$.

For the random limits the same argument applies. On the far-right line the leading-term lower bounds hold uniformly in all phases, and $\mathbb{E}|Z_\omega^{(j)}(s)|^2 = \sum_n (\log n)^{2j} n^{-2\Re s} < \infty$. The differential-polynomial identities in the limit and the same logarithmic inequality prove all the random moment bounds. □

7.3.3 Random transversality and a linear strip bound

Lemma 7.3.5 (No exceptional boundary and bounded mean intensity)**.** *On each compact substrip of $\Re s > 1/2$, the random functions $Q_{m,\omega}$ have almost surely no zero on any prescribed line segment, and almost surely no multiple zero in any prescribed compact rectangle. The same assertions hold for $V_{m,\omega}$. For a rectangle D in such a substrip,*

$$\mathbb{E}\#\{Q_{m,\omega} = 0 \text{ in } D\} \le C_m \operatorname{area}(D), \qquad \mathbb{E}\#\{V_{m,\omega} = 0 \text{ in } D\} \le C_m \operatorname{area}(D), \tag{7.3.10}$$

where the constants depend on the compact real-part interval but not on the height or location of D in that interval.

Proof. For fixed s, five prime summands in $Q_{m,\omega}(s)$ have a bounded planar density, uniformly for the real part in a compact interval. Their characteristic function is a product of five factors $J_0(b_p|u|)$, with $b_p = (\log p)^m p^{-\Re s}$ bounded away from zero and infinity on that interval. The bound $|J_0(x)| \le C(1+|x|)^{-1/2}$ makes this product integrable in two dimensions. Fourier inversion gives the asserted density, and convolution with the independent remainder preserves the bound. The five-prime sum also has uniformly bounded support.

The pair $(Q_{m,\omega}, Q'_{m,\omega})$ has a bounded density in four dimensions. Use ten primes, with $\lambda_p = \log p$. For every $(u,v) \in \mathbb{C}^2$, at least nine of the ten numbers $|u - \lambda_p v|$ are at least $c\|(u,v)\|$: two simultaneous violations would contradict invertibility of the corresponding pair of distinct linear forms when c is sufficiently small. The characteristic function is consequently $O((1+\|(u,v)\|)^{-9/2})$, which is integrable in dimension four.

These density estimates imply transversality by a mesh argument. First restrict to the event that the required derivatives on a compact neighborhood have supremum at most M. A zero on a line segment forces $|Q| \le CMh$ at a mesh point of spacing h. The probability is $O_M(h^{-1}h^2) \to 0$. A multiple zero in a rectangle forces $|Q| \le CMh^2$ and $|Q'| \le CMh$ at a square-mesh point. The four-dimensional density bound gives probability $O_M(h^{-2}h^6) \to 0$. Then let $M \to \infty$, using almost sure analyticity on the neighborhood. The same reasoning applies to any fixed level $Q = z$. For V_0 use the countable levels $P = 2\pi i k$; for $m \ge 1$ use the nowhere-vanishing factor Z_ω^m.

We give a direct expected-count bound, including the infinitely many levels in V_0. Split $Q = S + R$, with S the five-prime sum. On the fixed substrip, $|S| \le A$, $|S'| \le A'$, and the planar density of S is bounded by C. Conditional on the remaining phases, for $0 < \eta < 1$ and any target z,

$$\mathbb{E}_S\big[|Q'|^2 \mathbf{1}_{\{|Q-z|<\eta\}}\big] \le C(1+|R'|^2)\eta^2 \mathbf{1}_{\{|z-R|\le A+1\}}.$$

The constants are uniform in s, and $\mathbb{E}|R'(s)|^2$ is uniformly bounded by the convergent prime square sum. For the single target $z = 0$ this proves the required estimate on the approximate zero-density integral. For the targets $z = 2\pi i k$, only a bounded number of integers k can satisfy $|2\pi i k - R| \le A + 1$, independent of R. Summing the same bound over k therefore remains $O(\eta^2)$.

For a deterministic analytic function having no boundary zero, its number of zeros in D is the limit, as $\eta \downarrow 0$, of

$$\frac{1}{\pi\eta^2} \int_D \mathbf{1}_{\{|Q(s)-z|<\eta\}} |Q'(s)|^2 \, dA(s).$$

This follows from the local change of variables at simple zeros (and holds with multiplicity more generally). For the lattice of levels, sum the displayed integrals over k; only finitely many levels meet the image of the compact closure of D. Fatou's lemma and the conditional estimate give (7.3.10). Stationarity under imaginary translation makes all constants independent of the ordinate. □

7.3.4 Passage to the arithmetic endpoint

Let D be a fixed rectangle in $\Re s > 1/2$. By Lemma 7.3.5, its boundary almost surely contains no zero of $V_{m,\omega}$. Zero counting is therefore almost surely a continuous functional of the limiting analytic function, by Rouché's theorem. Equations (7.3.8) and (7.3.9) imply both weak convergence and convergence of every fixed positive moment of the lift counts in D. Uniform integrability is essential here; weak convergence alone would not justify this assertion.

In particular, with $D_\epsilon = (1, 1+\epsilon) \times (0, h)$,

$$\limsup_{T\to\infty} \frac{1}{T} \int_0^T \#\{P^{(m)} = 0 : \ 1 < \Re s < 1+\epsilon, \ t < \Im s < t+h\}\, dt \le C_m h\epsilon. \tag{7.3.11}$$

The arithmetic count is bounded by the lift count because of (7.3.7); this remains true for $m = 0$ although the lift includes additional levels. Enlarging the strip slightly proves the same inequality when either vertical edge is closed. A fixed initial height interval is harmless, since all zeros there are bounded by those of a holomorphic lift on a compact set.

For comparison, fix $\gamma > 1$. Absolute convergence gives a compact prime-phase hull for $P^{(m)}$ on $\Re s > 1$. Its members are

$$\sum_p (-\log p)^m \omega_p p^{-s}.$$

They have uniform local zero bounds on every compact rectangle: a contrary sequence would have a compact-uniform convergent subsequence with a nonzero analytic limit and unbounded zero counts, contradicting Rouché's theorem. The limit cannot be identically zero because its far-right leading coefficient is $(-\log 2)^m \omega_2$. The finite-phase equidistribution argument, compact approximation, and the boundary and multiple-zero null sets in Lemma 7.3.5 therefore imply convergence of counts in fixed rectangles to the random counts, with every moment. Equivalently, bounded zero-count observables are Riemann integrable on the compact phase torus and their time averages converge to Haar averages. This is the classical interior almost-periodic mechanism; it is not a new endpoint argument.

All arithmetic and random zeros in $\Re s > 1$ lie to the left of one fixed R_m. Indeed, for sufficiently large R_m the first prime term dominates the sum of all the others, uniformly in phases and heights. Thus $X_{m,h}$ differs from its count restricted to $\Re s > 1+\epsilon$ only if a zero occurs in the thin strip. Markov's inequality and (7.3.11) show that the upper asymptotic probability of this difference is $O_m(h\epsilon)$. For the random functions the same probability bound follows from (7.3.10). Letting first $T \to \infty$ at fixed ϵ, and then $\epsilon \downarrow 0$, proves (7.3.3). All powers of $X_{m,h}$ are uniformly integrable by domination by the lift count in the fixed compact rectangle $[1, R_m] \times [0, h]$ and Lemma 7.3.4. This proves (7.3.4).

Almost surely all random zeros are simple. The same truncation argument consequently gives the identical local laws and moments for distinct arithmetic zeros. The random mean zero measure is stationary under imaginary translations, so its mass in a height h rectangle equals h times its mass in height one. Put $c_m = \mathbb{E} X^\omega_{m,1} < \infty$.

To pass from sliding windows to counts up to height T, positivity and Fubini suffice; no uniform short-interval estimate is needed. For unit windows the integral from 0 to T counts every zero with ordinate in $(1, T)$ with weight one, while zeros in the two end windows receive a weight between zero and one. Consequently

$$\int_0^{T-1} X_{m,1}(t)\, dt \le N_m(T), \qquad N_m(T) - N_m(1) \le \int_0^T X_{m,1}(t)\, dt.$$

The first-moment convergence now proves (7.3.2). The identical argument with a thin vertical strip proves (7.3.5).

Finally $c_m > 0$. Let $a_p = (\log p)^m$ and choose the unique $S_m > 1$ satisfying $a_2 2^{-S_m} = \sum_{p\geq 3} a_p p^{-S_m}$. The frozen-phase series $a_2 2^{-s} - \sum_{p\geq 3} a_p p^{-s}$ has a simple zero at S_m, since the derivative of its normalization by $a_2 2^{-s}$ is strictly positive there. Translate the zero to height $1/2$. On a small disk around it, a sufficiently long finite phase specification controls the absolutely convergent tail uniformly. Rouché's theorem then places a zero in the disk for a cylinder set of phases of positive Haar probability. Thus $\mathbb{E}X^{\omega}_{m,1} > 0$, completing both theorems.

Corollary 7.3.6 (Endpoint ordinates and macroscopic rectangles)**.** *For every fixed $x \in [0,1]$,*

$$\frac{N_m(xT)}{N_m(T)} \longrightarrow x.$$

The convergence is uniform in $x \in [0,1]$. The same conclusion holds for distinct zeros. More generally, if $I \subset (1, R_m)$ is an interval whose left endpoint may equal one, and $0 \leq a < b < \infty$ are fixed, the number of zeros with real part in I and ordinate in (aT, bT), divided by T, converges to $(b-a)$ times the random mean zero mass in $I \times (0,1)$.

Proof. For each fixed $x > 0$ the first assertion follows from $N_m(xT) = c_m xT + o(T)$. Monotonicity and a finite mesh in $[0,1]$ upgrade pointwise convergence to uniform convergence, with the interval next to zero controlled by monotonicity. The rectangle assertion follows from the interior count law and endpoint truncation exactly as above, followed by subtraction of the counts up to aT and bT. □

Corollary 7.3.7 (A boundary formula for the density constant)**.** *Let $Q = Q_{m,\omega}$ and define $J_m(\sigma) = \mathbb{E}\log|Q(\sigma)|$ for $\sigma > 1/2$. Then J_m is continuously differentiable, with locally Lipschitz first derivative, and*

$$c_m = \frac{-\log 2 - J_m'(1)}{2\pi} = \frac{-\log 2 - \mathbb{E}\,\mathrm{Re}\{Q'(1)/Q(1)\}}{2\pi}. \tag{7.3.12}$$

The expectation in the last expression is absolutely integrable. It also has a finite-prime approximation with an explicit convergence order. Put

$$Q_M = \sum_{p\leq M} (-\log p)^m \omega_p/p, \qquad D_M = \sum_{p\leq M} (-\log p)^{m+1}\omega_p/p, \qquad \delta_M = M^{-1/6}(\log M)^{m/3}.$$

Then, as $M \to \infty$,

$$c_m = \frac{1}{2\pi}\left[-\log 2 - \mathbb{E}\,\mathrm{Re}\,\frac{D_M\overline{Q_M}}{|Q_M|^2 + \delta_M^2}\right] + O_m\Big(M^{-1/6}(\log M)^{m/3}\Big). \tag{7.3.13}$$

The finite expectation is a finite-dimensional torus integral; no claim about the computational complexity of evaluating that integral is made.

Proof. The bounded planar density and the finite second moment of Q make $\mathbb{E}|\log|Q(s)||$ locally bounded. They also make J_m continuous: the negative logarithmic tail is uniformly integrable by the planar small-ball bound, the positive logarithmic tail is uniformly integrable by the local second-moment bound, and $Q(\sigma)$ varies continuously in mean square. The distributional identity $\Delta \log|Q| = 2\pi \sum_{Q(z)=0} \delta_z$, followed by Fubini and imaginary-translation stationarity, shows that J_m'' is 2π times the mean zero measure per unit height. Lemma 7.3.5 bounds this measure by a constant multiple of Lebesgue measure on compact real-part intervals. Consequently J_m has a locally Lipschitz first derivative.

The five-prime decomposition in the proof of that lemma gives, uniformly on a compact substrip,

$$\mathbb{E}\left[\frac{|Q'|}{|Q|}\mathbf{1}_{\{|Q|<\eta\}}\right] \leq C\eta, \qquad 0 < \eta < 1.$$

Indeed, conditional on the remaining phases, $|Q'| \le A' + |R'|$, and integration of $|z|^{-1}$ against the bounded planar density on the disk $|z| < \eta$ is $O(\eta)$. Away from that disk use $|Q|^{-1} \le \eta^{-1}$ and $\mathbb{E}|Q'| < \infty$. This proves absolute integrability and justifies the derivative formula $J_m'(\sigma) = \mathbb{E}\,\mathrm{Re}(Q'/Q)(\sigma)$, first as a distributional identity, either by truncation or by integrating along real segments and using Fubini. The expectation on the right is continuous: truncate away from $Q = 0$, use the mean-square continuity of Q, Q' and their uniform moment bounds, and then remove the truncation using the displayed $O(\eta)$ estimate. Thus both continuous representatives agree at every $\sigma > 1/2$, in particular at $\sigma = 1$.

For sufficiently large σ the first prime term strictly dominates all the others. Averaging first over ω_2 gives exactly $J_m(\sigma) = m \log\log 2 - \sigma \log 2$ there, by the mean-value formula for the logarithm on a circle. Integrating J_m'' from one to such a real part proves (7.3.12).

For completeness the approximation error can be controlled without a small-denominator assumption. For $0 < \delta < 1$, the same conditional density argument gives

$$\mathbb{E}\left|\frac{Q'}{Q} - \frac{Q'\overline{Q}}{|Q|^2 + \delta^2}\right| \le C_m \delta.$$

On $|Q| \le 1$ this follows by integrating $\delta^2/(|z|(|z|^2 + \delta^2))$ against a bounded planar density; the polar integral is $O(\delta)$. On $|Q| > 1$ the contribution is $O(\delta^2 \mathbb{E}|Q'|)$. The map $z \mapsto \overline{z}/(|z|^2 + \delta^2)$ has supremum at most $1/(2\delta)$ and Lipschitz constant at most C/δ^2. Writing $Q = Q(1)$ and $D = Q'(1)$, Cauchy–Schwarz therefore yields

$$\mathbb{E}\left|\frac{D\overline{Q}}{|Q|^2 + \delta^2} - \frac{D_M\overline{Q_M}}{|Q_M|^2 + \delta^2}\right| \le C_m \left(\frac{M^{-1/2}(\log M)^{m+1}}{\delta} + \frac{M^{-1/2}(\log M)^m}{\delta^2}\right).$$

Here orthogonality of prime phases and comparison with the integer tails give $\mathbb{E}|Q - Q_M|^2 \le C_m M^{-1}(\log M)^{2m}$ and $\mathbb{E}|D - D_M|^2 \le C_m M^{-1}(\log M)^{2m+2}$; also $\mathbb{E}|D_M|^2$ is bounded uniformly in M. Choosing $\delta = \delta_M$ proves (7.3.13). □

What remains open in this section. The moment limits are for each *fixed* window height h. They do not give the asymptotic growth of $\mathrm{Var}(X^{\omega}_{m,h})$ as $h \to \infty$, a central limit theorem, GUE statistics, or a height-effective error term. No conclusion is made about a chosen branch of prime zeta inside $\Re s < 1$, and none of the arguments addresses the Riemann hypothesis. The present quantitative estimate is linear in the boundary-strip width after taking the height limit, not a joint uniform estimate for arbitrary $\epsilon = \epsilon(T)$.

7.4 Fixed-complexity adaptive information and convex-noise geometry

The exact-real recovery phenomenon of Krieg–Novak–Ullrich [60] allows the individual measurements to become arbitrarily complicated. Earlier results in this series replaced their continuous queries by convex polynomials, with no common degree bound. We prove that this lack of a degree bound is essential: with any fixed degree, fewer measurements than the dimension cannot give arbitrarily accurate recovery, even under completely unrestricted adaptive selection rules. The underlying obstruction applies to every uniformly definable finite-parameter family, rather than only to polynomials.

A separate transfer identifies the geometry controlling adversarial noise for bounded convex queries. On an arbitrary convex body the limiting recovery error is an explicit extremal-radius problem. On a norm ball its first-order small-noise coefficient is exactly the Jung constant of the normed space. The sharp variation estimate for bounded convex functions used in this transfer is due to Kwon [98]; we do not claim that estimate as new.

7.4.1 A dimension obstruction for a fixed definable family

Fix an o-minimal expansion $\mathcal{R}$ of the ordered real field; "definable" permits real parameters in this one structure. The semialgebraic case suffices for all polynomial conclusions below. We use three standard facts: definable choice; the dimension bound $\dim f(A) \le \dim A$ for a definable map; and the small-frontier inequality

$$\dim(\overline{A} \setminus A) < \dim A \qquad (A \neq \varnothing \text{ definable}). \tag{7.4.1}$$

These are standard consequences of cell decomposition; see [96].

Let $K \subset \mathbb{R}^m$ be nonempty, compact and definable, and let $P \subset \mathbb{R}^p$ be a nonempty definable parameter set. A uniformly definable scalar query family is one definable map

$$F : P \times K \longrightarrow \mathbb{R}.$$

An n-query algorithm selects $a_j \in P$ as an arbitrary function of the preceding exact answers and observes $y_j = F(a_j, x)$. The coefficient selection rules and decoder are not assumed to be continuous, measurable or definable. Write

$$e_n(K, F) = \inf_{A} \sup_{x \in K} \|x - A(x)\|_2,$$

where the infimum runs over all such deterministic adaptive algorithms. Queries may be padded with any fixed member of the nonempty family, so "at most n" is equivalent to allowing n queries.

Theorem 7.4.1 (Uniformly definable information cannot compress dimension)**.** *If $n < \dim K$, then*

$$e_n(K, F) > 0. \tag{7.4.2}$$

The positive constant can depend on the entire family F, its parameter domain P, and K. No continuity assumption on the individual queries is needed.

Proof. Suppose instead that $e_n(K, F) = 0$. For every $0 < \varepsilon < 1$ there is an algorithm whose worst-case error is at most ε. This statement can be written as a finite first-order game:

$$\begin{aligned} &\exists a_1 \in P\ \forall y_1 \in \mathbb{R}\ \cdots\ \exists a_n \in P\ \forall y_n \in \mathbb{R}\ \exists z \in \mathbb{R}^m\ \forall x \in K : \\ &\qquad \left[\bigwedge_{j=1}^{n} F(a_j, x) = y_j\right] \implies \|x - z\|_2^2 \le \varepsilon^2. \end{aligned} \tag{7.4.3}$$

Impossible answer histories cause no problem: the antecedent is then false, and all later existential choices can be arbitrary. Conversely, a winning strategy for this formula is exactly an adaptive recovery algorithm with the asserted error. Thus the use of the quantified formula does not restrict the algorithms being considered.

Starting with the first existential quantifier, apply definable choice to the set of parameters for which the remaining quantified formula holds. Substitute the selected parameter into the remaining formula and repeat. This constructs a winning strategy simultaneously for all $\varepsilon \in (0, 1)$ in which every choice is definable in ε and the earlier answers. In particular the resulting decoder is a definable map

$$\Phi : (0, 1) \times \mathbb{R}^n \longrightarrow \mathbb{R}^m.$$

One may assign fixed values outside any intermediate winning domain; these extensions are definable. Form its output family

$$A = \{(\varepsilon, \Phi(\varepsilon, y)) : 0 < \varepsilon < 1,\ y \in \mathbb{R}^n\}.$$

It is definable, and $\dim A \le n+1$. For every fixed $x \in K$ and every ε, feed x to this strategy and let $y(\varepsilon, x)$ be its actual answer vector. Its winning property gives

$$\|\Phi(\varepsilon, y(\varepsilon, x)) - x\|_2 \le \varepsilon.$$

Consequently

$$\{0\} \times K \subset \overline{A} \setminus A.$$

Applying (7.4.1), we obtain

$$\dim K \le \dim(\overline{A} \setminus A) < \dim A \le n+1,$$

so $\dim K \le n$, a contradiction. The case $n = 0$ follows from the same proof with no query variables. □

Why arbitrary decision rules do not evade the proof. The theorem does not assume that the actual algorithm supplied by an adversary is definable. It uses the stronger statement that, if any winning finite-horizon strategy exists for each accuracy, the definable finite-horizon game has a uniformly definable winning strategy. It is this replacement strategy, not the originally supplied one, whose output family is subjected to dimension theory. The parameter set must remain fixed; a union of polynomial families of all degrees is not a single finite-parameter definable family.

Corollary 7.4.2 (A common degree bound restores the full query count)**.** *Fix integers $m \ge 1$, $D \ge 1$ and $n < m$. For adaptive real polynomial queries of degree at most D, there is a constant $c_{m,n,D} > 0$ such that the worst-case Euclidean recovery error on B_2^m is at least $c_{m,n,D}$. On RB_2^m it is at least $Rc_{m,n,D}$. Hence every such n-query algorithm on all of $\mathbb{R}^m$ has infinite worst-case error.*

The exact minimum number of degree-at-most-D queries required for arbitrary-precision recovery of a ball, or for uniformly bounded-error recovery of $\mathbb{R}^m$, is therefore m. The same count holds if the queries must also be globally convex polynomials.

Proof. Write a polynomial of degree at most D in the monomial basis, using its coefficient vector in $\mathbb{R}^{\binom{m+D}{D}}$ as parameter. This is a single semialgebraic family, so Theorem 7.4.1 gives the positive unit-ball constant. Replacing x by Rx preserves the degree bound, and rescaling the decoder divides the error by R. This proves the scaling assertion and the infinite-error conclusion. Restricting to convex polynomials cannot weaken the lower bound. The m coordinate queries are affine, globally convex, and recover the vector exactly. □

Theorem 7.4.3 (Algebraicity and decidability of fixed-degree adaptive error)**.** *For fixed integers $m, n \ge 0$ and $D \ge 1$, with $m \ge 1$, the optimal worst-case Euclidean error on B_2^m of n adaptive polynomial queries of degree at most D is a real algebraic number. It can in principle be computed exactly by quantifier elimination over the real closed field of real algebraic numbers. The same assertion holds with the additional restriction of global convexity. For $n < m$ the number is strictly positive; for $n \ge m$ it is zero. No practical complexity bound for this decision procedure is asserted.*

Proof. Use the coefficient parameterization from the preceding corollary. The winning formula (7.4.3), including the unit-ball condition and the squared Euclidean error bound, is a first-order formula with polynomial equalities and inequalities and rational coefficients. Real quantifier elimination therefore computes a semialgebraic description over $\mathbb{Q}$ of the set E of accuracies $\varepsilon \ge 0$ for which a winning strategy exists. The set E is upward closed, nonempty (the zero decoder gives error one), and its infimum is the optimal error. Endpoints of one-dimensional

semialgebraic sets over $\mathbb{Q}$ are real algebraic numbers, and they can be isolated effectively. The infimum need not be attained for this argument. Global convexity of a polynomial is expressed by the additional first-order condition

$$\forall x, v \in \mathbb{R}^m : \quad v^{\mathsf{T}} \nabla^2 p(x) v \geq 0.$$

Thus it can be included in the same quantifier-elimination procedure. The zero/positive dichotomy follows from Corollary 7.4.2 and coordinate queries. □

Corollary 7.4.4 (Fixed architectures and necessary complexity growth)**.** *Let K be a compact definable set of dimension d. Scalar neural networks of any fixed finite architecture with ReLU activations form a semialgebraic query family. Scalar networks of a fixed finite architecture with logistic activations form a family definable in the real exponential field. For either family, fewer than d adaptive queries have a strictly positive worst-case recovery error on K, even with unrestricted real weights and unrestricted adaptive decision rules.*

In particular, any sequence of fewer-than-m polynomial-query algorithms whose errors on B_2^m tend to zero must have maximum query degrees tending to infinity along every subsequence with vanishing error. Unbounded coefficients alone cannot substitute for this growth.

Proof. A fixed ReLU network is a finite composition of affine maps and $u \mapsto \max(0, u)$, hence is semialgebraic jointly in weights and input. Logistic activation $1/(1 + e^{-u})$ is definable in the real exponential field, which is o-minimal; see [97]. Apply Theorem 7.4.1. If the polynomial conclusion failed, some degree bound D would hold along a vanishing-error subsequence, contradicting Corollary 7.4.2. □

The fixed-architecture conclusion concerns recovery from scalar observations of an unknown input, not the expressive power of a network decoder supplied with the entire input. It does not settle the unrestricted continuous or real-analytic two-query question in $\mathbb{R}^3$, and it does not contradict logarithmic query counts when polynomial degrees or network architectures may grow.

7.4.2 Convex noise as an extremal-radius problem

Let $K \subset \mathbb{R}^m$ be a compact convex body with nonempty interior, and fix any norm $\|\cdot\|$ for the reconstruction loss. Let $\mathcal{C}(K)$ consist of the continuous convex functions on K with range in $[-1, 1]$. Observations have the form

$$y_j = f_j(x) + \xi_j, \qquad |\xi_j| \leq \delta,$$

where $0 < \delta < 1$ and every selected f_j obeys the range restriction on all of K. Let e_n^{ad} and e_n^{non} be the deterministic worst-case recovery errors for adaptive and nonadaptive algorithms respectively. For a nonempty compact set S put

$$\mathrm{rad}(S) = \inf_{z \in \mathbb{R}^m} \sup_{x \in S} \|x - z\|.$$

The centers are allowed to lie anywhere in the ambient space.

Lemma 7.4.5 (Convex indistinguishability criterion)**.** *For $x, y \in K$ and $0 < \delta < 1$, the following conditions are equivalent:*

$$|f(x) - f(y)| \leq 2\delta \quad \text{for every } f \in \mathcal{C}(K); \tag{7.4.4}$$

$$y - (1 - \delta)x \in \delta K, \qquad x - (1 - \delta)y \in \delta K. \tag{7.4.5}$$

In testing the first condition, restrictions of globally convex hinge functions suffice.

Proof. If $y = (1-\delta)x + \delta z$ with $z \in K$, convexity and the range restriction give

$$f(y) - f(x) \le \delta(f(z) - f(x)) \le 2\delta.$$

The reversed containment gives the reverse inequality. For the converse, suppose $y-(1-\delta)x \notin \delta K$. Strict separation supplies a linear functional u such that

$$u(y) - (1-\delta)u(x) > \delta h_K(u), \qquad h_K(u) = \max_{z\in K} u(z).$$

Set $a = h_K(u) - u(x)$. Here $a > 0$: if $a = 0$, the displayed strict inequality would imply $u(y) > h_K(u)$, impossible for $y \in K$. The globally convex function

$$f(z) = -1 + \frac{2(u(z) - u(x))_+}{a}$$

has range in $[-1, 1]$ on K, satisfies $f(x) = -1$, and has $f(y) - f(x) > 2\delta$. The other failed containment is treated by interchanging x and y. This is also a direct proof of the part of Kwon's sharp variation estimate needed here. □

Theorem 7.4.6 (Exact limiting error on every convex body)**.** *Define*

$$\mathcal{R}_K(\delta) = \max\left\{\mathrm{rad}(S) : \varnothing \ne S \subset K \text{ compact}, \quad S - (1-\delta)S \subset \delta K\right\}. \tag{7.4.6}$$

Then

$$\inf_{n\ge 0} e_n^{\mathrm{ad}} = \inf_{n\ge 0} e_n^{\mathrm{non}} = \mathcal{R}_K(\delta). \tag{7.4.7}$$

For every $\varepsilon > 0$, a finite nonadaptive family of globally convex hinge queries attains error at most $\mathcal{R}_K(\delta)+\varepsilon$. The same infimum holds if queries are restricted to globally convex real-analytic functions, or to globally convex polynomials of unrestricted degree.

Proof. The hyperspace of nonempty compact subsets of K, equipped with the Hausdorff metric, is compact. The radius is continuous (indeed one-Lipschitz), and the containment in (7.4.6) is closed under Hausdorff limits. Thus the maximum exists.

If S satisfies that containment, the lemma shows that the range of every admissible query on S has width at most 2δ. Against an arbitrary adaptive algorithm, answer each selected query by the midpoint of its maximum and minimum on S. This answer is within δ of the true answer for every $x \in S$, so the same transcript remains possible for all of S. The worst-case error is at least $\mathrm{rad}(S)$, proving the lower bound.

For the upper bound, fix $\varepsilon > 0$ and consider the compact hyperspace subset

$$\mathcal{B} = \{S \subset K : \ S \ne \varnothing \text{ compact}, \ \mathrm{rad}(S) \ge \mathcal{R}_K(\delta) + \varepsilon\}.$$

Every $S \in \mathcal{B}$ fails the containment, so by the lemma there is a globally convex hinge f_S, with range in $[-1, 1]$ on K, for which

$$\max_S f_S - \min_S f_S > 2\delta.$$

For a fixed continuous function the oscillation is continuous on the hyperspace. The sets where these strict inequalities hold therefore form an open cover of $\mathcal{B}$. Choose a finite subcover, corresponding to $f_1, \dots, f_N$, and query all of them. For any possible answer vector y, the feasible set

$$C_y = \{x \in K : |f_i(x) - y_i| \le \delta, 1 \le i \le N\}$$

is compact and nonempty. Each f_i has oscillation at most 2δ on C_y, so $C_y \notin \mathcal{B}$. Outputting a Chebyshev center of C_y gives error less than $\mathcal{R}_K(\delta) + \varepsilon$. If $\mathcal{B}$ is empty, no query is needed for this accuracy.

For the regularity variants it suffices to replace each hinge by arbitrarily close uniformly approximating globally convex polynomials on K, while retaining the range bound. Here is a one-dimensional construction, applied to the affine argument of the hinge. Approximate the hinge uniformly on a compact interval by a smooth globally convex function h. Approximate the continuous function $\sqrt{h''}$ on that interval by a polynomial q, and integrate q^2 twice, matching the value and derivative of h at one point. The resulting polynomial is globally convex and uniformly approximates h on the interval. After composition with the linear functional it is globally convex on $\mathbb{R}^m$. If its approximation error to a range-normalized hinge is at most η, division by $1+\eta$ puts its range back in $[-1,1]$ on K and keeps the approximation error tending to zero. For each bad set choose an approximation close enough to retain its strict oscillation inequality; the same compact subcover argument then applies. These polynomials are also globally real analytic. □

Geometry rather than pairwise half-distance. The lower bound is an entire-set obstruction. A common answer must be compatible with every point of S, and the reconstruction error is its circumradius, which can strictly exceed half its diameter. This is why the cube calculation alone does not determine Euclidean-ball noise constants. Equation (7.4.7) also concerns the infimum over all finite query counts: it does not say that adaptivity never improves a fixed finite-query error.

7.4.3 The Jung constant is the exact small-noise coefficient

Let $X = (\mathbb{R}^m, \|\cdot\|)$ be any finite-dimensional normed space and let B_X be its closed unit ball. Use the same norm for the loss. Define the Jung constant in the radius/diameter convention by

$$J(X) = \sup\{\mathrm{rad}(S) : S \subset X \text{ nonempty compact, } \mathrm{diam}(S) \le 1\}.$$

This convention has $J(\ell_\infty^m) = 1/2$ and $J(\ell_2^m) = \sqrt{m/(2m+2)}$. In the theorem below put $e_\infty(\delta) = \inf_n e_n^{\mathrm{ad}} = \inf_n e_n^{\mathrm{non}}$ for bounded convex queries on B_X.

Theorem 7.4.7 (Jung law for adversarial convex information)**.** *For every $0 < \delta < 1$,*

$$\frac{J(X)\delta}{1+J(X)\delta} \le e_\infty(\delta) \le \frac{2J(X)\delta}{2-\delta}. \tag{7.4.8}$$

In particular,

$$\lim_{\delta\downarrow 0} \frac{e_\infty(\delta)}{\delta} = J(X). \tag{7.4.9}$$

The same result holds for globally convex polynomials of unrestricted degree and for globally convex real-analytic queries.

Proof. Let $S \subset B_X$ satisfy the containment in (7.4.6), and put $q = 1-\delta$. For $x, y \in S$,

$$\|x - qy\| \le \delta, \qquad \|y - qx\| \le \delta.$$

Subtracting these vectors and using the triangle inequality gives

$$(1+q)\|x-y\| \le 2\delta.$$

Thus $\mathrm{diam}(S) \le 2\delta/(2-\delta)$, and the upper bound follows from the definition of $J(X)$ and Theorem 7.4.6.

For the lower bound take any compact S_0 of diameter at most one and radius r, and translate a Chebyshev center to zero. Then $\|x\| \le r$ for $x \in S_0$. Set

$$a = \frac{\delta}{1+\delta r}, \qquad S = aS_0.$$

Since $ar < 1$, the set S lies in B_X. For $x, y \in S_0$,

$$\|ax - (1-\delta)ay\| \le a(\|x-y\| + \delta\|y\|) \le \delta.$$

Hence S is admissible in (7.4.6), giving $e_\infty(\delta) \ge ar$. Let $r \uparrow J(X)$. The two bounds have the same first-order coefficient as $\delta \downarrow 0$. The regularity variants follow from Theorem 7.4.6. □

Theorem 7.4.8 (The exact Euclidean convex-noise floor)**.** *For range-normalized convex queries on the Euclidean unit ball in $\mathbb{R}^m$, with Euclidean reconstruction loss, the exact unlimited-query minimax error is*

$$e_\infty(\delta) = \frac{\delta}{\sqrt{\delta^2 + 2(1-\delta)(m+1)/m}}, \qquad 0 < \delta < 1. \tag{7.4.10}$$

A centered regular m-simplex realizes the extremal ambiguity. The formula holds for the infimum over finite adaptive counts and for the infimum over finite nonadaptive counts, including the globally convex polynomial and real-analytic subclasses. In particular,

$$e_\infty(\delta) = \sqrt{\frac{m}{2m+2}}\,\delta + O_m(\delta^2).$$

For $m > 1$ and every $0 < \delta < 1$, this is strictly larger than the two-antipodal-point obstruction $\delta/(2-\delta)$.

Proof. Put $q = 1 - \delta$. For the lower bound let $u_0, \dots, u_m$ be a centered regular simplex on the unit sphere, so $\langle u_i, u_j\rangle = -1/m$ for $i \ne j$ and $\sum_i u_i = 0$. Set

$$r = \frac{\delta}{\sqrt{1 + q^2 + 2q/m}}, \qquad S = \{ru_0, \dots, ru_m\}.$$

For distinct vertices, $\|ru_i - qru_j\|_2 = \delta$; for equal vertices that norm is $\delta r \le \delta$. Thus S is admissible in Theorem 7.4.6, and its circumradius is r.

For the upper bound let $S \subset B_2^m$ be any nonempty compact set satisfying $S - qS \subset \delta B_2^m$. Write c and R for its Euclidean Chebyshev center and radius. If $R = 0$ there is nothing to prove. The contact-point characterization of a smallest enclosing ball, followed by Carathéodory's theorem, gives $2 \le N \le m+1$ contact points

$$x_i = c + Ru_i \in S, \quad \|u_i\|_2 = 1, \qquad \alpha_i > 0, \quad \sum_i \alpha_i = 1, \quad \sum_i \alpha_i u_i = 0.$$

The contact characterization follows by separation: if zero were outside the convex hull of the contact directions, moving the center a small distance toward a separating direction would strictly decrease the enclosing radius.

For each fixed i, multiply $\|x_i - qx_j\|_2^2 \le \delta^2$ by α_j and sum over $j \ne i$. Expansion, using $\sum_j \alpha_j u_j = 0$, gives

$$\begin{aligned}(1-\alpha_i)\delta^2\|c\|_2^2 + R^2(1 + q^2 - \alpha_i\delta^2) \\ + 2\delta R(1 - \delta\alpha_i)\langle c, u_i\rangle \le (1-\alpha_i)\delta^2.\end{aligned} \tag{7.4.11}$$

Multiply this inequality by $\alpha_i/(1-\delta\alpha_i)$ and sum over i. These weights cancel the center cross term exactly. Put

$$U=\sum_i \frac{\alpha_i}{1-\delta\alpha_i}, \qquad V=\sum_i \frac{\alpha_i^2}{1-\delta\alpha_i}=\frac{U-1}{\delta}, \qquad A=U-V, \quad B=(1+q^2)U-\delta^2 V.$$

Here $A>0$ because $N\ge 2$, and $B>0$. Equation (7.4.11) becomes

$$R^2 B \le \delta^2 A(1-\|c\|_2^2) \le \delta^2 A.$$

The expressions simplify to

$$A=\frac{1-qU}{\delta}, \qquad B=q(1+q)U+\delta.$$

The function A/B is decreasing in U, since its derivative is $-2q/(\delta B^2)<0$. Convexity of $a\mapsto a/(1-\delta a)$ and $N\le m+1$ yield

$$U\ge \frac{N}{N-\delta}\ge \frac{m+1}{m+1-\delta}.$$

At the last lower bound for U, direct substitution gives

$$\frac{A}{B}\le \frac{1}{1+q^2+2q/m}, \qquad R^2\le \frac{\delta^2}{1+q^2+2q/m}.$$

This matches the regular-simplex construction and proves the exact formula via Theorem 7.4.6. Since $1+q^2+2q/m=\delta^2+2(1-\delta)(m+1)/m$, it is the stated expression. The small-noise expansion follows immediately. For $m>1$, its squared denominator is strictly smaller than $(2-\delta)^2=1+q^2+2q$, proving the strict comparison. □

Status and remaining problems. The fixed-definable-family theorem and the convex-noise radius formula are new results claimed by this manuscript, with complete proofs above. The bounded-convex variation estimate and the classical Jung constants are existing ingredients. Quantitative lower bounds for $c_{m,n,D}$, finite-query attainment of the convex-noise infimum, and the unrestricted two-real-analytic-query problem in $\mathbb{R}^3$ remain open here. None of these unresolved statements is counted as a solved problem.

7.5 Irregular sampling and the scalar Gaussian tail boundary

The earlier matrix construction requires a quantitative upper bound on the density of the selected terminal times. In one dimension that restriction can be removed completely. The reason is geometric: the poles of a scalar Blaschke product lie on the negative real axis, so a contour with nonzero imaginary part can approach that axis without crossing a pole. A narrower contour absorbs the whole Blaschke product using only the sum of its positive zeros. No regular variation or separation of the terminal times is needed.

Let $\mathcal{N}$ be an infinite subset of the positive integers. Conditionally on $(A,V)\in\mathbb{R}\times[0,\infty)$ let the X_i be independent $N(A,V)$ variables, and put $S_n=\sum_{i=1}^n X_i$. We observe the separate population laws of the S_n, $n\in\mathcal{N}$. Choose a sufficiently large constant E and define

$$L_1(v)=\log(E+v), \qquad L_{j+1}(v)=\log L_j(v), \qquad W_{m,a}(v)=\frac{\sqrt{v}}{L_1(v)\cdots L_{m-1}(v)L_m(v)^a}.$$

Let $\mathcal{C}_{m,a}$ consist of all joint mixing laws for which $\mathbb{E}\exp\{cW_{m,a}(V)\}<\infty$ for some $c>0$. There is no restriction on dependence between A and V.

Theorem 7.5.1 (Complete scalar design–tail classification)**.** *For every fixed integer $m \ge 1$, every $a \ge 0$, and every infinite terminal set $\mathcal{N}$, the selected partial-sum laws identify every mixing law in $\mathcal{C}_{m,a}$ if and only if*

$$a \le 1 \quad \text{or} \quad \sum_{n\in\mathcal{N}} \frac{1}{n} = \infty. \tag{7.5.1}$$

When both conditions fail, two distinct absolutely continuous witnesses can have the same mean marginal, the same variance marginal, and all the same mixed polynomial moments. All these moments are finite. The witnesses satisfy $\mathbb{E}\exp\{cW_{m,a}(V)\} < \infty$ for every $c > 0$, and can obey any prescribed lower bound $V \ge b_0 > 0$.

The positive alternatives in (7.5.1) were proved in the preceding manuscripts. The new assertion is the negative alternative for *every* convergent harmonic subseries, including highly irregular ones. We give the positive argument as well so that the classification is self-contained. This is a closure of a gap in the preceding manuscripts, not an assertion that a named external conjecture has been solved. The relevant classical moment theory includes de Jeu's quasi-analytic weights [79].

7.5.1 A flat Laplace transform with arbitrary summable zeros

For large R, write

$$P_m(R) = \prod_{j=1}^{m} \log_j R, \qquad D_{m,p}(R) = P_m(R)(\log_m R)^p, \qquad p > 0.$$

These expressions are only used for sufficiently large arguments; elsewhere they can be extended smoothly and positively.

Lemma 7.5.2 (Scalar logarithmic regularizer)**.** *For every $m \ge 1$ and $p > 0$ there is a function Q, holomorphic off the negative real axis, such that $Q(\overline{w}) = \overline{Q(w)}$, $\operatorname{Re} Q(w) \ge 0$ for $\operatorname{Re} w \ge 0$, and*

$$\operatorname{Re} Q(w) \ge \frac{c|w|}{D_{m,p}(|w|)} \qquad \left(|w| \ge R_0, \quad |\arg w| \le \frac{\pi}{2} + \frac{c}{P_m(|w|)}\right). \tag{7.5.2}$$

Every fixed derivative of Q has at most polynomial growth in $|w| + |w|^{-1}$ on any slightly narrower such sector.

Proof. Choose K sufficiently large, use principal logarithms, and put

$$J_1(w) = \log(K + w), \qquad J_{j+1}(w) = \log(K + J_j(w)), \qquad Q(w) = \frac{w}{J_m(w)^p}.$$

All the logarithms have positive real part on the right half-plane. Their arguments have the sign of $\arg w$. More explicitly, $|\arg J_1(w)| \le |\arg w|/\log K$, since $\operatorname{Re} J_1 \ge \log K$ and $|\operatorname{Im} J_1| = |\arg(K + w)| \le |\arg w|$; iteration gives $|\arg J_m(w)| \le |\arg w|/(\log K)^m$. Enlarging K therefore makes $p|\arg J_m(w)| \le |\arg w|$ there. Thus the argument of $Q(w)$ lies between zero and the argument of w, proving nonnegativity. For bounded real θ, successive expansions of the logarithms give, uniformly together with their first angular derivatives,

$$|J_m(Re^{i\theta})| \sim \log_m R, \qquad \arg J_m(Re^{i\theta}) = \frac{\theta}{P_m(R)}(1 + o(1)).$$

Near $|\theta| = \pi/2$ the subtraction of $p \arg J_m$ therefore leaves an angular margin of a positive constant times $1/P_m(R)$. Allowing a sufficiently small overshoot of $|\theta|$ preserves half this margin.

Multiplying the resulting cosine bound by $|Q(w)| \asymp R/(\log_m R)^p$ proves (7.5.2). Away from the two sector endpoints the cosine has a larger lower bound. The derivative assertion follows either by differentiating the displayed formulas or by Cauchy's formula on smaller sectors. All branch cuts stay on the negative real axis because each nested logarithm maps the upper half-plane into itself and is positive on the positive real axis; reflection handles the lower half-plane. □

Lemma 7.5.3 (Summable zeros with a near-critical tail)**.** *Let $0 < \lambda_j \le M$ be distinct, with $\Lambda = \sum_j \lambda_j < \infty$. For every $m \ge 1$ and $p > 0$ there is a nonzero real Schwartz function f supported in $[0, \infty)$ whose Laplace transform vanishes at every λ_j and whose integral is zero. For each integer $k \ge 0$ there are positive constants C_k, c_k such that*

$$|f^{(k)}(v)| \le C_k \exp\{-c_k W_{m,1+p}(v)\}, \qquad v \ge 0. \tag{7.5.3}$$

Proof. The product

$$B(z) = \prod_j \frac{z - \lambda_j}{z + \lambda_j}$$

converges locally uniformly in the right half-plane, is nonzero identically, and has modulus at most one there. It continues meromorphically off zero with poles only at $-\lambda_j$. On the imaginary axis, away from zero, its modulus is one. Put

$$F(z) = B(z) \exp\{-\sqrt{z} - Q(1/z)\}, \qquad \operatorname{Re} z > 0.$$

The principal square root and Lemma 7.5.2 imply $|F(z)| \le e^{-\operatorname{Re}\sqrt{z}}$. In particular F is bounded and is not identically zero.

We first justify ordinary Fourier inversion. Logarithmic differentiation of B, followed by differentiation of each term, gives

$$\left| \left(\frac{B'}{B} \right)^{(r)} (iy) \right| \le C_r \Lambda |y|^{-r-2}, \qquad y \ne 0.$$

Indeed, each difference of inverse powers can be integrated over $[-\lambda_j, \lambda_j]$, where the distance to iy is at least $|y|$. The recursion for $B^{(k)}/B$ supplies polynomial bounds for every derivative of $B(iy)$ in $|y| + |y|^{-1}$. Equation (7.5.2) supplies damping $\exp\{-c/(|y| D_{m,p}(1/|y|))\}$ as $y \to 0$, while $e^{-\sqrt{z}}$ supplies $e^{-c\sqrt{|y|}}$ at infinity. Consequently $F(iy)$, extended by zero at $y = 0$, is a Schwartz function flat at zero. Define

$$f(v) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{iyv} F(iy)\, dy.$$

It is real, Schwartz, and nonzero. Shifting the line to $x + i\mathbb{R}$, $x > 0$, proves that $f(v) = 0$ for $v < 0$: the bound $\operatorname{Re}\sqrt{x + iy} \ge \sqrt{|y|/2}$ makes the integrals uniformly bounded before multiplication by e^{xv}, and then $x \to \infty$. The same contour identity, or Cauchy's formula followed by Fubini, proves $\int_0^\infty e^{-zv} f(v)\, dv = F(z)$ for $\operatorname{Re} z > 0$. It also gives $\int f = \lim_{z \downarrow 0} F(z) = 0$.

Here is the new estimate that removes all density assumptions on the zero set. If $z = -\eta + iy$, $r = |y|$, and $0 \le \eta \le r/2$, then

$$\begin{aligned} \log |B(z)| &= \frac{1}{2} \sum_j \log \left(1 + \frac{4\eta\lambda_j}{(\lambda_j - \eta)^2 + r^2} \right) \\ &\le C\eta \sum_j \frac{\lambda_j}{\lambda_j^2 + r^2} \le \frac{C\Lambda\eta}{r^2}. \end{aligned} \tag{7.5.4}$$

The first inequality uses $(\lambda-\eta)^2+r^2\ge c(\lambda^2+r^2)$ when $\eta\le r/2$. Choose a smooth positive function $\eta(r)$ such that, near zero,

$$\eta(r)=\frac{\varepsilon r}{D_{m,p}(1/r)}. \tag{7.5.5}$$

Extend it to a sufficiently small positive constant for large r, with bounded derivative and $\eta(r)\le r/2$. The constant $\varepsilon>0$ may depend on Λ,m,p. Since $D_{m,p}(R)\ge P_m(R)$ eventually, every inverse point $(-t\eta(r)+iy)^{-1}$, $0\le t\le 1$, lies in the sector of (7.5.2) for small r, and has modulus comparable to $1/r$. Equations (7.5.4) and (7.5.5) show that choosing ε small enough absorbs the whole product into half the Q damping. Thus uniformly along this homotopy, for small r,

$$|F(-t\eta(r)+iy)|\le C\exp\left\{-\frac{c}{rD_{m,p}(1/r)}\right\}.$$

For large r the estimate is $Ce^{-c\sqrt{r}}$; on the remaining compact set it is bounded. These latter bounds follow directly from (7.5.4), the boundedness of $Q(1/z)$ there, and the square-root factor.

Deform the upper and lower imaginary half-axes separately to $z=-\eta(|y|)+iy$. This deformation never crosses a negative real pole or a branch cut. The connecting arcs at zero vanish by the displayed flatness estimate, and the arcs at infinity vanish by square-root damping. The line element has bounded modulus because η' is bounded. For $v\ge 0$ the deformed integral therefore has, near zero, absolute integrand bounded by

$$C\exp\left\{-c\left(\frac{vr}{D_{m,p}(1/r)}+\frac{1}{rD_{m,p}(1/r)}\right)\right\}.$$

Uniformly in small r and large v, the expression in parentheses is at least $cW_{m,1+p}(v)$. On $\frac14\log v\le\log(1/r)\le\log v$ all the fixed-depth logarithms are comparable with the corresponding logarithms of v, and $(vr+r^{-1})\ge 2\sqrt{v}$ proves this assertion. If $\log(1/r)<\frac14\log v$, the first term is at least $v^{3/4}$ divided by logarithmic factors. If $\log(1/r)>\log v$, $r^{-1}/D_{m,p}(1/r)$ is increasing and is at least $v/D_{m,p}(v)$. Both are larger than the required bound. For r bounded away from zero, $\eta(r)$ has a positive lower bound and gives exponential decay in v. Reserving half the damping for integrability proves (7.5.3) for $k=0$. Differentiating the inverse integral k times introduces only the factor z^k and gives the same conclusion for every fixed k. □

Proof of Theorem 7.5.1. For the negative assertion put $\lambda_n=(2n)^{-1}$ and choose $0<p<a-1$. Let f be the preceding lemma's function and choose a nonzero real even $h\in C_c^\infty(\{t:1<|t|<2\})$. Define

$$g(a,v)=\frac{1}{2\pi}\int e^{-ita}h(t)t^2f(vt^2)\,dt.$$

This is a nonzero real Schwartz function supported on $v\ge 0$. Its Fourier–Laplace transform is

$$\int e^{ita-sv}g(a,v)\,da\,dv=h(t)F(s/t^2)\quad(t\ne0,s>0),$$

and is zero at $t=0$. At every observation argument $s=t^2/(2n)$ it vanishes. Its v marginal and all joint polynomial moments vanish because its partial Fourier transform is zero near $t=0$. Its a marginal vanishes because $\int f=0$. The normalized positive and negative parts of g are therefore the desired distinct mixing probabilities.

To check the quantitative tail, differentiate under the t integral and integrate by parts there. Every resulting term is a bounded coefficient times $v^jf^{(j)}(vt^2)$ on $1\le|t|\le2$. Equation (7.5.3) absorbs the polynomial factors and gives, for every integer M, constants $C_M,c_M>0$ with

$$|g(a,v)|\le C_M(1+|a|)^{-M}e^{-c_MW_{m,1+p}(v)}.$$

Since $W_{m,1+p}(v)/W_{m,a}(v) \to \infty$, the required integrability holds for every $c > 0$. Translation of v by b_0 multiplies the transform by $e^{-b_0 s}$ and preserves all zeros, equal marginals, equal moments, and tail properties.

For completeness, suppose two candidate laws agree and let ρ be their difference. For fixed $t \neq 0$ the bounded holomorphic function $H_t(s) = \int e^{ita-sv}\, d\rho(a,v)$ has zeros $t^2/(2n)$, $n \in \mathcal{N}$. If $\sum 1/n = \infty$, the half-plane Blaschke condition forces $H_t \equiv 0$. Indeed, map the half-plane to the disk and apply Jensen's formula; the zeros of a nonzero bounded holomorphic function must satisfy $\sum_j \operatorname{Re} z_j/(1+|z_j|^2) < \infty$.

If $a \leq 1$, let τ be the sum of the two variance marginals. The tail assumption gives

$$\left(\int v^k\, d\tau(v)\right)^{1/(2k)} \leq Ck \log k \cdots \log_{m-1} k\, (\log_m k)^a$$

for large k. One obtains this bound by maximizing $y^{2k} \exp\{-cy/[L_1(y^2)\cdots L_{m-1}(y^2)L_m(y^2)^a]\}$, with $y = \sqrt{v}$. The Stieltjes–Carleman series diverges. Since the zeros of H_t accumulate at zero, Taylor expansion from the positive real axis successively shows that every derivative of H_t at zero vanishes; all derivatives exist by the moment bound. The resulting finite complex measure in v, dominated by τ, has every moment zero. Apply moment determinacy to τ plus and minus its real and imaginary parts to conclude it is zero. Thus $H_t \equiv 0$ again. Continuity includes $t = 0$, and Fourier–Laplace uniqueness gives $\rho = 0$ in both cases. □

Remark 7.5.4 (Why the same proof does not settle the matrix gap). The scalar inverse variable $1/z$ has nonzero imaginary part all along the contour. In dimension greater than one, an inverse quadratic form $u^T T^{-1} u$ can be negative real on a shifted matrix contour even though T is nonsingular. A scalar Blaschke factor composed with this form then has genuine pole surfaces. The proof above does not cross or remove those surfaces. The exact irregular-design classification for full matrix covariance mixtures therefore remains open in this manuscript.

7.6 Exact sparse designs for every fixed stable shape

The 223-page manuscript classified full partial-sum observations for fixed stable shape, but stated its sparse sampling criteria only as sufficient. The next theorem proves their necessity as well. The construction transfers the Gaussian Blaschke method to nonreal zero sets and, at index one, to a frequency-dependent imaginary shift.

Fix $0 < \alpha \leq 2$ and $-1 \leq \beta \leq 1$, and use the stable parameterization

$$\widehat{K}_{a,b}(t) = e^{ita - b\psi_{\alpha,\beta}(t)}, \qquad \psi_{\alpha,\beta}(t) = \begin{cases} |t|^\alpha[1 - i\beta \tan(\pi\alpha/2) \operatorname{sgn} t], & \alpha \neq 1, \\ |t|[1 + i(2\beta/\pi) \operatorname{sgn} t \log |t|], & \alpha = 1, \end{cases}$$

with $\psi(0) = 0$, $a \in \mathbb{R}$, and $b \geq 0$. This is Nolan's 1-parameterization [99]. Draw (A, B) from an arbitrary joint mixing law and conditionally draw the X_i independently from $K_{A,B}$. Only the separate laws of $S_n = \sum_{i=1}^n X_i$, $n \in \mathcal{N}$, are supplied.

Theorem 7.6.1 (Sharp stable sampling classification). *Let $\mathcal{N} \subset \mathbb{N}$. Identification of every Borel mixing law on $\mathbb{R} \times [0, \infty)$ holds under exactly the following conditions:*

1. *If $\alpha \neq 1$, it holds if and only if*
$$\sum_{n \in \mathcal{N}} n^{-|1-\alpha|} = \infty. \tag{7.6.1}$$

2. *If $\alpha = 1$ and $\beta \neq 0$, it holds if and only if*
$$\sum_{n \in \mathcal{N}} \frac{1}{1 + (\log n)^2} = \infty. \tag{7.6.2}$$

3. If $\alpha = 1$ and $\beta = 0$, no terminal set identifies every mixing law, even $\mathcal{N} = \mathbb{N}$.

In either of the first two cases, whenever the displayed series converges the ambiguity has absolutely continuous witnesses with the same coordinate marginals and all the same mixed polynomial moments, all finite. Any fixed constraint $B \geq b_0 > 0$ can be imposed.

The Gaussian case $\alpha = 2$ and the sufficient directions are inherited. The new results are the converse for the other power-law indices and the converse for the skew index-one family. No mixing over α or β is allowed. The observation problem belongs to the exchangeable partial-sum framework of Lin [7]; we make no claim that this exact sparse-design statement was a named conjecture there.

7.6.1 A boundary-regular Blaschke construction

Lemma 7.6.2 (Three families of zero sets)**.** *Suppose the distinct points a_j belong to one of the following three classes:*

1. *$a_j = c\lambda_j$, where $\operatorname{Re} c > 0$, $0 < \lambda_j \leq M$, and $\sum_j \lambda_j < \infty$;*
2. *$a_j = c\lambda_j$, where $\operatorname{Re} c > 0$, $\lambda_j \geq c_0 > 0$, $\lambda_j \to \infty$, and $\sum_j \lambda_j^{-1} < \infty$;*
3. *$a_j = 1 + iy_j$, where $|y_j| \to \infty$ and $\sum_j (1 + y_j^2)^{-1} < \infty$.*

Finite sets are also permitted. There is a nonzero bounded holomorphic function B on the right half-plane with these zeros, $|B| \leq 1$, and unimodular boundary values on the imaginary axis except possibly at zero in the first case. In the first case every boundary derivative grows at most polynomially in $|y| + |y|^{-1}$; in the other cases it grows at most polynomially in $1 + |y|$.

Proof. All three assumptions imply the half-plane Blaschke condition $\sum \operatorname{Re} a_j/(1 + |a_j|^2) < \infty$. The usual disk Blaschke product after the map $(z - 1)/(z + 1)$ gives

$$B(z) = \prod_j \gamma_j \frac{z - a_j}{z + \overline{a_j}}, \qquad |\gamma_j| = 1,$$

with the standard convergence normalization. The same product converges locally through each part of the imaginary axis away from its possible accumulation point zero; its boundary modulus is one. These assertions can also be checked directly by expanding each normalized factor for small a_j, large ray points a_j, or large vertical points, respectively.

Here are explicit derivative estimates, which will justify all subsequent inversions. Write $Q_B = B'/B$. On the imaginary axis

$$Q_B(z) = \sum_j \left(\frac{1}{z - a_j} - \frac{1}{z + \overline{a_j}} \right).$$

For each $r \geq 0$, an individual derivative is at most

$$\frac{C_r \operatorname{Re} a_j}{[(\operatorname{Re} a_j)^2 + (y - \operatorname{Im} a_j)^2]^{(r+2)/2}} \quad \text{at } z = iy.$$

For $a_j = c\lambda_j$, the denominator inside brackets is bounded below by a positive constant times $y^2 + \lambda_j^2$. In the first case summing gives a bound $C_r(\sum \lambda_j)|y|^{-r-2}$. In the second case it gives a uniform constant, since each summand is at most $C_r \lambda_j^{-r-1}$ and $\lambda_j \geq c_0$. In the vertical case use

$$\frac{1}{1 + (y - y_j)^2} \leq \frac{2(1 + y^2)}{1 + y_j^2}$$

to bound every derivative of Q_B by $C_r(1+y^2)$. Recursion from $B' = Q_B B$, together with $|B(iy)| = 1$, proves the asserted polynomial derivative bounds. The convergence of the displayed derivative series is uniform on the specified compact boundary intervals, so these differentiations are legitimate. □

Proof of Theorem 7.6.1. The data for S_n/n are values of the mixing Fourier–Laplace transform at

$$(t, z_n(t)), \qquad z_n(t) = n\psi_{\alpha,\beta}(t/n).$$

For $\alpha \neq 1$ these points are

$$z_n(t) = |t|^\alpha(1 - i\kappa \operatorname{sgn} t)n^{1-\alpha}, \qquad \kappa = \beta \tan(\pi\alpha/2).$$

Their Blaschke sum diverges exactly when (7.6.1) holds. For $\alpha = 1$, put $k = 2\beta/\pi$. The points are

$$z_n(t) = |t| + ikt \log|t| - ikt \log n,$$

whose Blaschke sum diverges exactly under (7.6.2) when $k \neq 0$. The bounded-holomorphic uniqueness argument from the preceding theorem, followed by Fourier–Laplace uniqueness, proves both sufficient directions without any moment assumption.

For necessity first suppose $\alpha \neq 1$ and the series converges. Let B_+ be the preceding lemma's Blaschke product with zeros

$$a_n = (1 - i\kappa)n^{1-\alpha}, \qquad n \in \mathcal{N}.$$

These are the first family if $\alpha > 1$ and the second family if $\alpha < 1$. For $t > 0$ and $\operatorname{Re} s > 0$ define

$$F_t(s) = B_+(s/t^\alpha) \exp\{-\sqrt{s/t^\alpha} - 1/\sqrt{s/t^\alpha}\}. \tag{7.6.3}$$

It is bounded and holomorphic in s, is not identically zero, and vanishes at every $s = z_n(t)$.

If $\alpha = 1$, $k \neq 0$, and the logarithmic series converges, use instead the product with zeros $a_n = 1 - ik \log n$, the third family of the lemma, and define

$$F_t(s) = B_+(s/t - ik \log t) \exp\{-\sqrt{s/t} - 1/\sqrt{s/t}\}, \qquad t > 0. \tag{7.6.4}$$

The shift inside B_+ is purely imaginary, so boundedness on the right half-plane is preserved. At an observation argument, $z_n(t)/t - ik \log t = 1 - ik \log n$, proving the required zeros. The regularizing factor is deliberately *not* shifted: this ensures flatness at $s = 0$ and will preserve the mean marginal.

In both cases set $F_{-t}(s) = \overline{F_t(\bar{s})}$ for $t > 0$ and choose a nonzero real even $h \in C_c^\infty(\{t : 1 < |t| < 2\})$. For t in this compact support, Lemma 7.6.2 and the chain rule show that every derivative in t and y of $F_t(iy)$ has at most polynomial factors before multiplication by

$$\exp\{-c\sqrt{|y|} - c/\sqrt{|y|}\}.$$

Thus $h(t)F_t(iy)$ is Schwartz in (t, y), compactly supported in t, and flat at $y = 0$. In the ray case this uses the lemma's possible polynomial singularity at zero; in the vertical case the imaginary shift ranges over a fixed compact interval and the same estimates remain uniform. Define

$$G(t, b) = \frac{h(t)}{2\pi} \int_{\mathbb{R}} e^{iyb} F_t(iy)\, dy, \qquad g(a, b) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ita} G(t, b)\, dt.$$

The function g is real, nonzero, and Schwartz. To see the support in $b \geq 0$, for each fixed t shift the s contour to $x + i\mathbb{R}$. The Blaschke factor remains bounded by one, including after the pure

imaginary shift, while the square-root factor gives an integrable bound independent of $x \geq 0$. Letting $x \to \infty$ forces the inverse transform to vanish for $b < 0$. Fourier inversion and the same elementary half-plane contour argument give

$$\int e^{ita-sb} g(a,b)\, da\, db = h(t)F_t(s), \qquad \operatorname{Re} s > 0.$$

Every selected observation transform is therefore zero.

The b marginal and all mixed moments of g vanish because $G(t,b)$ is zero near $t = 0$. Its a marginal vanishes as well: boundedness of the Blaschke factor and the unshifted $e^{-1/\sqrt{s/t^{\alpha}}}$ or $e^{-1/\sqrt{s/t}}$ force $F_t(s) \to 0$ as $s \downarrow 0$. Dominated convergence in the Schwartz inverse transform justifies passing to this limit. Normalizing the positive and negative parts of g produces the claimed witnesses. Replacing B by $B + b_0$ multiplies their signed transform by $e^{-b_0 s}$ and preserves all assertions.

Finally, for the symmetric Cauchy case take independent standard symmetric Cauchy variables $Z_0, Z_1, \ldots$ and constants $b, c > 0$. The sequence $X_j = cZ_0 + bZ_j$ and an iid Cauchy sequence of scale $b + c$ have identical partial-sum laws for every n but different joint laws: their pairwise differences have scales $2b$ and $2(b + c)$, respectively. Thus no terminal subset suffices. This last counterexample is recalled and is not counted as new. □

Example 7.6.3 (Different sparse clocks distinguish the stable shapes)**.** For $n_j \asymp j^q$, $q \geq 1$, and $\alpha \neq 1$, the exact condition is $q|1 - \alpha| \leq 1$. For skew index one, the same polynomial clock always identifies, because $\sum_j (\log j)^{-2} = \infty$. If instead $n_j \asymp \exp(j^q)$ with $q > 0$, skew index one identifies exactly when $q \leq 1/2$, whereas every $\alpha \neq 1$ fails for every such q. These conclusions are direct consequences of Theorem 7.6.1, not separate solved problems.

7.7 Finite-height spectral observation and sharp noisy order detection

Two gaps remained after exact threshold inversion: the transition was an infinite-height observable, and positive noise prevented unconditional identification of the atom count. We give an explicit finite-height experiment under a quantitative recurrence assumption, and determine the noise scale at which a separated weak atom can be certified. These are newly formulated problems for the nonlinear observation below. The number-resolution exponent $2m - 2$ is already familiar in linear super-resolution, including Liu–Ammari [100]; our claim is its sharp transfer, including arbitrary small masses, to nonlinear minimum observations. Quantitative Kronecker approximation is classical [101]; no optimal recurrence exponent is claimed.

Fix $0 < a < b < 1$ and set

$$S_\mu(x) = \int_a^b x^r \, d\mu(r), \qquad T_\mu(C) = \min_{x>0} \frac{1 + Cx}{S_\mu(x)}, \qquad \mu \in \mathcal{P}[a,b]. \tag{7.7.1}$$

Here $\mathcal{P}_k[a,b]$ denotes the probabilities with at most k atoms. A list $\mathcal{C}$ is one-sided if its settings are distinct and belong to $(0, a/(1-a))$. Constants may depend on the fixed support interval, setting list, and specified atom bound, but not on unknown atom masses, separations, or phases.

7.7.1 The exact scale for detecting one additional atom

Theorem 7.7.1 (Sharp nonlinear model-order boundary)**.** *Fix $m \geq 2$ and any $2m-2$ one-sided settings $\mathcal{C}$. Let $\mu = \sum_{j=1}^m w_j \delta_{r_j}$, where $r_{j+1} - r_j \geq \Delta > 0$ and $w_j \geq w > 0$. Then*

$$\inf_{\nu \in \mathcal{P}_{m-1}[a,b]} \max_{C \in \mathcal{C}} |T_\mu(C) - T_\nu(C)| \geq c_m w \Delta^{2m-2}. \tag{7.7.2}$$

The joint dependence on w and Δ is sharp. For every $0 < w \le 1/m$ and every sufficiently small $\Delta > 0$, there are such an m-atomic μ and an $(m-1)$-atomic probability ν with

$$\sup_{C \in I} |T_\mu(C) - T_\nu(C)| \le K_{m,I} w \Delta^{2m-2} \tag{7.7.3}$$

for each fixed compact setting interval $I \subset (0, \infty)$. The constants are uniform as $w \downarrow 0$.

Thus adversarial threshold noise δ permits uniformly reliable rejection of every smaller positive spectrum when $\delta < cw\Delta^{2m-2}$; noise $\delta \ge Cw\Delta^{2m-2}$ can make an m-atom and an $(m-1)$-atom spectrum indistinguishable. This is a deterministic testing boundary, not a claim that a vanishingly small extra atom can be counted at fixed noise.

We first record explicitly the nonlinear-to-linear step used in the proof. It also explains why the observation settings need not locate the unobserved minimizers.

Lemma 7.7.2 (Smooth tests against a sparse competitor)**.** *Fix $n \ge 1$ and $2n$ one-sided settings. For $\alpha \in \mathcal{P}_n[a,b]$, $\beta \in \mathcal{P}[a,b]$, and $\varphi \in C^{2n}[a,b]$,*

$$\left| \int \varphi \, d(\beta - \alpha) \right| \le K \|\varphi\|_{C^{2n}} \max_i |T_\beta(C_i) - T_\alpha(C_i)|. \tag{7.7.4}$$

The constant does not depend on separation or masses in α.

Proof. For any positive profile S, put $H = S - xS'$. Its unique contact at setting C satisfies

$$C = \frac{S'(x)}{H(x)}, \qquad \frac{dC}{dx} = \frac{S''(x)S(x)}{H(x)^2} < 0, \qquad L_C(x) = \frac{1 + Cx}{T(C)}.$$

The line L_C majorizes S and is tangent at that contact. Moreover $a/(1-a) \le Cx \le b/(1-b)$. Hence, for the fixed settings, all contacts lie in a compact interval strictly above 1. On this interval $S, H, -S''$ are uniformly bounded above and below by positive constants. Consecutive contacts, tangent slopes, and the distances from their tangent-line intersections to either contact therefore have uniform positive lower bounds. For the last assertion, the strict concavity gap between contacts $v > w$ is at least $\inf(-S'')(v-w)^2/2$; divide it by the bounded tangent-slope difference.

Write δ for the threshold discrepancy. For small δ, the two sets of tangent-line coefficients and their intersections are within $O(\delta)$. The contacts belonging to setting i for both profiles lie in the same interval between neighboring tangent intersections, with uniform margins. At the first profile's contact $S_\alpha - S_\beta \ge -K\delta$, and at the second it is at most $K\delta$. An intermediate point has absolute difference at most $K\delta$. Taking logarithms gives $2n$ uniformly separated points u_i in a compact positive interval, where

$$\left| \int e^{u_i r} \, d(\beta - \alpha)(r) \right| \le K\delta.$$

Add the exact sample at $u_0 = 0$, using equal total masses.

Here is the requisite uniform positive interpolation argument. Pad the support of α to n nodes t_j, allowing repetitions. Interpolate φ with multiplicity two at each node by $R\varphi$ in the span of the first $2n$ exponentials among $e^{u_i r}$. Divided-difference interpolation matrices extend continuously across node collisions and are always invertible: a nonzero sum of $2n$ distinct real exponentials has at most $2n-1$ zeros counted with multiplicity, by repeated conjugation, differentiation, and Rolle's theorem. Compactness therefore bounds the interpolant coefficients and its C^{2n} norm by $K\|\varphi\|_{C^{2n}}$.

The interpolation residual q of the remaining exponential vanishes to even multiplicity at every t_j. With one choice of sign,

$$q(r) \ge c \prod_{j=1}^{n} (r - t_j)^2.$$

Indeed the quotient is a continuous divided difference, including collisions, and cannot vanish by the same zero count for $2n+1$ exponentials. Compactness gives the uniform lower bound. The ordinary Hermite remainder bound now shows that $R\varphi \pm K\|\varphi\|_{C^{2n}} q$ bracket φ everywhere and agree with it on the support of α. Integrating against the positive measure β proves (7.7.4) from the $2n+1$ exponential discrepancies. Large δ are absorbed by increasing K, using the trivial bound $2\|\varphi\|_\infty$. □

Proof of Theorem 7.7.1. For a competitor ν pad its support to $m-1$ nodes s_j and put $Q(r) = \prod_{j=1}^{m-1}(r - s_j)^2$. Its C^{2m-2} norm is bounded uniformly. Each open ball of radius $\Delta/2$ around an s_j contains at most one of the m true atoms. At least one true atom therefore has distance at least $\Delta/2$ from every s_j. Hence

$$\int Q \, d\mu \ge w(\Delta/2)^{2m-2}, \qquad \int Q \, d\nu = 0.$$

Lemma 7.7.2, with sparse side ν, gives the lower bound, uniformly over all competitors.

For the matching example, let $n = m - 1$, choose an interior r_0, and put the m atoms at $r_0 + j\Delta$, $0 \le j \le n$. Give the final atom mass w and each of the first n atoms mass $(1-w)/n \ge w$. Let p be the monic orthogonal polynomial of degree n for this probability. It exists because the moment matrix up to degree n is positive definite. Its n roots are simple and in the interior of the convex hull of the support: otherwise the product of its sign-change factors has degree less than n, and its product with p has a constant sign on the support, contradicting orthogonality.

The Gaussian quadrature measure ν on those roots has positive weights and agrees with μ through degree $2n-1$. To verify this without an existence citation, divide any polynomial of degree at most $2n-1$ by p and interpolate its remainder on the roots. Orthogonality makes the integral equal to the corresponding weighted sum. The resulting weight at root s_i equals the integral of the square of its degree-$(n-1)$ Lagrange polynomial, so it is positive; the weights sum to one.

Hermite interpolation of any C^{2n} function at these roots gives

$$\left| \int \varphi \, d(\mu - \nu) \right| \le \frac{\|\varphi^{(2n)}\|_\infty}{(2n)!} \int p(r)^2 \, d\mu(r). \tag{7.7.5}$$

The monic orthogonal polynomial minimizes this squared norm. Comparing with $\prod_{j=0}^{n-1}(r - r_0 - j\Delta)$ yields

$$\int p^2 \, d\mu \le w(n!)^2 \Delta^{2n}.$$

For x in a fixed compact subset of $(0, \infty)$ apply (7.7.5) to $\varphi(r) = x^r$. Its derivatives are uniformly bounded, so $|S_\mu(x) - S_\nu(x)| \le K w \Delta^{2n}$. All contacts for $C \in I$ lie in one such compact interval, and the denominators S are uniformly positive there. Comparing the two minimum quotients gives (7.7.3). □

Corollary 7.7.3 (Certified order from noisy data)**.** *Suppose the truth has $m \le M$ atoms and observe $2M-2$ fixed one-sided thresholds with errors at most δ. Return the smallest $k \ge 1$ for which some member of $\mathcal{P}_k[a,b]$ is consistent with those errors. This information-theoretic procedure returns $k = m$ whenever $m = 1$, or whenever $m \ge 2$ and $2\delta < c_m w \Delta^{2m-2}$. Conversely, at noise of order $w\Delta^{2m-2}$ even the entire threshold curve on a compact setting interval cannot uniformly distinguish m from $m-1$ atoms.*

Proof. The truth is feasible. Every smaller feasible measure would have threshold discrepancy at most 2δ, contradicting (7.7.2) on a sublist of $2m-2$ settings. For the converse use the midpoint of the two curves constructed in (7.7.3). No complexity assertion is made for the nonlinear feasibility search. □

7.7.2 A finite-height replacement for the transition oracle

Write the known largest frequency as $\ell > 0$, to distinguish it from the support endpoint b. Suppose $\mu = \sum_{j=1}^{m} w_j \delta_{r_j}$ and consider the controlled family

$$f_{A,C}(z) = -1 + A\sum_{j=1}^{m} w_j e^{\ell r_j z + i\phi_j} + Ce^{\ell z + i\phi_*}. \tag{7.7.6}$$

The control amplitudes A, C, the frequency ℓ, and the top phase ϕ_* are known. The smaller frequencies, weights and phases need not be known. An experiment at (A, C) reports all zero ordinates in $0 < t < H$ exactly. This is a finite-height observation model, not an algorithmic claim about computing a certified zero list.

Assume the frequency vector $\omega = (\ell r_1, \dots, \ell r_m, \ell) \in \mathbb{R}^d$, $d = m+1$, obeys the specified quantitative condition

$$|k \cdot \omega| \geq \gamma \|k\|_\infty^{-\tau} \quad (0 \neq k \in \mathbb{Z}^d), \qquad \gamma > 0, \quad \tau \geq 0. \tag{7.7.7}$$

In particular, its entries are rationally independent.

Theorem 7.7.4 (Finite amplitude grid and finite zero height)**.** *Fix a compact setting interval $I \subset (0, \infty)$. There are known constants $A_- < A_+$, $h_0, R, K > 0$, depending only on a, b, I, ℓ, m, τ, with the following property. For $0 < h < h_0$ take an equally spaced amplitude grid in $[A_-, A_+]$ with spacing in $[h/2, h]$, including its endpoints, and observe (7.7.6) at its $O(h^{-1})$ settings. Declare a grid amplitude accepted if one of its observed zero ordinates satisfies*

$$\operatorname{dist}(\ell t + \phi_*, \pi + 2\pi\mathbb{Z}) \leq R\sqrt{h}. \tag{7.7.8}$$

If

$$H \geq 3 + K\gamma^{-1}h^{-(d+\tau)}, \tag{7.7.9}$$

there is an accepted amplitude, and the least accepted one obeys

$$T_\mu(C) - Kh \leq \widehat{T}_h(C) \leq T_\mu(C) + h. \tag{7.7.10}$$

The result is uniform in $C \in I$, all positive masses, and all smaller phases. It does not assume that a qualitative density property can be recognized from a finite list.

Lemma 7.7.5 (An elementary uniform recurrence bound)**.** *Under (7.7.7), every target in $(\mathbb{R}/2\pi\mathbb{Z})^d$ is approached within coordinate distance η by $t\omega$ in every interval of length at least $K_{d,\tau}\gamma^{-1}\eta^{-(d+\tau)}$, for $0 < \eta < 1$.*

Proof. Choose a nonnegative smooth bump supported in the coordinate cube of radius one, scale it to radius η, and periodize around the target. Its mean is $c\eta^d > 0$ and its Fourier coefficients obey $|\widehat{q}(k)| \leq C_N \eta^d (1 + \eta\|k\|)^{-N}$ for every fixed N. Averaging $q(t\omega)$ on an interval of length L differs from its mean by at most

$$\frac{2}{L}\sum_{k \neq 0} \frac{|\widehat{q}(k)|}{|k \cdot \omega|} \leq \frac{K}{\gamma L}\eta^{-\tau}.$$

Here take $N > d + \tau$ and compare the lattice sum with its scaled integral. For the stated L, with a sufficiently large constant, the average is positive. Some point of the orbit therefore meets the bump support. The bound is independent of the interval's start. □

Proof of Theorem 7.7.4. All transition values lie in a fixed positive compact interval: $T(C) \le 1 + C$ by $x = 1$, while $S(x) \le \max(x^a, x^b)$ gives a uniform positive lower bound. Choose A_-, A_+ with positive margins around that interval.

Let x_* be the contact at $T = T_\mu(C)$ and $\sigma_* = (\log x_*)/\ell$. The phase-aligned function

$$g(z) = -1 + TS_\mu(e^{\ell z}) - Ce^{\ell z}$$

has a double zero at σ_*. Indeed $TS(x_*) = 1 + Cx_*$ and $TS'(x_*) = C$. Its second derivative is

$$g''(\sigma_*) = -\ell^2 T \int r(1-r)x_*^r \, d\mu(r) < 0.$$

The contact bound $a/(1-a) \le Cx_* \le b/(1-b)$ makes its magnitude uniformly bounded below, while the third derivative is uniformly bounded in a fixed complex neighborhood. Consequently, for a fixed small ρ_0 and $|z - \sigma_*| = \rho \le \rho_0$,

$$|g(z)| \ge c\rho^2. \tag{7.7.11}$$

The same Taylor estimate ensures that g has exactly two zeros, counted with multiplicity, inside that circle.

Choose a grid amplitude $A_0 \in [T, T + h]$. Apply Lemma 7.7.5, with $\eta = h$, to target phases $(-\phi_1, \dots, -\phi_m, \pi - \phi_*)$. For the height in (7.7.9), some $t_0 \in [1, H - 1]$ meets that target. On the fixed neighborhood of σ_*,

$$|f_{A_0,C}(z + it_0) - g(z)| \le K_0 h.$$

Choose a fixed R_0 large enough that $cR_0^2 > K_0$ and use the circle of radius $R_0\sqrt{h}$ in (7.7.11). For small h, Rouche's theorem gives two zeros in that disk for $f_{A_0,C}(z + it_0)$. Their original ordinates lie in $(0, H)$, and their top phases differ from π by at most $h + \ell R_0\sqrt{h}$. Taking $R = 1 + \ell R_0$ proves acceptance of A_0 and the upper bound in (7.7.10).

Conversely, let any accepted amplitude have zero $\sigma + it$, put $x = e^{\ell\sigma}$ and write its top angle as $\pi + v$ with $|v| \le R\sqrt{h}$. The zero equation and the triangle inequality imply

$$AS_\mu(x) \ge |1 - Cxe^{i(\pi+v)}|.$$

For $y = Cx$,

$$|1 - ye^{i(\pi+v)}|^2 = (1+y)^2 - 2y(1 - \cos v) \ge (1+y)^2 - yv^2 \ge (1+y)^2(1 - v^2/4).$$

Thus $A \ge (1 - R^2h/4)(1 + Cx)/S_\mu(x) \ge (1 - R^2h/4)T$. The uniform upper bound on T proves the lower bound in (7.7.10). □

The mechanism involves two quadratic effects: the critical aligned function has a double zero, while the missing-arc deficit is quadratic in the angular distance from π. They balance, so phase recurrence at accuracy h yields transition accuracy h, although the nearby zeros themselves move on the larger scale $\sqrt{h}$.

Corollary 7.7.6 (Conditional finite-height spectral resolution)**.** *Suppose the truth has at most m atoms, and run the finite-height experiment at $2m - 1$ fixed one-sided C settings. A minimum-distance fit in $\mathcal{P}_m[a, b]$ satisfies*

$$W_1(\widehat{\mu}, \mu) \le Kh^{1/(2m-1)}. \tag{7.7.12}$$

For fixed Diophantine parameters this yields $O(H^{-1/((d+\tau)(2m-1))})$ at $O(H^{1/(d+\tau)})$ amplitude settings per C, with constants depending on γ. Using $2m$ settings and allowing any positive consistent competitor instead gives the blind radius $O(h^{1/(2m)})$. For order certification up to a known bound M, the analogous $2M - 2$ setting experiment succeeds if $h < cw\Delta^{2m-2}$.

Proof. The finite-threshold comparison of the preceding paper [91] gives $W_1(\alpha,\beta) \le K \max_i |T_\alpha(C_i) - T_\beta(C_i)|^{1/(2m-1)}$ for two at most m-atomic probabilities, with $2m-1$ one-sided settings. Its elementary proof uses the separated near-zeros from Lemma 7.7.2: the transform difference is a sum of at most $2m$ real exponentials, so its values at $2m$ separated points, including 0, uniformly control its first $2m-1$ moments. A polynomial with roots at the at most $2m-2$ sign changes of the CDF difference then gives $W_1^{2m-1} \le K$ times the moment error. For one sparse side and an arbitrary positive side, the same proof uses (7.7.4) and at most $2m-1$ CDF sign changes, giving exponent $1/(2m)$. Apply these bounds to the threshold errors from Theorem 7.7.4. Minimum-distance fitting at most doubles them. The final assertion follows from Corollary 7.7.3. □

Proposition 7.7.7 (Why rational independence alone supplies no height)**.** *Even for one unknown frequency with support interval $[a,b] = [1/3, 2/3]$, no uniform finite observation height can guarantee detection of the arc in (7.7.8) over all rationally independent frequency pairs. More precisely, fix $\ell = C = 1$, zero phases, and any finite H. There is an irrational r arbitrarily close to $1/2$ for which, simultaneously for every $A \in [1,4]$, the function $-1 + Ae^{rz} + e^z$ has no zero in $0 \le \operatorname{Im} z \le H$ whose ordinate is within $\pi/4$ of $\pi + 2\pi\mathbb{Z}$. Nevertheless its transition $T_{\delta_r}(1)$ lies near 2, in the interior of this amplitude interval.*

Proof. At $r = 1/2$, substitute $u = e^{z/2}$. The quadratic $u^2 + Au - 1$ has two nonzero real roots for every $A \in [1,4]$. Consequently every zero has $\operatorname{Im} z \in 2\pi\mathbb{Z}$. Uniformly for $A \in [1,4]$ and r near $1/2$, all zeros lie in one fixed vertical strip: the constant dominates far left and e^z dominates far right. On the compact part of that strip satisfying the stated height and arc restrictions, the modulus at $r = 1/2$ has a positive minimum, uniformly also in A. Continuity preserves that positive lower bound for all sufficiently close irrational r. Finally $T_{\delta_r}(1) = r^{-r}(1-r)^{-(1-r)} \to 2$. □

The finite-height theorem therefore closes a conditional observation bridge, and the final proposition shows precisely why its quantitative hypothesis cannot simply be omitted. The height exponent above is only an explicit sufficient bound. Neither an optimal hitting-time exponent nor a uniform quantitative assertion for an infinite prime vector is claimed. The model-order theorem closes a different, sharp deterministic problem; its exponent is a transfer of known super-resolution geometry, not a newly discovered universal exponent.

7.8 Gamma deconvolution closes the remaining McKay SD gap

The remaining strictness question in the generalized McKay self-decomposability problem has a different solution from the finite moment tests of the preceding manuscript. Dividing out the gamma factor suggested by the tail produces an explicit signed density. A Sturm identity forces this density to change sign. This gives both a sharp gamma-factor obstruction and the missing strict SD boundary for every admissible parameter. We then determine the singular cost of a tilt-uniform HCM power just above the critical Bessel order. The general McKay questions are posed in [3, Section 4.3]; our gamma-factor and critical-power questions are newly formulated transfer problems.

7.8.1 Notation and previously established inputs

Scale $a = 1$ initially. Let

$$\alpha = \mu + \tfrac{1}{2}, \qquad q = \nu - \mu - 1, \qquad M = 2\alpha + q = \mu + \nu, \qquad A = \alpha + q = \nu - \tfrac{1}{2},$$

and write

$$J(u) = \int_0^\infty x^{\nu-1} e^{-ux} I_\mu(x)\, dx, \qquad u > 1.$$

The density is $x^{\nu-1}e^{-bx}I_\mu(x)/J(b)$, where $\alpha > -1/2$, $M > 0$, and $b > 1$. The preceding signed-spectral construction [76, Section 4] proves that every such law is ID. For $q > 0$, it supplies a positive smooth even kernel K with

$$-\frac{J'(u)}{J(u)} = \int_0^\infty e^{-ux}K(x)\,dx, \quad K(0) = M, \quad e^{-x}K(x) \longrightarrow A, \quad \frac{K'(x)}{K(x)} \longrightarrow 1. \tag{7.8.1}$$

Its SD boundary is

$$\beta_{\alpha,q} = \sup_{x\geq 0} \frac{K'(x)}{K(x)} < \infty : \qquad X \text{ is SD} \iff b \geq \beta_{\alpha,q}. \tag{7.8.2}$$

For $q \leq 0$ the law is GGC, hence SD, for every admissible tilt. These classifications and kernel identities are inputs, not new results counted in this section. In contrast, strictness of (7.8.2) for all $q > 0$ was not established previously.

7.8.2 An exact signed inverse for the critical gamma quotient

We use the usual entire Kummer function ${}_1F_1(a;c;z)$, with $c > 0$. The next sign lemma includes every positive noninteger q; it does not rely on a numerical zero count.

Lemma 7.8.1 (A forced negative interval)**.** *For every $q > 0$ and $c > 0$, the real function $y(z) = {}_1F_1(-q;c;z)$ is strictly negative on a nonempty open subinterval of $(0,\infty)$.*

Proof. Kummer's differential equation [6, Section 13.2] has the self-adjoint form

$$(z^c e^{-z}y'(z))' + qz^{c-1}e^{-z}y(z) = 0, \qquad y(0) = 1.$$

The left boundary flux is zero, because $c > 0$ and y' is bounded near zero. If y were positive on the whole positive axis, then for $z \geq 1$

$$y'(z) = -qe^z z^{-c}\int_0^z t^{c-1}e^{-t}y(t)\,dt \leq -Ce^z z^{-c}$$

for some $C > 0$. Integrating forces y to become negative, a contradiction. At its first zero the displayed integral is strictly positive, so the derivative there is strictly negative. Thus the first zero is followed by an open negative interval. □

Theorem 7.8.2 (Critical gamma extraction is possible exactly at $q \leq 0$)**.** *Assume $\alpha > 0$, $M > 0$, and $A > 0$, and let $b > a > 0$. The generalized McKay law admits a decomposition*

$$X \overset{\mathrm{d}}{=} G + Y, \qquad G \sim \mathrm{Gamma}(A, \textit{rate } b-a), \qquad Y \geq 0, \qquad G \perp Y,$$

if and only if $q \leq 0$. No infinite-divisibility assumption is made about the possible remainder Y.

For $a = 1$, a signed inverse Laplace density of $G_0(s) := s^A J(1+s)$ is

$$g(x) = c_\alpha \frac{\Gamma(M)}{\Gamma(\alpha)}x^{\alpha-1}e^{-2x}{}_1F_1(-q;2\alpha;2x), \qquad x > 0, \qquad c_\alpha = \frac{2^{1/2-\alpha}}{\Gamma(\alpha+1/2)}. \tag{7.8.3}$$

In the positive case the remainder has density $e^{-(b-1)x}g(x)/G_0(b-1)$.

Proof. The Euler integral for $I_{\alpha-1/2}$ [6, Section 10.32] gives

$$I_{\alpha-1/2}(x) = c_\alpha x^{\alpha-1/2}e^x \frac{1}{B(\alpha,\alpha)}\int_0^1 e^{-2xt}t^{\alpha-1}(1-t)^{\alpha-1}\,dt.$$

Consequently

$$J(1+s) = c_\alpha \Gamma(M)(s+2)^{-M} {}_2F_1\left(M, \alpha; 2\alpha; \frac{2}{s+2}\right). \tag{7.8.4}$$

For $s > 0$, termwise integration of the Kummer series is justified absolutely since $2/(s+2) < 1$, and gives

$$\begin{aligned}
&\int_0^\infty e^{-sx} x^{\alpha-1} e^{-2x} {}_1F_1(-q; 2\alpha; 2x)\, dx \\
&\qquad = \Gamma(\alpha)(s+2)^{-\alpha} {}_2F_1\left(-q, \alpha; 2\alpha; \frac{2}{s+2}\right) \\
&\qquad = \Gamma(\alpha) s^A (s+2)^{-M} {}_2F_1\left(M, \alpha; 2\alpha; \frac{2}{s+2}\right).
\end{aligned}$$

The last equality is Euler's transformation; its exponent is $2\alpha - (-q) - \alpha = A$. Comparison with (7.8.4) proves (7.8.3). The classical real-axis Kummer expansion, or its elementary polynomial formula at nonnegative integral q, shows that g is integrable at infinity: its algebraic part is $O(x^{-A-1})$, and its other part has exponential decay times a fixed power. It is integrable at zero since $\alpha > 0$. All exponentially tilted versions are absolutely integrable directly from the preceding series calculation as well.

The Laplace transform of the putative remainder is necessarily

$$\frac{J(b+s)}{J(b)} \left(\frac{b-1+s}{b-1}\right)^A = \frac{G_0(b-1+s)}{G_0(b-1)}.$$

Uniqueness of Laplace transforms of finite signed measures makes its only possible inverse the stated tilted density. When $q \le 0$ all coefficients of ${}_1F_1(-q; 2\alpha; 2x)$ are nonnegative and its constant coefficient is one. The inverse is positive, has total mass one, and gives the required independent convolution. When $q > 0$, Lemma 7.8.1 makes this inverse negative on an open interval, ruling out any probability remainder, including one with atoms. Rescaling x proves the statement for general a. □

The positive half of Theorem 7.8.2 also follows from the endpoint atom in the previously known Thorin measure. Its new content is the explicit inverse and the obstruction at $q > 0$ for an *arbitrary* nonnegative remainder, a stronger statement than failure of a GGC or ID remainder. The obstruction is independent of the exponential tilt b/a.

7.8.3 Universal strictness of the SD boundary

Theorem 7.8.3 (The remaining SD strictness question is closed)**.** *For every $q > 0$, $\alpha > -1/2$, and $2\alpha + q > 0$,*

$$1 < \beta_{\alpha,q} < \infty.$$

The supremum is attained at a finite positive point. In particular, for every admissible excess-power pair there is a nonempty interval

$$1 < b/a < \beta_{\alpha,q}$$

of generalized McKay laws that are ID but are not SD. Together with the earlier GGC classification, this proves the exact dichotomy:

$$\textit{the entire admissible tilt family is SD} \quad \Longleftrightarrow \quad q \le 0.$$

Proof. The kernel limit already gives $\beta \geq 1$. Suppose $\beta = 1$. Then $k(x) = e^{-x}K(x)$ is nonincreasing, tends to A, and satisfies $k(x) \geq A$. If $\alpha < 0$, this contradicts $k(0) - A = \alpha$. If $\alpha = 0$, monotonicity and equal endpoint values force k to be constant, whereas $k'(0) = K'(0) - K(0) = -M < 0$ because K is even.

It remains to treat $\alpha > 0$. For any fixed $b > 1$, the Lévy spectral function of X is $e^{-bx}K(x)$. Since $k \geq A$, subtracting the Lévy density of Gamma$(A, \text{rate } b-1)$ leaves the nonnegative Lévy density

$$\frac{e^{-(b-1)x}}{x}\{k(x) - A\}.$$

It is integrable in the Lévy sense: near zero $k - A$ is bounded, and at infinity the exponential tilt and boundedness of k suffice. The corresponding positive ID variable is an independent remainder in the gamma decomposition excluded by Theorem 7.8.2. This contradiction proves $\beta > 1$ for every remaining parameter, with no restriction to integer q or to a cut-sign interval. Finally K'/K is continuous, is zero at the origin, and tends to one. Its value above one therefore attains its global supremum at a finite positive point. Restoring a gives the stated tilt boundary and the dichotomy. □

What this resolves and what it does not. Theorem 7.8.3 supplies the missing strictness in the negative-cut intervals with $\alpha > 1$; it replaces rather than assumes termination of the finite moment hierarchy. The proof does not assert that that particular hierarchy always terminates. The exact numerical boundary remains the explicit spectral supremum (7.8.2); no elementary formula for its maximizer is claimed. The full ID classification was proved in the preceding work and is not counted again.

7.8.4 The singular cost of a tilt-uniform HCM power

Write $R_\mu(z) = I_{\mu+1}(z)/I_\mu(z)$ and, for $0 < \theta < \pi/2$, put

$$B_\mu(y, \theta) = \frac{\operatorname{Im}\{e^{i\theta}R_\mu(ye^{i\theta})\}}{\sin\theta}, \qquad C_\mu(\theta) = \sup_{y>0} B_\mu(y, \theta).$$

The exact power test from the preceding papers, recalled in [91, Bessel section], is

$$X^d \text{ is HCM} \iff d > 2 \quad \text{and} \quad b/a \geq C_\mu(\pi/d). \tag{7.8.5}$$

It is independent of the monomial exponent in the density. For $\mu > 1/2$, define the least tilt-uniform power by

$$d_{\text{unif}}(\mu) = \inf\{d > 2 : C_\mu(\pi/d) = 1\}.$$

The previous manuscript proved that this number is finite, but did not determine its singular behavior as $\mu \downarrow 1/2$. Because the HCM class is preserved under positive powers at least one, feasible values of d form an upper interval. Equivalently, the standard HCM composition rule is that $f(x^c)$ is HCM when f is HCM and $0 < c \leq 1$; monomial density factors do not affect this property [75].

Theorem 7.8.4 (A logarithm–double-logarithm uniform-power boundary)**.** *As $\varepsilon \downarrow 0$, with $L = \log(1/\varepsilon)$,*

$$d_{\text{unif}}(1/2 + \varepsilon) = L + 2\log L - 2 + o(1). \tag{7.8.6}$$

More precisely, if $d \to \infty$, $\varepsilon \geq 0$, and $\varepsilon e^d/d^2 \to \lambda \in [0, \infty)$, then

$$e^d\{C_{1/2+\varepsilon}(\pi/d) - 1\} \longrightarrow 2(e^{-2} - \lambda)_+. \tag{7.8.7}$$

Thus the algebraic order perturbation suppresses the exponentially small half-order resonance at the scale $\varepsilon \asymp d^2 e^{-d}$.

Proof. We give the uniform estimates needed to control the supremum; pointwise expansions at fixed y would not suffice.

Step 1: the exact half-order resonance. Put $k = \cot\theta$ and $t = 2y\sin\theta$. Since $R_{1/2}(z) = \coth z - 1/z$,

$$B_{1/2}(y,\theta) - 1 = \frac{\cos t - k\sin t - e^{-kt}}{\cosh(kt) - \cos t}. \tag{7.8.8}$$

For $0 < t \le \pi$ the numerator is nonpositive, since $[e^{kt}(\cos t - k\sin t)]' = -(1+k^2)e^{kt}\sin t \le 0$. For $t \ge \pi$, uniformly as $k \to \infty$,

$$B_{1/2} - 1 = 2e^{-kt}(\cos t - k\sin t) + O((1+k)e^{-2kt}).$$

The first positive maximum of $e^{-kt}(\cos t - k\sin t)$ occurs at $t_k = \pi + 2\arctan(1/k)$, has trigonometric factor exactly one, and subsequent positive maxima are smaller by successive factors $e^{-2\pi k}$. This follows by differentiating the displayed function; its derivative is $e^{-kt}\{-2k\cos t + (k^2-1)\sin t\}$. Therefore

$$C_{1/2}(\theta) - 1 \sim 2e^{-\pi k - 2}, \tag{7.8.9}$$

$$W(\theta) := \sup_{y>0} y^2(B_{1/2}(y,\theta) - 1) \sim \frac{\pi^2 k^2}{2} e^{-\pi k - 2}. \tag{7.8.10}$$

To justify the second supremum, observe that $y^2 = t^2(1+k^2)/4$. For $t \ge \pi + k^{-1/2}$ the absolute value of the main term after multiplication by t^2 is $O(ke^{-\pi k - \sqrt{k}})$, uniformly for large k. On the remaining interval $t^2 = \pi^2(1+o(1))$. The weighted error is exponentially smaller. This proves (7.8.10), and the point $t = t_k$ attains its asymptotic value. At $\theta = \pi/d$,

$$\pi k = d + O(d^{-1}), \qquad W(\pi/d) = \tfrac{1}{2}d^2 e^{-d-2}(1+o(1)), \qquad y_k \sim d/2.$$

Step 2: an estimate uniform in order and angle. For some fixed $\theta_1 < \pi/2$ and $Y_0 < \infty$, uniformly for $y \ge Y_0$, $0 \le \theta \le \theta_1$, and sufficiently small $\varepsilon \ge 0$,

$$B_{1/2+\varepsilon}(y,\theta) - B_{1/2}(y,\theta) = -\frac{\varepsilon(1+\varepsilon)}{2y^2} + O(\varepsilon y^{-3}). \tag{7.8.11}$$

Here zero angle means the continuous extension. Indeed the standard Bessel ratio expansion, uniform on a closed sector and on a small complex order disk around $1/2$, is

$$R_\mu(z) = 1 - \frac{\mu + 1/2}{z} + \frac{4\mu^2 - 1}{8z^2} + O(z^{-3}).$$

Cauchy's estimate in the order parameter bounds the difference of the two remainders by $O(\varepsilon|z|^{-3})$. For real orders this difference is real on the positive real axis. A Cauchy estimate in a slightly wider sector therefore bounds the imaginary part after multiplication by $e^{i\theta}$ by $O(\varepsilon\theta y^{-3})$. Dividing by $\sin\theta$ is thus uniformly legitimate. The z^{-1} difference contributes a real number, while the z^{-2} term contributes exactly the term in (7.8.11). These uniform Bessel asymptotics are standard; see [6, Section 10.40]. On each fixed bounded positive y interval, $B_{1/2+\varepsilon}(y,\theta)$ tends uniformly as $(\varepsilon,\theta) \to (0,0)$ to

$$D_{1/2}(y) = \coth y - y\,\mathrm{csch}^2 y < 1.$$

The power series at zero supplies the same strict upper bound near zero. In particular no bounded y can cause a failure of the uniform-power criterion in this limit.

Step 3: inversion at the least uniform power. Fix $\delta > 0$ and set $d_\pm = L + 2\log L - 2 \pm \delta$. Then

$$\frac{2W(\pi/d_\pm)}{\varepsilon} \longrightarrow e^{\mp\delta}.$$

For $d = d_+$ choose a fixed $Y \ge Y_0$ large enough that the $O(1/Y)$ relative error in (7.8.11) is smaller than the gap between $e^{-\delta}$ and one. For all $y \ge Y$,

$$y^2(B_{1/2+\varepsilon}(y, \pi/d_+) - 1) \le W(\pi/d_+) - \frac{\varepsilon}{2}(1 - O(1/Y)) < 0$$

for small ε. Step 2 excludes the remaining bounded interval. The limit as $y \to \infty$ is one, so $C_{1/2+\varepsilon}(\pi/d_+) = 1$. For $d = d_-$ evaluate at the first positive peak $y_k \sim d_-/2$. Equations (7.8.10) and (7.8.11) make the weighted difference strictly positive. Hence d_- is infeasible. The upper-interval property of feasible powers gives $d_- \le d_{\mathrm{unif}} \le d_+$. Letting $\delta \downarrow 0$ proves (7.8.6).

Step 4: the crossover. At the same first peak, (7.8.11) gives

$$e^d(B_{1/2+\varepsilon}(y_k, \pi/d) - 1) \longrightarrow 2e^{-2} - 2\lambda.$$

The limit at infinity gives the additional lower bound zero. For the upper bound, choose a fixed large Y for which the order perturbation in (7.8.11) is nonpositive. The bounded interval is excluded as before, and the region $2y\sin(\pi/d) \le \pi$ has $B_{1/2} \le 1$. For any fixed $\eta > 0$, the region $2y\sin(\pi/d) \ge \pi(1+\eta)$ has positive half-order excess $O(de^{-(1+\eta)d+O(1/d)}) = o(e^{-d})$, and the order perturbation is nonpositive. In the intervening region,

$$\frac{d}{2}(1 + o(1)) \le y \le \frac{d}{2}(1+\eta)(1 + o(1)).$$

Using (7.8.9) and (7.8.11), the scaled excess there has limsup at most $2e^{-2} - 2\lambda/(1+\eta)^2$. Take the maximum with zero and then let $\eta \downarrow 0$. This proves (7.8.7). □

Research status. The two structurally distinct advances are the gamma-deconvolution obstruction, which closes universal SD strictness, and the sharp uniform-HCM critical law. Formula (7.8.7) refines the second advance rather than constituting another named problem. The cited identities and HCM closure rules are classical. We make no priority claim for the separate Ismail–Miller conjecture discussed in the October 2026 Bessel preprint; none of our proofs uses it.

7.9 Complete replication-cap curves for analytic populations

The preceding manuscript established uncapped analytic-population rates. Here we determine how much replication is actually necessary to attain them. The new lower bound works after every adaptive history, including histories that nearly reveal an individual's endpoint.

An individual has latent $U \in [0,1]$ with density w. Conditional on U, its observations are independent uniforms on $[0, 2^U]$. A policy uses at most B scalar observations, with at most Q observations of any one individual, where $1 \le Q \le B$. Write $R^{\mathrm{ad}}_{B,Q}$ and $R^{\mathrm{fix}}_{B,Q}$ for minimax expected ordinary-interval Wasserstein loss, allowing respectively adaptive policies and predetermined designs. For fixed positive L, τ, ν, let

$$\mathcal{W}_\phi(L) = \left\{ w : \ w \ge \tfrac{1}{2}, \ \int_0^1 w = 1, \ w \text{ is one-periodic}, \quad |\widehat{w}(k)| \le Le^{-\phi(|k|)} \ (k \ne 0) \right\}.$$

Periodicity is a density constraint; the transportation cost does not identify the two endpoints.

Theorem 7.9.1 (The full cap curve)**.** *For either $\phi(k) = \tau k^\nu$ or $\phi(k) = \tau e^{\nu k}$, put respectively*

$$K_B = (\log B)^{1/\nu}, \qquad K_B = \log\log B.$$

As $B \to \infty$, uniformly over integers $1 \le Q \le B$,

$$R^{\mathrm{ad}}_{B,Q} \asymp R^{\mathrm{fix}}_{B,Q} \asymp \frac{1}{\sqrt{B}} \max\left\{\log K_B, \sqrt{K_B/Q}\right\}. \tag{7.9.1}$$

Constants may depend on the fixed class parameters, but not on B, Q. Consequently the least order of replication cap permitting the uncapped rate is

$$Q_{\mathrm{crit}} \asymp K_B/(\log K_B)^2.$$

This statement concerns a rate threshold; it does not assert a sharp leading constant or adaptation to an unknown smoothness envelope.

For the stretched-exponential class the two terms are $\log\log B/\sqrt{B}$ and $(\log B)^{1/(2\nu)}/\sqrt{BQ}$. For the double-exponential class they are $\log\log\log B/\sqrt{B}$ and $\sqrt{\log\log B/(BQ)}$. We count the two class-specific full curves as two internally formulated targets. The cap threshold is their consequence, not a third independent conjecture. The uncapped term was proved previously [91]; it is retained explicitly to make the new comparison checkable.

7.9.1 A design attaining the curve

Set $a = \log 2$. The log maximum of r observations is distributed as $Y_r = U - \mathrm{Exp}(ar)$. Thus

$$A_{r,k}(Y_r) = e^{-2\pi i k Y_r}\left(1 - \frac{2\pi i k}{ar}\right)$$

is unbiased for $\widehat{w}(k)$, with second moment at most $C\{1 + (k/r)^2\}$. Also $Y_1 + 1/a$ estimates the population mean with bounded variance. For any two densities in the class, Fourier expansion of their CDF difference gives

$$W_1(w,v) \le |\mu(w) - \mu(v)| + C\left(\sum_{1\le|k|\le K} \frac{|\widehat{w}(k) - \widehat{v}(k)|^2}{k^2}\right)^{1/2} + CLe^{-\phi(K)}. \tag{7.9.2}$$

Indeed the CDF difference has mean $\mu(v) - \mu(w)$ and nonzero Fourier coefficients $(\widehat{w}(k) - \widehat{v}(k))/(2\pi i k)$; apply Parseval and $\sum_{k>K} k^{-2} \le C$. This mean term is necessary for interval, as opposed to circular, transportation.

Use a quarter of the budget to estimate the mean. Divide frequencies into dyadic bands ending at $k \le K$, and choose $r_k = \min(Q, k)$. For n_k fresh individuals in this band its contribution to the squared Fourier error in (7.9.2) is at most Ca_k/n_k, where

$$a_k = k^{-1}\{1 + (k/r_k)^2\}, \qquad S = \sum_{k \text{ dyadic} \le K} \sqrt{r_k a_k}.$$

Choose $n_k = \lceil B\sqrt{a_k/r_k}/(4S)\rceil$. The cost is at most $B/4 + \sum r_k \le B/4 + CK$ and the total variance at most CS^2/B. Moreover

$$S \asymp \begin{cases} \log(eK), & Q \ge K, \\ \log(eQ) + \sqrt{K/Q}, & Q < K, \end{cases} \qquad S \le C\max\{\log(eK), \sqrt{K/Q}\}.$$

Project the estimated mean and Fourier vector onto the vectors attained by $\mathcal{W}_\phi(L)$, in the norm in (7.9.2). The class is compact in the uniform norm by its summable Fourier envelope, so a measurable approximate minimizer suffices. Projection costs at most a factor two. Choose a dyadic $K \asymp K_B$ large enough that $\phi(K) \ge \log B$. Since $K = o(B)$, rounding is harmless and proves the upper bound in (7.9.1).

7.9.2 A boundary-vanishing perturbation family

Put $\chi(u) = \sin^2(\pi u)$. For $K \le k \le 2K$ define

$$\Phi_{k,c}(u) = \frac{\chi(u)\sin(2\pi k u)}{2\pi k}, \qquad \Phi_{k,s}(u) = -\frac{\chi(u)\cos(2\pi k u)}{2\pi k}, \qquad \varphi_{k,*} = \Phi'_{k,*}.$$

Both endpoints of each Φ vanish, so $\int \varphi = 0$. The $d = 2(K+1)$ functions φ are uniformly bounded and have frequencies between $K-1$ and $2K+1$. Each frequency receives contributions from at most a fixed number of them. Consequently, for a sufficiently small fixed $c > 0$, the family

$$w_\beta = 1 + \sum_j \beta_j \varphi_j, \qquad |\beta_j| \le b_K := cK^{-2}e^{-\phi(2K+1)} \tag{7.9.3}$$

lies in $\mathcal{W}_\phi(L)$ and satisfies $1/2 \le w_\beta \le 3/2$. Furthermore

$$\|\Phi(u)\|_2 \le C\chi(u)/\sqrt{K} \le C(1-u)/\sqrt{K}. \tag{7.9.4}$$

The last boundary factor is what prevents highly informative exceptional histories from invalidating the cap lower bound.

Lemma 7.9.2 (Information with a cap, after arbitrary histories)**.** *For (7.9.3), every adaptive policy has Fisher-information trace at most CBQ/K, uniformly in β.*

Proof. We grant the statistician a stronger observation. For each individual expose a rate-one Poisson process on $[0,\lambda] \times [0,U]$, where exposure time λ may be allocated predictably from the observed history. Total exposure is at most aB and each individual receives at most aQ. A block of length a simulates an original observation: use its largest second coordinate when the block is nonempty, and an independent $-\operatorname{Exp}(a)$ otherwise. The result is $U - \operatorname{Exp}(a)$, the logarithm of a uniform observation. Thus a lower bound for this stronger experiment applies to the original one.

After exposure λ and largest observed mark b (take $b = 0$ if none), the posterior density is proportional to $w_\beta(u)e^{-\lambda u}\mathbf{1}_{[b,1]}(u)$. Define

$$Z(b) = \int_b^1 w_\beta(u)e^{-\lambda u}\,du, \quad H(b) = \int_b^1 \Phi'(u)e^{-\lambda u}\,du, \quad s(b) = H(b)/Z(b).$$

Write $D = 1 - b$. Integration by parts and (7.9.4) give

$$H(b) = -e^{-\lambda b}\Phi(b) + \lambda\int_b^1 e^{-\lambda u}\Phi(u)\,du, \qquad \|H(b)\|_2 \le CDe^{-\lambda b}/\sqrt{K}.$$

On the other hand $Z(b) \ge ce^{-\lambda b}(1 - e^{-\lambda D})/\lambda$, with its continuous value at $\lambda = 0$. Therefore

$$\|s(b)\|_2 \le C(1+\lambda D)/\sqrt{K} \le C(1+\lambda)/\sqrt{K}. \tag{7.9.5}$$

The conditional intensity of a new record at $y > b$ per unit exposure is $T(y) = Z(y)/Z(b)$, whose parameter score is $s(y) - s(b)$. Marks below b have parameter-independent intensity and zero score. Thus the conditional information trace per unit exposure is

$$\int_b^1 T(y)\|s(y) - s(b)\|_2^2\,dy \le \frac{C(1+\lambda)^2}{K}\int_b^1 T(y)\,dy \le \frac{C(1+\lambda)}{K}.$$

The last integral is the posterior mean of $U - b$, at most $C\min(1, \lambda^{-1}) \le C'/(1+\lambda)$, because w_β is bounded above and below. Since $\lambda \le aQ$ and total exposure is at most aB, integration proves the bound. The compensator formula already includes the information in intervals with no points. Conditional score increments are martingale differences; hence adaptive choices of the next individual do not create additional terms. Randomized policies can be conditioned on their independent random seed. □

7.9.3 Turning information into interval transportation

Use independent cosine-squared priors on $[-b_K, b_K]$. If J is the joint prior-and-data Fisher matrix, Lemma 7.9.2 gives

$$\operatorname{tr} J \le C\{BQ/K + K^5 e^{2\phi(2K+1)}\}. \tag{7.9.6}$$

We use the following elementary absolute-loss information inequality: for any vector v and estimator T,

$$\mathbb{E}|T - v^T\beta| \ge \tfrac{1}{4}\sqrt{v^T J^{-1} v}. \tag{7.9.7}$$

For clarity, extend the joint density by zero outside the prior cube. Translating its parameter argument by hz changes total variation by at most $|h|\sqrt{z^T J z}/2$. The original target and the translated target differ by $hv^T z$, and their risks under the respective densities are equal by change of variables. Integrating their losses over the common part, then choosing $|h| = (z^T J z)^{-1/2}$, proves the lower bound $|v^T z|/(4\sqrt{z^T J z})$. Optimizing z proves (7.9.7). The prior vanishes quadratically at its boundary, justifying the zero extension and differentiation.

Here the CDF is exactly $u + \Phi(u)^T\beta$. The one-dimensional identity $W_1 = \int_0^1 |F - \widehat{F}|$ therefore gives Bayes risk at least $c\int \sqrt{\Phi(u)^T J^{-1}\Phi(u)}\,du$. Write $\Phi = \chi v$. The sine-cosine pairs imply $\|v(u)\|_2^2 \asymp K^{-1}$ independently of u. Hence

$$\sqrt{v^T J^{-1} v} \ge \frac{\|v\|_2^2}{\sqrt{v^T J v}}.$$

Weighted Jensen with weight χ, followed by Fourier orthogonality, shows

$$\int_0^1 \sqrt{\Phi^T J^{-1}\Phi}\,du \ge \frac{c/K}{\sqrt{\int_0^1 \chi(u) v(u)^T J v(u)\,du}} \ge \frac{c}{\sqrt{\operatorname{tr} J}}.$$

Indeed $\int \chi = 1/2$, $0 \le \chi \le 1$, and $\int vv^T$ is diagonal with entries at most C/K^2. We have proved

$$R^{\mathrm{ad}}_{B,Q} \ge c\{BQ/K + K^5 e^{2\phi(2K+1)}\}^{-1/2}. \tag{7.9.8}$$

Choose $K \asymp K_B$ sufficiently small that $\phi(2K+1) \le (\log B)/8$. This is possible in both cases: for the double-exponential envelope one may take $K = \lfloor (2\nu)^{-1}\log(\log B/(8\tau e^{\nu}))\rfloor$. Then $K^5 e^{2\phi(2K+1)} \le K^5 B^{1/4} = o(B/K)$, uniformly for $Q \ge 1$. Equation (7.9.8) becomes $R^{\mathrm{ad}}_{B,Q} \ge c\sqrt{K_B/(BQ)}$.

7.9.4 The uncapped term and completion

We record the earlier uncapped information argument to specify the second lower-bound input. Use perturbations $w_\beta = 1 + \sqrt{2}\sum_{k=1}^{K}(\beta_{k,c}\cos(2\pi k u) + \beta_{k,s}\sin(2\pi k u))$, with $|\beta_{k,*}| \le cK^{-2}e^{-\phi(K)}$. They have uniformly bounded Lipschitz constants. Under the constant baseline the posterior of an individual is a truncated exponential $q \propto e^{-aru}\mathbf{1}_{[b,1]}$. If q_Y is its posterior after the next raw observation, direct integration gives

$$\mathbb{E}_0\|q_Y - q\|_2^2 = a/2, \qquad \|q\|_2^2 \operatorname{Var}_q(U) \le C.$$

For verification, putting $D = 1 - b$, $\lambda = ar$, $Z_\lambda = (1 - e^{-\lambda D})/\lambda$, and $\mu = \lambda + a$,

$$\mathbb{E}_0\|q_Y\|_2^2 = \frac{1 + e^{-\mu D}}{2Z_\lambda} + \frac{a}{2Z_\lambda}\int_0^D e^{-\lambda v}\{1 + e^{-\mu(D-v)}\}\,dv = \frac{a}{2} + \frac{1 + e^{-\lambda D}}{2Z_\lambda}.$$

Also $\|q\|_2^2 = O(\lambda + D^{-1})$ and $\mathrm{Var}_q(U) = O(\min(D^2, \lambda^{-2}))$. For $z = \int w_\beta q$, $z_Y = \int w_\beta q_Y$, Bessel's inequality bounds the conditional score trace by

$$C\mathbb{E}_0\|q_Y - q\|_2^2 + C\|q\|_2^2\mathbb{E}_0|z_Y - z|^2 \le C.$$

Here z, z_Y are bounded away from zero and infinity, and the second expectation is at most $C\,\mathrm{Var}_q(U)$: use the uniform Lipschitz bound, an independent coupling of q_Y with q, and $\mathbb{E}_0 q_Y = q$. Adding martingale score increments gives trace CB, even without a cap. The joint matrix thus has trace $C\{B + K^5 e^{2\phi(K)}\}$.

For the centered CDF the parameter vector is $(\sqrt{2}\sin(2\pi ku), -\sqrt{2}\cos(2\pi ku))/(2\pi k)$. Centering multiplies the integrated error by at most two. Apply (7.9.7), then average J over simultaneous frequency rotations. The map $J \mapsto \sqrt{v^T J^{-1} v}$ is convex, since it is $\sup_z |v^T z|/\sqrt{z^T J z}$. Rotation averaging over $R(u) = \mathrm{diag}(R_{2\pi ku})_{k=1}^K$, where R_θ is planar rotation through θ, leaves blocks $j_k I_2$. Cauchy–Schwarz gives

$$\sum_{k=1}^K (k^2 j_k)^{-1} \ge \frac{(\sum_{k=1}^K k^{-1})^2}{\sum_{k=1}^K j_k}.$$

Thus every capped or uncapped procedure has risk at least $c\log(eK)/\sqrt{B + K^5 e^{2\phi(K)}}$. Taking $K \asymp K_B$ with $\phi(K) \le (\log B)/8$ gives $c\log K_B/\sqrt{B}$. Combining this with (7.9.8) and the predetermined upper design completes the proof of Theorem 7.9.1.

What has and has not been transferred. The cap penalty comes from a boundary-vanishing high-frequency family and a conditional Poisson-information bound. It is not obtained by substituting a smoothness cutoff into an uncapped answer. The result allows arbitrary adaptive allocations and arbitrarily small admissible caps. It does not settle unknown-smoothness adaptation, nonperiodic analytic classes, binomial-mixture budget optimization, or a universal claim about every analytic inverse problem.

7.10 Remaining obstructions and what the new theorems do not close

Several ambitions that motivated the work remain open. Listing them explicitly prevents a restricted theorem from being mistaken for a solution of a broader problem.

1. **Prime-zeta fluctuations at growing scales.** We obtain all moments for each fixed window. A variance asymptotic as the window height tends to infinity, a central limit theorem, an effective error in T, and universality of pair correlations are not proved. The zeros counted are those of $P^{(m)}$ in $\Re s > 1$, not those of the Riemann zeta function in its critical strip.

2. **Unrestricted continuous or analytic two-query recovery.** The definable obstruction fixes one finite-parameter query family. It does not apply to the union of all polynomial degrees or to all analytic functions. The unrestricted two-query question in three dimensions is not declared solved.

3. **General matrix designs at the sharp tail boundary.** The arbitrary convergent-design construction is scalar. The previous fixed-dimensional matrix results for regular sparse designs remain distinct; an arbitrary matrix-design extension is not inferred by substituting a quadratic form into a scalar Blaschke product.

4. **Robust computation and optimal constants.** The robust population estimator uses a finite predictive cover and attains minimax order. A practical polynomial-time implementation, sharp risk constants, and the best universal exponent constant in the coefficient inequality are separate questions. Ordinary MLE robustness is not proved.

5. **Finite-height arithmetic reconstruction.** The spectral bridge assumes finitely many frequencies, a known top frequency and phase, a quantitative Diophantine inequality, and exact finite zero lists. It does not give a finite-height reconstruction algorithm for all primes or a new equivalence to the Riemann hypothesis. Its height exponent is not asserted optimal.

6. **Further Bessel boundaries.** The SD strictness theorem uses the signed spectral kernel established in the preceding manuscripts. It does not prove termination of every earlier finite-moment obstruction algorithm, nor an elementary formula for the SD maximizer. Rechecking that inherited kernel remains part of any external review of the full program.

7. **Unknown analytic regularity.** The population cap curves assume one fixed periodic Fourier envelope. Adaptation to its unknown parameters and nonperiodic analytic classes is not supplied by the present proof.

The strongest case for external verification is the coefficient theorem: its exact target is published, its dependence on the classical Hahn bound is short, and its remaining finite calculation is reproducible from this TEX file. The gamma-factor obstruction and definable-game theorem offer separate compact arguments to scrutinize. The larger statistical and analytic classifications require their own expert assessment. We regard these as research claims ready for such scrutiny, not as a certification of historical priority or journal level.

Part VIII

Geometry and Rigidity of Inverse Observation

Adaptive Width, Blind Zero Recovery, and Sharp Boundaries in Inverse Probability

Source manuscript: 16 October 2026.

Source abstract. We investigate twelve targets arising from a program on observation and recovery. The principal geometric result is a universal lower bound $e_n(X) \geq \mathrm{UW}_{2^n-1}(X)/2$ for adaptive continuous scalar information, with completely unrestricted selection and decoding rules. Together with the upper bound of Krieg, Novak, and Ullrich, it determines the exact number of measurements in every power-of-two dimension and leaves at most one measurement undecided in other dimensions. A second geometric transfer quantifies the topological and polynomial-degree cost of two-query recovery in dimension three. For binomial populations, an explicit rational linear program attains the previously established optimal contamination curve without knowing the contamination level; the same curve holds for adversaries that inspect the whole clean sample. For prime-supported Dirichlet series, one-sided exact zeros determine unknown positive coefficient magnitudes and relative phases, and the zero hull is conjugate to a compact torus with an explicitly identified dynamical spectrum. Further results close the arbitrary-design matrix Gaussian tail criterion, retain stable nonidentifiability inside every positive compact scale interval, resolve individual spectral parameters from nonlinear minima, and improve finite-zero experiments by adaptation and calibration. Finally, an exact gamma extraction boundary exhibits an exponential separation between arbitrary and infinitely divisible remainders. Classical ingredients, previous unpublished dependencies, and unresolved stronger questions are identified throughout. These are twelve scoped targets, including related refinements, not twelve named historical conjectures.

Reading guide and claim accounting

This is a sequel to the 223-page integrated observation-and-recovery manuscript and its subsequent papers, most immediately the 49-page *Exact Recovery Laws and a Coefficient Conjecture* [105]. We do not re-count that paper's coefficient conjecture or its statistical and zero-density results. The common question here is how the geometry of an observation map determines what can be recovered, at what precision, and with what computation. The mechanisms differ across the six branches; no theorem asserting that all six models are equivalent is intended.

The externally posed target is the dimension-dependent lower-bound question in [60, Section 3]. The new bound answers that request and gives exact counts for $m = 2^k$; it does not settle the optimal count in every dimension. Most other targets are extensions, gap closures, or new questions formulated within this research program. In particular, the large-order gamma asymptotic refines the exact capacity theorem and should not be advertised as a second independent historical conjecture. A search of the cited primary literature is not an exhaustive priority determination.

No.	Target	Result and scope	Proof
1	Unrestricted adaptive query lower bounds	Logarithmic necessity; exact query count in power-of-two dimensions. External question, partially resolved in general dimension.	8.1.2, 8.1.4

No.	Target	Result and scope	Proof
2	Cost of a tame first query	Explicit degree–error and fiber-topology obstructions for two queries; nonsharp degree exponent.	8.1.5, 8.1.6
3	Efficient robust binomial recovery	Polynomial-bit-time proper estimator with the previous sharp rate and unknown contamination.	8.2.1
4	Full-sample adversarial corruption	Matching upper and lower rates with a pathwise cap on replacements.	8.2.2
5	Blind prime-series inversion	One-sided zeros determine positive magnitudes and relative phases, up to the necessary scalar ambiguity.	8.3.2
6	Complete zero-process structure	Torus conjugacy, complete outside rigidity, and exact balanced-prime eigenfrequencies.	8.3.4
7	Arbitrary matrix Gaussian designs	Exact design–tail classification for every convergent or divergent harmonic subseries.	8.4.1
8	Compact stable scales	Sparse failure persists in every positive compact scale interval for fixed index below one.	8.5.1
9	Individual nonlinear spectral parameters	Sharp mass, separation, and noise exponents, with matching examples.	8.6.1
10	Adaptive noisy-zero experiments	Logarithmically many amplitude settings, improved sufficient height, and top-frequency calibration.	8.6.3
11	Maximal probability gamma factor	Exact one-point Kummer criterion on the first excess-power line at the tail rate.	8.7.3
12	Size of the gamma-capacity separation	Exponential unrestricted deficit versus algebraic ID deficit; refinement of target 11.	8.7.4

Evidence and dependencies. All twelve rows have proofs in this manuscript, including proofs of the new reductions. The geometric branch invokes classical width, Betti, and triangulation theorems; the statistical branch explicitly states the inverse estimate and lower bounds imported from the previous unpublished paper. The improved recurrence exponent in the spectral branch is an application of an established filling theorem. The prime-series inverse and gamma-capacity arguments do not rely on the earlier unpublished spectral constructions. Numerical calculations, where mentioned, are not substitutes for proofs. Automated cross-review found and prompted editorial and hypothesis clarifications; it does not constitute human refereeing or machine-checked formal verification.

What is not claimed. There is no proof here of the Riemann hypothesis, no recovery of unknown prime frequencies from nontrivial zeta zeros, no unrestricted two-query solution in dimension three, and no growing-window prime-zero central limit theorem. Theorems about exact infinite zero data do not by themselves provide stable finite-data algorithms. Publication-level importance and priority require assessment beyond this manuscript.

8.1 A logarithmic lower bound for unrestricted adaptive continuous information

We connect adaptive information to the multiplicative dimension bound for Urysohn widths. The resulting lower bound applies to arbitrary continuous measurements, with no definability, polynomial-degree, Lipschitz, or continuity assumption on the selection rule or decoder. It answers the request for a dimension-dependent lower bound explicitly posed by Krieg–Novak–Ullrich [60, Section 3]. It determines the exact query count when the dimension is a power of two. The three-dimensional two-query problem remains unresolved.

8.1.1 Width of an adaptive decision tree

For a nonempty compact metric space (X, d) and integer $d_0 \geq 0$, let

$$\mathrm{UW}_{d_0}(X) = \inf_{\mathcal{U}} \max_{U \in \mathcal{U}} \operatorname{diam} U,$$

where the infimum is over finite relatively open covers of X of multiplicity at most $d_0 + 1$. Multiplicity means the maximum number of cover members containing the same point. Distances on subsets are always inherited from X. In particular, the subscript is the *dimension* of an approximating complex, not the allowed cover multiplicity. This is the usual Urysohn-width convention [102, Section 1].

We recall the following width inequality. The special case needed here also has a short covering proof, which we include to make clear that singular and disconnected fibers cause no difficulty.

Lemma 8.1.1 (Width of a scalar map)**.** *If $f : X \to \mathbb{R}$ is continuous and X is compact, then for every integer $d_0 \geq 0$,*

$$\mathrm{UW}_{2d_0+1}(X) \leq \sup_{t \in f(X)} \mathrm{UW}_{d_0}(f^{-1}(t)). \tag{8.1.1}$$

More generally, for a continuous map to a metrizable space of covering dimension at most q, the subscript on the left can be replaced by $(q+1)(d_0+1)-1$.

Proof. This is the multiplicative width inequality of Gromov, in the form proved in [102, Theorem 2.1]. Here is its covering argument. Choose w strictly above the supremum on the right. A finite cover of a compact fiber of mesh less than w and multiplicity at most $d_0 + 1$ extends to a cover with the same bounds on the inverse image of a neighborhood of its target point. Indeed, take a finite closed shrinking of the fiber cover, then sufficiently small ambient open neighborhoods of the shrinking sets. Compactness preserves the mesh bound and keeps every originally empty $(d_0 + 2)$-fold intersection empty. Their union contains the fiber, so it contains the inverse image of a sufficiently small target neighborhood, again by compactness.

Refine the resulting cover of $f(X)$ to a finite open cover of multiplicity at most $q + 1$. Over each member use its corresponding fiber-neighborhood cover. The combined cover of X has mesh less than w and multiplicity at most $(q + 1)(d_0 + 1)$. Let w decrease to the supremum. For scalar targets, $q = 1$. □

An adaptive n-query procedure on X chooses a continuous real function at each stage, based arbitrarily on all earlier answers. It outputs a point of a metric space Z containing X isometrically. The selection rules and the output map may be discontinuous and need not be measurable. Write $e_n(X; Z)$ for its optimal worst-case error. This notation concerns exact measurements and deterministic algorithms.

Theorem 8.1.2 (The adaptive width bound)**.** *For every compact metric space $X \subset Z$ and every integer $n \geq 0$,*

$$e_n(X; Z) \geq \frac{1}{2} \mathrm{UW}_{2^n-1}(X). \tag{8.1.2}$$

The conclusion remains valid if a query is required to be continuous only on the compact feasible set determined by the preceding answers. Thus it also holds under the usual, stronger requirement that every selected query be continuous on the entire original input space.

Proof. Fix one procedure with worst-case error at most r. We prove by induction on its maximum remaining depth n that $\mathrm{UW}_{2^n-1}(X) \leq 2r$. With no remaining query the output is

one point $z \in Z$. All inputs lie in its radius-r ball; therefore $\operatorname{diam} X \le 2r$, and the one-set cover gives $\mathrm{UW}_0(X) \le 2r$.

For $n \ge 1$, let $f : X \to \mathbb{R}$ be the first query. Each nonempty fiber $X_t = f^{-1}(t)$ is compact. After its answer t is fixed, the remaining procedure on X_t has at most $n-1$ queries and error at most r. Every later query restricts to a continuous function on the corresponding compact feasible subset. The induction hypothesis gives

$$\mathrm{UW}_{2^{n-1}-1}(X_t) \le 2r \quad (t \in f(X)).$$

Apply Lemma 8.1.1 with $d_0 = 2^{n-1} - 1$. Its left subscript is $2(2^{n-1}-1)+1 = 2^n - 1$, as required. Padding a shorter branch with constant queries is harmless. No regularity in the parameter t is used: the width lemma combines covers whose existence has been proved separately on each fiber, not the possibly discontinuous choices made by the original procedure. Finally take the infimum over procedures. □

Corollary 8.1.3 (Vector-valued rounds)**.** *Suppose round j observes a continuous vector with q_j real coordinates, selected adaptively from the preceding rounds. For s rounds the worst-case error is at least*

$$\frac{1}{2}\,\mathrm{UW}_{\prod_{j=1}^{s}(q_j+1)-1}(X).$$

Proof. Use the general target-dimension version of Lemma 8.1.1 in the same induction, starting with cover multiplicity one at the leaves. The multiplicities are multiplied by $q_j + 1$ at the successive levels. □

8.1.2 Consequences for the open query-count problem

Let $N(m)$ be the minimum number of adaptive continuous scalar measurements sufficient to approximate every vector of $\mathbb{R}^m$ with any prescribed positive uniform error. One can fix any norm; finite-dimensional norm equivalence leaves $N(m)$ unchanged.

Theorem 8.1.4 (Logarithmic necessity and exact power-of-two counts)**.** *For every integer $m \ge 1$,*

$$\lceil \log_2(m+1) \rceil \le N(m) \le \lceil \log_2 m \rceil + 1. \tag{8.1.3}$$

In particular,

$$N(2^k) = k + 1 \qquad (k \ge 0). \tag{8.1.4}$$

More quantitatively, on the cube $[-R, R]^m$ with maximum-norm loss,

$$e_n([-R,R]^m; \mathbb{R}^m) = R \quad \textit{whenever } 2^n \le m. \tag{8.1.5}$$

Thus below this threshold adaptivity and unrestricted continuous nonlinear queries do not improve on the zero-output procedure at all.

Proof. The Lebesgue covering theorem for the cube implies

$$\mathrm{UW}_{m-1}([-R,R]^m, \|\cdot\|_\infty) = 2R.$$

For the lower bound, a cover of mesh less than $2R$ cannot have a member meeting opposite faces in any coordinate; the covering theorem therefore forces multiplicity at least $m+1$. For the upper bound, coordinate projection to $m-1$ coordinates has fibers of diameter $2R$ and gives the asserted width; equivalently use arbitrarily fine covers of the projected cube. Width is nonincreasing in its subscript. When $2^n \le m$, Theorem 8.1.2 consequently gives error at least R, and the zero output gives error at most R.

If an n-query procedure had bounded uniform error on all of $\mathbb{R}^m$ with $2^n \le m$, restricting it to cubes of arbitrarily large R would contradict (8.1.5). Hence $2^n > m$, which is precisely the lower integer bound in (8.1.3). The upper bound is the construction of Krieg–Novak–Ullrich [60, Theorem 1, version 2]. At $m = 2^k$ the lower and upper bounds both equal $k + 1$. □

What is and is not closed. The September 2026 version of [60, Section 3] explicitly asks for lower bounds beyond the Borsuk–Ulam bound $N(m) \ge 2$. Theorem 8.1.4 supplies a logarithmic lower bound and closes the exact-count question in every power-of-two dimension. For a dimension strictly between consecutive powers of two the bounds in (8.1.3) differ by one. In particular they give $2 \le N(3) \le 3$: this argument does not solve the two-continuous-query problem in dimension three. The width inequality used as an ingredient is classical; the claimed contribution is its application to an unrestricted adaptive decision tree and the resulting query bounds.

8.1.3 A quantitative cost for topologically tame first measurements

The general multiplicative-width bound loses information in dimension three. A second theorem of Balitskiy–Berdnikov retains the first homology of the first-query fibers and gives an explicit obstruction there. For a compact metric space X and a continuous scalar function f, put

$$\beta(f; X) = \sup_{t \in f(X)} \dim_{\mathbb{Q}} H_1(f^{-1}(t); \mathbb{Q}).$$

Call f *triangulable* if there is a homeomorphism from a finite polyhedron to X under which f is piecewise linear, allowing also a homeomorphic change of its scalar target coordinate. The metric on the polyhedron is the pullback of the metric of X. It need not be the linear Euclidean metric of a chosen geometric realization.

Theorem 8.1.5 (Two queries require fiber topology)**.** *Suppose $f : X \to \mathbb{R}$ is triangulable and $\beta = \beta(f; X) < \infty$. Every adaptive procedure which uses f as its first measurement and an arbitrary continuous scalar second measurement has error at least*

$$\frac{\mathrm{UW}_2(X)}{2(2\beta + 3)}. \tag{8.1.6}$$

The decoder and the choice of second query are unrestricted. For the Euclidean ball $B_2^3(R)$ this implies the explicit lower bound

$$\frac{R}{\sqrt{3}(2\beta + 3)}. \tag{8.1.7}$$

Proof. The bounded-topology waist theorem [102, Theorem 3.16, with one-dimensional target] gives a fiber $F = f^{-1}(t)$ such that

$$\mathrm{UW}_1(F) \ge \frac{\mathrm{UW}_2(X)}{2\beta + 3}.$$

Its hypothesis is a piecewise linear map from a metric polyhedron; triangulating f and pulling back the metric verifies that hypothesis without changing any width, any fiber topology, or any reconstruction distance. A homeomorphic scalar-coordinate change does not change the fibers. On F the second selected query is one continuous scalar function. Either the definition of width by maps to graphs, or the $n = 1$ case of Theorem 8.1.2, shows that its reconstruction error is at least $\mathrm{UW}_1(F)/2$. This proves (8.1.6).

For (8.1.7), the ball contains the cube $[-R/\sqrt{3}, R/\sqrt{3}]^3$. Restriction of covers and the Lebesgue covering theorem give $\mathrm{UW}_2(B_2^3(R)) \ge 2R/\sqrt{3}$. Here Euclidean mesh is at least maximum-norm mesh, so the cube lower bound applies unchanged. □

Theorem 8.1.6 (An explicit polynomial-degree barrier)**.** *Let $D \geq 1$. On $B_2^3(R)$, every two-query procedure whose first query is a real polynomial of degree at most D and whose second selected query is continuous has worst-case Euclidean error at least*

$$\frac{R}{\sqrt{3}\{2D(2D-1)^2+3\}}. \tag{8.1.8}$$

In particular, for $0 < \varepsilon < R/(3\sqrt{3})$, error at most ε requires

$$D \geq \left(\frac{R/(\sqrt{3}\varepsilon)-3}{8}\right)^{1/3}. \tag{8.1.9}$$

No degree or definability restriction is imposed on the second query. The exponent $1/3$ in this necessary degree growth is not claimed sharp.

Proof. A continuous semialgebraic real-valued function on a compact semialgebraic set admits a finite triangulation as a function; this is the semialgebraic case of Shiota's theorem [104]. Hence the first query restricted to the ball is triangulable. Milnor's Betti estimate, including the compact ball truncation proved in the proof of his Theorem 2 [103], gives

$$\sum_j \dim_{\mathbb{Q}} H_j(\{x : p(x) = t\} \cap B_2^3(R); \mathbb{Q}) \leq D(2D-1)^2.$$

The estimate also covers singular levels; compact semialgebraic triangulation identifies the singular and Čech versions of Betti numbers here. An identically constant polynomial has fiber the whole ball at its attained value, whose first Betti number is zero, and can be handled separately. Thus $\beta(p; B_2^3(R)) \leq D(2D-1)^2$. Substitution in (8.1.7) proves (8.1.8). Finally $2D(2D-1)^2 \leq 8D^3$, giving (8.1.9). □

Corollary 8.1.7 (Necessary topology growth for analytic two-query recovery)**.** *Let $f : \mathbb{R}^3 \to \mathbb{R}$ be real analytic. If a two-query procedure beginning with f has uniform error at most $\varepsilon > 0$ on all of $\mathbb{R}^3$, then for every $R > 0$,*

$$\sup_t \dim_{\mathbb{Q}} H_1(f^{-1}(t) \cap B_2^3(R); \mathbb{Q}) \geq \frac{R}{2\sqrt{3}\varepsilon} - \frac{3}{2}. \tag{8.1.10}$$

Consequently, a fixed analytic first query whose truncated fibers have uniformly sublinear first-Betti growth cannot start a two-query bounded-error recovery procedure.

Proof. The restriction of f to any compact ball is a continuous subanalytic function and is triangulable by [104]. Its fiber Betti numbers are uniformly finite on this ball, since the triangulation is finite. Apply (8.1.7) and rearrange. If the left side were $o(R)$, the resulting inequality would fail for large R. □

Accounting and limitations. There are two principal new transfers here: the universal adaptive width bound, and the quantitative topological/degree obstruction for two queries. The power-of-two counts, vector-valued-round bound, polynomial-degree estimate, and analytic topology-growth requirement are consequences with distinct interpretations, not independent solutions of four unrelated conjectures. The original width and Betti-number theorems are explicitly credited. Neither the exact asymptotic degree cost, nor arbitrary real-analytic two-query recovery in dimension three, nor the optimal count in every dimension has been settled by these arguments.

8.2 Polynomial-time recovery under unknown adversarial contamination

The preceding manuscript established the full Huber-contamination curve for unrestricted binomial populations, but its estimator searched a potentially exponential finite cover. This section removes that computational obstruction. A second result proves the same curve when an adversary observes the entire clean sample before choosing which reports to replace. The contamination proportion is not supplied to the algorithm. The two advances concern computation and the corruption model, respectively; the clean and Huber statistical rates themselves are prior results [105].

For a probability measure G on $[0,1]$, put

$$p_G(j) = \int_0^1 \binom{t}{j} x^j (1-x)^{t-j}\, dG(x), \qquad 0 \le j \le t.$$

On the count alphabet, $W_{1,t}$ denotes Wasserstein distance for cost $|i-j|/t$. Set $\log(e/0) = +\infty$ and define

$$r_{N,t}(\varepsilon) = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(eN)}} + \frac{1}{\sqrt{N}} + \varepsilon + \frac{1}{\sqrt{t\log(e/\varepsilon)}}\right\}. \tag{8.2.1}$$

All unspecified constants in this section are absolute.

Theorem 8.2.1 (Polynomial-time robust population recovery)**.** *There is a single algorithm taking the positive integers N, t and N reported counts, and returning a finitely supported probability measure $\widehat{G}$ with rational support and weights, with the following properties.*

1. *Its bit complexity is polynomial in the length of the explicit sample input, and it does not use a contamination proportion.*
2. *If the reports are independent with common law $R = (1-\varepsilon)p_G + \varepsilon Q$, where Q is arbitrary and $0 \le \varepsilon \le 1/4$, then*
$$\sup_{G,Q} \mathbb{E}_R W_1(\widehat{G}, G) \le C r_{N,t}(\varepsilon).$$

The rate is minimax optimal. For $t \le N$ the algorithm is a rational linear program with $O(Nt + t^2)$ variables and polynomially many constraints. For $t > N$ it returns the empirical distribution of the normalized reports.

Theorem 8.2.2 (Full-sample adversarial replacements)**.** *Let $J_1, \dots, J_N$ be independent with law p_G. After seeing all these counts, an adversary, possibly randomized, may replace at most $\lfloor \varepsilon N \rfloor$ of them by arbitrary elements of $\{0, \dots, t\}$. Write $\mathcal{A}_N(\varepsilon)$ for this class of replacement rules. Then*

$$\inf_{\widetilde{G}} \sup_{G} \sup_{A \in \mathcal{A}_N(\varepsilon)} \mathbb{E} W_1\big(\widetilde{G}(A(J_1, \dots, J_N)), G\big) \asymp r_{N,t}(\varepsilon) \tag{8.2.2}$$

for $0 \le \varepsilon \le 1/4$. The algorithm in Theorem 8.2.1 attains the upper bound, simultaneously for every ε, without knowing the adversary's replacement budget.

Efficient fitting by polynomial approximation, interval discrepancies, and linear or dynamic programming has substantial precedent [106]. Binomial population recovery by optimization is also established [107, 13]. The contribution here is the specific simultaneous control of a low-frequency transport metric and a computationally accessible high-frequency discrepancy. This preserves the $N^{-1/2}$ term while achieving the full inverse-problem rate. Neither interval dynamic programming nor polynomial-time linear programming is being claimed as new.

8.2.1 Hahn truncation makes the separating tests computable

Let $\pi(dx) = dx/(\pi\sqrt{x(1-x)})$ be the arcsine law and let $\rho_j = p_\pi(j) > 0$. Define $h_k(x) = T_k(2x-1)$ and the conditional-expectation operator

$$Qh(j) = \mathbb{E}_\pi[h(X) \mid J = j].$$

The classical beta–binomial orthogonal-polynomial calculation [70] is recorded in the form needed here.

Lemma 8.2.3 (Uniform low-degree likelihood-ratio approximation)**.** *For every G and $0 \le m \le t$ there is a polynomial $P_{G,m}$ of degree at most m such that the signed vector $\widetilde{p}_{G,m}(j) = \rho_j P_{G,m}(j)$ has total mass one and*

$$\sum_{j=0}^{t} |p_G(j) - \widetilde{p}_{G,m}(j)| \le \sqrt{2}\, t \exp\{-m^2/(4t)\} \quad (m < t). \tag{8.2.3}$$

For $m = t$ the approximation is exact. Positivity of the truncated vector is not asserted or needed.

Proof. The arcsine orthogonality relations are $\int h_k\, d\pi = 0$ and $\int h_k h_l\, d\pi = \frac{1}{2}\mathbf{1}_{k=l}$ for positive indices. The beta–binomial posterior shows that Qh_k is a polynomial of degree k in j. The self-adjoint operator Q^*Q preserves polynomial degree, and on degree k its leading coefficient is multiplied by

$$\lambda_{t,k} = \prod_{l=0}^{k-1} \frac{t-l}{t+1+l}.$$

Indeed $\mathbb{E}[X^k \mid J = j] = (j+1/2)_{\overline{k}}/(t+1)_{\overline{k}}$, and the leading coefficient of $\mathbb{E}[J^k \mid X = x]$ is $(t)_{\underline{k}}$. Self-adjointness and orthogonality consequently give $Q^*Qh_k = \lambda_{t,k}h_k$ and

$$\sum_j \rho_j |Qh_k(j)|^2 = \lambda_{t,k}/2.$$

Expanding the degree-t polynomial $\binom{t}{j}x^j(1-x)^{t-j}/\rho_j$ in the Chebyshev basis gives the exact identity

$$\frac{p_G(j)}{\rho_j} = 1 + 2\sum_{k=1}^{t} a_k(G) Qh_k(j), \qquad a_k(G) = \int h_k\, dG, \quad |a_k(G)| \le 1.$$

Truncate this expansion at m. Every retained nonconstant term has zero ρ-mean, proving the mass assertion. Further,

$$\log \lambda_{t,k} = \sum_{l<k} \log\left(1 - \frac{2l+1}{t+1+l}\right) \le -\frac{k^2}{2t}.$$

Cauchy–Schwarz and the triangle inequality therefore bound the ℓ_1 tail by $\sqrt{2}\sum_{k=m+1}^{t} e^{-k^2/(4t)}$, which implies the stated bound. □

For an integer $s \ge 1$, let $\mathcal{I}_s$ consist of subsets of $\{0, \dots, t\}$ that are unions of at most s intervals of consecutive integers; the empty set is included. For probability vectors define

$$D_s(p, q) = \max_{A \in \mathcal{I}_s} |p(A) - q(A)|.$$

Lemma 8.2.4 (A computable surrogate for predictive total variation)**.** *Suppose the approximation errors in Lemma 8.2.3 are at most β for every mixing law, and put $s = m + 1$. Then*

$$\|p_G - p_H\|_{\mathrm{TV}} \le D_s(p_G, p_H) + 4\beta. \tag{8.2.4}$$

Moreover $D_s(p, q)$ is computable using $O(st)$ arithmetic operations, and its epigraph has a linear-program formulation with $O(st)$ auxiliary variables and constraints.

Proof. The difference of the truncated likelihood ratios is a polynomial of degree at most m. Unless it vanishes identically, its positive entries on the integer grid form a union A of at most $m + 1$ intervals, since a nonzero polynomial has at most m real roots. This generous bound also covers isolated zeros separating positive grid entries. The truncated difference has total mass zero, so its half ℓ_1 norm equals its sum over A. The two approximation errors change the half ℓ_1 norm by at most β and the sum over A by at most 2β. This proves even the slightly sharper bound with 3β.

For completeness, interval maximization is a longest-path problem in an acyclic graph. Process the grid entries from left to right. A state records the current position, the number of intervals started (at most s), and whether the preceding entry was selected. Selecting a new entry adds its signed weight; changing from unselected to selected starts one interval. There are $O(st)$ states and edges. Run this recurrence for the weights $p_j - q_j$ and for their negatives, and take the larger terminal value. To express a bound on that value linearly, introduce one potential per reachable state, fix the initial potential to zero, and impose

$$V_{\mathrm{next}} \ge V_{\mathrm{previous}} + \text{edge weight}, \qquad V_{\mathrm{terminal}} \le d$$

for every edge and terminal state. Such potentials exist precisely when every path has weight at most d. Duplicating the construction for the two signs gives the epigraph $D_s(p, q) \le d$. □

8.2.2 The rational linear program and simultaneous fitting

Assume first $2 \le t \le N$; bounded exceptional cases can use the empirical normalized reports, with a change of the universal constant. Choose the following integers and rational threshold:

$$\begin{aligned} M &= Nt, & L &= \lceil \log_2(8tN) \rceil, \\ m &= \min\{t, \lceil \sqrt{4tL} \rceil\}, & s &= m + 1, \\ L_0 &= \lceil \log_2(2(t+2)) \rceil, & b_N &= \frac{\lceil \sqrt{N(2s+1)L_0} \rceil}{N}. \end{aligned}$$

Integer logarithms and ceiling square roots are computed by integer comparisons. Let q denote the empirical distribution of the reports. On the support grid $\{l/M : 0 \le l \le M\}$ minimize

$$F(G) = W_{1,t}(p_G, q) + (D_s(p_G, q) - b_N)_+. \tag{8.2.5}$$

The decision variables are the nonnegative mixing weights, summing to one. On the ordered count alphabet,

$$W_{1,t}(p_G, q) = \frac{1}{t} \sum_{j=0}^{t-1} \left| \sum_{i=0}^{j} (p_G(i) - q_i) \right|.$$

Use ordinary epigraph variables for these absolute values and the longest-path formulation of Lemma 8.2.4, with $d = b_N + z$ and $z \ge 0$, for the remaining term. This is a linear program with rational coefficients. The binomial probability at grid point l/M has bit length $O(t \log(M + 1))$. There are $M + 1 = Nt + 1$ mixing variables and $O(st + t) = O(t^2)$ further variables and constraints, apart from the polynomially sized dense coefficient matrix. Standard rational linear

programming therefore finds an optimal rational vertex in polynomial bit time. Since $t \le N$ on this branch, the bound is polynomial in the explicit sample input length. No finite cover is enumerated and no contamination parameter is used.

Lemma 8.2.5 (Simultaneous transport and variation control)**.** *For the estimator obtained from* (8.2.5)*, in either the Huber model or the replacement model of Theorem 8.2.2,*

$$\mathbb{E}W_{1,t}(p_{\widehat{G}}, p_G) \le C(\varepsilon + N^{-1/2}), \tag{8.2.6}$$

$$\mathbb{E}\|p_{\widehat{G}} - p_G\|_{\mathrm{TV}} \le C\left(\varepsilon + \frac{t^{1/4}\log(eN)^{3/4}}{\sqrt{N}}\right). \tag{8.2.7}$$

The expectations and constants are uniform over all adversaries in the replacement model.

Proof. First consider replacements, and let q^0 be the empirical law of the clean sample. Deterministically $\|q - q^0\|_{\mathrm{TV}} \le \varepsilon$ and $W_{1,t}(q, q^0) \le \varepsilon$. Define

$$S = \max_{A \in \mathcal{I}_s} |q^0(A) - p_G(A)|, \qquad Z = W_{1,t}(q^0, p_G).$$

There are at most $(t+2)^{2s}$ interval unions, by specifying ordered pairs of endpoints and allowing empty padding. Hoeffding's inequality and our choice of b_N give

$$\Pr\{S > b_N + u\} \le e^{-2Nu^2}, \qquad \mathbb{E}(S - b_N)_+ \le C/\sqrt{N}. \tag{8.2.8}$$

Also $\mathbb{E}Z \le 1/(2\sqrt{N})$, by integrating the standard deviation of each empirical CDF coordinate. Round the latent variable under G to the nearest grid point to obtain a comparator G_*. A common-uniform coupling of t Bernoulli variables gives

$$W_1(G_*, G) \le 1/M, \qquad \|p_{G_*} - p_G\|_{\mathrm{TV}} \le t/M = 1/N.$$

Consequently

$$F(G_*) \le 2(\varepsilon + N^{-1}) + Z + (S - b_N)_+.$$

Optimality yields $W_{1,t}(p_{\widehat{G}}, q) \le F(G_*)$ and $D_s(p_{\widehat{G}}, q) \le b_N + F(G_*)$. The triangle inequalities now prove (8.2.6) and

$$\mathbb{E}D_s(p_{\widehat{G}}, p_G) \le C(\varepsilon + b_N + N^{-1/2}).$$

The choice of m and Lemma 8.2.3 make its error $\beta \le 1/N$, since $L \ge \log(8tN)$. Apply Lemma 8.2.4. Finally, $m + 1 \le C\sqrt{t\log(eN)}$ and $L_0 \le C\log(eN)$ give $b_N \le Ct^{1/4}\log(eN)^{3/4}/\sqrt{N}$.

For independent Huber reports, define S and Z with the empirical law q and its mean R instead. The same two bounds hold, and $\|R - p_G\|_{\mathrm{TV}}, W_{1,t}(R, p_G) \le \varepsilon$ replace the deterministic comparison with q^0. Every other inequality is unchanged. □

8.2.3 Transfer to the latent population and sharpness

We state explicitly the quantitative inverse estimate from the preceding manuscript [105] that supplies this transfer. It holds for all probability measures G, H and integers $1 \le u \le k \le t/2$:

$$W_1(G, H) \le \frac{C}{k} + Ce^{4u^2/t}W_{1,t}(p_G, p_H) + \frac{Ce^{2k^2/t}}{u}\|p_G - p_H\|_{\mathrm{TV}}. \tag{8.2.9}$$

Here is its mechanism, to make clear that no property of the estimator is implicit in the transfer. Approximate each 1-Lipschitz test function by degree-k and degree-u polynomials P_k, P_u with uniform errors $O(1/k), O(1/u)$ and bounded derivatives. The degree-t Bernstein coefficients of

a degree-d polynomial have norm at most $Ce^{2d^2/t}$ times its uniform norm for $d \le t/2$. Applied to the derivative of P_u, this bounds the normalized-grid Lipschitz constant of its coefficients by $Ce^{4u^2/t}$. The coefficients of $P_k - P_u$ have size at most $Ce^{2k^2/t}/u$. Testing these two pieces by $W_{1,t}$ and TV proves (8.2.9). The coefficient estimate follows from the Cauchy coefficient formula for $(1+z)^t P(z/(1+z))$, the Bernstein complex-polynomial bound, and the binomial saddle-point estimate; these details are proved in [105, Two-scale inversion lemma]. In particular, the new algorithm does not require a stronger coefficient conjecture.

Upper bounds in Theorems 8.2.1 and 8.2.2. For $t \le N$, write $L_N = \log(eN)$ and $L_\varepsilon = \log(e/\varepsilon)$. In (8.2.9), take

$$u = \lfloor \sqrt{t}/16 \rfloor, \qquad k = \left\lfloor \min\{t/2, \sqrt{at \min(L_N, L_\varepsilon)}\} \right\rfloor, \qquad a = 1/64.$$

For sufficiently large t, these satisfy $1 \le u \le k$; bounded t are harmless because the loss is at most one and the rate contains $1/t$. The approximation term is bounded by the sum of the three inverse terms in (8.2.1). By Lemma 8.2.5, the low-frequency term is $O(\varepsilon + N^{-1/2})$. The contamination contribution to the last term is bounded by

$$\frac{C\varepsilon e^{2aL_\varepsilon}}{\sqrt{t}} \le \frac{C}{\sqrt{tL_\varepsilon}},$$

with the zero-contamination convention. Its sampling contribution is at most

$$\frac{C(eN)^{2a} L_N^{3/4}}{t^{1/4}\sqrt{N}} \le \frac{C}{\sqrt{tL_N}},$$

because $t \le N$ and $(eN)^{2a} N^{-1/4} L_N^{5/4}$ is uniformly bounded. The proof's test degree depends on ε, but the estimator does not.

When $t > N$, return the empirical law of the normalized reports. In the replacement experiment it is within ε in W_1 of the clean normalized empirical law. Coupling a binomial count to its latent parameter costs at most $1/(2\sqrt{t})$, while the latent empirical law has expected W_1 error at most $1/(2\sqrt{N})$. Thus the risk is at most $\varepsilon + 1/(2\sqrt{t}) + 1/(2\sqrt{N})$. For Huber reports, comparison with R and then p_G gives the same bound. This is $O(r_{N,t}(\varepsilon))$ when $t > N$. □

Lower bound for full-sample replacements. The identity replacement rule gives the clean lower bound

$$c \min\{1, t^{-1} + (t \log(eN))^{-1/2} + N^{-1/2}\}.$$

For reference, the first term follows from matching the first t moments of two population laws. The second follows by taking $G_k^{\pm} = (1 \pm h_k/2)\pi$, whose W_1 distance is $\asymp 1/k$ and whose predictive TV is at most $(2\sqrt{2})^{-1} e^{-k^2/(4t)}$, and setting $k \asymp \sqrt{t \log(eN)}$. If this degree exceeds t, the first term absorbs it. Endpoint mixtures with weights differing by $\asymp N^{-1/2}$ give the last term.

Let $q = \lfloor \varepsilon N \rfloor \ge 1$ and set $\eta = q/(16N)$. The TV inverse modulus proved in the preceding manuscript is

$$\sup_{\|p_G - p_H\|_{\mathrm{TV}} \le \eta} W_1(G, H) \asymp \min\{1, t^{-1} + (t \log(e/\eta))^{-1/2} + \eta\}. \tag{8.2.10}$$

Its lower bounds are the same Chebyshev perturbations just described, together with endpoint mass perturbations and exact moment matching; thus this use does not assume an unproved statistical equivalence. Choose a pair attaining at least a fixed positive fraction of the lower bound in (8.2.10), and independently maximally couple N observations from p_G and p_H. The

number K of unequal coordinates has expectation at most $N\eta = q/16$, so $\Pr\{K > q\} \le 1/16$. In the G experiment the adversary samples the coupled H vector conditional on the observed G vector. If $K \le q$ it makes exactly those replacements; otherwise it leaves the sample unchanged. This is an admissible randomized adversary for every realized sample. Its output law is within TV $1/16$ of $p_H^{\otimes N}$. Compare it to the identity adversary in the H experiment. Le Cam's two-point inequality gives a lower bound of an absolute positive constant times $W_1(G, H)$.

Since $q/N \ge \varepsilon/2$ whenever $q \ge 1$, equation (8.2.10) supplies the contamination terms in (8.2.1), with absolute constant changes inside logarithms. If $q = 0$, then $\varepsilon < 1/N$, so both contamination terms are already bounded by the clean terms. Combining the lower bounds proves (8.2.2). For the Huber theorem the sharp lower bound was established in [105], using intersection of Huber neighborhoods at predictive TV at most $\varepsilon/(1-\varepsilon)$. □

8.2.4 What remains separate

Polynomial time here means a bit-complexity guarantee for a rational linear program; no claim of nearly linear runtime or demonstrated large-data practicality is made. The ordinary unmodified nonparametric MLE has not been proved to attain the robust curves by this argument. The algorithm uses both terms in (8.2.5); replacing it by pure predictive TV fitting would lose the needed low-frequency control.

The sharper universal coefficient inequality

$$|C(t, m, j)| \overset{?}{\le} 2^{m^2/t}$$

was also investigated but is not proved here. If c_* denotes the least exponent constant for $|C(t, m, j)| \le e^{c_* m^2/t}$ uniformly, the previous constant-one theorem and the degree-t middle coefficient give only

$$\log 2 \le c_* \le 1.$$

Indeed at even $m = t$ and $j = t/2$, the coefficient magnitude is $\binom{2t}{t}/\binom{t}{t/2} = 2^{t-1/2}(1 + o(1))$. Numerical support for the stronger bound is not a proof and is not counted as an additional closure.

8.3 Blind recovery and complete rigidity of prime-series zero processes

An exact density does not decide whether a zero process forgets its past. In the present non-Gaussian model it does not. We prove two stronger statements. First, one half of a zero divisor determines both the positive coefficient magnitudes and their relative phases, even when the magnitudes were not supplied to the observer. Second, the entire zero process is topologically conjugate to an explicitly specified compact-group rotation. This identifies its complete dynamical spectrum and yields persistent, recurrent correlations of local zero counts. No growing-window central limit theorem is inferred from these statements.

The inverse theorem below concerns a prescribed prime support. It does not recover unknown primes from the nontrivial zeros of the Riemann zeta function. Its objects are zeros of a prime-supported Dirichlet series in its half-plane of absolute convergence.

8.3.1 The class and its frozen boundary

Fix a real number σ_0 and positive numbers a_p, indexed by the primes, such that

$$\sum_p a_p p^{-\sigma} < \infty \quad (\sigma > \sigma_0), \qquad \sum_{p \ge 3} \frac{a_p}{a_2}(2/p)^S = 1 \quad \text{for some } S > \sigma_0. \tag{8.3.1}$$

The second equation has at most one solution, because its left side is strictly decreasing. For $x = (x_p)_{p\ge 3} \in X := \prod_{p\ge 3} \mathbb{T}$, put

$$f_{a,x}(s) = a_2 2^{-s} + \sum_{p\ge 3} a_p x_p p^{-s}, \qquad D_{a,x} = \sum_{f_{a,x}(\rho)=0} \operatorname{ord}_\rho(f_{a,x})\, \delta_\rho$$

on $\mathcal{H} = \{\Re s > \sigma_0\}$. Thus a common phase has already been removed. The unnormalized series $\sum_p a_p \omega_p p^{-s}$ gives the same divisor with $x_p = \omega_p/\omega_2$.

The topology on divisors is vague convergence of locally finite point measures. The map $x \mapsto f_{a,x}$ is continuous for locally uniform convergence: on $\Re s \ge \alpha > \sigma_0$, discard a uniformly small absolutely summable tail and use continuity of the finitely many remaining coordinates. No member is identically zero, by its leading term at $+\infty$. Consequently

$$x \longmapsto D_{a,x} \quad \text{is continuous.} \tag{8.3.2}$$

For completeness, local uniform convergence to a nonzero analytic function preserves the numbers of zeros in finitely many small disks around its zeros, by Rouché's theorem, and gives no other zeros on the remaining compact set. This proves vague convergence, including multiplicities. The same argument applies after restriction to any open subdomain of $\mathcal{H}$.

Lemma 8.3.1 (Boundary recurrence and a coefficient-free decoder)**.** *Every zero of $f_{a,x}$ lies in $\Re s \le S$. In each of the two directions $\Im s \to \pm\infty$, there are zeros whose real parts tend to S. If $\rho_n = \sigma_n + it_n$ is any sequence of zeros with $\sigma_n \to S$, then, for each prime $p \ge 3$,*

$$x_p = -\lim_{n\to\infty} \exp(it_n \log(p/2)). \tag{8.3.3}$$

This decoder uses the known frequencies and the observed right edge, but not the unknown a_p.

Proof. Divide the series by $a_2 2^{-s}$ and write $b_p(\sigma) = (a_p/a_2)(2/p)^\sigma$. At a zero,

$$1 + \sum_{p\ge 3} b_p(\sigma) x_p e^{-it\log(p/2)} = 0.$$

The triangle inequality gives $\sum b_p(\sigma) \ge 1$, and hence $\sigma \le S$. Taking real parts gives the exact nonnegative identity

$$\sum_{p\ge 3} b_p(\sigma) \left| x_p e^{-it\log(p/2)} + 1 \right|^2 = 2\left(\sum_{p\ge 3} b_p(\sigma) - 1 \right). \tag{8.3.4}$$

As $\sigma \to S$, the right side tends to zero and each individual weight tends to a positive number. Equation (8.3.3) follows.

For existence, use the frozen point $x_p^* = -1$. The real function f_{a,x^*} vanishes at S, and

$$f'_{a,x^*}(S) = D_a := \sum_{p\ge 3} a_p p^{-S} \log(p/2) > 0. \tag{8.3.5}$$

The derivative series converges since $S > \sigma_0$. The zero is simple. For every small $\varepsilon > 0$, a disk about S, contained in $\mathcal{H}$ and of radius less than ε, has one zero of every $f_{a,u}$ for u in a suitable open neighborhood of x^*. This follows from uniform convergence and Rouché's theorem.

The flow

$$(\Phi_t x)_p = x_p e^{-it\log(p/2)} \tag{8.3.6}$$

is minimal on X. Indeed, every finite set of the relative frequencies is rationally independent: an integer relation among $\log(p/2)$ would give an equality of a product of odd-prime powers

and a power of 2, so all exponents vanish. The finite-dimensional Kronecker theorem therefore gives a dense orbit on every finite coordinate projection, hence on X. Returns to a nonempty open set occur in both time directions with bounded gaps. One elementary proof of the last assertion covers the compact group by finitely many translates of that set along a dense orbit; translating this finite cover works from every starting point.

Finally, $f_{a,x}(s+it) = 2^{-it} f_{a,\Phi_t x}(s)$. The preceding returns transfer the zero near S to arbitrarily large positive and negative heights of $f_{a,x}$. Choosing radii tending to zero proves the assertion. At most one such zero can lie exactly on $\Re s = S$: equality in (8.3.4) at two different heights would make both $\log(3/2)$ and $\log(5/2)$ integer multiples of a common nonzero frequency, contrary to their rational independence. Thus, if desired, all sufficiently chosen zeros approach the edge from its left. □

8.3.2 A blind inverse theorem from one half of the divisor

Theorem 8.3.2 (Unknown magnitudes and phases from one-sided zeros)**.** *Let a, b satisfy (8.3.1) on the same half-plane, and let $x, y \in X$. Suppose their zero divisors agree on*

$$\{s : \Re s > \sigma_0,\ \Im s < A\}$$

for some finite A. Then

$$b_p = c a_p \quad \text{for every prime } p, \qquad y = x, \qquad c > 0. \tag{8.3.7}$$

The same conclusion holds with $\Im s > A$ in place of $\Im s < A$. More generally, agreement need only hold on a common open vertical strip containing the two right edges and on one of these half-planes. In unnormalized notation, two such series with the same one-sided divisor differ by a nonzero constant.

Proof. The supremum of the real parts of the one-sided observed zeros is the corresponding edge, by Lemma 8.3.1. Thus the two edges coincide, say at S. Applying the coefficient-free decoder to their common zeros gives $x = y$.

For any $u \in X$, minimality supplies a sequence $t_n \to -\infty$ with $\Phi_{t_n} x \to u$. On each compact subset of $\mathcal{H}$, the translated original divisors agree for all sufficiently large n, since its translates then lie below A. By (8.3.2), their limits agree. Hence

$$D_{a,u} = D_{b,u} \quad \text{for every } u \in X. \tag{8.3.8}$$

In the strip version this equality is needed and obtained only on that strip, which still contains a disk about S.

Fix an odd prime p and change only its frozen phase, setting $u_p(\theta) = -e^{i\theta}$ and $u_q(\theta) = -1$ for $q \neq p$. The implicit-function theorem at the simple frozen zero gives a unique analytic zero $\rho_{a,p}(\theta)$ near S. Its derivative is

$$\rho'_{a,p}(0) = i \frac{a_p p^{-S}}{D_a}. \tag{8.3.9}$$

There is an analogous zero for b. Equation (8.3.8), applied for real θ near zero, identifies the two zeros and thus their derivatives. It follows that $b_p/a_p = D_b/D_a =: c$ for every odd prime. The common edge equation then gives $b_2 = c a_2$ as well. This proves (8.3.7). Positive heights are identical with the sign of the returning sequence reversed. □

Theorem 8.3.3 (The support of the zero law identifies the magnitudes)**.** *Let μ_a be the law of $D_{a,x}$ for Haar-distributed $x \in X$. For two admissible positive coefficient sequences, the following are equivalent:*

1. $b_p = ca_p$ *for all primes and some* $c > 0$*;*
2. $\mu_a = \mu_b$*;*
3. $\operatorname{supp} \mu_a = \operatorname{supp} \mu_b$*.*

Thus the full zero law, and even its topological support, determines every ratio a_p/a_2*. Knowledge of only the mean zero intensity is not claimed to suffice.*

Proof. The image $\mathcal{Z}_a = \{D_{a,x} : x \in X\}$ is compact by (8.3.2), and it is exactly the support of μ_a. To check the latter equality, a neighborhood of any image point has a nonempty open preimage in X, hence positive Haar measure. If the two supports agree, choose any divisor in their common image. Theorem 8.3.2, even with the full divisor available, gives proportional coefficients. The other implications are immediate. □

There is also an explicit reconstruction within the recovered graph. The decoder labels every divisor in $\mathcal{Z}_a$ by its phase x. In that graph let $\rho_p(\theta)$ be the frozen zero with only the p-th odd phase changed, and let $\rho_{\mathrm{all}}(\theta)$ be the frozen zero when every odd phase equals $-e^{i\theta}$. Uniform absolute convergence permits the latter simultaneous change. Then

$$\boxed{\frac{a_p}{a_2} = (p/2)^S \frac{\rho_p'(0)}{\rho_{\mathrm{all}}'(0)} \quad (p \geq 3).} \tag{8.3.10}$$

Indeed, the denominator is $i\sum_{q\geq 3} a_q q^{-S}/D_a = ia_2 2^{-S}/D_a \neq 0$. This construction uses the exact support, including its limit points; it is not an estimator from finitely many observed zeros.

For $a_p = (\log p)^m$, $m \geq 0$, condition (8.3.1) holds on $\Re s > 1$, because the edge sum diverges as $\sigma \downarrow 1$ and tends to zero as $\sigma \to \infty$. Consequently the stationary zero laws of the fixed derivatives $P^{(m)}$ are pairwise distinct. Equality of two such laws would force $(\log p/\log 2)^m = (\log p/\log 2)^n$ for every odd prime, and hence $m = n$. This assertion concerns the entire law; it does not prove that their scalar density constants are distinct.

8.3.3 Complete spatial rigidity and the exact dynamical spectrum

Theorem 8.3.4 (A compact torus is encoded in the zeros)**.** *For a fixed admissible* a*, the map* $x \mapsto D_{a,x}$ *is a homeomorphism from* X *onto* $\mathcal{Z}_a$*. Under vertical translation of zero configurations it conjugates the flow* (8.3.6)*. The same injectivity, and a continuous inverse onto its image, hold for the restriction of the divisor to either one-sided open half-plane, or to a one-sided open strip containing* S*.*

In particular, for every bounded Borel region $B \Subset \mathcal{H}$*, the complete configuration in* B *is a measurable function of the configuration outside* B*. Its conditional law given the outside configuration is a point mass. In fact the entire configuration is already determined by its sufficiently remote past.*

With Haar phases, the dynamical system is ergodic and has simple pure point spectrum. Its additive eigenfrequency group is exactly

$$\begin{aligned}\Lambda_0 &= \left\{ \sum_{p\geq 3} k_p \log(p/2) : k_p \in \mathbb{Z} \text{ and finitely many } k_p \neq 0 \right\} \\ &= \{\log(r/s) : r, s \in \mathbb{N},\ \Omega(r) = \Omega(s)\}.\end{aligned} \tag{8.3.11}$$

Here $\Omega(n)$ *counts prime factors with multiplicity. Every frequency displayed occurs, not merely a subset of them.*

Proof. Injectivity follows already from Lemma 8.3.1; continuity was proved above. A continuous injection from a compact space to a Hausdorff space is a homeomorphism onto its image. The same proof uses the open restricted domains and their one-sided decoders. The covariance identity $f_{a,x}(s+it) = 2^{-it} f_{a,\Phi_t x}(s)$ proves conjugacy.

For the outside assertion choose A below all heights in B. Restriction to $\Im s < A$ is measurable from the outside data and its continuous inverse recovers x. The full divisor, and hence its restriction to B, is then a measurable function of x.

The characters $\chi_k(x) = \prod_{p\geq 3} x_p^{k_p}$ form a complete orthonormal basis of $L^2(X)$: cylinder functions are dense, and finite-dimensional Fourier polynomials are dense on each cylinder. Under the flow,

$$\chi_k(\Phi_t x) = e^{-it\lambda_k}\chi_k(x), \qquad \lambda_k = \sum_{p\geq 3} k_p \log(p/2).$$

Unique factorization makes $k \mapsto \lambda_k$ injective. In particular, the only invariant character is constant, proving ergodicity, and every eigenspace is one-dimensional. Completeness proves that there is no continuous spectral component. Conjugacy transfers every character to a measurable observable of the zeros, so none of the displayed eigenfrequencies is lost.

To obtain the last expression in (8.3.11), insert $k_2 = -\sum_{p\geq 3} k_p$. Splitting the positive and negative exponents produces integers r, s with equal total prime-factor counts. Conversely, the prime exponents of any such ratio give these k_p. Cancellation of a common divisor preserves the equality of the two counts. This proves the exact formula. □

The restriction in (8.3.11) is substantive. For example, $\log 3$ is absent, whereas $\log(3/2)$ occurs. The zero process forgets the common phase, and this removes all unbalanced prime characters. It does not forget any further phase information. The spectrum in this theorem is the spectrum of all square-integrable observables of the process; it is not a claim that each local counting observable has a nonzero coefficient at every listed frequency.

Proposition 8.3.5 (Exact recovery after sparse deletions)**.** *Fix a and retain only some of the zeros below height 0. If the number deleted between heights $-T$ and 0 is $o(T)$, then the retained zeros still determine x and the full divisor. The same conclusion holds if one directly assumes that the retained zeros contain a sequence approaching S.*

Proof. For each fixed $\varepsilon > 0$, the Rouché neighborhood and bounded-gap returns in Lemma 8.3.1 give at least $c_\varepsilon T - O(1)$ distinct zeros in $S - \varepsilon < \Re s \leq S$, $-T < \Im s < 0$. To see distinctness, use separated returning windows, with spacing greater than the fixed disk diameter. Deleting $o(T)$ points cannot remove all these zeros, or all but finitely many of them. A diagonal choice gives a retained sequence whose real parts tend to S. The decoder then recovers x. Multiplicities do not affect the argument, since the Rouché disk has one simple zero. No inserted false zeros are allowed in this statement. □

8.3.4 What the classification says about fluctuations

For a real, centered $G \in L^2(\mu_a)$, identify it with a function on X and write $G = \sum_{k\neq 0} g_k \chi_k$ in L^2. Orthogonality gives the absolutely and uniformly convergent covariance expansion

$$C_G(t) := \mathbb{E}[G(D)G(D-it)] = \sum_{k\neq 0} |g_k|^2 e^{it\lambda_k}. \tag{8.3.12}$$

Here $D - it$ is the vertically shifted configuration, with the sign chosen consistently with (8.3.6); replacing t by $-t$ would give the same real covariance. In particular,

$$\lim_{H\to\infty} \frac{1}{H} \int_0^H C_G(t)\, dt = 0, \tag{8.3.13}$$

$$\lim_{H\to\infty} \frac{1}{H} \int_0^H |C_G(t)|^2\, dt = \sum_{k\neq 0} |g_k|^4 > 0 \quad \text{if } G \neq 0. \tag{8.3.14}$$

Furthermore, there are $t_j \to \infty$ for which $C_G(t_j) \to C_G(0) = \operatorname{Var} G$. To prove these facts, first truncate the absolutely convergent sum. Nonzero distinct frequencies have zero time average; this proves the two limits, with the tails controlled uniformly. Simultaneous approximation of the finitely many remaining phases makes their cosines tend to one along arbitrarily large times. An increasing sequence of truncations proves the recurrence statement. Thus the stationary zero process is not weakly mixing and does not have decaying correlations for every local observable.

This last assertion can be witnessed by actual local zero counts, not just by a decoder depending on the whole divisor. Choose $\sigma_0 < \alpha < S < \beta$ so that the mean zero measure has no mass on $\Re s = \alpha$. Such α exist outside a countable set. There are no zeros on $\Re s = \beta$. On this fixed strip, counts in bounded-height windows are uniformly bounded over x: otherwise compactness, locally uniform convergence, and Rouché's theorem would give unboundedly many zeros of a nonzero limit on a compact neighborhood. Stationarity therefore gives a finite intensity $0 < c_{a,\alpha} < \infty$, with positivity from the frozen-zero neighborhood. For any $0 < h < 1/c_{a,\alpha}$, the count

$$X_h(x) = D_{a,x}\big((\alpha, \beta) \times (0, h)\big)$$

has mean strictly between zero and one and is an integer-valued bounded random variable, hence has positive variance. It obeys (8.3.12)–(8.3.14) and recurrent maximal covariance. Horizontal boundary lines have zero mean mass: by stationarity such an atom at one height would occur at every height, contradicting local finiteness of the mean measure.

For an ordinary, fixed-phase prime series the same covariance is an arithmetic translation average. The finite torus Fourier argument proves unique ergodicity of (8.3.6). A bounded function with a Haar-null discontinuity set has the same time average on every orbit: sandwich its upper and lower regularizations by continuous functions whose integrals differ arbitrarily little. The count X_h is discontinuous only when a zero meets the boundary of the rectangle, a Haar-null event by the preceding choice of its edges. Its product with a fixed translate has the same property. Consequently, for every fixed real τ and every $x \in X$,

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T [X_h(\Phi_t x) - c_{a,\alpha} h][X_h(\Phi_{t+\tau} x) - c_{a,\alpha} h]\, dt = C_{X_h}(\tau). \tag{8.3.15}$$

Thus persistent correlations apply to the actual translates of $P^{(m)}$, as well as to a formal random model. The order of limits is explicit: $T \to \infty$ first for each fixed lag, followed by a study of that limiting covariance as the lag grows.

Finally, for the integrated centered observable $A_H = \int_0^H G(D - it)\, dt$, Parseval gives the exact identity

$$\operatorname{Var} A_H = \sum_{k\neq 0} |g_k|^2 \frac{4 \sin^2(H\lambda_k/2)}{\lambda_k^2}. \tag{8.3.16}$$

The equality follows first for Fourier polynomials and then in L^2, since integration over $[0, H]$ is a bounded operator of norm at most H. In particular $\operatorname{Var} A_H = o(H^2)$ by dominated convergence. Determining a sharper growth law requires the actual small-frequency weights $|g_k|^2$ near zero. A dense pure point frequency group can accumulate at zero, so the absence of a continuous spectral component alone neither rules out a central limit theorem nor proves bounded number variance.

8.3.5 Relation to earlier work and unresolved questions

The qualitative freezing of Euler-product prime phases near the rightmost critical-point edge goes back to Arias de Reyna and van de Lune [47]. The preceding manuscripts developed its quantitative prime-series versions. We use the elementary nonnegative defect behind that mechanism, but the blind theorem also recovers *unknown positive magnitudes* by differentiating the phase-labeled zero graph. That additional inverse step, the one-sided formulation, and the exact identification of the zero system are the claims made here.

There is a different and much more developed rigidity theory for Gaussian analytic functions. Sodin's form of Calabi rigidity [108] identifies Gaussian zero laws from their first intensity in the appropriate setting. Our coefficients have fixed magnitudes and independent uniform phases, and the proof above uses the full topological support of the zero law, not its first intensity. Feldheim's stationary Gaussian zero results [50] likewise concern a different probability structure. Favorov's pure-point Fourier results [109] concern real zero sets of Dirichlet series with bounded spectrum. The dynamical spectrum in (8.3.11) is not identified with the distributional Fourier transform of an individual unweighted zero measure.

These are internally formulated inverse and fluctuation problems; we do not identify them with a named external open conjecture. The following stronger requests remain open here: recovery from first intensity alone; quantitative stable blind recovery from noisy finite zero data; a growing-window variance asymptotic for the endpoint prime-zeta count; and a central limit theorem or a rigorous contrary normalization theorem. In particular, recurrence of local correlations is not a proof that every possible central limit theorem fails.

8.4 Every sparse matrix design at the quasi-analytic tail boundary

The previous two manuscripts left a specific obstruction: a scalar Blaschke denominator, composed with an inverse quadratic form, acquires pole hypersurfaces inside the matrix contour deformation. The following construction removes that obstruction. The numerator still depends on the quadratic form, but its compensating denominator depends on the *individual matrix eigenvalues*. This distinction permits arbitrary convergent harmonic subseries, without regularity of their counting function.

Conditionally on $(A,\Sigma)\in\mathbb{R}^d\times\mathbb{S}_d^+$, let X_j be independent $N_d(A,\Sigma)$ vectors and put

$$\mathcal{O}_n=(X_1,\dots,X_{d-1},X_1+\cdots+X_n),\qquad n\ge d.$$

Only the separate population laws of these experiments are given. For sufficiently large E, define

$$L_1(v)=\log(E+v),\quad L_{j+1}(v)=\log L_j(v),\qquad W_{m,a}(v)=\frac{\sqrt{v}}{L_1(v)\cdots L_{m-1}(v)L_m(v)^a}.$$

Let $\mathcal{C}_{m,a}^{(d)}$ be the class of mixing laws satisfying $\mathbb{E}e^{cW_{m,a}(\operatorname{tr}\Sigma)}<\infty$ for some $c>0$, with no assumption on the mean marginal or its dependence on the covariance.

Theorem 8.4.1 (Complete matrix design–tail classification)**.** *Fix $d,m\ge 1$, $a\ge 0$, and an infinite set $\mathcal{N}\subset\{d,d+1,\dots\}$. The selected experiments identify every mixing law in $\mathcal{C}_{m,a}^{(d)}$ if and only if*

$$a\le 1\quad\text{or}\quad\sum_{n\in\mathcal{N}}\frac{1}{n}=\infty. \tag{8.4.1}$$

If both conditions fail, there are distinct absolutely continuous witnesses with identical complete mean and covariance marginals and identical finite mixed moments of every order. Their covariance tails satisfy the defining exponential integrability for every $c > 0$. *Any prescribed lower bound* $\Sigma \succeq b_0 I$, $b_0 > 0$, *can be imposed.*

The positive directions and the negative direction for regular sparse matrix designs are inherited. The new statement is the matrix converse for *every* convergent harmonic subseries. This closes an internal gap, rather than a named external conjecture. Classical quasi-analytic moment theory supplies the positive tail criterion [79]; general differential-operator obstructions to infinite-mixture identification are discussed in [8].

8.4.1 A denominator on the spectrum rather than the quadratic form

Let $a_j > 0$, $a_j \to \infty$, and $\Lambda = \sum_j a_j^{-1} < \infty$. Put

$$E_0(z) = \prod_j (1 - z/a_j), \quad L(z) = \sum_j \log(1 + z/a_j), \quad J(R) = \frac{1}{2} \sum_j \log(1 + R^2/a_j^2). \tag{8.4.2}$$

The product is entire and the principal logarithm sum is holomorphic off the negative real axis. Both converge locally uniformly on their respective domains. We use the elementary bounds

$$\log |E_0(z)| \le J(|z|) + \Lambda(-\operatorname{Re} z)_+, \tag{8.4.3}$$

$$\operatorname{Re} L(z) \ge J(|z|) \qquad (\operatorname{Re} z \ge 0), \tag{8.4.4}$$

$$\operatorname{Re} L(z) \ge J(|z|) - 2\Lambda(-\operatorname{Re} z)_+ \qquad (|\operatorname{Re} z| \le |z|/2). \tag{8.4.5}$$

For the first bound, square each factor and use $\log(1+x) \le x$ with $x = 2a_j(-\operatorname{Re} z)_+/(a_j^2 + |z|^2)$. The second follows directly from the squared factors of $1 + z/a_j$. For the third, the possible negative correction inside the logarithm has magnitude at most $1/2$; use $\log(1 - x) \ge -2x$ and sum. In particular, no quantitative density bound on the a_j is used.

For large R let

$$P_m(R) = \prod_{j=1}^{m} \log_j R, \qquad D(R) = P_m(R)(\log_m R)^p, \qquad p > 0.$$

Choose K sufficiently large and define

$$J_1^*(w) = \log(K + w), \quad J_{j+1}^*(w) = \log(K + J_j^*(w)), \qquad Q(w) = w/J_m^*(w)^p.$$

Principal logarithms are used. These functions are holomorphic off the negative real axis, with no additional cuts or denominator zeros. The following sector facts will be used:

$$\operatorname{Re} Q(w) \ge 0 \quad (\operatorname{Re} w \ge 0), \qquad \operatorname{Re} Q(w) \ge \frac{c|w|}{D(|w|)} \quad \left(|w| \ge R_0, \quad |\arg w| \le \frac{\pi}{2} + \frac{c}{P_m(|w|)}\right). \tag{8.4.6}$$

For clarity, these estimates follow from

$$|J_m^*(Re^{i\theta})| \sim \log_m R, \qquad \arg J_m^*(Re^{i\theta}) = \frac{\theta}{P_m(R)}(1 + o(1)),$$

uniformly with first angular derivatives for bounded θ. The denominator subtracts an angular margin of order $1/P_m(R)$ near $\theta = \pm\pi/2$. On the right half-plane its argument has the same sign as that of w and, by increasing K, satisfies $p|\arg J_m^*(w)| \le |\arg w|$. These observations prove (8.4.6). Fixed derivatives satisfy polynomial bounds on slightly narrower sectors by Cauchy's formula. At zero, $Q(w) = b_1 w - b_2 w^2 + O(w^3)$ with $b_1, b_2 > 0$.

Lemma 8.4.2 (Matrix realization with arbitrary summable reciprocal zeros)**.** *Let $h \in C_c^\infty(\mathbb{R}^d \setminus \{0\})$ be nonzero, real and even. For a sufficiently large constant c_0, the function*

$$\mathcal{H}(u,T) = h(u)E_0(u^T T^{-1}u)\exp\{-2\operatorname{tr} L(c_0T^{-1}) - \operatorname{tr} Q(T^{-1}) - \operatorname{tr} T^{1/2}\}, \quad \operatorname{Re} T \succ 0, \tag{8.4.7}$$

is the Fourier–Laplace transform of a nonzero real Schwartz function $g(a,\Sigma)$ supported on $\Sigma \succeq 0$. Its two coordinate marginals and all joint polynomial moments vanish. For every $M \geq 0$ there are positive C_M, c_M such that

$$|g(a,\Sigma)| \leq C_M(1+|a|)^{-M}e^{-c_M W_{m,1+p}(\operatorname{tr}\Sigma)}. \tag{8.4.8}$$

Proof. All matrix functions are principal holomorphic functions. We verify the positive tube estimate, the boundary realization, and the contour estimate, including the location of the possible poles.

Positive tube. Write $R = T^{-1}$. If $\operatorname{Re} T \succ 0$ then $\operatorname{Re} R = T^{-*}(\operatorname{Re} T)T^{-1} \succ 0$. For an accretive matrix,

$$\|R\| \leq C_d \sum_{i=1}^{d} |\lambda_i(R)|. \tag{8.4.9}$$

To prove this, triangularize R unitarily. Positivity of its Hermitian part implies $|R_{ij}| \leq 2\sqrt{\operatorname{Re}\lambda_i \operatorname{Re}\lambda_j}$ for the strictly upper triangular entries. The Frobenius norm gives (8.4.9). Let $U = \sup_{u\in\operatorname{supp} h}|u|^2$. Taking $c_0 \geq 2UC_d d$ gives $|u^T Ru| \leq (c_0/2)\max_i|\lambda_i(R)|$. Since $\operatorname{Re}(u^T Ru) \geq 0$, equations (8.4.3)–(8.4.4) show that one copy of $\operatorname{Re}\operatorname{tr} L(c_0R)$ absorbs the entire numerator. The other copy is nonnegative. Furthermore, (8.4.9) and (8.4.6) imply

$$\operatorname{Re}\operatorname{tr} Q(R) \geq c\|R\|/D(\|R\|) - C$$

when $\|R\|$ is large. For $T = X + iV$, $X \succ 0$, write $T^{1/2} = B + iC$ with real symmetric matrices B, C. The principal square root is accretive, $B^2 - C^2 = X \succ 0$, and $BC + CB = V$. Thus $\|C\| \leq \|B\|$ and $\|V\| \leq 2(\operatorname{tr} B)^2$. Consequently

$$|\mathcal{H}(u, X+iV)| \leq Ce^{-c\sqrt{\|V\|}}, \qquad X \succ 0, \tag{8.4.10}$$

uniformly in X and in the compact u support.

A boundary neighborhood estimate. For invertible real symmetric V, put $r = r(V) = \{\operatorname{tr}(V^{-2})\}^{-1/2}$. Then $\min_i|\lambda_i(V)|/\sqrt{d} \leq r \leq \min_i|\lambda_i(V)|$ and $|\nabla r| \leq 1$. Choose a smooth positive η with bounded derivative, sufficiently small upper bound, equal to a fixed positive constant for all sufficiently large r, and

$$\eta(r) = \varepsilon r/D(1/r) \quad \text{for small } r. \tag{8.4.11}$$

Consider $T = iV + Z$, where $\|Z\| \leq C_1\eta(r)$ and C_1 is any fixed constant. The inverse norm is $O(1/r)$ by a Neumann series. Every eigenvalue of T stays within $C_1\eta(r)$ of the spectrum of the normal matrix iV. Grouping overlapping spectral disks shows that some inverse eigenvalue has modulus at least c_d/r; all inverse eigenvalues have arguments within $O(\eta(r)/|\lambda_i(V)|)$ of $\pm\pi/2$. Their Q terms have nonnegative real parts and at least one supplies $c/(rD(1/r))$ damping. To justify nonnegativity also for bounded and small inverse eigenvalues, use compactness away from zero and the expansion $Q(w) = b_1w - b_2w^2 + O(w^3)$ at zero; a sufficiently small upper bound on η gives $\operatorname{Re} Q((-t\eta + i\lambda)^{-1}) \geq 0$ for real $\lambda \neq 0$. For the complex neighborhood used here, an eigenvalue of T has the form $x + iy$, with $|x| \leq C_1\eta(r)$ and $|y| \geq |\lambda_i(V)|/2$. If $|(x+iy)^{-1}|$ is large, the angular margin in (8.4.6) applies, since $\eta(r)/|y| \leq c/P_m(1/|y|)$ after reducing ε. In a compact annulus, strict positivity at the imaginary axis and compactness apply. Finally, for large $|y|$, direct Taylor expansion gives

$$\operatorname{Re} Q((x+iy)^{-1}) = \frac{b_2 + b_1x}{y^2} + O(|y|^{-3}),$$

uniformly for small bounded x. Reducing the fixed upper bound of η makes this positive too. These three ranges justify the nonnegativity assertion for every eigenvalue in the neighborhood.

In this neighborhood

$$(-\operatorname{Re}(u^T T^{-1} u))_+ \le C\eta(r)/r^2, \qquad (-\operatorname{Re}\lambda_i(T^{-1}))_+ \le C\eta(r)/r^2.$$

Choose c_0 still larger if necessary. The largest inverse eigenvalue then makes $J(c_0|\lambda_i|)$ dominate $J(|u^T T^{-1} u|)$. Equations (8.4.3) and (8.4.5) therefore bound the logarithm of the numerator and its L regularizer together by

$$C\Lambda\eta(r)/r^2 = C\Lambda\varepsilon/(rD(1/r)). \tag{8.4.12}$$

For sufficiently small ε, the Q damping absorbs this quantity. This is the step that eliminates every design-density assumption. At large $\|V\|$ the square-root trace retains its $c\sqrt{\|V\|}$ damping, up to a constant. Cauchy's formula on these neighborhoods gives the same damping for each fixed matrix derivative, times powers of $1/r$, $D(1/r)$, and $1 + \|V\|$. For u derivatives, apply the scalar Cauchy formula to E_0 on circles of radius one and use the chain rule. Enlarging c_0 absorbs the change in J; the additional negative-real-part error is bounded by $C\Lambda$. All polynomial factors are absorbed by the damping. Thus $H(u, V) = \mathcal{H}(u, iV)$, extended by zero at $\det V = 0$, is Schwartz, compactly supported in u.

Inversion and cone support. In orthonormal coordinates on $\mathbb{S}_d$, with $D_d = d(d+1)/2$, define

$$g_u(\Sigma) = (2\pi)^{-D_d} \int e^{i\operatorname{tr}(V\Sigma)} H(u, V)\, dV, \quad g(a, \Sigma) = (2\pi)^{-d} \int e^{-iu^T a} g_u(\Sigma)\, du.$$

These are Schwartz functions. If the matrix line is shifted to $X + i\mathbb{S}_d$, $X \succ 0$, the inverse integral equals $e^{-\operatorname{tr}(X\Sigma)} g_u(\Sigma)$, by the Cauchy–Riemann equations and integration by parts, first starting at $X = \epsilon I$ and then using (8.4.10) to let $\epsilon \downarrow 0$. The same estimate bounds these integrals uniformly in X. If Σ has a negative eigenvalue, choose $X_0 \succ 0$ with $\operatorname{tr}(X_0\Sigma) < 0$ and let $X = tX_0$, $t \to \infty$. This proves cone support. Fourier inversion now yields the required Laplace identity.

The contour and its poles. Deform iV to

$$T(V) = -\eta(r(V))I + iV.$$

Throughout the homotopy, every matrix is normal with nonreal eigenvalues. Its inverse eigenvalues avoid the negative real axis. Consequently the L factors have no poles or cuts on the swept region. Equivalently,

$$e^{-2\operatorname{tr} L(c_0 T^{-1})} = \prod_j \det(I + c_0 T^{-1}/a_j)^{-2};$$

its poles require an individual eigenvalue of T to equal $-c_0/a_j$, which cannot occur. In contrast, a denominator depending on $u^T T^{-1} u$ could vanish there. The numerator E_0 is entire.

For rigor, write $V = W + tI/\sqrt{d}$, $\operatorname{tr} W = 0$, and fix W. Break the real t axis at the finitely many roots of $\det(W + tI/\sqrt{d}) = 0$. On each open interval, deform the scalar contour to $t + i\sqrt{d}\,\eta(r(W + tI/\sqrt{d}))$. Every eigenvalue has nonzero imaginary part throughout the vertical strip swept out by this contour, so all the preceding functions are holomorphic. The short joining segments vanish at finite endpoints by the $\exp\{-c/(rD(1/r))\}$ estimate and at infinity by the square-root estimate. The Jacobian has modulus at most $1 + \sqrt{d}\sup|\eta'|$. These uniform integrable bounds justify Fubini in W and prove the full contour identity.

Put $v = \operatorname{tr}\Sigma$. Along the deformed contour, the absolute inverse integrand for small r is bounded by

$$C\exp\left\{-c\sqrt{\|V\|} - c\left(\frac{vr}{D(1/r)} + \frac{1}{rD(1/r)}\right)\right\}.$$

The expression in parentheses is at least $cW_{m,1+p}(v)$ uniformly for large v. Indeed, on $\frac{1}{4}\log v \le \log(1/r) \le \log v$ the fixed-depth logarithms are comparable with those of v, and $vr + r^{-1} \ge 2\sqrt{v}$. Below this range the first term is at least $v^{3/4}$ divided by logarithmic factors; above it the second is at least $v/D(v)$. Both dominate the required bound. For r bounded away from zero, $\eta(r)$ has a positive lower bound and gives exponential decay in v. Reserve half the damping for integration in V and integrate by parts in u to obtain (8.4.8).

Reality follows by conjugation and evenness. The transform is nonzero because varying $T = tI$ avoids the discrete zeros of E_0. Its vanishing near $u = 0$ gives the covariance marginal and every joint polynomial moment. Finally $\mathcal{H}(u, \epsilon I) \to 0$ as $\epsilon \downarrow 0$, by the positive-tube Q estimate; dominated convergence gives the zero mean marginal. □

Proof of Theorem 8.4.1. The conditionally independent blocks of lengths $1, \dots, 1, n-d+1$ yield exactly the mixing-transform arguments

$$(u, T), \qquad T = C + uu^T/(2n), \quad C \succeq 0, \operatorname{rank} C \le d-1. \tag{8.4.13}$$

This follows by projecting the weighted frequency matrix onto the vector of square roots of these block lengths and its orthogonal complement. If $T \succ 0$ on this surface, the determinant lemma gives $u^T T^{-1} u = 2n$.

When $a > 1$ and $\sum 1/n < \infty$, apply Lemma 8.4.2 with $a_j = 2n_j$ and $0 < p < a - 1$. Its transform vanishes at all positive definite arguments in (8.4.13). Singular arguments follow by perturbing the square weighted frequency matrix to an invertible one and using continuity. Normalize the positive and negative parts of g to obtain the two mixing laws. Their marginals and moments coincide; the tail assertion follows from $W_{m,1+p}/W_{m,a} \to \infty$. Translation by $b_0 I$ imposes the covariance lower bound without changing any transform zero.

For completeness, let ρ be the difference of two candidate laws with identical experiments. Fix $u \neq 0$ and a rank-$d-1$ matrix $C \succeq 0$ with u not orthogonal to its kernel. The bounded holomorphic function

$$H(z) = \int e^{iu^T A - \operatorname{tr}(C\Sigma) - z u^T \Sigma u}\, d\rho$$

vanishes at $z = 1/(2n)$, $n \in \mathcal{N}$. If the harmonic subseries diverges, the half-plane Blaschke condition gives $H \equiv 0$. If $a \le 1$, the covariance envelope instead yields

$$(\mathbb{E}(\operatorname{tr}\Sigma)^k)^{1/(2k)} \le Ck \log k \cdots \log_{m-1} k\, (\log_m k)^a,$$

so the Stieltjes–Carleman series diverges. Taylor expansion at zero along the accumulating zeros makes every moment of the associated complex measure vanish; moment determinacy of a common positive dominating measure implies that measure is zero. Thus $H \equiv 0$ again. For arbitrary $T \succ 0$, take $z = (u^T T^{-1} u)^{-1}$ and $C = T - zuu^T$ to recover the full transform. Continuity at $u = 0$ and matrix Fourier–Laplace uniqueness give $\rho = 0$. □

8.5 Compact scale does not repair sparse stable observation below index one

We next ask whether a bounded scale interval eliminates the sparse sampling counterexamples. For stable index greater than one, it does. For index less than one, it does not, even on an arbitrarily short positive scale interval. The distinction comes from whether the Laplace observation points accumulate at zero or escape to infinity. The classical background is completeness of exponential systems and Müntz theory [110]; we prove the smooth signed-density construction explicitly.

Fix $0 < \alpha < 1$, $-1 \le \beta \le 1$, and let

$$\widehat{K}_{a,b}(t) = \exp\{ita - b|t|^{\alpha}(1 - i\kappa \operatorname{sgn} t)\}, \qquad \kappa = \beta \tan(\pi\alpha/2).$$

Conditionally on (A, B), the X_j are independent with this kernel. We observe only the separate laws of $S_n = X_1 + \cdots + X_n$.

Theorem 8.5.1 (Exact compact-scale stable sampling criterion)**.** *For any fixed $0 < b_0 < b_1 < \infty$ and terminal set $\mathcal{N} \subset \mathbb{N}$, these observations identify every mixing law on $\mathbb{R} \times [b_0, b_1]$ if and only if*

$$\sum_{n \in \mathcal{N}} n^{-(1-\alpha)} = \infty. \tag{8.5.1}$$

When the series converges there are distinct absolutely continuous witnesses, with scale support in the interior of $[b_0, b_1]$, having the same two coordinate marginals and the same finite mixed polynomial moments of every order. Their signed difference is smooth, compactly supported in scale, and Schwartz in location.

No mixing over the index or skewness is permitted. The unrestricted scale criterion was already proved in the preceding manuscript; the new claim is the persistence of its converse inside every prescribed compact positive scale interval.

Lemma 8.5.2 (Smooth compact support with summable ray zeros)**.** *Suppose $z_j = c\lambda_j$, $\operatorname{Re} c > 0$, $\lambda_j \to \infty$, and $\sum_j \lambda_j^{-1} < \infty$. For every nonempty bounded open real interval I there is a nonzero real $f \in C_c^{\infty}(I)$ whose bilateral Laplace transform F satisfies $F(z_j) = F(\overline{z}_j) = F(0) = 0$ for every j. Finite zero sets are allowed as well.*

Proof. Put $r_j = |z_j|$. For real x,

$$(1 + x^2)\left|\frac{\sin 4x}{4x}\right|^2 \le 1. \tag{8.5.2}$$

If $|x| \ge 1/\sqrt{15}$, use $|\sin 4x| \le 1$. Otherwise $|4x| \le 4/\sqrt{15}$ and the alternating Taylor bound gives $\sin u/u \le 1 - u^2/7$ on this interval. Squaring shows $|\sin 4x/(4x)|^2 \le 1 - x^2$, proving the assertion. Moreover

$$|(1 - iy/z_j)(1 - iy/\overline{z}_j)| \le 1 + y^2/r_j^2.$$

This follows by squaring: the difference from $(1 + y^2/r_j^2)^2$ is $-4y^2(\operatorname{Im} z_j)^2/r_j^4$.

Choose J_0 so large that $A = 8\sum_{j>J_0} r_j^{-1}$ is arbitrarily small, and form the entire product

$$G(s) = \prod_{j>J_0} (1 - s/z_j)(1 - s/\overline{z}_j)\left\{\frac{\sinh(4s/r_j)}{4s/r_j}\right\}^2.$$

It converges locally uniformly, since its deviations from one are $O(r_j^{-1})$ on compact sets. It is real on the real axis and $G(0) = 1$. Equation (8.5.2) gives $|G(iy)| \le 1$. Its exponential type is at most A: the product of the hyperbolic sine factors has type at most A, while

$$\log \prod_j (1 + |s|/r_j)^2 = o(|s|).$$

For the last assertion, remove finitely many factors and bound the remaining logarithms by $2|s|\sum_{j>J} r_j^{-1}$, then send $J \to \infty$.

The elementary bounded-axis form of the Paley–Wiener theorem now realizes G as the transform of a compactly supported distribution on $[-A, A]$. One may apply the usual theorem with $A + \epsilon$ and then let $\epsilon \downarrow 0$: the bound on the imaginary axis and the Phragmén–Lindelöf

principle give $|G(s)| \le e^{(A+\epsilon)|\operatorname{Re} s|}$ up to a constant. Multiply $G(s)$ by the Laplace transform of a nonzero real smooth bump supported in $[-\epsilon, \epsilon]$, by $s\prod_{j\le J_0}(1 - s/z_j)(1 - s/\overline{z}_j)$, and by e^{-bs}, where b is the midpoint of I. In real space these operations are smoothing, finitely many differentiations, and translation. By first making A and then ϵ sufficiently small, the resulting nonzero smooth function is supported in I. The resulting transform has all the prescribed zeros and the factor s makes its integral zero. □

Proof of Theorem 8.5.1. For a fixed nonzero t, the data for S_n/n evaluate the mixing Fourier–Laplace transform at

$$(t, s_n(t)), \qquad s_n(t) = |t|^{\alpha}(1 - i\kappa \operatorname{sgn} t)n^{1-\alpha}.$$

Their half-plane Blaschke sum diverges precisely under (8.5.1), proving sufficiency without any scale bound.

For necessity apply Lemma 8.5.2 to $z_n = (1 - i\kappa)n^{1-\alpha}$, choosing the support of f in a small interval strictly inside (b_0, b_1). Choose a real nonzero even function $h \in C_c^{\infty}(\mathbb{R} \setminus \{0\})$ supported sufficiently close to $\{-1, 1\}$ that

$$G(t, b) = h(t)|t|^{\alpha} f(b|t|^{\alpha})$$

has its b support inside (b_0, b_1) for every t in that support. Set $g(a, b) = (2\pi)^{-1} \int e^{-ita} G(t, b)\, dt$. It is real and nonzero, smooth, compactly supported in b, and Schwartz in a. Its transform is

$$\int e^{ita - sb} g(a, b)\, da\, db = h(t)F(s/|t|^{\alpha}).$$

For positive t the observation argument is a prescribed zero z_n, and for negative t its conjugate is a prescribed zero. The scale marginal and every mixed polynomial moment vanish because G is zero near $t = 0$. The location marginal vanishes because $F(0) = 0$. Normalize the positive and negative parts of g to finish. □

Remark 8.5.3 (The contrasting index range and remaining targets). For $1 < \alpha \le 2$ and fixed skewness, a compact upper bound on B makes the transform entire in s. For every infinite terminal set the points $s_n(t)$ converge to zero, so the identity theorem identifies the mixing law. This elementary positive comparison is not counted as a third new target. At index one with nonzero skewness, the points instead run along a vertical line with imaginary parts proportional to $\log n$; compact-scale identification then requires a different entire-function sampling analysis, not the preceding ray-zero argument. Neither that classification nor joint mixing over unknown stable shapes is claimed here. Quantitative finite-data stability at the matrix quasi-analytic boundary also remains open.

8.6 Adaptive zero observation and sharp recovery of individual spectral atoms

The earlier spectral results determined an atom-count boundary and constructed a finite-height transition experiment on a uniform amplitude grid. We address two different targets here. First, a noisy zero test can be turned into an adaptive comparison oracle, including calibration of the previously known largest frequency and its phase. Second, the nonlinear threshold data determine individual nodes and masses with the sharp separation exponents, not merely a Wasserstein error. The latter exponents are classical in linear super-resolution [111, 112]; our contribution is their transfer, with an explicit weak-mass dependence, to the nonlinear minimum observations. The improvement of the recurrence exponent uses the classical optimal filling-time theorem [113], and is not claimed as a new theorem in Diophantine approximation.

Throughout this section fix $0 < a < b < 1$, and write

$$S_\mu(x) = \int_a^b x^r \, d\mu(r), \qquad T_\mu(C) = \min_{x>0} \frac{1 + Cx}{S_\mu(x)}, \qquad \mu \in \mathcal{P}[a, b]. \tag{8.6.1}$$

A one-sided setting list consists of distinct points of $(0, a/(1-a))$. Constants below may depend on the fixed support interval, setting list, and atom bound. They do not depend on the minimum mass, node separation, or noise level.

8.6.1 Recovery of individual masses and locations

Theorem 8.6.1 (Sharp local parameter recovery from nonlinear minima)**.** *Fix $m \geq 2$ and $2m-1$ one-sided settings C_i. Suppose $\mu = \sum_{j=1}^m w_j \delta_{r_j}$, with increasing nodes, $r_{j+1} - r_j \geq \Delta > 0$, and $w_j \geq w > 0$. Let ν be any probability on $[a, b]$ with at most m atoms, and put*

$$\delta = \max_i |T_\nu(C_i) - T_\mu(C_i)|, \qquad D_j = \prod_{k \neq j} |r_j - r_k|, \qquad \kappa_j = \sum_{k \neq j} |r_j - r_k|^{-1}.$$

There are constants $c, K > 0$ such that, whenever

$$\delta \leq cw\Delta^{2m-1}, \tag{8.6.2}$$

ν has exactly m atoms, which can be written in increasing order as $\nu = \sum_j v_j \delta_{s_j}$, and

$$|s_j - r_j| \leq \frac{K\delta}{w_j D_j^2}, \tag{8.6.3}$$

$$|v_j - w_j| \leq \frac{K(1 + \kappa_j)\delta}{D_j^2}. \tag{8.6.4}$$

In particular, the uniform bounds are

$$\max_j |s_j - r_j| \leq \frac{K\delta}{w\Delta^{2m-2}}, \qquad \max_j |v_j - w_j| \leq \frac{K\delta}{\Delta^{2m-1}}.$$

Their joint dependence on δ, w, Δ is sharp: for every $0 < w \leq 1/(2m)$, every sufficiently small Δ, and $0 < \delta \leq cw\Delta^{2m-1}$, there are two m-atomic probabilities, both with minimum mass at least w and minimum separation at least Δ, whose complete threshold curves on any fixed compact interval $I \subset (0, \infty)$ differ by at most δ, while one node moves at least $c_I\delta/(w\Delta^{2m-2})$ and one mass changes by at least $c_I\delta/\Delta^{2m-1}$.

The threshold required for a stable assignment of individual atoms is of order $w\Delta^{2m-1}$. This is distinct from the larger $w\Delta^{2m-2}$ noise scale at which the preceding paper could reject every smaller positive model. Counting the atoms can therefore be easier than resolving their individual locations.

Lemma 8.6.2 (Uniform smooth-test conversion with two sparse sides)**.** *For probabilities α, β having at most m atoms each, and any $p \in C^{2m-1}[a, b]$, one has*

$$\left| \int p \, d(\alpha - \beta) \right| \leq K \|p\|_{C^{2m-1}} \max_{1 \leq i \leq 2m-1} |T_\alpha(C_i) - T_\beta(C_i)|. \tag{8.6.5}$$

No separation or mass bound is needed in this lemma.

Proof. Here are the geometric and interpolation details of the conversion. For a profile S, put $H = S - xS'$. Strict concavity gives a unique contact $x(C)$, satisfying

$$C = \frac{S'(x)}{H(x)}, \qquad \frac{dC}{dx} = \frac{S''(x)S(x)}{H(x)^2} < 0, \qquad \frac{a}{1-a} \le Cx \le \frac{b}{1-b}.$$

The tangent line is $L_C(x) = (1 + Cx)/T(C)$. At our fixed one-sided settings, all contacts lie in a fixed compact interval above 1; $S, H, -S''$ are bounded above and below there. Consequently distinct contacts have a uniform positive separation. The intersection of two consecutive tangent lines lies strictly between their contacts, at a uniform positive distance from both: this follows by integrating the uniform lower bound for $-S''$ and dividing by the bounded difference of their slopes. For the two end contacts, close the corresponding brackets by fixed external endpoints strictly outside the common compact contact interval, choosing the lower external endpoint still larger than 1.

Write δ for the maximum threshold difference. For small δ, the two tangent-line systems, and their consecutive intersections, differ by $O(\delta)$. For each setting, the two contacts lie between the corresponding neighboring intersections, with uniform margins. At the α contact, $S_\alpha - S_\beta \ge -K\delta$, and at the β contact the reverse inequality holds. An intermediate point therefore has $|S_\alpha - S_\beta| \le K\delta$. The margins imply that the $2m - 1$ points obtained in this way have a uniform separation. Their logarithms, together with $u_0 = 0$, give $2m$ separated points in a fixed compact set, with $u_i > 0$ for $i > 0$, such that

$$\left| \int e^{u_i r} \, d(\alpha - \beta)(r) \right| \le K\delta.$$

The sample at 0 is exactly zero.

Interpolate p on the union of the supports by a linear combination of the $2m$ functions $e^{u_i r}$; pad the node list, if necessary, with repeated nodes and impose the corresponding derivatives. In divided-difference coordinates the interpolation matrix extends continuously across every collision. Every limiting matrix is invertible, since a nonzero sum of $2m$ real exponentials with distinct exponents has at most $2m - 1$ zeros counted with multiplicity. The latter assertion follows inductively from Rolle's theorem after multiplication by the first exponential's reciprocal. Compactness bounds the interpolation coefficients by $K\|p\|_{C^{2m-1}}$. Integrating the interpolant proves the claim. For large δ increase K and use $|\int p \, d(\alpha - \beta)| \le 2\|p\|_\infty$. □

Upper bounds in Theorem 8.6.1. Write the nodes of ν as $s_1, \dots, s_q$, $q \le m$, padding to m by repetitions for the following polynomial test. For each j use

$$Q_j(r) = \prod_{k \ne j} (r - r_k) \prod_{\ell=1}^{m} (r - s_\ell).$$

Its degree is $2m - 1$ and its C^{2m-1} norm is uniformly bounded. Lemma 8.6.2 gives

$$w_j D_j \prod_{\ell=1}^{m} |r_j - s_\ell| \le K\delta. \tag{8.6.6}$$

If no node of ν is within $\Delta/4$ of r_j, the left side is at least $w\Delta^{2m-1}/4^m$. Thus, for sufficiently small c in (8.6.2), each of the m disjoint $\Delta/4$ neighborhoods contains a node. There are exactly m distinct nodes, matched in increasing order. For $k \ne j$,

$$|r_j - s_k| \ge |r_j - r_k| - \Delta/4 \ge \tfrac{3}{4}|r_j - r_k|.$$

Substitution into (8.6.6) gives (8.6.3).

For the masses put

$$P_j(r)=\frac{\prod_{k\neq j}(r-r_k)(r-s_k)}{\prod_{k\neq j}(r_j-r_k)(r_j-s_k)}.$$

Its C^{2m-1} norm is at most K/D_j^2, so

$$|w_j-v_jP_j(s_j)|\le K\delta/D_j^2. \tag{8.6.7}$$

By taking c smaller, the node bound ensures $|s_j-r_j|\le c_0\Delta$ with an arbitrarily small fixed c_0. On this short segment all the factors defining P_j stay away from zero, its logarithmic derivative is bounded by $K\kappa_j$, and

$$\tfrac12\le P_j(s_j)\le 2,\qquad |P_j(s_j)-1|\le K\kappa_j|s_j-r_j|.$$

Moreover $\delta/D_j^2\le cw\Delta\le cw_j$. Thus (8.6.7) first gives $v_j\le Kw_j$, and then (8.6.3) and (8.6.7) give (8.6.4). □

Matching examples in Theorem 8.6.1. Choose nodes $r_j(0)=r_0+2(j-1)\Delta$ in the interior of $[a,b]$. Give the left endpoint mass $2w$ and divide the remaining mass equally among the other $m-1$ nodes. All masses are at least $2w$. For distinct moving nodes define

$$B_j=\prod_{k\neq j}(r_j-r_k)^{-2},\qquad A_j=-2B_j\sum_{k\neq j}(r_j-r_k)^{-1},$$

and solve the smooth ordinary differential equations

$$\dot w_j=A_j,\qquad \dot r_j=B_j/w_j. \tag{8.6.8}$$

For $0\le s\le cw\Delta^{2m-1}$, a direct bootstrap gives node separations between fixed positive multiples of Δ, displacements at most $\Delta/4$, masses at least w, and $w_1(s)\asymp w$. Indeed, while these bounds hold, $|\dot w_j|\le K\Delta^{-(2m-1)}$ and $|\dot r_j|\le K/(w\Delta^{2m-2})$. Integrating gives strict versions of the assumed bounds for a sufficiently small c, so standard continuation extends the solution throughout the interval.

The partial-fraction identity for $\prod_j(z-r_j)^{-2}$, or the leading coefficient of a Hermite interpolant, gives for every $f\in C^{2m-1}[a,b]$

$$\sum_j\{A_jf(r_j)+B_jf'(r_j)\}=f[r_1,r_1,\dots,r_m,r_m]. \tag{8.6.9}$$

Here the right side is the confluent divided difference of order $2m-1$; its absolute value is at most $\|f^{(2m-1)}\|_\infty/(2m-1)!$. This identity proves that the flow preserves every moment through degree $2m-2$, including total mass, while its moment of degree $2m-1$ has derivative 1. For $f(r)=x^r$ on any fixed compact x interval, (8.6.9) is uniformly bounded. Therefore

$$\sup_x|S_{\mu_s}(x)-S_{\mu_0}(x)|\le Ks,\qquad \sup_{C\in I}|T_{\mu_s}(C)-T_{\mu_0}(C)|\le K_Is.$$

The second estimate follows by evaluating the two quotients at each other's minimizing contacts, which all lie in a fixed compact interval with positive denominators.

At the leftmost node all terms in the sum defining A_1 have the same sign. Along the bootstrap interval,

$$\dot w_1\ge c\Delta^{-(2m-1)},\qquad \dot r_1\ge c/(w\Delta^{2m-2}).$$

Take $s=\delta/K_I$, decreasing the allowed constant in (8.6.2) if necessary. Integrating these two inequalities proves both matching lower bounds, while the minimum separation remains at least Δ. □

8.6.2 A self-calibrating noisy comparison oracle

Consider the controlled exponential polynomial

$$f_{A,C}(z) = -1 + A\sum_{j=1}^{m} w_j e^{\ell r_j z + i\phi_j} + Ce^{\ell z + i\phi_*}. \tag{8.6.10}$$

The control amplitudes A, C are known, but ℓ, ϕ_*, and all the smaller phases may be unknown. Assume only a known compact bound $0 < \ell_- \le \ell \le \ell_+$. A zero observation at height H means a complete list, with multiplicity, of the zero ordinates in $(0, H)$, each perturbed by at most ϵ. There are no unreported or spurious zeros. This is an observation model; it does not assert that a certified zero list can be computed at negligible cost.

Put $d = m + 1$ and $\omega = (\ell r_1, \dots, \ell r_m, \ell)$. It is enough to assume a finite collection of nonresonance inequalities at the resolution of the intended experiment:

$$|k \cdot \omega| \ge \gamma \|k\|_\infty^{-\tau} \quad (0 < \|k\|_\infty \le K_0/h), \qquad 0 < \gamma \le 1, \quad \tau > d - 1. \tag{8.6.11}$$

The fixed constant K_0 below depends on d and the frequency bounds. In particular, the theorem does not require infinitely many inequalities or even global rational independence.

Theorem 8.6.3 (Adaptive calibrated recovery from noisy zero ordinates)**.** *Fix $I \Subset (0, \infty)$. There are known constants $A_- < A_+$, $h_0, \epsilon_0, K_0, K, H_0 > 0$, depending only on $a, b, m, I, \ell_-, \ell_+, \tau$, with the following property. Let $0 < h < h_0$, $0 \le \epsilon < \epsilon_0$, and assume (8.6.11). At height*

$$H \ge H_0 + K\gamma^{-1}h^{-\tau}, \tag{8.6.12}$$

one calibration experiment at $A = 0$, and at most $K \log(3/h)$ adaptively selected amplitudes in $[A_-, A_+]$ for each $C \in I$, suffice to construct

$$|\widehat{T}(C) - T_\mu(C)| \le K(h + \epsilon^2). \tag{8.6.13}$$

The result is uniform in all phases and positive masses. With a known top frequency and phase, omit the calibration experiment. The smaller phases are not reconstructed by this procedure.

Compared with the preceding uniform-grid construction, the number of amplitude experiments changes from $O(h^{-1})$ to $O(\log(1/h))$, while the sufficient height changes from $O(\gamma^{-1}h^{-(d+\tau)})$ to $O(\gamma^{-1}h^{-\tau})$. The second improvement comes from using an established optimal recurrence estimate. The statement does not claim an optimal lower bound for the height needed to observe zeros.

Lemma 8.6.4 (Finite-order filling bound)**.** *Under (8.6.11), every target on $(\mathbb{R}/2\pi\mathbb{Z})^d$ is approached within coordinate distance h by $t\omega$ in every interval of length $K\gamma^{-1}h^{-\tau}$.*

Proof. Apply the truncated filling-time theorem of Dumas and Fischler [113, Theorem 1], which recovers the optimal theorem of Berti, Biasco, and Bolle. On the unit torus, a unit direction α satisfying $|k \cdot \alpha| \ge \gamma_0 \|k\|_2^{-\tau}$ up to order $(1 + d^2 d!)/\eta$ fills to Euclidean accuracy η in time at most $C(d, \tau)\gamma_0^{-1}\eta^{-\tau}$. Use $\alpha = \omega/\|\omega\|_2$ and $\eta = h/(2\pi)$. Norm equivalence and the fixed bounds on $\|\omega\|_2$ are absorbed in K_0, K. Decrease the normalized Diophantine constant by a fixed factor, if necessary, so that the nonempty-class condition in that theorem holds; this is possible for every $\tau > d - 1$ and changes only K. Translating the initial point proves the assertion for every interval. □

Lemma 8.6.5 (Calibration with bounded ordinate errors)**.** *For sufficiently large fixed H_0 and small fixed ϵ_0, the $A = 0$ list yields $\widehat{\ell}, \widehat{\phi}_*$ such that*

$$|\widehat{\ell} - \ell| \le K\epsilon/H, \qquad \operatorname{dist}\big(\widehat{\ell}\widetilde{t} + \widehat{\phi}_* - (\ell t + \phi_*), 2\pi\mathbb{Z}\big) \le K\epsilon$$

for every $t \in [0, H]$ and every reported $\widetilde{t}$ with $|\widetilde{t} - t| \le \epsilon$.

Proof. At $A = 0$ the ordinates form an arithmetic progression of spacing $2\pi/\ell$. If $\epsilon < \pi/(4\ell_+)$, their order is preserved. Let $\widetilde{t}_1, \widetilde{t}_N$ be the first and last entries and set

$$\widehat{\ell} = \frac{2\pi(N-1)}{\widetilde{t}_N - \widetilde{t}_1}, \qquad \widehat{\phi}_* = -\widehat{\ell}\widetilde{t}_1 \pmod{2\pi}.$$

For $H \geq H_0$ the true first-to-last span is comparable to H. Perturbing its two endpoints by at most ϵ proves the frequency estimate. Since $\ell t_1 + \phi_*$ is an integral multiple of 2π, the phase difference for any t is bounded by $|\widehat{\ell} - \ell|H + K\epsilon$, as required. □

The step that permits adaptive amplitude selection is stronger than the existence of a successful grid point near the transition.

Lemma 8.6.6 (A uniform one-sided comparison test)**.** *There is a known $R > 0$ such that the following rule applies to every $A \in [A_-, A_+]$. Accept A if its noisy zero list contains $\widetilde{t}$ satisfying*

$$\operatorname{dist}(\widehat{\ell}\widetilde{t} + \widehat{\phi}_*, \pi + 2\pi\mathbb{Z}) \leq R\sqrt{h} + K\epsilon. \tag{8.6.14}$$

At the height (8.6.12):

1. *every $A \geq T_\mu(C)$ is accepted;*
2. *every accepted A satisfies $A \geq T_\mu(C) - K(h + \epsilon^2)$.*

Proof. All thresholds lie in a fixed positive compact interval: evaluate at $x = 1$ for the upper bound, and use $S(x) \leq \max(x^a, x^b)$ for the lower bound. Choose $[A_-, A_+]$ containing this interval with positive margins. Write $T = T_\mu(C)$, let x_* be its minimizing contact, and set

$$g_A(z) = -1 + AS_\mu(e^{\ell z}) - Ce^{\ell z}, \qquad \sigma_* = (\log x_*)/\ell.$$

The function g_T has a double zero at σ_*, and

$$g_T''(\sigma_*) = -\ell^2 T \int r(1-r)x_*^r \, d\mu(r) \leq -c < 0.$$

All third derivatives are uniformly bounded in a fixed complex neighborhood. Hence, on a sufficiently small circle of radius ρ, $|g_T(z)| \geq c\rho^2$, and g_T has exactly two zeros there, counting multiplicity.

If $A > T$, the strictly concave function $F_A(x) = AS_\mu(x) - 1 - Cx$ has two real positive zeros. All these zeros, for $A \in [T, A_+]$, lie in a fixed compact interval: F_A is uniformly negative near 0 and near infinity. On this interval $F_A'' \leq -c$, and $\max F_A \geq (A - T)S(x_*) \geq c(A - T)$. The upper quadratic bound for a concave function of curvature at most $-c$ now gives

$$|F_A'(x)| \geq c\sqrt{A - T}$$

at either root. The corresponding derivative of g_A has the same lower bound, and its second derivative is uniformly bounded.

Lemma 8.6.4, applied to the phase target $(-\phi_1, \ldots, -\phi_m, \pi - \phi_*)$, gives $t_0 \in [1, H-1]$ such that, uniformly in the fixed complex neighborhood of these real roots,

$$|f_{A,C}(z + it_0) - g_A(z)| \leq K_1 h. \tag{8.6.15}$$

We show that the perturbed function has a zero within imaginary distance $K\sqrt{h}$ of the real line, uniformly for every $A \geq T$.

For $0 \leq A - T \leq Dh$, compare directly with g_T on a circle of radius $R_0\sqrt{h}$. Choose R_0 so large that its quadratic lower bound dominates both (8.6.15) and the $O(Dh)$ difference

$g_A - g_T$. Rouche's theorem gives two zeros inside. Here D is the fixed constant chosen next. For $A - T > Dh$, use a circle about a real simple root of g_A of radius $R_1 h/\sqrt{A-T}$. Taylor's formula gives the lower bound

$$|g_A(z)| \geq \left(cR_1 - \frac{KR_1^2}{D} \right) h.$$

First choose R_1 so that $cR_1 > 2K_1$, then D large enough to make this bound exceed $K_1 h$. Linear-term dominance also shows that g_A has exactly one zero in this disk; Rouche's theorem supplies the perturbed zero. Its distance is at most $(R_1/\sqrt{D})\sqrt{h}$. Finally choose R_0 for the first case. Both cases are valid for all sufficiently small h.

After translating back, the resulting zero lies in $(0, H)$ and its top phase is within $K\sqrt{h}$ of π. The calibration lemma and the ordinate error prove acceptance with a suitable R. This proves the first assertion.

Conversely, any accepted entry comes from a true zero whose top phase equals $\pi + v$ modulo 2π, with $|v| \leq K(\sqrt{h} + \epsilon)$. Put $x = e^{\ell \Re z}$ and $y = Cx$. The zero equation and the triangle inequality give

$$AS_\mu(x) \geq |1 - ye^{i(\pi+v)}|, \qquad |1 - ye^{i(\pi+v)}|^2 \geq (1+y)^2 - yv^2 \geq (1+y)^2(1 - v^2/4).$$

For small h, ϵ, it follows that $A \geq (1 - K(h+\epsilon^2))(1+Cx)/S_\mu(x) \geq T - K(h+\epsilon^2)$. This proves the second assertion. □

Proof of Theorem 8.6.3. Calibrate once, and start with a known interval $[L, U]$ containing all possible thresholds. At each step test its midpoint A. If rejected, replace L by A; the first assertion of Lemma 8.6.6 guarantees $A < T$. If accepted, replace U by $\min(U, A + E)$, where $E = K(h + \epsilon^2)$ is the known bound in its second assertion. The true threshold remains in the interval. Its width after J steps is at most

$$2^{-J}(U_0 - L_0) + 2E.$$

Taking $J = \lceil \log_2((U_0 - L_0)/h) \rceil$ makes the first term at most h, with $J \leq K \log(3/h)$ for small h. The midpoint of the final interval satisfies (8.6.13). No monotonicity of the finite zero-list acceptance function is assumed or needed. □

Corollary 8.6.7 (Individual spectral resolution from finite noisy zeros)**.** *Run the adaptive experiment at $2m-1$ fixed one-sided settings, and fit any at most m-atomic probability to the resulting threshold vector within its guaranteed error. If the truth has minimum mass w, separation Δ, and $h + \epsilon^2 \leq cw\Delta^{2m-1}$, then after ordering the atoms its location and mass errors are at most*

$$\frac{K(h+\epsilon^2)}{w\Delta^{2m-2}}, \qquad \frac{K(h+\epsilon^2)}{\Delta^{2m-1}},$$

respectively. The sharper D_j, κ_j bounds in Theorem 8.6.1 also apply.

Proof. The truth is a feasible fit, and the threshold discrepancy between the truth and any such fit is at most $K(h + \epsilon^2)$. Apply Theorem 8.6.1. □

The quadratic occurrence of ordinate noise in this construction comes from the quadratic angular deficit at the antipodal phase. It is a proved robustness bound for this experiment, not a general minimax lower bound for all uses of a noisy zero divisor. The calibration experiment requires the ability to set $A = 0$, and the finite nonresonance hypothesis remains an external quantitative assumption. Neither arbitrary missing zeros nor a uniform infinite-prime analogue is covered.

8.7 Gamma extraction has two different sharp boundaries

The preceding paper excluded extraction of the full tail gamma factor from generalized McKay laws with positive excess power. Exclusion of one factor does not determine the largest factor that can actually be extracted. We solve that optimization problem on the first excess-power line, allowing an arbitrary probability remainder. The answer is strictly different from the answer with an infinitely divisible (ID) remainder. Their deficits from the tail shape have respectively exponential and algebraic size at large Bessel order.

These are new transfer questions, not named conjectures in [3]. All arguments in this section are independent of the earlier unpublished signed-spectral constructions. We use only an elementary Laplace transform, uniqueness of Laplace transforms, the Lévy–Khintchine representation on the positive half-line, and standard identities for Kummer's function [6, 75].

8.7.1 The family and the two capacities

Fix $\kappa > 1/2$ and $b > 1$, and let $X_{\kappa,b}$ have density proportional to

$$x^{\kappa-1/2}e^{-bx}I_{\kappa-3/2}(x), \qquad x > 0.$$

This is the $q = 1$ generalized McKay line, with $\mu = \kappa - 3/2$ and $\nu = \mu + 2$. Expanding the positive I_μ series, integrating term by term, and applying the gamma duplication formula gives

$$\int_0^\infty x^{\kappa-1/2}e^{-bx}I_{\kappa-3/2}(x)\,dx = \frac{2^{\kappa-1/2}\Gamma(\kappa)}{\sqrt{\pi}}\frac{b}{(b^2-1)^\kappa}.$$

Consequently

$$L_{\kappa,b}(s) = \frac{b+s}{b}\left(\frac{b^2-1}{(b+s)^2-1}\right)^\kappa, \qquad s \geq 0. \tag{8.7.1}$$

The range $\kappa > 1/2$ covers the full standard parameter range $\mu > -1$ considered here, including Bessel orders $-1 < \mu \leq -1/2$.

For a rate $\rho > 0$, let $h_{\mathrm{P}}(\rho)$ be the supremum of shapes $h \geq 0$ for which

$$X_{\kappa,b} \overset{\mathrm{d}}{=} G_{h,\rho} + Y, \qquad G_{h,\rho} \sim \mathrm{Gamma}(h, \text{rate } \rho), \qquad Y \geq 0, \qquad G_{h,\rho} \perp Y.$$

The convention at $h = 0$ is a point mass at zero. Define $h_{\mathrm{ID}}(\rho)$ by additionally requiring Y to be ID. The admissible shapes form intervals starting at zero: a smaller gamma factor is obtained by putting the remaining independent gamma factor into Y. We first calculate the ID capacity at every rate.

Proposition 8.7.1 (All-rate ID extraction diagram)**.** *Put $\lambda = \rho - b$. Then*

$$h_{\mathrm{ID}}(\rho) = \begin{cases} 0, & \lambda < -1, \\ z_\lambda^{-1-\lambda}\{\kappa(1+z_\lambda^2) - z_\lambda\}, & -1 \leq \lambda < 0, \\ 2\kappa - 1, & \lambda \geq 0, \end{cases} \tag{8.7.2}$$

where

$$z_\lambda = \frac{-\lambda + \sqrt{\lambda^2 + 4\kappa^2(1-\lambda^2)}}{2\kappa(1-\lambda)}.$$

Every displayed maximum is attained. In particular

$$h_{\mathrm{ID}}(b-1) = \kappa - \frac{1}{4\kappa}. \tag{8.7.3}$$

For unrestricted probability remainders one also has $h_{\mathrm{P}}(\rho) = 0$ when $\rho < b-1$ and $h_{\mathrm{P}}(\rho) = 2\kappa - 1$ when $\rho \geq b$.

Proof. Logarithmic differentiation of (8.7.1) gives the positive Lévy density

$$\frac{e^{-bx}}{x}\{2\kappa\cosh x-1\},\qquad x>0.$$

Subtracting the gamma Lévy density shows, by uniqueness, that an ID remainder exists exactly when

$$h\le\inf_{x\ge0}e^{\lambda x}(2\kappa\cosh x-1).$$

The infimum is zero for $\lambda<-1$, while for $\lambda\ge0$ the function is nondecreasing and its minimum is $2\kappa-1$. For $-1\le\lambda<0$, its derivative vanishes at exactly one positive point. With $z=e^{-x}$, the equation is

$$\kappa(1-\lambda)z^2+\lambda z-\kappa(1+\lambda)=0.$$

Its relevant root is $z_\lambda\in(0,1)$; at $\lambda=-1$ it equals $1/(2\kappa)$. The derivative changes from negative to positive there, proving (8.7.2). The residual Lévy density is nonnegative at equality and is integrable in the Lévy sense, proving attainment.

If $\rho<b-1$, choose t with $\rho<t<b-1$. Formula (8.7.1) gives $\mathbb{E}e^{tX}<\infty$, whereas $\mathbb{E}e^{tG_{h,\rho}}=\infty$ for every $h>0$. The pathwise inequality $G_{h,\rho}\le G_{h,\rho}+Y$ rules out extraction. Finally $L_{\kappa,b}(s)\asymp s^{-(2\kappa-1)}$ at infinity. The prospective remainder transform would grow without bound if $h>2\kappa-1$, which is impossible for a probability transform. The ID construction attains this upper bound when $\rho\ge b$. □

8.7.2 A global sign problem reduces to one evaluation

Put $\delta_{\min}=\max(0,1-\kappa)$. For $\delta_{\min}<\delta<1$ set $c=\kappa+\delta-1>0$ and define

$$g_{\kappa,\delta}(x)=\frac{x^{c-1}e^{-2x}}{\Gamma(c)}B_{\kappa,\delta}(2x),\tag{8.7.4}$$

$$B_{\kappa,\delta}(z)={}_1F_1(\delta-1;c;z)+\frac{z}{2c}{}_1F_1(\delta;c+1;z).\tag{8.7.5}$$

Termwise Laplace integration and Kummer's transformation yield

$$\int_0^\infty e^{-sx}g_{\kappa,\delta}(x)\,dx=s^{-\delta}(s+1)(s+2)^{-\kappa},\qquad s>0.\tag{8.7.6}$$

For example one can invert separately $s^{1-\delta}(s+2)^{-\kappa}$ and $s^{-\delta}(s+2)^{-\kappa}$; their inverses are respectively $x^{c-1}{}_1F_1(\kappa;c;-2x)/\Gamma(c)$ and $x^c{}_1F_1(\kappa;c+1;-2x)/\Gamma(c+1)$. All tilted signed densities are absolutely integrable. For $0<\delta<1$, the real-axis Kummer expansion gives

$$g_{\kappa,\delta}(x)\sim\frac{2^{-\kappa}}{\Gamma(\delta)}x^{\delta-1}\quad(x\to\infty).\tag{8.7.7}$$

Lemma 8.7.2 (A one-point positivity criterion)**.** *If $\kappa>1/2$ and $\delta_{\min}<\delta<\min(1,\kappa)$, then*

$$g_{\kappa,\delta}(x)\ge0\ \text{for all}\ x>0\quad\Longleftrightarrow\quad B_{\kappa,\delta}(2(\kappa-\delta))\ge0.\tag{8.7.8}$$

If equality holds on the right, the only zero of the density is $x=\kappa-\delta$, and it is a zero of order exactly two.

Proof. Write $p=1-\delta\in(0,1)$ and $y(z)={}_1F_1(-p;c;z)$. The derivative identity for Kummer's function gives $y'(z)=-(p/c){}_1F_1(1-p;c+1;z)<0$ for $z>0$. Moreover

$$B=y-\frac{z}{2p}y',\qquad zy''+(c-z)y'+py=0.$$

At a zero of B, one has $y = zy'/(2p) < 0$. Substitution into the differential equation gives the exact crossing rule

$$B'(z) = \frac{y'(z)}{4p}\{2(\kappa-\delta) - z\} \quad \text{whenever } B(z) = 0. \tag{8.7.9}$$

Consequently every zero below $2(\kappa-\delta)$ is a strict downward crossing and every zero above it is a strict upward crossing. If a zero is exactly at that point, differentiating Kummer's equation once gives

$$B'(z) = 0, \qquad B''(z) = -\frac{y'(z)}{4p} > 0. \tag{8.7.10}$$

The function is positive near zero because $B(0) = 1$, and is positive for all sufficiently large z by (8.7.7). Any negative component must therefore start with a downward crossing below $2(\kappa-\delta)$ and end with an upward crossing above it. The double-zero rule excludes a negative component having that point as an endpoint. Thus a negative component exists exactly when the value at that point is negative. If the value is zero, no other zero is compatible with the crossing directions and the positive endpoint signs. Equation (8.7.10) proves that the remaining zero has order two. □

Theorem 8.7.3 (Exact capacity with an arbitrary remainder)**.** *For each $\kappa > 1/2$ there is a unique number*

$$\max(0, 1-\kappa) < \delta_*(\kappa) < \frac{1}{4\kappa}$$

satisfying

$${}_1F_1(\delta-1; \kappa+\delta-1; 2(\kappa-\delta)) + \frac{\kappa-\delta}{\kappa+\delta-1}\,{}_1F_1(\delta; \kappa+\delta; 2(\kappa-\delta)) = 0. \tag{8.7.11}$$

The endpoint-rate capacity is attained and equals

$$h_{\mathrm{P}}(b-1) = \kappa - \delta_*(\kappa) > \kappa - \frac{1}{4\kappa} = h_{\mathrm{ID}}(b-1). \tag{8.7.12}$$

It is independent of b. For $h = \kappa - \delta_*(\kappa)$ the unique probability remainder has density

$$\frac{e^{-(b-1)x} g_{\kappa,\delta_*}(x)}{(b-1)^{-\delta_*} b (b+1)^{-\kappa}}, \qquad x > 0.$$

This density is strictly positive except for its double zero at $x = h$. Every shape $h_{\mathrm{ID}}(b-1) < h \le h_{\mathrm{P}}(b-1)$ therefore gives an independent probability remainder which is not ID.

Proof. The transform of a putative remainder after extracting shape $h = \kappa - \delta$ and rate $b-1$ is the normalization of $s^{-\delta}(s+1)(s+2)^{-\kappa}$ translated by $b-1$. Uniqueness of finite signed Laplace transforms and (8.7.6) show that extraction is equivalent to $g_{\kappa,\delta} \ge 0$ and is independent of the tilt. When $\kappa > 1$, at $\delta = 0$ the inverse simplifies to

$$g_{\kappa,0}(x) = \frac{e^{-2x} x^{\kappa-2}}{\Gamma(\kappa)}(\kappa - 1 - x). \tag{8.7.13}$$

It is negative at $x = \kappa$, so continuity excludes all sufficiently small positive δ. Extraction of any $h > \kappa$ would imply extraction of κ and is likewise impossible. When $1/2 < \kappa \le 1$, the general near-zero shape bound gives $h \le 2\kappa - 1$, or $\delta \ge 1 - \kappa$. At equality the unnormalized candidate transform has the expansion

$$s^{\kappa-1}(s+1)(s+2)^{-\kappa} = 1 - \frac{2\kappa-1}{s} + O(s^{-2}).$$

Its derivative is positive for large s, impossible for a translated Laplace transform. Fixing such a large s and using continuity in δ excludes a neighborhood above $\delta_{\min} = 1 - \kappa$ as well.

Set $\delta_0 = 1/(4\kappa)$. Proposition 8.7.1 gives an ID remainder at δ_0. Its Lévy density is

$$\frac{\kappa e^{-(b-1)x}}{x}\left(e^{-x} - \frac{1}{2\kappa}\right)^2.$$

Its continuous density is strictly positive at every $x > 0$. Here is a direct justification of this strictness. Split its Lévy measure at a sufficiently small $\eta > 0$. The sum of jumps at most η has mean tending to zero with η and thus has positive probability below $x/2$. The compound Poisson part from jumps above η has, on the event of exactly two jumps, a density strictly positive on $(2\eta, \infty)$: its one-jump density is positive except at one point. Their convolution gives a positive continuous lower bound in a neighborhood of x, if $\eta < x/8$. This proves the assertion also at a specified point, not merely almost everywhere. In particular the right side of (8.7.8) is strictly positive at δ_0. By continuity and the lemma, some $\delta < \delta_0$ is already feasible.

Feasibility is upward closed in δ, by convolution with an additional gamma factor. A neighborhood above $\delta_{\min}$ has just been excluded. On the remaining compact interval ending at δ_0, feasibility is closed by continuity of the one-point test, so it has a least point $\delta_* \in (\delta_{\min}, \delta_0)$. At that point the test equals zero; otherwise continuity would allow a smaller value. The one-point lemma gives the stated unique double zero. If $\delta > \delta_*$, then

$$g_{\kappa,\delta}(x) = \frac{1}{\Gamma(\delta - \delta_*)}\int_0^x (x-y)^{\delta-\delta_*-1} g_{\kappa,\delta_*}(y)\,dy > 0.$$

This strict inequality follows from positivity of the integrand on an interval near zero. Thus (8.7.11) has no second solution in $(\delta_{\min}, \delta_0)$, and all larger δ are feasible. The explicit density and its normalization follow from (8.7.6). Finally the exact ID capacity rules out an ID remainder above it. □

8.7.3 Exponential versus algebraic distance from the tail shape

Theorem 8.7.4 (Sharp large-order separation)**.** *As $\kappa \to \infty$,*

$$\delta_*(\kappa) \sim \frac{2^{\kappa-1}\kappa^{\kappa-1}e^{-2\kappa}}{\Gamma(\kappa)} \sim \frac{1}{2\sqrt{2\pi\kappa}}\left(\frac{2}{e}\right)^{\kappa}. \tag{8.7.14}$$

In contrast, $\kappa - h_{\mathrm{ID}}(b-1) = 1/(4\kappa)$ exactly. Thus

$$\frac{\kappa - h_{\mathrm{P}}(b-1)}{\kappa - h_{\mathrm{ID}}(b-1)} \sim \sqrt{\frac{2\kappa}{\pi}}\left(\frac{2}{e}\right)^{\kappa} \longrightarrow 0.$$

This is a quantitative refinement of the preceding capacity theorem, not a second historical conjecture.

Proof. Let

$$f_\kappa(y) = \frac{e^{-2y}y^{\kappa-1}}{\Gamma(\kappa)}, \quad g_0 = f_\kappa' + f_\kappa, \quad A_\kappa = -g_0(\kappa) = \frac{e^{-2\kappa}\kappa^{\kappa-2}}{\Gamma(\kappa)}.$$

For $\delta > 0$, (8.7.6) gives the fractional-integral identity

$$g_{\kappa,\delta}(x) = \frac{x^\delta}{\Gamma(1+\delta)}g_0(x) + \frac{1}{\Gamma(\delta)}\int_0^x (x-y)^{\delta-1}\{g_0(y) - g_0(x)\}\,dy. \tag{8.7.15}$$

The subtraction makes the integral regular at its upper limit. We need the estimate

$$J_\kappa := \int_0^\kappa \frac{g_0(y) - g_0(\kappa)}{\kappa - y}\,dy = \frac{2^{1-\kappa}}{\kappa}(1 + O(\kappa^{-1})). \tag{8.7.16}$$

To prove it, split at $3\kappa/4$. On the first interval, integration by parts in the f'_κ term gives

$$\int_0^{3\kappa/4} \frac{g_0(y)}{\kappa - y}\, dy = \frac{f_\kappa(3\kappa/4)}{\kappa/4} + \int_0^{3\kappa/4} f_\kappa(y) \left\{ \frac{1}{\kappa - y} - \frac{1}{(\kappa - y)^2} \right\} dy.$$

The measure $2^\kappa f_\kappa(y)dy$ is Gamma$(\kappa, 2)$, with mean $\kappa/2$ and variance $\kappa/4$. Taylor expansion of the denominators on $y \le 3\kappa/4$, together with the gamma exponential tail bound, evaluates the last integral as $2^{-\kappa}\{2/\kappa + O(\kappa^{-2})\}$; the boundary term is exponentially smaller. Such a tail bound follows directly from Markov's inequality and the gamma moment generating function. On $[3\kappa/4, \kappa]$, the divided difference of g_0 is at most the supremum of $|g_0'|$ there. The explicit formula for f_κ and Stirling's formula bound this contribution by $2^{-\kappa}e^{-c\kappa}$ times a fixed polynomial in κ, for a fixed $c > 0$. Finally the subtracted constant on $[0, 3\kappa/4]$ contributes only $g_0(\kappa)\log 4$, again exponentially smaller than $2^{-\kappa}/\kappa$. This proves (8.7.16).

Set $d_\kappa = A_\kappa/J_\kappa$. For every fixed $C > 0$, the same split in (8.7.15) gives uniformly for $0 \le \delta \le Cd_\kappa$, with $x = \kappa - \delta$,

$$g_{\kappa,\delta}(\kappa - \delta) = -A_\kappa(1 + o(1)) + \delta J_\kappa(1 + o(1)). \tag{8.7.17}$$

For completeness, on $y \le 3\kappa/4$ the kernel changes from $(\kappa - y)^{-1}$ by the uniform relative amount $O(\delta \log \kappa + \delta/\kappa) = o(1)$. On the remaining interval the subtracted divided difference is bounded by $\sup |g_0'|$, and the integral of v^δ, including its logarithmic derivative in δ, is bounded by a polynomial in κ and $\log \kappa$. The exponential tail estimate above therefore remains negligible uniformly. Also $g_0(\kappa - \delta)/g_0(\kappa) = 1 + o(1)$, $x^\delta/\Gamma(1 + \delta) = 1 + o(1)$, and $1/\Gamma(\delta) = \delta(1 + o(1))$. These observations prove (8.7.17) without interchanging an uncontrolled limit with a supremum.

For each fixed $\varepsilon \in (0, 1)$, the one-point test is negative at $(1 - \varepsilon)d_\kappa$ and positive at $(1 + \varepsilon)d_\kappa$ for all sufficiently large κ. The exact capacity theorem brackets δ_* between these two values. Hence $\delta_* \sim d_\kappa$. Inserting (8.7.16) and then Stirling's formula gives (8.7.14). □

Scope and remaining gaps. The new exact boundary concerns arbitrary probability remainders on the first excess-power line and the slowest allowable gamma rate. The all-rate ID formula is an elementary supporting classification; the exponential law quantifies a large separation which the ID criterion misses. Neither the unrestricted capacity for every noninteger excess power nor its full intermediate-rate diagram is settled here. An attempted transfer to universal SD strictness for the squared-Bessel McKay family remains open. In particular, no termination claim is made for the earlier finite moment hierarchy. The noncentral chi-square HCM classification is already proved in [5] and is not counted as our new result.

8.8 Remaining problems and the limits of the transfers

The strongest external step is the universal adaptive-width bound. For example it gives $N(4) = 3$, $N(8) = 4$, and $N(16) = 5$; for dimension three the bound remains $2 \le N(3) \le 3$. The scalar-fiber recursion reaches UW_3 after two queries, which vanishes on a three-dimensional compact polyhedron. Thus that recursion alone cannot decide the three-dimensional case. The Betti-sensitive theorem adds a real restriction, but analytic functions can have increasingly complicated fibers on expanding balls. A necessary topology-growth condition is not a contradiction.

The statistical advance changes computation and the corruption model, not the previously proved minimax order. Its linear program is polynomial in the explicit input length but can be large. A nearly linear algorithm, a practical implementation with comparable constants, and a

robustness theorem for the unmodified nonparametric MLE remain separate problems. The best universal Chebyshev–Bernstein exponent is also unresolved: the available interval remains $\log 2 \le c_* \le 1$.

The prime-series inverse theorem identifies all positive weights on a prescribed frequency support, using infinitely many exact zeros in one height direction. Its proof combines phase freezing with the derivative of a frozen simple zero. A stability theorem for finitely many noisy zeros would need effective recurrence and conditioning bounds that are absent here. For fluctuations, the exact pure-point variance expansion makes the missing task explicit: control the total Fourier weight at very small nonzero frequencies. Recurrent local correlations do not alone prohibit a central limit theorem.

The matrix construction closes a qualitative arbitrary-design gap, but it supplies no uniform inverse modulus near the quasi-analytic boundary. In stable models, compact scale support sharply distinguishes indices below and above one; skew index one leads to a different entire-function sampling problem, still open here. Mixing over unknown stable indices or skewness is not covered by the fixed-shape theorems.

The finite-zero spectral experiment assumes complete zero lists, controlled amplitudes including $A = 0$, and quantitative finite-order nonresonance. Missing or spurious zeros, efficient certified fitting, an optimal lower bound on the necessary height, and an infinite-prime version require further work. The classical recurrence theorem improves a sufficient height; it does not prove that this height is optimal for the inverse zero problem.

The gamma-capacity calculation resolves the first excess-power line at the slowest allowable gamma rate. Arbitrary noninteger excess powers, the unrestricted intermediate-rate capacity diagram, universal strict self-decomposability for the squared-Bessel family, and termination of the earlier finite moment hierarchy remain unresolved. The scalar crossing rule that reduces global positivity to one evaluation is the specific structure behind the present exact solution.

Research status. The manuscript records twelve attained targets at the proof-draft level and explicitly preserves the stronger questions above. Its claims have undergone automated cross-review, with no blocking gap identified in the final review round. This status is narrower than independent external validation, and none of the target counts is a claim to an Annals acceptance or a major-prize-level result.

Part IX

Sharp Boundaries for Adaptive Recovery and Mixture Identification

Homogeneous Information, Transport Corruption, Stable Mixtures, and Spectral Edge Laws

Source manuscript: 16 October 2026.

Source abstract. We develop thirteen scoped targets extending an observation-and-recovery program. Positive homogeneity of the first continuous measurement forces at least $\lceil \log_2 m \rceil + 1$ queries for bounded-error recovery on $\mathbb{R}^m$, while the fully positively homogeneous class still has logarithmic complexity. Every finite adaptive budget has a strictly positive gap above the exact Euclidean convex-noise floor in dimension at least two. For a bounded Hilbert operator, the limiting adaptive continuous error is exactly its essential norm; spectral bounds also prove optimal exponential and double-exponential recovery orders. A new predictive transport modulus yields the full binomial-population minimax curve under a hard budget of individual trial flips, with a polynomial-time estimator. Compact-scale skew stable mixtures at index one are classified by the completeness radius of logarithmic terminal times. For symmetric zero-location stable mixtures with continuously unknown index and scale, a mild two-sided scale moment makes every infinite terminal design identifying, while every finite design fails. Further results quantify frequency-relation losses and zero intensities at spectral edges, make nonlinear spectral fitting constructive, and determine the strict intermediate-rate gamma factor region and its quartic separation from infinitely divisible remainders. Classical theorems, prior unpublished dependencies, related refinements, and unresolved stronger questions are explicitly separated. The target count is not a count of historical conjectures solved.

Research scope and thirteen targets

This paper continues the 223-page integrated manuscript and its sequels, most immediately the 39-page *Geometry and Rigidity of Inverse Observation* [115]. It contains new arguments and extensions rather than a reprint of that manuscript. The common question is how restrictions on observations change identifiability, optimal precision, and computational recovery. The sections use different models; a general equivalence between them is not asserted.

The restricted-information question in [60, Section 3] is externally posed. Our homogeneous result answers its order-of-growth aspect and several exact cases; the exact count in every dimension remains open. The remaining rows are quantitative transfers or open questions within this research program. Closely related consequences are grouped: for example, the two gamma asymptotic regimes occupy one row, and spectral decay corollaries are grouped with the operator theorem. No claim is made that these thirteen rows are thirteen independent major conjectures.

No.	Target	Result and limitation	Proof
1	Homogeneous information complexity	Logarithmic upper and lower counts; exact low-dimensional and hybrid cases. General exact count retains a gap.	9.1.2, 9.1.3
2	Attainment of the convex-noise floor	Every fixed finite adaptive budget has a uniform positive gap in dimension $m \geq 2$. The gap's sharp rate is not obtained.	9.1.6

No.	Target	Result and limitation	Proof
3	Infinite-dimensional adaptive recovery	Essential norm is the exact limiting error; two-sided spectral bounds prove sharp decay orders.	9.2.1
4	Predictive transport inversion	Sharp binomial-mixture inverse modulus, with logarithm $\log(e + 1/(\eta\sqrt{t}))$.	9.3.1
5	Individual-trial adversarial flips	Full sharp minimax curve and polynomial-time estimator under a known hard flip budget.	9.3.2
6	Frequency relations and edge loss	Explicit positive relation matrix and fourth-order remainder for the local boundary loss.	9.4.1
7	Dimension visible in zero intensity	Sharp edge intensity and leading constant; compatible hulls reveal rational rank.	9.4.2, 9.4.4
8	Compact-scale skew index one	Exact completeness-radius criterion, including smooth joint counterexamples.	9.5.1
9	Jointly mixed stable index and scale	Every infinite design identifies under the stated moments; every finite design has counterexamples.	9.6.1
10	Constructive nonlinear spectral fitting	Threshold inversion and a finite moment reconstruction under separation and small noise.	9.7.1
11	Controlled critical zero counts	A local near-threshold intensity law; no claim of a uniform finite-height discrepancy bound.	9.7.3
12	Intermediate-rate gamma factors	Attained continuous strictly monotone boundary, strictly above the ID boundary at every interior rate.	9.8.2
13	Singular gamma-capacity gaps	Exact quartic laws at the central rate and lowest Bessel order; refinement of row 12.	9.8.3, 9.8.4

How to interpret the results. The new arguments are supplied in full for their stated models. Some inputs remain prior unpublished results: the adaptive-width recursion, the exact convex-noise floor, the Bernstein approximation estimates and clean binomial lower bounds, the previous spectral parameter stability bound, and the endpoint gamma capacity. Those dependencies are identified where used. The KNU construction, Euclidean widths, Beurling–Malliavin theorem, moment-determinate projection uniqueness, and classical moment reconstruction are credited as existing tools. Checked primary sources support these comparisons; the literature search does not certify historical priority of every transfer.

There is no proof here of the Riemann hypothesis, no determination of all primes from a rightmost zeta zero, and no unrestricted solution of the two-query problem in dimension three. Exact population laws and exact real measurements are mathematical observation models, not claims of finite-precision experimental feasibility. Proof review within this research process is not external peer review or formal verification.

9.1 Homogeneous first measurements and the necessity of infinitely many noisy convex queries

The two results in this section address different losses of information. In the exact-real model, positive homogeneity of the first query costs a full dimension in a way that survives arbitrary subsequent adaptive choices. This gives a stronger lower count than the unrestricted continuous bound in every dimension that is not a power of two. In the bounded-noise model, the exact Euclidean convex-information floor from the earlier manuscript is a genuine infinite-query limit: in dimension at least two, every fixed finite query budget has a strictly larger minimax error, even with unrestricted adaptivity.

9.1.1 Large level patches and the homogeneous-query question

A continuous function $f : \mathbb{R}^m \to \mathbb{R}$ is positively homogeneous of degree $a > 0$ if $f(rx) = r^a f(x)$ for all $r > 0$. This convention does not require oddness. It includes norms and even-degree homogeneous polynomials. The distinction is relevant to the homogeneous-information question in [60, Section 3].

Krieg and Kritzer study homogeneous full information maps and recovery maps, and show that on cones a finite-error procedure using homogeneous measurements can be rescaled to arbitrarily small error at unchanged cost [114, Propositions 1 and 2]. The positive-scaling proof of their second statement applies to positive homogeneity as well. Here we quantify the number of individually continuous, adaptively selected measurements; their joint transcript need not be homogeneous. We do not resolve their general optimality question for homogeneous algorithms [114, Open Problem 1].

We use the adaptive-width bound from the preceding manuscript [115]: on a compact metric space X, every q-query procedure using arbitrary continuous scalar measurements has error at least $\mathrm{UW}_{2^q-1}(X)/2$. Here UW_d is the infimum of the mesh of finite open covers of multiplicity at most $d+1$. Its short proof iterates the classical inequality

$$\mathrm{UW}_{2d+1}(X) \le \sup_t \mathrm{UW}_d(f^{-1}(t))$$

for a continuous scalar map on a compact metric space; see [102, Theorem 2.1]. Compact fibers and arbitrary selection rules are allowed. The Lebesgue covering theorem gives $\mathrm{UW}_{d-1}([-h,h]^d, \|\cdot\|_\infty) = 2h$.

Lemma 9.1.1 (A large-patch obstruction)**.** *Let $f : \mathbb{R}^m \to \mathbb{R}$ be continuous. Suppose there are levels of f containing compact sets X_R and continuous bijections $\gamma_R : [-1,1]^d \to X_R$ such that*

$$\|\gamma_R(u) - \gamma_R(v)\|_2 \ge R\|u - v\|_\infty, \qquad R \longrightarrow \infty. \tag{9.1.1}$$

Any n-query procedure beginning with f and using arbitrary continuous subsequent queries has infinite uniform recovery error whenever $2^{n-1} \le d$.

Proof. The inverse of γ_R is continuous and R^{-1}-Lipschitz by (9.1.1). Pulling a cover of X_R back to the cube shows $\mathrm{UW}_{d-1}(X_R) \ge 2R$. After the first answer is fixed, the remaining procedure has $n-1$ queries on X_R. Since $2^{n-1} - 1 \le d - 1$, the adaptive-width bound gives error at least R. Let R tend to infinity. □

Theorem 9.1.2 (A homogeneous first query forces the stronger count)**.** *Let $m \ge 2$. If the first query is continuous and positively homogeneous of any degree $a > 0$, uniformly bounded-error recovery on $\mathbb{R}^m$ requires*

$$n \ge \lceil \log_2 m \rceil + 1. \tag{9.1.2}$$

Every later query may be an arbitrary continuous function, with no homogeneity, degree, or definability restriction. In particular, two queries cannot suffice in $\mathbb{R}^3$ under this first-query assumption.

Proof. Suppose first that f is not identically zero. Change its sign and rotate coordinates so that $f(e_m) > 0$. By continuity, on a sufficiently small cube $C = [-h,h]^{m-1}$ the function $v \mapsto f(v,1)$ is bounded above and below by strictly positive constants. Put

$$b(v) = f(v,1)^{-1/a}, \qquad \gamma_\lambda(v) = \lambda b(v)(v,1), \qquad \lambda > 0.$$

Positive homogeneity gives $f(\gamma_\lambda(v)) = \lambda^a$. The map γ_λ is injective because its inverse is the ratio map $x \mapsto x'/x_m$. On its image, x_m is between two positive constant multiples of λ, and $\|x'\|_2 \le C\lambda$. Consequently, for $x = \gamma_\lambda(v)$ and $y = \gamma_\lambda(w)$,

$$\|v - w\|_2 \le \frac{\|x' - y'\|_2}{x_m} + \frac{\|y'\|_2 |x_m - y_m|}{x_m y_m} \le \frac{C}{\lambda}\|x - y\|_2.$$

Rescaling C to $[-1,1]^{m-1}$ verifies Lemma 9.1.1 with $d = m-1$ and an expansion factor tending to infinity. Thus $2^{n-1} \le m-1$ is impossible. Since the left side is an integer, its negation is exactly $n \ge \lceil \log_2 m \rceil + 1$. If $f \equiv 0$, the first query contributes no information; the unrestricted adaptive-width bound on expanding m-cubes gives at least the same conclusion. □

Corollary 9.1.3 (Logarithmic complexity within the homogeneous class)**.** *Let $N_{\mathrm{hom}}(m)$ denote the minimum number of continuous, positively degree-one homogeneous adaptive scalar queries sufficient for arbitrarily accurate uniform recovery on $\mathbb{R}^m$. For $m \ge 2$,*

$$\lceil \log_2 m \rceil + 1 \le N_{\mathrm{hom}}(m) \le \min\{m, \lceil \log_2 m \rceil + 2\}. \tag{9.1.3}$$

Hence positive homogeneity preserves logarithmic query complexity. The count is exactly two for $m = 2$ and exactly three for $m = 3$. For larger dimensions these bounds generally retain a one-query gap.

Proof. The lower bound is Theorem 9.1.2; coordinate queries give the upper bound m. For the other upper bound, first query $r = \|x\|_2$. If $r = 0$, output zero. Otherwise apply the Krieg–Novak–Ullrich continuous algorithm [60, Theorem 1] to $u = x/r$, at accuracy ε/r, using at most $\lceil \log_2 m \rceil + 1$ further queries. Each particular continuous query g selected by that algorithm, restricted to the unit sphere, can be implemented by

$$H_g(x) = \begin{cases} \|x\|_2 g(x/\|x\|_2), & x \ne 0, \\ 0, & x = 0. \end{cases}$$

It is positively homogeneous of degree one. It is continuous at zero because g is bounded on the compact sphere. Dividing its answer by the already known r recovers the intended answer $g(u)$. The selected function g may depend on r and on previous answers, as allowed by the adaptive model. Multiplying the final approximation to u by r gives error at most ε. □

Why the sign convention matters. If every admissible query instead satisfies $f(rx) = rf(x)$ for every real r, exactly m queries are necessary and sufficient. Indeed, freeze the n continuous odd functions selected along the all-zero transcript. For $n < m$, Borsuk–Ulam applied to their joint map on S^{m-1} gives a common zero u. Every point ru, $r > 0$, then follows that same transcript by homogeneity, so its recovery error is unbounded. Coordinate measurements give the matching upper bound. Thus the logarithmic conclusion above uses the larger class allowing positive homogeneity without oddness, which includes the norm and intensity measurements motivating [60, Section 3].

Corollary 9.1.4 (An exact hybrid count in shifted dyadic dimensions)**.** *If only the first query is required to be positively homogeneous and all subsequent queries may be arbitrary continuous functions, the minimum count in dimension $m = 2^k + 1$ is exactly $k + 2$ for $k \ge 0$.*

Proof. The lower bound follows from (9.1.2). For the upper bound first read the last coordinate, and then apply [60, Theorem 1] to the remaining 2^k coordinates with $k + 1$ continuous queries. The first coordinate query is homogeneous. No homogeneity is asserted for the later queries in this hybrid model. □

Other large-patch classes. The same lower count (9.1.2) holds when the first query is a real polynomial or a continuous quasiconvex function. Here the required $(m-1)$-dimensional level graphs were established in the previous two-query obstruction, and the new step is their use with an arbitrary number of remaining queries. Briefly, for a polynomial, choose a direction where its leading homogeneous term is nonzero. On each transverse ball its sufficiently distant positive tail is strictly monotone, so a sufficiently high level is a graph over that ball. Orthogonal projection is a one-Lipschitz inverse. For a quasiconvex function, either a maximal level contains a halfspace, or a proper convex sublevel set contains an arbitrarily large ball. An upper boundary graph in a supporting direction gives the same one-Lipschitz inverse. These are level sets of the first query, so the later queries remain unrestricted.

9.1.2 A finite-query gap above the Euclidean convex-noise floor

Fix $m \ge 2$ and $0 < \delta < 1$. On the unit Euclidean ball B, an admissible query is a continuous convex function with range in $[-1, 1]$. Its observed value may differ from its true value by at most δ. Write e_n^{ad} for the optimal worst-case Euclidean recovery error after at most n adaptive queries, allowing arbitrary selection rules and decoders. Put

$$q = 1 - \delta, \qquad r_* = \frac{\delta}{\sqrt{1 + q^2 + 2q/m}}. \tag{9.1.4}$$

The exact unlimited-query value proved in the earlier manuscript is $\inf_n e_n^{\mathrm{ad}} = r_*$ [105]. The issue here is whether any fixed finite budget attains this infimum.

Let $u_0, \dots, u_m$ be the vertices of a centered regular simplex on the unit sphere, with $\langle u_i, u_j \rangle = -1/m$ for $i \ne j$. For $U \in SO(m)$ put $S_U = \{r_* U u_i : 0 \le i \le m\}$. For distinct vertices $x, y \in S_U$,

$$y = qx + \delta z, \qquad z \in \partial B. \tag{9.1.5}$$

Convexity and the range bound imply $\operatorname{osc}_{S_U} f \le 2\delta$ for every admissible query.

Lemma 9.1.5 (No query saturates an open set of simplex rotations)**.** *Let f be convex on the open unit ball with $-1 \le f \le 1$. Then*

$$Z_f = \{U \in SO(m) : \operatorname{osc}_{S_U} f = 2\delta\}$$

is closed and has empty interior.

Proof. A finite convex function is continuous in the open ball, so Z_f is closed. Suppose it had nonempty interior. It is the union over finitely many ordered pairs $i \ne j$ of the closed sets on which $f(r_* U u_j) - f(r_* U u_i) = 2\delta$. The Baire property of the locally compact rotation group implies that one ordered pair has this equality on a nonempty open set V.

Fix $U \in V$ and write $x = r_* U u_i$, $y = r_* U u_j$, and $z = (y - qx)/\delta$. The range bound and convexity imply $f(y) - f(x) \le \delta(1 - f(x)) \le 2\delta$, so equality forces $f(x) = -1$ and $f(y) = -1 + 2\delta$. This argument uses only interior values: replace z by a point just short of it on the segment from x, apply convexity, and pass to the endpoint limit.

Take a supporting affine function $\ell(w) = b + v \cdot w$ for f at y. Since $\ell \le f \le 1$ on the open ball, continuity of ℓ gives $\ell(z) \le 1$. Also $\ell(x) \le -1$. The identity $y = qx + \delta z$ and $\ell(y) = -1 + 2\delta$ force $\ell(x) = -1$ and $\ell(z) = 1$. Thus the nonzero linear functional v attains its maximum over the unit ball at z, whence $v = cz$ for some $c > 0$.

For all rotations in V, the corresponding vertex $r_* U u_i$ is a minimum point of f. The map $U \mapsto U u_i$ is open onto the sphere, so these points contain a relatively open sphere patch around x. Since $\ell \le f = -1$ on this patch and $\ell(x) = -1$, the linear functional v has a local maximum

on that sphere patch at x. Consequently v is parallel to x. But $z = (r_*/\delta)U(u_j - qu_i)$ is not parallel to x when $m \geq 2$: the distinct regular-simplex directions u_i, u_j are then not collinear. This contradiction proves the lemma. □

Theorem 9.1.6 (Every finite adaptive budget has a strict positive gap)**.** *For every $m \geq 2$, $0 < \delta < 1$, and finite integer $n \geq 0$, there is $c_{m,\delta,n} > 0$ such that*

$$e_n^{\mathrm{ad}} \geq r_* + c_{m,\delta,n} > r_*. \tag{9.1.6}$$

In particular, the exact unlimited-query convex-noise floor cannot be attained at any finite query count, even by taking an infimum over all adaptive algorithms with that count. The statement also applies to any subclass of these queries, including globally convex analytic functions and globally convex polynomials of arbitrary degree.

Proof. The case $n = 0$ is immediate from $e_0^{\mathrm{ad}} = 1 > r_*$. Fix $n \geq 1$ and choose $r_* < \rho < 1$. All convex functions bounded by one in absolute value on the open unit ball are uniformly Lipschitz on ρB, with Lipschitz constant

$$L = \frac{2}{1-\rho}.$$

Their restrictions to ρB form a compact family $\mathcal{F}$ in the uniform norm. To verify closedness, take an arbitrary sequence and use the same bounds on every smaller concentric ball, followed by a diagonal subsequence. Its limit is convex and bounded by one on the open ball, and restricts to the prescribed uniform limit on ρB. Thus Lemma 9.1.5 applies also to limits in this compact family.

Equip $SO(m)$ with the distance $\|U - V\|_{\mathrm{op}}$. For $a > 0$, write $E_a = \{V \in SO(m) : \|V - I\|_{\mathrm{op}} \leq a\}$ and set

$$\kappa(a) = \min_{f \in \mathcal{F},\, U \in SO(m)} \max_{V \in E_a} \{2\delta - \mathrm{osc}_{S_{UV}} f\}. \tag{9.1.7}$$

The minimized function is continuous in (f, U), and the parameter space is compact. Lemma 9.1.5 says that its value at every (f, U) is positive, because UE_a contains a nonempty open set. Therefore $\kappa(a) > 0$ for every $a > 0$.

We construct a common noisy transcript for an entire ball of simplex rotations. Start with any rotation center and radius $a_0 = 1$. Define deterministic positive radii by

$$a_j = \min\left\{\frac{a_{j-1}}{2}, \frac{\kappa(a_{j-1}/2)}{4Lr_*}\right\}, \qquad \sigma_n = \min_{1 \leq j \leq n} \frac{\kappa(a_{j-1}/2)}{4} > 0.$$

At stage j, the previous answers fix the selected query f_j. Use (9.1.7) to find a rotation within distance $a_{j-1}/2$ of the current center at which its oscillation is at most $2\delta - \kappa(a_{j-1}/2)$. Answer by the midpoint of its maximum and minimum on that simplex. Every vertex value is then within $\delta - \kappa(a_{j-1}/2)/2$ of the answer. If the rotation changes by at most a_j, a vertex moves by at most $r_* a_j$ and its query value changes by at most $Lr_* a_j$. The same answer is consequently compatible, with slack at least $\kappa(a_{j-1}/2)/4$, with every simplex in the new rotation ball. That ball lies in the previous one by the triangle inequality, so every earlier answer and its slack are preserved. This induction uses no continuity of the query-selection rule: at each stage the transcript has already fixed one query.

After n stages, choose any rotation U in the retained ball. Every query value on S_U is within $\delta - \sigma_n$ of its corresponding answer. Increase the simplex radius by

$$h = \min\left\{\frac{\rho - r_*}{2}, \frac{\sigma_n}{2L}\right\} > 0.$$

All enlarged vertices remain in ρB, and each recorded query value changes by at most $Lh \le \sigma_n/2$. Thus the same transcript is possible for every vertex of the regular simplex $\{(r_*+h)Uu_i\}_{i=0}^m$. Its circumradius is r_*+h, so every decoder has worst-case error at least that value. The number h depends only on m, δ, n and the fixed choice of ρ, not on the algorithm. This proves (9.1.6). □

The one-dimensional exception. For $B = [-1,1]$ put $r = \delta/(2-\delta)$. The two fixed convex queries

$$f_+(x) = -1 + \frac{2(x+r)_+}{1+r}, \qquad f_-(x) = -1 + \frac{2(r-x)_+}{1+r}$$

have range in $[-1,1]$ and attain the exact floor r. Indeed, if $x < y$ are compatible with the same two noisy answers and $x \ge -r$, the first query implies $y - x \le \delta(1+r) = 2r$. If $y \le r$, the second gives the same bound. In the only remaining case $x < -r < r < y$, the first query difference already exceeds 2δ, which is impossible. Every feasible set therefore has diameter at most $2r$, and its interval midpoint gives error at most r. The universal indistinguishable pair $\{-r, r\}$ gives the matching lower bound. Thus the finite-query obstruction is genuinely dimensional.

Status and remaining scope. The homogeneous-first lower bound and logarithmic homogeneous-class construction address the externally posed restricted-information question. The exact hybrid count is a consequence, not a closure of the unrestricted KNU problem. The finite-query gap settles the attainment question left open by the preceding exact convex-noise formula. Its positive constant is established by compactness; neither its sharp dependence on n nor a sharp finite-query complexity law is claimed. Unrestricted two-query recovery in $\mathbb{R}^3$, and the exact positive-homogeneous count in higher dimensions, remain open.

9.2 The exact compactness barrier for adaptive Hilbert-space recovery

The earlier adaptive-width bound supplies a missing converse to the known exponential speed-up for Hilbert operators. Arbitrary continuous adaptive information cannot remove the essential-norm obstruction. For compact operators with regularly decaying singular values, the same argument proves that the exponential speed-up has the optimal order, including its exponent for algebraic decay.

Let $S : H \to G$ be a bounded linear operator between real Hilbert spaces. Measurements are arbitrary continuous real functions on the closed unit ball B_H, chosen adaptively; the decoder is unrestricted. Write $e_n(S)$ for the infimum of their worst-case errors after n exact measurements. Define the approximation numbers by

$$a_j(S) = \inf_{\operatorname{rank} R < j} \|S - R\|, \qquad j \ge 1,$$

and the essential norm by $\|S\|_{\mathrm{ess}} = \inf_{K \text{ compact}} \|S - K\|$. These conventions make a_j the j-th singular value in the compact case. In finite dimension extend the sequence by zeros.

Theorem 9.2.1 (Two-sided spectral bounds and the exact error floor)**.** *For every $n \ge 1$,*

$$\max\left\{\sqrt{\frac{1+2^{-n}}{2}}\, a_{2^n}(S), \quad a_{2^{n+1}}(S)\right\} \le e_n(S) \le a_{2^{n-1}+1}(S). \tag{9.2.1}$$

Consequently

$$\lim_{n\to\infty} e_n(S) = \|S\|_{\mathrm{ess}}. \tag{9.2.2}$$

Thus recovery to arbitrarily small uniform error by finitely many adaptive continuous measurements is possible if and only if S is compact. This statement permits discontinuous selection rules and decoders; imposing their continuity is not part of its proof.

The upper exponential index comes from the existing finite-dimensional construction of Krieg–Novak–Ullrich [60]. Their Corollary 11 already gives a Hilbert-operator upper bound with index 2^{n-2}. The improvement of that index here is a direct use of their sharper finite-dimensional construction and is not the main novelty. The contribution is the converse and the exact essential-norm floor. The Euclidean width constants used below are classical [116], and the adaptive-width mechanism is from our preceding manuscript [115].

Lemma 9.2.2 (The two Euclidean widths used in the proof)**.** *For the Euclidean unit ball B_2^m,*

$$\mathrm{UW}_{m-1}(B_2^m) = \sqrt{2(1+1/m)} \quad (m \geq 2), \qquad \mathrm{UW}_d(B_2^m) = 2 \quad (0 \leq d \leq m/2 - 1).$$

Proof. These are the $p = 2$ case of Gournay's Theorem 1.4 [116], with his width-dimension convention converted to the diameter convention used here. Explicitly, his statement says that below $\sqrt{2(1+1/m)}$ the width-dimension is m, and at that threshold a map to an $(m-1)$-complex exists. It also says that at every diameter threshold less than 2 the width-dimension is strictly larger than $m/2 - 1$. The upper bound 2 follows from the one-set cover. Open versus non-strict fiber thresholds give the same infimum. No new Euclidean width computation is claimed. □

Proof of Theorem 9.2.1. We recall the width principle in the exact form required: for any compact metric input X embedded isometrically in the output space, an n-query adaptive continuous procedure of error r satisfies

$$\mathrm{UW}_{2^n-1}(X) \leq 2r. \tag{9.2.3}$$

Indeed, a zero-query leaf has diameter at most $2r$. For a continuous scalar query, the multiplicative fiber-width inequality bounds $\mathrm{UW}_{2d+1}(X)$ by the supremum of the UW_d of its compact fibers. Induction on remaining depth gives (9.2.3). It does not require continuity of the selected family as the preceding answers vary.

For an m-dimensional subspace $V \subset H$, set $c_V = \inf_{x \in V, \|x\|=1} \|Sx\|$. If $c_V > 0$, the image $S(B_V)$ is a compact ellipsoid. Pulling back covers along $S|_V$ and using $\|Sx - Sy\| \geq c_V \|x - y\|$ shows

$$\mathrm{UW}_d(S(B_V)) \geq c_V \, \mathrm{UW}_d(B_2^m).$$

Every algorithm on B_H restricts to one on B_V and, through the continuous inverse $S|_V^{-1}$, to one on this ellipsoid. Apply (9.2.3). On Hilbert spaces the min–max identity

$$\sup_{\dim V = m} \inf_{x \in V, \|x\|=1} \|Sx\| = a_m(S) \tag{9.2.4}$$

holds also for noncompact operators. It follows from the spectral theorem for S^*S: a subspace on which its spectrum is bounded below by c^2 supplies the lower estimate, while removing a finite-dimensional spectral subspace above c^2 supplies a rank approximation of error at most c. Taking limits at spectral endpoints gives (9.2.4). If $c_V = 0$, the inequality being asserted for that subspace is simply zero and needs no inverse.

If $m > \dim H$, the associated approximation number is zero and the lower bound is trivial; restrict (9.2.4) to $m \leq \dim H$ otherwise. Choose $m = 2^n$ for the first width in Lemma 9.2.2, and $m = 2^{n+1}$ for the second. Taking the supremum over V gives both lower bounds in (9.2.1).

For the upper bound put $k = 2^{n-1}$. Choose a linear operator R of rank at most k with $\|S - R\| \leq a_{k+1}(S) + \eta$. Its range is a Euclidean space of dimension at most k. The construction

in [60, Theorem 1] approximates every vector in that space to any prescribed positive error using at most $\lceil \log_2 k \rceil + 1 = n$ continuous adaptive measurements. Composing each selected measurement with R gives admissible continuous measurements on B_H. Recovering Rx within η yields error at most $a_{k+1}(S) + 2\eta$ for Sx. Let $\eta \downarrow 0$. A rank-zero case is immediate.

Finally $a_j(S) \downarrow \|S\|_{\mathrm{ess}}$. One inequality follows because finite-rank operators are compact; the other follows by approximating a compact operator in norm by finite-rank operators on Hilbert spaces. The second lower bound and the upper bound in (9.2.1) now converge to the same number, proving (9.2.2). Compact operators form a norm-closed ideal, so essential norm zero is equivalent to compactness. □

Corollary 9.2.3 (Optimal exponential and double-exponential orders)**.** *If $a_j(S) \asymp j^{-s}$ for $s > 0$, then*

$$e_n(S) \asymp 2^{-sn}.$$

If $a_j(S) \asymp \exp(-cj^{\beta})$ with $c, \beta > 0$, then

$$\log\log \frac{1}{e_n(S)} = \beta n \log 2 + O(1).$$

Writing $n(\varepsilon)$ for the least query count with error at most ε, the corresponding complexities are

$$n(\varepsilon) = s^{-1} \log_2(1/\varepsilon) + O(1), \qquad n(\varepsilon) = \beta^{-1} \log_2 \log(1/\varepsilon) + O(1),$$

respectively, as $\varepsilon \downarrow 0$.

Proof. Substitute the two spectral envelopes in (9.2.1) and take logarithms. Replacing an infimal error by an attained procedure is unnecessary: for the upper complexity choose an index whose infimal error is strictly smaller than ε, which changes the displayed query counts by at most a constant. □

For example the diagonal map $(x_j) \mapsto (j^{-s} x_j)$ on ℓ_2 has algebraic optimal error $a_{n+1} \asymp n^{-s}$ under nonadaptive continuous information, by the classical Borsuk–Ulam argument and coordinate projection. Its unrestricted adaptive error has exactly exponential order. Conversely the identity on an infinite-dimensional Hilbert space has $e_n = 1$ for every finite n: the second lower bound is one, and output zero attains it.

This is one operator-theoretic transfer with several consequences, not separate conjecture closures for every spectral envelope. It does not give the exact finite-n error for general ellipsoids or a corresponding essential-norm theorem for arbitrary Banach operators. It assumes exact real-valued measurements with unrestricted computational precision; deterministic noisy information obeys different entropy bounds [88].

9.3 The transport inverse law and adversarial corruption of individual trials

The geometry of corruption changes when a budget counts individual Bernoulli flips rather than entirely replaced reports. The relevant predictive discrepancy is then transportation distance, and its sharp inverse curve differs from the previously established total-variation curve. We prove that curve and obtain a matching, polynomial-time minimax theorem for a strong adversary acting on individual trials. Unlike the preceding whole-report theorem, the algorithm here is given a valid upper bound on the flip budget.

For a mixing law $G \in \mathcal{P}([0,1])$, write

$$p_G(j) = \int \binom{t}{j} x^j (1-x)^{t-j}\, dG(x), \qquad j = 0, \dots, t.$$

Let $W_{1,t}$ be Wasserstein distance on this count alphabet with normalized cost $|i-j|/t$. Define

$$\omega_t^{\mathrm{tr}}(\eta) = \sup\{W_1(G,H) : W_{1,t}(p_G, p_H) \le \eta\}, \qquad L_t(\eta) = \log\left(e + \frac{1}{\eta\sqrt{t}}\right),$$

with $L_t(0) = +\infty$.

Theorem 9.3.1 (The sharp transportation inverse modulus)**.** *Uniformly for integers $t \ge 1$ and $0 \le \eta \le 1/4$,*

$$\omega_t^{\mathrm{tr}}(\eta) \asymp \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{tL_t(\eta)}} + \eta\right\}. \tag{9.3.1}$$

The constants are absolute. In particular the logarithm depends on $\eta\sqrt{t}$, rather than just on η.

The observation model for the second theorem is as follows. Draw $X_1, \dots, X_N$ independently from G, and, conditional on them, observe t independent Bernoulli trials for each individual. A strong adversary sees the entire clean trial array and may flip at most $q = \lfloor \varepsilon N t \rfloor$ bits in total. Only the resulting counts are passed to the estimator. Equivalently, the adversary can change clean counts J_i into reports J_i' subject to

$$\sum_{i=1}^{N} |J_i' - J_i| \le q.$$

Every such count change can be implemented by exactly that many bit flips. Allowing the adversary to see the order of the clean bits adds only randomization conditional on the counts, which is already allowed. Denote this class of adversaries by $\mathcal{A}_{N,t}^{\mathrm{bit}}(\varepsilon)$.

Theorem 9.3.2 (The full bit-corruption minimax curve)**.** *For $0 \le \varepsilon \le 1/4$, define*

$$r_{N,t}^{\mathrm{bit}}(\varepsilon) = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(eN)}} + \frac{1}{\sqrt{N}} + \varepsilon + \frac{1}{\sqrt{tL_t(\varepsilon)}}\right\}. \tag{9.3.2}$$

Then

$$\inf_{\widehat{G}} \sup_{G} \sup_{A \in \mathcal{A}_{N,t}^{\mathrm{bit}}(\varepsilon)} \mathbb{E} W_1(\widehat{G}(A(J)), G) \asymp r_{N,t}^{\mathrm{bit}}(\varepsilon).$$

The upper bound is attained by a rational linear program whose bit complexity is polynomial in the explicit sample input length. The algorithm is supplied with q or a rational upper bound ε; no adaptation to an unknown bit-flip budget is claimed.

Wasserstein contamination, modulus-based lower bounds, and computationally motivated projection estimators are established topics [117]. Coordinated Wasserstein perturbations and their finite-sample minimax analysis were studied by Chao and Dobriban [118] for location, regression, and density estimation. The new claims are the sharp binomial-mixture transport modulus and the matching latent-population theorem under a hard post-sample bit budget. The Hahn operator and interval-discrepancy linear program are classical or prior ingredients, as specified below.

9.3.1 A discrete energy estimate for the hidden oscillations

Put $\pi(dx) = dx/(\pi\sqrt{x(1-x)})$, $\rho_j = p_\pi(j)$, $h_k(x) = T_k(2x-1)$, and

$$Qh(j) = \mathbb{E}_\pi[h(X) \mid J = j], \qquad \lambda_{t,k} = \prod_{l=0}^{k-1} \frac{t-l}{t+1+l}.$$

The beta–binomial orthogonal-polynomial diagonalization [70] gives

$$Qh_k \text{ is a degree-}k \text{ polynomial}, \qquad \sum_j \rho_j (Qh_k(j))^2 = \lambda_{t,k}/2, \qquad \lambda_{t,k} \le e^{-k^2/(2t)}. \tag{9.3.3}$$

These identities follow directly from the posterior beta moments: Q^*Q is self-adjoint, preserves polynomial degree, and has leading coefficient multiplier $\lambda_{t,k}$ on degree k. Arcsine orthogonality proves the first two identities, and $\log(1-u) \le -u$ proves the last one.

Lemma 9.3.3 (Transportation attenuation of a Hahn mode)**.** *For $1 \le k \le t$, let $G_k^\pm = (1 \pm h_k/2)\pi$. Then*

$$W_1(G_k^+, G_k^-) \asymp k^{-1}, \qquad W_{1,t}(p_{G_k^+}, p_{G_k^-}) \le \frac{e^{-k^2/(4t)}}{k}. \tag{9.3.4}$$

The extra factor $1/k$ in the second inequality is the feature needed for the transport inverse law.

Proof. Consider the reversible birth-and-death generator

$$\begin{aligned}(\mathcal{L}f)(j) = {} & (j+1/2)(t-j)[f(j+1) - f(j)] \\ & + j(t-j+1/2)[f(j-1) - f(j)],\end{aligned}$$

with the vanishing endpoint terms omitted. Detailed balance follows from the beta–binomial ratio ρ_{j+1}/ρ_j. The generator preserves polynomial degree and has leading multiplier $-k^2$ on degree k. Since the polynomials Qh_k are orthogonal under ρ, self-adjointness gives $\mathcal{L}Qh_k = -k^2 Qh_k$. Its Dirichlet form is

$$\mathcal{E}(f,g) = -\langle f, \mathcal{L}g\rangle_\rho = \sum_{j=0}^{t-1} \rho_j (j+1/2)(t-j)[f(j+1) - f(j)][g(j+1) - g(j)].$$

If f is 1-Lipschitz for the normalized count metric, then $\mathcal{E}(f,f) \le 2$, since its consecutive differences are at most $1/t$ and $(j+1/2)(t-j) \le 2t^2$. Thus

$$|\langle f, Qh_k\rangle_\rho| = k^{-2}|\mathcal{E}(f, Qh_k)| \le k^{-1}\sqrt{\mathcal{E}(f,f)}\sqrt{\lambda_{t,k}/2} \le \frac{\sqrt{\lambda_{t,k}}}{k}.$$

The predictive signed difference is exactly $\rho_j Qh_k(j)$. Kantorovich duality and (9.3.3) prove the second assertion. For the first, use $x = (1+\cos\theta)/2$. The signed population CDF has magnitude $|\sin(k\theta)|/(\pi k)$, and hence

$$W_1(G_k^+, G_k^-) = \frac{1}{2\pi k}\int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \asymp k^{-1}.$$

Uniform lower and upper constants follow by integrating on $[\pi/4, 3\pi/4]$, with finitely many small k treated directly. □

9.3.2 Polynomial testing and proof of the modulus

We use the following quantitative approximation ingredient, proved in [105] and used in the preceding computational paper [115]. For each 1-Lipschitz $f:[0,1]\to\mathbb{R}$ and integer $d\ge 1$ there is a degree-d polynomial P_d with

$$\|P_d-f\|_\infty\le C/d,\qquad \|P_d'\|_\infty\le C.$$

If a polynomial P of degree at most $d\le t/2$ has degree-t Bernstein coefficients b_j, then

$$\max_j|b_j|\le Ce^{2d^2/t}\|P\|_\infty,\qquad t\max_j|b_{j+1}-b_j|\le Ce^{4d^2/t}\|P'\|_\infty. \tag{9.3.5}$$

The second bound follows by differentiating the Bernstein expansion and applying the first at degree $t-1$. The first follows from the Cauchy coefficient contour for $(1+z)^tP(z/(1+z))$, with the Bernstein complex-polynomial inequality and binomial saddle-point normalization. We do not re-count these approximation bounds as new.

Proof of Theorem 9.3.1. Testing the two populations with the polynomial P_k and its Bernstein coefficients gives

$$W_1(G,H)\le C/k+Ce^{4k^2/t}W_{1,t}(p_G,p_H),\qquad 1\le k\le t/2. \tag{9.3.6}$$

There is also the elementary estimate

$$W_1(G,H)\le W_{1,t}(p_G,p_H)+t^{-1/2}, \tag{9.3.7}$$

since a normalized binomial count can be coupled to its parameter with mean absolute error at most $1/(2\sqrt{t})$. If $\eta\sqrt{t}\ge 1$, equation (9.3.7) proves the required upper bound. Otherwise choose

$$k=\left\lfloor \min\{t/2,\sqrt{atL_t(\eta)}\}\right\rfloor,\qquad a=1/128.$$

At $\eta=0$ take $k=\lfloor t/2\rfloor$. For sufficiently large t these are admissible; bounded t are absorbed by the $1/t$ term. Writing $x=\eta\sqrt{t}\in(0,1)$, the quantity $x(e+1/x)^{4a}\sqrt{\log(e+1/x)}$ is uniformly bounded. Thus the second term of (9.3.6) is $O((tL_t(\eta))^{-1/2})$, while $1/k$ is bounded by that term plus C/t. This proves the upper bound, with clipping at loss one.

For the lower bound, endpoint mixing laws differing by mass η have both population and predictive transport distance η. The pair $G^{\pm}_{t+1}$ has identical predictive laws by moment orthogonality, and population distance $\asymp 1/t$. If $0<\eta\sqrt{t}<1$, put $k=\lceil\sqrt{8tL_t(\eta)}\rceil$. When $k\le t$, Lemma 9.3.3 gives

$$W_{1,t}(p_{G_k^+},p_{G_k^-})\le\frac{e^{-2L_t(\eta)}}{\sqrt{8tL_t(\eta)}}\le\eta,$$

and the latent separation is $\asymp(tL_t(\eta))^{-1/2}$. If $k>t$, this inverse term is $O(1/t)$ and exact moment matching suffices. If $\eta\sqrt{t}\ge 1$, it is instead $O(\eta)$. Combining these lower bounds proves (9.3.1). □

9.3.3 A decontamination linear program

For a signed vector z on $\{0,\dots,t\}$ of total mass zero, let

$$D_s(z)=\max_{A\in\mathcal{I}_s}\left|\sum_{j\in A}z_j\right|,$$

where $\mathcal{I}_s$ consists of unions of at most s consecutive integer intervals. We will need a property that holds even when z is not a difference of model laws.

Lemma 9.3.4 (Polynomial tests from interval discrepancies)**.** *If $b(j)$ is a polynomial of degree at most k on the count grid, and $s \ge k+1$, then*

$$\left|\sum_j b(j) z_j\right| \le 2 \max_j |b(j)| D_s(z). \tag{9.3.8}$$

Proof. Subtract the minimum of b and use the layer-cake formula. Every superlevel set $\{j : b(j) > u\}$ is a union of at most $k+1$ intervals, because a degree-k nonconstant polynomial has at most k real roots at each level. The integral has length at most $2 \max |b|$. A constant polynomial gives zero because z has total mass zero. □

Suppose $2 \le t \le N$. Put

$$M = Nt, \qquad m = \min\{t, \lceil \sqrt{4t\lceil \log_2(8tN) \rceil} \rceil\}, \quad s = m+1,$$

and choose the rational threshold

$$b_N = \frac{\lceil \sqrt{N(2s+1)\lceil \log_2(2(t+2)) \rceil} \rceil}{N}.$$

Let q' be the empirical law of the reported counts. Jointly minimize

$$W_{1,t}(p_H, \widetilde{q}) + (D_s(p_H - \widetilde{q}) - b_N)_+ \tag{9.3.9}$$

over all probability laws H supported on $\{l/M : 0 \le l \le M\}$ and all probability vectors $\widetilde{q}$ on the count alphabet, subject to

$$W_{1,t}(\widetilde{q}, q') \le \varepsilon. \tag{9.3.10}$$

Return the minimizing $\widehat{G} = H$. Here $\widetilde{q}$ is a fitted clean empirical law, not a binomial-mixture law.

This is a polynomial-size rational linear program. Wasserstein constraints and objectives on an ordered finite grid are sums of absolute cumulative discrepancies. The epigraph of D_s is a longest-path dynamic program over states recording the position, the number of selected intervals, and whether the last point was selected. Introducing one potential for each state, and one inequality per edge, expresses its epigraph with $O(st)$ variables and constraints, as proved in [115]. The additional variable $\widetilde{q}$ and constraint (9.3.10) add only $O(t)$ variables and constraints. The support grid has $Nt+1$ points, with binomial probabilities of bit length $O(t \log(Nt+1))$. Standard rational linear programming therefore gives polynomial bit complexity when $t \le N$. If $t > N$, or if $\varepsilon\sqrt{t} \ge 1$, the estimator instead returns the empirical law of the normalized reports; this branch is also polynomial time. Bounded exceptional values are harmless.

Upper bound in Theorem 9.3.2. Let q^0 be the unobserved clean empirical law. The hard flip budget implies $W_{1,t}(q^0, q') \le \varepsilon$ deterministically. Put

$$S = D_s(q^0 - p_G), \qquad Z = W_{1,t}(q^0, p_G), \qquad A_N = 2/N + Z + (S - b_N)_+.$$

The number of interval unions is at most $(t+2)^{2s}$. Hoeffding's inequality and the rational choice of b_N give

$$\mathbb{E}(S - b_N)_+ \le C/\sqrt{N}, \quad \mathbb{E}S \le b_N + C/\sqrt{N}, \quad \mathbb{E}Z \le 1/(2\sqrt{N}), \quad b_N \le \frac{Ct^{1/4}\log(eN)^{3/4}}{\sqrt{N}}.$$

Round G to the support grid, obtaining G_*. Common-uniform Bernoulli coupling gives $\|p_{G_*} - p_G\|_{\mathrm{TV}} \le t/M = 1/N$. The feasible comparator (G_*, q^0) has objective at most A_N. Thus the optimizer satisfies

$$W_{1,t}(p_{\widehat{G}}, \widetilde{q}) \le A_N, \qquad D_s(p_{\widehat{G}} - \widetilde{q}) \le b_N + A_N, \tag{9.3.11}$$

$$W_{1,t}(\widetilde{q}, q^0) \le 2\varepsilon, \qquad \mathbb{E}A_N \le C/\sqrt{N}. \tag{9.3.12}$$

All these comparisons hold uniformly over the post-sample adversary.

To invert these bounds, split a Lipschitz test approximation into $P_u + (P_k - P_u)$, with $u \asymp \sqrt{t}$ and $u \le k \le \min(m, t/2)$. The low-degree Bernstein coefficient vector has normalized-grid Lipschitz constant at most an absolute constant. The residual coefficient vector has supremum norm at most $Ce^{2k^2/t}/u$ and normalized-grid Lipschitz constant at most $Ce^{4k^2/t}$, by (9.3.5). Each coefficient vector is a polynomial in the count index of degree at most k, as follows from the falling-factorial representation of monomials.

Decompose the predictive signed difference as

$$p_{\widehat{G}} - p_G = (p_{\widehat{G}} - \widetilde{q}) + (\widetilde{q} - q^0) + (q^0 - p_G).$$

Test the low-degree vector using transportation distance throughout. For the residual vector use Lemma 9.3.4 on the first and third summands, and its Lipschitz bound on the middle summand. After taking expectations and the Lipschitz dual supremum, this yields

$$\mathbb{E}W_1(\widehat{G}, G) \le \frac{C}{k} + C(\varepsilon + N^{-1/2}) + \frac{Ce^{2k^2/t}}{\sqrt{t}}(b_N + N^{-1/2}) + Ce^{4k^2/t}\varepsilon. \tag{9.3.13}$$

This hybrid estimate is why ordinary transport projection alone is not substituted for (9.3.9).

Take $u = \lfloor \sqrt{t}/16 \rfloor$ and

$$k = \left\lfloor \min\{t/2, \sqrt{at \min(\log(eN), L_t(\varepsilon))}\} \right\rfloor, \qquad a = 1/128.$$

For large enough t these choices have $1 \le u \le k \le m$; bounded t are absorbed by the $1/t$ term. The sampling contribution in (9.3.13) is at most $C(t\log(eN))^{-1/2}$, since $t \le N$ and $(eN)^{2a}N^{-1/4}\log(eN)^{5/4}$ is bounded. The last term is $O((tL_t(\varepsilon))^{-1/2})$ by the same elementary logarithmic inequality as in the modulus proof. The approximation term has the required two inverse scales plus $1/t$.

On either empirical-output branch, the risk is at most $\varepsilon + (2\sqrt{t})^{-1} + (2\sqrt{N})^{-1}$, by the deterministic transport budget and binomial coupling. If $t > N$, the second term is absorbed by $N^{-1/2}$; if $\varepsilon\sqrt{t} \ge 1$, it is absorbed by ε. This proves the claimed upper bound everywhere. □

Lower bound in Theorem 9.3.2. No flips are permitted as an adversarial choice, so the established clean lower bound supplies $t^{-1} + (t\log(eN))^{-1/2} + N^{-1/2}$, clipped at one. Its three elementary constructions are respectively exact moment matching, the Chebyshev perturbations with $k \asymp \sqrt{t\log(eN)}$, and endpoint mixing weights differing by $\asymp N^{-1/2}$; see [105].

Let $q = \lfloor \varepsilon Nt \rfloor \ge 1$ and set $\eta = q/(16Nt)$. Choose a pair from the lower-bound constructions in Theorem 9.3.1, with predictive transportation distance at most η and latent separation at least a fixed positive multiple of the modulus curve (using whichever term is largest). Independently couple each pair of counts optimally in transport distance. The expected total integer displacement is at most $Nt\eta = q/16$. By Markov's inequality the total displacement exceeds q with probability at most $1/16$. In the first experiment the adversary samples the coupled second count vector conditional on its observed vector. If its total displacement is at most q, it applies these changes; otherwise it leaves the sample untouched. This obeys the hard

budget for every realization. Its output distribution is within TV 1/16 of the clean second experiment. Le Cam's inequality therefore gives a fixed positive multiple of the latent separation as a lower bound on maximum expected loss.

For $q \geq 1$, $q/(Nt) \geq \varepsilon/2$, so scaling the modulus argument by the absolute factor 1/32 changes its curve only by absolute constants. If $q = 0$, then $\varepsilon < 1/(Nt)$, and both contamination terms are bounded by the clean terms because $L_t(\varepsilon) \geq c \log(eN)$ for an absolute $c > 0$. This proves the complete lower bound. □

9.3.4 An unbounded separation between two corruption geometries

For comparison, corruption of an ε fraction of whole counts has the prior minimax curve

$$\min\{1, t^{-1} + (t \log(eN))^{-1/2} + N^{-1/2} + \varepsilon + (t \log(e/\varepsilon))^{-1/2}\}.$$

Set $N = t^3$ and $\varepsilon = (t \log t)^{-1/2}$, along integer t tending to infinity. The whole-report rate is $\asymp (t \log t)^{-1/2}$, whereas the bit-budget rate is $\asymp (t \log \log t)^{-1/2}$. Their ratio is therefore

$$\asymp \sqrt{\frac{\log t}{\log \log t}} \longrightarrow \infty.$$

This comparison uses the same numerical fraction ε but different resource accounting: the second adversary may distribute its flips across many individuals. It is not a claim that equal sets of allowable corrupted data have different risks.

Scope and unresolved targets. The transport inverse theorem and the sharp bit-budget theorem are two related new targets, not two previously named conjectures. Their proof uses earlier clean-binomial lower bounds and Bernstein approximation estimates, which remain explicitly cited dependencies. The stronger Chebyshev–Bernstein exponent $\log 2$, robustness of the unmodified NPMLE, and simultaneous optimal adaptation to an unknown bit budget remain unresolved here. The unknown-budget adaptation previously proved for whole-report replacements cannot simply be imported: that proof controls a total-variation contamination error, whereas cheap individual flips may produce a much larger total-variation error.

9.4 Relations, edge loss, and the dimension visible in a zero intensity

The preceding blind inverse theorem requires independent relative frequencies. Here we determine what happens when that condition fails, including the exact leading loss of the rightmost zero boundary. A second result computes a sharp edge-intensity constant in the independent finite-frequency case. It shows that a first intensity can reveal the number of hidden terms even though identification of all their magnitudes from that intensity remains open in this work.

We use the finite exponential sum

$$F_\theta(s) = 1 - \sum_{j=1}^{N} a_j e^{-\lambda_j s + i\theta_j}, \qquad 0 < \lambda_1 < \cdots < \lambda_N, \quad a_j > 0. \tag{9.4.1}$$

There is a unique real S such that

$$\sum_j a_j e^{-\lambda_j S} = 1. \qquad b_j = a_j e^{-\lambda_j S}, \quad B = \operatorname{diag}(b_1, \ldots, b_N), \quad M = \sum_j b_j \lambda_j > 0. \tag{9.4.2}$$

The point S is the frozen right boundary, and $F_0'(S) = M$. Write $s_+(\theta) = \sup\{\Re\rho : F_\theta(\rho) = 0\}$. All zeros lie in a fixed vertical strip, uniformly in θ: the constant dominates far to the right, and the term of largest frequency dominates far to the left.

9.4.1 An exact criterion and a sharp quadratic loss caused by relations

Let

$$\mathcal{L} = \{k \in \mathbb{Z}^N : k \cdot \lambda = 0\}, \qquad H = \overline{\{t\lambda \bmod 2\pi : t \in \mathbb{R}\}}.$$

The standard character description of a compact torus gives

$$H = \{\eta \bmod 2\pi : k \cdot \eta \in 2\pi\mathbb{Z} \text{ for all } k \in \mathcal{L}\}. \tag{9.4.3}$$

It is a connected subtorus. Its Lie algebra is $V = \{v \in \mathbb{R}^N : k \cdot v = 0 \text{ for all } k \in \mathcal{L}\}$, and $\lambda \in V$. Choose linearly independent rows of an integer matrix K spanning the real relation space; put $q = \operatorname{rank} K$. Locally near 0, the invariant transverse phase coordinate is $\xi = K\theta \in \mathbb{R}^q$, with no reduction modulo 2π needed.

Theorem 9.4.1 (The relation matrix determines the leading edge loss)**.** *For every phase vector,*

$$s_+(\theta) = S \quad \Longleftrightarrow \quad \theta \in H. \tag{9.4.4}$$

Assume $q \geq 1$. The matrix

$$\mathcal{A} = K(B^{-1} - \mathbf{1}\mathbf{1}^{\mathsf{T}})K^{\mathsf{T}} \tag{9.4.5}$$

is positive definite. For θ sufficiently close to H, use the local lift of its transverse coordinate near 0. Then

$$\boxed{S - s_+(\theta) = \frac{1}{2M}\,\xi^{\mathsf{T}}\mathcal{A}^{-1}\xi + O(\|\xi\|^4).} \tag{9.4.6}$$

The left side is a real-analytic, even function of ξ in that neighborhood. In particular every sufficiently small nonzero relation-phase defect creates a strictly positive zero-free gap. If there are no relations, then $s_+(\theta) = S$ for all phases.

This is a statement about the characteristic roots themselves. For a delay-difference characteristic function of the form (9.4.1), with $\sum a_j = 1$ and hence $S = 0$, it gives the exact leading leftward displacement of the spectral boundary. It does not invoke an unstated equivalence between a root condition and stability in a particular history space.

Proof. At a zero $s = S - \delta + it$, set $\eta_j = \theta_j - t\lambda_j \bmod 2\pi$. The zero equation implies

$$\sum_j b_j e^{\lambda_j \delta}|e^{i\eta_j} - 1|^2 = 2\left(\sum_j b_j e^{\lambda_j \delta} - 1\right), \qquad \delta \geq 0. \tag{9.4.7}$$

Thus a sequence of zeros with $\delta \to 0$ has $\eta \to 0$ in the torus. Since $\eta \in \theta + H$, this implies $\theta \in H$. Conversely, if $\theta \in H$, the translation hull contains F_0, whose zero at S is simple. Locally uniform approximation and Rouché's theorem put zeros of F_θ arbitrarily close to S. This proves (9.4.4). It also proves strict loss away from H: compactness of the coset in the left side of (9.4.7) prevents a sequence with loss tending to zero.

We now calculate the loss, rather than merely its sign. Give $\mathbb{R}^N$ the inner product $\langle u, v\rangle_B = u^{\mathsf{T}}Bv$. Let P be the orthogonal projection onto V, and choose the unique small representative $z \in V^{\perp_B}$ of the phase coset. Put

$$p = P\mathbf{1}, \quad c = \|p\|_B^2 > 0, \qquad \eta_0(z) = z - \frac{\langle \mathbf{1}, z\rangle_B}{c}\,p, \qquad W = V \cap \{w : \langle \mathbf{1}, w\rangle_B = 0\}.$$

Here $c > 0$ because $\langle \mathbf{1}, \lambda\rangle_B = M > 0$ and $\lambda \in V$. The vector $\eta_0(z)$ has weighted mean zero, lies in $z + V$, and is B-orthogonal to W. Therefore

$$Q(z) := \|\eta_0(z)\|_B^2 = \|z\|_B^2 + \frac{\langle \mathbf{1}, z\rangle_B^2}{c}. \tag{9.4.8}$$

Every small phase in the coset $z + H$ has a unique local expression

$$\eta = \eta_0(z) + w + u\lambda, \qquad w \in W, \quad u \in \mathbb{R}.$$

Apply the real-analytic implicit-function theorem to

$$1 - \sum_j b_j \exp\{\lambda_j \delta + i[\eta_0(z) + w + u\lambda]_j\} = 0. \tag{9.4.9}$$

At the origin, derivatives in (δ, u) are $-M$ and $-iM$, so the two real equations have a unique analytic solution $\delta = \delta(z, w)$, $u = u(z, w)$. Complex conjugation shows that $\delta(-z, -w) = \delta(z, w)$ and $u(-z, -w) = -u(z, w)$. Weighted mean zero removes the first-order terms in δ and u. Expanding the real equation to second order gives

$$\delta(z, w) = \frac{Q(z) + \|w\|_B^2}{2M} + O((\|z\| + \|w\|)^4). \tag{9.4.10}$$

In particular its Hessian in w is positive definite. There is a unique local minimizing point $w_*(z)$, analytic by another implicit-function theorem, and $w_*(z) = O(\|z\|^3)$. It follows that

$$\min_w \delta(z, w) = Q(z)/(2M) + O(\|z\|^4). \tag{9.4.11}$$

This local minimum is the actual edge loss. To check this point, (9.4.7) forces every zero in a sufficiently thin edge strip to have a small phase representative. Those phases are exactly parametrized by (9.4.9). Conversely every zero produced there is simple, since its s derivative is close to M, and belongs to a member of the phase hull $z + H$. Minimality of translation on this hull and Rouché's theorem show that its real part is a limit of real parts of actual zeros. Since (9.4.11) tends to zero with z, zeros outside the thin strip cannot maximize the real part. This proves the assertion.

It remains to write Q in invariant coordinates. Set $G = KB^{-1}K^{\mathsf{T}}$ and $h = K\mathbf{1}$. Then

$$z = B^{-1}K^{\mathsf{T}}G^{-1}\xi, \quad \|z\|_B^2 = \xi^{\mathsf{T}}G^{-1}\xi, \quad \langle \mathbf{1}, z\rangle_B = h^{\mathsf{T}}G^{-1}\xi, \quad c = 1 - h^{\mathsf{T}}G^{-1}h.$$

The Sherman–Morrison identity gives $Q(z) = \xi^{\mathsf{T}}(G - hh^{\mathsf{T}})^{-1}\xi$. For a direct check of positivity, Cauchy–Schwarz and $\sum b_j = 1$ give $B^{-1} - \mathbf{1}\mathbf{1}^{\mathsf{T}} \succeq 0$, with kernel spanned by $b = (b_j)$. If $K^{\mathsf{T}}v$ were a nonzero multiple of b, taking its dot product with λ would contradict $K\lambda = 0$ and $b \cdot \lambda = M > 0$. Thus the restriction (9.4.5) is positive definite. Since the coordinates z and ξ are linearly equivalent, substitution in (9.4.11) proves (9.4.6). □

For example, take $N = 2$, $(\lambda_1, \lambda_2) = (1, 2)$, $b_1 = u$, $b_2 = 1 - u$ and $K = (2, -1)$. Then $M = 2 - u$, and

$$S - s_+(\theta) = \frac{u(1-u)}{2(2-u)^3}(2\theta_1 - \theta_2)^2 + O((2\theta_1 - \theta_2)^4). \tag{9.4.12}$$

This agrees with expansion of the explicit quadratic equation in e^{-s}, but the matrix theorem applies to arbitrary rank and arbitrary finitely many integer relations.

9.4.2 The exact first-intensity law at an independent frozen edge

Now suppose $N \geq 2$ and $\lambda_1, \dots, \lambda_N$ are linearly independent over $\mathbb{Q}$. For fixed θ, let $n_\theta(\varepsilon, T)$ count zeros, with multiplicity, in $S - \varepsilon < \Re s < S$, $0 < \Im s < T$. Let $V_d = \pi^{d/2}/\Gamma(1 + d/2)$ denote the volume of the Euclidean unit ball in dimension d.

Theorem 9.4.2 (Edge intensity with its leading constant)**.** *For all sufficiently small $\varepsilon > 0$ and every θ,*

$$\lim_{T\to\infty} \frac{n_\theta(\varepsilon, T)}{T} = c(\varepsilon), \tag{9.4.13}$$

where

$$\boxed{c(\varepsilon) = \frac{MV_{N-1}(2M)^{(N-1)/2}}{(2\pi)^N \sqrt{\prod_{j=1}^N b_j}}\, \varepsilon^{(N-1)/2}(1 + O(\varepsilon)).} \tag{9.4.14}$$

The same $c(\varepsilon)$ is the expected zero count per unit height in that strip under independent uniform phases.

Proof. We compute the mean count by a local parametrization of every zero in a thin edge strip. Identity (9.4.7) puts all its zero phases in a small neighborhood of 0. Write $v = (\eta_1, \dots, \eta_{N-1})$. Solve the real and imaginary parts of

$$1 - \sum_j b_j e^{\lambda_j \delta + i\eta_j} = 0$$

for (δ, η_N) as analytic functions of v. The relevant Jacobian at zero has absolute determinant $Mb_N > 0$. Complex conjugation implies that $\delta(v)$ is even and $\eta_N(v)$ is odd. Their leading terms are

$$\eta_N(v) = -\frac{1}{b_N} \sum_{j<N} b_j v_j + O(\|v\|^3), \tag{9.4.15}$$

$$\delta(v) = \frac{1}{2M} v^{\mathsf{T}} C v + O(\|v\|^4), \quad C = \operatorname{diag}(b_1, \dots, b_{N-1}) + \frac{1}{b_N}(b_i b_j)_{i,j<N}. \tag{9.4.16}$$

The positive definite matrix satisfies

$$\det C = \frac{\prod_{j=1}^N b_j}{b_N^2} \tag{9.4.17}$$

by the determinant lemma and $\sum b_j = 1$.

Consider the map from local phase coordinates and height to original phases,

$$(v, t) \longmapsto \Theta(v, t) = \eta(v) + t\lambda \pmod{2\pi}, \qquad 0 < t < 1.$$

Each preimage gives exactly the zero $s = S - \delta(v) + it$ of $F_{\Theta(v,t)}$. Conversely every zero in the thin strip and that height interval gives such a preimage, since its phase has the unique small representative provided above. These zeros are simple: their s derivatives tend uniformly to M as the strip shrinks. The ordinary change-of-variables formula with multiplicities, applied on finitely many coordinate charts of the torus, therefore gives the expected count as

$$\frac{1}{(2\pi)^N} \int_{\{v:0<\delta(v)<\varepsilon\}} J(v)\, dv, \qquad J(v) = \left|\det[\partial_{v_1}\eta, \dots, \partial_{v_{N-1}}\eta, \lambda]\right|. \tag{9.4.18}$$

This formula also follows by partitioning the domain into locally injective charts; its Jacobian is nonzero for small v, so the usual finite-dimensional area formula counts all preimages correctly. Equation (9.4.15) yields

$$J(0) = \lambda_N + \sum_{j<N} (b_j/b_N)\lambda_j = M/b_N.$$

Oddness of η makes J even, so $J(v) = M/b_N + O(\|v\|^2)$.

Since C is positive definite, the sublevel set in (9.4.18) has $\|v\| = O(\sqrt{\varepsilon})$ and volume

$$\frac{V_{N-1}(2M\varepsilon)^{(N-1)/2}}{\sqrt{\det C}}(1+O(\varepsilon)).$$

For a direct error bound, use polar coordinates after the linear change $y = C^{1/2}v$ in (9.4.16); each radial boundary is $\sqrt{2M\varepsilon}(1+O(\varepsilon))$, uniformly in direction. Combining this with (9.4.17) and the expansion of J proves (9.4.14) for the expected count.

It remains to justify the deterministic limit, rather than replacing it by the random model. The finite-dimensional phase flow is uniquely ergodic by rational independence. Counts in any fixed compact rectangle are uniformly bounded over phases: a failure of boundedness, followed by compactness of the phase torus and locally uniform convergence, would contradict the finiteness of zeros of a nonzero analytic limit in a slightly larger compact neighborhood. No limit function is identically zero because its constant term dominates at $+\infty$.

The count in $(S-\varepsilon, S)\times(0,1)$ is continuous in phase except when a zero meets its boundary. The above parametrization shows that the vertical boundary $S-\varepsilon$ has zero mean mass: the level set $\delta(v)=\varepsilon$ is a smooth hypersurface for small positive ε. The boundary at S corresponds to $v=0$ and has measure zero as $N\ge 2$. Horizontal boundaries also have zero mean mass by stationarity and local finiteness. Thus this bounded counting function is Riemann integrable on the phase torus. Unique ergodicity, with continuous upper and lower approximations, identifies its time average on every orbit with its Haar integral. Integrating unit-height counts over starting heights in $[0,T]$ differs from $n_\theta(\varepsilon,T)$ by $O(1)$, using the same uniform local bound at the two endpoints. This proves (9.4.13). □

Corollary 9.4.3 (The first intensity identifies the hidden term count)**.** *Within this independent-frequency finite model, the intensity of real parts near its right support endpoint determines the number $N+1$ of terms:*

$$N = 1 + 2\lim_{\varepsilon\downarrow 0}\frac{\log c(\varepsilon)}{\log\varepsilon}.$$

After N and S are known, it also determines the combination $M^{(N+1)/2}/\sqrt{\prod b_j}$. Neither fact by itself identifies the individual frequencies or magnitudes.

The case $N=1$ is distinct. Its zeros lie exactly on the line S with intensity $\lambda_1/(2\pi)$, so the first intensity has an atom at the edge rather than the continuous edge law above. The corollary counts the constant term as well as the N exponentials.

Theorem 9.4.4 (The intensity reads rational rank on a compatible hull)**.** *Allow integer frequency relations again, and suppose $\theta\in H$. Put $r=\dim V=\dim_{\mathbb{Q}}\operatorname{span}_{\mathbb{Q}}\{\lambda_1,\dots,\lambda_N\}$. Assume $r\ge 2$. Let $\mathfrak{v}_H$ be the Euclidean covolume in V of the full lattice $V\cap(2\pi\mathbb{Z}^N)$, let b_V be the Euclidean orthogonal projection of b onto V, and put*

$$W=\{w\in V : b\cdot w=0\},\qquad B_W=E^{\mathsf{T}}BE,$$

where the columns of E are any Euclidean orthonormal basis of W. Then the vertical mean edge count exists for every $\theta\in H$, and

$$\boxed{c_H(\varepsilon)=\frac{MV_{r-1}(2M)^{(r-1)/2}}{\mathfrak{v}_H\,\|b_V\|\sqrt{\det B_W}}\,\varepsilon^{(r-1)/2}(1+O(\varepsilon)).} \tag{9.4.19}$$

Thus a non-atomic first intensity at the compatible edge determines the rational rank r, rather than the number of terms in a possibly redundant frequency representation. For $r=1$, the edge intensity is instead an atom of mass $1/T_0$, where T_0 is the least positive period of $t\lambda \bmod 2\pi$; a sufficiently thin strict strip to its left has no zeros.

Proof. Since V is a rational subspace, its displayed lattice has full rank r, and Haar measure on H is Euclidean volume divided by $\mathfrak{v}_H$. Also $b_V \neq 0$ because $b \cdot \lambda = M > 0$. The flow is uniquely ergodic on H; rational independence in its own character group is exactly the definition of its orbit closure.

For small $w \in W$, solve

$$1 - \sum_j b_j \exp\{\lambda_j \delta + i(w_j + u\lambda_j)\} = 0$$

for (δ, u). The same two-variable implicit-function argument as in (9.4.9) gives

$$\delta(w) = \frac{w^{\mathsf{T}} B w}{2M} + O(\|w\|^4), \qquad u(-w) = -u(w), \quad \delta(-w) = \delta(w).$$

All edge-zero phases on H are parametrized this way by the defect identity. The map $(w, t) \mapsto w + u(w)\lambda + t\lambda \bmod 2\pi$ has constant Euclidean Jacobian from $W \times \mathbb{R}$ to V, namely

$$J_H = |\mathrm{proj}_{W^\perp \cap V}\, \lambda| = \frac{b_V \cdot \lambda}{\|b_V\|} = \frac{M}{\|b_V\|}.$$

Indeed, adding multiples of the last column λ to the other columns does not change the determinant. The area formula on H therefore gives the expected edge count in a unit height interval as $J_H/\mathfrak{v}_H$ times the Euclidean volume in W of $0 < \delta(w) < \varepsilon$. Its leading ellipsoid volume is $V_{r-1}(2M\varepsilon)^{(r-1)/2}/\sqrt{\det B_W}$ with relative error $O(\varepsilon)$. Uniform local zero bounds, null boundary sets, and unique ergodicity give the deterministic vertical mean exactly as in Theorem 9.4.2.

If $r = 1$, write $\lambda_j = n_j \nu$ with positive integers n_j having greatest common divisor 1. Then $T_0 = 2\pi/\nu$. On a compatible hull the function is a translate of a polynomial in $e^{-\nu s}$ with positive magnitudes and frozen phases. Its edge zero is simple and is the only zero on that edge modulo iT_0, by equality in the triangle inequality. Finitely many polynomial roots give a positive gap to all other real parts. This proves the atomic assertion.

For consistency, when $r = N$ one has $\mathfrak{v}_H = (2\pi)^N$, $b_V = b$, and $\det B_W = (\prod b_j)/\|b\|^2$. The last identity follows by choosing the unit normal $b/\|b\|$ and using $(b/\|b\|)^{\mathsf{T}} B^{-1} (b/\|b\|) = 1/\|b\|^2$. Equation (9.4.19) then reduces to (9.4.14). □

9.4.3 An exact obstruction to recovery with unrestricted support

Proposition 9.4.5 (Unknown support can be invisible even in exact data)**.** *An exact zero divisor on $\Re s > 0$, at every height, need not determine an exponential polynomial when its frequency support is allowed to change. This failure occurs with strictly positive coefficient magnitudes and distinct positive real frequencies.*

Proof. For $0 < c < 1$ set

$$f(s) = 1 - 2e^{-s}, \qquad g_c(s) = (1 - 2e^{-s})(1 - ce^{-s}) = 1 - (2 + c)e^{-s} + 2ce^{-2s}.$$

All extra zeros of g_c lie on $\Re s = \log c < 0$. Thus the two divisors agree exactly on $\Re s > 0$, although the functions are not constant multiples and have different supports and magnitudes. Both have the form (9.4.1), the last coefficient of g_c being represented by phase π. The relation $(2, -1) \cdot (1, 2) = 0$ has invariant phase defect $2 \cdot 0 - \pi \neq 0 \bmod 2\pi$, explaining why its triangle bound is not the observable right edge. □

This obstruction has zero statistical distance and zero measurement error. Increasing the observation height or reducing ordinate noise cannot remove it without a support or factor restriction. It is an elementary counterexample delimiting the target, not a claimed new named impossibility theorem.

9.4.4 Literature comparison and remaining targets

The closure of real parts of zeros and its dependence on rational relations have a long history. Avellar and Hale [119] relate the spectral boundary to delay equations. Sepulcre [68] gives a pointwise polygon criterion in the independent case and discusses discontinuity with respect to the delays. We do not claim those qualitative principles as new. The specific quantitative target here is the explicit positive matrix and fourth-order remainder in (9.4.6).

The averaged logarithm for (9.4.1) is a restriction of the Ronkin function of an affine linear polynomial. Lundqvist [120] computes second derivatives in three variables using elliptic and hypergeometric functions. Our edge calculation is a local, all-finite-dimensions formula with an explicit constant and an interpretation as identification of term count. The proof is self-contained; historical priority beyond these checked sources is not asserted. It concerns a simple frozen edge, whereas the spectral observation section concerns a controlled double-zero transition.

The original growing-window variance and central-limit questions remain unresolved here. So does full magnitude recovery from first intensity alone when there are at least four terms. The blind inverse from the preceding manuscript still concerns exact infinite data on prescribed independent support; no finite-noise blind algorithm is claimed by this section.

9.5 The skew index-one boundary is a completeness-radius problem

The previous manuscript settled compact-scale sampling below stable index one and left the skew index-one case open. Here the sampling criterion changes its analytic nature: the observation points travel along a vertical line, and the exact answer is the completeness radius of the logarithms of the terminal times. The radius is a classical object. The result below transfers it to joint location–scale mixing, including the construction of a single joint signed density when the radius is finite.

Fix $\beta \neq 0$, $|\beta| \le 1$, put $k = 2\beta/\pi$, and use the stable index-one kernels

$$\widehat{K}_{a,b}(t) = \exp\{ita - b|t|(1 + ik\,\mathrm{sgn}(t)\log|t|)\}, \qquad a \in \mathbb{R},\ b > 0.$$

Conditionally on (A, B), the X_j are independent with this law; $S_n = X_1 + \cdots + X_n$. Only the separate population laws of the selected sums are supplied. The location and scale may be dependent.

For a locally finite set $\Lambda \subset \mathbb{R}$, define

$$R(\Lambda) = \sup\{a > 0 : \mathrm{span}\{e^{i\lambda x} : \lambda \in \Lambda\} \text{ is dense in } L^2(-a, a)\}. \tag{9.5.1}$$

The supremum of the empty set is zero. Completeness on an interval implies completeness on smaller intervals, and translation of the interval does not affect completeness. Multiplication of Λ by a nonzero constant preserves whether R is infinite.

For reference, the Beurling–Malliavin theorem gives

$$R(\Lambda) = \pi D_{\mathrm{eff}}(\Lambda), \tag{9.5.2}$$

where D_{eff} is the supremum of $a \ge 0$ for which there are disjoint bounded intervals I_j with

$$\sum_j \frac{|I_j|^2}{1 + \mathrm{dist}(0, I_j)^2} = \infty, \qquad \#(\Lambda \cap I_j) \ge a|I_j| \quad \text{for every } j.$$

This is the effective, or exterior, density convention used in [121, Section 1.1]. Equation (9.5.2) and its density interpretation are classical and are not claimed as new.

Theorem 9.5.1 (Exact compact-scale index-one criterion)**.** *Fix $0 < b_0 < b_1 < \infty$ and a terminal set $\mathcal{N} \subset \mathbb{N}$. The selected sums identify every Borel mixing law on $\mathbb{R} \times [b_0, b_1]$ if and only if*

$$R(\{\log n : n \in \mathcal{N}\}) = \infty, \tag{9.5.3}$$

equivalently, the effective density of that logarithmic set is infinite. If the radius is finite, there are distinct absolutely continuous witnesses with scale support strictly inside (b_0, b_1), equal coordinate marginals, and equal finite mixed polynomial moments of every order. Their signed difference is smooth, compactly supported in scale, and Schwartz in location.

The interval width does not enter the criterion. This is because location frequencies are unrestricted and can rescale a compact annihilator into any positive scale interval. The theorem does not assert that a finite-radius system is incomplete on every short interval at one fixed frequency.

9.5.1 From completeness to joint identification

Lemma 9.5.2 (Finite measures and smooth annihilators)**.** *For a locally finite real set Λ, the following hold.*

1. *If $R(\Lambda) = \infty$, every compactly supported finite complex measure ν satisfying $\int e^{i\lambda x}\, d\nu(x) = 0$ for all $\lambda \in \Lambda$ is zero.*
2. *If $R(\Lambda) < \infty$, there is a nonzero complex $f \in C_c^\infty(\mathbb{R})$ whose bilateral Laplace transform F satisfies $F(1 - i\lambda) = 0$ for all $\lambda \in \Lambda$. The support of f may be translated to have any prescribed center.*

Proof. For the first assertion, convolve ν with a smooth compactly supported approximate identity. Each convolution is a compactly supported L^2 function with the same Fourier zeros. Completeness on a large enough interval forces it to vanish; letting the smoothing scale tend to zero gives $\nu = 0$. The sign in the Fourier transform does not matter: a function with $\int g(x)e^{i\lambda x}dx = 0$ is orthogonal after conjugation to the given exponential system.

For the second, choose $a > R(\Lambda)$. Noncompleteness supplies a nonzero $L^2(-a, a)$ function q with the required Fourier zeros, with conjugation if needed. Smoothing by a nonzero compact bump preserves the zeros and does not annihilate q: the two entire Fourier transforms have a nonzero product. The resulting smooth compact function q_0 gives $f(x) = e^x q_0(x)$, since

$$F(1 - i\lambda) = \int e^{i\lambda x} q_0(x)\, dx = 0.$$

Translation multiplies F by a nonvanishing exponential, preserving all zeros and allowing any support center. □

Proof of Theorem 9.5.1. For $t > 0$, the mixing transform of S_n/n is evaluated at

$$(t, s_n(t)), \qquad s_n(t) = t + ikt \log t - ikt \log n. \tag{9.5.4}$$

Suppose the radius in (9.5.3) is infinite, and let ρ be the signed difference of two candidates. For fixed $t > 0$, define a complex measure on $x = tB$ by pushing forward

$$e^{itA - tB - iktB \log t}\, d\rho(A, B).$$

It is supported in $[tb_0, tb_1]$ and its Fourier transform vanishes at every $k \log n$ by (9.5.4). The first part of Lemma 9.5.2 makes it zero. Undoing the bounded, nonvanishing weight on this compact interval shows

$$\int e^{itA}\, d\rho(A, dB) = 0.$$

Conjugation gives negative t and continuity gives $t = 0$. Fourier uniqueness in A, after restriction to arbitrary Borel sets in B, then proves $\rho = 0$.

For necessity let $\Lambda = \{k \log n : n \in \mathcal{N}\}$ and choose f, F from the second part of the lemma. Write the initial support of f as an interval of length $2a$. Choose t_0 large enough that $2a/t_0 < b_1 - b_0$, then translate f to be centered at $t_0(b_0 + b_1)/2$. Let $h \in C_c^\infty((0, \infty))$ be real and nonzero with support sufficiently close to t_0 that the support of $f(tb)$ lies strictly inside $b_0 < b < b_1$ for all $t \in \operatorname{supp} h$. For $t > 0$ set

$$G(t, b) = h(t)\partial_b\{tf(tb)e^{iktb \log t}\}, \qquad G(-t, b) = \overline{G(t, b)}, \tag{9.5.5}$$

and extend by zero near $t = 0$. Define $g(a, b) = (2\pi)^{-1} \int e^{-ita} G(t, b)\, dt$. It is real, nonzero, smooth, compactly supported in b, and Schwartz in a. Nonzeroness follows because a nonzero compactly supported function of b cannot have identically zero derivative. Integration by parts, with no boundary terms, gives for $t > 0$

$$\int e^{ita - sb} g(a, b)\, da\, db = h(t) s F(s/t - ik \log t). \tag{9.5.6}$$

At (9.5.4), its last argument is $1 - ik \log n$, a zero of F. Negative t follows by conjugation. The derivative in (9.5.5) makes the location marginal zero; vanishing of G near $t = 0$ makes the scale marginal and all mixed polynomial moments zero. Normalizing the positive and negative parts of g yields the asserted witnesses. □

9.5.2 An explicit change of sampling threshold

Corollary 9.5.3 (Stretched-exponential terminal times)**.** *Let $p > 0$ and $n_j = \lceil \exp(j^p) \rceil$, discarding any repeated initial terms. In the compact-scale index-one model above,*

$$\{\mathcal{L}(S_{n_j}) : j \geq 1\} \text{ identifies every mixing law} \quad \Longleftrightarrow \quad p < 1.$$

The unrestricted-scale criterion in the preceding manuscript is $p \leq 1/2$. Thus $1/2 < p < 1$ is a genuine interval of designs repaired by compact scale support, whereas geometric and faster designs are not repaired.

Proof. The successive differences of e^{j^p} tend to infinity for every $p > 0$, so only finitely many repetitions can occur. Also

$$\log n_j = j^p + O(e^{-j^p}).$$

If $p < 1$, the logarithmic set has $\asymp R^{1/p}$ points in $[0, R]$. A nonzero entire function of finite exponential type has only $O(R)$ zeros in such a disk, by Jensen's formula centered at a point where it is nonzero. Hence no nonzero compactly supported L^2 function can have Fourier zeros at all these points, and $R(\{\log n_j\}) = \infty$.

If $p \geq 1$, the logarithmic sequence is uniformly separated after discarding finitely many terms. Any uniformly separated real sequence has finite effective density. Indeed, if the separation is $s > 0$, an interval I contains at most $1 + |I|/s$ points. For a fixed $a > 2/s$, intervals with $\#(\Lambda \cap I) \geq a|I|$ have uniformly bounded length. Apart from finitely many intervals near zero, assign such intervals to distinct points they contain; separation implies that the sum of $(1 + \operatorname{dist}(0, I)^2)^{-1}$ is finite. They therefore cannot form the long family in the definition of D_{eff}. Finite modifications do not change this conclusion. Equation (9.5.2) and Theorem 9.5.1 prove failure. Finally, the unrestricted series is $\sum_j [1 + (\log n_j)^2]^{-1} \asymp \sum_j j^{-2p}$, which diverges exactly when $p \leq 1/2$. □

9.6 Unknown stable index: every infinite design, no finite design

The next result allows the stability index itself to vary continuously. It concerns zero-location symmetric stable kernels, not arbitrary location or skewness mixing. A logarithmic survival transform converts the nonlinear stable mixture into a random-coefficient projection problem, and a nonvanishing gamma factor can then be deconvolved. The projection uniqueness argument is classical in spirit [122]; the new assertions concern its exact realization inside the stable-mixture model and the matching finite-design counterexamples after inverse-gamma deconvolution.

Fix $0 < \alpha_0 < \alpha_1 \le 2$. Draw (α, B) with $\alpha \in [\alpha_0, \alpha_1]$, $B > 0$, and conditionally let X_j be independent with characteristic function

$$\mathbb{E}(e^{itX_j} \mid \alpha, B) = e^{-B|t|^\alpha}.$$

Again write $S_n = \sum_{j=1}^n X_j$ and observe only its separate population laws. "Zero location" does not assert existence of a finite mean for the stable observations.

Theorem 9.6.1 (Sharp finite-versus-infinite threshold for unknown index)**.** *Consider the class of all mixing laws just described for which*

$$\mathbb{E}(B^\eta + B^{-\eta}) < \infty \quad \text{for some } \eta > 0. \tag{9.6.1}$$

The selected population laws $\{\mathcal{L}(S_n) : n \in \mathcal{N}\}$ identify every joint law of (α, B) in this class if and only if $\mathcal{N}$ is infinite.

For every finite terminal set and every integer $L \ge 0$, failure has two distinct absolutely continuous witnesses whose index support lies in any prescribed nonempty open subinterval of (α_0, α_1). The witnesses have the same index marginal, the same conditional moments of $\log B$ through degree L, and satisfy (9.6.1) for every $\eta > 0$.

The positive theorem includes all compact positive scale supports. The finite-design witnesses below have unbounded scale support; no finite-design nonidentifiability claim under a fixed compact scale restriction is made.

9.6.1 A nonlinear mixture becomes a family of linear projections

Let E be an independent exponential random variable of mean one, and put

$$R = 1/\alpha, \qquad Q = -R \log B, \qquad C = Q + R \log E. \tag{9.6.2}$$

Then for every real x,

$$\mathbb{P}(C - R \log n > x) = \mathbb{E}e^{-nBe^{\alpha x}} = \mathbb{E}e^{ie^x S_n}. \tag{9.6.3}$$

Thus the characteristic function of each observed stable sum supplies the entire survival function of one linear projection of (C, R). No realization of E is observed or required; it is a mathematical device for describing this deterministic transform of the population data.

Identification from every infinite design. Condition (9.6.1) is equivalent, up to a constant, to an exponential moment of $|\log B|$. Since R is bounded above and below and $\mathbb{E}e^{c|\log E|} < \infty$ for $0 < c < 1$, there is $\epsilon > 0$ with

$$\mathbb{E}e^{\epsilon(|C|+|R|)} < \infty.$$

For each integer $j \ge 0$ the projection moment is the polynomial

$$\mathbb{E}(C - R\ell)^j = \sum_{q=0}^{j} \binom{j}{q} (-\ell)^q \mathbb{E}(C^{j-q} R^q). \tag{9.6.4}$$

The observations give this polynomial at every distinct value $\ell = \log n$, $n \in \mathcal{N}$. For an infinite terminal set, these values determine every coefficient. Hence two candidates with equal observations have the same mixed moments of (C, R). The exponential moment makes their joint moment generating functions analytic near zero, so moment equality identifies the joint law. This polynomial projection argument is the familiar moment-determinate form of sparse Cramér–Wold uniqueness; it is not a new general projection theorem.

It remains to undo the noise in (9.6.2). For every real t and bounded measurable φ,

$$\mathbb{E}\{\varphi(R)e^{itC}\} = \mathbb{E}\{\varphi(R)\Gamma(1+itR)e^{itQ}\}. \tag{9.6.5}$$

The gamma function has no zeros, and on the compact interval of possible R its reciprocal at $1 + itR$ is bounded for fixed t. Replacing φ by $\varphi/\Gamma(1+itR)$ recovers $\mathbb{E}\{\varphi(R)e^{itQ}\}$. Fourier uniqueness therefore recovers the joint law of (Q, R), and the invertible map $\alpha = 1/R$, $B = e^{-Q/R}$ recovers the desired mixing law. □

9.6.2 Finite projection annihilators survive gamma deconvolution

Lemma 9.6.2 (An inverse-gamma realization)**.** *Let $J \Subset (0,\infty)$ be an open interval, let $q \in C_c^\infty(J)$ be real and nonzero, and let $\ell_1, \dots, \ell_M$ be finitely many real numbers. For any integer $K \geq 1$ define*

$$\widehat{g}_Q(t,r) = \frac{(-it)^K e^{-t^2}}{\Gamma(1+itr)} \prod_{j=1}^{M} (\partial_r - it\ell_j) q(r), \qquad g_Q(v,r) = \frac{1}{2\pi} \int e^{-itv} \widehat{g}_Q(t,r)\, dt. \tag{9.6.6}$$

Then g_Q is a nonzero real smooth integrable function, compactly supported in $r \in J$. For every fixed $a, b \geq 0$ and $H > 0$,

$$|\partial_v^a \partial_r^b g_Q(v,r)| \leq C_{a,b,H} e^{-H|v|}. \tag{9.6.7}$$

For every r, its v moments of degrees below K vanish. After adding independent noise $r \log E$ to v, its signed joint density becomes

$$g_C(c,r) = \partial_c^K \prod_{j=1}^{M} (\partial_r + \ell_j \partial_c)\{\gamma(c) q(r)\}, \qquad \widehat{\gamma}(t) = e^{-t^2}. \tag{9.6.8}$$

Every signed projection $c - r\ell_j$ of g_C is zero.

Proof. The reciprocal gamma function is entire. For a fixed real y, vertical Stirling asymptotics, uniformly for r in a compact positive interval, give

$$|\Gamma(1 + i(x+iy)r)^{-1}| \leq C_y (1+|x|)^{C_y} e^{\pi r|x|/2}.$$

The same bound with extra polynomial and logarithmic factors holds after any fixed number of r derivatives. These estimates follow from the uniform gamma asymptotic in [6, Equation 5.11.9] and Cauchy's formula; bounded x is harmless because $1/\Gamma$ is entire. The Gaussian factor has modulus $e^{-x^2+y^2}$, so all resulting horizontal-line integrals converge absolutely and uniformly in r, together with their derivatives. Shifting the t contour to $\operatorname{Im} t = -H$ for $v > 0$ and to $\operatorname{Im} t = H$ for $v < 0$ proves (9.6.7). The joining vertical segments vanish by Gaussian decay. Compact support in r follows directly from the formula, as does smoothness across its support boundary.

Conjugation symmetry of (9.6.6) proves reality. The function is nonzero: for fixed real $t \neq 0$ the nonzero constant-coefficient differential operator $\prod_j (\partial_r - it\ell_j)$ cannot annihilate a nonzero compactly supported smooth q, since its solutions are real analytic. The reciprocal gamma

factor has no zeros for real t and $r > 0$. The factor t^K makes all the stated conditional moments vanish.

Conditionally on r, the Fourier transform of $r \log E$ is $\Gamma(1 + itr)$. Multiplication cancels the denominator in (9.6.6), yielding exactly (9.6.8); all convolutions are justified by the exponential decay. Finally the kernel $e^{it(c-r\ell_j)}$ is annihilated by $\partial_r + \ell_j \partial_c$. Integration by parts in (9.6.8) therefore makes each of its projection characteristic functions zero, proving the last assertion. □

Finite-design failure in Theorem 9.6.1. For a finite terminal set take $\ell_j = \log n_j$. Choose $J \Subset (1/\alpha_1, 1/\alpha_0)$ so that its reciprocal lies in the prescribed index subinterval, and apply Lemma 9.6.2 with $K > L$. The signed density g_Q has total mass zero, so its nonzero positive and negative parts have the same positive mass. Normalize them and push forward by

$$(v, r) \longmapsto (\alpha, B) = (1/r, e^{-v/r}).$$

This is a smooth invertible map with nonzero Jacobian. It gives distinct absolutely continuous mixing probabilities with the required index support. Equation (9.6.7), with arbitrary H, implies $\mathbb{E}(B^\eta + B^{-\eta}) < \infty$ for every $\eta > 0$. The conditional zero moments in v show that the index marginals and the conditional moments of $\log B = -v/r$ through degree L agree, interpreted as equal finite signed moment measures in r.

After the independent auxiliary noise is added, the difference of the laws of (C, R) is the normalized signed density g_C in (9.6.8). Every selected projection $C - R \log n_j$ has zero signed law. The survival identity (9.6.3) therefore gives equality of the characteristic functions of S_{n_j} at every positive argument. Symmetry gives the negative arguments, and normalization gives the value at zero. Thus all selected stable sum laws agree, completing the proof. □

Classical inputs, new scope, and remaining gaps. The completeness-radius theorem, sparse moment-determinate projection uniqueness, and the general differential-annihilator principle are classical or established methods; see [121, 122, 8]. We do not count them as newly proved conjectures. The two new targets here are the exact compact-scale skew index-one classification, and the finite-versus-infinite classification with a continuously mixed stable index. They refine different models: the former allows arbitrary location dependence but fixes the skewness and index, whereas the latter fixes location zero and symmetry but mixes the index. Identification with index, skewness, and location all unknown is not settled. Neither theorem claims quantitative recovery from noisy finite samples. The finite counterexamples in the unknown-index model do not have compact scale support, and that stronger support-constrained question remains open.

9.7 Constructive threshold inversion and the density of transition zeros

Two issues remain after sharp stability of an inverse map: how to compute the inverse without a nonlinear search, and how many usable zeros survive near a transition. We resolve these separately. An adaptive scalar minimization converts nonlinear threshold observations to ordinary moments, after which a positive generalized-eigenvalue construction recovers the atoms. A second theorem computes an explicit zero-density constant at a double-root transition. The latter is different from a simple rightmost frozen edge: it observes a shrinking antipodal phase window at a perturbed double zero, and its exponent is $(m + 1)/2$, rather than the simple-edge exponent.

The eigenvalue construction is classical Gaussian quadrature; see, for example, Blekherman et al. [123]. The claim here is the noisy adaptive reduction from nonlinear minima and its

combination with sharp parameter bounds. The zero-density proof uses an area formula, within the general non-Gaussian Rice framework [124]; all local coordinates and constants needed here are computed explicitly below.

Fix $0 < a < b < 1$, and write

$$S_\mu(x) = \int_a^b x^r \, d\mu(r), \qquad T_\mu(C) = \min_{x>0} \frac{1 + Cx}{S_\mu(x)}, \qquad \mu \in \mathcal{P}[a, b]. \tag{9.7.1}$$

9.7.1 A constructive inverse from adaptive threshold queries

Theorem 9.7.1 (Adaptive reduction to a positive matrix problem)**.** *Fix $m \geq 2$. Suppose $\mu = \sum_{j=1}^m w_j \delta_{r_j}$, the increasing nodes have separation at least Δ, and $w_j \geq w > 0$. An oracle returns, at a requested C in the fixed compact interval $J \subset (0, \infty)$ specified below, a real number $Y(C)$ satisfying $|Y(C) - T_\mu(C)| \leq \varepsilon$. There are $c, K > 0$, depending on a, b, m, such that if*

$$0 < \varepsilon \leq c w \Delta^{2m-1}, \tag{9.7.2}$$

the following procedure uses at most $Km \log(3/\varepsilon)$ oracle calls and returns positive weights $\widehat{w}_j$ summing to one and ordered nodes $\widehat{r}_j \in [a, b]$ with

$$\max_j |\widehat{r}_j - r_j| \leq \frac{K\varepsilon}{w\Delta^{2m-2}}, \qquad \max_j |\widehat{w}_j - w_j| \leq \frac{K\varepsilon}{\Delta^{2m-1}}. \tag{9.7.3}$$

The procedure consists of scalar interval contractions, formation of two $m \times m$ Hankel matrices, and a symmetric generalized eigenvalue problem. It requires no nonconvex search over atomic measures and no semidefinite feasibility assumption.

This is an exact-arithmetic algorithmic statement with a quantitative oracle-error guarantee. It does not claim that unguarded computations in a fixed machine precision remain accurate when atoms collide. The separation and mass assumptions provide the explicit positive matrix margins used in the proof.

Lemma 9.7.2 (Noisy threshold-to-profile conversion)**.** *For every $x \in [1, e]$, $S_\mu(x)$ can be estimated to error $K\varepsilon$ with $K \log(3/\varepsilon)$ threshold queries in a fixed compact J. The constants are uniform over all probabilities $\mu \in \mathcal{P}[a, b]$ and $x \in [1, e]$.*

Proof. Strict concavity of S gives a unique contact x_C at each setting. With $H = S - xS'$, the contact equation is

$$C = \frac{S'(x_C)}{H(x_C)}, \qquad \frac{dC}{dx} = \frac{S''(x)S(x)}{H(x)^2} < 0, \qquad \frac{a}{1-a} \leq Cx_C \leq \frac{b}{1-b}.$$

Choose, for example,

$$J = \left[\frac{a}{2e(1-a)}, \frac{2b}{1-b} \right].$$

Every contact setting $C_x = S'(x)/(S(x) - xS'(x))$ for $x \in [1, e]$ lies in the interior of J. On the common compact range of contacts for $C \in J$, the functions $S, H, -S''$ have uniform positive upper and lower bounds. Thus $-K \leq dx_C/dC \leq -c < 0$.

The tangent majorization identity gives

$$S(x) = \min_{C \in J} g_x(C), \qquad g_x(C) = \frac{1 + Cx}{T(C)}, \qquad g_x'(C) = \frac{x - x_C}{T(C)(1 + Cx_C)}. \tag{9.7.4}$$

Consequently g_x decreases up to C_x and increases afterwards, and

$$c|C - C_x| \leq |g_x'(C)| \leq K|C - C_x|. \tag{9.7.5}$$

The observed value $(1+Cx)/Y(C)$ has error at most $\eta = K_0\varepsilon$, since all true thresholds have uniform positive lower bounds on J.

Maintain an interval $[L,U]$ containing C_x, of width q, and query its two trisection points C_1, C_2. If the observed difference $\widetilde{g}(C_2) - \widetilde{g}(C_1)$ exceeds 2η, retain $[L, C_2]$; if it is less than -2η, retain $[C_1, U]$. Unimodality makes both decisions valid. In the remaining case the true difference has modulus at most 4η. If $C_x \le C_1$, integration of (9.7.5) gives $g(C_2) - g(C_1) \ge cq^2/18$; the analogous bound holds when $C_x \ge C_2$. Thus, while $q^2 > 144\eta/c$, the ambiguous case certifies $C_x \in (C_1, C_2)$, and we retain that middle interval. Each step contracts by at least a factor $2/3$. Stop when $q^2 \le 144\eta/c$. One final midpoint query, together with the upper bound in (9.7.5), estimates $g(C_x) = S(x)$ to $O(\eta)$. The number of queries is $O(\log(3/\varepsilon))$. □

Construction and proof of Theorem 9.7.1. Set $\zeta = 1/(2m)$ and $t_j = e^{\zeta r_j}$. For $0 \le k \le 2m-1$ the numbers

$$c_k = S_\mu(e^{k\zeta}) = \sum_j w_j t_j^k$$

are ordinary power moments. Set $\widehat{c}_0 = 1$ exactly, and apply Lemma 9.7.2 at the other $2m-1$ points to get $|\widehat{c}_k - c_k| \le K\varepsilon$. Form

$$\widehat{H}_0 = (\widehat{c}_{i+j})_{0\le i,j<m}, \qquad \widehat{H}_1 = (\widehat{c}_{i+j+1})_{0\le i,j<m}. \tag{9.7.6}$$

Let $L = e^{\zeta a}/2$ and $U = 2e^{\zeta b}$. The true matrices factor as $H_0 = V\,\mathrm{diag}(w_j)V^\top$ and $H_1 = V\,\mathrm{diag}(w_j t_j)V^\top$, where $V_{ij} = t_j^i$. Coefficients of the Lagrange polynomials at the t_j bound $\|V^{-1}\| \le K\Delta^{-(m-1)}$. Therefore

$$H_0 \succeq cw\Delta^{2m-2}I, \quad H_1 - LH_0 \succeq cw\Delta^{2m-2}I, \quad UH_0 - H_1 \succeq cw\Delta^{2m-2}I. \tag{9.7.7}$$

The enlarged interval supplies fixed positive endpoint margins. The perturbation of each matrix has norm at most $K\varepsilon$. Under (9.7.2), all three inequalities remain strict for the estimated matrices. These are directly checkable matrix certificates for the construction.

On the polynomial space of degree at most $m-1$, put $\langle p, q\rangle = \widehat{L}(pq)$, where $\widehat{L}(t^k) = \widehat{c}_k$, and define the self-adjoint operator M by

$$\langle Mp, q\rangle = \widehat{L}(tpq).$$

Its coefficient matrix is $\widehat{H}_0^{-1}\widehat{H}_1$; Cholesky reduction makes it a symmetric eigenvalue problem. The matrix certificates put every eigenvalue in (L,U). For a polynomial of degree at most $m-2$, one has $Mp = tp$ exactly. Hence $M^k 1 = t^k$ for $0 \le k < m$, and 1 is a cyclic vector. Self-adjointness and cyclicity imply that the m eigenvalues $\widehat{t}_j$ are distinct and the spectral weights $\widehat{w}_j = |\langle 1, v_j\rangle|^2$ are strictly positive, where v_j are orthonormal eigenvectors in this inner product. Their sum is $\langle 1, 1\rangle = 1$.

The spectral measure matches all estimated moments through degree $2m-1$. For degrees at most $2m-2$, split $k = i+j$ with $0 \le i, j < m$ and use $\langle M^i 1, M^j 1\rangle = \widehat{c}_{i+j}$. At degree $2m-1$, use

$$\langle MM^{m-1}1, M^{m-1}1\rangle = \widehat{L}(t^{2m-1}).$$

This proves the claim without appealing to a separate truncated moment existence theorem.

We include the stability step to make the algorithmic claim self-contained. For two at most m-atomic probabilities whose moments through degree $2m-1$ differ by $O(\varepsilon)$, any polynomial of that degree with bounded coefficients has integral discrepancy $O(\varepsilon)$. At a true node t_j apply this to

$$Q_j(t) = \prod_{k\ne j}(t - t_k)\prod_{\ell=1}^{m}(t - \widehat{t}_\ell).$$

If no estimated node lies within a quarter of the minimum true separation, its integral has modulus at least $cw\Delta^{2m-1}$, contradicting (9.7.2). Thus the nodes match, and division by the other factors gives $|\widehat{t}_j - t_j| \le K\varepsilon/(w_j\Delta^{2m-2})$. For the weights test

$$P_j(t) = \frac{\prod_{k\neq j}(t-t_k)(t-\widehat{t}_k)}{\prod_{k\neq j}(t_j-t_k)(t_j-\widehat{t}_k)}.$$

Its coefficients are $O(\Delta^{-(2m-2)})$, and near t_j its derivative is $O(\Delta^{-1})$. The test first yields $\widehat{w}_j \asymp w_j$ and then $|\widehat{w}_j - w_j| \le K\varepsilon/\Delta^{2m-1}$. Finally take $\log \widehat{t}_j/\zeta$ and project each result onto $[a,b]$. The logarithm is uniformly Lipschitz on $[L,U]$, and projection cannot increase the location error. The estimated nodes remain distinct in the stated regime. This proves (9.7.3) and the query bound. □

The separation exponents in (9.7.3) are sharp for this observation model by the positive moment-preserving examples in the preceding paper [115]. The present advance is constructive attainment through an adaptive threshold oracle, not a second claim to the already proved exponents. For rational moment estimates, the matrix certificates and the characteristic polynomial $\det(\widehat{H}_1 - t\widehat{H}_0)$ also permit interval-certified real-root calculations. No theorem about optimal bit complexity of the complete zero-observation experiment is asserted.

9.7.2 An explicit density law at the double-root transition

Fix $m \ge 1$ and distinct $r_1,\dots,r_m \in (a,b)$, positive weights summing to one, $C>0$, and $\ell>0$. Consider

$$f_{A,\phi}(z) = -1 + A\sum_{j=1}^{m} w_j e^{\ell r_j z + i\phi_j} + Ce^{\ell z + i\phi_*}. \tag{9.7.8}$$

Write $T = T_\mu(C)$ and let x_* be its contact. Define

$$\begin{aligned} &s_* = S_\mu(x_*), \quad a_j = T w_j x_*^{r_j}, \quad y = Cx_*, \\ &a_0 = 1 + y = \sum_j a_j, \quad B = \sum_j a_j r_j(1-r_j), \quad D = \frac{y}{1+y}. \end{aligned} \tag{9.7.9}$$

Both B and D are positive, and the contact equation gives $\sum_j a_j r_j = y$. Let $|\mathbb{S}^{m-1}|$ be the surface area of the unit sphere in $\mathbb{R}^m$, with $|\mathbb{S}^0| = 2$.

Fix $\alpha>0$ and $R>0$. At amplitude $A_h = T + \alpha h$, call a zero $z=\sigma+it$ marked when

$$\operatorname{dist}(\ell t + \phi_*, \pi + 2\pi\mathbb{Z}) < R\sqrt{h}. \tag{9.7.10}$$

Let $N_h(H;\phi)$ count marked zeros in $0<t<H$, with multiplicity and without restricting their real parts.

Theorem 9.7.3 (Double-root caustic intensity)**.** *For independent Haar phases ϕ, there is an intensity ρ_h such that $\mathbb{E}N_h(H;\phi) = H\rho_h$ for every $H>0$. As $h\downarrow 0$,*

$$\rho_h = \mathcal{K}_{\mu,C,\ell}(\alpha,R)h^{(m+1)/2}(1+O(\sqrt{h})), \tag{9.7.11}$$

where the explicitly positive constant is

$$\mathcal{K}_{\mu,C,\ell}(\alpha,R) = \frac{\ell|\mathbb{S}^{m-1}|D}{(2\pi)^{m+1}m\sqrt{Ba_0\prod_{j=1}^m a_j}}\mathcal{I}_m(\alpha,R), \tag{9.7.12}$$

$$\mathcal{I}_m(\alpha,R) = \int_{-R}^{R}(2\alpha s_* + Dv^2)^{(m-2)/2}[2\alpha s_* + (m+1)Dv^2]\,dv. \tag{9.7.13}$$

If $\ell r_1, \dots, \ell r_m, \ell$ are rationally independent, then for every fixed choice of phases and every sufficiently small fixed h,

$$\lim_{H\to\infty} \frac{N_h(H;\phi)}{H} = \rho_h. \tag{9.7.14}$$

The constants in the error term may depend on the fixed spectrum, α, and R. The theorem is not uniform as a weight tends to zero, as $\alpha \downarrow 0$, or as frequencies become resonant.

Proof. We give localization, the zero-count formula, and the constant calculation separately. All zeros of the family with A in a fixed compact interval lie in a uniform vertical strip: the constant term dominates sufficiently far left, and the largest exponential dominates sufficiently far right. For a marked zero put $x = e^{\ell\sigma}$ and write its top phase as $\pi + v$, with $|v| < R\sqrt{h}$. The zero equation gives

$$A_h S(x) \geq |1 + Cxe^{iv}| \geq (1 - Kh)(1 + Cx).$$

The quotient $(1 + Cx)/S(x)$ has a unique strict minimum at x_*, with positive second derivative there. Thus every marked zero satisfies $x - x_* = O(\sqrt{h})$. The total magnitude of the smaller terms exceeds the magnitude of their resultant by only $O(h)$. The identity

$$\sum_j q_j |e^{i\theta_j} - e^{i\psi}|^2 = 2\left(\sum_j q_j - \left|\sum_j q_j e^{i\theta_j}\right|\right), \qquad e^{i\psi} = \frac{\sum_j q_j e^{i\theta_j}}{|\sum_j q_j e^{i\theta_j}|},$$

with $q_j = A_h w_j x^{r_j}$, shows that every smaller phase is $O(\sqrt{h})$ modulo 2π. The resultant is $1 + Cxe^{iv}$ and has argument $O(\sqrt{h})$, so all marked zeros are localized in this single phase chart. Strict positivity of the fixed weights is used here.

Use the coordinate $u = \ell\sigma - \log x_*$ and let

$$F_h(u, \theta, v) = -1 + (1 + \alpha h/T) \sum_j a_j e^{r_j u + i\theta_j} - ye^{u+iv}.$$

With $q = (\sqrt{a_j})_{j=1}^m$, write the weighted phases uniquely as

$$(\sqrt{a_j}\theta_j)_{j=1}^m = Dvq + t + \xi \frac{q}{\sqrt{a_0}}, \qquad t \perp q. \tag{9.7.15}$$

The linear imaginary part of F_h is $\sqrt{a_0}\xi$. The implicit-function theorem therefore eliminates ξ on $\Im F_h = 0$, with $\xi = O(h)$ on the localized region. The measure factor in this elimination is

$$d\theta_1 \cdots d\theta_m \, \delta_0(\Im F_h) = \frac{1 + O(\sqrt{h})}{\sqrt{a_0 \prod_j a_j}} \, dt. \tag{9.7.16}$$

Here and below δ_0 is shorthand for integration over the regular level set with the coarea Jacobian, not a probabilistic independence assertion.

Set $u = \sqrt{h}\, U$, $t = \sqrt{h}\, Z$, and $v = \sqrt{h}\, V$. Uniformly with two derivatives on the bounded localized set, after the imaginary part has been eliminated,

$$h^{-1} \Re F_h = \alpha s_* - \tfrac{1}{2} BU^2 - \tfrac{1}{2}|Z|^2 + \tfrac{1}{2} DV^2 + O(\sqrt{h}). \tag{9.7.17}$$

Let $b \in q^\perp$ have coordinates $b_j = \sqrt{a_j}(r_j - D)$. Then

$$h^{-1/2} \partial_u F_h = -BU + i(b \cdot Z - DV) + O(\sqrt{h}), \qquad |b|^2 = D - B. \tag{9.7.18}$$

Here $\partial_u F_h$ is the original partial derivative with the phases held fixed, evaluated on the imaginary-zero surface; it is not the total derivative after eliminating ξ. The last identity follows from $\sum a_j r_j = y$ and $\sum a_j r_j^2 = y - B$. In particular $D \geq B$.

Because $\alpha > 0$, the limiting real-zero surface

$$BU^2 + |Z|^2 = 2\alpha s_* + DV^2$$

has a uniformly nonzero gradient in (U, Z) for $|V| \le R$. The actual surfaces are uniformly regular perturbations of these spheres after the linear change $Q = (\sqrt{B}U, Z)$. Radial parameterization and the implicit-function theorem give uniform C^1 error $O(\sqrt{h})$ in their surface measures and all the bounded integrands below. This verifies the regularity needed for the coarea calculation, including the case $m = 1$ of a two-point sphere.

We justify the zero-count identity being used. For a holomorphic function, the real Jacobian of its value map is the squared modulus of its complex derivative. The area formula, followed by averaging its level-counting function over a small disk about zero, consequently counts its zeros with multiplicity. On a compact zero window the count is uniformly bounded for this compact phase family, also after subtracting a sufficiently small complex constant: locally uniform limits cannot be identically zero, so Rouché's theorem bounds the number of zeros. This uniform bound permits averaging over phases before taking the disk radius to zero. On the localized region just described the combined map $(u, \theta, v) \mapsto (\Re F_h, \Im F_h)$ is a submersion. Its coarea formula thus gives the limit. The exceptional phases for which a localized zero is multiple have Haar measure zero by Sard's theorem applied to projection of this regular zero manifold onto phase space. These observations also justify indicators by continuous approximation when their boundary has zero coarea measure.

Phase translation by $(\ell r_1 t, \dots, \ell r_m t, \ell t)$ preserves Haar measure. Thus the expected marked count per unit height is

$$\rho_h = \frac{\ell}{(2\pi)^{m+1}} \int_{|v|<R\sqrt{h}} \int |\partial_u F_h|^2 \, \delta_0(\Re F_h)\delta_0(\Im F_h) \, du \, d\theta \, dv. \tag{9.7.19}$$

The factor ℓ is ℓ^2 from the complex derivative and ℓ^{-1} from $d\sigma = du/\ell$. Stationarity proves the exact identity $\mathbb{E}N_h(H) = H\rho_h$.

Substitute (9.7.16)–(9.7.18) into (9.7.19). The power of h is $h^{(m+1)/2}$: the variables (u, t, v) contribute $h^{(m+1)/2}$, the real level constraint contributes h^{-1}, and the derivative squared contributes h. The limiting integral at a fixed V is

$$\int_{\mathbb{R}\times\mathbb{R}^{m-1}} [B^2U^2 + (b \cdot Z - DV)^2]\delta_0\left(\alpha s_* - \frac{BU^2 + |Z|^2 - DV^2}{2}\right) dU \, dZ.$$

Put $Q = (\sqrt{B}U, Z)$ and $R_V^2 = 2\alpha s_* + DV^2$. Radial integration and spherical symmetry evaluate this integral as

$$\frac{|\mathbb{S}^{m-1}|}{\sqrt{B}} R_V^{m-2} \left[\frac{(B + |b|^2)R_V^2}{m} + D^2V^2\right] = \frac{|\mathbb{S}^{m-1}|D}{m\sqrt{B}} R_V^{m-2}[2\alpha s_* + (m+1)DV^2].$$

This gives (9.7.12)–(9.7.13). The uniform surface perturbation already proved gives the stated relative $O(\sqrt{h})$ remainder. Positivity is immediate.

Finally assume rational independence. The phase flow is uniquely ergodic on the full torus, by finite-dimensional Fourier approximation. The marked count in a unit-height window is a bounded function of the initial phases. Its discontinuities can occur only at a zero on the horizontal boundary or at the two angular boundaries in (9.7.10). Horizontal boundaries have zero mean mass by stationarity and local finiteness; the coarea formula above gives zero mean mass to either angular boundary. Thus the function has Haar-null discontinuity set. Continuous upper and lower approximations give its time average on every orbit. Integrating the sliding unit-window count differs from $N_h(H)$ by a uniformly bounded endpoint contribution. This proves (9.7.14). □

Corollary 9.7.4 (A deletion and insertion scale for marked zeros)**.** *Under the rational-independence hypothesis, consider the reduced record consisting only of the marked ordinates. At $A_h = T+\alpha h$ it contains $\rho_h H+o(H)$ entries. If an adversary deletes at most ηH genuine entries and inserts at most ηH arbitrary entries, the test "more than ηH reported marks" distinguishes this record from an empty genuine marked record for all sufficiently large H whenever $2\eta < \rho_h$. Conversely, when $2\eta > \rho_h$, these two reduced records can be made identical within those budgets for all sufficiently large H. In particular, the critical density of corruptions is of order $h^{(m+1)/2}$, with leading constant $\mathcal{K}_{\mu,C,\ell}(\alpha, R)/2$ for this reduced-record comparison.*

Proof. The nonempty genuine record retains at least $\rho_h H - \eta H + o(H)$ marks, whereas the empty genuine record can acquire at most ηH. This proves the first assertion. For the converse, keep half the genuine marks in the nonempty record and insert those same marks into the empty record. Each budget is $\rho_h H/2 + o(H)$. □

Empty marked records do occur below the transition: the triangle inequality used in the proof gives $A \geq (1 - K_R h)T$ for any marked zero, so amplitudes $A < T - Kh$ have none for a sufficiently large fixed K. The corollary deliberately concerns the reduced record after discarding the other zeros. It is not an impossibility theorem for an observer allowed to exploit the entire zero divisor. Nor does (9.7.14) provide a uniform finite height for its asymptotic to become accurate; quantitative recurrence information is still needed for that stronger request.

The two results close distinct internal targets: constructive inversion of noisy nonlinear thresholds, and an explicit local zero-density law with its marked-record corruption scale. They do not resolve finite-data blind recovery of all phases, optimal zero-observation height, or a general algorithm robust to missing and spurious zeros without a density budget.

9.8 The entire intermediate-rate gamma region and its quartic gap

The preceding paper determined the largest endpoint-rate gamma factor on the first excess-power McKay line. It left open what happens when the rate is moved inside the two original gamma rates. We prove that the unrestricted factor boundary is continuous and strictly monotone, that it lies strictly above the ID-remainder boundary at every intermediate rate, and that its boundary remainder has finitely many interior zeros. A local singularity analysis gives the exact quartic separation when the central rate is approached. The same analytic construction resolves the singular limit at the lowest Bessel order. These are new internal transfer questions, not additional named conjectures from the external source [3].

9.8.1 An analytic inverse valid at every intermediate rate

Fix $\kappa > 1/2$, $b > 1$, and write $M = 2\kappa - 1$. We use the same McKay line as in the preceding paper:

$$L(s) = \frac{b+s}{b}\left(\frac{b^2-1}{(b+s)^2-1}\right)^{\kappa}. \tag{9.8.1}$$

For an extracted gamma factor of shape h and rate ρ, put $c = b - \rho$. The intermediate-rate interval is $0 < c < 1$. The two capacities, with an arbitrary nonnegative remainder and an ID remainder respectively, are denoted $H_{\mathrm{P}}(c)$ and $H_{\mathrm{ID}}(c)$. A positive normalizing constant and a translation of the Laplace variable reduce the prospective remainder transform to

$$F_{c,h}(w) = w^h \frac{w+c}{((w+c)^2-1)^{\kappa}}. \tag{9.8.2}$$

For $0 \le h < M$, write $\eta = M - h > 0$ and define coefficients by

$$(1+cz)\{1+2cz+(c^2-1)z^2\}^{-\kappa} = \sum_{n=0}^{\infty} a_n(c) z^n. \tag{9.8.3}$$

Its inverse on the positive axis is

$$g_{c,h}(x) = \sum_{n=0}^{\infty} \frac{a_n(c) x^{\eta+n-1}}{\Gamma(\eta+n)}, \qquad x > 0. \tag{9.8.4}$$

This series and every fixed derivative converge locally uniformly. Indeed Cauchy's estimate gives a geometric bound on a_n, and the gamma denominator dominates it on every compact x interval. The actual probability remainder, if it exists, has density $e^{-\rho x} g_{c,h}(x)/F_{c,h}(\rho)$.

Lemma 9.8.1 (Uniform positivity at both ends)**.** *For $0 \le c < 1$ and $0 \le h < M$,*

$$g_{c,h}(x) = \frac{x^{M-h-1}}{\Gamma(M-h)}(1+O(x)), \qquad x \downarrow 0, \tag{9.8.5}$$

$$g_{c,h}(x) = \frac{2^{-\kappa}(1-c)^h}{\Gamma(\kappa)} e^{(1-c)x} x^{\kappa-1}(1+O(x^{-1})), \qquad x \to \infty. \tag{9.8.6}$$

The estimates are uniform when (κ, c, h) ranges over a compact subset of the displayed strict parameter domain. The large-x estimate also holds uniformly as $h \uparrow M$. On compact subintervals of $(0,\infty)$ the inverse extends continuously to $h = M$, after removing the atom at zero that appears at that value. Consequently positivity of $g_{c,h}$ on all of $(0,\infty)$ is an open condition in the parameters whenever it is strict everywhere and $h < M$.

Proof. The first assertion follows from (9.8.4). To justify the second and the Laplace inversion, let f_c be the positive inverse of $(w+c)/((w+c)^2-1)^{\kappa}$. The Bessel integral gives

$$f_c(x) = \frac{\sqrt{\pi}}{2^{\kappa-1/2}\Gamma(\kappa)} e^{-cx} x^{\kappa-1/2} I_{\kappa-3/2}(x).$$

For an integer $N > h$, let $\gamma = N - h > 0$. Then

$$g_{c,h}(x) = \frac{d^N}{dx^N}\left\{ \frac{1}{\Gamma(\gamma)} \int_0^x (x-y)^{\gamma-1} f_c(y)\, dy \right\}.$$

Near zero the expression in braces has order $x^{M+N-h-1}$; its derivatives through order $N-1$ vanish at zero when $h < M$. Taking its Laplace transform therefore gives (9.8.2). The series inversion agrees first for large real w and then by uniqueness everywhere the tilted transform is defined.

The standard differentiated Bessel expansion gives $f_c(x) = 2^{-\kappa} e^{(1-c)x} x^{\kappa-1}/\Gamma(\kappa)$ times a full inverse-power expansion. Write the fractional integral with $v = x - y$, split it at $v = x/2$, and expand $(x-v)^{\kappa-1}$ on the first part. Integration against $e^{-(1-c)v} v^{\gamma-1}/\Gamma(\gamma)$ supplies the leading factor $(1-c)^{-\gamma}$. The other part is exponentially smaller, together with every fixed derivative. Differentiating N times supplies $(1-c)^N$. This proves (9.8.6), with uniform errors on the stated compact sets, also if h tends to M. At that limiting value the small-x initial term represents an atom at zero; it does not affect this calculation on $x > 0$. Positivity is uniform near the two ends and persists on the intervening compact interval by continuity, proving the last assertion. □

9.8.2 Strict gain at every intermediate rate

The ID capacity was computed previously by minimizing the Lévy kernel. We recall its elementary expression:

$$H_{\mathrm{ID}}(c) = \min_{x \geq 0} e^{-cx}(2\kappa \cosh x - 1), \qquad 0 \leq c \leq 1. \tag{9.8.7}$$

For $0 < c \leq 1$ the minimum is attained at a unique positive point and is strictly smaller than M. These assertions follow by putting $z = e^{-x}$ in the derivative equation:

$$\kappa(1+c)z^2 - cz - \kappa(1-c) = 0.$$

At $c = 1$ its value is $\kappa - 1/(4\kappa)$. At $c = 0$ it equals M. These formulas are background, not newly counted results.

Theorem 9.8.2 (The unrestricted factor region)**.** *For every $\kappa > 1/2$ and every $0 < c < 1$,*

$$0 < H_{\mathrm{ID}}(c) < H_{\mathrm{P}}(c) < M. \tag{9.8.8}$$

The supremum defining $H_{\mathrm{P}}(c)$ is attained. The function $c \mapsto H_{\mathrm{P}}(c)$ is continuous and strictly decreasing on $[0,1]$, with the endpoint values

$$H_{\mathrm{P}}(0) = M, \qquad H_{\mathrm{P}}(1) = \kappa - \delta_*(\kappa),$$

where the endpoint Kummer root δ_* was determined in the preceding paper [115]. Every boundary remainder for $0 < c < 1$ has a real analytic density on $(0,\infty)$, strictly positive near both endpoints, with a nonempty finite set of interior zeros, each of positive even order. For $H_{\mathrm{ID}}(c) < h \leq H_{\mathrm{P}}(c)$ the unique remainder is a probability law which is not ID.

Proof. The difference between the original and extracted gamma Lévy densities is

$$\frac{e^{-\rho x}}{x}\{e^{-cx}(2\kappa \cosh x - 1) - h\}.$$

At $h = H_{\mathrm{ID}}(c)$ it is nonnegative and vanishes at just one positive point. Its ID law has density strictly positive at every $x > 0$. To see strict positivity directly, split the jumps at $\epsilon > 0$. The small-jump sum has mean tending to zero with ϵ. The event of exactly two larger jumps has a density strictly positive on $(2\epsilon, \infty)$, since the one-jump density is positive except at one point. Convolution with the small-jump law gives a positive continuous lower bound near any prescribed $x > 0$. Lemma 9.8.1 now permits a small increase of h, proving the strict lower inequality.

As $w \to \infty$, the candidate transform has expansion

$$F_{c,h}(w) = w^{-(M-h)}\{1 - Mc/w + O(w^{-2})\}.$$

For $h > M$ it is unbounded; for $h = M$ it approaches its positive limit from below and has positive derivative for all sufficiently large w. Neither behavior is possible for a translated probability Laplace transform. The admissible set is closed: if h_j are admissible and tend to h, their normalized transforms converge pointwise to the normalized expression (9.8.2), which is continuous at the origin. The continuity theorem for Laplace transforms gives a probability limit. Thus a largest shape exists and is strictly less than M.

At this largest shape the density cannot be positive everywhere, by the openness assertion in Lemma 9.8.1. It is analytic and positive near zero and infinity. Its zeros therefore lie in a compact interior interval, are isolated, and are finite in number. Nonnegativity forces each multiplicity to be even.

To prove strict monotonicity, take rates $b-1 \le \rho_1 < \rho_2 < b$. For every $h > 0$,

$$G_{h,\rho_1} \overset{\mathrm{d}}{=} G_{h,\rho_2} + Z,$$

where Z is compound Poisson with Lévy density $h(e^{-\rho_1 x} - e^{-\rho_2 x})/x > 0$. Its one-jump density is strictly positive on $(0,\infty)$. If the maximal factor at ρ_1 is extracted, convolution of its remainder with Z makes the resulting remainder strictly positive everywhere at ρ_2: the old remainder has positive mass arbitrarily close to zero, and the one-jump term suffices. The same endpoint stability argument permits a strict increase in shape at ρ_2. The known endpoint theorem supplies the required positive density near zero also when $\rho_1 = b-1$. At $\rho_2 = b$, the maximum M is strictly larger by the inequality already proved.

Closedness of admissible parameters gives upper semicontinuity. For lower semicontinuity at an interior rate, fix a shape strictly below the maximum. Its remainder is the convolution of a maximal remainder with a positive gamma density, hence is positive everywhere. Lemma 9.8.1 preserves this shape at nearby rates. At $c = 1$, monotonicity together with closedness gives one-sided continuity. At $c = 0$, the squeeze $H_{\mathrm{ID}}(c) \le H_{\mathrm{P}}(c) \le M$ and $H_{\mathrm{ID}}(c) \to M$ gives continuity. Finally uniqueness of the Lévy representation excludes an ID remainder for every shape above (9.8.7). □

This theorem determines the topology and strict ordering of the entire rate region. It does not pretend to give an elementary formula for the intermediate boundary. In particular, multiple interior contacts are not excluded away from the singular regimes analyzed next.

9.8.3 A double-root bifurcation at the central rate

It is useful to remove the singular gamma prefactor. Define

$$\mathcal{B}(\eta, c, x) = \eta + \sum_{n\ge 1} \frac{a_n(c) x^n}{(1+\eta)_{n-1}}. \tag{9.8.9}$$

Then $g_{c,h}(x) = x^{\eta-1}\mathcal{B}(\eta,c,x)/\Gamma(1+\eta)$. The function $\mathcal{B}$ is analytic jointly near $\eta = c = x = 0$. At $c = \eta = 0$ it is strictly positive for $x > 0$, since $a_{2j}(0) = (\kappa)_j/j!$ and $a_{2j+1}(0) = 0$.

Theorem 9.8.3 (The exact quartic cost of imposing ID)**.** *For fixed $\kappa > 1/2$, as $c \downarrow 0$,*

$$M - H_{\mathrm{P}}(c) = \frac{M^2}{4\kappa} c^2 - \frac{M^4(\kappa+1)}{192\kappa^3} c^4 + O(c^6), \tag{9.8.10}$$

$$M - H_{\mathrm{ID}}(c) = \frac{M^2}{4\kappa} c^2 - \frac{M^4}{192\kappa^3} c^4 + O(c^6). \tag{9.8.11}$$

Consequently

$$H_{\mathrm{P}}(c) - H_{\mathrm{ID}}(c) = \frac{M^4}{192\kappa^2} c^4 + O(c^6). \tag{9.8.12}$$

For all sufficiently small positive c, the maximal probability remainder has exactly one zero on $(0,\infty)$; it is a double zero at

$$x_c = \frac{M}{2\kappa} c + O(c^3).$$

Both $M - H_{\mathrm{P}}(c)$ and x_c/c are analytic functions of c^2 near zero after the indicated leading factor is removed.

Proof. The first coefficients in (9.8.3) are

$$a_1 = -Mc, \quad a_2 = \kappa(1 + Mc^2), \quad a_3 = -\kappa(M+2)c + O(c^3), \quad a_4 = \kappa(\kappa+1)/2 + O(c^2).$$

Set $\eta = c^2 A$ and $x = cy$. Parity of $a_n(c)$ implies that $\mathcal{B}(c^2 A, c, cy)/c^2$ is analytic in $t = c^2, A, y$ near $t = 0$ and equals

$$Q(t, A, y) = A - My + \kappa y^2 + tR(A, y) + O(t^2),$$
$$R(A, y) = \kappa(M - A)y^2 - \frac{\kappa(M + 2)}{2} y^3 + \frac{\kappa(\kappa + 1)}{12} y^4.$$

The double-root equations $Q = Q_y = 0$ at $t = 0$ have solution

$$A_0 = \frac{M^2}{4\kappa}, \qquad y_0 = \frac{M}{2\kappa}.$$

Their Jacobian in (A, y) has determinant $2\kappa > 0$. The analytic implicit-function theorem gives unique analytic functions $A(t), y(t)$ near this solution. Moreover

$$A'(0) = -R(A_0, y_0) = -\frac{M^4(\kappa + 1)}{192\kappa^3}.$$

We verify that this local double root is the global factor boundary. For x bounded away from zero, positivity at $c = \eta = 0$, continuity on compact intervals, and the uniform large-x estimate in Lemma 9.8.1 exclude any other zero. For small x the convergent series gives, uniformly for $\eta \asymp c^2$,

$$\mathcal{B} = \eta - Mcx + \kappa x^2 + O(c^2 x^2 + cx^3 + x^4 + \eta x^2).$$

For x/c sufficiently small the positive η term dominates; for x/c sufficiently large and x still small the positive κx^2 term dominates. Thus only a fixed compact y interval remains. There $Q(0, A_0, y) = \kappa(y - y_0)^2$, so zeros are confined to a small neighborhood of y_0, where $Q_{yy} > 0$ and $Q_A > 0$. The double-root curve is consequently nonnegative everywhere, and decreasing A makes it negative at its unique minimum. It is exactly the capacity boundary. This also proves uniqueness and order two of the contact. Expansion of $c^2 A(c^2)$ yields (9.8.10).

For the ID boundary minimize $e^{-cx}(2\kappa \cosh x - 1)$ at $x = cy$. Its expansion is

$$M + c^2(\kappa y^2 - My) + c^4 \left(\frac{\kappa y^4}{12} - \kappa y^3 + \frac{My^2}{2} \right) + O(c^6).$$

The leading minimizer is y_0, and its $O(c^2)$ correction does not affect the fourth-order value. Substitution gives the coefficient $M^4/(192\kappa^3)$ for the minimum, proving (9.8.11). Subtraction gives (9.8.12). □

9.8.4 The singular lowest-order limit

Corollary 9.8.4 (The fourth-order gap at $\kappa = 1/2$)**.** *Fix $0 < c \le 1$ and let $M = 2\kappa - 1 \downarrow 0$. Then*

$$M - H_{\mathrm{P}}(c) = \frac{c^2}{2} M^2 - \frac{c^2}{2} M^3 + \left(\frac{c^2}{2} - \frac{c^4}{16} + \frac{c^6}{24} \right) M^4 + O(M^5),$$
$$M - H_{\mathrm{ID}}(c) = \frac{c^2}{2} M^2 - \frac{c^2}{2} M^3 + \left(\frac{c^2}{2} - \frac{c^4}{24} + \frac{c^6}{24} \right) M^4 + O(M^5).$$

In particular,

$$H_{\mathrm{P}}(c) - H_{\mathrm{ID}}(c) = \frac{c^4}{48}(2\kappa - 1)^4 + O((2\kappa - 1)^5).$$

At the endpoint rate $c = 1$ this is the sharp singular gap left undetermined by the preceding Kummer characterization.

Proof. Use the same analytic series with $x = My$ and $\eta = M^2 A$. At $M = 0$ the leading equation is $A - cy + y^2/2 = 0$, with double root $y = c$, $A = c^2/2$. The positive-axis inverse at $M = h = 0$, away from its atom at zero, is $e^{-cx} I_1(x) > 0$, since the transform is $(w+c)/\sqrt{(w+c)^2 - 1}$. The global exclusion and the analytic implicit-function argument from the preceding proof apply again. For fixed $c < 1$, the large-x expansion remains uniform in this limit. At $c = 1$, put $\delta = \kappa - h$ and define the positive Euler convolution

$$k(x) = \frac{1}{\Gamma(\kappa)\Gamma(\delta)} \int_0^x e^{-2u} u^{\kappa-1} (x-u)^{\delta-1} \, du.$$

Its Laplace transform is $w^{-\delta}(w+2)^{-\kappa}$. Since $\kappa + \delta - 1 = \eta > 0$, one has $k(0) = 0$, and (9.8.2) has inverse $g_{1,h} = k' + k$. Both κ and δ stay in a compact positive interval about $1/2$. Splitting the convolution with a smooth cutoff at $u \asymp x/2$ and expanding on the side $u < x/2$ gives uniformly

$$k(x) = \frac{2^{-\kappa}}{\Gamma(\delta)} x^{\delta-1}(1 + O(x^{-1})), \qquad k'(x) = O(x^{\delta-2}).$$

Indeed the moments of $e^{-2u} u^{\kappa-1}$ are uniformly finite. On the other part set $v = x - u$ before differentiating; the factor $v^{\delta-1}$ is uniformly integrable and the exponential factor makes this part and its derivative exponentially small. Derivatives of the cutoff have the same bound. Thus $g_{1,h}$ has a uniformly positive algebraic tail, without any singular lower hypergeometric parameter. No root can escape to infinity.

Here are the coefficients to make the expansion reproducible. With $\kappa = (1 + M)/2$, the normalized equation has the form

$$A - cy + y^2/2 + M q_1(y) + M^2 q_2(A, y) + O(M^3),$$

where

$$q_1(y) = \frac{1 + c^2}{2} y^2 - \frac{c}{2} y^3, \quad q_2(A, y) = \left(\frac{c^2}{2} - \frac{A}{2} \right) y^2 - \left(\frac{3c}{4} + \frac{c^3}{6} \right) y^3 + \left(\frac{1}{16} + \frac{c^2}{4} \right) y^4.$$

At $y = c$, $A = c^2/2$, one has $q_1 = c^2/2$ and $q_1' = c - c^3/2$. The next minimum coefficient is $q_2 - (q_1')^2/2 = -c^2/2 + c^4/16 - c^6/24$. This gives the first expansion. Expanding the elementary ID minimum in the same variables changes $c^4/16$ to $c^4/24$, with the other coefficients unchanged. Subtraction proves the last assertion. □

What remains unresolved. The strict factor-region theorem and the singular quartic laws are proved here. They do not provide a one-point test at every intermediate rate, nor a formula for arbitrary noninteger excess power. Attempts to transfer gamma deconvolution to universal SD strictness of the squared-Bessel McKay family did not yield a complete sign proof. Termination of the earlier finite moment hierarchy likewise remains unproved. These unresolved targets are recorded rather than counted as closures.

9.9 What the transfers change, and what remains open

The main geometric transfer is from finite-dimensional fiber width to an exact infinite-dimensional obstruction. The preceding paper supplied a lower count on a compact input; the present operator theorem shows that even arbitrary adaptive continuous information cannot recover the noncompact part of a Hilbert operator. Homogeneity adds a different restriction: one large level patch already forces the stronger count. These results do not settle the unrestricted three-dimensional case, the exact homogeneous count in every dimension, finite-query ellipsoid

constants, or an essential-norm theorem for general Banach operators. The strict finite-budget convex-noise gap is nonquantitative; its sharp decay in the query count remains an independent problem.

The statistical transfer changes the discrepancy induced by the adversary. Whole-report replacement is governed by total variation, whereas individual flips are governed by transport. The resulting logarithmic inverse scales can differ by an unbounded factor, as the explicit example in Section 9.3 shows. This is a change in the corruption model, not an improvement over the optimal rate for the same model. The present linear program uses the flip budget. Simultaneous adaptation to an unknown budget, robustness of the ordinary NPMLE, nearly linear computation, and the optimal Bernstein exponent $\log 2$ remain unresolved.

The two stable-mixture classifications close distinct gaps. The skew index-one case allows arbitrary location–scale dependence but fixes the shape. The unknown-index case fixes symmetry and location zero but allows the shape and scale to mix jointly. Its finite-design witnesses have unbounded scale, even though every two-sided scale moment is finite. Compact-scale finite-design failure in that latter model, fully unknown location and skewness, and quantitative noisy-sample inversion are not settled. The survival identity is a useful bridge to random linear projections; it does not turn a classical projection theorem into a newly discovered general principle.

In the spectral branch, rational relations create an observable boundary defect and change the dimension read by an edge intensity. Without a support restriction, even exact zero divisors on an entire half-plane can fail to identify the function, as the explicit factor counterexample demonstrates. The finite controlled experiment and its critical intensity concern a different, double-zero geometry. Neither local intensity law gives a quantitative ergodic discrepancy rate by itself. Full recovery of all magnitudes from first intensity alone, finite-noise blind prime-series inversion, optimal observation height, and growing-window variance or central limit laws remain open here.

The gamma results extend the endpoint calculation across the whole intermediate-rate interval at the structural level. They also identify the exact local cost of insisting on an infinitely divisible remainder. An elementary closed formula or a one-point positivity test at every interior rate is not obtained. Arbitrary excess powers, universal strict self-decomposability for the squared-Bessel family, and termination of the earlier moment hierarchy remain unresolved.

Research status. The thirteen-row accounting describes the stated results and their related refinements. It does not certify thirteen external open-problem closures, or a particular journal or prize level. Independent automated cross-checks within this research process examined every branch, and the final text includes corrections and explicit scope restrictions from those checks. External mathematical review, a broader priority search, and further development are still needed before publication claims.

Part X

Observation Budgets and Hidden Parameters

Noncompact Recovery, Replication Thresholds, and Blind Spectral Identification

Source manuscript: 16 October 2026.

Source abstract. We study thirteen scoped questions about the information required to recover hidden geometric, probabilistic, and spectral structure. Smooth strongly convex measurements with uniformly bounded Hessians retain logarithmic adaptive query complexity in the dimension. For the noncompact embedding $\ell_p \hookrightarrow \ell_q$, $p < q$, the optimal continuous-information errors have orders $n^{-(1/p-1/q)}$ nonadaptively and $2^{-n(1/p-1/q)}$ adaptively. This gives a sharp boundary to the preceding Hilbert compactness theorem. In latent Bernoulli populations, unequal replication yields the optimal $B^{-1/3}$ total-budget rate, while honest uncertainty under an unknown corruption budget has an unavoidable clean-data diameter. For stable mixtures, removing all scale moments changes separate-terminal identification to an exact exponential-completeness criterion; two joint replicates nevertheless identify arbitrary continuously mixed index and scale. Six replicates suffice when location and skewness are also mixed and correlated. One-sided zero sets identify the unknown internally rationally independent support of a finite exponential sum; finitely many selected simple zeros also give a local analytic inverse at each such support. Near-resonance creates an explicit finite-record obstruction to uniform stability. Further results give a sharp inverse for arbitrary positive spectra, a Diophantine criterion for optimal marked-zero observation height, a complete two-rate gamma factor region, and recovery of a boundary remainder from its factor-exchange response. Full proofs, classical inputs, prior unpublished dependencies, and remaining open questions are distinguished. The thirteen questions are not thirteen named historical conjectures.

Scope, principal mechanisms, and claim accounting

This manuscript continues the 223-page *Observation and Recovery in Arithmetic and Probability: Prime Flows, Rare Events, and Infinite Mixtures* and the subsequent papers, most immediately the 41-page *Sharp Boundaries for Adaptive Recovery and Mixture Identification*. We do not re-count their results. In particular, the exact power-of-two query count, Hilbert essential-norm floor, fixed-design binomial corruption curve, prescribed-support zero inverse, and single-rate gamma capacity are starting points rather than new closures here.

The organizing question is what changes when the observation resource is changed. A norm measurement makes subsequent smooth codes strongly convex without changing their information. A support-localizing query makes an unstable sparse moment inverse usable in an infinite sequence space. Joint replicates retain a common latent variable that separate terminal laws lose. At an exponential-sum edge, shared zeros force two initially unknown frequency tori to be the same integral graph. On a gamma factor boundary, directional response becomes a Laplace transform of a positive measure. These are distinct mathematical mechanisms; no assertion of a universal equivalence between the models is intended.

External question versus internal extensions. Krieg–Novak–Ullrich explicitly ask for complexity bounds when measurements are convex, homogeneous, or analytic [60, Section 3]. The smooth uniformly curved construction here answers the order-of-growth part of the convex question and leaves a one-query gap in general dimension. Our noncompact sequence embedding

gives a fully quantified family for the Banach-space adaptive comparison discussed in their Section 4, not a general solution of that comparison. The other questions in the table were formulated within this observation-and-recovery program. Some are sharp transfers of established approximation or number-theoretic principles; they are not advertised as newly invented classical tools.

No.	Target	Result and remaining scope	Proof
1	Smooth strongly convex information	Logarithmic count with fixed two-sided Hessian bounds; exact general count remains open.	10.1.1
2	Quantitative approach to a noise floor	Explicit exponential adaptive and inverse-linear nonadaptive lower gaps; matching rates not claimed.	10.1.3
3	Noncompact Banach recovery	Sharp algebraic and exponential errors for all $\ell_p \hookrightarrow \ell_q$, $p < q$; finite-bit obstruction is a consequence.	10.2.1
4	Total Bernoulli observation budget	Optimal unequal-replication design and a lower bound allowing adaptive allocation.	10.3.1
5	Unknown corruption and honest uncertainty	Clean-law diameter lower bound for confidence sets honest over a specified corruption range.	10.3.2
6	Unknown frequency support from zeros	One-sided zero-set rigidity and a finite local inverse for internally independent positive supports.	10.4.1, 10.4.4
7	Near-resonant observation delay	Exact first-arrival asymptotic and uniform noisy-record impossibility in a two-frequency family.	10.4.5, 10.4.6
8	Stable terminal designs without moments	Necessary and sufficient infinite completeness radius; matching smooth signed witnesses.	10.5.1
9	Replication of continuously mixed stable laws	Two observations are necessary and sufficient for index–scale mixing; five/six suffice for additional correlated parameters.	10.6.1, 10.7.2
10	Continuous positive spectra	Sharp $\log\log(1/\delta)/\log(1/\delta)$ inverse modulus and finite convex reconstruction.	10.8.1
11	Optimal marked-zero observation height	Order h^{-1} holds exactly for bounded partial quotients in the one-frequency model.	10.8.4
12	Simultaneous gamma extraction	Complete two-rate region, curved and atomic faces, and root-free boundary remainders.	10.9.2, 10.9.3
13	Recovery from factor-exchange response	The full rate-response determines the boundary remainder and both residual exponents.	10.9.4

What the proof status means. Each table row is proved for its stated model below. Classical width and completeness theorems, finite-dimensional continuous recovery, multilevel estimation, moment matching, and stable subordination are credited where used. Selected results also use the previous unpublished convex-first obstruction, convex-noise floor, and binomial transport modulus; these dependencies are identified explicitly. Separate proof reviews within this research process are checks for gaps, not external peer review or machine-checked formal certification. A bounded search of primary literature does not establish universal priority.

The exact-real query results do not imply finite-bit algorithms. Population-law identification does not give stable finite-sample estimation automatically. The zero-set theorem concerns finite exponential sums with explicit independence assumptions, not the Riemann zeta function. The unrestricted value $N(3)$, the general analytic-information problem, unrestricted infinite-support blind inversion, and minimal replication for fully mixed stable parameters remain open in this work.

10.1 Smooth convex information and quantitative finite-query obstructions

We address two questions that were not settled by the earlier query-count or noise-floor results. First, does convexity of every measurement destroy the logarithmic advantage of adaptive continuous information? It does not: logarithmically many measurements suffice even when all measurements are smooth and their Hessians lie between two fixed positive multiples of the identity. Second, how rapidly can a finite number of bounded convex measurements approach the exact convex-noise floor? We give uniform exponential lower bounds for adaptive budgets and inverse-linear lower bounds for nonadaptive budgets. These are quantitative obstructions; matching convergence rates are not claimed.

10.1.1 Logarithmic recovery with uniformly curved smooth queries

In this subsection measurements are exact real numbers. An algorithm may choose its next function arbitrarily from the previous answers, and its decoder may be discontinuous. All errors are uniform over the full space $\mathbb{R}^m$.

Write $N_{\mathrm{cvx}}(m)$ for the minimum query count allowing arbitrarily accurate recovery when all selected functions are continuous and convex. Let $N_{\mathrm{uc}}(m)$ denote the corresponding count for the smaller class

$$\mathcal{C}_m = \{F \in C^\infty(\mathbb{R}^m) : \tfrac{1}{2} I \preceq D^2F(x) \preceq \tfrac{3}{2} I \text{ for every } x \in \mathbb{R}^m\}. \tag{10.1.1}$$

The range of a query is not bounded in this exact-information model. The curvature bounds in (10.1.1) are independent of the requested accuracy, the input, and the stage of the algorithm.

Theorem 10.1.1 (Convexity and smoothness preserve logarithmic complexity)**.** *For every $m \geq 2$,*

$$\lceil \log_2 m \rceil + 1 \leq N_{\mathrm{cvx}}(m) \leq \min\{m, \lceil \log_2 m \rceil + 2\}, \tag{10.1.2}$$

$$\lceil \log_2 m \rceil + 1 \leq N_{\mathrm{uc}}(m) \leq \lceil \log_2 m \rceil + 2. \tag{10.1.3}$$

In particular, both counts divided by $\log_2 m$ tend to one.

The convex class is one of the restricted-information questions posed in [60, Section 3]. The new assertion is the upper bound, including its uniform-curvature strengthening. The first-query quasiconvex obstruction in the preceding manuscript [125] gives the displayed lower bounds. Thus we do not claim an exact count in every dimension.

We give the details of the smooth construction, since uniform smooth approximation of a continuous information function would not suffice: the membership tests below use exact zero values.

Lemma 10.1.2 (Smooth membership and finite smooth codes)**.** *For every closed $C \subset \mathbb{R}^d$ there is a nonnegative $g \in C^\infty(\mathbb{R}^d)$ with $g^{-1}(0) = C$. If $K_1, \dots, K_s$ are pairwise disjoint compact sets, there is a compactly supported smooth function taking the constant value i on K_i, for $1 \leq i \leq s$.*

Proof. Cover the open complement of C by a countable family of open balls whose closures miss C, and choose nonnegative compactly supported smooth functions ϕ_j in those balls such that their positivity sets cover the complement. Choose positive a_j so small that

$$a_j \max_{|\alpha| \leq j} \|D^\alpha \phi_j\|_\infty \leq 2^{-j}.$$

For each derivative order the tail series converges uniformly. Consequently $g = \sum_j a_j \phi_j$ is smooth, nonnegative, zero on C, and positive off C. This argument also handles an empty complement by taking $g = 0$. For the second assertion choose disjoint bounded open neighborhoods of the compact sets and smooth bump functions ψ_i supported in these neighborhoods and equal to one near K_i. Then $\sum_i i\psi_i$ is the required function. □

Proof of Theorem 10.1.1. We construct the upper bound for (10.1.1). Set $Q(x) = \|x\|_2^2/2$ and first read $Q(x)$. This determines $r = \|x\|_2$. For $r = 0$ no further information is needed. Otherwise all later construction choices may depend on this already recorded radius.

The colored partition in [60, Section 2], applied in $\mathbb{R}^{m-1}$ and scaled to accuracy ε, has the following properties: its cells have Euclidean diameter at most ε; there are m colors; distinct cells of the same color have a fixed positive separation; and the partition is locally finite. We use the closures of its cells. Only finitely many such closures meet $[-r, r]^{m-1}$. For each color take the union of these finitely many closed cells, intersected with $[-r, r]^{m-1}$, and call the resulting compact set E_a. The sets E_a, $1 \le a \le m$, cover that cube. A point on a boundary may belong to several colors; this causes no difficulty.

Lemma 10.1.2 supplies an exact smooth zero-set test for each union $\bigcup_{a \in J} E_a$. Regard that test as a function of $x' = (x_1, \dots, x_{m-1})$ and multiply by a smooth cutoff in $\mathbb{R}^m$ equal to one on the ball of radius $r + 1$. On the known sphere this is still an exact membership test and is now compactly supported and smooth. Binary search among the m colors uses at most $\lceil \log_2 m \rceil$ such tests and selects a color a with $x' \in E_a$.

For this color, enumerate its compact cell pieces as $K_1, \dots, K_s$. They are pairwise disjoint. A smooth code $c(x')$ takes the integer value i on K_i by Lemma 10.1.2. Fix a smooth bijection $h : \mathbb{R} \to (0, 1)$, for example $h(t) = \frac{1}{2} + \pi^{-1} \arctan t$. The last intended measurement is $c(x') + h(x_m)$, again multiplied by a cutoff equal to one on the known sphere. Its integer part identifies K_i, and its fractional part determines x_m exactly. Any representative of K_i then approximates x' within ε.

It remains to implement each intended compactly supported smooth function g using an admissible query. Its Hessian has a finite global bound M_g. Choose $\alpha_g > 0$ with $\alpha_g M_g \le 1/2$ and read

$$F_g = Q + \alpha_g g.$$

This belongs to (10.1.1). Since $Q(x)$ is already known exactly, the answer determines $g(x)$ exactly by subtraction and division by α_g. The argument uses no lower bound on α_g and no robustness to rounding. The total count is at most $1 + \lceil \log_2 m \rceil + 1$. The convex upper bound m is given separately by coordinate queries, which need not belong to the uniformly curved class. The earlier convex-first lower bound applies to both classes and completes the proof. □

What this does not establish. The construction contains smooth functions flat on open sets and is not an analytic or polynomial construction. The subtraction of two exact real measurements is essential; the theorem does not imply the same performance with a fixed amount of observation noise. In particular it neither closes the unrestricted two-query problem in $\mathbb{R}^3$ nor the analytic-information question.

10.1.2 A quantitative obstruction at the convex-noise floor

We now return to continuous convex queries $f : B_2^m \to [-1, 1]$, each observed with adversarial additive error at most δ, where $m \ge 2$ and $0 < \delta < 1$. Let e_n^{ad} and e_n^{non} be the optimal worst-case Euclidean errors for adaptive and nonadaptive n-query procedures, respectively. Selection rules

and decoders remain arbitrary. Put

$$q = 1 - \delta, \qquad r = \frac{\delta}{\sqrt{1 + q^2 + 2q/m}}, \qquad \rho = \frac{1 + r}{2}, \qquad L = \frac{2}{1 - \rho}. \tag{10.1.4}$$

The earlier exact floor is $\inf_n e_n^{\mathrm{ad}} = r$ [105]. The preceding manuscript established strict inequality $e_n^{\mathrm{ad}} > r$ for each finite n by compactness [125]. The next theorem replaces that existence argument with quantitative estimates.

Theorem 10.1.3 (Quantitative finite-budget gaps)**.** *There are constants $C_{\mathrm{ad}}, C_{\mathrm{non}} > 0$ and $0 < \theta < 1$, depending only on m and δ, such that for every $n \ge 1$,*

$$e_n^{\mathrm{ad}} \ge r + C_{\mathrm{ad}} \theta^{n-1}, \qquad e_n^{\mathrm{non}} \ge r + \frac{C_{\mathrm{non}}}{n}. \tag{10.1.5}$$

The constants are given explicitly below after choosing one generic skew-symmetric matrix. Consequently reaching error $r + \varepsilon$ requires at least a constant multiple of $\log(1/\varepsilon)$ adaptive queries, and at least a constant multiple of $1/\varepsilon$ nonadaptive queries, for sufficiently small ε.

The second inequality does not prove a separation between the two models: no matching adaptive upper bound is asserted. General noisy continuous information and its entropy-number description are studied in [88]; the additional convexity and bounded-range constraints here create the positive floor around which we quantify finite-budget behavior.

Let $u_0, \dots, u_m$ be regular-simplex directions, with $\|u_i\|_2 = 1$ and $\langle u_i, u_j \rangle = -1/m$ for $i \ne j$. Choose a skew-symmetric matrix A with $\|A\|_{\mathrm{op}} \le 1$ and

$$\omega := \min_{i \ne j} |u_j \cdot A u_i| > 0.$$

Such an A exists: for each noncollinear pair u_i, u_j the corresponding linear functional on the space of skew-symmetric matrices is nonzero, and finitely many proper hyperplanes cannot cover that space. A rational matrix outside these hyperplanes can be chosen and then normalized. No optimization of this choice is required. Define the positive constants

$$\begin{aligned} d &= \sqrt{2 + 2/m}, & v_0 &= \frac{\delta}{rd}, & b &= \frac{v_0 r^2 \omega}{2\delta}, \\ \eta_0 &= \min\left\{\delta, \frac{v_0^3 r^2 \omega^2}{8\delta L^2}\right\}, & T &= \min\left\{1, \frac{b}{2Lr}\right\}, & B_0 &= \frac{2}{b\delta q}, \\ P &= m(m+1), & c &= \min\left\{\frac{\eta_0}{T}, \frac{1}{PB_0}\right\}. \end{aligned} \tag{10.1.6}$$

Lemma 10.1.4 (Short near-saturation intervals)**.** *Let f be convex and bounded between -1 and 1 on the open unit ball. Fix $U \in SO(m)$ and put $x_i(t) = rU \exp(tA) u_i$ for $-T \le t \le T$. For $0 < \eta \le \eta_0$ and each ordered pair $i \ne j$, the set*

$$E_{ij}(f, \eta) = \{t \in [-T, T] : f(x_j(t)) - f(x_i(t)) \ge 2\delta - \eta\}$$

has diameter at most $B_0 \eta$.

Proof. At a time in this set write $x = x_i(t)$, $y = x_j(t)$ and $z = (y - qx)/\delta$. Equation (10.1.4) gives $\|z\|_2 = 1$. Let $\ell(w) = a + v \cdot w$ be a supporting affine function for f at y. It satisfies $\ell \le f$ on the open ball and $\ell \le 1$ on its closure. With $E = f(x) - \ell(x) \ge 0$ and $D = 1 - \ell(z) \ge 0$, the affine identity $\ell(y) = q\ell(x) + \delta\ell(z)$ gives

$$\delta(f(x) + 1) + qE + \delta D = 2\delta - (f(y) - f(x)) \le \eta. \tag{10.1.7}$$

In particular,

$$f(x) \le -1 + \eta/\delta, \qquad \ell(x) \ge -1 - \eta/q, \qquad D \le \eta/\delta. \tag{10.1.8}$$

Every supporting slope at a point of ρB_2^m has norm at most L, by comparing its affine support with the range bound along segments of length approaching $1-\rho$. Also

$$v \cdot (y - x) = \ell(y) - \ell(x) \ge f(y) - f(x) \ge 2\delta - \eta \ge \delta,$$

so $\|v\|_2 \ge v_0$. Since $a + \|v\|_2 \le 1$,

$$\left\| \frac{v}{\|v\|_2} - z \right\|_2 \le \sqrt{\frac{2D}{\|v\|_2}} \le \sqrt{\frac{2\eta}{\delta v_0}}.$$

The simultaneous rotation preserves its skew coupling:

$$z \cdot x_i'(t) = \frac{r^2}{\delta}\, u_j \cdot A u_i.$$

The choice of η_0 therefore implies

$$|v \cdot x_i'(t)| \ge b, \qquad \operatorname{sgn}(v \cdot x_i'(t)) = \operatorname{sgn}(u_j \cdot A u_i). \tag{10.1.9}$$

Indeed, the directional error is at most $Lr\sqrt{2\eta/(\delta v_0)} \le b$, whereas the absolute main term is at least $2b$.

Take two times $t < s$ in $E_{ij}(f, \eta)$ and let $\Delta = s - t$. If the sign in (10.1.9) is positive, use the supporting affine function at time t; if negative, use the one at time s and run the comparison backwards. Since $\|x_i''(t)\|_2 \le r$, Taylor's formula for the curve and (10.1.8) yield in either case

$$b\Delta - \frac{Lr}{2}\Delta^2 \le \eta\left(\frac{1}{\delta} + \frac{1}{q}\right) = \frac{\eta}{\delta q}.$$

Our choice $\Delta \le 2T \le b/(Lr)$ makes the left side at least $b\Delta/2$. Thus $\Delta \le B_0\eta$, as asserted. □

Corollary 10.1.5 (A uniform linear oscillation deficit)**.** *For every such f, every $U \in SO(m)$, and $0 < a \le T$, there is $V \in SO(m)$ with $\|V - I\|_{\mathrm{op}} \le a$ such that*

$$\operatorname{osc}_{\{rUVu_i\}_{i=0}^m} f < 2\delta - ca.$$

Proof. Apply Lemma 10.1.4 on $[-a, a]$ with $\eta = ca \le \eta_0$. Each of the P near-saturation sets lies in an interval of length at most $B_0 ca$. Their union has length at most $PB_0 ca \le a$, so it cannot cover $[-a, a]$. At a point outside the union every ordered difference is below $2\delta - ca$. Set $V = \exp(tA)$; then $\|V - I\|_{\mathrm{op}} \le |t|\|A\|_{\mathrm{op}} \le a$. □

Proof of Theorem 10.1.3. For the adaptive bound define

$$\theta = \min\left\{\frac{1}{2}, \frac{c}{8Lr}\right\}, \qquad C_{\mathrm{ad}} = \min\left\{\frac{\rho - r}{2}, \frac{cT}{16L}\right\}, \qquad C_{\mathrm{non}} = \min\left\{\frac{\rho - r}{2}, \frac{cT}{4L}\right\}. \tag{10.1.10}$$

Maintain a ball of simplex rotations all compatible with the same recorded noisy transcript. Start with any center and radius T, using operator-norm distance on $SO(m)$. If the current radius is a, the transcript fixes the next selected function f. Corollary 10.1.5, applied within distance $a/2$ of the current center, provides a simplex on which the oscillation is below $2\delta - ca/2$. Answer with the midpoint of the maximum and minimum on that simplex. Every vertex then has noise slack at least $ca/4$.

Retain the ball of radius θa about the new center. It is contained in the old ball, and a vertex moves by at most $r\theta a$. Since the query is L-Lipschitz on ρB_2^m, the answer remains

feasible throughout the new ball with slack at least $ca/8$. Previously recorded answers retain their earlier slack by nesting. After n stages the radius is $T\theta^n$ and every answer has slack at least $cT\theta^{n-1}/8$.

Choose any retained simplex and increase its radius by

$$h = \min\left\{\frac{\rho - r}{2}, \frac{cT\theta^{n-1}}{16L}\right\} \geq C_{\mathrm{ad}}\theta^{n-1}.$$

Every enlarged vertex remains in ρB_2^m and all query values change by at most Lh, so the same transcript is feasible for all these vertices. Their circumradius is $r + h$. More explicitly, for every decoded point w, averaging squared distances over the centered regular simplex gives

$$\frac{1}{m+1}\sum_i \|(r+h)Uu_i - w\|_2^2 = (r+h)^2 + \|w\|_2^2.$$

Thus some vertex has error at least $r + h$. The construction is valid for arbitrary, even discontinuous, query-selection rules because each query is fixed only after its transcript is fixed.

For n fixed nonadaptive functions, put $\eta = cT/n$ and use the single rotation interval $[-T, T]$. The union of all their nP near-saturation sets has length at most $nPB_0\eta \leq T < 2T$. One rotation consequently gives oscillation below $2\delta - \eta$ for every function at once. Choose their midpoint answers and inflate by

$$h = \min\left\{\frac{\rho - r}{2}, \frac{\eta}{4L}\right\} \geq C_{\mathrm{non}}/n.$$

The same simplex argument proves the second bound. Solving these inequalities for n proves the stated necessary query complexities. □

Open targets and failed extensions. The unrestricted exact value $N(3)$ remains unresolved here. The available width recursion reaches index $2^2 - 1 = 3$, which gives no positive width obstruction on a three-dimensional cube; replacing that recursion by a stronger assertion for scalar targets has not been justified. For analytic first queries, the large-level graphs available for polynomials or convex functions need not follow from analyticity alone. Neither an unrestricted two-query construction nor an analytic impossibility proof is claimed. For the noisy problem we have quantified the earlier strict gap, but have not proved that the exponential adaptive bound or inverse-linear nonadaptive bound is sharp.

10.2 Sharp adaptive recovery of a noncompact sequence embedding

The Hilbert-space compactness criterion from the preceding paper [125] does not extend to arbitrary Banach operators. We prove this with a complete error-rate calculation for the canonical embedding $I_{p,q} : \ell_p \to \ell_q$, where $1 \leq p < q \leq \infty$. The input is the closed unit ball B_p with its ℓ_p topology. For $q = \infty$ we may take the output space to be either c_0 or ℓ_∞, with its supremum norm. Write

$$\alpha = \frac{1}{p} - \frac{1}{q} > 0,$$

with $1/\infty = 0$. Let e_n^{ad} and e_n^{non} denote infimal worst-case errors using at most n continuous real-valued measurements, adaptively and nonadaptively, respectively. The decoder and the dependence of later queries on earlier answers are unrestricted. All observations in these two definitions are exact real numbers.

Theorem 10.2.1 (Noncompact recovery and its exact decay exponents)**.** *For every $n \ge 1$,*

$$e_n^{\mathrm{ad}}(I_{p,q}) \ge \tfrac{1}{2}2^{-\alpha n}, \qquad e_n^{\mathrm{non}}(I_{p,q}) \ge (n+1)^{-\alpha}.$$

For every integer $s \ge 1$,

$$e_{2s}^{\mathrm{non}}(I_{p,q}) \le s^{-\alpha}, \qquad e_{\lceil \log_2 s \rceil + 3}^{\mathrm{ad}}(I_{p,q}) \le s^{-\alpha}.$$

The nonadaptive upper bound can be attained with all queries 1-Lipschitz on B_p. Consequently

$$e_n^{\mathrm{non}}(I_{p,q}) \asymp n^{-\alpha}, \qquad e_n^{\mathrm{ad}}(I_{p,q}) \asymp 2^{-\alpha n}. \tag{10.2.1}$$

In particular, a bounded noncompact linear operator can have uniformly vanishing error under continuous nonlinear information. Adaptivity changes the optimal algebraic order into an exponential one, with the same exponent α.

The macroscopic-width behavior of finite-dimensional ℓ_p balls in the ℓ_q metric is classical, and was studied by Tsukamoto [126]. The finite-dimensional adaptive compression used below is the construction of Krieg–Novak–Ullrich [60]. The matching adaptive converse uses the preceding adaptive-width argument. Our contribution here is the explicit infinite-sequence encoding, its adaptive localization, and the resulting two-sided noncompact-operator rate theorem. The general Banach-space comparison asked in [60, Section 4] is not settled by this one family of operators.

10.2.1 A finite sparse code for an infinite sequence

For $a > 0$, define the soft threshold

$$(T_a x)_j = \operatorname{sgn}(x_j)(|x_j| - a)_+.$$

If $x \in B_p$ and $a = s^{-1/p}$, the sequence $T_a x$ has at most s nonzero coordinates and $\|T_a x\|_p \le 1$. Moreover,

$$\|x - T_a x\|_q \le s^{-\alpha}. \tag{10.2.2}$$

For finite q, this follows from $\min\{|x_j|, a\}^q \le a^{q-p}|x_j|^p$ and summation; for $q = \infty$ it is the coordinatewise bound a.

Define the $2s$ moment queries

$$L_l(x) = \sum_{j \ge 1} j^{-l}(T_a x)_j, \qquad l = 0, \dots, 2s-1. \tag{10.2.3}$$

These are finite sums at every $x \in \ell_p$ and are continuous in the ℓ_p topology. Indeed, for any fixed x, all sufficiently large coordinates have magnitude less than $a/2$. On a sufficiently small ℓ_p neighborhood of x, all those coordinates remain below a, so the displayed sum is locally a fixed finite sum of continuous functions. On B_p the stronger bound

$$|L_l(x) - L_l(z)| \le \|T_a x - T_a z\|_1 \le (2s)^{1-1/p}\|x - z\|_p$$

holds: the difference is supported on at most $2s$ coordinates, and soft thresholding contracts the ℓ_p distance. Dividing each query by $(2s)^{1-1/p}$ makes it 1-Lipschitz, without losing any information because the responses are exact.

The map in (10.2.3) determines every s-sparse sequence uniquely. If two such sequences have the same moments, their difference is supported on distinct integers $j_1, \dots, j_r$, with $r \le 2s$. Its first r zero moments at the distinct nodes $1/j_1, \dots, 1/j_r$ form a square Vandermonde system, whose determinant is nonzero. All its coefficients therefore vanish. An unrestricted decoder returns the unique sparse sequence when it exists, and zero otherwise. Equation (10.2.2) proves the nonadaptive upper bound.

10.2.2 Localizing the code before adaptive compression

Direct approximation of the moment vector is insufficient: its inverse need not have a uniform modulus over all integer supports. One additional continuous query removes exactly that obstruction.

Choose a continuous function $\psi : [0,\infty) \to [0,1]$ that is zero on $[0,1/2]$ and one on $[1,\infty)$. Query

$$H_a(x) = \sum_{j\ge 1} j\,\psi(|x_j|/a). \tag{10.2.4}$$

The same local-finiteness argument proves continuity on ℓ_p. Set $M = \max\{1, \lceil H_a(x)\rceil\}$. If $|x_j| \ge a$, then $H_a(x) \ge j$; consequently $T_a x$ is supported in $\{1,\dots,M\}$.

For each recorded M, consider the compact set

$$K_{s,M} = \{y \in \mathbb{R}^M : \|y\|_p \le 1,\ |\operatorname{supp} y| \le s\}$$

and its moment map $V_M : K_{s,M} \to \mathbb{R}^{2s}$, $(V_M y)_l = \sum_{j=1}^{M} j^{-l} y_j$. The preceding Vandermonde argument shows injectivity. More quantitatively, since there are finitely many supports of size at most $2s$ in $\{1,\dots,M\}$, there is a finite $C_{s,M} > 0$ such that

$$\|y - z\|_q \le C_{s,M}\|V_M y - V_M z\|_\infty \qquad (y, z \in K_{s,M}). \tag{10.2.5}$$

For each support this follows by a left inverse of its full-column-rank Vandermonde matrix, and one takes the maximum of the finitely many operator norms. The output vectors are padded by zeros when M varies.

Fix any additional output tolerance $\eta > 0$. After observing M, apply the KNU procedure to the $2s$-dimensional vector

$$Z_M(x) = V_M\big((T_a x)_1, \dots, (T_a x)_M\big)$$

with accuracy $\eta/(2C_{s,M})$ in the maximum norm. For each fixed history and hence fixed M, this vector and every query composed with it are continuous functions of x on the entire input space. The number of additional measurements is at most $\lceil \log_2(2s)\rceil + 1 = \lceil \log_2 s\rceil + 2$. Given its approximation $\widehat{Z}$, choose a minimizer $\widehat{y} \in K_{s,M}$ of $\|V_M y - \widehat{Z}\|_\infty$. Such a minimizer exists by compactness. Since the true $T_a x$ is feasible, (10.2.5) yields $\|\widehat{y} - T_a x\|_q \le \eta$. Combining with (10.2.2) gives error at most $s^{-\alpha} + \eta$. Taking the infimum over $\eta > 0$ proves the adaptive upper bound. The discontinuity of M as a function of the recorded answer is permitted by the model; it is not being used as a continuous query.

10.2.3 Matching lower bounds

For nonadaptive information, restrict to the first $m = n + 1$ coordinates. Borsuk–Ulam on the ℓ_p unit sphere gives antipodal $x, -x$ with identical data. Every common answer makes error at least $\|x\|_q \ge m^{-\alpha}\|x\|_p = m^{-\alpha}$.

For the adaptive bound we recall, and justify, the width input. Let $\mathrm{UW}_d(X)$ be the infimum of the mesh of finite open covers of a compact metric space X with multiplicity at most $d+1$. The scalar fiber inequality [102, Theorem 2.1] is

$$\mathrm{UW}_{2d+1}(X) \le \sup_{t} \mathrm{UW}_d(f^{-1}(t)) \quad (f : X \to \mathbb{R} \text{ continuous}).$$

If an n-query procedure has error r, a terminal feasible set has diameter at most $2r$. Applying this inequality inductively to the fixed-history continuous queries gives

$$\mathrm{UW}_{2^n-1}(X) \le 2r. \tag{10.2.6}$$

No continuity of the choices across histories is needed.

Take $m = 2^n$ and $X = B_p^m$, with its ℓ_q metric. The classical width-dimension inequality recalled in [126, Eq. (2)] implies $\mathrm{UW}_{m-1}(B_p^m, \ell_p) \geq 1$. Since $\|u - v\|_q \geq m^{-\alpha}\|u - v\|_p$ in $\mathbb{R}^m$, pulling back the same covers gives

$$\mathrm{UW}_{m-1}(B_p^m, \ell_q) \geq m^{-\alpha}.$$

Equation (10.2.6) proves $r \geq \frac{1}{2}m^{-\alpha}$. This completes the proof of Theorem 10.2.1.

10.2.4 What finite precision removes

The standard unit vectors have pairwise distance $2^{1/q}$ for finite q, and one for $q = \infty$. Thus $I_{p,q}$ is noncompact. The following consequence isolates the role of exact real information; it is an elementary finite-codebook obstruction, not a separate claim of historical novelty.

Proposition 10.2.2 (Every finite transcript alphabet has error one)**.** *Take the output space to be ℓ_q for $q < \infty$, or c_0 for $q = \infty$. If the complete transcript has a finite alphabet, then the smallest worst-case error for $I_{p,q}$ is exactly one. This remains true for randomized procedures with finitely many possible outputs conditional on each realization of their random seed, when the risk is worst-case expected norm error.*

Proof. Every finite codebook D in either output space satisfies

$$\liminf_{j\to\infty} \min_{z\in D} \|e_j - z\|_q \geq 1,$$

because its finitely many coordinate tails vanish uniformly, and $\|e_j - z\|_q \geq |1 - z_j|$. This proves the deterministic lower bound. For a randomized procedure condition on its seed ω, let D_ω be its finite output codebook, and apply Fatou's lemma to the nonnegative quantities $\min_{z\in D_\omega} \|e_j - z\|_q$. Their limiting inferior is at least one for every seed. Hence the supremum of expected errors over the inputs e_j is at least one. The algorithm returning zero has error at most one and attains the bound. In the randomized case all maps needed for the stated expectations are assumed measurable. □

The c_0 qualification matters: an arbitrary codeword in ℓ_∞ need not have vanishing coordinates. The proposition does not assert the same exact constant for unrestricted ℓ_∞ codewords. For instance $p = 1, q = 2$ gives $e_n^{\mathrm{ad}} \asymp 2^{-n/2}$ with exact continuous real observations, while every fixed finite bit budget has error one. This is a statement about observation models, not a finite-precision compression algorithm.

10.3 Optimal replication budgets and the price of honest robustness

We study two changes to the observation model. First, the experimenter may distribute a total budget of Bernoulli observations among individuals, including by an adaptive rule. We determine the minimax population recovery rate for this experiment. Second, with a fixed rectangular experiment, we ask whether confidence sets can adapt to the actual number of corrupted bits. A sharp obstruction already occurs at one fixed, uncorrupted population. These are distinct questions: the second result does not rule out adaptation of point estimators.

10.3.1 A total-budget theorem, including adaptive allocation

Let G be an arbitrary probability law on $[0,1]$. Each fresh individual has an independent parameter $P \sim G$, and its observations are conditionally independent Bernoulli(P) trials. At every stage a sampling policy may introduce a fresh individual or revisit any earlier individual. The choice may depend arbitrarily on all previous choices and observations and on private randomization. The number of trials is at most an integer B almost surely. Write

$$\mathfrak{R}_B = \inf_{\mathcal{S},\widehat{G}} \sup_{G} \mathbb{E}_{G,\mathcal{S}} W_1(\widehat{G}, G),$$

where the infimum includes all such policies and estimators. Introducing an individual has no cost beyond the trials assigned to it.

Theorem 10.3.1 (The unrestricted Bernoulli-budget law)**.** *There are absolute positive constants c, C such that, for every integer $B \geq 1$,*

$$cB^{-1/3} \leq \mathfrak{R}_B \leq CB^{-1/3}. \tag{10.3.1}$$

The upper bound is attained by a predetermined heterogeneous design and a finite linear program of size polynomial in B. The lower bound allows fully adaptive allocation, arbitrary revisits, and arbitrary randomization. No smoothness, finite-support assumption, or separation condition on G is imposed.

The multilevel idea is classical. Bujok, Hambly and Reisinger [127] use increasing Bernoulli replication levels to estimate fixed functionals of a latent population. Giles, Nagapetyan and Ritter [9] develop multilevel estimation of distribution functions under regularity assumptions. Adaptive sampling of unknown coins is studied by Lee and Valiant [11] for estimating the proportion of coins on either side of a prescribed gap. Here the target is the entire arbitrary mixing law in W_1, and the lower bound covers all adaptive designs. The theorem combines an integrated indicator-variance calculation with an adaptive information lower bound. We do not claim that either multilevel simulation or adaptive coin sampling originates here. The primary sources checked do not state the exact unrestricted population theorem (10.3.1); this is a bounded literature comparison, not an assertion that every possible precedent is excluded.

Upper bound. Put $t_l = 2^l$, for $0 \leq l \leq L$, and $T = 2^L$. For every level use an independent cohort of N_l fresh individuals. At level zero observe one trial from each individual. At level $l \geq 1$ observe t_l trials and let A_l be their full average and C_l their first-half average. For each individual define the random function

$$D_l(x) = \mathbf{1}_{\{A_l \leq x\}} - \mathbf{1}_{\{C_l \leq x\}}, \qquad 0 \leq x \leq 1.$$

Let G_t be the law of a normalized t-trial count under G, write F_t for its distribution function, and set

$$\widetilde{F}(x) = \widehat{F}_1(x) + \sum_{l=1}^{L} \frac{1}{N_l} \sum_{i=1}^{N_l} D_{l,i}(x).$$

The expectation telescopes: $\mathbb{E}\widetilde{F} = F_T$. Conditional on P, the full average minus the first-half average has mean zero and variance $P(1-P)/t_l$. Consequently

$$\int_0^1 \mathrm{Var}(D_l(x))\, dx \leq \mathbb{E}|A_l - C_l| \leq \frac{1}{2\sqrt{t_l}}. \tag{10.3.2}$$

For the level-zero indicator the integrated variance is at most $1/4$. Independence of the cohorts, followed by Cauchy–Schwarz on $[0,1]$, gives

$$\mathbb{E}\|\widetilde{F} - F_T\|_{L^1} \leq C \left(\sum_{l=0}^{L} \frac{2^{-l/2}}{N_l} \right)^{1/2}. \tag{10.3.3}$$

Let $S_L = \sum_{l=0}^{L} 2^{l/4}$ and choose

$$N_l = \left\lfloor \frac{B 2^{-3l/4}}{2 S_L} \right\rfloor .$$

Choose L so that T is between two fixed positive multiples of $B^{2/3}$. For all sufficiently large B, every expression inside the floor is at least two. Indeed its minimum occurs at $l = L$ and is bounded below by a constant times B/T. Thus rounding loses at most a factor two and

$$\sum_l N_l t_l \le B/2, \qquad \sum_l \frac{2^{-l/2}}{N_l} \le \frac{4 S_L^2}{B} \le \frac{C T^{1/2}}{B}.$$

All quantities in $\widetilde{F}$ are constant between consecutive points of the grid $\{0, 1/T, \dots, 1\}$. Project $\widetilde{F}$ in L^1 onto distribution functions supported on this grid. This is a finite linear program: the grid values must be nondecreasing, between zero and one, with terminal value one. Since F_T is feasible, any minimizer $\widehat{F}$ satisfies

$$\|\widehat{F} - F_T\|_{L^1} \le 2 \|\widetilde{F} - F_T\|_{L^1}.$$

Let $\widehat{G}$ be its probability law. Coupling P to its normalized T-trial count gives $W_1(G, G_T) \le 1/(2\sqrt{T})$. The identity between one-dimensional W_1 and integrated distribution function distance now yields

$$\mathbb{E} W_1(\widehat{G}, G) \le C \left(T^{-1/2} + \frac{T^{1/4}}{\sqrt{B}} \right) \le C B^{-1/3}.$$

The finitely many smaller B are covered by increasing C and using a fixed estimate. The grid has at most $B + 1$ points for large B, and all linear-program data are rational with polynomial bit length. This also proves the computational assertion. □

Lower bound, including adaptive policies. Fix an integer m and set

$$c_j = \frac{1}{4} + \frac{j - 1/2}{2m}, \quad h = \frac{1}{8m}, \quad G_v = \frac{1}{m} \sum_{j=1}^{m} \delta_{c_j + h v_j}, \qquad v \in \{-1, 1\}^m.$$

The two possible locations in each cell are separated from those in all other cells and belong to $[1/4, 3/4]$. Monotone coupling gives

$$W_1(G_v, G_w) = \frac{2h}{m} d_H(v, w). \tag{10.3.4}$$

Enlarge the experiment: reveal the cell of every individual and allow the learner to choose any cell for its next observation. In this oracle experiment, querying cell j returns an independent Bernoulli$(c_j + h v_j)$ observation. It is more informative than the original experiment restricted to these populations: a fresh person in the original model is simulated by choosing a uniform cell and remembering its label, and subsequent visits query that cell.

Let $\mathbb{P}_v$ be the complete transcript law of any adaptive policy in this enlarged experiment, including its random seed. Let T_j be the number of queries to cell j. If $v^{(j)}$ changes only sign j, the chain rule for relative entropy gives

$$D(\mathbb{P}_v \| \mathbb{P}_{v^{(j)}}) \le C h^2 \mathbb{E}_v T_j. \tag{10.3.5}$$

Indeed, conditional on an identical history the policy uses the same choice kernel under both laws. Only a query to cell j has a different response distribution, and $D(\mathrm{Ber}(p) \| \mathrm{Ber}(q)) \le (p - q)^2 / [q(1 - q)] \le C h^2$ here. A stopping policy can be padded by uninformative symbols, so

the same chain rule applies to a hard budget. Summing (10.3.5) over v, j, using $\sum_j T_j \le B$, and applying Pinsker's inequality and Jensen's inequality proves

$$\frac{1}{m2^m}\sum_{v,j}\|\mathbb{P}_v - \mathbb{P}_{v^{(j)}}\|_{\mathrm{TV}} \le \sqrt{CB/m^3}.$$

Choose $m = \lceil KB^{1/3}\rceil$ for a sufficiently large absolute K, making the right side at most $1/2$.

Decode an arbitrary estimate $\widehat{G}$ to a nearest member $G_{\widehat{v}}$ of the finite family. The triangle inequality and (10.3.4) imply $W_1(\widehat{G}, G_v) \ge (h/m)d_H(\widehat{v}, v)$. For each coordinate, pairing the two neighboring hypotheses shows that the average probability of a sign error is at least half of one minus their total variation distance. Averaging over all coordinates therefore gives

$$2^{-m}\sum_v \mathbb{E}_v d_H(\widehat{v}, v) \ge m/4.$$

The average Wasserstein risk is at least $h/4 \ge cB^{-1/3}$. The oracle experiment was easier than the original one, so its lower bound proves the required assertion. □

Comparison with equal replication. The earlier fixed-replication population theorem gives, after optimizing $Nt \le B$ with the same t for every individual, rate $(B\log(eB))^{-1/4}$. The improvement in Theorem 10.3.1 is a change in the power of the budget, made possible by using different depths. It is not an improvement of that earlier theorem within its own observation class. Adaptivity itself is unnecessary for attaining the new minimax order.

10.3.2 Honest confidence sets cannot learn the actual corruption budget

We return to N independent individuals, each observed t times. An adversary may, after observing the entire clean array, flip at most $q = \lfloor \bar{\varepsilon}Nt\rfloor$ bits, where $0 \le \bar{\varepsilon} \le 1/4$ is an upper bound. Equivalently, counts J_i may be changed to J_i' with $\sum_i |J_i - J_i'| \le q$. The upper bound is known; the actual number of changed bits is not. Define

$$L_t(u) = \log\left(e + \frac{1}{u\sqrt{t}}\right), \quad L_t(0) = +\infty,$$

$$r_{N,t}(u) = \min\left\{1, \frac{1}{t} + \frac{1}{\sqrt{t\log(eN)}} + \frac{1}{\sqrt{N}} + u + \frac{1}{\sqrt{tL_t(u)}}\right\}. \tag{10.3.6}$$

A random set $\mathcal{C}$ of mixing laws is honest at level $1-\alpha$ if, for every G and every allowed adversary A,

$$\mathbb{P}_{G,A}\{G \in \mathcal{C}\} \ge 1 - \alpha.$$

All sets and events below are assumed measurable; auxiliary randomization in a confidence procedure is permitted.

Theorem 10.3.2 (Sharp honest diameter at a fixed clean population)**.** *Let $0 < \alpha \le 1/8$ and let $\mathcal{C}$ be honest for the hard bit budget above. For the fixed arcsine law $\pi(dx) = dx/(\pi\sqrt{x(1-x)})$, with no corruption at all,*

$$\mathbb{P}_{\pi,\mathrm{id}}\left\{\mathrm{diam}_{W_1}(\mathcal{C}) \ge c\, r_{N,t}(\bar{\varepsilon})\right\} \ge 1 - 2\alpha - \frac{1}{8}. \tag{10.3.7}$$

Here $c > 0$ is absolute and does not depend on $N, t, \bar{\varepsilon}$. Conversely, for each fixed α there are honest confidence sets whose diameter is everywhere at most $C_\alpha r_{N,t}(\bar{\varepsilon})$. Thus the rate in (10.3.7) is sharp even at this one clean law.

The distinction between adaptive point estimation and adaptive honest confidence is established in robust statistics. Luo and Gao [128] obtain sharp adaptive confidence phenomena for Huber contamination. Cho, Luo and Gao [129] study a single binomial proportion under Huber contamination. The claim here concerns a nonparametric binomial mixing law, a hard budget of individual bit changes, the complete transport inverse curve, and a fixed clean arcsine center. It is not a claim that the general obstruction to robust confidence adaptation is new.

Lemma 10.3.3 (Centered oscillatory alternatives)**.** *Put* $h_k(x) = T_k(2x-1)$ *and* $G_{k,a} = (1 + ah_k)\pi$*, where* $k \geq 1$ *and* $0 < a \leq 1/2$*. There is an absolute* $c > 0$ *such that*

$$W_1(G_{k,a}, \pi) \geq ca/k.$$

Write p_G *for the* t*-trial predictive count law and* $\rho = p_\pi$*. For* $k \leq t$*,*

$$\chi^2(p_{G_{k,a}}, \rho) = a^2\lambda_{t,k}/2, \qquad \lambda_{t,k} = \prod_{l=0}^{k-1} \frac{t-l}{t+1+l} \leq e^{-k^2/(2t)}, \tag{10.3.8}$$

$$W_{1,t}(p_{G_{k,a}}, \rho) \leq \frac{a}{k} e^{-k^2/(4t)}. \tag{10.3.9}$$

Here $W_{1,t}$ *uses count cost* $|i-j|/t$*. For* $k > t$ *the two predictive laws are identical.*

Proof. These are the centered forms of the oscillatory bounds in [125]; we include their derivation. The change of variable $x = (1 + \cos\theta)/2$ gives

$$W_1(G_{k,a}, \pi) = \frac{a}{2\pi k} \int_0^\pi |\sin(k\theta)| \sin\theta \, d\theta \asymp a/k.$$

For a uniform lower bound, restrict to $[\pi/4, 3\pi/4]$ and sum complete periods of $|\sin(k\theta)|$; the remaining finitely many small k have positive integrals. The posterior operator $Qh(j) = \mathbb{E}_\pi[h(P) \mid J = j]$ satisfies $\|Qh_k\|^2_{L^2(\rho)} = \lambda_{t,k}/2$. This classical beta–binomial diagonalization [70] also follows by observing that the self-adjoint operator Q^*Q preserves polynomial degree and has leading multiplier $\lambda_{t,k}$ on degree k, and then using arcsine orthogonality. Since $p_{G_{k,a}}/\rho = 1 + aQh_k$, this proves (10.3.8). Moreover

$$\log \lambda_{t,k} \leq -\sum_{l=0}^{k-1} \frac{1+2l}{t+1+l} \leq -\frac{k^2}{2t}.$$

For completeness, the reversible birth-and-death generator with rates $(j+1/2)(t-j)$ upward and $j(t-j+1/2)$ downward has eigenfunction Qh_k with eigenvalue $-k^2$. Its Dirichlet form is

$$\mathcal{E}(f,g) = \sum_{j=0}^{t-1} \rho_j (j+1/2)(t-j)(f(j+1) - f(j))(g(j+1) - g(j)).$$

For a function f with normalized count Lipschitz constant one, $\mathcal{E}(f,f) \leq 2$. Cauchy–Schwarz for this form implies $|\langle f, Qh_k\rangle_\rho| \leq \sqrt{\lambda_{t,k}}/k$. Kantorovich duality proves (10.3.9). If $k > t$, each binomial probability is a polynomial of degree t and is orthogonal to h_k under π, proving the last assertion. □

Proof of Theorem 10.3.2. We use a two-experiment observation. Suppose an alternative H, after an allowed attack, has data law within total variation τ of the clean data law at π. Honesty at H and at π implies

$$\mathbb{P}_{\pi,\mathrm{id}}\{\pi, H \in \mathcal{C}\} \geq 1 - 2\alpha - \tau. \tag{10.3.10}$$

Randomization can be appended to the data without increasing total variation. On this event the diameter is at least $W_1(\pi, H)$.

First we produce clean alternatives for the three sampling terms. Taking $k = t + 1$ and $a = 1/2$ gives identical predictive laws and separation at least c/t. For the logarithmic sampling term, take $k = \lceil \sqrt{16t \log(eN)} \rceil$ and $a = 1/2$ if $k \le t$. The product chi-squared identity gives

$$1 + \chi^2(p_{G_{k,a}}^{\otimes N}, \rho^{\otimes N}) = (1 + a^2 \lambda_{t,k}/2)^N,$$

so their total variation distance is less than $1/8$, with separation at least $c/\sqrt{t \log(eN)}$. If $k > t$, this desired term is bounded by C/t, and the preceding exact alternative suffices. Finally take $k = 1$ and $a = 1/(8\sqrt{N})$. The same product identity gives total variation less than $1/8$ and separation at least $c/\sqrt{N}$.

For the corruption terms assume first that $q \ge 1$ and put $\eta = q/(16Nt)$. We claim that there is an alternative H with

$$W_{1,t}(p_H, \rho) \le \eta, \qquad W_1(H, \pi) \ge c \max\left\{ \frac{1}{t}, \eta, \frac{1}{\sqrt{tL_t(\eta)}} \right\}. \tag{10.3.11}$$

It is enough to construct an alternative separately for each term and select the largest. The $1/t$ term again uses $k = t + 1$. For the linear term take $H = (1 - \eta)\pi + \eta\delta_1$. Its predictive distance is at most η, and its population distance is exactly $\eta/2$. For the remaining term set $x = \eta\sqrt{t}$. If $x \ge 1$ the linear term already dominates it. Otherwise take $k = \lceil \sqrt{16tL_t(\eta)} \rceil$ and $a = 1/2$. If $k \le t$, (10.3.9) implies

$$W_{1,t}(p_{G_{k,a}}, \rho) \le \frac{(e + 1/x)^{-4}}{8\sqrt{tL_t(\eta)}} \le \eta,$$

and the latent separation has the desired order. If $k > t$, the $1/t$ alternative dominates this term up to an absolute constant. This proves (10.3.11).

Couple independently each count from p_H to a target count from ρ by an optimal transportation coupling. The expected sum of absolute count changes is at most $Nt\eta = q/16$. Given the clean counts from H, an adversary samples the conditional target counts. It implements the changes when their total is at most q and otherwise leaves the array unchanged. Every count change is implemented by the same number of bit flips. To cover procedures that observe the full array, add the required ones uniformly among the zero positions, or remove them uniformly among the one positions. Conditional on its count, a clean row is uniform among strings with that count; this symmetric operation produces a uniform row with the target count. Thus the untruncated coupling produces exactly the clean full-array law at π, as well as its count law. The product chi-squared comparisons above likewise apply to full arrays, since conditional row arrangements carry no additional information about G. Markov's inequality shows that the resulting output law differs from $\rho^{\otimes N}$ in total variation by at most $1/16$. Hence (10.3.10) applies.

When $q \ge 1$, $q/(Nt) \ge \bar{\varepsilon}/2$, so η and $\bar{\varepsilon}$ differ by at most a fixed factor. In particular $L_t(\eta) \le L_t(\bar{\varepsilon}) + \log 32 \le CL_t(\bar{\varepsilon})$. When $q = 0$, $\bar{\varepsilon} < 1/(Nt)$, so the linear corruption term is bounded by $N^{-1/2}$ and $L_t(\bar{\varepsilon}) \ge c \log(eN)$; the other corruption term is then absorbed by the clean logarithmic term. Thus in every case one of the alternatives constructed above has separation at least $cr_{N,t}(\bar{\varepsilon})$ and data total variation at most $1/8$. Equation (10.3.10) proves the lower bound.

For the upper bound, the earlier hard-bit-budget theorem [125] supplies an estimator with uniform expected loss at most $Cr_{N,t}(\bar{\varepsilon})$. The closed Wasserstein ball of radius $Cr_{N,t}(\bar{\varepsilon})/\alpha$ about that estimator is honest by Markov's inequality, and has diameter at most twice this radius, or one if smaller. This proves the fixed-confidence matching assertion. No optimal dependence on α is claimed. □

For example, let $N = t^3$ and $\bar{\varepsilon} = t^{-1/2}$, with $t \to \infty$. The clean point-estimation rate is of order $(t \log t)^{-1/2}$, whereas every confidence set honest for this vanishing upper corruption

budget has diameter at least $ct^{-1/2}$ with probability bounded away from zero at the clean law π. The ratio diverges like $\sqrt{\log t}$. This is an obstruction to honest adaptation to the actual budget zero, even at a fixed population; it is not an impossibility theorem for adaptive point estimators.

10.3.3 A remaining allocation problem under bit corruption

For clarity, the clean adaptive-budget theorem does not close its malicious-bit analogue. There is nevertheless a rigorous upper bound for a predetermined design. If an adversary may change at most $\lfloor \varepsilon B \rfloor$ observed bits after sampling, the multilevel construction above, with its maximum level chosen using ε, has worst-case expected loss

$$C \min\{1, B^{-1/3} + \varepsilon^{2/3}\}. \tag{10.3.12}$$

To see this, a flip in a level-l observation changes the integrated absolute value of its full-average indicator by $1/t_l$, and that of its half-average indicator by at most $2/t_l$. Thus it changes $\widetilde{F}$ in L^1 by at most $3/(N_l t_l)$. With the allocation above this is bounded by $CT^{1/4}/B$ uniformly over levels. Projection and the triangle inequality therefore give

$$\mathbb{E} W_1(\widehat{G}, G) \le C\{T^{-1/2} + (B^{-1/2} + \varepsilon)T^{1/4}\}.$$

Choosing a dyadic T of order $(B^{-1/2} + \varepsilon)^{-4/3}$ proves (10.3.12); when this order is bounded, a fixed estimate suffices. The rounding argument is valid for large B, since this T is at most a constant times $B^{2/3}$.

This bound is not claimed to be minimax, and we do not count it as a third closed target. The exact corrupted adaptive-allocation law and adaptation of point estimates to an unknown hard bit budget remain open in this manuscript. Neither is resolved by Theorem 10.3.2.

10.4 Unknown frequency supports and the resonance barrier to observation

The preceding inverse theorem prescribed the frequency support. The first result below removes that prescription for finite sums whose frequencies are internally rationally independent. A second result shows why exact identification does not supply a uniform finite-height recovery guarantee: a nearly rational relation can hide a fixed change of coefficients and of the true right boundary. We compute both the finite-window ambiguity and the first height at which a shrinking right-edge strip becomes visible.

These are two self-posed inverse-observation targets. The qualitative dependence of spectral boundaries on rational relations is classical [119, 68]; it is not counted again here. Nor is the first theorem a solution of the general common-zero problem for exponential polynomials. Its much stronger observation hypothesis and its positive real, internally independent frequencies permit a direct proof without a common-factor conjecture.

10.4.1 One-sided zeros identify an unknown independent support

Consider

$$F(s) = 1 - \sum_{j=1}^{N} a_j e^{-\lambda_j s + i\theta_j}, \qquad G(s) = 1 - \sum_{k=1}^{M} d_k e^{-\mu_k s + i\psi_k}, \tag{10.4.1}$$

where all magnitudes are strictly positive and $0 < \lambda_1 < \cdots < \lambda_N$, $0 < \mu_1 < \cdots < \mu_M$. Neither the two supports nor their sizes are assumed known or equal. Let S_F, S_G be the unique real

solutions of

$$\sum_j a_j e^{-\lambda_j S_F} = 1, \qquad \sum_k d_k e^{-\mu_k S_G} = 1.$$

Theorem 10.4.1 (Unknown-support rigidity from a right-hand zero set)**.** *Suppose each of the two frequency lists in (10.4.1) is linearly independent over $\mathbb{Q}$. Suppose there is a real $\sigma_0 < \min(S_F, S_G)$ such that*

$$\{s : \Re s > \sigma_0,\ F(s) = 0\} = \{s : \Re s > \sigma_0,\ G(s) = 0\}. \tag{10.4.2}$$

Equality of sets suffices; multiplicities need not be supplied. Then $N = M$ and, after an ordering of the terms,

$$\lambda_j = \mu_j, \qquad a_j = d_j, \qquad \theta_j = \psi_j \pmod{2\pi}.$$

Thus $F = G$ identically. In particular the observation determines the number of terms, the positive frequencies, and all complex coefficients within this model class.

The normalization by the constant term and the positivity of all frequencies are substantive. Multiplication by a nonzero exponential preserves every zero, so a normalization fixing the lowest frequency is necessary. The assumption that the observed half-plane contains both right boundaries prevents an empty-record ambiguity.

Lemma 10.4.2 (The joint hull is forced to be an integral graph)**.** *Under the hypotheses of Theorem 10.4.1, one has $S_F = S_G =: S$. Moreover $N = M$, and there is a matrix $K \in \mathrm{GL}_N(\mathbb{Z})$ such that*

$$\mu = K\lambda, \qquad \psi = K\theta \pmod{2\pi}. \tag{10.4.3}$$

For every $x \in \mathbb{T}^N$, the phase-frozen pair (F_x, G_{Kx}) has the same zero set in $\Re s > \sigma_0$. Here F_x replaces θ by x, and similarly for G.

Proof. The triangle inequality excludes zeros to the right of S_F. Rational independence makes $t\lambda$ dense modulo 2π, so vertical translates of F approach F_0. Its zero at S_F is simple, with derivative $\sum a_j \lambda_j e^{-\lambda_j S_F} > 0$. Rouché's theorem therefore places real parts of zeros of F arbitrarily close to S_F. The same applies to G. The observed zero sets thus have right endpoints S_F and S_G, proving their equality. Put

$$b_j = a_j e^{-\lambda_j S}, \qquad c_k = d_k e^{-\mu_k S}, \qquad \sum b_j = \sum c_k = 1.$$

Let $\mathcal{H}$ be the closure in $\mathbb{T}^{N+M}$ of the one-parameter subgroup $t(\lambda, \mu)$, and let $\mathcal{C} = (\theta, \psi) + \mathcal{H}$. Every pair of functions with phases in $\mathcal{C}$ is a locally uniform limit of common vertical translates of (F, G). Such a limit pair still has the same zero set in the observed half-plane. Indeed, a zero of one limit function is approximated by zeros of its approximants inside an arbitrarily small disk contained in the half-plane; those are zeros of the other approximants by (10.4.2). Passing to the limit makes the disk center a zero of the other limit function. Reverse the roles for the converse. No limit function is identically zero, since its constant term dominates as $\Re s \to +\infty$.

The first projection of $\mathcal{C}$ is all of $\mathbb{T}^N$. Consequently it contains some $(0, y)$. The corresponding first function vanishes at S, so the second one does also. Hence $\sum_k c_k e^{iy_k} = 1$. Equality in the triangle inequality, with all $c_k > 0$ and their sum one, forces $y = 0$. In particular $(0, 0) \in \mathcal{C}$, so $\mathcal{C} = \mathcal{H}$. The same argument shows that the kernel of the first projection $\mathcal{H} \to \mathbb{T}^N$ is trivial. The second projection is likewise surjective and has trivial kernel.

Both projections are therefore continuous isomorphisms of compact groups. The resulting isomorphism $\mathbb{T}^N \to \mathbb{T}^M$ is given by an integer matrix: each of its component characters is an integer character of $\mathbb{T}^N$. Applying the same argument to the inverse gives integer matrices inverse to each other. Thus $N = M$, the matrix K is unimodular, and $\mathcal{H}$ is its graph. Differentiating the identity $t\mu = K(t\lambda) \bmod 2\pi$ at $t = 0$ yields $\mu = K\lambda$. Membership of (θ, ψ) in the graph gives the phase identity. The limit-pair argument above proves the last assertion. □

Proof of Theorem 10.4.1. Use the notation and matrix supplied by Lemma 10.4.2. In the Laurent polynomial ring $\mathcal{R} = \mathbb{C}[z_1^{\pm 1}, \dots, z_N^{\pm 1}]$ set

$$P(z) = 1 - \sum_j b_j z_j, \qquad Q(z) = 1 - \sum_k c_k z^{K_k}, \tag{10.4.4}$$

where K_k is the k-th row of K. We first show that P divides Q in $\mathcal{R}$.

For z complex and near $\mathbf{1}$, the equation

$$\mathcal{F}(u, z) = 1 - \sum_j b_j z_j e^{-\lambda_j u} = 0$$

has a unique small holomorphic solution $u = \rho(z)$, since $\partial_u \mathcal{F}(0, \mathbf{1}) = \sum b_j \lambda_j > 0$. For $z_j = e^{ix_j}$ and small real x, the corresponding root $S + \rho(z)$ lies in the observed half-plane. The graph lemma gives

$$1 - \sum_k c_k z^{K_k} e^{-\mu_k \rho(z)} = 0.$$

The expression on the left is holomorphic near $\mathbf{1}$ and vanishes on an open part of the product of unit circles. Successive one-variable identity theorems, or the substitution $z = e^{iw}$, show that it vanishes identically in a complex neighborhood. If $P(z) = 0$ in that neighborhood, then the uniqueness of the implicit root gives $\rho(z) = 0$. Therefore $Q(z) = 0$ on an open part of the affine hyperplane $P(z) = 0$.

Multiply Q by a Laurent monomial to clear all negative exponents. Substitute $z_N = (1 - \sum_{j<N} b_j z_j)/b_N$ in the resulting polynomial. It vanishes on an open set and hence identically. Polynomial division by the linear polynomial P proves divisibility. The clearing monomial is a unit in $\mathcal{R}$, so $P \mid Q$ in that ring. This argument includes $N = 1$, when the hyperplane is a single point and ordinary division by a linear factor applies.

The polynomial P is irreducible in $\mathcal{R}$: it is linear and irreducible in the ordinary polynomial ring, and it divides no monomial, so localization by the coordinate variables preserves its primality. Likewise $1 - \sum c_k z_k$ is irreducible. The unimodular substitution $z_k \mapsto z^{K_k}$ is an automorphism of $\mathcal{R}$, so Q is irreducible as well. Consequently

$$Q(z) = \gamma z^{v} P(z) \tag{10.4.5}$$

for some $\gamma \neq 0$ and $v \in \mathbb{Z}^N$; the units of a Laurent polynomial ring are precisely the nonzero monomials.

Evaluate exponent vectors by the real linear functional $k \mapsto k \cdot \lambda$. The smallest frequency in the support of Q is 0, because the others are the strictly positive numbers μ_k. The smallest frequency in the right side of (10.4.5) is $v \cdot \lambda$, because those of P are $0, \lambda_1, \dots, \lambda_N$. It follows that $v \cdot \lambda = 0$. Rational independence gives $v = 0$, and the constant coefficient gives $\gamma = 1$. Thus the rows of K are the standard basis vectors in some order, and the corresponding c_k and b_j coincide. Equation (10.4.3) now identifies frequencies and phases; normalization at the common S identifies the original magnitudes. This proves the theorem. □

Corollary 10.4.3 (Zero-density exact discrepancies cannot hide a support)**.** *Keep the model and independence hypotheses of Theorem 10.4.1, with $\sigma_0 < \min(S_F, S_G)$. Let $D(T)$ count the distinct points in the symmetric difference of their zero sets satisfying $\Re s > \sigma_0$ and $|\Im s| \le T$. The conclusion of the theorem still holds if exact equality is replaced by $D(T) = o(T)$. Finite changes or exact discrepancies of zero vertical density therefore cannot conceal a different support.*

Proof. We prove that $D(T) = o(T)$ already forces equality of the two zero sets throughout the observed half-plane. This step does not require rational independence. Suppose, to the

contrary, that z_0 is a zero of F and not of G there. Choose a closed disk of radius $r > 0$ about z_0 contained in the half-plane, with no zero of G in the disk and no zero of F on its boundary. For every phase pair in a sufficiently small relative open neighborhood U of (θ, ψ) in its joint translation hull, Rouché's theorem preserves an F zero in the disk and the absence of any G zero there.

Translation on the joint hull is uniquely ergodic for its Haar measure: its characters have nonzero time frequencies except for the characters trivial on that hull. A continuous nonnegative function supported in U with positive Haar integral therefore shows that the set E_T of times $t \in [0, T]$ whose translated phase lies in U has $|E_T| \geq cT$ for all large T. For each such time there is an unmatched zero of F in the translated disk centered at $z_0 + it$. A fixed unmatched zero can account for a set of such times of length at most $2r$. Consequently

$$|E_T| \leq 2rD(T + C) = o(T),$$

where C is fixed. These counts are finite because the zeros of each finite sum are locally finite and confined to a fixed vertical strip. This is a contradiction. Interchanging F and G excludes the opposite kind of discrepancy. Exact equality now follows, and Theorem 10.4.1 applies. □

Theorem 10.4.4 (A finite local inverse at each independent support)**.** *Fix one model F from Theorem 10.4.1, its term count N, and a half-plane $\Re s > \sigma_0$ with $\sigma_0 < S_F$. Parametrize nearby models by the real vector*

$$p = (\lambda_1, \dots, \lambda_N, \Re\alpha_1, \Im\alpha_1, \dots, \Re\alpha_N, \Im\alpha_N), \qquad \alpha_j = a_j e^{i\theta_j}.$$

One can select at most $3N$ distinct simple zeros of F in that half-plane and pairwise disjoint disks about them such that their locally matched zero coordinates determine p real-analytically on a neighborhood of the base model. If $\Psi(p)$ is the vector of these matched complex zero coordinates, this neighborhood can be chosen so that

$$\|p - q\| \leq C_F \|\Psi(p) - \Psi(q)\|. \tag{10.4.6}$$

The nearby candidate models have the same term count and ordered positive frequencies; they need not themselves be rationally independent. The selected disks, neighborhood, and constant C_F depend on the base model. No uniform conditioning over all independent supports is asserted.

Proof. We first show that the derivatives of all observed simple-zero coordinates separate the tangent space. Write a tangent vector as (ν_j, β_j) with $\nu_j \in \mathbb{R}$ and $\beta_j \in \mathbb{C}$. The corresponding function derivative is

$$H(s) = -\sum_j (\beta_j - \alpha_j \nu_j s) e^{-\lambda_j s}.$$

The derivative of a simple zero ρ is $-H(\rho)/F'(\rho)$. Suppose it vanishes at every simple zero in the observed half-plane. Thus H vanishes at all those zeros.

For every real phase x sufficiently close to zero, the frozen function F_x has a unique simple root $\rho(x)$ near S_F, still strictly inside the observed half-plane. Rational independence supplies a sequence $t_n \to +\infty$ with $\theta - t_n\lambda \to x$ modulo 2π. Arbitrarily large times can be used in every phase neighborhood because the translation flow is recurrent and dense. Locally uniform convergence and Rouché's theorem give unique simple roots $\rho_n \to \rho(x)$ of $F(s + it_n)$ in a small disk about $\rho(x)$. In particular $\rho_n + it_n$ is an observed simple zero of the base function, so $H(\rho_n + it_n) = 0$.

On every fixed compact set of s, division by t_n gives the locally uniform limit

$$\frac{H(s + it_n)}{t_n} \longrightarrow i \sum_j a_j \nu_j e^{ix_j} e^{-\lambda_j s}.$$

Thus this last sum vanishes at $s = \rho(x)$ for every small real x. Normalize at S_F, putting $b_j = a_j e^{-\lambda_j S_F}$. The same implicit-root complexification used in the preceding theorem shows that the linear polynomial $\sum b_j \nu_j z_j$ vanishes on an open part of the hyperplane $1 - \sum b_j z_j = 0$. It is therefore a scalar multiple of that affine linear polynomial. Its constant term is zero, so that scalar and all its coefficients vanish: $\nu_j = 0$ for every j.

With the frequency part removed, repeat the same limiting argument without dividing $H(s + it_n)$ by t_n. Its limit is $-\sum \beta_j e^{-i\theta_j} e^{ix_j} e^{-\lambda_j s}$. The resulting linear polynomial on the same hyperplane has coefficients $\beta_j e^{-i\theta_j} e^{-\lambda_j S_F}$ and zero constant term. The identical argument gives $\beta_j = 0$. This proves separation of the full $3N$-dimensional real tangent space.

Starting with that tangent space, select a simple-zero derivative which is nonzero on the current common kernel, and intersect with its kernel. Each selection lowers the dimension by at least one. After at most $3N$ distinct zeros, the joint derivative has trivial kernel. Choose disjoint small disks around the selected zeros. Rouché's theorem and the implicit function theorem define their unique matched roots for all parameters in a common neighborhood, and these roots depend real-analytically on the parameters.

The Jacobian $A = D\Psi(p_0)$ at the base point is injective. Some $3N$ real coordinate rows form an invertible square minor, so the inverse function theorem gives a real-analytic local inverse using those coordinates. If $s_* > 0$ is the smallest singular value of A, shrink to a convex parameter ball on which $\|D\Psi - A\| \le s_*/2$. Integration along a segment gives

$$\|\Psi(p) - \Psi(q)\| \ge \|A(p - q)\| - (s_*/2)\|p - q\| \ge (s_*/2)\|p - q\|.$$

This is (10.4.6) with $C_F = 2/s_*$. It also explains why neither a prescribed finite height nor a uniform constant follows from this local theorem. □

The global inverse observes complex zero locations throughout a half-plane, allowing the zero-density exception just proved. The finite local inverse instead fixes the term count and matches zeros inside specified disks. A right endpoint or the distribution of real parts alone can fail to identify coefficients [68]. The general factorization and common-zero questions are discussed in [130]; in particular its general irreducible common-zero statement carries the Schanuel-conjecture assumption. No such assumption or result is used in the proof above. Historical priority for this precise restricted inverse statement remains to be assessed beyond the checked primary sources.

10.4.2 A sharp resonance escape law and finite-record ambiguity

Fix $c \in I = [1/4, 1/2]$ and a small positive detuning d. Define

$$g_{c,d}(s) = 1 - (2 + c)e^{-s} + 2ce^{-(2+d)s}. \tag{10.4.7}$$

For irrational d, the frequencies $(1, 2 + d)$ are rationally independent. Their separation is at least one, and all nonconstant coefficient magnitudes lie in the fixed interval $[1/2, 5/2]$. Let $S_{c,d}$ solve

$$(2 + c)e^{-S_{c,d}} + 2ce^{-(2+d)S_{c,d}} = 1. \tag{10.4.8}$$

For irrational d, this is the actual right zero boundary. Put $u_c = (2 + c)e^{-S_{c,0}} \in (0, 1)$ and

$$\kappa_c = \left[\frac{2(2 - u_c)^3}{u_c(1 - u_c)}\right]^{1/2}. \tag{10.4.9}$$

Theorem 10.4.5 (Resonance escape at a precisely determined height)**.** *Let $T_{c,d}$ be the infimum of positive ordinates of zeros of $g_{c,d}$ in the strict strip $S_{c,d}-d<\Re s<S_{c,d}$. Uniformly for $c\in I$, as $d\downarrow 0$ through irrational values,*

$$dT_{c,d}=\pi-\kappa_c\sqrt{d}+O(d). \tag{10.4.10}$$

In particular $T_{c,d}\sim\pi/d$. This is a first-arrival law, not an average-density assertion or an effective ergodic theorem.

Proof. Write $S=S_{c,d}$ and

$$u=(2+c)e^{-S},\qquad 1-u=2ce^{-(2+d)S},\qquad M_d=u+(2+d)(1-u).$$

All these quantities vary smoothly on a compact neighborhood of $I\times\{0\}$; u and $1-u$ are uniformly positive. At a zero $s=S-x+it$ in the strip, define its small phases, whenever they exist, by

$$\eta_1=-t\pmod{2\pi},\qquad \eta_2=\pi-(2+d)t\pmod{2\pi}.$$

The exact triangle-defect identity is

$$ue^x|e^{i\eta_1}-1|^2+(1-u)e^{(2+d)x}|e^{i\eta_2}-1|^2=2\{ue^x+(1-u)e^{(2+d)x}-1\}.$$

For $0<x<d$ it forces the unique small representatives $\eta_1,\eta_2=O(\sqrt{d})$, uniformly. Consequently their relation coordinate

$$\xi=2\eta_1-\eta_2=dt-\pi\pmod{2\pi} \tag{10.4.11}$$

is also $O(\sqrt{d})$.

Parametrize all small zero phases by setting $\eta_1=A$, $\eta_2=2A-\xi$ and solving

$$1-ue^{x+iA}-(1-u)e^{(2+d)x+i(2A-\xi)}=0 \tag{10.4.12}$$

for (x,A). The real and imaginary derivatives in those two variables at zero are $-M_d$ and $-(2-u)$, respectively. The implicit-function theorem gives analytic functions $x_d(\xi)$ and $A_d(\xi)$ on a fixed neighborhood, uniformly in c,d. Conjugation makes x_d even and A_d odd. Expanding gives

$$A_d(\xi)=\frac{1-u}{2-u}\xi+O(\xi^3), \tag{10.4.13}$$

$$x_d(\xi)=C_d\xi^2+O(\xi^4),\qquad C_d=\frac{u(1-u)}{2M_d(2-u)^2}>0. \tag{10.4.14}$$

For example, the quadratic numerator is $u[(1-u)/(2-u)]^2+(1-u)[u/(2-u)]^2$; division by $2M_d$ gives the displayed coefficient. All remainders are uniform.

For sufficiently small positive d, the condition $0<x_d(\xi)<d$ is therefore equivalent in this chart to

$$0<|\xi|<r_d,\qquad r_d=\sqrt{d/C_d}\,(1+O(d))=\kappa_c\sqrt{d}+O(d^{3/2}).$$

The second equality uses smooth dependence on d and $C_0=u_c(1-u_c)/(2(2-u_c)^3)$. The defect identity proves that this chart includes every zero in the strip, not just a selected local branch.

The first possible positive-height group of such zeros has $dt=\pi+\xi$, with $|\xi|<r_d$. The phase equation $\eta_1=A_d(\xi)=-t \bmod 2\pi$ is equivalent to

$$\xi+dA_d(\xi)=2\pi nd-\pi,\qquad n\in\mathbb{Z}. \tag{10.4.15}$$

The function on the left is strictly increasing, with derivative $1+O(d)$. As n increases, the right side advances in steps of exactly $2\pi d$. Since $r_d \asymp \sqrt{d} \gg d$, there is an admissible first lattice point in the interval corresponding to $-r_d < \xi < r_d$, and its solution satisfies

$$\xi = -r_d + O(d).$$

If a lattice point lands exactly on the excluded boundary, use the next one, which changes this bound by at most $O(d)$. For small d this first point has negative ξ, so the excluded value $\xi = 0$ cannot interfere. Equation (10.4.11) excludes all smaller positive heights outside this first group; its previous possible group is centered at $-\pi/d$. Every solution of (10.4.15) gives an actual zero, since both phase equations and (10.4.12) then hold. Thus $dT_{c,d} = \pi - r_d + O(d)$, proving (10.4.10). □

The same detuned family yields a quantitative obstruction to noisy inversion even when its two frequencies are supplied to the observer. Let

$$\rho_n = \log 2 + 2\pi i n, \qquad \mathcal{C}_n = \{s : \Re s > 0, |\Im s - 2\pi n| < \pi\}.$$

Proposition 10.4.6 (Uniform ambiguity of finite noisy zero records)**.** *There are absolute constants $k, K > 0$ such that, for every $c \in I$ and integer $L \geq 0$, if $d(1+L) \leq k$, then $g_{c,d}$ has exactly one zero $\rho_{n,c,d}$ in each $\mathcal{C}_n$, $|n| \leq L$, and no other zeros in their union. The zeros are simple and satisfy*

$$|\rho_{n,c,d} - \rho_n| \leq Kd(1+|n|). \tag{10.4.16}$$

Moreover, uniformly in that regime,

$$\rho_{n,c,d} = \rho_n + \frac{c}{2-c}\, d\rho_n + O(d^2(1+|n|)^2). \tag{10.4.17}$$

For $c_- = 1/4$ and $c_+ = 1/2$, the two models have a fixed positive coefficient separation and a fixed positive separation of their right boundaries as $d \downarrow 0$, yet their records in these cells admit a pairing at distance at most $2Kd(1+L)$.

Proof. At $d = 0$ the exact factorization is

$$g_{c,0}(s) = (1 - 2e^{-s})(1 - ce^{-s}). \tag{10.4.18}$$

The second factor has all its zeros in $\Re s = \log c < 0$. In $\Re s \geq 0$ its modulus is at least $1 - c \geq 1/2$. On $\Re s = 0$, the first factor has modulus at least one; on the horizontal sides of a cell it is $1 + 2e^{-\Re s}$. Choose a fixed large right boundary R on which the constant term dominates every $g_{c,d}$ for all small $d \geq 0$. The modulus of $g_{c,0}$ is bounded below by a fixed positive constant on all four sides of these rectangles.

For $s = \sigma + it$, $\sigma \geq 0$, the elementary integral bound $|e^{-ds} - 1| \leq d|s|$ gives

$$|g_{c,d}(s) - g_{c,0}(s)| \leq 2cde^{-2\sigma}(\sigma + |t|) \leq Kd(1+|t|). \tag{10.4.19}$$

If k is sufficiently small, Rouché's theorem on each rectangle proves that it contains exactly one zero, counted with multiplicity. Constant-term domination excludes zeros to its right. This proves the exact count and simplicity, and also excludes zeros on the horizontal or left boundaries.

On a fixed disk of radius at most $1/4$ around ρ_n, the factorization (10.4.18) gives $|g_{c,0}(s)| \geq k_0|s - \rho_n|$, uniformly in c, n. Applying Rouché on the circle of radius $K_0 d(1+|n|)$, with K_0 large and then k small, proves (10.4.16).

For the expansion put $s = \rho_n + z$. Periodicity of the integer-frequency factors makes

$$g_{c,d}(\rho_n + z) = 1 - \frac{2+c}{2}e^{-z} + \frac{c}{2}e^{-2z}e^{-d\rho_n - dz}.$$

Use $z = O(d(1+|n|))$ in its Taylor expansion. Its linear terms are $(1-c/2)z - (c/2)d\rho_n$, and the remaining terms are $O(d^2(1+|n|)^2)$. Division by the uniformly positive $1 - c/2$ proves (10.4.17).

The magnitudes $(2+c, 2c)$ differ by a fixed amount for c_- and c_+. At $d = 0$, the left side of (10.4.8), for fixed S, is strictly increasing in c and strictly decreasing in S. Thus $S_{c_+,0} > S_{c_-,0}$. Smooth dependence on d preserves half that positive separation for all sufficiently small d. Finally pair zeros with the same index and apply (10.4.16) twice. □

Corollary 10.4.7 (No uniform consistency without a resonance margin)**.** *Suppose an observer receives all zeros in the cells $|n| \le L$, each with arbitrary complex location error at most ϵ, and is given the exact frequencies $(1, 2+d)$, the atom count two, and the fixed phases $(0, \pi)$. There is a constant $\gamma > 0$ such that, whenever $Kd(1+L) \le \epsilon$ and d is sufficiently small and irrational, some common reported record is consistent with both c_- and c_+. Every estimator therefore makes coefficient error at least $1/8$, or right-boundary error at least γ, on one of these two models, when estimating the respective target. For any finite observation schedule $L = L(\epsilon)$, uniform consistency over this internally independent model class is impossible as $\epsilon \downarrow 0$.*

Proof. Report the midpoint of each pair $(\rho_{n,c_-,d}, \rho_{n,c_+,d})$. The preceding pairing bound puts this record within the required error of either truth, without any missing or inserted zero. The two first coefficient magnitudes differ by $1/4$, so any single estimate has error at least $1/8$ for one truth. Their right boundaries have a uniform positive separation, giving the analogous γ lower bound. For each positive ϵ and finite $L(\epsilon)$ one can choose an irrational positive $d < \epsilon/[K(1+L(\epsilon))]$ satisfying all the smallness conditions. This proves the uniform inconsistency assertion. □

There is no contradiction with Theorem 10.4.1: the finite records are close, not equal, while the infinite exact records eventually reveal the distinct models. Nor does the corollary rule out exact finite-data identification in infinite precision. The obstruction concerns noise uniformly over supports arbitrarily close to an integer relation. Ordinary pairwise frequency separation does not control this effect. The detuning $d = \lambda_2 - 2\lambda_1$ is the relevant small quantity.

10.4.3 Dependencies, scope, and unresolved problems

The unknown-support theorem extends the earlier prescribed-support inverse result by introducing the joint-hull graph argument and the Laurent-hyperplane rigidity step. Its proof repeats the elementary frozen-edge argument only as a dependency, not as a new target. The zero-density corollary and the finite local inverse are dependent extensions of that first target, not extra target counts. The local inverse is compatible with the nonuniform finite-noise obstruction: its conditioning and selected observation heights may deteriorate as the support approaches a rational relation. The escape law and finite-noise obstruction form a single second target: they quantify the observation cost of approaching a rational relation. The exact factorization at zero detuning was already an elementary obstruction in the preceding manuscript; the first-arrival asymptotic, its explicit square-root correction, and the uniform noisy-record conclusion are the new assertions here.

Avellar–Hale and the discussion in Sepulcre [119, 68] already establish that small frequency changes can radically change the spectral boundary. Our additional claims concern how long the change can remain unobserved and what finite noisy zero data can distinguish. These

are proved for the stated two-frequency family, not for all near-resonant exponential sums. A general quantitative inverse theorem with unknown independent support, a sharp dependence on a suitable Diophantine margin, and extension to infinite Dirichlet supports remain open in this work. No assertion about the zeros of the Riemann zeta function follows from these finite-sum results.

10.5 Without scale moments, terminal sampling has a different boundary

The preceding manuscript proved that every infinite collection of separate sum laws identifies a continuously mixed symmetric stable index and scale, provided the scale has a positive and a negative moment. We now remove that assumption and obtain a different exact criterion. We then change the observation experiment: two jointly observed replicates recover the entire mixing law without any scale moment. The latter result extends to correlated location, scale, index, and skewness with six replicates.

These are two internal research targets: the necessity of the scale moment assumption, and finite replication with continuously varying stable parameters. They are not named conjectures in the external literature. Classical exponential completeness, positive stable subordination, and transform uniqueness are acknowledged inputs. Historical priority of the specific mixture statements is not certified.

Fix $0 < \alpha_0 < \alpha_1 \le 2$. In the first two parts, conditional on $(\alpha, B) \in [\alpha_0, \alpha_1] \times (0, \infty)$, the variables X_j are independent with characteristic function

$$\mathbb{E}(e^{itX_j} \mid \alpha, B) = e^{-B|t|^\alpha}. \tag{10.5.1}$$

The law of (α, B) is otherwise unrestricted. Put $S_n = X_1 + \cdots + X_n$. Separate laws of selected S_n do not include the joint laws of observations from a common latent draw.

For a locally finite real set Λ, let

$$\mathcal{R}(\Lambda) = \sup\{a > 0 : \operatorname{span}\{e^{i\lambda x} : \lambda \in \Lambda\} \text{ is dense in } L^2(-a, a)\}.$$

The supremum of the empty set is zero. The classical Beurling–Malliavin theorem identifies this radius with $\pi D_{\text{eff}}(\Lambda)$ in the effective-density convention of [121, Section 1.1].

Theorem 10.5.1 (Exact moment-free terminal criterion)**.** *For $\mathcal{N} \subset \mathbb{N}$, the separate population laws $\{\mathcal{L}(S_n) : n \in \mathcal{N}\}$ identify every mixing law in (10.5.1) if and only if*

$$\mathcal{R}(\{\log n : n \in \mathcal{N}\}) = \infty. \tag{10.5.2}$$

If this radius is finite, there are distinct absolutely continuous witnesses with index supported inside any prescribed nonempty open subinterval of (α_0, α_1). They have the same index marginal and the same conditional moments of $\log B$ of every nonnegative integer order. All absolute log-scale moments are finite, but the witnesses can both be chosen to satisfy

$$\mathbb{E}B^\epsilon = \mathbb{E}B^{-\epsilon} = \infty \quad \text{for every } \epsilon > 0.$$

Their signed difference is smooth after the change of coordinates $(\alpha, B) \mapsto (1/\alpha, -\log B/\alpha)$, compact in the first coordinate and Schwartz in the second.

The geometric design $n_j = 2^j$ illustrates the change: its logarithms are uniformly separated and have finite completeness radius, so it fails in the unrestricted class. The preceding positive-and-negative scale-moment theorem identifies the mixing law from this same infinite design [125]. Having every log-scale moment, even with those moments supplied as extra conditional data, does not repair the failure proved here.

10.5.1 A projection representation requiring no moments

Let E be an independent mean-one exponential variable and define

$$R = 1/\alpha, \qquad Q = -R \log B, \qquad C = Q + R \log E. \tag{10.5.3}$$

For every real x and positive integer n,

$$\mathbb{E} e^{ie^x S_n} = \mathbb{E} e^{-nBe^{\alpha x}} = \mathbb{P}(C - R \log n > x). \tag{10.5.4}$$

This is an identity between population transforms, not an assumption that the auxiliary variable E is observed. The selected sum laws are equivalent to the indicated projection laws. In particular, positive frequencies determine the symmetric characteristic functions at negative frequencies as well.

The map from the joint law of (Q, R) to that of (C, R) is injective on finite complex measures, not only on probabilities. For a fixed real t, conditional Fourier transformation multiplies the complex measure in r by

$$\mathbb{E} E^{itr} = \Gamma(1 + itr). \tag{10.5.5}$$

This factor never vanishes and has a bounded reciprocal on the compact interval $[1/\alpha_1, 1/\alpha_0]$. Division for each fixed t, followed by Fourier uniqueness, proves the assertion. No moment of Q, C, or B is used.

We record two immediate consequences of the definition of completeness radius. If $\mathcal{R}(\Lambda) = \infty$, a compactly supported finite complex measure whose Fourier transform vanishes on Λ must be zero: convolution with a compact smooth approximate identity gives compactly supported L^2 annihilators, which vanish by completeness on sufficiently large intervals. Conversely, if the radius is finite, there is a nonzero complex $f \in C_c^\infty(\mathbb{R})$ with

$$\int e^{-i\lambda y} f(y)\, dy = 0 \quad (\lambda \in \Lambda). \tag{10.5.6}$$

Indeed choose an L^2 annihilator on an interval larger than the radius and convolve it with a nonzero compact smooth bump. The convolution is nonzero because its entire Fourier transform is a product of two nonzero entire functions. Conjugation changes the Fourier sign if needed, and translation of f preserves all its zeros.

Proof of Theorem 10.5.1. Suppose first that (10.5.2) holds. Let σ be the difference of the induced laws of (C, R) for two candidates. For fixed $t \neq 0$, the compact complex measure

$$\nu_t(dr) = \int e^{itc}\, \sigma(dc, dr)$$

has Fourier zeros at $-t \log n$ by (10.5.4). Scaling or reflecting frequencies preserves whether the completeness radius is infinite. Thus $\nu_t = 0$. Continuity gives the same conclusion at $t = 0$, and Fourier uniqueness after restriction to arbitrary Borel sets in r gives $\sigma = 0$. Equation (10.5.5) recovers the law of (Q, R) and hence of (α, B).

Now assume the radius is finite and choose f satisfying (10.5.6) for $\Lambda = \{\log n : n \in \mathcal{N}\}$. Let J be a compact nondegenerate interval strictly inside the prescribed reciprocal-index interval. Choose $t_0 > 0$ sufficiently large, and translate f, so that $\operatorname{supp} f/t_0$ lies strictly inside J. Choose a nonzero real $h \in C_c^\infty((0, \infty))$ with support sufficiently close to t_0 that $\operatorname{supp} f/t \subset \operatorname{int} J$ throughout that support. For $t > 0$ define

$$G_C(t, r) = h(t) t f(tr), \qquad G_Q(t, r) = \frac{G_C(t, r)}{\Gamma(1 + itr)}. \tag{10.5.7}$$

Extend both functions to negative t by conjugation and by zero near $t = 0$. They are smooth with compact support in (t, r). Their inverse Fourier transforms in t, denoted $g_C(c, r)$ and $g_Q(q, r)$, are real, nonzero, compactly supported in $r \in J$, and Schwartz in their other coordinate, uniformly in r. Equation (10.5.5) shows that adding $R \log E$ to the signed measure $g_Q(q, r)\, dq\, dr$ gives exactly $g_C(c, r)\, dc\, dr$.

For $t > 0$ and every selected n,

$$\int e^{-itr \log n} G_C(t, r)\, dr = h(t) \int e^{-iy \log n} f(y)\, dy = 0.$$

Negative t follows by conjugation. Consequently every selected projection $C - R \log n$ has zero signed law, and (10.5.4) gives zero signed stable sum law. Since G_Q vanishes in a neighborhood of $t = 0$,

$$\int q^j g_Q(q, r)\, dq = 0 \quad (r \in J,\ j = 0, 1, 2, \ldots). \tag{10.5.8}$$

The positive and negative parts have the same nonzero total mass. Normalize them to probabilities and push forward by $\alpha = 1/r$, $B = e^{-q/r}$. Their index marginals and all conditional log-scale moments agree by (10.5.8); their absolute log-scale moments are finite by the uniform Schwartz bound.

Finally mix each witness, with the same fixed positive weight, with one common probability having a smooth index density in the allowed interval and, independently in (Q, R) coordinates, a Q density proportional to $\exp\{-(1 + q^2)^{1/4}\}$. This common law has all absolute log-scale moments and has infinite positive and negative scale moments of every positive order. Both modified witnesses therefore have these properties, while all observational equalities, conditional moment equalities, and nonzeroness of their difference are preserved. □

10.6 Two joint replicates recover an arbitrary index–scale mixture

The change from separate sum laws to a joint replicated law supplies information that the preceding sparse designs discard. The following result permits arbitrary continuous mixing and does not bound the number of mixture components. General finite-mixture grouped-sample results, such as [31, 131], address a different scope.

Theorem 10.6.1 (Exact replication threshold at zero location)**.** *For the class* (10.5.1)*, the joint population law of* (X_1, X_2) *identifies every Borel mixing law of* (α, B) *without any moment assumption. More strongly, the map*

$$\mu \longmapsto \left\{ \int e^{-b(|u|^\alpha + |v|^\alpha)}\, d\mu(\alpha, b) : u, v \in \mathbb{R} \right\} \tag{10.6.1}$$

is injective on finite complex measures on $[\alpha_0, \alpha_1] \times (0, \infty)$*. One replicate does not identify all probability mixtures on any nondegenerate index interval. Thus two is the exact minimal number of joint replicates for this class.*

10.6.1 A triangular dilation lemma

Lemma 10.6.2 (Recovery from integer dilations)**.** *Let* $0 < a < b < \infty$*, and let* c_m *be bounded Borel functions on* $[a, b]$ *satisfying* $c_1 = 1$ *and* $\sup_{\alpha \in [a,b]} |c_m(\alpha)| \le C m^p$ *for some fixed* C, p*. If a finite complex measure* η *on* $[a, b]$ *satisfies*

$$\sum_{m \ge 1} \int c_m(\alpha) e^{-m\alpha d}\, d\eta(\alpha) = 0 \quad (d > 0), \tag{10.6.2}$$

then $\eta = 0$.

Proof. Let $D_m(\alpha) = m\alpha$ and form the locally finite complex measure on $[a, \infty)$

$$M_\eta = \sum_{m \geq 1} (D_m)_*(c_m \eta).$$

Only finitely many terms meet any bounded interval, and $\int e^{-dx}\, d|M_\eta|(x) < \infty$ for every $d > 0$ by the polynomial coefficient bound. Equation (10.6.2) says that its Laplace transform is zero. For completeness, fix $d_0 > 0$, multiply the measure by $e^{-d_0 x}$, and push forward by $y = e^{-x}$. Its moments of every nonnegative integer order vanish. Polynomial density in the continuous functions on $[0, e^{-a}]$ gives the zero measure, so $M_\eta = 0$.

On $[a, 2a)$ only $m = 1$ contributes, hence $\eta = 0$ there. If η is already zero on $[a, L)$, then on $[a, 2L)$ every term with $m \geq 2$ comes from $\alpha < L$ and is zero. Thus $M_\eta = 0$ forces $\eta = 0$ on $[a, 2L)$. Iteration reaches past b after finitely many steps. This proves the lemma and also describes a recursive recovery of η from M_η. □

10.6.2 The two-replicate inversion

Proof of Theorem 10.6.1. It suffices to prove that a finite complex measure in the kernel of (10.6.1) is zero. Push that measure to (Q, R) coordinates by (10.5.3), writing it as ρ. Let E_1, E_2 be independent mean-one exponential variables and set

$$Y_j = Q + R \log E_j, \qquad j = 1, 2.$$

The induced finite complex measure of (Y_1, Y_2) has upper-orthant values

$$\int \exp\{-B(e^{\alpha x} + e^{\alpha y})\}\, d\mu(\alpha, B), \tag{10.6.3}$$

which vanish by (10.6.1). Upper orthants determine finite complex measures: rectangle values follow by inclusion and exclusion, and their unbounded endpoints by finite-variation limits. Thus the induced measure of (Y_1, Y_2) is zero.

Fix a real t. Fourier-weight its first coordinate by e^{itY_1} and project to $D = Y_2 - Y_1$. This gives the zero complex measure on the real line. Write

$$\nu_t(dr) = \int e^{itq}\, \rho(dq, dr).$$

The density of that weighted difference is

$$T_t(d) = \int \frac{\Gamma(2 + itr)}{r} \frac{e^{d/r}}{(1 + e^{d/r})^{2+itr}}\, d\nu_t(r). \tag{10.6.4}$$

To verify the formula, set $E_1 = z$ and $E_2 = ze^{d/r}$ in the two exponential densities. The Jacobian is $ze^{d/r}/r$, while the Fourier weight contributes $e^{itq} z^{itr}$. Integration in $z > 0$ gives (10.6.4). Its integrand is continuous in d, locally uniformly in the compact positive r interval. Hence T_t is continuous and vanishes everywhere, not merely almost everywhere.

For $d > 0$, expand the denominator at $e^{-d/r}$ to obtain

$$e^{itd} T_t(d) = \sum_{m \geq 1} \int \frac{\Gamma(2 + itr)}{r} \frac{(-1)^{m-1} (2 + itr)_{m-1}}{(m-1)!} e^{-md/r}\, d\nu_t(r). \tag{10.6.5}$$

The cancellation of the factor e^{-itd} is exact. For $|tr| \le T$, the coefficients obey the elementary uniform bound

$$\frac{|(2+itr)_{m-1}|}{(m-1)!} = m \prod_{j=2}^{m} \left(1 + \frac{t^2 r^2}{j^2}\right)^{1/2} \le m \exp\left\{\frac{T^2}{2} \sum_{j=2}^{\infty} j^{-2}\right\}.$$

Consequently the series can be integrated absolutely for every $d > 0$. Push ν_t forward by $\alpha = 1/r$ and multiply it by the bounded nonvanishing function $\alpha\Gamma(2 + it/\alpha)$, obtaining a finite complex measure η_t on $[\alpha_0, \alpha_1]$. Equation (10.6.5) has exactly the form (10.6.2), with

$$c_m(\alpha) = \frac{(-1)^{m-1}(2 + it/\alpha)_{m-1}}{(m-1)!}.$$

Lemma 10.6.2 yields $\eta_t = 0$, hence $\nu_t = 0$. This holds for every real t. Fourier uniqueness in q after restriction to arbitrary Borel sets in r gives $\rho = 0$ and then $\mu = 0$. No moment differentiation or analytic continuation in the unbounded location coordinate Q has been used.

To prove failure with one replicate, choose $\alpha_0 \le \alpha_a < \alpha_b \le \alpha_1$ and put $\delta = \alpha_a/\alpha_b \in (0, 1)$. The classical positive stable variable V has Laplace transform $\mathbb{E}e^{-sV} = e^{-s^\delta}$; see [132, Equation (1.1)]. Its existence also follows directly from the Lévy measure $\delta x^{-1-\delta} dx/\Gamma(1-\delta)$, since its Laplace exponent is s^δ. The variable is strictly positive because its Laplace transform tends to zero at infinity. Therefore

$$e^{-|t|^{\alpha_a}} = \mathbb{E}e^{-V|t|^{\alpha_b}}.$$

The mixing laws $(\alpha, B) = (\alpha_a, 1)$ and $(\alpha, B) = (\alpha_b, V)$ are distinct and give the same law of one observation. This classical subordination obstruction establishes minimality; it is not itself claimed as a new result. □

The inversion above also proves identification of arbitrary location–scale mixtures of the reflected Gumbel kernel $Q + R \log E$ from two replicates, with R in a compact positive interval. Equivalently it identifies continuously mixed Weibull shape and rate from a joint pair of conditionally independent lifetimes. These are transform-equivalent consequences of the same proof, not separately counted discoveries. Transform transfer is a classical methodological theme; compare [133].

10.7 Contrasts and anchors recover correlated stable parameters

The complex-measure formulation in Theorem 10.6.1 is essential for recovering dependence on additional latent parameters. The following extensions make no independence assumption between those latent coordinates.

Corollary 10.7.1 (Five replicates with arbitrary symmetric location)**.** *Conditional on (A, α, B), let X_j be independent with characteristic function $e^{itA - B|t|^\alpha}$, where $A \in \mathbb{R}$, $\alpha \in [\alpha_0, \alpha_1]$, and $B > 0$. The joint law of five observations identifies every joint mixing law of (A, α, B), without moments of any coordinate. Five is a sufficient bound; its minimality is not asserted.*

Proof. Set $D_1 = X_2 - X_3$ and $D_2 = X_4 - X_5$. Given the latent draw, these are independent symmetric stable observations with exponent coefficient $2B$, independent also of X_1. For a difference ρ of two mixing laws giving the same joint observations and a fixed real t, define

$$\sigma_t(d\alpha, dB) = \int e^{itA - B|t|^\alpha} \rho(dA, d\alpha, dB).$$

The joint transform of (X_1, D_1, D_2) shows that

$$\int e^{-2B(|u|^\alpha+|v|^\alpha)}\, d\sigma_t = 0 \quad (u, v \in \mathbb{R}).$$

Theorem 10.6.1, applied after the invertible change $B \mapsto 2B$, gives $\sigma_t = 0$. The multiplier $e^{-B|t|^\alpha}$ is positive everywhere, so it can be removed by restricting first to bounded B intervals and then taking their union. Thus the Fourier transform in A of ρ restricted to each Borel set in (α, B) is zero. Fourier uniqueness proves $\rho = 0$. □

For varying skewness we specify the parametrization, since a location coordinate changes under different stable conventions. Put

$$\psi_\alpha(t) = \begin{cases} \tan(\pi\alpha/2)\,\mathrm{sgn}(t)|t|^\alpha, & \alpha \neq 1,\ 2, \\ -(2/\pi)t\log|t|, & \alpha = 1, \\ 0, & \alpha = 2, \end{cases} \qquad \psi_\alpha(0) = 0. \tag{10.7.1}$$

Use the standard stable kernels

$$\widehat{K}_{A,\alpha,B,\beta}(t) = \exp\{itA - B|t|^\alpha + iB\beta\psi_\alpha(t)\}. \tag{10.7.2}$$

Here B is the coefficient of $|t|^\alpha$ and $\beta \in [-1, 1]$. At $\alpha = 2$ impose $\beta = 0$, since skewness is otherwise a redundant, unobservable Gaussian parameter. The convention is allowed to be discontinuous in α at one; the proof needs only continuity in the observation frequencies for each fixed latent parameter, which holds.

Corollary 10.7.2 (Six replicates with all four stable parameters mixed)**.** *Let $0 < \alpha_0 < 2$ and let the conditional kernels be (10.7.2), with $A \in \mathbb{R}$, $\alpha \in [\alpha_0, 2]$, $B > 0$, and $\beta \in [-1, 1]$, subject to $\beta = 0$ at $\alpha = 2$. The joint population law of six conditionally independent observations identifies every Borel joint mixing law of (A, α, B, β). No coordinate moment and no latent-coordinate independence is required. Six is a sufficient bound, not an asserted minimal bound.*

Proof. Keep X_1, X_2 as anchors and set $D_1 = X_3 - X_4$, $D_2 = X_5 - X_6$. The function ψ_α is odd. Thus, conditional on all latent coordinates, D_1, D_2 are independent symmetric stable observations of coefficient $2B$, independent of the two anchors.

Let ρ be the difference of two candidate mixing laws. For fixed real p, q, integrate ρ over (A, β) with weight

$$e^{i(p+q)A - B(|p|^\alpha+|q|^\alpha)+iB\beta(\psi_\alpha(p)+\psi_\alpha(q))}.$$

This gives a finite complex measure on (α, B). The joint transform with the two contrasts places it in the kernel of (10.6.1), so it is zero by Theorem 10.6.1. Remove the strictly positive real factor $e^{-B(|p|^\alpha+|q|^\alpha)}$ as in the preceding proof.

To make the ensuing conditional argument simultaneous in the frequencies, dominate both (α, B) marginals by their sum λ and disintegrate ρ with respect to λ, obtaining finite signed conditional measures $\rho_{\alpha,B}$ on (A, β). First intersect the full-measure sets furnished by the countably many rational pairs (p, q). On that common set the identities extend to all real (p, q) by bounded convergence and frequency continuity. Hence, for almost every fixed (α, B),

$$\int e^{i(p+q)A+iB\beta(\psi_\alpha(p)+\psi_\alpha(q))}\, d\rho_{\alpha,B}(A, \beta) = 0 \quad (p, q \in \mathbb{R}). \tag{10.7.3}$$

Fix such a pair (α, B) with $\alpha < 2$ and a nonzero $u = p + q$. As p varies strictly between 0 and u when $u > 0$ (with reversed signs when $u < 0$), the values

$$B\{\psi_\alpha(p) + \psi_\alpha(u - p)\}$$

contain a nondegenerate interval. For $\alpha \neq 1$, this follows because $\tan(\pi\alpha/2) \neq 0$ and the second derivative of $p^\alpha + (u-p)^\alpha$ is $\alpha(\alpha-1)\{p^{\alpha-2} + (u-p)^{\alpha-2}\} \neq 0$. For $\alpha = 1$, the second derivative of $p\log p + (u-p)\log(u-p)$ is $1/p + 1/(u-p) > 0$.

For each fixed u, the function

$$z \longmapsto \int e^{iuA+iz\beta}\, d\rho_{\alpha,B}(A,\beta)$$

is entire because $|\beta| \leq 1$. By (10.7.3) it vanishes on that real interval, hence everywhere. Continuity also supplies $u = 0$. Fourier uniqueness in (A,β) gives $\rho_{\alpha,B} = 0$. If $\alpha = 2$, the imposed condition $\beta = 0$ reduces (10.7.3) directly to Fourier uniqueness in A. Therefore all conditional signed measures vanish and $\rho = 0$. □

Scope, previous work, and remaining barriers. The two main targets are the exact unrestricted-scale terminal criterion and the exact two-replicate threshold for zero-location symmetric mixtures. The five- and six-replicate results are extensions of the second mechanism and are not counted as separate independent closures. The classical BM theorem and the one-observation subordination identity are not new results. The consulted primary sources include the BM theorem [121], grouped finite-mixture identification [31, 131], and transform-based transfers [133]; they do not constitute an exhaustive priority search.

The lower bound $\alpha \geq \alpha_0 > 0$ is used essentially in the displayed triangular inversion. If indices approach zero, integer dilations have no positive initial interval from which to begin; this proof does not settle that case. Minimal replication with unknown location or skewness remains open, as does quantitative stability under noisy estimated characteristic functions. The finite-stage dilation recursion is an exact uniqueness mechanism after Laplace inversion; it is not a claim that statistical inversion is well conditioned. Nor do the results identify all four parameters from separate sum marginals: the anchor argument explicitly uses joint grouped observations.

10.8 Continuous spectra and arithmetic observation costs

The finite-atom theory leaves two different questions unanswered. What can a threshold curve determine when no finite bound on the number of frequencies is available? When can a single experiment near a transition be observed in a height of the best possible order? We answer the first question for arbitrary probability spectra, and the second for one hidden frequency. The first answer is a sharp logarithmic modulus, attained by a finite convex reconstruction. The second is an arithmetic classification of the observation height.

Neither underlying classical mechanism is claimed as a new principle. Polynomial approximation and moment matching are standard tools for Wasserstein recovery; see Han, Jiao, and Weissman [134]. The same factorial inverse scale also appeared for a different kernel in the earlier integrated manuscript [1]. Likewise, the equivalence between optimal covering of a circle rotation and bounded partial quotients belongs to classical Diophantine approximation; Beck and Chen [135] describe its continuous version and extensions. The results below establish the nonlinear reductions needed for these mechanisms to answer the two present observation problems.

10.8.1 An inverse without a bound on spectral complexity

Fix $0 < a < b < 1$, and let $\mathcal{P}[a,b]$ be all Borel probability measures on $[a,b]$. Define

$$S_\mu(x) = \int_a^b x^r\, d\mu(r), \qquad T_\mu(C) = \min_{x>0} \frac{1+Cx}{S_\mu(x)}, \qquad J = \left[\frac{a}{2e(1-a)}, \frac{2b}{1-b}\right]. \tag{10.8.1}$$

Write $d_J(\mu,\nu)=\|T_\mu-T_\nu\|_{L^\infty(J)}$ and

$$\Omega_J(\delta)=\sup\{W_1(\mu,\nu):\mu,\nu\in\mathcal{P}[a,b],\ d_J(\mu,\nu)\le\delta\}.$$

The Wasserstein distance uses the ordinary distance between exponents.

Theorem 10.8.1 (Sharp nonparametric threshold inverse)**.** *There are constants $c,K,\delta_0>0$, depending only on a,b, such that, for $0<\delta<\delta_0$,*

$$c\,\frac{\log\log(1/\delta)}{\log(1/\delta)}\le\Omega_J(\delta)\le K\,\frac{\log\log(1/\delta)}{\log(1/\delta)}. \tag{10.8.2}$$

In particular the exact threshold curve on J determines every probability spectrum, including continuous spectra. The lower bound holds even though the entire curve, rather than finitely many settings, is supplied.

If an oracle returns $Y(C)$ with $|Y(C)-T_\mu(C)|\le\varepsilon$ at each requested $C\in J$, a finite algorithm attains the upper bound in (10.8.2), with δ replaced by ε. It uses at most

$$K\frac{\log^2(1/\varepsilon)}{\log\log(1/\varepsilon)}$$

oracle calls and a linear feasibility problem with at most K/ε nonnegative variables. There is no atom-count, minimum-mass, or separation assumption.

The computational assertion uses exact arithmetic and bounded adversarial oracle errors. Its grid size is polynomial in inverse accuracy, not in the number of bits of accuracy. It therefore does not assert optimal computational complexity.

Lemma 10.8.2 (Uniform contact geometry)**.** *For every $\mu\in\mathcal{P}[a,b]$,*

$$S_\mu(x)=\min_{C\in J}\frac{1+Cx}{T_\mu(C)}\quad(1\le x\le e). \tag{10.8.3}$$

Consequently $d_J(\mu,\nu)\le\delta$ implies $|S_\mu(x)-S_\nu(x)|\le K\delta$ on $[1,e]$. Moreover, for any prescribed $x\in[1,e]$, an oracle of error ε estimates $S_\mu(x)$ to error $K\varepsilon$ using $K\log(3/\varepsilon)$ queries.

Proof. All derivatives of S may be taken under the integral. In particular $S'>0$, $S''<0$, and $H=S-xS'>0$. The unique minimizer x_C solves

$$C=\frac{S'(x_C)}{H(x_C)},\qquad\frac{dC}{dx}=\frac{S''(x)S(x)}{H(x)^2}<0,\qquad\frac{a}{1-a}\le Cx_C\le\frac{b}{1-b}.$$

The last inequality and the compactness of J put every x_C in a common compact subset of $(0,\infty)$. On this compact set, $S,H,-S''$, and $T(C)$ have uniform positive lower and upper bounds. The contact setting $C_x=S'(x)/H(x)$ for $x\in[1,e]$ lies in the interior of J. Every affine function $(1+Cx)/T(C)$ majorizes $S(x)$, with equality at x_C, proving (10.8.3). The assertion for two profiles follows by comparing the two minima; the uniform lower bound on T controls inversion of T.

For completeness the noisy scalar minimization is explicit. For $g_x(C)=(1+Cx)/T(C)$, differentiation of the contact formula gives

$$g_x'(C)=\frac{x-x_C}{T(C)(1+Cx_C)},\qquad c|C-C_x|\le|g_x'(C)|\le K|C-C_x|.$$

The observed g_x has error at most $\eta=K_0\varepsilon$. Maintain an interval $[L,U]$ containing C_x, and compare its two trisection points $C_1<C_2$. Define the observed difference as $\widetilde{g}_x(C_2)-\widetilde{g}_x(C_1)$.

If it is greater than 2η, discard the right third; if less than -2η, discard the left third. These decisions follow from unimodality. Otherwise the true difference has modulus at most 4η. When the interval width q satisfies $q^2 > 144\eta/c$, integration of the lower derivative bound shows that a minimizer outside the middle third would give a difference at least $cq^2/18 > 4\eta$ in modulus. Thus in the ambiguous case retain the middle third. Stop at $q^2 \le 144\eta/c$ and query the midpoint. The upper derivative bound gives error $K\eta$ in the minimum. Each step contracts by at least $2/3$, proving the query bound. □

Lemma 10.8.3 (Exponential samples control Wasserstein distance)**.** *For $k \ge 2$ and $\mu, \nu \in \mathcal{P}[a,b]$, put*

$$E_k = \max_{0 \le l \le k} \left| \int e^{lr/k} \, d(\mu - \nu)(r) \right|.$$

Then

$$W_1(\mu, \nu) \le \frac{K}{k} + E_k (Kk)^k. \tag{10.8.4}$$

Proof. Let $v = (e^{r/k} - e^{a/k})/d_k$, where $d_k = e^{b/k} - e^{a/k} \asymp k^{-1}$. This maps $[a,b]$ onto $[0,1]$, and its inverse $r_k(v) = k \log(e^{a/k} + d_k v)$ has a uniformly bounded derivative. For $0 \le j \le k$, expansion by the binomial theorem yields

$$\left| \int v(r)^j \, d(\mu - \nu)(r) \right| \le E_k d_k^{-j} (1 + e^{a/k})^j \le E_k (Kk)^j.$$

If f is 1-Lipschitz and $f(a) = 0$, the function $g(v) = f(r_k(v))$ is uniformly bounded and Lipschitz. There exists a polynomial $p(v) = \sum_{j=0}^{k} p_j v^j$ with

$$\|g - p\|_\infty \le K/k, \qquad \sum_{j=0}^{k} |p_j| \le K^k. \tag{10.8.5}$$

Here is a direct justification of the two bounds together. The even periodic function $g((1 + \cos\theta)/2)$ has a uniform Lipschitz constant. Convolve it with the normalized nonnegative trigonometric kernel

$$J_n(\theta) = c_n \left(\frac{\sin(n\theta/2)}{\sin(\theta/2)} \right)^4, \qquad n = \lfloor k/2 \rfloor + 1,$$

where normalization means integral one on $[-\pi, \pi]$. Elementary bounds on the sine give $c_n \asymp n^{-3}$ and $\int |\theta| J_n(\theta) \, d\theta \le K/n$. The convolution is an even trigonometric polynomial of degree $2(n-1) \le k$, with approximation error K/k and uniformly bounded cosine coefficients. Writing it as a polynomial in v uses the Chebyshev polynomials $T_j(2v - 1)$. Their recurrence bounds the sum of absolute monomial coefficients by K^j, which proves (10.8.5). Integrating the approximation and using the moment bounds gives (10.8.4), after enlarging K. Taking the supremum over 1-Lipschitz f is the Kantorovich–Rubinstein formula on an interval. □

Proof of Theorem 10.8.1. Lemma 10.8.2 gives $E_k \le K\delta$. With $L = \log(1/\delta)$ and $k = \lfloor L/(4 \log L) \rfloor$, the second term of (10.8.4) is smaller than $1/k$ for all sufficiently small δ. This proves the upper bound.

For the lower bound let $d = b - a$, and define two probabilities by the even and odd parts of

$$\mu_k - \nu_k = 2^{1-k} \sum_{j=0}^{k} (-1)^j \binom{k}{j} \delta_{a + jd/k}. \tag{10.8.6}$$

The signed mass has total zero and both signs have mass one. For any real u,

$$\int e^{ur} \, d(\mu_k - \nu_k)(r) = 2^{1-k} e^{au} (1 - e^{ud/k})^k.$$

On any fixed bounded u interval its modulus is at most $(K/k)^k$. The contact bounds in the preceding lemma put every minimizer for $C \in J$ and every probability spectrum in a fixed compact x interval. The profile there is uniformly positive. Comparing the two objective functions at each other's minimizers therefore gives

$$d_J(\mu_k, \nu_k) \le (K_1/k)^k.$$

The signed distribution function between the jth and $(j+1)$st nodes has modulus $2^{1-k}\binom{k-1}{j}$, by the alternating binomial identity. The one-dimensional transport formula consequently gives the exact value

$$W_1(\mu_k, \nu_k) = d/k.$$

Choose the least sufficiently large integer k for which $(K_1/k)^k \le \delta$. Elementary comparison of $k\log(k/K_1)$ with L gives $k \le KL/\log L$ as $\delta \downarrow 0$. This proves the lower bound at every sufficiently small δ, rather than only along a sequence.

For reconstruction choose the preceding upper-bound value of k, now using $L = \log(1/\varepsilon)$. Estimate $S_\mu(e^{l/k})$, $1 \le l \le k$, by Lemma 10.8.2; set the zeroth sample equal to one. Take a grid $\mathcal{G}_\varepsilon \subset [a,b]$ of mesh at most ε including the endpoints. Solve the linear constraints

$$q_g \ge 0, \quad \sum_{g\in\mathcal{G}_\varepsilon} q_g = 1, \quad \left|\sum_{g\in\mathcal{G}_\varepsilon} q_g e^{lg/k} - \widehat{S}_l\right| \le K_2\varepsilon \quad (1 \le l \le k).$$

They are feasible: rounding every point of the true measure to a nearest grid point changes each exponential sample by at most $K\varepsilon$, uniformly in $l \le k$. Choose K_2 larger than this constant plus the profile-estimation constant. Every feasible probability $\widehat{\mu} = \sum q_g \delta_g$ has $E_k(\widehat{\mu}, \mu) \le K\varepsilon$ and hence satisfies the desired Wasserstein bound. The grid has $O(\varepsilon^{-1})$ points and the query count is $O(k\log(1/\varepsilon))$. □

Thus replacing a finite atomic prior by arbitrary positive spectra changes the stability class. Positivity still permits recovery, but there is no uniform power-law inverse on the enlarged class. This is an answer about a nonlinear threshold observation operator, not a closure of a named conjecture about arbitrary Laplace inversion.

10.8.2 The exact arithmetic obstruction to a short zero experiment

Fix $0 < r < 1$, $C, \ell, \alpha, R > 0$. Set

$$x_* = \frac{r}{C(1-r)}, \quad T = \frac{1 + Cx_*}{x_*^r}, \quad A_h = T + \alpha h, \qquad f_h(z) = -1 + A_h e^{\ell r z + i\phi} + Ce^{\ell z + i\phi_*}. \tag{10.8.7}$$

A zero $z = \sigma + it$ is *marked* if

$$\operatorname{dist}(\ell t + \phi_*, \pi + 2\pi\mathbb{Z}) \le R\sqrt{h}. \tag{10.8.8}$$

Let $H_r(h)$ be the infimum of heights H such that, for *every* pair of phases (ϕ, ϕ_*), there is a marked zero with $0 \le t \le H$. The real coordinate is unrestricted. For an irrational r, write

$$\mathcal{R}_N(r) = \sup_{\theta\in\mathbb{R}/\mathbb{Z}} \min_{0\le n<N} \|nr - \theta\|_{\mathbb{R}/\mathbb{Z}}, \qquad Q_r(\eta) = \min\{N \ge 1 : \mathcal{R}_N(r) \le \eta\}.$$

Theorem 10.8.4 (Marked-zero height and bounded partial quotients)**.** *For fixed irrational r and the other fixed parameters in (10.8.7), there are $c, K, h_0 > 0$ such that*

$$\frac{2\pi}{\ell}\bigl(Q_r(Kh) - 3\bigr) \le H_r(h) \le \frac{2\pi}{\ell}\bigl(Q_r(ch) + 3\bigr) \qquad (0 < h < h_0). \tag{10.8.9}$$

Consequently the following are equivalent:

1. *r has bounded continued-fraction partial quotients;*
2. *there is $K_r < \infty$ such that $H_r(h) \le K_r/h$ for all sufficiently small h.*

In this case $c_r/h \le H_r(h) \le K_r/h$ for small h. If the partial quotients are unbounded, then $\limsup_{h\downarrow 0} hH_r(h) = \infty$.

This theorem concerns the presence of a marked zero at one fixed critical-scale amplitude. It is not a lower bound for every algorithm that may use several amplitudes, unmarked zeros, or other statistics. It is stronger than an asymptotic density statement because the height bound is uniform over the initial phases. The arithmetic criterion itself is classical; the essential bridge is that the effective target for the rotation has width h, despite the $\sqrt{h}$ marking window.

Lemma 10.8.5 (Quadratic phase cancellation)**.** *Put $t_n = (\pi - \phi_* + 2\pi n)/\ell$ and $\kappa_n = \phi + r(\pi - \phi_*) + 2\pi nr$ modulo 2π. For sufficiently small h, the following statements hold, uniformly over the phases and the integer n:*

1. *Every marked zero satisfies $|t - t_n| \le R\sqrt{h}/\ell$ and $\operatorname{dist}(\kappa_n, 2\pi\mathbb{Z}) \le Kh$ for its nearest top-phase crossing t_n.*
2. *If $\operatorname{dist}(\kappa_n, 2\pi\mathbb{Z}) \le ch$, then a marked zero exists with $|t - t_n| \le R\sqrt{h}/\ell$.*

Proof. At a zero put $x = e^{\ell\sigma}$ and $v = \ell(t - t_n)$, so $|v| \le R\sqrt{h}$. The zero equation reads

$$A_h x^r e^{i(\kappa_n + rv)} = 1 + Cxe^{iv}. \tag{10.8.10}$$

For all $x > 0$,

$$|1 + Cxe^{iv}| \ge (1 - Kv^2)(1 + Cx).$$

Thus $(1 + Cx)/x^r \le T + Kh$. This objective tends to infinity at both ends, has its unique minimum at x_*, and has strictly positive second derivative in the variable $u = \log(x/x_*)$ at $u = 0$. It follows that $|u| \le K\sqrt{h}$. Let $y_* = Cx_*$. Taking the argument in (10.8.10) near $u = v = 0$ gives

$$\kappa_n = \arg(1 + y_* e^u e^{iv}) - rv \pmod{2\pi}.$$

The coefficient of v at $u = 0$ is $y_*/(1 + y_*) - r = 0$. More explicitly, smoothness and oddness in v give an absolute bound $K(|u||v| + |v|^3)$ for the right side. This is at most Kh, proving the necessary condition.

For the sufficient condition use the complex coordinate $z = (\log x_* + \zeta)/\ell + it_n$ and choose the representative $|\kappa_n| \le ch$. The function in this coordinate is

$$F_{h,\kappa}(\zeta) = -1 + (T + \alpha h)x_*^r e^{r\zeta + i\kappa} - y_* e^{\zeta}.$$

Write $s_* = x_*^r$ and $B = r > 0$; the contact identities give, uniformly for bounded complex Z,

$$h^{-1} F_{h,0}(\sqrt{h}Z) = \alpha s_* - \tfrac{1}{2} BZ^2 + O(\sqrt{h}).$$

The limiting polynomial has two distinct real roots $\pm d$, where $d = \sqrt{2\alpha s_*/B}$. Choose a radius $0 < \rho < \min(R, d)/4$ around d. Its boundary contains no limiting root; on it $|F_{h,0}(\sqrt{h}Z)| \ge c_0 h$ for small h, and Rouché's theorem gives one zero in the disk. The perturbation obeys $|F_{h,\kappa} - F_{h,0}| \le K_0|\kappa|$ there. Choose $c < c_0/(2K_0)$. A second application of Rouché gives a zero of $F_{h,\kappa}$ in the same disk. Its imaginary coordinate has modulus less than $\rho\sqrt{h} < R\sqrt{h}$, so it is marked and has the claimed height proximity. □

Proof of (10.8.9). Absorb the conversion from angles to turns into c, K. The numbers $\kappa_n/(2\pi)$ form a translate of the rotation orbit nr modulo one. The spacing of the crossing times is $L_* = 2\pi/\ell$. Take $N = Q_r(ch)$ consecutive crossings, starting with the first one after $R\sqrt{h}/\ell$. One of their phase mismatches is at most ch. By Lemma 10.8.5 it supplies a marked zero in $[0, (N+3)L_*]$, for all initial phases. This proves the upper bound.

Conversely fix $\phi_* = \pi$, so crossings occur at nL_*. For h small, every marked zero in $[0, H]$ is associated to one of the at most $\lfloor H/L_* \rfloor + 3$ consecutive crossings whose distance from this interval is at most $R\sqrt{h}/\ell$. As ϕ varies, the phase offset ranges over the whole circle. If every such offset supplies a marked zero, the necessary condition in Lemma 10.8.5 says those consecutive rotation points form a Kh-net. Translation of the starting index does not change their covering radius. Therefore $Q_r(Kh) \le \lfloor H/L_* \rfloor + 3$, proving the lower bound by taking the infimum over admissible H. □

Lemma 10.8.6 (Classical rotation covering criterion)**.** *For irrational r, $Q_r(\eta) \ge 1/(2\eta)$. Moreover, $Q_r(\eta) = O_r(\eta^{-1})$ as $\eta \downarrow 0$ if and only if the continued-fraction partial quotients of r are bounded.*

Proof. The union of N arcs of radius η has length at most $2N\eta$, which proves the lower bound. Let p_j/q_j denote the continued-fraction convergents. We use their elementary identities

$$q_{j+1} = a_{j+1}q_j + q_{j-1}, \qquad \left| r - \frac{p_j}{q_j} \right| < \frac{1}{q_j q_{j+1}}, \qquad (p_j, q_j) = 1.$$

For completeness the approximation inequality follows by writing $r = (p_j\xi + p_{j-1})/(q_j\xi + q_{j-1})$, where the next complete quotient satisfies $\xi > a_{j+1}$, and using the determinant $|p_jq_{j-1} - p_{j-1}q_j| = 1$.

If all $a_j \le A$, then $q_{j+1} \le (A+1)q_j$. The first q_j orbit points lie within $1/q_{j+1}$ of the uniform q_j-point grid, since multiplication by p_j permutes the residue classes modulo q_j. Hence

$$\mathcal{R}_{q_j}(r) \le \frac{1}{2q_j} + \frac{1}{q_{j+1}} \le \frac{3}{2q_j}.$$

Choose the first $q_j \ge 3/(2\eta)$. Its size is at most $3(A+1)/(2\eta)$, up to the immaterial initial convergents, proving the required upper bound.

If the partial quotients are unbounded, some ratios q_{j+1}/q_j tend to infinity. Put $N_j = \lfloor q_{j+1}/8 \rfloor$. For $0 \le n < N_j$,

$$\left\| nr - \frac{np_j}{q_j} \right\| < \frac{1}{8q_j}.$$

Thus all these points lie within $1/(8q_j)$ of the uniform q_j-grid. The midpoint of any grid gap has distance at least $3/(8q_j)$ from the orbit segment. For $\eta_j = 1/(4q_j)$ it follows that $Q_r(\eta_j) > N_j$, and $\eta_j Q_r(\eta_j) \to \infty$ along the chosen subsequence. This excludes a bound of order η^{-1}. □

Combining the lemma with (10.8.9) proves the remaining assertions of Theorem 10.8.4. In particular, if $hH_r(h)$ were bounded for all sufficiently small h, the lower bridge would imply a bound for $\eta Q_r(\eta)$ at every small η by setting $h = \eta/K$. The unbounded-partial-quotient case therefore gives the stated limsup, not merely failure of one proposed constant.

These two investigations extend the earlier finite-dimensional theory in different directions: a sharp inverse for unrestricted positive spectra, and an exact criterion for optimal-height marked zero detection in the one-frequency model. Higher-dimensional height optimality, recovery from incomplete unmarked divisors, and joint phase-frequency estimation from finitely many noisy zeros remain open here.

10.9 Simultaneous gamma factors and reconstruction from boundary response

The earlier endpoint extraction theorem involved one gamma factor. We now classify simultaneous extraction at both original gamma rates, including the atomic face where the total possible shape is attained. The curved face has a qualitatively different property: none of its remainders has any nontrivial convolution root. A second investigation turns factor geometry into an inverse problem. The infinitesimal cost of exchanging gamma factors at varying rates determines the entire boundary remainder. These are internally formulated transfer problems; they are not claimed as additional named conjectures in [3], or as a priority determination for all related factorization literature.

10.9.1 The complete simultaneous extraction diagram

Fix $\kappa > 1/2$, $b > 1$, put $\lambda = b - 1$ and $M = 2\kappa - 1$, and let X have the previously studied McKay transform

$$L_X(s) = \frac{b+s}{b}\left\{\frac{b^2-1}{(b+s)^2-1}\right\}^{\kappa}. \tag{10.9.1}$$

Gamma variables below are parametrized by shape and rate. We ask for all $h_-, h_+ \geq 0$ for which independent nonnegative variables satisfy

$$X \stackrel{\mathrm{d}}{=} G_{h_-,\lambda} + G_{h_+,\lambda+2} + R. \tag{10.9.2}$$

A shape zero means the constant zero. Define residual exponents $a = \kappa - h_-$ and $d = \kappa - h_+$. A prospective remainder has transform

$$L_R(s) = \frac{F_{a,d}(\lambda+s)}{F_{a,d}(\lambda)}, \qquad F_{a,d}(u) = (u+1)u^{-a}(u+2)^{-d}. \tag{10.9.3}$$

In particular, d need not be positive: a fast-rate factor can have shape larger than κ.

We first recall the scalar root from the endpoint theorem, with the details needed for its extension and differentiation. For $\eta = a + d - 1 > 0$ set

$$g_{a,d}(x) = \frac{x^{\eta-1}e^{-2x}}{\Gamma(\eta)} B_{a,d}(2x), \tag{10.9.4}$$

$$B_{a,d}(z) = {}_1F_1(a-1;\eta;z) + \frac{z}{2\eta}\,{}_1F_1(a;\eta+1;z). \tag{10.9.5}$$

Its Laplace transform is $F_{a,d}(u)$. This follows by termwise integration for large u and Kummer's transformation, and then by uniqueness. The series has at most exponential growth and is real analytic for $x > 0$. The Kummer identities used here and below are recorded in [6].

Lemma 10.9.1 (The scalar ingredient and its exact slope)**.** *For every $d > 1/2$ there is a unique number*

$$A(d) \in (\max(0, 1-d),\ 1/(4d))$$

such that $g_{a,d} \geq 0$ exactly when $a \geq A(d)$, among $a > 0$ with $a + d > 1$. It is the unique root in the displayed interval of

$$B_{a,d}(2(d-a)) = 0. \tag{10.9.6}$$

At $a = A(d)$ the density $g = g_{A(d),d}$ has precisely one zero, at $x_* = d - A(d)$, and this zero has order two. The function A is real analytic on $(1/2, \infty)$ and

$$A'(d) = -\frac{\displaystyle\int_0^{x_*} e^{-2y} g(x_* - y)\, dy/y}{\displaystyle\int_0^{x_*} g(x_* - y)\, dy/y}, \qquad -1 < A'(d) < -e^{-2x_*} < 0. \tag{10.9.7}$$

The root and one-point criterion are previous inputs. The analytic tradeoff and formula (10.9.7) *are established here.*

Proof. For completeness, the one-point argument is short. When $0 < a < 1$, put $p = 1 - a$ and $y(z) = {}_1F_1(-p; \eta; z)$. Then $y'(z) < 0$ for $z > 0$, since $y' = -p\,{}_1F_1(a; \eta + 1; z)/\eta$. Kummer's differential equation gives the full identity

$$B' = \frac{B}{2} + \frac{2(d-a) - z}{4p} y'.$$

Thus a zero before $2(d-a)$ crosses downward, whereas a zero after it crosses upward. A zero at that point satisfies $B'' = -y'/(4p) > 0$. If $d \le a$, downward crossing is impossible and B is positive. If $d > a$, positivity at both ends implies that $B \ge 0$ everywhere exactly when $B(2(d-a)) \ge 0$; equality leaves just the stated double zero. For $a \ge 1$, both terms in (10.9.5) are positive, so there is no obstruction.

Here positivity at infinity, uniformly on compact subsets of $a, d > 0$, $a + d > 1$, follows from an elementary convolution. Put

$$k_{a,d}(x) = \frac{1}{\Gamma(a)\Gamma(d)} \int_0^x e^{-2v} v^{d-1} (x - v)^{a-1}\, dv.$$

Since $k_{a,d}(0) = 0$, $g_{a,d} = k'_{a,d} + k_{a,d}$. Splitting the integral smoothly near $v = x/2$ and expanding on its first part gives, also after differentiation,

$$g_{a,d}(x) \sim \frac{x^{a+d-2}}{\Gamma(a+d-1)} \quad (x \downarrow 0), \qquad g_{a,d}(x) \sim \frac{2^{-d} x^{a-1}}{\Gamma(a)} \quad (x \to \infty). \tag{10.9.8}$$

On the other part, use $x - v$ as variable; its contribution and fixed derivatives are exponentially small. These estimates are uniform on compact strict parameter sets. Strict positivity everywhere is therefore open in those parameters.

The logarithmic derivative of (10.9.3) has signed Lévy kernel

$$\frac{e^{-(\lambda+1)x}}{x} \{a e^{x} + d e^{-x} - 1\}. \tag{10.9.9}$$

At $a = 1/(4d)$, $d > 1/2$, this is nonnegative and has just one positive zero. The associated ID density is strictly positive everywhere: split its jumps at a small ϵ, use the positive two-jump density on $(2\epsilon, \infty)$, and use the fact that the small-jump sum has mean tending to zero. This gives a positive lower bound near any prescribed positive point. Endpoint stability now allows a strict decrease of a.

The lower endpoint is inadmissible with a nonempty inadmissible neighborhood. For $1/2 < d < 1$, at $a = 1 - d$ the transform expands as $F_{a,d}(u) = 1 + (1 - 2d)/u + O(u^{-2})$ and increases for large u. For $d = 1$, $a = 0$ gives $1 - 1/(u + 2)$. For $d > 1$, at $a = 0$ the inverse on $x > 0$ is

$$\frac{e^{-2x} x^{d-2}}{\Gamma(d-1)} \left(1 - \frac{x}{d-1}\right),$$

which is negative on an interval. Continuity preserves the corresponding obstruction. Increasing a convolves the inverse with $x^{\epsilon-1}/\Gamma(\epsilon)$; after tilting this is gamma convolution. Consequently the admissible set is an upper interval, its lower endpoint is attained by the continuity theorem for Laplace transforms, and above that endpoint its density is strictly positive. The one-point test proves existence, uniqueness and the double-zero description of $A(d)$.

At a contact point, differentiating the preceding convolution at $\epsilon = 0$ gives

$$\partial_a g_{a,d}(x_*) = \int_0^{x_*} g_{a,d}(x_* - y)\, dy/y,$$
$$\partial_d g_{a,d}(x_*) = \int_0^{x_*} e^{-2y} g_{a,d}(x_* - y)\, dy/y.$$

The integrals converge: the double zero gives $O(y^2)$ at $y = 0$, and (10.9.8) is integrable at the other endpoint. To see the derivative directly, divide the convolution by ϵ and use $1/\Gamma(\epsilon) \sim \epsilon$; its constant term is zero at the contact. Analytic dependence in (10.9.4) makes this one-sided calculation the ordinary derivative. The first integral is positive. Since $g_x(x_*) = 0$, the implicit function theorem applied to $g_{a,d}(d - a) = 0$ proves analyticity and (10.9.7). The strict bounds follow because the quotient is an average of e^{-2y} over $0 < y < x_*$. □

Extend A to $d \le 1/2$ by $A(d) = 1 - d$, and define

$$I(d) = \begin{cases} 1 - d, & d \le 1/2, \\ 1/(4d), & d > 1/2. \end{cases} \tag{10.9.10}$$

Theorem 10.9.2 (Two-rate extraction, including its atomic face)**.** *For $h_-, h_+ \ge 0$, the factorization (10.9.2) exists if and only if*

$$\kappa - h_- \ge A(\kappa - h_+). \tag{10.9.11}$$

Its remainder is ID if and only if the same inequality holds with A replaced by I. Equivalently, for $0 \le h_+ \le M$, the largest admissible slow-rate shape is

$$h_-^{\max}(h_+) = \begin{cases} \kappa - A(\kappa - h_+), & 0 \le h_+ < M/2, \\ M - h_+, & M/2 \le h_+ \le M. \end{cases} \tag{10.9.12}$$

The curved part is analytic, with derivative strictly between -1 and 0. It joins the line segment continuously with slope -1.

The total-shape bound $h_- + h_+ \le M$ is attained precisely on the line segment in (10.9.12). There the remainder is compound Poisson and has atom

$$\mathbb{P}(R = 0) = \frac{\lambda^a (\lambda + 2)^{1-a}}{\lambda + 1}, \qquad a = \kappa - h_- \ge 1/2. \tag{10.9.13}$$

On the curved part the remainder has no atom, is not ID, and its density has exactly one interior double zero.

Proof. Since the original transform has order s^{-M} at infinity, every feasible pair satisfies $h_- + h_+ \le M$, or $a + d \ge 1$. Also $a \ge 0$: otherwise the moment generating function of the remainder, computed from (10.9.2), tends to zero as its argument increases to λ, although it must be at least one. The case $a = 0$, $d \ge 1$, is excluded by the explicit negative inverse or increasing transform in the preceding proof. Thus $a > 0$ is necessary.

If $a + d > 1$ and $d > 1/2$, Lemma 10.9.1 gives exactly $a \ge A(d)$. If $d \le 1/2$, then $a + d \ge 1$ forces $a \ge 1 - d \ge d$. The function $ae^x + de^{-x} - 1$ is nonnegative for $x \ge 0$: its derivative is positive for $x > 0$ when $d \le a$, and its value at zero is $a + d - 1$. This argument permits negative d and proves ID sufficiency.

On the boundary $a + d = 1$, the expansion $F_{a,d}(u) = 1 + (2a - 1)/u + O(u^{-2})$ excludes $a < 1/2$. For $a \ge 1/2$ the preceding nonnegative kernel applies. Its value at zero vanishes, so

the Lévy density in (10.9.9) is integrable at zero and at infinity. The law is compound Poisson, and the limit of its Laplace transform at infinity is (10.9.13).

For the full ID criterion minimize $ae^{x} + de^{-x}$ over $x \geq 0$. If $d \leq a$, its minimum is $a + d$; if $d > a > 0$, its minimum is $2\sqrt{ad}$. Together with $a + d \geq 1$ these are exactly $a \geq I(d)$. Necessity follows from uniqueness of the Lévy representation, since (10.9.9) is integrable with the usual Lévy weight and is the inverse logarithmic derivative.

Translating the parameter inequalities gives (10.9.12). Its curved derivative is $A'(\kappa - h_{+})$. As $d \downarrow 1/2$, the two bounds on $A(d)$ squeeze it to $1/2$, hence $x_{*} = d - A(d) \to 0$. Formula (10.9.7) then squeezes $A'(d)$ to -1. The remaining assertions follow from the strict scalar inequalities and the double-zero result. □

10.9.2 Root obstruction on the curved factor boundary

The distinction between the two boundary faces is stronger than ID versus non-ID. Here a convolution root means a probability law Q with $Q^{*n} = \mathcal{L}(R)$ for an integer $n \geq 2$.

Theorem 10.9.3 (No gamma factors and no convolution roots)**.** *Every remainder on the curved boundary of Theorem 10.9.2 has no nonzero gamma factor at any finite positive rate, and has no convolution root of any integer order $n \geq 2$. This assertion includes possible roots on the whole real line. In contrast, all remainders on the linear face have convolution roots of every order, since they are compound Poisson.*

Proof. Let $f(x) = e^{-\lambda x} g_{a,d}(x)/F_{a,d}(\lambda)$ be a curved boundary density, let x_{*} be its unique zero, and put $\eta = a + d - 1 > 0$. If $R = G_{h,\rho} + Y$, with $h > 0$, then Y has support arbitrarily close to zero because R does. The part of its convolution with the gamma density coming from $Y \leq x_{*}/2$ has a strictly positive lower bound throughout a neighborhood of x_{*}. This contradicts continuity and $f(x_{*}) = 0$. This argument is valid for every $0 < \rho < \infty$; it makes no claim about a formal gamma factor at an infinite rate.

We next verify the absolute continuity needed for the root argument. For any $t > 0$, the only candidate Laplace transform for a positive convolution power is $L_R(s)^{t}$. At large $u = s + \lambda$ write

$$(1 + z)^{t}(1 + 2z)^{-dt} = \sum_{j \geq 0} c_j(t) z^{j}.$$

Its candidate inverse is

$$q_t(x) = F_{a,d}(\lambda)^{-t} e^{-\lambda x} \sum_{j \geq 0} \frac{c_j(t) x^{t\eta + j - 1}}{\Gamma(t\eta + j)}. \tag{10.9.14}$$

Cauchy's bound on c_j proves local convergence of this series and of its derivatives on $x > 0$, and gives at most exponential growth. Termwise integration for sufficiently large s gives $L_R(s)^{t}$. If this transform belongs to a probability law, uniqueness after a sufficiently large exponential tilt forces that law to have density (10.9.14). Thus every putative positive power has a real analytic density, with no atom, and a positive leading term at zero.

If a root of order n existed, its support would lie in $[0, \infty)$: any positive mass on $(-\infty, 0)$ would give positive mass there for the sum of n independent copies. Its Laplace transform must be $L_R^{1/n}$, so the preceding density is nonnegative and real analytic. It is nonzero and hence positive except at isolated points on $(0, \infty)$. Its two-fold convolution is strictly positive at every $x > 0$, by integration over $(0, x)$; the endpoint singularities are integrable since $\eta/n > 0$. The same is true of every higher convolution power. The resulting continuous density cannot equal f, which vanishes at x_{*}. This is a contradiction for every $n \geq 2$. □

10.9.3 The inverse theorem for gamma-rate response

Fix a curved boundary point $a = A(d)$ with $d > 1/2$, and denote its remainder transform by L_R and density by f. The original slow-rate factor has positive shape. It can therefore be reduced slightly, adding a gamma component back to the remainder, while a small new gamma factor at another rate is extracted. For a fixed $\rho \geq \lambda$ define the candidate transform

$$L_{t,\epsilon,\rho}(s) = L_R(s) \left(\frac{\lambda}{\lambda + s} \right)^t \left(\frac{\rho + s}{\rho} \right)^\epsilon . \tag{10.9.15}$$

Here $\epsilon \geq 0$ is the newly extracted shape, and $t \geq 0$ is the amount returned at the slow rate. This describes an actual exchange of factors of X for sufficiently small parameters.

Theorem 10.9.4 (The boundary response determines the remainder)**.** *For each finite $\rho \geq \lambda$ there is a real analytic local boundary $t = T_\rho(\epsilon)$ with $T_\rho(0) = 0$ such that (10.9.15) is a probability Laplace transform locally exactly when $t \geq T_\rho(\epsilon)$. Its derivative is*

$$T'_\rho(0) = \frac{\displaystyle\int_0^{x_*} e^{-\rho y} f(x_* - y)\, dy/y}{\displaystyle\int_0^{x_*} e^{-\lambda y} f(x_* - y)\, dy/y}. \tag{10.9.16}$$

Write $C(r) = T'_{\lambda+r}(0)$ for $r \geq 0$. Then C is the Laplace transform of the probability density

$$w(y) = \frac{g_{a,d}(x_* - y)}{yJ}, \qquad J = \int_0^{x_*} g_{a,d}(x_* - y)\, dy/y, \qquad 0 < y < x_*. \tag{10.9.17}$$

In particular C is strictly completely monotone, $C(0) = 1$, and

$$x_* = \lim_{n\to\infty} \{(-1)^n C^{(n)}(0)\}^{1/n}, \tag{10.9.18}$$

$$C(r) \sim \frac{g''_{a,d}(x_*)}{2J}\, r^{-2} \qquad (r \to \infty). \tag{10.9.19}$$

For the fixed known base rate λ, the entire response function C uniquely determines the remainder law. Within the present family it even recovers both residual exponents from the endpoint behavior of its uniquely determined density w:

$$\eta = 1 + \lim_{z\downarrow 0} \frac{\log w(x_* - z)}{\log z}, \qquad a = \frac{\eta + 1 - x_*}{2}, \qquad d = \frac{\eta + 1 + x_*}{2}. \tag{10.9.20}$$

These are exact-data uniqueness assertions, not numerical stability claims for analytic continuation or Laplace inversion.

Proof. We first establish that the local minimum controls positivity on the whole half-line. Put $r = \rho - \lambda$. Apart from its positive normalization, the inverse needed in (10.9.15) is that of

$$F_{a,d}(u) u^{-t} (u + r)^\epsilon.$$

At large u, expand $(1 + z)(1 + 2z)^{-d}(1 + rz)^\epsilon$ into a Taylor series and invert termwise, with gamma denominator of initial order $\eta + t - \epsilon > 0$. The same geometric-coefficient argument as in (10.9.14) gives a jointly real analytic inverse on $x > 0$, depending analytically on t, ϵ near zero. Its small-x leading coefficient is positive and has power $x^{\eta+t-\epsilon-1}$.

For $r > 0$, its large-x leading term is

$$\frac{2^{-d} r^\epsilon}{\Gamma(a + t)} x^{a+t-1}. \tag{10.9.21}$$

To justify uniformity in a neighborhood of $(t,\epsilon) = (0,0)$, start with $g_{a+t,d}$ and convolve it with $e^{-rx}x^{-\epsilon}/\Gamma(1-\epsilon)$, then apply $D+r$. This has the required transform. The initial value before differentiation is zero since $\eta + t - \epsilon > 0$. Splitting the convolution at half its argument shows that its leading factor is $r^{\epsilon-1}$; the part near the remote endpoint and its derivative are exponentially small. Applying $D+r$ gives (10.9.21), with a uniform relative $O(x^{-1})$ error. For $r = 0$ the inverse is exactly $g_{a+t-\epsilon,d}$ and (10.9.8) applies directly. Thus both ends remain strictly positive, uniformly for nearby parameters. The normalized inverses are integrable after tilting by $e^{-\lambda x}$ and have total mass one.

At the original density zero, adding a gamma shape of size t at rate λ gives derivative

$$N_\lambda = \int_0^{x_*} e^{-\lambda y} f(x_* - y)\, dy/y > 0.$$

The same convolution calculation as in Lemma 10.9.1 justifies the derivative; normalizing terms multiply $f(x_*) = 0$. Extracting shape ϵ at rate ρ gives derivative $-N_\rho$, where N_ρ is the corresponding integral in (10.9.16). This follows either by differentiating the inverse multiplier or by analytically continuing the gamma-convolution identity from negative ϵ.

Since $f''(x_*) > 0$, the implicit function theorem first supplies a unique critical point near x_* for the perturbed density. Its value $m(t,\epsilon)$ is analytic, with $m_t(0,0) = N_\lambda$ and $m_\epsilon(0,0) = -N_\rho$. A second application gives the unique local graph $m = 0$ and (10.9.16). Outside a fixed small neighborhood of x_*, positivity persists by the endpoint estimates and positivity on the remaining compact interval. Inside that neighborhood the critical point is the unique minimum. Thus $m \geq 0$ is exactly global nonnegativity, proving the claimed local factor boundary.

Substitution of $f(x) = e^{-\lambda x} g_{a,d}(x)/F_{a,d}(\lambda)$ in (10.9.16) gives $C(r) = \int_0^{x_*} e^{-ry} w(y)\, dy$. This density is positive on $(0, x_*)$, integrable at both ends, and has total mass one. The strict derivative inequalities and (10.9.18) now follow from differentiation and the limit of the L^n norms of a random variable whose essential supremum is x_*. At zero,

$$w(y) = \frac{g''_{a,d}(x_*)}{2J} y + O(y^2),$$

which gives (10.9.19) by a direct Laplace endpoint expansion. The singularity of w at x_* is integrable and contributes only an exponentially small error to that expansion.

Laplace-transform uniqueness determines w from C, and the moment limit determines its right support endpoint. Consequently (10.9.17) recovers $g_{a,d}$ up to a constant on $(0, x_*)$. Real analyticity extends that determination to $(0,\infty)$, and normalization after the known exponential tilt fixes the constant. More directly in the present family, (10.9.8) gives $w(x_* - z) \sim z^{\eta-1}/\{x_* J\Gamma(\eta)\}$. This proves the first identity in (10.9.20); the other two follow from $a+d = \eta+1$ and $d - a = x_*$. □

At the fast original rate, $\rho = \lambda + 2$, the response theorem reduces to $T'_{\lambda+2}(0) = -A'(d)$, so it includes the tradeoff slope of the two-rate diagram. Allowing all rates supplies strictly more information: it reveals a whole compactly supported measure, rather than a single directional derivative. The linear atomic face is excluded from this inverse theorem; it has no positive interior contact to which this argument could apply.

The simultaneous classification and the response inverse theorem are the two principal investigations of this section. Root failure is a structural consequence of the first, not an extra historical conjecture. Three-or-more-rate global regions, convexity of the unrestricted extraction set, and quantitative stability of recovery from finitely many noisy response values remain open here.

10.10 Where the next obstructions lie

The results identify several precise next steps. In continuous information, the missing one-query gaps and the unrestricted three-dimensional two-query case are not settled by the width recursion. Smooth exact zero-set tests cannot simply be replaced by analytic ones. For noisy convex information, a matching upper construction is needed before the displayed budget gaps can be called sharp rates. For Banach operators, the sequence-space example shows that compactness alone is the wrong general criterion, but it does not identify the complete operator class with vanishing adaptive error.

For mixtures, exact replication is much stronger than a collection of separate marginals. The two-replicate stable inversion uses a positive lower index bound and a triangular dilation recursion; it leaves the case of indices accumulating at zero, sharp statistical stability, and minimal replication for correlated location and skewness open. Honest uncertainty under corruption and adaptive point estimation are different questions. The confidence-set obstruction proved above should not be read as an impossibility theorem for every adaptive estimator.

For spectra, infinitely many exact zeros can carry an unknown frequency support while finite noisy records remain uniformly ambiguous near a rational relation. A useful quantitative extension must therefore state an arithmetic margin, not only pairwise frequency separation. The marked-zero height theorem resolves one frequency through circle rotations; multidimensional sharp covering criteria require additional Diophantine analysis. Finally, the gamma response inverse is exact but uses a full continuum of rates. A stable finite-rate analogue and global regions for three or more rates are not supplied here.

These remaining questions are recorded to prevent a proved restricted statement from silently becoming a stronger conjecture. The intended next research step is to attack those explicit obstructions, rather than to count further special cases of the present theorems as independent discoveries.

References

[1] KEIO, *Observation and Recovery in Arithmetic and Probability: Prime Flows, Rare Events, and Infinite Mixtures*, integrated working manuscript, 14 October 2026, 223 pp. AI-assisted, unpublished, and not externally peer reviewed. This is the external predecessor of the present compilation. Source: `KEIO_Integrated_Observation_and_Recovery.tex`. Part X cites its section on factorial smoothing and the sharp logarithmic modulus.

[2] A. Rainer, *Ultradifferentiable extension theorems: a survey*, author manuscript dated 30 December 2021, Theorem 3.6 and Corollary 3.12. https://www.tuwien.at/mg/arainer/wp-content/uploads/2025/03/survey-uet.pdf.

[3] Á. Baricz, Dhivya Prabhu K, S. Singh, and Antony Vijesh V, *Infinitely divisible modified Bessel distributions*, Pacific J. Math. **343** (2026), 261–313. doi:10.2140/pjm.2026.343.261. Preprint: arXiv:2406.17721. Published text: https://msp.org/pjm/2026/343-2/pjm-v343-n2-p01-p.pdf. The maximal ID-domain question is in Section 2.3; the multipart McKay subclass problem is in Section 4.3, pp. 282–283 (locators retained from Part III).

[4] L. Bondesson and T. Simon, *Stieltjes functions of finite order and hyperbolic monotonicity*, Trans. Amer. Math. Soc. **370** (2018), 4201–4222. doi:10.1090/tran/7123. arXiv:1604.05267. Part I specifically cites equation (7) of the 2016 preprint.

[5] D. S. P. Salazar, *A Bessel-zero obstruction to hyperbolic complete monotonicity of noncentral chi-square densities*, arXiv:2606.22066 (2026).

[6] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*, https://dlmf.nist.gov/. The consolidated citations refer to Sections 5.11, 10.9, 10.21, 10.27, 10.29, 10.32, 10.39, 10.40, 10.42, 13.2–13.4, 13.7, 15.4, 15.8, and 15.10. Specific equation locators retained from the source manuscripts: 5.11.9; 10.9.30 (Nicholson's integral); 10.21.15 (the infinite product for J_ν); and 15.8.1 (Euler's hypergeometric transformation). Parts I–III record access on 14 October 2026.

[7] J. Lin, *A negative answer to a problem of Aldous on determination of exchangeable sequences*, arXiv:1507.00678v1 (2015). The open normal-class question audited here occurs in Section 2.6, printed p. 17 (PDF page index 16); the exponential-moment result is Proposition 2.6.4. Published in *Electronic Journal of Probability* **21** (2016), paper 30, 1–26, doi:10.1214/16-EJP4403. The publisher PDF was not inspected in this audit.

[8] D. Le, N. Bariletto, A. Rinaldo, and N. Ho, *Partial Differential Equation Barriers to Identifiability in Infinite Mixture Models*, arXiv:2608.08597 (2026). Question and definition locations refer to the author manuscript dated 8 August 2026: the remark following Proposition 6 and Section 6, printed p. 13 (PDF page index 12); operator classes in Definitions 1 and 4.

[9] M. B. Giles, T. Nagapetyan and K. Ritter, Multilevel Monte Carlo approximation of distribution functions and densities, SIAM/ASA Journal on Uncertainty Quantification 3 (2015), 267–295. doi:10.1137/140960086. https://people.maths.ox.ac.uk/gilesm/files/juq_15.pdf.

[10] M. B. Giles, MLMC for nested expectations, in Contemporary Computational Mathematics: A Celebration of the 80th Birthday of Ian Sloan, Springer (2018), 425–442. https://people.maths.ox.ac.uk/gilesm/files/SLOAN80-056.pdf.

[11] J. C. H. Lee and P. Valiant, Uncertainty about uncertainty: Optimal adaptive algorithms for estimating mixtures of unknown coins, Proceedings of the 2021 ACM-SIAM Symposium on Discrete Algorithms, 414–433. doi:10.1137/1.9781611976465.25. https://www.ias.edu/sites/default/files/LeeValiant_Uncertainty.pdf. Preprint: https://arxiv.org/abs/1904.09228.

[12] C. H. J. Lee, Classic and Modern Challenges in Statistical Estimation, Ph.D. thesis, Brown University, May 2021, Chapter 4, especially pp. 65–67. https://cs.brown.edu/research/pubs/theses/phd/2021/chun.hin.jasper.lee.pdf.

[13] R. K. Vinayak, W. Kong, G. Valiant and S. Kakade, *Maximum likelihood estimation for learning populations of parameters*, Proceedings of the 36th International Conference on Machine Learning, PMLR **97** (2019), 6448–6457.
https://proceedings.mlr.press/v97/vinayak19a/vinayak19a.pdf. Theorem 3.2, Remark 3.2 and Theorem 3.3 appear on printed paper page 4; Lemma 4.2 and its following discussion appear on printed page 5. Remarks 3.2–3.3 distinguish the published replication-range MLE conjecture from the suggested localized moment-matching route; Theorem 3.3 gives the minimax lower bound (locators retained from Parts V–VI). Part VII additionally cites the Supplement, Remark 4.1, equation (23):
https://proceedings.mlr.press/v97/vinayak19a.html.

[14] C. Musco, C. Musco, L. Rosenblatt and A. V. Singh, *Sharper bounds for Chebyshev moment matching, with applications*, Proceedings of the 38th Conference on Learning Theory, PMLR **291** (2025), 4309–4358; author version arXiv:2408.12385v3, revised 18 May 2026.
https://arxiv.org/html/2408.12385v3. Theorem 20 and Section 6.2 are the version-specific comparison points. Proceedings record:
https://proceedings.mlr.press/v291/musco25a.html. The cited version's Section 6.2 records the coefficient and likelihood questions (Parts V–VI).

[15] I. Lau and J. Scarlett, Open problem: Is interaction necessary for order-optimal 1-bit mean estimation?, Proceedings of Machine Learning Research 336 (2026), 7123–7128.
https://proceedings.mlr.press/v336/lau26a.html.

[16] J. Lee and E. H. Kennedy, Minimax rates for learning smooth populations of parameters, arXiv:2610.04135v2, 9 October 2026. https://arxiv.org/abs/2610.04135. Part IV uses the earlier version arXiv:2610.04135v1, 2 October 2026, Theorem 1 and Remark 1, for the distinct smooth-density target: https://arxiv.org/html/2610.04135v1.

[17] V. Bürgin, J. Epperlein, and F. Wirth, *Remarks on the tail order on moment sequences*, Journal of Mathematical Analysis and Applications **512** (2022), no. 1, 126135.
doi:10.1016/j.jmaa.2022.126135. Locators refer to arXiv:2104.10572v3.

[18] W. A. J. Luxemburg and J. Korevaar, *Entire functions and Müntz–Szász type approximation*, Transactions of the American Mathematical Society **157** (1971), 23–37.
doi:10.1090/S0002-9947-1971-0281929-0. Theorem 5.2, p. 30, is used through the explicit restatement in [19, Theorem 21].

[19] M. Wellershoff, *Injectivity of sampled Gabor phase retrieval in spaces with general integrability conditions*, ETH Zürich SAM Research Report 2021-43, 18 December 2021. Author-hosted report. Theorem 21 explicitly restates W. A. J. Luxemburg and J. Korevaar, *Entire functions and Müntz–Szász type approximation*, Transactions of the American Mathematical Society **157** (1971), 23–37, Theorem 5.2, p. 30. The 2021 restatement, rather than the inaccessible original publisher PDF, was checked for the product construction used here.

[20] P. Borwein and T. Erdélyi, *Müntz spaces and Remez inequalities*, Bulletin of the American Mathematical Society (N.S.) **32** (1995), 38–42. arXiv:math/9501225.

[21] I. Belovas, R. Čepaitytė, and M. Sabaliauskas, *On the zero-free region and the distribution of zeros of the prime zeta function*, Analele Ştiinţifice ale Universităţii Ovidius Constanţa, Seria Matematică **33** (2025), no. 2, 27–44. doi:10.2478/auom-2025-0017. The conjectures cited here are in Section 3, p. 33; Conjectures 2 and 3 impose both $0 < \gamma < \sigma_0$ and $\gamma > 1$.

[22] J. M. Sepulcre and T. Vidal, *On the real projections of zeros of analytic almost periodic functions*, Carpathian Journal of Mathematics **38** (2022), no. 2, 489–501. Journal-hosted full text. Theorem 4.3 is on p. 494.

[23] B. Jessen and H. Tornehave, *Mean motions and zeros of almost periodic functions*, Acta Mathematica **77** (1945), 137–279. doi:10.1007/BF02392225. See the introductory Sections 25–26, pp. 165–166, and Theorems 5–7, pp. 182–185.

[24] L. Alpöge and R. Furman, *More than two thirds of the zeros of the Riemann zeta function are simple and on the critical line*, arXiv:2608.13637v2 (2026). https://arxiv.org/html/2608.13637v2.

[25] K. Ford and A. Zaharescu, *On the distribution of imaginary parts of zeros of the Riemann zeta function*, J. reine angew. Math. 579 (2005), 145–158. https://ford126.web.illinois.edu/wwwpapers/fz.pdf.

[26] K. Ford, K. Soundararajan and A. Zaharescu, *On the distribution of imaginary parts of zeros of the Riemann zeta function, II*, Math. Ann. 343 (2009), 487–505. https://arxiv.org/pdf/0805.2745.

[27] F. Çiçek and S. M. Gonek, *The uniform distribution modulo one of certain subsequences of ordinates of zeros of the zeta function*, Math. Proc. Cambridge Philos. Soc. 176 (2024), 593–608. https://arxiv.org/html/2310.10119v1.

[28] J. Arias de Reyna, *On the distribution (mod 1) of the normalized zeros of the Riemann zeta-function*, J. Number Theory 153 (2015), 37–53. https://arxiv.org/pdf/1403.7126.

[29] KEIO, *Exact Recovery, Optimal Replication, and Bessel Classification: Extensions of Observation and Recovery, with an Audit of Fifteen Published Questions*, research manuscript, 14 October 2026. Original edition: 42 pp. Reproduced as Part I of this compilation. AI-assisted and not externally peer reviewed.

[30] L. R. Haff, P. T. Kim, J.-Y. Koo, and D. St. P. Richards, Minimax estimation for mixtures of Wishart distributions, *Annals of Statistics* **39** (2011), no. 6, 3417–3440. doi:10.1214/11-AOS951; inspected electronic reprint: arXiv:1203.3342.

[31] R. A. Vandermeulen and C. Scott, *On the identifiability of mixture models from grouped samples*, arXiv:1502.06644 (2015). Theorem 4 in the inspected 16-page preprint concerns mixtures with finitely many components. The revised preprint is arXiv:1502.06644v2 (2022). As noted in Part V, the authors state that the work was subsumed and expanded in their Annals of Statistics article *An operator theoretic approach to nonparametric mixture models.*

[32] S. J. Yakowitz and J. D. Spragins, On the identifiability of finite mixtures, *Annals of Mathematical Statistics* **39** (1968), no. 1, 209–214. doi:10.1214/aoms/1177698520. The publisher full text was unavailable in this audit; the relevant finite-Gaussian result is proved independently in the text.

[33] O. Barndorff-Nielsen, Identifiability of mixtures of exponential families, *Journal of Mathematical Analysis and Applications* **12** (1965), no. 1, 115–121. doi:10.1016/0022-247X(65)90059-4. Full text not inspected; included as an unresolved priority check, not as support for an unverified theorem attribution.

[34] G. Blekherman and R. Sinn, *Maximum likelihood threshold and generic completion rank of graphs*, Discrete & Computational Geometry **61** (2019), 303–324. https://arxiv.org/abs/1703.07849. Inspected version: arXiv:1703.07849v2, 23 May 2017, https://arxiv.org/html/1703.07849v2. Cited locators: Definition 1.1, Theorems 2.1 and 2.7, and Corollary 2.11 (Part II). The question on bounding generic completion rank by maximum likelihood threshold is at the end of the Introduction (Part III).

[35] D. I. Bernstein, S. Dewar, S. J. Gortler, A. Nixon, M. Sitharam, and L. Theran, *Maximum likelihood thresholds via graph rigidity*, Annals of Applied Probability **34** (2024), no. 3, 3288–3319. doi:10.1214/23-AAP2039. https://arxiv.org/abs/2108.02185. Part III inspected author version https://arxiv.org/html/2108.02185v3.

[36] J. Weed and Q. Berthet, Estimation of smooth densities in Wasserstein distance, Proceedings of Machine Learning Research 99 (2019), 3118–3119 (extended abstract); full version arXiv:1902.01778. https://proceedings.mlr.press/v99/weed19a.html. https://arxiv.org/abs/1902.01778.

[37] M. G. Pavlides and J. A. Wellner, Nonparametric estimation of multivariate scale mixtures of uniform densities, Journal of Multivariate Analysis 107 (2012), 71–89. doi:10.1016/j.jmva.2012.01.001. https://sites.stat.washington.edu/jaw/JAW-papers/jaw-pavlides.jmva.12.pdf.

[38] A. K. H. Kim, G. Kur and A. Guntuboyina, Convergence rates for estimating multivariate scale mixtures of uniform densities, Electronic Journal of Statistics 19 (2025), 3771–3834. doi:10.1214/25-EJS2426. Author version arXiv:2410.10251v2, revised 22 February 2026. https://arxiv.org/abs/2410.10251.

[39] Á. Baricz and Dhivya Prabhu K, *On an infinitely divisible distribution involving modified Bessel functions*, arXiv:2610.09779, 7 October 2026, Theorem 1.1.

[40] A. Prékopa, *On logarithmic concave measures and functions*, Acta Scientiarum Mathematicarum (Szeged) **34** (1973), 335–343. Author-hosted text, Theorem 6.

[41] M. Allain, S. Aslan, W. Coene, S. Dirksen, J. Dong, J. Flamant, M. Iwen, F. Krahmer, T. van Leeuwen, O. Melnyk, A. Menzel, A. P. Mosk, V. Nikitin, P. Salanevich, G. Plonka, and M. Wellershoff, *Phasebook: a survey of selected open problems in phase retrieval*, Sampling Theory, Signal Processing, and Data Analysis **23** (2025), article 23. doi:10.1007/s43670-025-00117-0.

[42] R. Beinert, *Non-negativity constraints in the one-dimensional discrete-time phase retrieval problem*, Information and Inference **6** (2017), 213–224. arXiv:1605.05482.

[43] M. Choulli, *Various stability estimates for the problem of determining an initial heat distribution from a single measurement*, Rivista di Matematica della Università di Parma (N.S.) **7** (2016), no. 2, 279–307. arXiv:1512.07421.

[44] A. Bounemoura, *Ergodization time for linear flows on tori via geometry of numbers*, arXiv:1506.06469 (2015), especially Theorem 1 and its account of preceding optimal bounds.

[45] F. Aurzada, *A short note on small deviations of sequences of i.i.d. random variables with exponentially decreasing weights*, Statistics & Probability Letters **78** (2008), no. 15, 2300–2307. Author manuscript: arXiv:0711.2576. Theorem 4 contains the geometric-weight small-deviation constant.

[46] L. V. Rozovsky, *Small deviation probabilities of weighted sums with fast decreasing weights*, Probability and Mathematical Statistics **35** (2015), no. 1, 161–178. Publisher abstract. This is background on the established small-deviation problem; no theorem from this paper is an unproved input to our argument.

[47] J. Arias de Reyna and J. van de Lune, *Some bounds and limits in the theory of Riemann's zeta function*, Journal of Mathematical Analysis and Applications **396** (2012), 199–214. https://doi.org/10.1016/j.jmaa.2012.06.017. Preprint https://arxiv.org/abs/1107.5134. Theorem 5.1 and Theorem 6.1 in the inspected preprint.

[48] J. J. Podesta, *A Function Used to Locate the Zeros of an Exponential Sum*, Monatshefte für Mathematik **210** (2026), 223–268, https://doi.org/10.1007/s00605-026-02174-6. The modular-ordinate density question and its deferred solution are stated in Section 1. Part II refers to the same Section 1 formulation as a modular-ordinate existence question.

[49] D. Lenz, T. Spindeler, and N. Strungaru, *Pure point diffraction and almost periodicity*, Israel Journal of Chemistry **64** (2024), e202300158. https://doi.org/10.1002/ijch.202300158; https://arxiv.org/abs/2312.12825.

[50] N. Feldheim, *Zeroes of Gaussian analytic functions with translation-invariant distribution*, Israel Journal of Mathematics **195** (2013), 317–345. https://doi.org/10.1007/s11856-012-0130-0; https://arxiv.org/abs/1105.3929. Theorem 1 and Remark 2.3.

[51] N. D. Feldheim, *Variance of the number of zeroes of shift-invariant Gaussian analytic functions*, arXiv:1309.2111v5 (2016). https://arxiv.org/abs/1309.2111. Published as *Variance of the number of zeroes of stationary Gaussian analytic functions*, Israel Journal of Mathematics **227** (2018), 753–792. Theorem 1 and Section 2.

[52] N. D. Feldheim, research statement, author-hosted document (2017), p. 1, paragraph on the unresolved atomic-spectrum density problem. https://u.cs.biu.ac.il/~feldhen/CV%20and%20stat/Naomi_short_res_2017.pdf.

[53] KEIO, *Observation Geometry and Exact Recovery: Replication Rank, Thorin Measures, and Zero Patterns*, research manuscript, 14 October 2026. Original edition: 56 pp. Reproduced as Part II of this compilation. AI-assisted and not externally peer reviewed. Part III cites the theorem "A rank-completing sequence of partial-sum experiments" as the predecessor of its exact terminal criterion.

[54] C.-H. Rhee and P. W. Glynn, Unbiased estimation with square root convergence for SDE models, Operations Research 63 (2015), 1026–1043. https://web.stanford.edu/~glynn/papers/2015/RheeG15.html.

[55] J. B. Garnett, *Bounded Analytic Functions*, revised first edition, Graduate Texts in Mathematics 236, Springer, New York, 2007. https://doi.org/10.1007/0-387-49763-3.

[56] Y. Peled, *On the d-rigidity phase transition in random graphs*, arXiv:2605.25711v3, revised 24 August 2026. https://arxiv.org/pdf/2605.25711v3. Section 7, item 3, p. 30, explicitly records the remaining global-rigidity route to the sparse-random-graph likelihood conjecture.

[57] N. Feldheim, A. Haimi, G. Koliander and J. L. Romero, *Hyperuniformity and non-hyperuniformity of zeros of Gaussian Weyl–Heisenberg Functions*, arXiv:2406.20003v3 (8 June 2026), https://arxiv.org/abs/2406.20003v3.

[58] W. Banaszczyk, *New bounds in some transference theorems in the geometry of numbers*, Mathematische Annalen **296** (1993), 625–635. doi:10.1007/BF01445125.

[59] D. Aggarwal and N. Stephens-Davidowitz, *An improved constant in Banaszczyk's transference theorem*, arXiv:1907.09020 (2019). Theorems 1.1 and 1.2.

[60] D. Krieg, E. Novak and M. Ullrich, *How many continuous measurements are needed to learn a vector?*, Proc. Amer. Math. Soc., doi:10.1090/proc/17658; arXiv:2412.06468v2, September 22, 2026. https://arxiv.org/html/2412.06468v2. Part IX specifically cites Theorem 1 and Corollary 11 of arXiv:2412.06468v2.

[61] KEIO, *Binomial Mixture Likelihood and Exact Observation Thresholds*, research manuscript, 14 October 2026. Original edition: 39 pp. Reproduced as Part III of this compilation. AI-assisted and not externally peer reviewed.

[62] B. G. Lindsay, *The geometry of mixture likelihoods: a general theory*, Annals of Statistics **11** (1983), no. 1, 86–94. https://doi.org/10.1214/aos/1176346059.

[63] S. G. Bobkov and F. Götze, *Exponential integrability and transportation cost related to logarithmic Sobolev inequalities*, Journal of Functional Analysis **163** (1999), no. 1, 1–28. https://doi.org/10.1006/jfan.1998.3326.

[64] E. Clarkson, Probability of error for detecting a change in a parameter and Bayesian Fisher information, Journal of the Optical Society of America A 37 (2020), 174–181. https://arxiv.org/html/1902.00099v1.

[65] A. Dyachenko and D. Karp, *Ratios of the Gauss hypergeometric functions with parameters shifted by integers: more on integral representations*, arXiv:2108.03595 (2021), especially Lemma 3 and Theorem 3, https://arxiv.org/abs/2108.03595.

[66] J. Stoyanov and G. D. Lin, *Hardy's condition in the moment problem for probability distributions*, Theory of Probability and Its Applications **57** (2013), 699–708. https://doi.org/10.1137/S0040585X9798631X.

[67] G. D. Lin, *Recent developments on the moment problem*, Journal of Statistical Distributions and Applications **4** (2017), article 5. https://arxiv.org/abs/1703.01027.

[68] J. M. Sepulcre, *On the Result of Invariance of the Closure Set of the Real Projections of the Zeros of an Important Class of Exponential Polynomials*, Journal of Function Spaces **2016** (2016), Article 3605690, 9 pp. https://doi.org/10.1155/2016/3605690.

[69] M. Wellershoff, *Phase retrieval of entire functions and its implications for Gabor phase retrieval*, Journal of Functional Analysis **286** (2024), no. 11, 110403. https://arxiv.org/abs/2202.03733.

[70] P. Diaconis, K. Khare and L. Saloff-Coste, *Gibbs sampling, exponential families and orthogonal polynomials*, Statistical Science **23** (2008), no. 2, 151–178. https://doi.org/10.1214/07-STS252. Author copy: https://arxiv.org/pdf/0808.3852. Section 4.1, especially Propositions 4.1–4.2, contains the classical beta–binomial spectral calculation used here.

[71] L. Feng and L. H. Dicker, *Approximate nonparametric maximum likelihood for mixture models: A convex optimization approach to fitting arbitrary multivariate mixing distributions*, Computational Statistics & Data Analysis **122** (2018), 80–91. https://doi.org/10.1016/j.csda.2018.01.006. Author version: https://arxiv.org/abs/1606.02011.

[72] J. Dedecker, A. Fischer and B. Michel, Improved rates for Wasserstein deconvolution with ordinary smooth error in dimension one, arXiv:1404.0646v2, revised 4 March 2015. https://arxiv.org/abs/1404.0646.

[73] J. Rousseau and C. Scricciolo, Wasserstein convergence in Bayesian and frequentist deconvolution models, arXiv:2309.15300v1, 26 September 2023. https://arxiv.org/abs/2309.15300.

[74] C. Scricciolo, Adaptive minimax multivariate L^1-deconvolution: harmonic-mean rates under noise filtering, and a 1-Wasserstein equivalence, arXiv:2610.11760v1, 8 October 2026. https://arxiv.org/abs/2610.11760.

[75] L. Bondesson, *Generalized Gamma Convolutions and Related Classes of Distributions and Densities*, Lecture Notes in Statistics 76, Springer, New York, 1992, especially Chapter 5, https://doi.org/10.1007/978-1-4612-2948-3.

[76] KEIO, *Transport, Spectral Positivity, and Sharp Observation Boundaries: Near-linear Mixture Likelihood, Full McKay Domains, and Critical Sampling Budgets*, research manuscript, 14 October 2026. Reproduced as Part IV of this compilation. AI-assisted and not externally peer reviewed. Cited locator: original Section 4.

[77] P. Bosch, *HCM property and the half-Cauchy distribution*, Probab. Math. Statist. **35** (2015), 191–200, https://arxiv.org/abs/1402.1059.

[78] T. Sjödin, *The Power Problem for Generalized Gamma Convolutions (GGC) and Related Questions*, arXiv:2601.04038 (2026), https://arxiv.org/abs/2601.04038.

[79] M. de Jeu, Determinate multidimensional measures, the extended Carleman theorem and quasi-analytic weights, *Annals of Probability* **31** (2003), 1205–1227. doi:10.1214/aop/1055425776. arXiv:math/0111019.

[80] Y. Wu and P. Yang, Optimal estimation of Gaussian mixtures via denoised method of moments, *Annals of Statistics* **48** (2020), 1981–2007. https://doi.org/10.1214/19-AOS1873; https://arxiv.org/abs/1807.07237.

[81] E. Fricain and P. Lefèvre, *L^2-Müntz spaces as model spaces*, author manuscript, 22 September 2017. https://pro.univ-lille.fr/fileadmin/user_upload/pages_pros/emmanuel_fricain/Muntz-model.pdf.

[82] L. Gaillard and P. Lefèvre, *Lacunary Müntz spaces: isomorphisms and Carleson embeddings*, Annales de l'Institut Fourier **68** (2018), no. 5, 2215–2251. https://doi.org/10.5802/aif.3207.

[83] P. Maréchal, F. Triki, and W. C. Simo Tao Lee, *Regularization of the inverse Laplace transform by Mollification*, arXiv:2304.08057, 2023. https://arxiv.org/abs/2304.08057.

[84] KEIO, *Adaptive Analytic Information and Sharp Recovery Boundaries*, research manuscript, 14 October 2026. Original edition: 52 pp. Reproduced as Part V of this compilation. AI-assisted and not externally peer reviewed. Cited locators: original Theorems 6.2 and 6.4.

[85] Y. E. Tuzcu, *Analysis of the prime zeta function zeros placement patterns*, Master's thesis, Vilnius University, 2026, p. 16. University-hosted full text.

[86] E. C. Titchmarsh, revised by D. R. Heath-Brown, *The Theory of the Riemann Zeta-Function*, second edition, Clarendon Press, Oxford, 1986, Chapter VII.

[87] S. M. Gonek, *Three lectures on the Riemann zeta-function*, arXiv:math/0401126, 2004, Lecture II, Sections 3–4. arXiv:math/0401126.

[88] D. Krieg, E. Novak, L. Plaskota and M. Ullrich, *Noisy nonlinear information and entropy numbers*, J. Fourier Anal. Appl. **32** (2026), article 36, doi:10.1007/s00041-026-10249-z. https://arxiv.org/abs/2510.23213.

[89] H. Queffélec and R. Zarouf, *On Bernstein's inequality for polynomials*, Analysis and Mathematical Physics (2019); arXiv:1903.10801. https://arxiv.org/abs/1903.10801.

[90] A. Eftekhari, J. Tanner, A. Thompson, B. Toader, and H. Tyagi, *Sparse non-negative super-resolution—simplified and stabilised*, arXiv:1804.01490, version 2 (2019). https://arxiv.org/abs/1804.01490.

[91] KEIO, *Boundary Laws for Observation and Recovery: Prime-Zeta Counting, Global Likelihood Optimality, and Sharp Inverse Thresholds*, research manuscript, 15 October 2026. Original edition: 48 pp. Reproduced as Part VI of this compilation. AI-assisted and not externally peer reviewed. Part VII cites the sections on the complete binomial-mixture likelihood transition, blind spectral certification from nonlinear transition thresholds, Bessel distributions, and analytic populations.

[92] H. Dette, New bounds for Hahn and Krawtchouk polynomials, *SIAM Journal on Mathematical Analysis* **26** (1995), 1647–1659. doi:10.1137/S0036141093255661. Preprint arXiv:math/9406223, Corollary 3.6(b). https://arxiv.org/abs/math/9406223.

[93] M. Chen, C. Gao and Z. Ren, A general decision theory for Huber's ε-contamination model, *Electronic Journal of Statistics* **10** (2016), 3752–3774. doi:10.1214/16-EJS1216. https://arxiv.org/abs/1511.04144.

[94] B. Bagchi, *The statistical behaviour and universality properties of the Riemann zeta-function and other allied Dirichlet series*, Ph.D. thesis, Indian Statistical Institute, Calcutta, 1981. Institutional repository record.

[95] A. Laurinčikas, *Limit theorems for the Matsumoto zeta-function*, Journal de Théorie des Nombres de Bordeaux **8** (1996), no. 1, 143–159. Journal-hosted full text.

[96] M. Coste, *An Introduction to O-Minimal Geometry*, Dottorato di Ricerca in Matematica, Dipartimento di Matematica, Università di Pisa, Istituti Editoriali e Poligrafici Internazionali, Pisa, 2000 (lecture notes dated 1999).

[97] A. J. Wilkie, *O-minimal structures*, Astérisque **326** (2009), Séminaire Bourbaki, Exp. No. 985, 131–142. https://www.numdam.org/article/AST_2009__326__131_0.pdf.

[98] J. Kwon, *A universal bound on the variations of bounded convex functions*, J. Convex Anal. **24** (2017), no. 1, 67–73; arXiv:1401.2104.

[99] J. P. Nolan, *Stable Distributions: Models for Heavy Tailed Data*, Chapter 1, author manuscript processed 30 January 2018, Definition 1.8 and equation (1.6). Author's chapter PDF.

[100] P. Liu and H. Ammari, *A mathematical theory of super-resolution and two-point resolution*, Forum of Mathematics, Sigma **12** (2024), e83. doi:10.1017/fms.2024.72.

[101] S. M. Gonek and H. L. Montgomery, *Kronecker's approximation theorem*, Indagationes Mathematicae **27** (2016), 506–523. doi:10.1016/j.indag.2016.02.002.

[102] A. Balitskiy and A. Berdnikov, *Waist of maps measured via Urysohn width*, arXiv:2009.04558v2, July 25, 2021; Theorems 2.1 and 3.16. https://arxiv.org/abs/2009.04558.

[103] J. Milnor, *On the Betti numbers of real varieties*, Proc. Amer. Math. Soc. **15** (1964), 275–280, doi:10.1090/S0002-9939-1964-0161339-9.

[104] M. Shiota, *Piecewise linearization of real-valued subanalytic functions*, Trans. Amer. Math. Soc. **312** (1989), no. 2, 663–679, doi:10.1090/S0002-9947-1989-0958902-3.

[105] KEIO, *Exact Recovery Laws and a Coefficient Conjecture*, research manuscript, 16 October 2026. Reproduced as Part VII of this compilation. AI-assisted and not externally peer reviewed. Cited sections concern the constant-one Chebyshev–Bernstein bound, two-scale inversion and robust binomial population recovery, and the exact Euclidean convex-noise floor.

[106] S.-O. Chan, I. Diakonikolas, R. A. Servedio and X. Sun, Efficient density estimation via piecewise polynomial approximation, *Proceedings of the 46th Annual ACM Symposium on Theory of Computing* (2014), 604–613. doi:10.1145/2591796.2591848. https://arxiv.org/abs/1305.3207.

[107] K. Tian, W. Kong and G. Valiant, Learning populations of parameters, *Advances in Neural Information Processing Systems* **30** (2017). https://arxiv.org/abs/1709.02707.

[108] M. Sodin, *Zeros of Gaussian analytic functions*, Mathematical Research Letters **7** (2000), 371–381. arXiv:math/0007030.

[109] S. Favorov, *Generalized Fourier quasicrystals, almost periodic sets, and zeros of Dirichlet series*, arXiv:2311.02728 (2023).

[110] J. M. Almira, Müntz type theorems I, *Surveys in Approximation Theory* **3** (2007), 152–194. arXiv:0710.3570.

[111] D. Batenkov, G. Goldman, and Y. Yomdin, *Super-resolution of near-colliding point sources*, Information and Inference **10** (2021), 515–572. arXiv:1904.09186.

[112] R. Katz, N. Diab, and D. Batenkov, *On the accuracy of Prony's method for recovery of exponential sums with closely spaced exponents*, arXiv:2302.05883v2 (2024). arXiv:2302.05883.

[113] H. S. Dumas and S. Fischler, *Filling times for linear flow on the torus with truncated Diophantine conditions: a brief review and new proof*, arXiv:2112.11786 (2021). arXiv:2112.11786.

[114] D. Krieg and P. Kritzer, *Homogeneous algorithms and solvable problems on cones*, arXiv:2311.15767v2, March 3, 2024, Propositions 1–2 and Open Problem 1. https://arxiv.org/abs/2311.15767.

[115] KEIO, *Geometry and Rigidity of Inverse Observation*, research manuscript, 16 October 2026. Original edition: 39 pp. Reproduced as Part VIII of this compilation. AI-assisted and not externally peer reviewed. Part IX cites original Section 1 on logarithmic lower bounds for unrestricted adaptive continuous information, and the sections on polynomial-time population recovery under unknown adversarial contamination, spectral observation, and endpoint-rate gamma extraction.

[116] A. Gournay, *Width of l^p balls*, arXiv:0711.3081v2 (2008), Theorem 1.4. https://arxiv.org/abs/0711.3081.

[117] Z. Liu and P.-L. Loh, Robust W-GAN-based estimation under Wasserstein contamination, *Information and Inference* **12** (2023), 312–362. doi:10.1093/imaiai/iaac020. https://arxiv.org/abs/2101.07969.

[118] P. Chao and E. Dobriban, *Minimax Statistical Estimation under Wasserstein Contamination*, arXiv:2308.01853v3 (2025). https://arxiv.org/abs/2308.01853.

[119] C. E. Avellar and J. K. Hale, *On the zeros of exponential polynomials*, Journal of Mathematical Analysis and Applications **73** (1980), no. 2, 434–452. doi:10.1016/0022-247X(80)90289-9.

[120] J. Lundqvist, *An explicit calculation of the Ronkin function*, Annales de la Faculté des Sciences de Toulouse, Série 6, **24** (2015), no. 2, 227–250. Journal-hosted article.

[121] N. Makarov and A. Poltoratski, *Beurling–Malliavin theory for Toeplitz kernels*, Invent. Math. **180** (2010), 443–480; https://arxiv.org/abs/math/0702497. Part IX specifically cites Section 1.1 of the 2007 preprint.

[122] J. A. Cuesta-Albertos, R. Fraiman, and T. Ransford, A sharp form of the Cramér–Wold theorem, *Journal of Theoretical Probability* **20** (2007), 201–209. doi:10.1007/s10959-007-0060-7.

[123] G. Blekherman, M. Kummer, C. Riener, M. Schweighofer, and C. Vinzant, *Generalized eigenvalue methods for Gaussian quadrature rules*, Annales Henri Lebesgue **3** (2020), 1327–1341. doi:10.5802/ahl.62.

[124] D. Armentano, J.-M. Azaïs, and J. R. León, *On a general Kac–Rice formula for the measure of a level set*, arXiv:2304.07424v4 (2025). arXiv:2304.07424.

[125] KEIO, *Sharp Boundaries for Adaptive Recovery and Mixture Identification*, research manuscript, 16 October 2026. Original edition: 41 pp. Reproduced as Part IX of this compilation. AI-assisted and not externally peer reviewed.

[126] M. Tsukamoto, *Macroscopic dimension of the ℓ^p-ball with respect to the ℓ^q-norm*, J. Math. Kyoto Univ. 48 (2008), 445–454; arXiv:0801.4548. https://arxiv.org/abs/0801.4548.

[127] K. Bujok, B. M. Hambly and C. Reisinger, Multilevel simulation of functionals of Bernoulli random variables with application to basket credit derivatives, *Methodology and Computing in Applied Probability* **17** (2015), 579–604. doi:10.1007/s11009-013-9380-5. https://arxiv.org/abs/1211.0707.

[128] Y. Luo and C. Gao, Adaptive robust confidence intervals, *Annals of Statistics* **54** (2026), 1128–1152. doi:10.1214/25-AOS2574. https://arxiv.org/abs/2410.22647.

[129] M. Cho, Y. Luo and C. Gao, *Robust confidence intervals for a binomial proportion: local optimality and adaptivity*, arXiv:2509.05568v2 (2026). https://arxiv.org/abs/2509.05568.

[130] A. Fornasiero and G. Terzo, *A note on exponential polynomials*, Ricerche di Matematica **74** (2025), 301–308; published online 4 July 2024. doi:10.1007/s11587-024-00871-8.

[131] S. Bonhomme, K. Jochmans, and J.-M. Robin, *Non-parametric estimation of finite mixtures from repeated measurements*, J. R. Stat. Soc. Ser. B **78** (2016), 211–229, https://doi.org/10.1111/rssb.12110.

[132] T. Simon, *Comparing Fréchet and positive stable laws*, Electron. J. Probab. **19** (2014), paper 16, 1–25, https://doi.org/10.1214/EJP.v19-3058; https://arxiv.org/html/1310.1888v2.

[133] M. R. Powers and J. Xu, *A criterion for extending continuous-mixture identifiability results*, J. Stat. Theory Appl. **24** (2025), 515–533, https://doi.org/10.1007/s44199-025-00124-z.

[134] Y. Han, J. Jiao, and T. Weissman, *Local moment matching: A unified methodology for symmetric functional estimation and distribution estimation under Wasserstein distance*, Proceedings of the 31st Conference on Learning Theory, PMLR **75** (2018), 3189–3221. Publisher's article.

[135] J. Beck and W. W. L. Chen, *Generalization of a density theorem of Khinchin and diophantine approximation*, Journal de théorie des nombres de Bordeaux **35** (2023), no. 2, 511–542. doi:10.5802/jtnb.1255.